



COST & MANAGEMENT ACCOUNTING

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COST & MANAGEMENT ACCOUNTING

C U R R I C U L U M

COST & MANAGEMENT ACCOUNTING

MANAGEMENT ACCOUNTING FUNDAMENTALS: An Introduction, Objectives of Management Accounting, Cost Accounting, Best Practices in Costing System, Practical Applications of Financial, Cost and Management Accounting, Management Accounting, Essential Features of Management Accounting, Management Accounting Information and Their Use, Practical Implications of Management Accountancy, Cost Management for Competitive Advantage, Decision-Making Process in a Firm, Planning/Directing/Controlling, Some Innovative Management Accounting Practices, Management Accounting as Profession, Limitations of Management Accounting

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MANAGEMENT ACCOUNTING FUNDAMENTALS

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INTRODUCTORY CASELET

Cost Accounting Transformation at Toyota

Toyota, one of the largest automobile manufacturers in the world, is renowned for its efficiency and lean manufacturing principles. However, in the early 2000s, Toyota faced rising production costs, particularly due to increasing global competition and raw material price fluctuations. To maintain its competitive edge, Toyota decided to overhaul its cost accounting practices, focusing on reducing production costs while maintaining high-quality standards. The company implemented advanced cost accounting methods to monitor costs across its vast production network.

Toyota adopted Activity-Based Costing (ABC), which allowed it to identify and assign overhead costs more accurately to different production activities. This shift was crucial in pinpointing areas of inefficiency across the production line. Additionally, the company integrated Target Costing, a practice where the target price of a product was determined based on market conditions, and production costs were adjusted to ensure profitability. This method was particularly useful as Toyota aimed to produce vehicles that were both affordable and profitable in emerging markets like India and Brazil.

With the integration of advanced cost accounting systems, Toyota was able to reduce production costs by optimizing resource allocation and eliminating waste. The detailed cost information provided by ABC enabled Toyota to streamline operations, while Target Costing helped maintain competitive pricing. As a result, Toyota not only improved its profit margins but also maintained its position as a global leader in quality and cost-efficiency, demonstrating the value of precise cost management in sustaining competitive advantage.

**QUESTIONS**

1. In what ways could an organization benefit from choosing the right costing method, and how might this decision impact both short-term profitability and long-term sustainability?

(Hint: Consider how different costing methods align with the nature of a business's operations and contribute to accurate cost measurement, which aids in decision-making and competitive positioning.)

2. How did Toyota utilize Activity-Based Costing (ABC) to improve its cost management practices, and what specific benefits did the company gain from this approach?

(Hint: Focus on how ABC helped Toyota identify and allocate overhead costs more accurately and the resulting impact on resource allocation and cost reduction.)

**LEARNING OBJECTIVES**

After reading this unit, you will be able to:

- Understand the primary goals of management accounting and its role in enhancing decision-making processes within an organization.
- Identify and analyze best practices in costing systems that lead to improved accuracy and efficiency in cost management.
- Examine the key concepts and frameworks of management accounting
- Analyze the types of information generated by management accounting and how they are utilized for strategic planning and operational efficiency.
- Understand the significance of cost management strategies in achieving competitive advantage
- Explore the stages of the decision-making process within a firm, focusing on how management accounting contributes to each phase.
- Explore the evolving role of management accounting as a profession.
- Identify the limitations and challenges of management accounting practices

1.1 AN INTRODUCTION

Accounting is often described as the “language” that businesses use to communicate with stakeholders. In 2014, Warren Buffet advised a 17-year-old intern at an investment firm, stating that accounting is the “language of business.” Like any language, it must be learned before it can be fully understood. This analogy emphasizes that accounting allows businesses to convey financial information to users in a structured, universally understood format.

Example: Consider Radha, who wants to invest in a company. She reviews the company’s financial statements, which are written in the “language” of accounting. By analyzing these statements, Radha can make an informed decision on whether to invest, as the reports clearly communicate the company’s profitability, assets, liabilities, and overall financial health.

Branches of Accounting: Accounting (**Figure 1.1**) is broadly divided into three main branches:

1. Financial Accounting
2. Cost Accounting
3. Management Accounting

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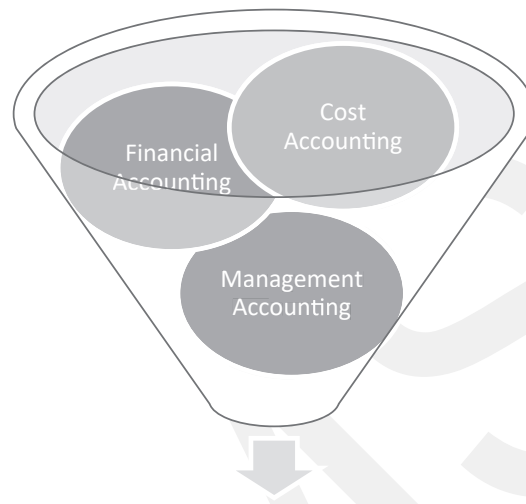


Figure 1.1 Branches of Accounting.

1. **Financial Accounting:** This branch focuses on preparing and presenting financial statements, such as the income statement, balance sheet, and cash flow statement, which are used primarily by external stakeholders.

Example: Radha's investment firm prepares financial statements at the end of each financial year. These statements are shared with investors, like Radha, to provide a clear picture of the firm's financial position and performance.

2. **Cost Accounting:** Cost accounting deals with calculating the cost of production and is crucial for pricing decisions, controlling expenses, and managing inventory.

Example: Shyam's store introduces a new product line. To determine the selling price, Shyam calculates the total production cost, which includes raw materials, labor, and overheads. He then adds a profit margin to set a competitive selling price.

3. **Management Accounting:** Management accounting provides internal reports and information to help management make decisions regarding planning, controlling, and evaluating business operations.

Example: Radha's investment firm uses management accounting to create monthly performance reports. These reports help the firm's managers assess which investment strategies are successful and which require adjustments.

Core Concepts of Management Accounting Management accounting combines two essential elements: 'management' and 'accounting'. While management focuses on running the organization in alignment with strategic goals, accounting evaluates and reports the organization's performance. Management accounting helps in:

- ❑ **Stewardship:** Monitoring the use of company resources.
- ❑ **Control:** Ensuring operations are aligned with objectives.
- ❑ **Auditing:** Reviewing processes to confirm compliance and accuracy.

Example: Shyam's store uses management accounting to set daily sales targets for employees. By tracking sales performance each day, Shyam can identify trends and take corrective actions if targets are not met.

1.1.1 OBJECTIVES OF MANAGEMENT ACCOUNTING

Running a business isn't just about selling products — it's about making informed decisions every step of the way. That's where management accounting comes in. It provides crucial insights that guide planning, control, and decision-making. Unlike traditional financial accounting, which focuses on the past, management accounting looks forward, helping businesses steer in the right direction.

Management accounting isn't static. It evolves with changing business needs. What was relevant a decade ago may not be today, and new challenges demand new solutions. But no matter how it changes, the core objectives of management accounting remain the same. Let's explore them in detail:

1. **Budgeting, Planning, and Forecasting:** Before launching a new product or service, every business needs a plan. This is where management accounting shines. It helps businesses create budgets, plan for future growth, and predict upcoming revenue and expenses.

Example: Imagine Timber Craft Furniture is planning to launch a new line of eco-friendly wooden chairs. Using management accounting, the company forecasts demand, estimates raw material costs, and calculates expected sales revenue. This helps Timber Craft Furniture determine how much to spend on marketing, raw materials, and production, ensuring that every rupee is allocated wisely.

2. **Performance Evaluation:** It's one thing to have a plan — but are you actually following it? Management accounting tracks performance, not just for the entire organization, but also for departments, teams, and individual processes.

Example: The production team at Timber Craft Furniture was tasked with producing 500 wooden chairs a week, but recent reports show they are only making 450 chairs per week. Management accounting helps identify the cause — perhaps it's machine downtime, raw material delays, or a lack of trained workers. This insight allows the company to take corrective action and boost production.

3. **Result Comparisons:** How do you know if your company is performing well? By comparing it to something! Management accounting allows companies to compare results with industry benchmarks, past performance, or the performance of different branches or departments.

Example: Timber Craft Furniture has two production units, one in Delhi and one in Mumbai. Management accounting reveals that the Delhi unit produces chairs 10% faster than the Mumbai unit. This prompts management to investigate. It turns out the Delhi unit has newer equipment, which leads the company to consider upgrading Mumbai's machinery as well.

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- 4. Enhancing Efficiency and Effectiveness:** Every company wants to do more with less. Management accounting identifies inefficiencies and helps businesses streamline processes to get better results.

Example: Timber Craft Furniture notices that workers are spending too much time carrying raw materials from the storage room to the assembly line. Management accounting reveals that by moving raw material storage closer to the production floor, they can save 20 minutes per worker each shift. This small change results in a major increase in productivity.

- 5. Capital Investment Evaluation:** Big investments, like buying new equipment, require careful analysis. Management accounting helps businesses determine if an investment is worthwhile by using techniques like payback period, NPV (Net Present Value), and IRR (Internal Rate of Return).

Example: Timber Craft Furniture is considering purchasing an automated wood-cutting machine that costs ₹50 lakh but promises to reduce production time by 30%. Management accounting evaluates the payback period—how long it will take for the cost savings to cover the initial investment. If the machine can pay for itself in 3 years, it might be a smart move. But if the payback period is too long, they may reconsider.

- 6. Product Mix and Pricing:** Not every product has the same profit margin. Management accounting helps identify which products are most profitable and how to price them.

Example: Timber Craft Furniture sells wooden chairs, coffee tables, and bookshelves. A review of product profitability reveals that wooden chairs bring in a 30% profit margin, while bookshelves only generate 10%. Based on this, Timber Craft Furniture decides to prioritize wooden chair production and cut back on bookshelf production.

- 7. Outsourcing Considerations:** Should you make it yourself or buy it from someone else? Management accounting helps decide whether to produce something in-house or outsource it to an external supplier.

Example: Timber Craft Furniture currently produces the cushions for its chairs in-house. But after a cost-benefit analysis, management accounting reveals that buying ready-made cushions from a supplier would cost 15% less than making them. Based on this insight, the company outsources cushion production, freeing up its workers to focus on assembling the chairs.

- 8. Resource Allocation:** When resources are limited, it's crucial to use them wisely. Management accounting guides decisions on where to allocate money, time, and labor.

Example: Timber Craft Furniture has ₹10 lakh to invest. Should they spend it on new marketing campaigns or upgrading factory equipment? Management accounting calculates the potential returns of both options. If upgrading the factory leads to higher production (and profits), it makes sense to prioritize equipment upgrades.

- 9. Strategic Decision Support:** Big decisions, like launching new products or entering new markets, require clear, data-backed insights. Management accounting helps by providing relevant data and projections.

Example: Timber Craft Furniture is thinking of launching its wooden chairs in international markets. Before making a decision, management accounting analyzes factors like export taxes, foreign exchange rates, and shipping costs. Based on this analysis, they discover that entering the international market could increase profits by 25%, but it also exposes them to risks like currency fluctuations. Armed with this information, the company can make a more informed choice.

- 10. Profitability Assessment:** Not all products, services, or divisions are equally profitable. Management accounting shows where the profits are coming from and which areas need improvement.

Example: Timber Craft Furniture produces wooden chairs, coffee tables, and stools. Management accounting reveals that wooden chairs are the most profitable, while stools barely break even. As a result, the company decides to focus marketing efforts on wooden chairs and consider discontinuing the stools.

- 11. Cost Allocation and Transfer Pricing:** When one division of a company “sells” a product or service to another division, how do you price it? Management accounting establishes internal transfer prices to ensure fair and transparent cost-sharing.

Example: Timber Craft Furniture’s R&D department develops a new technique for making eco-friendly wooden finishes. This technique is then “sold” to the production department for implementation. Management accounting sets a transfer price to reflect the value of the R&D department’s efforts, ensuring that each department’s performance is measured accurately.



SELF-ASSESSMENT QUESTIONS

1. What is the primary purpose of cost accounting within the management accounting framework?
 - a. To create financial statements for shareholders.
 - b. To control and reduce production costs through budgetary control and variance analysis.
 - c. To forecast future market trends for the organization.
 - d. To provide tax-saving strategies for the organization.
2. What is one of the primary objectives of management accounting?
 - a. Preparing tax returns for the organization.
 - b. Assessing the profitability of external competitors.
 - c. Assisting in budgeting, planning, and forecasting to align with strategic objectives.
 - d. Determining the market value of the company’s stock.

1.2 COST ACCOUNTING

Cost accounting is a specialized branch of accounting that focuses on the classification, accumulation, assignment, and control of costs. According to the Chartered Institute of Management Accountants (CIMA), it involves

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establishing budgets, determining standard and actual costs for operations, processes, activities, or products, and analyzing variances, profitability, and the social use of funds. Cost accounting is defined as the classification, recording, and appropriate allocation of expenditure to determine the costs of products or services, as well as presenting data in a structured manner for management control and guidance.

Cost accounting provides the basis and information necessary for cost ascertainment, while costing refers to the arithmetical process of determining costs, using the data provided by cost accounting. Originally, cost accounting focused on determining product costs. However, with the advent of large-scale production, its scope has expanded to include providing information for cost control and cost reduction. These functions are essential for planning, decision-making, and management control. Cost accounting is no longer limited to manufacturing activities; it is now applied across various sectors, including service organizations, government bodies, agricultural operations, and extractive industries.

One of the primary roles of cost accounting is to guide the ascertainment of production costs. Additionally, it helps management identify both profitable and unprofitable activities, enabling them to focus on eliminating inefficient operations. Cost accounting provides crucial data for preparing estimates and tenders, and it highlights losses that occur in the form of idle time, spoilage, or scrap. It also supports inventory control through a perpetual inventory system, contributing to the preparation of interim financial statements. By using tools like budgetary control and standard costing, cost accounting aids in managing production costs and informs future production policies. Moreover, it reveals the relative efficiency of workers, facilitating effective wage determination.

Decision-Making Scenario: Cost Management at Shyam Ltd.

As the Cost Manager at Shyam Ltd., a company specializing in wooden furniture, your role is to classify, assign, and control costs for producing a custom-designed wooden chair. This exercise will guide you through a structured decision-making process.

Step 1: Classifying Costs

To begin, let's categorize the costs associated with producing the wooden chair:

Cost Category	Examples
Direct Material Costs	Wood, screws, varnish, paint, etc.
Direct Labor Costs	Wages for carpenters and machine operators
Overhead Costs	Factory rent, electricity, equipment depreciation, and supervisor salaries

Question for You

Which of the following would you classify as a **direct cost** for the wooden chair?

- ❑ (A) Wages of machine operators
- ❑ (B) Factory rent
- ❑ (C) Varnish used to finish the chair

Answer: (A) and (C)

Step 2: Accumulating and Recording Costs

Assume Shyam Ltd. plans to produce 100 chairs in a batch. The cost breakdown for the batch is as follows:

Cost Component	Amount (₹)
Direct Materials	₹50,000 (wood) + ₹10,000 (varnish) + ₹5,000 (screws) = ₹65,000
Direct Labor	₹40,000 (based on time sheets for carpenters and operators)
Overheads	₹30,000 (allocated factory rent, electricity, depreciation)

Total Production Cost: ₹65,000 (Materials) + ₹40,000 (Labor) + ₹30,000 (Overheads) = ₹1,35,000

Cost Per Chair: ₹1,35,000 ÷ 100 = ₹1,350 per chair

Step 3: Analyzing and Controlling Costs

After production, it was found that 10 chairs were defective due to poor varnish application and required rework. The rework costs are as follows:

Cost Component	Amount per Chair (₹)
Additional Labor	₹150
Extra Varnish	₹100

Total Rework Cost Per Chair: ₹150 + ₹100 = ₹250

Total Rework Cost for 10 Chairs: ₹250 × 10 = ₹2,500

Step 4: Managerial Decision-Making

The additional rework cost of ₹2,500 has increased the total production cost. Management is now evaluating two options to minimize future defects:

Option	Details
Option 1: Staff Training	Training cost of ₹5,000 to reduce defects by 80%
Option 2: Current Process	Continue with the existing process, incurring ₹2,500 rework cost per batch

Analysis

If training reduces defects by 80%, only 2 defective chairs (20% of 10) would need rework.

- ❑ **Rework Cost for 2 Chairs:** ₹250 × 2 = ₹500 per batch
- ❑ **Annual Rework Cost (10 Batches):** ₹500 × 10 = ₹5,000

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Conclusion

- ❑ **Option 1 (Training):** One-time cost of ₹5,000 eliminates recurring high rework costs, improving efficiency long-term. Reducing the rework cost to 5,000. Total cost incurred 10,000 out of which training cost (5,000) is only one time.
- ❑ **Option 2 (Current Process):** Continues to incur a rework cost of ₹2,500 per batch, which totals ₹25,000 annually.

Recommendation

Choose Option 1: Investing in staff training minimizes defects and reduces costs long-term, making it the more sustainable solution.

1.3 BEST PRACTICES IN COSTING SYSTEM

Best practices in a costing system focus on selecting and applying suitable costing methods and techniques, tailored to the specific nature of an organization's activities. These practices are essential for accurate cost measurement, cost control, and informed decision-making. Below are some best practices in costing system:

Choosing the Right Costing Method for the Business Type: The costing method should be aligned with the business's nature and production processes. For instance:

- ❑ **Job Costing:** Job costing is a method tailored for businesses that create products or deliver services based on specific customer requirements. Examples include printing presses, interior decorators, and repair centers. Each job is considered a unique cost unit, and all related expenses—such as materials, labor, and overhead—are tracked separately. This approach provides a comprehensive breakdown of the costs associated with completing individual jobs, facilitating precise billing and profitability evaluation. It is particularly advantageous for enterprises offering varied products or services where each job differs in scope and cost structure.
- ❑ **Contract Costing:** Contract costing is most applicable to businesses involved in long-term projects, such as building roads, bridges, buildings, or dams. In this system, the contract itself is treated as the cost unit. Costs are recorded and monitored throughout the project's duration, enabling businesses to assess profitability, regulate cash flows, and oversee project-specific expenditures effectively. This method typically incorporates interim progress reports to compare actual costs incurred with the estimated budget at various stages of the project.
- ❑ **Batch Costing:** Batch costing is used when products are produced in groups or batches, commonly seen in industries like toy manufacturing, automobile production, or garment making. In this system, each batch is assigned as a distinct cost unit. Costs are calculated for the entire batch and then distributed among the individual units within the batch. This method aids businesses in achieving tighter cost control, improving production schedules, and enhancing efficiency by identifying cost-effective methods for batch production.

- ❑ **Process Costing:** Process costing is well-suited for industries engaged in continuous production processes, such as chemical manufacturing, textile production, or food processing. Costs are accumulated for each department or production stage, and the total cost is divided by the number of units produced to derive the cost per unit. This approach supports large-scale production by ensuring expenses at every stage are captured, analyzing waste, reviewing efficiency, and maintaining cost control. It provides a detailed view of the total production cost.
- ❑ **Operating Costing:** Operating costing, a variant of process costing, is applied in industries where production involves a series of distinct operations, such as transportation or utilities. By segmenting production into individual operations, this system enables precise cost allocation and better control. It is especially advantageous in service-oriented businesses, as tracking costs for each operation ensures improved financial oversight and refined pricing strategies.

Using Historical and Standard Costing for Effective Monitoring

- ❑ **Historical Costing:** Historical costing involves recording the actual expenses incurred during production or operations, including costs for materials, labor, and overheads. These real-time expenditures are tracked throughout the process and analyzed once the operations conclude. By comparing the recorded actual costs with the budgeted figures, businesses can evaluate their financial performance and identify variances. This method provides an accurate account of what was spent, making it invaluable for post-operation analysis and pinpointing areas for cost improvement. Additionally, historical costing helps in formulating realistic budgets for future projects, ensuring better financial planning and control. **For example;** a garment manufacturer tracks actual expenses for producing a batch of shirts. If the actual cost is ₹50,000 and the budgeted cost was ₹45,000, the analysis can reveal why costs were higher—such as due to material price increases or production inefficiencies.
- ❑ **Standard Costing:** Standard costing involves setting pre-determined estimates for materials, labor, and overhead based on historical data or industry benchmarks. These standard costs serve as a reference point against which actual expenses are regularly compared. Any deviations, or variances, are analyzed to identify inefficiencies and implement corrective measures. This method acts as an effective performance measurement tool, enabling businesses to control costs by highlighting unexpected deviations. By providing a clear benchmark, standard costing promotes operational efficiency and helps organizations maintain financial discipline. For instance; An electronics company sets the standard cost for producing a gadget at ₹1,000 per unit. After production, the actual cost comes to ₹1,100 per unit. A variance of ₹100 indicates a potential inefficiency, such as higher labor costs or wastage of materials.

Adopting Marginal and Absorption Costing According to Business Needs

- ❑ **Absorption Costing:** Absorption costing is a method where both fixed costs (e.g., rent, depreciation) and variable costs (e.g., raw materials,

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direct labor) are included in the total cost of a product. This approach ensures that overhead costs are evenly distributed across all products, making it particularly useful in industries requiring precise cost allocation, such as manufacturing and construction. It is also widely used for external reporting, as mandated by accounting standards.

Example: A company produces **10,000 units** of a product.

Fixed Costs: ₹5,00,000

Variable Costs: ₹20 per unit

The total cost per unit under absorption costing is calculated as:

Total Cost per Unit = (Fixed Costs/Total Units) + Variable Cost

Total Cost per Unit = (₹5,00,000/10,000) + ₹20 = ₹70 per unit.

- ❑ **Marginal Costing:** Marginal costing focuses exclusively on variable costs, treating fixed costs as period expenses. This method proves particularly useful for decision-making related to pricing, profitability, and short-term operations, as it emphasizes the contribution margin, calculated as the difference between the selling price and the variable cost. For instance, a company produces a product with a selling price of ₹150, a variable cost of ₹100 per unit, and total fixed costs amounting to ₹50,000. If the company produces and sells 1,000 units, the contribution margin per unit is ₹150 – ₹100 = ₹50. The total contribution, therefore, is ₹50 × 1,000 = ₹50,000. Since this contribution covers the fixed costs, the company breaks even. Marginal costing is commonly applied in scenarios such as analyzing pricing decisions, determining whether to continue or discontinue the production of a specific product, and conducting break-even or profitability analyses.

Maintaining Uniform Costing Across Similar Units or Branches

In multi-branch or multi-unit organizations, implementing Uniform Costing promotes consistency in costing practices. This uniformity enables easier performance comparison across units and supports standardized cost reporting.

Automating and Integrating Costing Systems

Use technology to simplify costing processes. Integrating costing systems with other financial systems, such as inventory management, general ledger, and ERP systems, ensures accurate and seamless cost data flow throughout the organization. Automation helps minimize errors and improves the accuracy of data for timely decision-making.

Example: Radha Ltd. has decided to implement an ERP system to integrate cost accounting with inventory management and financial reporting. This system will automatically track inventory levels, production costs, and ledger entries.

Question for You

What are the benefits of integrating cost accounting with an ERP system?

- A. Reduces manual errors
- B. Provides real-time cost tracking

- C. Simplifies financial reporting
- D. All of the above

Answer

The correct answer is D (All of the above). Integration allows for real-time cost tracking, fewer manual errors, and automated reporting, which helps managers make timely and informed decisions.

Regularly Updating Cost Structures

Periodically review cost structures to reflect changes in production processes, materials, or market conditions. As costs change, ensure both direct and indirect costs are accurately updated to provide a realistic financial picture.

Example: Radha Ltd. notices a 30% rise in raw material costs for ceramic powder, which is a key input for production.

Question for You

What action should Radha Ltd. take in its costing system?

- A. Continue using old material costs
- B. Update direct material costs for ceramic powder
- C. Wait for market conditions to stabilize

Answer

The correct answer is B (Update direct material costs). Cost structures must be updated to reflect current prices to ensure the most accurate view of production costs. This helps in setting appropriate prices and identifying profitability.

Leveraging Costing Data for Management Decision-Making

Use the data from the costing system to make important management decisions. This data can help with pricing products, evaluating product performance, or improving operations. For example:

- ❑ **Marginal Costing:** Helps in deciding product prices by focusing on costs that change with production. For instance, the cost to make one extra cake is ₹150. Based on this, the bakery decides to price the cake at ₹200 to cover costs and make a profit.
- ❑ **Standard Costing:** Pinpoints areas where efficiency can be improved by comparing actual costs to expected costs. For example, the expected cost of ingredients for one cake is ₹120, but the actual cost turns out to be ₹140. This difference highlights inefficiencies, such as wastage or price increases, prompting the bakery to investigate and improve.

Establishing Clear Variance Analysis Guidelines

Regularly analyze variances between actual and standard costs. This practice not only highlights inefficiencies but also strengthens cost control by enabling management to take corrective action when necessary.

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Example: At the end of the quarter, Branch A reports the following:

- Standard cost per tile = ₹200
- Actual cost per tile = ₹230

Question for You

What is the variance per tile, and is it a favorable or unfavorable variance?

- A. ₹30 (Favorable)
- B. ₹30 (Unfavorable)
- C. ₹230 (Unfavorable)

Answer

The correct answer is B (₹30 Unfavorable). The actual cost (₹230) exceeds the standard cost (₹200), leading to an unfavorable variance of ₹30 per tile

1.3.1 PRACTICAL APPLICATIONS OF FINANCIAL, COST AND MANAGEMENT ACCOUNTING

Cost Accounting emerged to overcome the shortcomings of Financial Accounting in terms of internal reporting and management control. Each branch of accounting plays a unique and essential role within an organization. Below are the practical uses of Financial Accounting, Cost Accounting, and Management Accounting:

Practical Applications of Financial Accounting

- **External Reporting:** Financial accounting provides an accurate picture of a company's overall financial performance over a specified period, helping external stakeholders such as shareholders, creditors, and regulatory bodies. Through reports like profit and loss statements and balance sheets, businesses can evaluate their financial health. For example, **UrbanBites**, a growing restaurant chain, would prepare its **annual financial statements**, including profit and loss reports and balance sheets, to share with stakeholders. These reports help external parties assess the financial health of the company, influencing investment decisions or lending terms.
- **Asset Control:** Financial accounting ensures effective monitoring and control of company assets, helping management protect resources from misuse or misappropriation. By analyzing historical financial data, businesses can better understand long-term trends and improve planning. For example, **UrbanBites** could track the depreciation of kitchen equipment and assess the value of restaurant properties. This financial information helps in strategic decisions about upgrading equipment or expanding the business.
- **Regulatory Compliance:** Financial accounting helps organizations meet legal requirements under statutes like the Companies Act and Income Tax Act. It ensures transparency with regulatory authorities, helping businesses avoid fines or penalties. For **UrbanBites**, staying compliant with the **Income Tax Act** and **Food Safety Standards Act**

is crucial. Accurate financial accounting ensures that the company submits tax filings on time and meets food safety standards, helping avoid penalties.

- ❑ **Profitability Analysis:** Financial accounting offers a comprehensive overview of the company's profitability. This allows management to evaluate business success and make strategic decisions about growth, downsizing, or sustainability. For instance, **UrbanBites** could use financial reports to determine the profitability of each restaurant location or menu item. By assessing the data, the management can decide whether to continue investing in certain menu offerings or locations that are underperforming.

Practical Applications of Cost Accounting

- ❑ **Cost Control and Reduction:** Cost accounting enables management to monitor production costs, identify inefficiencies, and take steps to reduce expenses. This contributes to enhanced profitability by ensuring that resources are allocated and used efficiently. For **UrbanBites**, cost accounting could track the expenses related to food ingredients, labor, and overhead for each restaurant. Identifying cost-cutting opportunities, such as sourcing cheaper ingredients without compromising quality, would enhance profitability.
- ❑ **Product and Service Pricing:** By calculating the costs of individual products, jobs, or services, cost accounting allows businesses to set accurate prices, ensuring competitiveness and profitability in the marketplace. **UrbanBites** could use cost accounting to determine the cost of producing individual menu items, factoring in ingredients, labor, and overhead costs. This data helps set menu prices that ensure the restaurant remains competitive while maintaining healthy profit margins.
- ❑ **Internal Decision-Making:** Cost accounting provides detailed insights into the costs of various units or departments, empowering management to make informed decisions regarding product lines, resource allocation, and operational strategies. For **UrbanBites**, this could mean assessing which restaurant locations are more profitable and deciding whether to expand in certain cities or improve underperforming locations.
- ❑ **Budgeting and Forecasting:** By reviewing historical cost data, management can project future expenses and revenues more accurately, enabling more precise budgeting and helping to anticipate potential cost overruns. **UrbanBites** could use historical cost data to predict future food costs, labor expenses, and seasonal revenue trends. This allows for more accurate budgeting and planning for upcoming quarters.
- ❑ **Operational Efficiency:** Cost accounting highlights the relative efficiency of operations, processes, and machinery, assisting management in optimizing production and improving overall operational performance. **UrbanBites** could use this data to optimize its food preparation processes, reduce wait times, or improve staffing schedules during peak hours, thereby improving overall operational efficiency.

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Practical Applications of Management Accounting

- ❑ **Internal Reporting for Decision-Making:** Management accounting supplies crucial information for internal decision-making. It supports management in assessing operational costs, tracking performance, and making data-driven decisions to enhance efficiency. For **Urban Bites**, such reports could include daily sales data, inventory levels, and labor costs. This real-time information supports decision-making about menu changes, staffing adjustments, or promotional strategies.
- ❑ **Performance Evaluation:** Management accounting helps managers evaluate the performance of various divisions, departments, and products. It provides insights into operational efficiency and profitability, allowing management to focus on areas that require improvement or investment. For instance, **UrbanBites** could use management accounting to analyze the performance of a newly introduced dish across all restaurants, helping decide whether to expand the menu offering or remove it.
- ❑ **Resource Allocation:** Management accounting aids in resource allocation decisions, both financial and non-financial, across the organization. This is key to optimizing investments, determining the best product mix, and deciding on market entry or exit strategies. **UrbanBites** could use this to decide how to allocate marketing budgets across various locations or promotional campaigns to maximize their impact on sales and profitability.

Strategic Planning and Forecasting: Management accounting plays a vital role in long-term planning by generating forecasts, conducting scenario analysis, and preparing budgets to help the organization meet its strategic objectives. **UrbanBites** could use management accounting to forecast future trends in the restaurant industry, such as customer preferences shifting toward plant-based foods, and adjust its menu offerings and marketing strategies accordingly.

- ❑ **Cost-Benefit Analysis for Investment:** Management accounting supports capital investment decisions by evaluating the costs and expected returns of different options. This ensures that businesses choose projects that offer the most value. For **UrbanBites**, management may need to evaluate the costs and benefits of opening a new restaurant in a new city. By performing a cost-benefit analysis, they can assess whether the investment will yield a good return.
- ❑ **Risk Management and Predictive Analysis:** Management accounting utilizes tools like sensitivity analysis and risk assessment to help management anticipate and prepare for future risks and opportunities, strengthening decision-making processes. **UrbanBites** could use predictive analytics to assess the impact of economic downturns or changes in food supply prices, enabling proactive decision-making to mitigate risks.

Flexible Reporting: Unlike financial accounting, which focuses on periodic reports (e.g., annually or quarterly), management accounting provides real-time or frequent reports to enable continuous monitoring and more agile decision-making. **UrbanBites** could use this flexibility to track daily sales, customer feedback, or inventory usage, enabling quick adjustments in operations or marketing strategies.

While Financial Accounting, Cost Accounting, and Management Accounting serve distinct roles (as shown in **Table 1.1**), they are inter-related within an organization. Financial accounting is mainly used for external reporting and regulatory compliance, cost accounting is vital for managing internal costs, and management accounting provides critical data for internal decision-making and long-term planning. Together, these types of accounting ensure that businesses operate efficiently and make well-informed financial and operational decisions.

TABLE 1.1 COMPARISON BETWEEN FINANCIAL ACCOUNTING, COST ACCOUNTING AND MANAGEMENT ACCOUNTING

Aspect	Financial Accounting	Cost Accounting	Management Accounting
Meaning	Financial accounting involves systematically recording transactions to ascertain the business's profit or loss and financial position at year-end.	Cost accounting applies costing principles, techniques, and methods to determine total production costs, unit costs, cost control, and cost reduction.	Management accounting presents financial data in a manner useful for management in functions like planning, organizing, controlling, and decision-making.
Scope	Limited to financial transactions only.	Broader than financial accounting, covering additional cost-related transactions.	Broadest scope, encompassing all monetary transactions and managerial considerations.
Qualification	No specific qualifications required, though Chartered Accountants (CAs) often handle this in large organizations.	Typically handled by Cost Accountants qualified under ICWA standards.	Performed by senior management staff with expertise in accounting and management, often designated as Management Accountants.
Period	Records transactions and provides information for one financial year, revealed at year-end.	Usually collects and presents data for one year.	Focused on future-oriented data preparation, tailored to management's needs, often covering periods shorter than a year.
Usefulness	Useful for determining the business's profit or loss and financial position at the end of the year.	Useful for analyzing production costs, unit costs, cost control, and cost reduction.	Supports management in planning, organizing, controlling, and decision-making.
Legal Position	Mandatory under provisions of the Companies Act and Income Tax Act.	Not mandatory except for specific industries governed by Section 33-B provisions.	Entirely voluntary, prepared solely for management's benefit.

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Status	Financial accountants typically have lower organizational status.	Cost accountants generally hold a higher status than financial accountants.	Management accountants enjoy the highest status within the organization.
Accuracy of Amounts	Requires absolute accuracy in figures.	Requires a reasonable level of accuracy in cost-related figures.	Relies on approximate figures; high accuracy is less critical.



SELF-ASSESSMENT QUESTIONS

3. Which costing method is most suitable for businesses producing goods based on customer-specific requirements?
 - a. Process Costing
 - b. Job Costing
 - c. Batch Costing
 - d. Absorption Costing
4. What is the primary purpose of using standard costing in a business?
 - a. To ensure regulatory compliance
 - b. To track actual costs as they occur
 - c. To serve as a benchmark for performance measurement
 - d. To determine the total profitability of the organization

Activity

You are an accountant at a company that manufactures electronic gadgets. Recently, your team implemented a new costing system to improve efficiency and accuracy in product costing. Identify two costing methods that could be applied in this scenario. For each method, explain one advantage of using it in the context of electronic gadget manufacturing.

1.4 MANAGEMENT ACCOUNTING

Think of Management Accounting as a helpful guide for management. Its primary goal is to support decision-making by using the right tools and techniques. These tools—borrowed from areas like Cost Accounting, Financial Accounting, Economics, Mathematics, and Statistics—help managers plan and make decisions to reach specific business goals.

Here's How it Works: A Management Accountant collects, analyzes, and organizes data, making sure it's relevant to the problem at hand. They present this information in a way that helps management tackle everyday challenges, make informed decisions, and align actions with the organization's objectives. It's not just about crunching numbers; it's about ensuring the data is meaningful and useful for the situation.

Management Accounting goes beyond just tracking costs (which is the focus of Cost Accounting). It combines insights from multiple areas, including Financial Accounting, Statistics, and even Psychology, to help businesses maximize profits or minimize losses. It's not just about understanding where the money is going but also about planning for the future.

1.4.1 ESSENTIAL FEATURES OF MANAGEMENT ACCOUNTING

The following features represent the key characteristics of management accounting:

- i. **Management Accounting as a Decision-Making System:** Management accounting provides accounting information that assists management in policy formulation and day-to-day activities. While the management accountant does not make decisions, they provide crucial data that supports the decision-making process, organizing and presenting a wide range of relevant facts systematically.
- ii. **Futuristic Nature of Management Accounting:** Unlike financial accounting, management accounting is forward-looking. It focuses on future planning, as decisions are made concerning future actions. Management accounting provides selective and valuable information extracted from collected data to aid in this decision-making process.
- iii. **Selective Technique in Management Accounting:** The management accountant only considers the data from financial statements that are useful for decision-making purposes. They filter and communicate information that is most relevant to management when making decisions.
- iv. **Variable Analysis:** Management accounting helps analyze the causes of variations in profit compared to previous periods. It also assesses the impact of various factors on the company's profits and profitability.
- v. **No Fixed Formats for Information:** Management accounting presents information in a format that best supports management in decision-making, allowing flexibility in how data is communicated to meet specific business needs.



SELF-ASSESSMENT QUESTIONS

5. Which of the following is NOT a feature of Management Accounting?
 - a. Futuristic nature
 - b. Variable analysis
 - c. Selective technique in information presentation
 - d. Fixed formats for information
6. How does Management Accounting differ from Financial Accounting?
 - a. Management Accounting focuses solely on past data.
 - b. Management Accounting is primarily concerned with external reporting.
 - c. Management Accounting is forward-looking and aids in future planning.
 - d. Management Accounting uses only financial data.

1.5 MANAGEMENT ACCOUNTING INFORMATION AND THEIR USE

Management accounting plays a vital role as part of a business's overall information system. Managers need to make informed decisions about

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the allocation of scarce economic resources, and for this, they require reliable economic data. The management accounting system is responsible for providing this critical information through processes of data gathering and communication.

Key features of the management accounting information system align with those of other business information systems:

- ❑ **Identifying and capturing relevant data** (in this case, economic information).
- ❑ **Recording information** systematically.
- ❑ **Analyzing and interpreting collected data.**
- ❑ **Reporting insights** in a format tailored to managerial needs.

Management accounting is often seen as a service, where managers act as “clients.” The type of information required can be grouped into four major decision-making areas:

1. **Developing Objectives and Plans:** Managers set the business’s mission and objectives and devise strategies to achieve them. Management accounting aids this process by gathering relevant information and generating financial plans, helping evaluate and select appropriate strategies.
2. **Performance Evaluation and Control:** Management accounting supports performance review against predetermined criteria, often combining financial and non-financial metrics. Controls ensure actual outcomes align with plans, prompting investigation and corrective actions where deviations occur.
3. **Allocating Resources:** Given limited resources, managers must use them efficiently. Management accounting provides insights for decisions such as optimal output levels, product mixes, and investment choices.
4. **Determining Costs and Benefits:** Managers need to assess costs and benefits for decisions like launching products or discontinuing operations. Though precise quantification may be challenging, approximate data often proves invaluable.

A growing strategic focus has underscored the importance of non-financial measures, such as quality, innovation, and delivery times, in assessing long-term success. Management accountants now integrate these non-financial metrics into their reporting alongside traditional financial KPIs.

While management accounting seeks to influence behavior positively by improving decision quality, it may sometimes have unintended effects. Employees might prioritize measurable aspects over unmeasured but critical factors, potentially leading to suboptimal outcomes. For instance:

- ❑ Managers might manipulate measures (e.g., using outdated equipment to reduce depreciation costs) for short-term gains, compromising long-term benefits.

- ❑ Reward systems linked to metrics like profit or sales targets may incentivize manipulative practices, such as underreporting forecasts to secure easier goals.

Thus, management accountants must design comprehensive measures that address all performance aspects and remain vigilant against behaviors aimed at exploiting these systems rather than achieving genuine business objectives.

1.6 PRACTICAL IMPLICATIONS OF MANAGEMENT ACCOUNTANCY

Today, management accounting has become a central force in shaping business strategies and driving organizational growth. Beyond the preparation of financial statements, it serves as a vital support system for decision-making, planning, and control. Let's take a closer look at how management accounting directly impacts business operations.

1. **Crafting Agile Accounting Systems:** Think of accounting systems as the “nervous system” of a business. For the organization to function properly, its accounting, cost accounting, and tax systems must be well-designed, adaptable, and up-to-date. Management accountants play a crucial role in developing and upgrading these systems, ensuring they remain responsive to the ever-changing business landscape.

Example: Imagine a growing **e-commerce business** like Shyam's Retail. As sales increase, the company's accounting system needs to adapt to track new payment methods, multi-currency transactions, and inventory movements. Management accountants step in to design a system that automates these processes, ensuring Shyam's Retail stays efficient as it grows.

2. **Using Historical Data for Smart Forecasting:** Every good decision starts with solid information. Management accountants collect and store financial data from past business operations, enabling them to identify trends and predict future outcomes. This data-driven approach helps businesses plan their next moves with confidence.

Example: Radha's Food Products wants to predict demand for its newly launched organic snacks. Management accountants analyze past sales data, seasonal trends, and customer preferences. By examining patterns, they create a demand forecast that informs production schedules, marketing strategies, and supply chain decisions.

3. **Facilitating Communication Across the Company:** Good communication can turn a good plan into a great one. Management accountants bridge the gap between senior leadership, departments, and employees by clearly communicating organizational goals. Their reports provide clarity on everyone's role in achieving company objectives.

Example: When Shyam's Retail decides to expand to international markets, management accountants translate the big-picture strategy into actionable department-level plans. Each team (marketing,

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logistics, finance) gets clear instructions on their targets, deadlines, and resource allocation. This alignment ensures smooth execution of the expansion plan.

4. **Feedback and Control Systems:** Mistakes are inevitable, but how quickly they are detected and corrected makes all the difference. Management accountants set up **feedback systems** that track key performance indicators (KPIs) and highlight deviations from the plan. This system is known as “management by exception” — only significant issues require immediate attention.

Example: At Radha’s Food Products, the production team was expected to reduce waste by 10%, but halfway through the year, waste increased by 5%. The management accountant’s control system quickly flags this issue. A report identifies the cause (a faulty supplier batch) and suggests corrective action (switch suppliers). With timely intervention, production efficiency is restored.

5. **Data Analysis and Interpretation:** Data is powerful—but only if you know what it means. Management accountants analyze both financial and non-financial data, turning it into meaningful insights. These insights help senior management make informed decisions.

Example: When Shyam’s Retail notices a sudden dip in online sales, management accountants investigate the data. They discover that shipping delays are to blame. Their analysis recommends a shift to faster, more reliable shipping partners, which results in improved customer satisfaction and a rebound in sales.

6. **Supporting Management in Decision-Making:** When big decisions are on the table—like launching a new product or opening a new branch—management accountants are the “go-to” experts. They assess the financial impact of various options, identify risks, and suggest the most cost-effective course of action.

Example: Radha’s Food Products is considering launching a new line of gluten-free snacks. Management accountants evaluate costs, forecast demand, and calculate expected profits. Based on their analysis, the management team decides to proceed, confident that the new product will be profitable.

7. **Performance Evaluation for Accountability:** How do you know if employees are performing well? Management accountants track performance through clear, quantifiable metrics. This approach, known as “**management by objectives**” (MBO), ensures that employee performance is aligned with organizational goals.

Example: In Shyam’s Retail, each department is given specific targets—like reducing costs by 5% or increasing sales by 20%. Management accountants monitor performance against these objectives and prepare performance reports. This system motivates employees to stay focused and strive for excellence.

8. **Continuously Improving Analytical Techniques:** Management accounting never stands still. As new technologies emerge, management accountants constantly enhance their tools, methods, and

techniques to stay ahead of the competition. They introduce **modern analytics** like predictive modeling and data visualization, enabling smarter decisions.

Example: Shyam's Retail switches from basic spreadsheets to advanced **business intelligence (BI) software** for cost analysis. With the new system, management can see real-time insights on cost overruns, sales trends, and operational bottlenecks. This shift improves decision-making speed and accuracy.

1.7 COST MANAGEMENT FOR COMPETITIVE ADVANTAGE

In today's fast-paced global marketplace, survival isn't enough—businesses must strive to stand out. But how do companies achieve this edge over their rivals? The answer lies in strategic cost management, a powerful approach that combines strategic planning with cost control. Companies that master this practice can maintain profitability, enhance customer satisfaction, and gain a clear competitive advantage.

Let's explore some key factors that drive the need for strategic cost management in the modern business world.

1. **The Rise of New Distribution Channels:** Gone are the days when businesses had just one or two ways to reach customers. Today, the world is filled with multiple sales channels—e-commerce platforms, social media marketplaces, direct-to-consumer models, and more. This shift is especially evident in industries like **insurance**, where direct insurers have bypassed brokers and agents to reduce commission costs. By cutting out the middleman, direct insurers have gained a significant cost advantage, capturing more customers from traditional players.

Real-World Example: Think of how companies like Geico revolutionized the insurance industry by going direct-to-customer. Without agent commissions, they reduced operational costs, passed on savings to customers, and claimed a larger market share.

2. **Smarter, More Informed Consumers:** With the click of a button, customers can compare products, read reviews, and hunt for the best prices. Informed consumers know their worth—and they won't settle for less. This shift is especially visible in the financial services sector, where customers frequently switch banks, insurers, and investment firms for better deals. Businesses must now work harder to retain loyal customers while managing rising acquisition costs.

Real-World Example: Think about how online banking apps like Revolut and N26 have changed consumer expectations. Customers now demand lower fees, seamless user experiences, and instant payments. To stay competitive, traditional banks had to ramp up their technology, improve service quality, and revise their cost structures to retain customers.

3. **Pressure from Financial Markets:** Publicly listed companies face a unique challenge: the unblinking gaze of financial analysts and

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investors. Businesses are constantly evaluated on their productivity and cost efficiency, as these factors influence their market valuation. Companies that manage their costs better are seen as more stable, profitable, and “investor-friendly.”

Real-World Example: Take Tesla, for instance. Its cost management strategy, particularly in battery production and supply chain efficiency, has led to consistent investor interest. By optimizing production processes, Tesla keeps costs down, boosts margins, and maintains its appeal to financial analysts and shareholders.

4. **The Rise of “Category Killers”:** “Category killers” are companies that dominate a single product category with unmatched operational efficiency. Think of brands that specialize in one thing — but do it better, faster, and cheaper than anyone else. Their lean cost structures and streamlined processes force multi-product firms to rethink their strategies to avoid being left behind.

Real-World Example: Consider IKEA, a “category killer” in the furniture sector. IKEA’s flat-pack design reduces shipping and storage costs, enabling it to offer quality products at lower prices. This cost advantage has allowed IKEA to dominate its market, pushing smaller retailers to adapt or exit.

5. **Power of Research and Development (R&D):** Innovation isn’t just about launching new products; it’s also about making existing processes smarter. Companies that invest in R&D can develop more efficient production methods, use cheaper materials, or cut down on waste. This enables them to offer high-quality products at lower prices, a win-win for both the company and its customers.

Real-World Example: Apple is a master of R&D. From designing energy-efficient chips to sourcing more durable materials, Apple constantly refines its production process. This approach keeps product quality high, limits recalls, and drives down production costs, allowing for higher profit margins on devices like the iPhone and MacBook.

6. **Smarter Use of Resources:** When it comes to staying competitive, using resources wisely is non-negotiable. Companies that optimize their use of time, labor, and materials can keep costs under control while maintaining product quality. Tools like Activity-Based Costing (ABC) help businesses pinpoint which operations are driving costs and where improvements can be made.

Real-World Example: Amazon is a leader in optimizing resource utilization. Its use of robotics in warehouses has significantly reduced labor costs and increased processing speed. Robots handle repetitive tasks like sorting, packing, and moving inventory, allowing human workers to focus on more complex roles. This strategy improves efficiency, reduces delays, and cuts operational costs — all while enhancing customer satisfaction with faster deliveries.

By mastering cost management, companies don’t just “keep up”—they get ahead. Whether it’s adapting to new customer demands, responding to

market pressures, or handling competition from “category killers,” companies with strong cost strategies remain resilient and profitable.

So, the next time you hear about a company’s “competitive advantage,” remember: **it’s not just about what they do—it’s about how well they control their costs.**

1.8 DECISION-MAKING PROCESS IN A FIRM

Decision-making is a crucial process in management that involves choosing the best option among various alternatives. It serves as the foundation for key management functions like planning, organizing, directing, controlling, and staffing. Without effective decision-making, these functions cannot be carried out efficiently. Decision-making helps management select the most suitable course of action to achieve organizational objectives.

Decision-making is fundamental to the success of any organization, as it directly influences the management process. Each management function—whether it’s planning, organizing, directing, controlling, or staffing—relies on sound decision-making. Effective decision-making ensures the fulfillment of organizational goals, while poor or irrational decisions can lead to adverse outcomes.

Following a structured and logical decision-making process helps managers select the most appropriate course of action to address challenges, meet objectives, and guide the organization toward success.

The decision-making process in management follows a structured approach divided into three primary stages: **planning, directing, and controlling**. Each stage is crucial in ensuring that decisions are well thought out and effectively executed, aligning with the organization’s goals.

1.8.1 PLANNING

The initial and most important phase of decision-making is planning, which entails identifying and diagnosing the problem. A thorough understanding of the issue is key to finding an appropriate solution. This involves distinguishing between the actual root cause and any symptoms. For example, a decline in sales may be a symptom, while the underlying problem could be subpar product quality or weak marketing strategies. Gathering relevant information is essential at this stage, as it forms the basis for analyzing the problem from different perspectives. The manager must then assess the problem’s scope, long-term impact, and qualitative factors. Once analyzed, the next step is to develop alternative solutions, taking into account factors such as time, costs, and any obstacles to implementation. The planning phase lays the foundation for effective decision-making by defining the problem and outlining possible solutions.

Example: Eco-Tech Electronics has noticed a recent decline in sales of its energy-efficient air conditioners. Initially, the team assumes the problem is a lack of customer awareness, but deeper investigation reveals the true issue is a price increase that customers perceive as too steep

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compared to similar models from competitors. The manager needs to clearly define this problem before moving forward with potential solutions. Once the root cause is identified, they can explore several alternatives, such as adjusting the pricing strategy, launching a marketing campaign to highlight the long-term savings, or offering bundle deals with other products.

1.8.2 DIRECTING

After planning, the process moves into the directing phase, where the manager evaluates the alternative solutions and selects the most rational and effective one. The chosen solution must align with the organization's objectives and have the highest likelihood of success. Effective communication is critical at this stage, as the decision must be conveyed to all relevant stakeholders and team members involved in its implementation. Gaining their acceptance and cooperation is essential to ensure smooth execution. The manager's role in this phase is to guide and motivate the team to turn the decision from a theoretical plan into practical action. Leadership and the ability to convince the team of the decision's validity are key to successful implementation.

Example: After evaluating different options, the manager at Eco-Tech Electronics decides to introduce a **temporary promotional discount** on the air conditioners to make the price more competitive while emphasizing the long-term savings on energy bills. The next step is to communicate this decision clearly to the sales team, marketing department, and customer service. The manager ensures everyone understands the promotional strategy, why it was chosen, and how it aligns with the company's goal to boost sales without sacrificing product quality. The manager also ensures that the marketing team develops the right promotional materials and that the sales team is trained to effectively communicate the benefits of the promotion.

1.8.3 CONTROLLING

The final stage of the decision-making process is controlling, where the manager monitors the decision's implementation to ensure it is delivering the desired results. This involves establishing feedback mechanisms to assess the actual outcomes against the expected ones. Feedback is essential for identifying any discrepancies, and it allows the manager to take corrective action if needed. Tools such as reports, observations, and performance metrics are used to evaluate the effectiveness of the decision. The controlling phase is crucial for maintaining the quality and success of the decision, as it ensures that adjustments are made to address any issues and optimize performance. This stage closes the decision-making loop, ensuring that decisions are both implemented and effective in achieving organizational objectives.

Example: Once the promotional discount is launched, Eco-Tech Electronics begins tracking sales data, customer feedback, and the effectiveness of marketing campaigns. The manager looks for signs of improvement in both sales volume and customer perception. After a

month, the company notices an increase in sales, but customer feedback suggests that some customers are still uncertain about the product's long-term energy savings. In response, the manager decides to **revise the marketing materials** to better highlight the energy savings and provide more detailed comparisons with competitor products.



SELF-ASSESSMENT QUESTIONS

7. What is the primary purpose of the controlling stage in the decision-making process?
 - a. To monitor the decision's implementation and make adjustments.
 - b. To develop alternative solutions.
 - c. To plan and diagnose the problem.
 - d. To organize resources for decision-making.
8. Which of the following best describes the relationship between decision-making and other management functions?
 - a. Decision-making is independent of other management functions.
 - b. Decision-making directly influences planning, organizing, and controlling.
 - c. Decision-making is only important for planning and directing.
 - d. Decision-making is not essential to staffing and organizing.

1.9 SOME INNOVATIVE MANAGEMENT ACCOUNTING PRACTICES

Innovative management accounting practices can be developed by building on the traditional tools and techniques currently used, infusing modern technology, data analytics, and sustainability-focused strategies. Here are some innovative management accounting practices:

1. **Sustainability-Focused Financial Planning:** Incorporating sustainability into financial planning involves setting long-term goals that encompass not only profitability but also environmental and social impacts. This approach utilizes Environmental, Social, and Governance (ESG) metrics to guide sustainable decisions. For example, financial strategies would now include targets for reducing carbon emissions, exploring green financing options, and investing in renewable energy alongside traditional objectives such as revenue growth and profit margins.

Example: Tata Steel's Sustainability Drive

Tata Steel, a global steel manufacturer, integrates sustainability goals into its financial planning. The company aims to reduce its carbon footprint by adopting energy-efficient production processes and investing in green hydrogen as a clean energy source. This shift influences its financial strategy, requiring increased allocation of resources for sustainable R&D projects. Tata Steel also issues green bonds

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to raise capital for eco-friendly projects, aligning financial planning with its ESG (Environmental, Social, and Governance) objectives.

2. **AI-Powered Financial Statement Analysis:** AI can enhance financial statement analysis by using machine learning algorithms to detect patterns, anomalies, and risks that manual reviews might miss. AI platforms can automatically generate comparative and trend analyses from large datasets, highlighting areas such as declining profitability and offering recommendations based on past trends, making the data more insightful and actionable for management.

Example: HDFC Bank's AI-Driven Risk Analysis

HDFC Bank leverages AI-based tools to analyze financial statements and detect risks related to loan defaults. The AI platform analyzes large volumes of customer financial data and identifies unusual patterns, such as a sudden drop in a borrower's cash flow. This predictive capability helps the bank prevent bad loans and take pre-emptive actions, like offering restructuring options to customers. The system generates trend reports and identifies high-risk areas, enabling faster decision-making compared to manual reviews.

3. **Real-Time Decision-Making Solutions:** By integrating real-time data analytics and cloud-based dashboards, management can make quicker and more informed decisions. These tools allow for immediate simulations of various decision outcomes, such as investment strategies or market changes. For instance, dynamic dashboards provide real-time calculations of the potential effects of different decisions, empowering management to act on current data rather than waiting for periodic reports.

Example: Amazon's Real-Time Pricing Adjustments

Amazon's management relies on a real-time data analytics system that dynamically adjusts product prices. The company uses big data, AI, and cloud-based dashboards to track market demand, competitor prices, and stock availability in real time. Based on this analysis, the system can increase or decrease prices almost instantly. For example, during festive sales, if a product experiences a surge in demand, the system increases the price to maximize profits. This real-time decision-making boosts Amazon's pricing strategy and enhances overall profitability.

4. **Behavioral Insights in Control Techniques:** Behavioral analytics can be embedded in control techniques like budgetary control and standard costing to understand how human behavior affects financial performance. Analytics tools can track employee interactions with budgets and resources, identifying patterns of over-spending or under-spending. This allows management to adjust policies or provide targeted training to improve resource efficiency and financial outcomes.

Example: Google's Employee Budget Interaction Analysis

Google employs behavioral analytics to study how employees interact with departmental budgets. Using internal tracking tools,

management can identify instances where departments consistently under-spend or over-spend their budgets. If a marketing team frequently overspends, Google investigates the behavior driving this action—it may result from poor forecasting or delays in procurement. Based on the analysis, the finance department can revise budgetary control techniques or provide personalized training, promoting more efficient spending habits.

5. **Predictive Analytics for Statistical and Graphical Techniques:** Predictive analytics, when applied to statistical and graphical techniques, helps provide more accurate forecasting by analyzing historical data and market conditions. For instance, linear programming combined with machine learning can optimize production schedules and supply chain management by predicting future demand based on both historical and real-time data trends.

Example: Unilever's Demand Forecasting System

Unilever, a global consumer goods company, employs predictive analytics to forecast product demand for items like soaps, shampoos, and food products. By using machine learning models, Unilever analyzes past sales data, consumer trends, and market conditions to predict future demand. For example, during the COVID-19 pandemic, Unilever used predictive analytics to anticipate a surge in demand for hygiene products like sanitizers and disinfectants. This proactive approach helped the company adjust its production schedules and ensure availability at retail stores, reducing stockouts and boosting customer satisfaction.

6. **Automated, Interactive Reporting Systems:** Developing automated and interactive reporting tools ensures that management receives real-time, customizable insights. These systems enable management to filter reports by region, department, or timeframe, all while reducing manual input and enhancing accuracy. For example, interactive dashboards allow managers to receive instant alerts on performance metrics like cost overruns, making it easier to react quickly to issues.

Example: Reliance Industries' Interactive Dashboards

Reliance Industries employs interactive dashboards to provide its management with real-time financial performance insights. These dashboards, powered by Power BI and SAP Analytics, allow executives to filter reports by region, business division, or specific time periods. For example, senior management can track profitability for each of its subsidiaries, such as Reliance Retail or Jio Platforms, and drill down into detailed expense reports. If a particular store or product line is underperforming, alerts are triggered on the dashboard, prompting management to take corrective action immediately.

Activity

As a management accountant in a manufacturing company, you are tasked with recommending an innovative management accounting

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practice. Choose one of the following practices and explain how it could benefit your organization:

1. Sustainability-Focused Financial Planning
2. AI-Powered Financial Statement Analysis
3. Real-Time Decision-Making Solutions
4. Predictive Analytics

1.10 MANAGEMENT ACCOUNTING AS PROFESSION

Management accountant, often called the controller, plays a vital role in providing accounting information to aid in planning, controlling, and decision-making within an organization. The management accountant is responsible for collecting, processing, and reporting data that helps managers make informed decisions.

As a profession, management accounting encompasses a wide range of responsibilities, including:

1. **Active Participation in the Management Process:** Management accountants occupy key roles, educating executives on the use of control information and supporting them with accurate, relevant data for decision-making.
2. **Maintaining a Balanced Capital Structure:** They contribute to financial planning by advising on the appropriate mix of debt and equity, ensuring that the organization raises funds in the most cost-effective manner.
3. **Investment Advisory:** Management accountants guide firms or individuals on investment strategies, offering insights on how, where, and when to invest to ensure maximum returns.
4. **Conducting Financial Investigations:** They assist management in financial investigations related to mergers, acquisitions, or the analysis of profit declines and cost increases, providing critical insights for financial decisions.
5. **Long-term and Short-term Planning:** A management accountant's role is forward-looking, contributing to both long-term strategies and immediate decision-making processes by forecasting future business and economic trends.
6. **Supporting Decision-making:** They supply comprehensive and reliable financial data to help management make informed decisions on various business matters, such as pricing, capital investments, and product development.
7. **Establishing Control Systems:** Management accountants are responsible for preparing reports like budgets, variance analyses, and cash flow evaluations, which help organizations manage and control resources effectively.
8. **Designing Management Information Systems (MIS):** They create systems that ensure the timely flow of information to managers, enabling prompt and accurate decision-making.

9. **Stewardship and Accountability:** Management accountants set up frameworks for cost and financial accounting, preparing essential reports that support operational and financial decision-making.
10. **Corporate Planning and Financial Restructuring:** They play a key role in advising on mergers, acquisitions, and resource allocation, ensuring the efficient use of organizational resources.

If you're considering a career in management accounting, you're signing up for an exciting role where you get to make a real impact on a business's success. You'll use your skills to solve problems, guide decisions, and plan for the future—all while working closely with top executives.

It's not just about numbers; it's about strategy, creativity, and leadership. So, are you ready to take on this superhero role?



SELF-ASSESSMENT QUESTIONS

9. Which of the following is a key responsibility of management accountants in an organization?
 - a. Preparing tax returns
 - b. Handling day-to-day cash transactions
 - c. Filing regulatory compliance reports
 - d. Establishing control systems through reports like budgets and variance analyses
10. How do management accountants assist organizations in financial investigations?
 - a. By forecasting future economic trends
 - b. By preparing cash flow evaluations
 - c. By providing critical insights on mergers, acquisitions, and profit or cost changes
 - d. By filing financial reports with regulatory authorities

1.11 LIMITATIONS OF MANAGEMENT ACCOUNTING

The management accountant plays a crucial role in delivering reliable accounting data and other relevant information for managerial use. To be truly impactful in the decision-making process, this information must meet specific criteria: it should be relevant and accurate, consistent and comparable, clearly presented, timely, and tailored to the needs of different management levels. Despite these efforts, the information and reports provided by the management accountant still have certain limitations:

1. **Varied Interpretations of Terms:** In accounting, many terms have different meanings based on the context. For instance, the term "cost" can refer to historical cost, variable cost, opportunity cost, incremental cost, and others. These terms may seem similar but can have distinct meanings, leading to confusion, especially for those less familiar with them. To avoid misunderstandings, the management

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accountant should define these terms clearly in each context and use them consistently.

2. **Approximations:** Management accounting data often involves a level of approximation, particularly when reports are prepared quickly. The shorter the reporting timeframe, the higher the level of approximation involved. Even historical data might require approximations when full accuracy is too costly or time-consuming to achieve. It's important for management to be aware of these approximations and for the management accountant to apply them consistently.
3. **Incomplete Data:** Management accountants typically provide quantitative data, but business decisions often require both quantitative and qualitative information. For example, management accounting reports may not indicate how product quality is impacted by changes in materials or production methods. It's essential that management understands the limitations of numerical data and considers qualitative factors as part of their decision-making process.
4. **The Need for Proper Management Action:** While management accountants can present data in an accurate and accessible format, the figures themselves do not lead to action. Successful business outcomes depend on management's ability to understand and act on the provided data. If management misinterprets the data or fails to take action, the information becomes ineffective. Thus, the ultimate responsibility lies with management to use the data wisely.

These challenges emphasize the importance of clarity, consistency, and proactive management in utilizing management accounting data effectively.

1.12 SUMMARY

- ❑ Cost accounting focuses on classifying, accumulating, and controlling costs, providing crucial insights for budgeting, variance analysis, and profitability assessment.
- ❑ It aids in identifying profitable and unprofitable activities, guiding management in eliminating inefficiencies and optimizing operations across various sectors.
- ❑ Best practices in costing systems involve selecting appropriate methods (e.g., job, batch, process costing) and using historical and standard costing for effective monitoring and decision-making.
- ❑ The integration of technology and automation in costing systems enhances data accuracy, facilitates seamless cost management, and supports timely decision-making.
- ❑ Management accounting leverages cost accounting data to inform strategic planning, performance evaluation, and resource allocation, emphasizing the importance of tailored information for effective management decisions.
- ❑ Management accounting focuses on internal decision-making by providing relevant data, including cost information, performance metrics,

budgeting, and variance analysis, to support planning and control within an organization.

- ❑ Effective management accounting enhances decision-making through structured processes involving planning (identifying problems), directing (evaluating and choosing solutions), and controlling (monitoring implementation and outcomes).
- ❑ The role of management accountants extends beyond data reporting; they actively participate in management processes, maintain balanced capital structures, advise on investments, and design management information systems.
- ❑ Strategic cost management is essential for gaining competitive advantage, requiring businesses to adapt to market changes, optimize resource utilization, and leverage innovations in processes and product offerings.
- ❑ Despite the value of management accounting, challenges such as varied interpretations of terms, approximations in data, and the necessity for proactive management action highlight the importance of clarity and consistency in financial reporting and decision-making.

1.13 KEY WORDS

- ❑ **Cost Accounting:** A specialized branch of accounting that focuses on the classification, accumulation, and control of costs to assist in budgeting and decision-making.
- ❑ **Cost Control:** The process of monitoring and regulating expenses to enhance efficiency and profitability within an organization.
- ❑ **Standard Costing:** A costing technique that uses pre-determined costs as benchmarks to measure performance and identify variances for cost management.
- ❑ **Management Accounting:** A field of accounting that provides financial and non-financial information to support internal decision-making and strategic planning.
- ❑ **Variance Analysis:** The examination of differences between actual and expected financial performance, used to identify inefficiencies and improve cost management.
- ❑ **Budgeting:** The process of creating a financial plan that outlines expected revenues and expenditures over a specific period, aiding in resource allocation and financial control.
- ❑ **Activity-Based Costing (ABC):** A costing method that assigns overhead and indirect costs to specific activities, enabling more accurate product costing and better decision-making regarding pricing and profitability.
- ❑ **Strategic Cost Management:** A management approach that aligns an organization's cost management strategies with its business strategies to achieve a competitive advantage in the market.

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1.14 CLOSING CASELET**COST MANAGEMENT STRATEGY AT SOUTHWEST AIRLINES**

Southwest Airlines, established in 1967, is renowned for its low-cost business model. In the early 2000s, faced with rising fuel prices and intensified competition, Southwest recognized the need for a robust cost management strategy to sustain profitability while offering affordable fares. The airline leveraged management accounting techniques to analyze operational costs across departments, identifying key areas for savings and optimizing efficiency.

One of the pivotal strategies involved the implementation of stringent cost control measures and an innovative point-to-point transit model, which enhanced aircraft utilization and reduced layover times. Additionally, Southwest utilized a fuel hedging strategy to shield itself from volatile fuel prices, allowing it to maintain stable operating costs. By focusing on labor productivity and dynamic pricing models, the airline successfully managed to offer low fares while ensuring profitability, solidifying its position as a leader in the low-cost airline segment. As a result, Southwest consistently reported profits over decades, demonstrating the effectiveness of its cost management approach in navigating the competitive airline industry.

**QUESTIONS**

1. How did Southwest Airlines' use of management accounting techniques contribute to its competitive advantage in the airline industry, particularly during challenging economic times?
(**Hint:** Consider specific management accounting strategies that were implemented and their impact on operational efficiency and cost control.)
2. In what ways might Southwest Airlines' cost management strategies need to evolve in response to emerging trends in the airline industry, such as increasing environmental regulations and shifts in consumer preferences for sustainability?
(**Hint:** Think about external factors influencing the airline industry today, including sustainability initiatives, technological advancements, and customer expectations.)

1.15 HIGHER ORDER THINKING SKILLS QUESTIONS (HOTS)

1. How can the principles of cost accounting be applied to improve decision-making in a service-oriented organization compared to a manufacturing firm?
2. Analyze the potential impacts of adopting a standardized costing system across multiple branches of a retail business. What challenges might arise, and how could they be addressed?

3. How can the integration of advanced technologies, such as artificial intelligence and data analytics, enhance the effectiveness of management accounting practices in decision-making and performance evaluation?
4. In what ways can management accountants contribute to fostering a culture of accountability and transparency within an organization, and how might this impact overall business performance?
5. Considering the evolving nature of market dynamics, how should management accounting adapt its practices and reporting methods to provide relevant and timely information for strategic decision-making in a global context?

1.16 ANSWER KEY

SELF-ASSESSMENT QUESTIONS

Topic	Q. No.	Answer
An Introduction	1.	b. To control and reduce production costs through budgetary control and variance analysis.
	2.	c. Assisting in budgeting, planning, and forecasting to align with strategic objectives.
Best Practices in Costing System	3.	b. Job Costing
	4.	c. To serve as a benchmark for performance measurement
Management Accounting	5.	d. Fixed formats for information
	6.	c. Management Accounting is forward-looking and aids in future planning.
Decision-Making Process in a Firm	7.	a. To monitor the decision's implementation and make adjustments.
	8.	b. Decision-making directly influences planning, organizing, and controlling.
Management Accounting as Profession	9.	d. Establishing control systems through reports like budgets and variance analyses
	10.	c. By providing critical insights on mergers, acquisitions, and profit or cost changes

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U N I T

MATERIAL COST CONTROL

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- 2.3 Essential Features of Material Control Process
- 2.4 Responsibilities of the Purchase Department
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INTRODUCTORY CASELET

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Boeing's Material Control and Inventory Management

Boeing, one of the world's leading aerospace manufacturers, faced significant challenges in its material control and inventory management processes during the production of its 787 Dreamliner aircraft. The company's initial approach relied on traditional inventory management practices, which resulted in long lead times, excessive inventory costs, and challenges in coordinating with a global network of suppliers. With multiple suppliers providing thousands of parts and components, Boeing struggled with inventory accuracy and timely material availability, which led to production delays and increased costs.

To address these challenges, Boeing adopted Lean Manufacturing principles, specifically focusing on a Just-in-Time (JIT) inventory system to streamline its material control processes. The company implemented a Material Requirement Planning (MRP) system that allowed for precise tracking of material needs based on production schedules. A comprehensive Bill of Materials (BOM) was created for each aircraft, detailing the exact specifications and quantities of parts required. This approach helped Boeing better forecast its material requirements, reduce excess inventory, and improve supplier coordination.

As part of the JIT implementation, Boeing collaborated closely with suppliers to ensure timely delivery of materials, often utilizing a kanban system for inventory replenishment. By integrating suppliers into its planning processes, Boeing could maintain lower inventory levels while ensuring that critical components arrived just in time for production. The company also established strict wastage control measures to monitor and minimize material usage throughout the production process.

The implementation of Lean principles and the JIT inventory system led to significant improvements at Boeing. The company was able to reduce its inventory holding costs by approximately 30%, resulting in substantial savings. Additionally, production lead times decreased, enabling Boeing to ramp up production more efficiently and meet delivery schedules. The collaboration with suppliers improved the overall supply chain responsiveness, minimizing delays caused by material shortages. Ultimately, Boeing's material control enhancements contributed to the successful launch of the 787 Dreamliner, which became a cornerstone of its product line-up, showcasing the effectiveness of integrated material management strategies in a complex manufacturing environment. This case illustrates how effective material control can lead to substantial operational improvements and cost savings in a high-stakes industry.

INTRODUCTORY CASELET



QUESTIONS

1. What were the primary challenges Boeing faced in its material control and inventory management processes before adopting Lean Manufacturing principles?

(Hint: Consider the issues related to inventory levels, production delays, and supplier coordination in the context of the 787 Dreamliner project.)

2. How did the implementation of a Just-in-Time (JIT) inventory system impact Boeing's production efficiency and cost management?

(Hint: Think about the effects of reduced inventory levels, improved supplier relationships, and changes in production lead times as a result of these systems.)

**LEARNING OBJECTIVES**

After reading this unit, you will be able to:

- Understand the roles and responsibilities of the material control department.
- Identify the key elements of an effective material control process.
- Understand the procedures for issuing materials from inventory, including the documentation required.
- Understand different methods used to price materials.
- Understand the EOQ formula and its use in minimizing inventory costs.
- Learn about various approaches to controlling inventory.

2.1 INTRODUCTION

The term “material” refers to tangible commodities or products supplied to an organization, which are classified as either direct or indirect materials. Materials represent a significant portion of the total cost of a finished product, making accurate documentation and effective management of material costs essential.

Material cost control involves the systematic regulation of the procurement, storage, and usage of materials to maintain uninterrupted production while avoiding excessive investment in inventory. Proper material cost management minimizes losses and reduces material wastage that might otherwise go unnoticed, forming a critical part of materials management. The significance of materials is closely tied to the costs of idle time for workers and machinery, as well as the urgency of demands. If manufacturing workers, machinery, and customers could afford delays, there would be no need for materials to remain readily available, and inventories would not need to be maintained. However, it is highly inefficient to leave personnel and machinery idle, as the demands of modern life necessitate the availability of supplies before they are needed. Consequently, organizations must ensure timely transportation of supplies.

Since materials account for a considerable share of the production cost and are somewhat manageable, effective planning and oversight of inventory are crucial. Material cost control employs a systematic approach to determining order quantities, aiming to minimize procurement and storage costs without disrupting production or sales. Without adequate control, material costs can exceed economic limits, leading to excess capital tied up in surplus inventories, disruptions in production, and financial strain on the organization. Additionally, poor material management can lead to excessive consumption and waste, as an overabundance of resources may result in employee negligence.

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Example 1: An invoice for a consignment of chemicals X and Y provides the following information:

Particulars	Quantity	Rate (₹)	Amount (₹)
Chemical X	12,000 kgs.	₹12 per kg.	1,44,000
Chemical Y	9,000 kgs.	₹15 per kg.	1,35,000
Total			2,79,000

Additional costs:

□ Central Sales Tax @ 2% = ₹5,580

□ Railway Freight = ₹4,200

Total cost = ₹2,88,780

A shortage of 600 kgs. in Chemical X and 400 kgs. in Chemical Y is noticed due to normal breakages. A further provision of 2% is made for deterioration. You are required to determine the rate per kg of each chemical.

Solution Working:

1. Computation of Effective Quantity of Each Chemical Available for Use

Particulars	Chemical X (kg)	Chemical Y (kg)
Quantity purchased	12,000	9,000
Less: Shortage due to normal breakages	600	400
Balance	11,400	8,600
Less: Provision for deterioration (2%)	228	172
Effective quantity available	11,172	8,428

2. Statement Showing the Computation of Rate per kg of Each Chemical

Particulars	Chemical X (₹)	Chemical Y (₹)
Purchase price	1,44,000	1,35,000
Add: Central Sales Tax @ 2%	2,880	2,700
Add: Railway Freight (in the ratio of quantity purchased, i.e., 12:9)	2,400	1,800
Total Cost (A)	1,49,280	1,39,500
Effective Quantity (B)	11,172 kg	8,428 kg
Rate per kg (A ÷ B)	₹13.36	₹16.55

2.2 FUNCTIONS OF MATERIAL CONTROL DEPARTMENT

The Material Control Department plays a crucial role in ensuring efficient material management through the following key functions:

- 1. Ensuring Continuous Material Flow:** The department is responsible for providing a steady supply of required materials, parts, and components, ensuring that production processes remain efficient and uninterrupted.

2. **Optimizing Inventory Investment:** It focuses on minimizing investments in inventories while taking operational requirements into account, thereby freeing up capital for other business needs.
3. **Efficient Material Storage Management:** The department implements practices to store materials efficiently, protecting inventories from potential losses due to fire, theft, or mishandling, while also minimizing handling time and associated costs.
4. **Minimizing Surplus and Obsolete Inventory:** A primary function is to minimize surplus and obsolete things, thus decreasing waste and guaranteeing effective resource utilization.
5. **Dynamic Inventory Adjustments:** The department monitors and adjusts material levels in response to sales requirements and production schedules, ensuring that inventory aligns with actual operational needs.
6. **Collaborative Decision-Making:** While top management holds the ultimate responsibility for material control, the department engages in collaborative decision-making involving the production manager, controller, sales manager, and purchasing manager to address diverse perspectives and financial considerations.
7. **Coordinating Across Departments:** It coordinates with different departments to manage conflicting viewpoints regarding inventory levels, balancing the needs of sales, production, and financial management.
8. **Assigning Specific Control Responsibilities:** In many organizations, the department is tasked with the specific responsibility of material control, particularly within the purchasing department, ensuring effective management of procurement processes in line with overall inventory strategies.

2.3 ESSENTIAL FEATURES OF MATERIAL CONTROL PROCESS

In every successful business, especially in manufacturing and production, material control plays a crucial role. It ensures the right materials are available at the right time, in the right quantity, and at the right cost. But how exactly does it work? Let's explore its key features in a way that's easy to understand.

1. **Ensuring Materials Are Always Available:** Imagine production coming to a sudden halt because raw materials are out of stock. This is exactly what material control seeks to avoid. The system ensures that essential materials are continuously available so that production never stops. Whether it's spare parts, raw ingredients, or essential tools, material control keeps everything in check, ensuring smooth operations.

How It Works: Regular monitoring of stock levels triggers timely reorders. This proactive approach avoids "stock-outs" and keeps production lines running without delays.

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- 2. Smart Purchasing at the Best Price:** Every business wants to cut costs, but not at the expense of quality. Material control systems ensure materials are purchased at the most competitive prices while maintaining high-quality standards. It's not just about finding the lowest price—it's about striking the right balance between price, quality, and supplier reliability.

How It Works: Companies develop strong supplier relationships, negotiate better rates, and use bulk-buying tactics to save on costs. Periodic supplier reviews ensure that only the most reliable vendors are retained.

- 3. Cutting Down on Waste:** No one likes waste—and businesses feel it most in their profit margins. Material control actively works to reduce wastage by tackling issues like overstocking, improper storage, and damage from prolonged storage. Controlling waste not only reduces costs but also supports sustainability efforts.

How It Works: Systems are put in place to ensure “First-In, First-Out” (FIFO) usage of materials, reducing the risk of obsolescence. Materials are stored in safe, controlled environments to avoid deterioration.

- 4. Keeping Track with Reliable Information:** Without proper information, it's hard to know how much stock you have, how much you need, or where it's being used. Material control emphasizes accurate record-keeping, allowing for better production planning, waste reduction, and pilferage detection. It's all about having clear, up-to-date information at your fingertips.

How It Works: Advanced software tracks inventory movement in real time. Every time materials are issued, returned, or ordered, it's recorded in a system, ensuring complete visibility of stock levels and movement.

- 5. On-Time Order Fulfillment:** Delivering orders on time boosts a company's reputation and builds trust with customers. Material control ensures production deadlines are met by maintaining a smooth flow of materials throughout the production process. If materials are delayed, production suffers—and so does the company's image.

How It Works: By syncing production schedules with material availability, companies ensure that everything needed for production is ready before it's required. Just-in-time (JIT) inventory methods are often used to achieve this.

2.4 RESPONSIBILITIES OF THE PURCHASE DEPARTMENT

The Purchasing Department plays a crucial role in acquiring the materials and supplies required to ensure smooth operations. Its main duties involve deciding what to buy, determining the right time for purchases, selecting the best suppliers, estimating the quantity to purchase, and securing the most favorable price.

- 1. Deciding What to Buy:** The department is responsible for identifying and sourcing the correct materials that adhere to the necessary quality standards, aligning with production requirements.
- 2. Determining the Right Time for Purchases:** It ensures that materials are obtained at the appropriate time to prevent production delays and maintain a seamless workflow.
- 3. Selecting the Best Suppliers:** The department thoroughly researches and chooses the most dependable sources for materials to guarantee consistent supply and quality.
- 4. Estimating the Quantity to Purchase:** It carefully evaluates and calculates the appropriate amount of materials to order, ensuring a balance between inventory levels and production needs, thereby minimizing costs.
- 5. Securing the Most Favorable Price:** The Purchasing Department strives to procure materials at the most cost-effective rates, while considering factors such as quality, delivery schedules, and supplier dependability.

2.5 MATERIALS ISSUE PROCESS

Materials Issue process involves several stages, from planning material requirements to issuing materials, authorizing requests, and monitoring wastage, all aimed at ensuring smooth operations and accurate inventory management as shown in **Figure 2.1**.

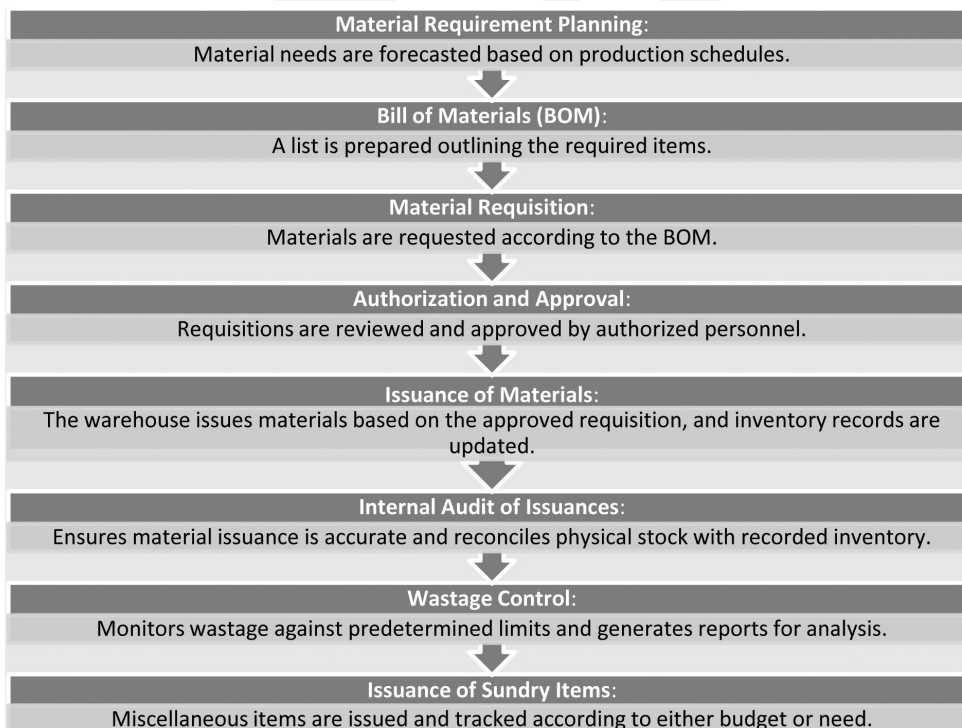


Figure 2.1 Material issue Process.

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The steps involved in the **materials issue process** begin with **Material Requirement Planning**, where material needs are systematically projected based on production schedules. A **Bill of Materials (BOM)** is prepared to estimate the required items, ensuring that all material needs are covered ahead of time.

In the **Material Requisition** stage, materials are requested in accordance with the quantities outlined in the BOM. It is essential that the materials issued do not exceed established standards. If additional quantities are required, a revised BOM must be authorized by a competent individual.

Next is **Authorization and Approval**, which guarantees that every requisition is evaluated by authorized personnel. Any requests that surpass the standard material limits must be documented and receive proper approval.

Issuance of Materials follows, where warehouse or store staff issue the materials based on the approved requisition, ensuring that records are updated in the inventory management system for tracking purposes.

The process includes an **Internal Audit of Issuances**, which ensures the correctness of material issuance. Any discrepancies, whether excess or shortage, are accounted for, and an internal audit reconciles physical stock with recorded figures to ensure consistency.

Wastage Control is another significant aspect, where predetermined wastage limits are set, and actual wastage is monitored. Any deviation from the standard must be thoroughly examined, and a wastage report is generated for analysis.

Finally, **Issuance of Sundry Items** covers the issuance of miscellaneous materials required by different departments. These items can either be requested based on a pre-planned budget or as needed, and their usage is carefully monitored to ensure effective management.

2.5.1 BILL OF MATERIALS

The Bill of Materials (BOM), also known as the Materials Specification List or Materials List, is a document that outlines the standard quantities of materials required for a specific operation or production unit. A comprehensive Materials List provides detailed descriptions and specifications of all materials needed for a task or manufacturing unit, including the exact quantities, enabling easy identification of any deviations from the standard requirements. Typically, this list is prepared in a standardized format by the Engineering or Planning Department. The number of copies created may vary depending on the specific requirements of the organization. The BOM is usually distributed to the Production Control Department, Store Department, and Cost/Accounting Department, while the Engineering or Planning Department retains a copy for reference. A performa of the Bill of Materials usually follows the format shown in **Figure 2.2** below.

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Job No. No.						
Department authorized..... Date.....						
Sl. No.	Code No.	Description	Qty. issued	Date of issue	Rate	Amount
Authorized by..... Received by..... Checked by..... Store Keeper's signature..... Cost clerk.....						

Figure 2.2 Bill of Materials.

The following describes the advantages of utilizing the Bill of Materials throughout all relevant departments:

Stores Department:

1. The Bill of Materials serves as an essential basis for generating material purchase requisitions.
2. It functions as approval for the issuance of the complete material requirements.
3. Clerical work is minimized, as the stores clerk can release all or part of the material requirements to users when the necessary details are included in the Bill of Materials.

Cost Accounts Department:

1. The Bill of Materials is utilized by the Cost Accounts Department to create estimates or budgets for the material costs associated with specific jobs, processes, or operations.
2. It can act as a tool for controlling excess material costs by pinpointing material variances and analyzing the reasons behind them.

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Production Control Department:

1. This department may use the Bill of Materials to effectively control material usage.
2. Its use saves time that would otherwise be spent preparing individual material requisitions.

Engineering or Planning Department: The Engineering or Planning Department is responsible for preparing the materials list in a standardized format. A copy of this list is then distributed to the Stores, Cost Accounts, and Production Control Departments

2.5.2 MATERIAL REQUISITION SLIP

The Material Requisition Slip, also known as a material requisition voucher, functions as the formal authorization to issue materials for use in the factory or its departments. Typically, it is initiated by the production department, and materials are withdrawn based on the material requisition list or Bill of Materials. If no materials list has been prepared, it is advisable for the planning department or the requesting department to prepare the material requisition notes. While a foreman's authority is usually sufficient, for high-cost materials, approval from a higher authority, such as the superintendent or works manager, is recommended before the requisition is submitted to the Stores. A copy of the material requisition is also sent to the Cost Accounting Department, and another copy is retained for office records.

A specimen form of the Material Requisition is provided below in **Figure 2.3**:

Material Requisition Note				
Work Order No.		No.		
Department		Date.....		
Item No.	Particulars	Quantity	Rate	Amount
Store-keeper the material		Workman receiving		Foreman S.L. Clerk

Figure 2.3 Material Requisition Slip.

**SELF-ASSESSMENT QUESTIONS**

1. Which department uses the Bill of Materials to create material cost estimates or budgets?
 - a. Stores Department
 - b. Cost Accounts Department
 - c. Production Control Department
 - d. Engineering or Planning Department

2. What is the purpose of the Material Requisition Slip?
 - a. To list the standard quantities of materials required for production
 - b. To reconcile physical stock with recorded inventory
 - c. To manage excess material costs by identifying material variances
 - d. To provide authorization for issuing materials for use in the factory

2.6 MATERIAL CONTROL TECHNIQUES

Material control techniques are essential tools used in organizations to ensure the efficient use, storage, and management of materials. These techniques aim to maintain an optimal balance between material availability and minimizing associated costs such as carrying and ordering costs. The following goals should be attained by scientific material management:

- i. Ensure an uninterrupted supply of essential materials, parts, and components to support seamless production.
- ii. Minimize investment in inventories while ensuring operational requirements are met.
- iii. Enable efficient storage of materials to safeguard inventory against risks such as fire or theft, while also reducing handling time and associated costs.
- iv. Keep excess and obsolete items at a minimum to avoid unnecessary stockpiling.

It may seem obvious that inventory control is effective when material levels are decreasing, but materials should be adjusted in both quantity and timing to align with sales demands and production schedules. While the ultimate responsibility for material control lies with top management, decisions in this area often require collaboration among the production manager, controller, sales manager, and purchasing manager. This coordinated approach is essential due to the financial impact and the need to balance various materials while considering the differing priorities of each department. For example, sales managers, purchasing executives, and production managers typically advocate for higher stock levels, though for different reasons, while the financial manager tends to favor minimizing material investments. In many organizations, material control is typically assigned as a specific responsibility of the purchasing department.

2.7 MATERIAL PRICING METHODS

When materials are issued to the production department, a challenge emerges regarding the pricing at which these materials should be charged. This difficulty arises because the same type of material may have been acquired in different batches at various times and prices. Consequently,

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the actual cost of the materials can differ across these purchases. To resolve this, it becomes essential to adopt a consistent method for determining the price at which the materials will be issued for production. There are several methods used for pricing of material shown in **Figure 2.4**.

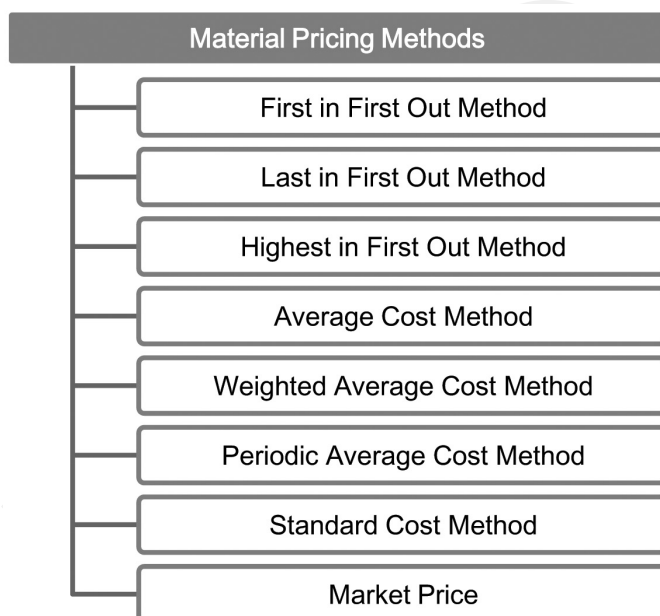


Figure 2.4 Material Pricing Methods.

2.7.1 FIRST IN FIRST OUT METHOD

The FIFO (First-In, First-Out) method is based on the premise that materials acquired first are the first to be issued. Under this approach, materials are issued according to the price of the earliest purchase until that batch is entirely depleted. In essence, materials are charged at the oldest recorded cost in the stores ledger, while the remaining inventory is valued at the most recent purchase prices. It's important to understand that FIFO is used for accounting purposes, and the physical flow of materials does not necessarily need to follow this sequence, although in practice, older materials are typically used before newer ones.

Advantages:

1. **Efficient Inventory Management:** The older stock is used first, which ensures the inventory comprises the most recent purchases, reducing the risk of obsolescence.
2. **Logical Approach:** It follows a natural order where older materials are generally used first.
3. **Easy to Understand:** This method is simple to grasp and implement.
4. **Supports Comparisons:** FIFO enables both inter-firm and intra-firm comparisons, making inventory valuation consistent.
5. **Consistent Valuation:** It aligns the valuation of inventory and the cost of finished goods with actual costs, offering a realistic perspective.

Disadvantages:

1. **Not Reflective of Current Prices:** The production costs may not reflect current market prices if there is a significant time gap between purchases and usage.
2. **Understated Production Costs in Rising Prices:** When prices are rising, production costs may be understated. However, a high inventory turnover can help reflect current prices quickly.
3. **Complex Calculations:** When multiple purchases are made at different prices, calculating the cost can become complicated.
4. **Difficulties with Material Returns:** Pricing material returns can be challenging under this method.
5. **Inflation Issues:** During inflationary periods, replacing materials can be problematic, as FIFO does not account for rising prices in replacements.
6. **Multiple Prices for a Single Issue:** Sometimes, more than one price needs to be applied for a single material issue, complicating the process.
7. **Complicated Cost Comparisons:** Comparing costs between production batches becomes difficult when different prices are applied to issues.

Example 1:

Let's assume the following transactions:

Receipts:

- ❑ 10th Nov: 400 kgs @ ₹4.80 per kg
- ❑ 15th Nov: 300 kgs @ ₹5.20 per kg

Issues:

- ❑ 18th Nov: Issue of 550 kgs
 - ◆ 400 kgs @ ₹4.80 per kg
 - ◆ 150 kgs @ ₹5.20 per kg

In this example, materials are received in two batches at different prices. When 550 kgs are issued, the FIFO method dictates that 400 kgs are issued at ₹4.80 per kg (the oldest batch), and the remaining 150 kgs are issued at ₹5.20 per kg (the next batch), ensuring that older materials are used first.

2.7.2 LAST IN FIRST OUT METHOD

This method's core premise posits that inventory is valued at the costs of the oldest acquisitions, while supplies issued for production are drawn from the most recent purchases. Known as the replacement cost method, it uses the current material cost to determine output and is an essential way for matching revenue to costs for determining income.

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Advantages:

1. It is straightforward and beneficial when transactions are minimal.
2. It can provide tax advantages.
3. This method is systematic and aligns current costs with current revenues more effectively.
4. It shows real income more accurately during inflationary periods.
5. It reduces unrealized inventory gains and losses, which stabilizes reported operating profits—especially in sectors of the economy where price swings can be severe.

Disadvantages:

1. The method can become complicated when material prices fluctuate widely.
2. More than one price may need to be used for a single issue.
3. Variations in batch costs make it harder to compare costs between firms or within the same firm.
4. Inventory adjustments are necessary during periods of falling prices.
5. Matching current costs with current revenues becomes difficult unless purchases and sales are of equal quantities.
6. Companies may time purchases to influence costs and alter reported income at will.
7. Changes in accounting methods may affect profit-sharing and bonuses, leading to confusion among employees.

Example 1:

Let's assume the following transactions:

Receipts:

- 12th Nov: 400 kgs @ ₹6.00 per kg
- 15th Nov: 300 kgs @ ₹5.50 per kg

Issues:

- 20th Nov: Issue of 550 kgs
 - ◆ 300 kgs @ ₹5.50 per kg (latest batch)
 - ◆ 250 kgs @ ₹6.00 per kg (remaining from the earlier batch)

In this example, under the LIFO method, the most recent purchase (₹5.50 per kg) is used first, followed by the earlier purchase (₹6.00 per kg) to fulfill the material issue. The remaining stock in inventory would be valued at the oldest price.

2.7.3 HIGHEST IN FIRST OUT METHOD

The most costly resources are issued first, and the remaining stock is evaluated at the lowest feasible price, according to the Highest-in First-out (HIFO) Method. Due to frequent fluctuations in material costs, this method

requires meticulous record-keeping. It is often used in scenarios involving monopoly products or cost-plus contracts. By undervaluing the remaining stock, this method creates hidden reserves that are not immediately visible in financial statements.

Advantages:

1. **Maximizes cost recognition:** By issuing the most expensive materials first, this method helps to recognize higher costs in financial statements, potentially reducing taxable income.
2. **Conservative valuation:** It values inventory at the lowest cost, providing a conservative estimate of stock on hand and minimizing the chance of overvaluation.
3. **Cost-plus contract efficiency:** This method is particularly useful for cost-plus contracts, ensuring the highest material costs are reflected for billing purposes.
4. **Hidden reserve creation:** The undervaluation of inventory contributes to creating hidden reserves, which can be used in future periods without affecting visible profits.

Disadvantages:

1. **Complex record-keeping:** Due to frequent material price fluctuations, this method demands detailed record maintenance, leading to increased administrative workload.
2. **Unrealistic inventory valuation:** Valuing inventory at the lowest cost may provide an inaccurate representation of the actual stock value, which may not align with market trends.
3. **Potential financial distortion:** The hidden reserve can distort the company's true financial position, potentially misleading stakeholders.
4. **Limited practical use:** This method is not widely used for regular accounting purposes as it may misrepresent the business's actual cost structure to external users.

Example 1:

Consider the following transactions:

Receipts:

- ❑ 10th Dec: 400 kgs @ ₹7.50 per kg
- ❑ 14th Dec: 600 kgs @ ₹5.00 per kg

Issues:

- ❑ 18th Dec: 500 kgs issued
- ❑ 400 kgs @ ₹7.50 per kg (most expensive batch)
- ❑ 100 kgs @ ₹5.00 per kg

In this scenario, under the FIFO method, the costliest materials (₹7.50 per kg) are issued first, followed by the cheaper materials (₹5.00 per kg). The remaining inventory is valued at the lower cost of ₹5.00 per kg.

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2.7.4 AVERAGE COST METHOD

The simple average method determines the average price of materials without considering the quantities involved. This approach is appropriate when prices remain relatively stable, stock purchases are made in equal quantities, or the overall stock value is relatively low. The average is calculated by dividing the total prices of the materials by the number of price entries, with a new average established with each new receipt of materials.

Example 1:

Receipts:

- ❑ 12th Nov: 300 kgs @ ₹6.00 per kg
- ❑ 16th Nov: 400 kgs @ ₹7.00 per kg

To calculate the simple average price:

$$(\text{₹}6.00 + \text{₹}7.00) / 2 = \text{₹}6.50 \text{ per kg}$$

In this example, two batches of materials are received at different prices—₹6.00 and ₹7.00 per kg. Using the simple average method, the average price is calculated as ₹6.50 per kg by adding the two prices and dividing by the number of price entries, without considering the quantities purchased.

2.7.5 WEIGHTED AVERAGE COST METHOD

In this method, the calculation of the average price takes into account both the total quantities and total costs. The weighted average price is determined after each purchase by adding the newly received quantity to the existing stock and incorporating the cost of the new purchase into the total stock cost. The total cost is then divided by the total quantity to derive the weighted average price. This approach helps to mitigate price fluctuations, simplifies calculations, and provides a more stable figure for inventory valuation.

Advantages

- ❑ It maintains logical consistency.
- ❑ Variations in prices do not influence material issuance or inventory levels.
- ❑ The values genuinely represent the actual costs.

Disadvantages

- ❑ It requires a significant amount of administrative work.
- ❑ The procedure is complicated and less convenient by frequent price variations.
- ❑ It might not be realistic because it doesn't fairly represent the true cost.

Example 1:

Consider the following transactions:

Receipts:

- ❑ 5th Jan: 200 kgs @ ₹4.00 per kg
- ❑ 10th Jan: 300 kgs @ ₹6.00 per kg

To calculate the weighted average price:

$$(\text{₹}4.00 \times 200 \text{ kg} + \text{₹}6.00 \times 300 \text{ kg}) / (200 \text{ kg} + 300 \text{ kg}) = \text{₹}5.20 \text{ per kg.}$$

In this example, two batches of materials are received at different prices—₹4.00 per kg for 200 kgs and ₹6.00 per kg for 300 kgs. The weighted average price is calculated by taking the total cost of each batch (₹4.00 × 200 kg and ₹6.00 × 300 kg), adding these amounts together, and then dividing by the total quantity received (200 kg + 300 kg). This results in a weighted average price of ₹5.20 per kg, which provides a more accurate reflection of the cost per unit considering both the quantities and prices of the materials received.

2.7.6 PERIODIC AVERAGE COST METHOD

Certain companies might determine material prices by averaging the costs of all receipts over a specified period, such as a week or a month, for use in the following period. Only the prices applicable to that particular timeframe are considered, which includes purchases made during that period and the closing stock.

Advantages

- ❑ Because it eliminates the need to compute the issue price following each receipt, it is simple to implement.
- ❑ This approach works particularly well for continuous process costing, in which the total cost of manufacturing big quantities of goods is equal to the sum of all individual orders.

Disadvantages

- ❑ The pricing of issues does not take into account notable price shifts that have occurred recently.
- ❑ It is not regarded as an exact cost method.
- ❑ It entails a considerable amount of clerical work.

Example:

Suppose the receipts during the month were priced at ₹3.00, ₹4.50, ₹5.25, and ₹6.75. The periodic simple average would be calculated as follows:

$$\text{Total Prices of Materials} = \text{₹}3.00 + \text{₹}4.50 + \text{₹}5.25 + \text{₹}6.75 = \text{₹}19.50$$

$$\text{Total Number of Prices} = 4$$

$$\text{Periodic Simple Average} = \text{₹}19.50 / 4 = \text{₹}4.88.$$

In this example, four batches of materials are received at different prices—₹3.00, ₹4.50, ₹5.25, and ₹6.75. The periodic simple average is calculated by summing the total prices (₹19.50) and dividing by the number of price points (4), resulting in an average price of ₹4.88 per unit. This method provides a straightforward way to determine the average material cost over the specified period.

2.7.7 STANDARD COST METHOD

For a predetermined period, the price for material concerns is predetermined taking into account variables like market trends and

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transportation expenses. Every material has a standard price, which should be checked on a regular basis. All problems and inventories are recorded at these standard rates. Standards come in two flavors: basic and contemporary. While the current standard modifies product pricing to match current market conditions, the basic standard stays constant over extended periods of time and represents the optimum price, assisting in forward planning. On the other hand, the fundamental standard makes it easier to analyze historical production cost trends. The purchase price variance account records any differences between the standard and actual costs.

Example: Imagine a textile company that relies on cotton as its primary raw material. The company sets a standard price for cotton at ₹400 per kg, based on market forecasts, historical prices, and expected transportation costs. However, due to a sudden rise in global demand, the company ends up purchasing cotton at ₹450 per kg. During the period, the company purchases 2,000 kg of cotton.

To record the transaction, the cost of materials is calculated using the standard price:

$$\begin{aligned}\text{Total Recorded Cost} &= \text{Standard Price} \times \text{Quantity Purchased} \\ &= ₹400 \times 2,000 = ₹8,00,000\end{aligned}$$

Next, the Purchase Price Variance (PPV) is calculated as the difference between the actual purchase cost and the standard cost:

$$\begin{aligned}\text{Purchase Price Variance} &= (\text{Actual Price} - \text{Standard Price}) \times \text{Quantity Purchased} \\ &= (₹450 - ₹400) \times 2,000 = ₹1,00,000\end{aligned}$$

This ₹1,00,000 unfavorable variance is recorded in the Purchase Price Variance Account, highlighting that the actual cost of materials exceeded the standard due to changes in market conditions.

Advantages:

- ❑ Since only quantities need to be recorded, accounting is made simpler.
- ❑ The adoption of a single rate minimizes inconsistencies.
- ❑ It helps determine purchasing efficiency; it indicates poor purchasing efficiency if actual costs are higher than standard costs, and vice versa.
- ❑ The method is easy to implement.
- ❑ It gives the costing system stability.

Disadvantages:

- ❑ It is only intended to be a target and does not represent actual or projected expenditures.
- ❑ Establishing a standard price can be challenging when prices experience frequent fluctuations.

2.7.8 MARKET PRICE

Materials can be valued according to market price on the basis of two methods show in **Figure 2.5**

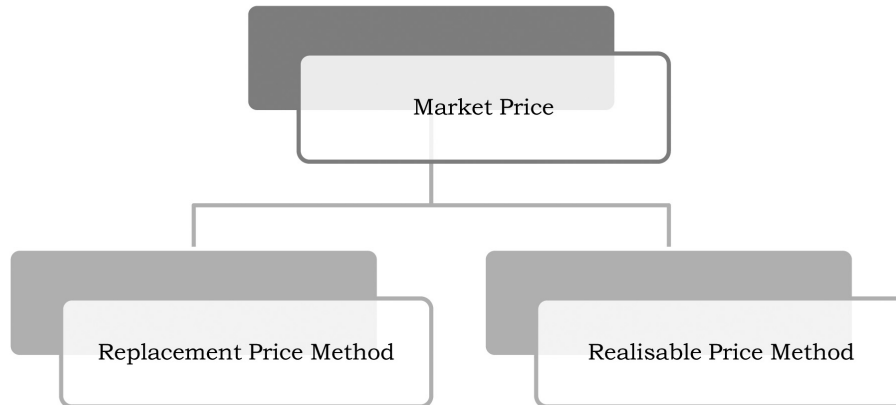


Figure 2.5 Methods under Market Price.

Replacement Price Method

The replacement price method determines the value of materials issued by referencing the cost of acquiring the same type of materials from the market at a specific time. This approach is especially beneficial for accurately calculating production costs and valuing material issues during times of rising prices, as it ensures that the cost of materials included in production reflects potential replacement costs at higher prices. The replacement price method is applicable to items used in manufacturing.

Example: A furniture manufacturing company relies on wood as a key raw material. To manage fluctuations in wood prices, the company uses the replacement price method to value issued materials. At the beginning of the month, the company has 100 cubic feet of wood in stock, purchased earlier at ₹400 per cubic foot. The current market price for wood rises to ₹450 per cubic foot, and the company issues 50 cubic feet for production. Under the replacement price method, the issue price per unit is set at ₹450. Hence, the total cost of issued wood is calculated as:

$$\text{Total Issue Cost} = 50 \text{ cubic feet} \times ₹450 = ₹22,500$$

The remaining 50 cubic feet of wood is still valued at the original purchase price of ₹400 per cubic foot.

$$\text{Value of Remaining Inventory} = 50 \text{ cubic feet} \times ₹400 = ₹20,000$$

- ❑ By using the current market price (₹450) for issued materials, the company ensures that the production cost reflects the replacement cost rather than historical costs. This is particularly useful in times of rising prices.
- ❑ The company does not account for the actual purchase price of the wood in stock (₹400), which could result in discrepancies when calculating inventory values.

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Advantages:

- ❑ **Accurate Measurement:** This method accurately measures results by matching current revenues with current costs.
- ❑ **Differentiation of Gains:** It provides a more transparent financial picture by differentiating between holding gains and operating gains.
- ❑ **Competitive Pricing:** The replacement cost makes it possible to calculate a reasonable and appealing selling price.

Disadvantages:

- ❑ **Market Price Dependency:** The method cannot establish a replacement price in the absence of a market price.
- ❑ **Accounting Complexity:** Due to the fact that it is not based on actual expenses, accounting may become confusing and complex.

Realizable Price Method

The realizable price is the market value at which the material intended for issuance can be sold. This price can vary, potentially being higher or lower than the initial purchase cost. Like the replacement price method, this approach allows the stores ledger to indicate profit or loss, offering valuable insights into the financial implications of inventory management.

**SELF-ASSESSMENT QUESTIONS**

3. A company uses the FIFO (First-In-First-Out) method for inventory valuation. Below is the information regarding purchases and issues of a particular material:

Date	Purchase Quantity	Purchase Price
Jan 1	100 units	₹5 per unit
Jan 15	150 units	₹6 per unit

On Jan 20, the company issues 120 units for production. What will be the cost of goods issued using the FIFO method?

- a. ₹600 b. ₹640
c. ₹620 d. ₹720

4. A company maintains the following records for material Y during a particular period:

Date	Purchases	Units Purchased	Purchase Price (per unit)
Jan 1	Opening Stock	200 units	₹10 per unit
Jan 10	Purchase	100 units	₹12 per unit
Jan 20	Purchase	150 units	₹14 per unit
Jan 30	Purchase	50 units	₹15 per unit

The company uses the Weighted Average Cost (WAC) method for inventory valuation. On Jan 25, it issues 180 units for production. What is the total cost of goods issued using the Weighted Average Cost method?

- | | |
|-----------|-----------|
| a. ₹2,120 | b. ₹2,130 |
| c. ₹2,178 | d. ₹2,278 |
5. Which disadvantage applies to the replacement price method?
- It simplifies accounting by recording only quantities.
 - It cannot establish prices without a market value.
 - It minimizes price fluctuations for material issues.
 - It reflects bulk purchase reductions in the pricing.

2.8 QUANTITY OF MATERIAL PROCUREMENT

Inventory management is like walking a tightrope—you need just the right balance to keep operations running smoothly without overspending. To achieve this, both top management and the materials department must work in harmony. While top management sets the financial boundaries, the materials department ensures that these resources are allocated efficiently across different items.

Let's explore some key factors that influence stock levels and how they contribute to effective inventory control.

- Lead Time for Deliveries:** Imagine you're waiting for a crucial material to arrive. Knowing how long it takes for deliveries to reach you helps you plan ahead. Misjudging lead time could lead to disruptions in production—a scenario no manager wants to face.
- Rate of Consumption:** How fast are materials being used? Understanding the consumption rate helps you anticipate when you'll need to restock. This insight ensures that production never comes to a halt due to unexpected shortages.
- Financial Requirements:** Inventory isn't just about materials; it's also about money. Ensuring that funds are available when needed prevents delays in procurement and helps avoid tying up too much working capital in inventory.
- Storage Considerations:** Storage isn't free, and it's not limitless. Costs for warehousing, maintaining ideal conditions, and handling materials can add up quickly. Efficient storage strategies can minimize these expenses while making the best use of available space.
- Deterioration and Shelf Life:** Not all materials last forever. Some may spoil, evaporate, or degrade over time. Knowing the lifespan of materials ensures you don't overstock items that might go to waste.
- Price Fluctuations:** Market prices rarely stay stable. Monitoring these changes allows you to make informed purchasing decisions. Buying at the right time can save costs and boost profitability.
- Seasonal Factors:** Some materials are cheaper or more readily available during certain seasons. Planning purchases around these trends can give you a financial edge and ensure uninterrupted production.
- Risk of Obsolescence:** Technology and trends change quickly. Materials that are useful today may become outdated tomorrow. By

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assessing this risk, you can avoid overinvesting in items that might lose their relevance.

9. **Economic Order Quantity (EOQ):** EOQ is the sweet spot for order size—it minimizes the total cost of ordering and holding inventory. Think of it as the formula that helps you spend less while ensuring you always have what you need.
10. **Compliance with Regulations:** Government policies and statutory restrictions can impact how much of a material you can stock. Staying informed and compliant ensures smooth operations and avoids legal complications.

2.8.1 ECONOMIC ORDER QUANTITY

The ideal order size to reduce the overall expenses of placing and maintaining inventory is known as the Economic Order Quantity (EOQ). It represents the most efficient quantity of stock to purchase, balancing the costs of ordering and holding materials.

Factors Influencing EOQ:

Ordering Cost

Ordering cost refers to the expenses incurred during the process of purchasing materials or goods needed for production. These include costs such as tender invitations (soliciting bids from suppliers), preparing purchase orders (administrative work), salaries of procurement staff involved in managing the purchase process, transportation expenses for delivering goods from suppliers, and inspection costs to verify the quality of received materials. Managing these costs is crucial to ensure procurement efficiency without overspending.

Carrying Cost

Carrying cost, also known as holding cost, represents the expenses associated with storing and maintaining inventory over a period of time. It includes the cost of capital tied up in inventory (opportunity cost of funds invested), storage expenses like warehouse rent and utilities, insurance to safeguard against risks such as theft or damage, and costs related to inventory deterioration or obsolescence if products have a shelf life. Proper management of carrying costs helps businesses strike a balance between maintaining sufficient inventory to meet demand and minimizing excess storage expenses.

Formula: The formula for calculating EOQ is given by: $EOQ = \sqrt{\frac{2AO}{C}}$

Where:

- ❑ A = Annual usage units
- ❑ O = Ordering cost per order
- ❑ C = Annual carrying cost of one unit (calculated as the carrying cost percentage multiplied by the cost of one unit)

Assumptions Underlying EOQ:

The calculation of the Economic Order Quantity is based on several assumptions:

1. The ordering cost per order and the carrying cost per unit per annum are known and remain fixed.
2. The anticipated usage of materials in units is known and consistent.
3. The cost per unit of the material is constant and known.
4. The quantity of material ordered is received instantaneously, meaning the lead time is considered to be zero.

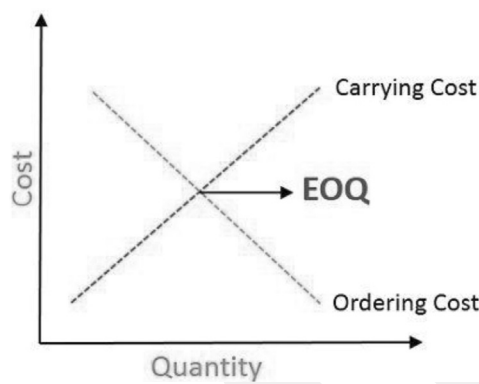


Figure 2.6 Graph of Economic Order Quantity.

Source: <https://businessjargons.com/wp-content/uploads/2020/06/EOQ-graph.jpg>

The graph (**Figure 2.6**) is a representation of the Economic Order Quantity (EOQ) model, where,

- ❑ **X-axis (Quantity):** This axis represents the quantity of inventory ordered each time.
- ❑ **Y-axis (Cost):** This axis represents the total costs associated with ordering and holding inventory.
- ❑ **Carrying Cost (Red Line):** This line shows the cost of holding or storing inventory. As the quantity of inventory increases, the carrying cost increases because more resources are needed to store larger quantities.
- ❑ **Ordering Cost (Blue Line):** This line represents the cost incurred for placing orders. As the quantity ordered increases, the frequency of placing orders decreases, leading to lower ordering costs. Hence, the ordering cost decreases as quantity increases.
- ❑ **EOQ (Economic Order Quantity):** The point where the carrying cost and ordering cost lines intersect represents the EOQ. At this point, the total cost (which is the sum of both ordering and carrying costs) is minimized. EOQ is the optimal order quantity that minimizes the total cost associated with inventory management.

Example 1: Calculate the Economic Order Quantity (EOQ) using the following details and determine how many orders should be placed in a year:

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- ❑ Annual consumption of materials: 8,000 kg
- ❑ Cost per order: ₹40
- ❑ Cost per kg of raw materials: ₹3
- ❑ Storage costs: 10% on average inventory

Solution:

We can calculate EOQ using the formula: $EOQ = \sqrt{\frac{2AO}{C}}$

Here:

A = 8,000 kg (annual consumption)

O = ₹40 (ordering cost per order)

C = 10% of ₹3 (storage cost per unit per year) = ₹0.30

Now, substituting these values into the EOQ formula:

$$EOQ = \sqrt{\frac{2 \times 8000 \times 40}{0.30}} = \sqrt{\frac{6,40,000}{0.30}} = \sqrt{21,33,333.33} = 1,460.99 \text{ kg}$$

So, the EOQ is approximately 1,461 kg.

Number of orders to be placed in a year:

Number of orders per year =

$$\frac{\text{Total consumption of materials per annum}}{EOQ} = \frac{8,000}{1,461} \approx 5.48$$

Thus, around **5 to 6 orders per year** would need to be placed.

**SELF-ASSESSMENT QUESTIONS**

6. What is the main objective of calculating the Economic Order Quantity (EOQ)?
 - a. To maximize the quantity of orders
 - b. To minimize total inventory costs
 - c. To minimize ordering costs only
 - d. To minimize the lead time of delivery
7. XYZ Company purchases 600 boxes of materials in each order, which constitutes a 4-month supply. The cost per box is ₹150, the ordering cost per order is ₹200, and the inventory carrying cost is 25% of the unit cost. Calculate the total annual cost of the current inventory policy.

a. ₹10,200	b. ₹11,850
c. ₹12,300	d. ₹13,500

2.9 MATERIAL MANAGEMENT AT STORES

Material management at stores encompasses the systematic processes of receiving, storing, and issuing materials to ensure their safe custody

until they are needed by production and other departments. Key components of material management at the store:

- ❑ **Storage Location:** Materials are stored in a designated area known as a “store,” organized for easy access and management.
- ❑ **Personnel:** The storekeeper manages the store department, overseeing the receipt, storage, and issuance of materials under the supervision of a store manager.
- ❑ **Requisition Management:** The storekeeper issues purchase requisitions to the purchasing department when materials reach the reorder level, ensuring timely stock replenishment.
- ❑ **Receiving Materials:** Materials received from the Goods Receiving or Inspection Department are checked against the Goods Received Note or Goods Inspection Report for accuracy.
- ❑ **Record Maintenance:** Accurate records of all transactions, including receipts, issues, and balances, are maintained for effective inventory management.
- ❑ **Storage Facilities:** Adequate and appropriate storage is provided for every item, ensuring suitable conditions to prevent damage.
- ❑ **Organization:** Materials are properly arranged in designated locations for easy access, improving operational efficiency.
- ❑ **Condition Management:** Materials are kept in good condition to avoid issues like pilferage, waste, and deterioration through regular inspections.
- ❑ **Issuance Process:** Materials are issued only against valid store requisitions, ensuring accountability in inventory management.
- ❑ **Stock Level Monitoring:** Regular checks are conducted to ensure stock levels remain within specified limits to prevent overstocking and stock-outs.
- ❑ **Physical Stock Taking:** Periodic physical counts are conducted to compare recorded balances and identify discrepancies.
- ❑ **Obsolete Item Identification:** Slow-moving and obsolete items are identified and reported to management for inventory optimization.
- ❑ **Equipment Accuracy:** Measuring equipment, such as balances and scales, is periodically verified for accuracy to ensure reliable measurements.
- ❑ **Cost Minimization:** Strategies are implemented to minimize handling, storage, and maintenance costs associated with material management.

2.10 INVENTORY CONTROL TECHNIQUES

The main objective of inventory control is to achieve maximum efficiency in production and sales while reducing the investment in inventory. Inventory includes stocks of raw materials, stores and consumables, work-in-progress, and finished goods. The techniques typically utilized for inventory control include the following shown in **Figure 2.7**:

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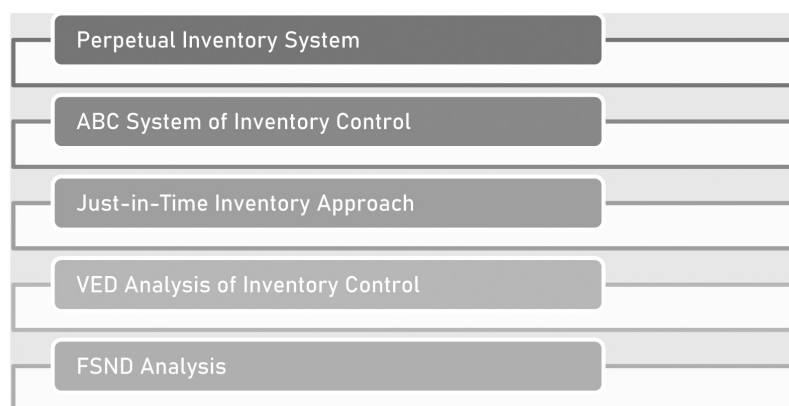


Figure 2.7 Inventory Control Techniques.

2.10.1 PERPETUAL INVENTORY SYSTEM

The perpetual inventory system is intended to aid in material control. It functions as a stock control method used by the stores department, utilizing a recording technique that ensures information about every receipt, issue, and the current stock balance is always accessible.

This system calculates the current balance by documenting each receipt and issue of materials through stock records. It is important to note that perpetual inventory is usually verified through a continuous stock-taking program. While perpetual inventory pertains to the record-keeping system, continuous stock-taking involves physically checking these records against the actual stock.

Components of the Perpetual Inventory System

1. **Comparison of Bin Cards and Stores Ledger Accounts:** Bin cards are maintained by the storekeeper, while the stores ledger account is managed by the stores accountant. Each inventory item is recorded in both locations simultaneously, and generally, the balances in these two records align. However, discrepancies may occur due to several reasons: omission of an item in either the bin card or the stores ledger account, incorrect posting of an item in either record, or arithmetic errors in calculating their balances. To ensure accurate figures, it is essential to frequently reconcile the balances of these two records.
2. **Physical Stock Verification:** Without a methodical process for physically verifying inventory, the perpetual inventory system is insufficient. It is necessary to physically verify the balances on the bin card or shops ledger account to ensure their accuracy. This verification can take place in two primary ways:
 - ❑ **Periodic Stock Verification:** Physical stock verification is usually conducted as part of this system on a recurring basis, perhaps once or twice a year. Using this method, the worth of the stock is ascertained on a specified day, typically at the end of the year, by physically counting the goods.

- ❑ **Continuous Stock Verification:** In order to guarantee that every inventory item gets examined more than once a year, this approach comprises the random counting and verification of a certain number of items every day of the year. Only on the day of verification is the storekeeper notified of which merchandise needs to be verified.

Advantages of the Perpetual Inventory System

1. Errors and fraudulent activities can be swiftly detected, aiding in prevention.
2. Enhanced management of all receipts and issues provides a comprehensive overview of stock quantities and values at any time.
3. Physical verification of stock is conducted on a planned and regular basis, ensuring uninterrupted production.
4. Simultaneous recording in the bin cards and stores ledger accounts serves as an internal check for prompt error identification.
5. Continuous comparison of actual stock with minimum and maximum levels helps maintain material investments at a minimum.
6. Losses from shrinkage, evaporation, accidents, fire, theft, and other factors can be easily detected.
7. Ongoing stock-taking encourages both the storekeeper and stores accountant to maintain accurate and up-to-date records.
8. Simplifies the preparation of periodic profit and loss accounts and balance sheets without requiring physical stock-taking.
9. Accurate stock data is readily available for settling insurance claims.
10. Helps mitigate issues associated with excessive stock, such as loss of interest on capital tied up in inventory, deterioration losses, and risks of obsolescence.
11. Distributing work throughout the year allows for the employment of full-time specialized personnel.

2.10.2 ABC SYSTEM OF INVENTORY CONTROL

The ABC analysis methodology is rooted in the need for effective material control, particularly in industrial environments where numerous parts and materials are involved in production.

The foundation of ABC analysis is the categorization of materials for focused control. This process entails assessing the monetary value (cost significance) of each material item in relation to the total cost and material value of the inventory. The core principle is that management should assess each stock item based on its usage, lead time, technical challenges, and its financial impact on the overall inventory investment. Consequently, high-value items are given careful scrutiny, whereas low-value items necessitate minimal resources and effort for inventory management. **Figure 2.8** shows the graph on ABC analysis, showing different categories.

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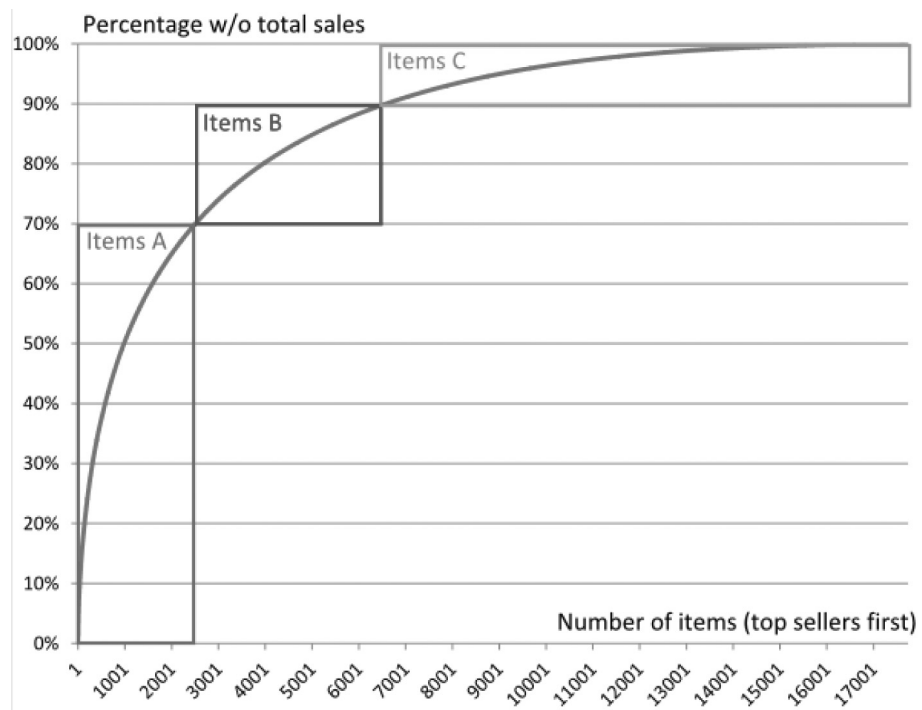


Figure 2.8 ABC Analysis Graph.

Categories of Items

1. 'A' Category Items:

- ❑ Let's say it affects 10% of the total things that the stores handle; these items are highly expensive, making up around 70% of the value of the entire inventory. This large expenditure could be the result of high costs, heavy use, or both.
- ❑ Effective control measures are essential for this category to avoid overstocking or shortages in production. The management of A items involves meticulous planning through budgeting for total material requirements. To maintain control, specific inventory levels are established, including **maximum, minimum, and reorder levels**.

2. 'B' Category Items:

- ❑ Say this category includes relatively less critical items, making up around **20% of the total items** but requiring only about **20% of the overall investment** in inventories.
- ❑ Given the fair sum involved, the control applied to these items does not need to be as rigorous as for A items. Orders for B category items can be placed after periodic reviews of their status.

3. 'C' Category Items:

- ❑ Say it represents about **70% of total items** but only requiring around **10% of the total inventory value**, these items do not necessitate constant control.

- ❑ Orders for products in the C category may be placed once or twice a year, contingent on consumption requirements. Reducing the cost of ordering and handling is the main objective here.

Steps in Conducting ABC Analysis

To implement ABC analysis effectively, the following steps shown in **Figure 2.9** should be undertaken:

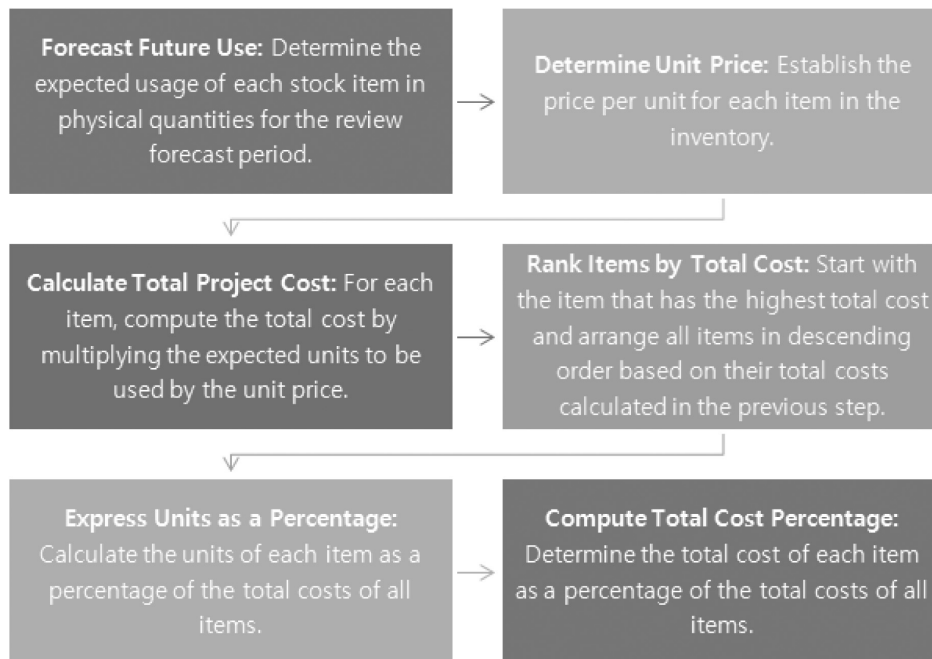


Figure 2.9 Steps in conducting ABC.

Advantages of ABC Analysis

- 1. Continuity in Production:** By ensuring that there is no risk of production interruptions due to a shortage of materials or stock, ABC Analysis promotes minimum investment in inventories. This strategic approach maintains the necessary levels of materials to support continuous production.
- 2. Lower Costs:** The overall costs associated with placing orders, receiving goods, and maintaining inventory are minimized, particularly when the ABC system is integrated with the determination of proper economic order quantities (EOQ). This optimization leads to more cost-effective inventory management.
- 3. Reduced Management Attention Required:** The ABC system allows management to focus their attention on a limited number of critical items rather than all items in inventory. This prioritization saves time and resources, allowing management to allocate their efforts more effectively.
- 4. Systematic Working:** The implementation of the ABC system enables much of the purchasing process to be organized in a

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systematic manner. This structure allows routine tasks to be delegated to subordinate staff, streamlining operations and enhancing productivity.

Example:

A company manufactures 10,000 different products and uses a variety of inventory items. The following data provides details regarding inventory holding and usage:

No. of Varieties of Inventory	% of Total Varieties	% Value of Inventory Holding (Average)	% Inventory Usage (in End-Product)
25	0.25%	60%	75%
500	5%	30%	15%
9,475	94.75%	10%	10%
Total	100%	100%	100%

Task:

Classify the items of inventory into A, B, and C categories using ABC Analysis with reasons.

Solution:

Step 1: Classify items into categories

Category A:

- ❑ Number of Items: 25
- ❑ Reason:
 - ◆ These items constitute only 0.25% of the total inventory varieties, the smallest portion.
 - ◆ However, they account for 60% of the inventory holding value, the largest proportion.
 - ◆ Their usage is significant, representing 75% of the total inventory usage in end-products.

Category B:

- ❑ Number of Items: 500
- ❑ Reason:
 - ◆ These items constitute 5% of the total inventory varieties, a moderate portion.
 - ◆ They account for 30% of the inventory holding value, requiring moderate investment.
 - ◆ Their usage in end-products is 15%, which is moderate.

Category C:

- ❑ Number of Items: 9,475
- ❑ Reason:

- ◆ These items make up the majority, 94.75% of the total inventory varieties.
- ◆ They account for only 10% of the inventory holding value, the smallest portion.
- ◆ Their usage in end-products is minimal, just 10%.

2.10.3 JUST-IN-TIME INVENTORY APPROACH

Just-In-Time (JIT) is a production approach designed to enhance a business's return on investment by minimizing in-process inventory and associated carrying costs. Unlike traditional accounting, which views inventory as a value-adding asset, JIT considers it a cost or waste. The core philosophy of the JIT inventory system is to ensure "the right material, at the right time, at the right place, and in the exact amount" without the reliance on safety stock.

Advantages of the Just-In-Time System:

1. **Strengthened Supplier Relationships:** A company operating without inventory places a high value on its relationships with suppliers. Any disruptions in the supply chain that lead to part shortages can negatively impact production, making dependable suppliers essential.
2. **Consistent Supply Intervals:** Supplies are delivered at regular intervals throughout the production day, ensuring that inventory levels are aligned with production demand. This synchronization allows for optimal inventory levels at all times, thereby reducing the need for storage facilities, as parts move directly from delivery to assembly.
3. **Lower Working Capital Requirements:** Maintaining minimal inventory results in reduced working capital needs, freeing up financial resources for other business priorities.
4. **Decreased Storage Space Needs:** JIT minimizes the necessity for extensive storage facilities by keeping inventory levels to a minimum, which optimizes space utilization.
5. **Reduced Risk of Obsolescence or Damage:** With lower inventory levels, the likelihood of stock becoming obsolete or damaged is significantly lessened, resulting in cost savings and improved operational efficiency.

Example 1: Automobile Manufacturer

A car manufacturer integrates the Just-in-Time (JIT) inventory system to streamline the supply of essential parts such as engines, tires, and seats. In this approach, parts are delivered to the assembly line daily, synchronized with production schedules. For instance, engines are delivered at 8:00 AM for vehicles scheduled for assembly that same day, while tires are restocked each evening based on daily consumption. This system minimizes warehouse space requirements and reduces holding costs, as inventory is not stored in bulk. However, it introduces a risk of production delays if suppliers fail to deliver on time.

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2.10.4 VED ANALYSIS OF INVENTORY CONTROL

VED stands for Vital, Essential, and Desirable, and this analysis is primarily utilized for managing spare parts within an organization. Spare parts are categorized based on their criticality to production as follows:

1. **Vital:** Vital spares are those whose stock-out, even for a brief period, can halt production for an extended duration, resulting in substantial stock-out costs. For instance, in a car assembly company, the engine is considered a vital part; without it, the assembly process cannot commence.
2. **Essential:** Essential items are those whose absence can be tolerated for only a few hours or a day. The cost of lost production during this time is significant, making these parts critical for continuing operations. An example in a car assembly company would be tires; without tires, the assembly of the car cannot be completed.
3. **Desirable:** Desirable spares are parts that are needed but whose absence for a week or longer will not cause a production halt. For example, a CD player in a car assembly company falls into this category, as its lack does not immediately affect the production process.

It's important to note that some spare parts, even though they may have a low monetary value, can be vital for production and require consistent monitoring. If an organization employs ABC analysis, these critical spares might not receive the necessary attention, underscoring the importance of VED analysis for effective spare parts management.

Example: VED Analysis in a Hospital Pharmacy

A hospital pharmacy employs VED (Vital, Essential, Desirable) analysis to categorize its inventory of medicines and medical supplies, ensuring optimal stock levels based on their criticality to patient care. Vital items, such as life-saving drugs like insulin, emergency medications like epinephrine, and surgical supplies like oxygen cylinders, are indispensable and must always be available. These items are given top priority, with high stock levels maintained and close monitoring for timely replenishment. Essential items, including antibiotics, analgesics, and intravenous fluids, are important but not immediately life-saving. These are stocked based on average consumption, with a moderate safety margin to prevent shortages during fluctuating demand. Desirable items, such as multivitamins, skin creams, and other non-critical supplies, are kept at lower stock levels since their absence has minimal impact on patient care.

By implementing VED analysis, the hospital ensures efficient resource allocation, prioritizing critical items to reduce the risk of shortages while lowering storage costs for non-essential supplies. This approach enhances decision-making and maintains a balance between cost efficiency and service quality. However, it also poses challenges, such as the risk of misclassifying items, which could disrupt patient care, and the need for reliable suppliers to ensure the uninterrupted availability of “Vital” and “Essential” items.

2.10.5 FSND ANALYSIS

FSND analysis is a technique utilized in inventory management that classifies items according to their movement and lifecycle. This strategy assists businesses in managing their inventory more efficiently by pinpointing which items need greater attention and which can be lowered in priority. The categories are as follows:

Fast-moving (F): Fast-moving items are those that sell rapidly and are in high demand. These items usually have a brief turnover period, indicating they are replenished frequently. Companies should ensure that these items are consistently in stock to satisfy customer demand and prevent lost sales. Effective inventory control and ongoing monitoring are crucial for the efficient management of fast-moving items.

Slow-moving (S): Slow-moving items have a longer turnover period and are not sold as frequently. Although they remain important to have available, businesses must analyze why these items are not selling quickly. Possible reasons could include overstocking, reduced demand, or shifts in consumer preferences. Companies might consider implementing promotional strategies or discounts to boost sales of slow-moving items.

New (N): New items are those that have been recently added to the inventory. These products require careful monitoring to evaluate their performance and market acceptance. Businesses should analyze sales trends and gather customer feedback to decide whether to continue stocking these items or make adjustments based on their performance.

Dead (D): Dead items are products that have not sold for a prolonged period and are essentially considered obsolete. These items occupy valuable inventory space and resources. Companies should explore strategies for liquidating dead stock, such as clearance sales or donations, to make room for more profitable items.

Benefits of FSND Analysis:

- ❑ FSND analysis aids in maintaining optimal stock levels by differentiating item types for better procurement decisions.
- ❑ Identifying dead and slow-moving items helps businesses lower carrying costs and minimize unnecessary inventory investments.
- ❑ Prioritizing fast-moving items ensures popular products are readily available, boosting customer satisfaction and sales.
- ❑ Analyzing new items regularly enables quick adaptation to market trends and consumer preferences, enhancing competitiveness.

Example: FSND Analysis in an Electronics Store

An electronics store employs FSND analysis to categorize its inventory, ensuring efficient management and better allocation of resources. Fast-moving items, such as smartphones, headphones, and chargers, are highly in demand and sell quickly. These items have a short turnover period and are restocked frequently to meet consistent customer demand.

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The store ensures they are always available to prevent losing sales opportunities. Slow-moving items, including advanced projectors, high-end gaming accessories, and niche gadgets, take longer to sell and require closer examination. The store analyzes potential reasons for their slower movement, such as reduced demand or customer preference changes, and may use promotional discounts to increase their sales.

New items, such as the latest models of smartwatches or recently launched home automation devices, are closely monitored for customer acceptance and sales performance. Feedback and sales data are collected to determine whether these products should be stocked further or adjusted based on market response. Dead items, such as outdated models of laptops or discontinued product lines, have been in the inventory for a prolonged period without selling. The store implements clearance sales or offers bundled discounts to liquidate these items and free up space for more profitable stock.

**SELF-ASSESSMENT QUESTIONS**

8. Which of the following is not an advantage of the Perpetual Inventory System?
 - a. Early detection of stock losses
 - b. Prevention of production interruptions
 - c. Reduced requirement for physical stock verification
 - d. Moral check on employees
9. The Just-in-Time (JIT) inventory approach emphasizes:
 - a. Holding large amounts of inventory to avoid stock-outs
 - b. Minimal in-process inventory to reduce carrying costs
 - c. Maximizing storage space to hold future orders
 - d. Increasing lead time to manage supplier relationships
10. VED analysis is primarily used to categorize:
 - a. Manufacturing materials based on their cost
 - b. Spare parts based on their criticality to production
 - c. Inventory based on movement speed
 - d. Products based on market demand

2.11 ADDITIONAL SOLVED PROBLEMS

1. The stock of material held on 1-4-2024 was 400 units @ ₹50 per unit. The following receipts and issues were recorded. You are required to prepare the Stores Ledger Account, showing how the values of issues would be calculated both through FIFO AND LIFO.

Receipts and Issues:

- ❑ 2-4-2024: Purchased 100 units @ ₹55 per unit
- ❑ 6-4-2024: Issued 400 units

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- ❑ 10-4-2024: Purchased 600 units @ ₹55 per unit
- ❑ 13-4-2024: Issued 400 units
- ❑ 20-4-2024: Purchased 500 units @ ₹65 per unit
- ❑ 25-4-2024: Issued 600 units

Solution:**FIFO Method**

FIFO (First In, First Out) assumes that the oldest stock is issued first.

Date	Description	Units	Rate (₹)	Amount (₹)	Balance (Units)	Balance Amount (₹)
1-4-2024	Opening Stock	400	50	20,000	400	20,000
2-4-2024	Purchase	100	55	5,500	500	25,500
6-4-2024	Issue	400	50	20,000	100 (500-400)	5,500
10-4-2024	Purchase	600	55	33,000	700	38,500
13-4-2024	Issue	400	55	22,000	300 (700-400)	16,500
20-4-2024	Purchase	500	65	32,500	800	49,000
25-4-2024	Issue	600	300@55 300@65	36,000	200	13,000

LIFO Method

LIFO (Last In, First Out) assumes that the most recent stock is issued first.

Date	Description	Units	Rate (₹)	Amount (₹)	Balance (Units)	Balance Amount (₹)
1-4-2024	Opening Stock	400	50	20,000	400	20,000
2-4-2024	Purchase	100	55	5,500	500	25,500
6-4-2024	Issue	400	100@55 300@50	20,500	100	5,000
10-4-2024	Purchase	600	55	33,000	700	38,000
13-4-2024	Issue	400	55	22,000	300	16,000
20-4-2024	Purchase	500	65	32,500	800	48,500
25-4-2024	Issue	600	500@65 100@55	38,000	200	10,500

2. Prepare a statement showing the pricing of issues, on the basis of
 - a. Simple Average and
 - b. Weighted Average methods

From the following information pertaining to material-X:

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- ❑ March 1: Purchased 50 units @ ₹20 each
- ❑ March 5: Purchased 100 units @ ₹22 each
- ❑ March 10: Issued 80 units
- ❑ March 15: Purchased 70 units @ ₹23 each
- ❑ March 20: Issued 60 units

Solution:**a. Simple Average Method:**

Stores Ledger A/c

Date	Receipts			Issues			Balance	
	Qty.	Price (₹)	Value (₹)	Qty.	Price (₹)	Value (₹)	Qty.	Value (₹)
March 1	50	20	1,000				50	1,000
March 5	100	22	2,200				150	3,200
March 10				80	21 (22 + 20)/2	1,680	70	1,520
March 15	70	23	1,610				140	3,130
March 20				60	21.67 (20 + 22 + 23)/3	1,300	80	1,830

b. Weighted Average Method:

Stores Ledger A/c

Date	Receipts			Issues			Balance	
	Qty.	Price (₹)	Value (₹)	Qty.	Price (₹)	Value (₹)	Qty.	Value (₹)
March 1	50	20	1,000				50	1,000
March 5	100	22	2,200				150	3,200
March 10				80	21.33 (3200/150)	1,706.67	70	1,493.33
March 15	70	23	1,610				140	3,103.33
March 20				60	22.17 (3,103.33/140)	1,330	80	1,773.33

3. A company produces a custom product that requires a part called "Beta." The following data has been collected for the year 2024:

1. Annual demand for Beta: 10,000 units
2. Ordering cost: ₹300 per order
3. Cost per unit of Beta: ₹500
4. Carrying cost: 18% per annum

The company has received an offer of a 5% quantity discount if it orders 5,000 units of Beta at a time. **Required:**

- Calculate the economic order quantity (EOQ).
- Provide advice on whether the company should accept the quantity discount offer.

Solution:

a. Calculation of Economic Order Quantity (EOQ): $EOQ = \sqrt{\frac{2AO}{C}}$

Here:

A = 10,000

O = 300 per order

C = $0.18 \times ₹500 = ₹90$ per unit per year

$$EOQ = \sqrt{\frac{2 \times 10,000 \times 300}{90}} = \sqrt{\frac{60,00,000}{90}} = \sqrt{66,666.67} \\ = 258.20 \text{ units} \approx 258 \text{ units}$$

b. Evaluation of Profitability of Different Options of Order Quantity

❑ When EOQ is ordered

Purchasing Cost	$10,000 \times 500$	50,00,000
Ordering Cost	$(10,000/258) \times 300$	11,628
Carrying Cost	$(258/2) \times 90$	11,610
Total Cost		50,23,238

❑ When quantity discount is accepted

Purchasing Cost	$10,000 \times 475$	47,50,000
Ordering Cost	$(10,000/5,000) \times 300$	600
Carrying Cost	$(5,000/2) \times 90$	2,25,000
Total Cost		49,75,600

Since the total cost with the discount (₹4,975,600) is lower than the cost with EOQ (₹5,023,238), the company **should accept the quantity discount offer.**

- The annual demand for a product is 5,000 units. Each unit costs ₹8, and the inventory carrying cost is 22% per annum. The cost for each order is ₹200.

Required:

- Calculate the Economic Order Quantity (EOQ).
- Determine the number of orders per year.
- Calculate the time interval between two consecutive orders.

Solution:

Given data:

Annual demand (A) = 5,000 units

Unit cost = ₹8

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Carrying cost (C) = 22% of unit cost = $0.22 \times ₹8 = ₹1.76$

Procurement cost (O) = ₹200 per order

a. Calculate the Economic Order Quantity (EOQ):

The formula for EOQ is:

$$EOQ = \sqrt{\frac{2AO}{C}}$$

$$EOQ = \sqrt{\frac{2 \times 5,000 \times 200}{1.76}} = \sqrt{\frac{20,00,000}{1.76}} = \sqrt{11,36,363.63} = 1,066$$

b. Determine the number of orders per year:

$$\text{Number of orders per year} = \frac{A}{EOQ} = \frac{5,000}{1,066} = 4.69 \text{ orders per year}$$

Thus, the company places approximately **5 orders per year**.

c. Calculate the time interval between two consecutive orders:

The time between two orders is calculated as:

$$\text{Time between orders} = \frac{\text{Number of working days in a year}}{\text{Number of orders per year}} = \frac{365}{5} = 73$$

Thus, the time interval between two consecutive orders is approximately 73 days.

5. Beta Corporation purchases its annual requirement of 48,000 units in eight installments. Each unit costs ₹2, and the ordering cost is ₹50 per order. The inventory carrying cost is estimated at 18% of the unit value.

Required:

1. Calculate the total annual cost of the existing inventory policy.
2. Determine how much money can be saved by using the Economic Order Quantity (EOQ) model.

Solution:

Given Data:

Annual demand (A) = 48,000 units

Cost per unit = ₹2

Ordering cost (O) = ₹50 per order

Carrying cost (C) = 18% of unit value = $0.18 \times ₹2 = ₹0.36$

Current order size = $48,000/8 = 6,000$ units per order

Step 1: Total Annual Cost of Existing Policy

Purchasing Cost	$48,000 \times 2$	96,000
Ordering Cost	8×50	400
Carrying Cost	$(6,000 / 2) \times 0.36$	1,080
Total Cost		97,480

Step 2: Calculation of EOQ

$$\text{EOQ} = \sqrt{\frac{2AO}{C}}$$

$$\text{EOQ} = \sqrt{\frac{2 \times 48,000 \times 50}{0.36}} = \sqrt{\frac{48,00,000}{0.36}} = \sqrt{1,33,33,333.33}$$

$$= 3,651.48 \approx 3,651 \text{ units}$$

Step 3: Total Annual Cost Using EOQ

Purchasing Cost	$48,000 \times 2$	96,000
Ordering Cost	$(48,000/3651) \times 50$	657.35
Carrying Cost	$(3,651 / 2) \times 0.36$	657.18
Total Cost		97,314.53

Step 4: Savings by Using EOQ

$$\begin{aligned} \text{Savings} &= \text{Total cost under existing policy} - \text{Total cost using EOQ} \\ &= ₹97,480 - ₹97,314.53 \\ &= ₹165.47 \end{aligned}$$

Thus, the company can save **₹165.47** annually by using the EOQ model.

6. The Green Thumb Nursery is deciding on the economic order quantity (EOQ) for two brands of plant fertilizer: Grow Max and Earth Care. The following information is collected:

Fertilizer	Grow Max	Earth Care
Annual Demand	2,500 bags	1,500 bags
Ordering Cost per Purchase Order	₹1,000	₹900
Carrying Cost per Bag per Year	₹500	₹450

Required:

- Compute EOQ for Grow Max and Earth Care.
- For the EOQ, what is the sum of the total annual relevant ordering costs and total annual relevant carrying costs for Grow Max and Earth Care?
- For the EOQ, compute the number of deliveries per year for Grow Max and Earth Care.

Solution:

$$\text{EOQ Formula: } \text{EOQ} = \sqrt{\frac{2AO}{C}}$$

i. Calculation of EOQ

For Grow Max:

$$\text{EOQ} = \sqrt{\frac{2 \times 2,500 \times 1,000}{500}} = \sqrt{\frac{50,00,000}{500}} = 100 \text{ bags}$$

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For Earth Care:

$$EOQ = \sqrt{\frac{2 \times 1,500 \times 900}{450}} = \sqrt{\frac{27,00,000}{450}} = \approx 77.46 \text{ bags} \approx 77 \text{ bags}$$

ii. Total Annual Relevant Costs

For Grow Max:

- Ordering Cost: $(2,500/100) \times ₹1,000 = 25 \times ₹1,000 = ₹25,000$
- Carrying Cost: $100/2 \times ₹500 = 50 \times ₹500 = ₹25,000$
- Total Cost: $₹25,000 + ₹25,000 = ₹50,000$

For Earth Care:

- Ordering Cost: $(1,500/77) \times ₹900 = 19.48 \times ₹900 \approx ₹17,532$
- Carrying Cost: $77/2 \times ₹450 = 38.5 \times ₹450 = ₹17,325$
- Total Cost: $₹17,532 + ₹17,325 = ₹34,857$

iii. Number of Deliveries per Year

For Grow Max:

Number of Deliveries = Annual Demand/EOQ = $2,500/100 = 25$ deliveries per year

For Earth Care:

Number of Deliveries = Annual Demand/EOQ = $1,500/77 \approx 19.48$ deliveries per year ≈ 19 deliveries per year

- 7.** The following information is extracted from the stores ledger for Material Y:

Date	Particulars	Units	Rate (₹)
Feb. 1	Purchase	150	₹3 per unit
Feb. 15	Purchase	200	₹4 per unit
Feb. 18	Issue to Job Z21	100	—
Feb. 20	Issue to Job Z22	120	—

Required:

- 1.** Complete the receipts and issues valuation by adopting the following methods:
 - i.** First-In-First-Out (FIFO)
 - ii.** Last-In-First-Out (LIFO)
 - iii.** Weighted Average Method
- 2.** Tabulate the values allocated to Job Z21, Job Z22, and the closing stock under the three methods mentioned above.

Solution:

- i.** First-In-First-Out (FIFO) Method

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Date	Particulars	Receipts	Issues	Balance
		Units	Rate	Value
Feb. 1	Purchase	150	₹3	₹450
Feb. 15	Purchase	200	₹4	₹800
Feb. 18	Issue to Job Z21	100	₹3	₹300
Feb. 20	Issue to Job Z22	120	₹3 × 50 = 150 ₹4 × 70 = 280	₹430

Summary for FIFO

- ❑ Material for Job Z21 = ₹300
- ❑ Material for Job Z22 = ₹150 + ₹280 = ₹430
- ❑ Closing Stock = 130 units @ ₹4 = ₹520

ii. Last-In-First-Out (LIFO) Method

Date	Particulars	Receipts	Issues	Balance
		Units	Rate	Value
Feb. 1	Purchase	150	₹3	₹450
Feb. 15	Purchase	200	₹4	₹800
Feb. 18	Issue to Job Z21	100	₹4	₹400
Feb. 20	Issue to Job Z22	120	₹4 × 100 = 400 ₹3 × 20 = 60	₹460

Summary for LIFO

- ❑ Material for Job Z21 = ₹400
- ❑ Material for Job Z22 = ₹400 + ₹60 = ₹460
- ❑ Closing Stock = 130 units @ ₹3 = ₹390

iii. Weighted Average Method

Calculation of Weighted Average Rate

$$\text{Weighted Average Rate} = \frac{(150 \times 3) + (200 \times 4)}{150 + 200} = \frac{1,250}{350} = \approx ₹3.57 \text{ per unit}$$

Date	Particulars	Receipts	Issues	Balance
		Units	Rate	Value
Feb. 1	Purchase	150	₹3	₹450
Feb. 15	Purchase	200	₹4	₹800
Feb. 18	Issue to Job Z21	100	₹3.57	₹357
Feb. 20	Issue to Job Z22	120	₹3.57	₹428.4

Summary for Weighted Average

- ❑ Material for Job Z21 = ₹357
- ❑ Material for Job Z22 = ₹428.4
- ❑ Closing Stock = ₹465.1

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Summary of Material Values for Jobs and Closing Stock

Particulars	FIFO (₹)	LIFO (₹)	Weighted Average (₹)
Material for Job Z21	300	400	357
Material for Job Z22	430	460	428.4
Total Material Cost	730	950	785.4
Closing Stock	520	390	465.1

2.12 SUMMARY

- ❑ The Material Control Department ensures continuous material flow, optimizes inventory investment, and coordinates across departments to balance production, sales, and financial needs.
- ❑ Essential features of material control include continuous availability, cost-efficient procurement, waste reduction, and comprehensive record-keeping.
- ❑ The Purchasing Department is responsible for deciding what and when to buy, selecting suppliers, estimating quantities, and securing favorable prices.
- ❑ The materials issue process involves planning, authorization, issuance, and internal audits to ensure material needs are met efficiently while minimizing wastage.
- ❑ The Bill of Materials (BOM) serves as a standardized schedule that specifies material requirements, aiding in material requisition, cost estimation, and production control.
- ❑ **FIFO Method (First-In, First-Out)** assumes that the oldest inventory is issued first, valuing remaining stock at the most recent purchase prices. It provides consistent inventory valuation but may understate production costs in rising price environments.
- ❑ **LIFO Method (Last-In, First-Out)** assumes that the most recently purchased materials are issued first, valuing inventory at older prices. It aligns current costs with revenues but can be complex in fluctuating price markets.
- ❑ EOQ is the optimal inventory order size that minimizes both ordering and carrying costs, ensuring efficient stock management.
- ❑ The EOQ is calculated using the formula $\sqrt{(2AO/C)}$, where A is annual usage, O is the ordering cost, and C is the carrying cost per unit.
- ❑ It assumes constant costs for ordering, carrying, and unit pricing, with materials received instantaneously, and a known, consistent usage rate.
- ❑ Key methods include Perpetual Inventory System (tracking stock in real-time), ABC analysis (categorizing items by value), and Just-in-Time (minimizing inventory to reduce waste).

2.13 KEY WORDS

1. **Bill of Materials (BOM):** A detailed list of all materials required for a specific job or production, specifying quantities and descriptions.

2. **Material Requisition Slip:** A formal document used to authorize the withdrawal of materials from inventory for use in production.
3. **FIFO (First-In, First-Out):** Inventory method where the oldest stock is issued first, ensuring remaining inventory is valued at recent purchase prices.
4. **LIFO (Last-In, First-Out):** Inventory method where the most recent stock is issued first, aligning current material costs with production.
5. **HIFO (Highest-In, First-Out):** Inventory method where the costliest materials are issued first, maximizing cost recognition in financial statements.
6. **Economic Order Quantity (EOQ):** The optimal quantity of inventory to order that minimizes total costs of ordering and holding.
7. **Ordering Cost:** Expenses associated with placing and receiving orders, such as transportation and inspection costs.
8. **Carrying Cost:** Costs incurred for storing and holding inventory, including capital, storage, and insurance expenses.
9. **ABC Analysis:** A method for categorizing inventory into groups (A, B, C) based on their value and impact on total inventory cost.
10. **Just-in-Time (JIT):** An inventory strategy focused on minimizing stock levels by receiving goods only as they are needed.

2.14 CLOSING CASELET

ABC ANALYSIS AT FORD MOTOR COMPANY

Ford Motor Company, one of the largest automobile manufacturers in the world, has long faced challenges associated with managing its extensive inventory of components and raw materials. With a diverse range of vehicles and numerous parts required for assembly, Ford needed an effective strategy to optimize its inventory management, reduce costs, and improve overall operational efficiency. To address these challenges, Ford implemented the **ABC analysis methodology**.

Ford began its ABC analysis by categorizing its inventory into three classes: **'A', 'B', and 'C'**. The **'A category'** items included high-cost components such as engines and transmission systems. Ford established rigorous control measures for these critical items, which included frequent reviews of stock levels, detailed tracking of usage patterns, and the establishment of precise reorder points to prevent any disruptions in the assembly line.

The adoption of ABC analysis at Ford led to several significant improvements in inventory management and operational efficiency. By prioritizing control over high-value items, Ford was able to reduce stock-outs of essential components, thereby enhancing production efficiency and ensuring timely vehicle delivery. The overall costs associated with inventory management decreased as a result of better alignment of ordering practices with actual usage rates. The streamlined processes allowed Ford's management to focus on critical items that directly impacted production quality and cost efficiency. Ultimately, ABC analysis proved instrumental

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in helping Ford achieve a leaner inventory system, optimize resource allocation, and maintain its competitive edge in the automotive industry.



QUESTIONS

1. What are the key benefits of implementing ABC analysis at Ford Motor Company, particularly in relation to high-value items?
(**Hint:** Consider how Ford managed its inventory of high-cost components and how this management impacted production efficiency and cost reduction.)
2. What steps Ford might have taken to implement the ABC analysis effectively.
(**Hint:** Think about the different percentages of total items and inventory value for categories A, B, and C, and how Ford tailored its control measures accordingly for each category.)

2.15 HIGHER ORDER THINKING SKILLS QUESTIONS (HOTS)

1. How might the choice of inventory valuation method (FIFO, LIFO, HIFO) impact a company's financial statements during periods of inflation or deflation? Discuss the implications for management decision-making and stakeholder perceptions.
2. Evaluate the effectiveness of the weighted average cost method compared to the standard cost method in managing inventory for a manufacturing company facing frequent price fluctuations. How can these methods influence strategic financial planning and operational efficiency?
3. How can the integration of Economic Order Quantity (EOQ) with modern supply chain technologies, such as automation and predictive analytics, transform inventory management efficiency?
4. In what ways could an over-reliance on the Just-in-Time (JIT) inventory system expose a company to vulnerabilities, particularly during global supply chain disruptions?
5. How does the ABC analysis method influence decision-making in resource allocation, and what potential challenges arise when applying this model to a business with rapidly fluctuating product demand?
6. From the following transactions, prepare separate stores ledger accounts using the following pricing methods: (i) FIFO and (ii) LIFO

Date	Transaction	Details
April 1	Opening balance	150 units @ Rs. 4 each
April 10	Received	400 units @ Rs. 5 each
April 15	Issued	300 units
May 5	Issued	200 units
May 10	Received	300 units @ Rs. 6 each
May 20	Issued	250 units
June 1	Received	200 units @ Rs. 5.50 each
June 15	Issued	200 units

7. The following transactions are recorded for a store item:

Date	Receipts (kg)	Rate per Unit (₹)	Issues (kg)
January 5	300	2.00	—
January 10	500	2.50	—
January 15	—	—	400
January 20	600	3.00	—
January 25	—	—	500
January 31	200	2.80	—

Prepare a Stores Ledger Account pricing the issues using the Weighted Average Method.

Calculate the weighted average rate after each receipt and use this rate to value the issued quantities.

8. The following information is extracted from the Stores Ledger:

Material Y

Opening Stock: Nil

Purchases:

- ☐ Jan 1: 120 units @ ₹3 per unit
- ☐ Jan 15: 80 units @ ₹4 per unit

Issues:

- ☐ Jan 18: 100 units for Job A1
- ☐ Jan 22: 50 units for Job A2

Required:

Complete the receipts and issues valuation by adopting the First-In-First-Out (FIFO), Last-In-First-Out (LIFO), and Weighted Average methods. Tabulate the values allocated to Job A1, Job A2, and the closing stock under each method. Additionally, discuss the advantages and disadvantages of each method and which one you would prefer.

9. ABC Ltd., a company engaged in manufacturing, requires a specific component for its production process and needs to manage its inventory efficiently. The cost of each component is ₹15, and the total annual requirement for the component is 2,400 units. The company incurs an ordering cost of ₹50 per order and a carrying cost equivalent to 12% of the unit cost per annum.

The purchase manager has suggested placing a single bulk order for the entire annual requirement of 2,400 units to minimize ordering costs. The supplier has also offered a 2% discount on the unit price if ABC Ltd. agrees to place such a bulk order.

The following tasks are required to analyze the situation:

- ☐ Calculate the Economic Order Quantity (EOQ) for the component.
- ☐ Determine the total annual cost of inventory under the EOQ model, including purchase cost, ordering cost, and carrying cost.

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- ❑ Calculate the total annual cost of inventory if the company follows the purchase manager's proposal and avails of the 2% discount on the unit price.
- ❑ Based on the results, recommend whether ABC Ltd. should adhere to the EOQ model or accept the supplier's bulk order proposal.

2.16 ANSWER KEYS**SELF-ASSESSMENT QUESTIONS**

Topic	Q. No.	Answer
Materials Issue Process	1.	d. Cost Accounts Department
	2.	b. To provide authorization for issuing materials for use in the factory
Material Pricing Methods	3.	c. 620
	4.	a. 2,120
Quantity of Material Procurement	5.	b. It cannot establish prices without a market value.
	6.	b. To minimize total inventory costs
	7.	b. 11,850
Inventory Control Techniques	8.	c. Reduced requirement for physical stock verification
	9.	b. Minimal in-process inventory to reduce carrying costs
	10.	b. Spare parts based on their criticality to production

2.17 SUGGESTED READINGS

- ❑ Articles: "The Impact of Inventory Valuation Methods on Financial Statements"
- ❑ Article: "Inventory Management Techniques: A Comprehensive Overview"
- ❑ Bragg, S. M. (2019). *Inventory Best Practices* (2nd ed.). Wiley
- ❑ Axsäter, S. (2015). *Inventory Control* (2nd ed.). Springer

2.18 REFERENCES

- ❑ Institute of Management Accountants (IMA): IMA Website
- ❑ Research Paper: "Inventory Valuation and Management: Evidence from the Retail Industry"
- ❑ Waters, D. (2021). *Inventory Control and Management* (2nd ed.). Wiley.

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U N I T

LABOR COST AND OVERHEAD COST CONTROL

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INTRODUCTORY CASELET

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Addressing High Labor Turnover at Acme Corporation

Acme Corporation faced a significant increase in labor turnover, leading to disruptions and increased costs. To address this, they conducted a thorough analysis to identify the root causes, which included dissatisfaction with compensation, poor working conditions, limited career growth, and unfair work practices.

To improve retention, Acme implemented several strategies. They adjusted compensation to market standards, upgraded factory facilities, introduced career development programs, and established fair work practices. Additionally, they focused on employee engagement through surveys, feedback sessions, and social events.

These initiatives resulted in a significant decrease in labor turnover, improved productivity, and enhanced the company's reputation. By prioritizing employee well-being and addressing the root causes of turnover, Acme demonstrated the importance of a comprehensive approach to retention.



QUESTIONS

1. Beyond the specific actions taken by Acme Corporation, what underlying principles or philosophies guided their approach to addressing high labor turnover?

(Hint: Consider the broader context of human resource management, employee motivation, and organizational culture.)

2. How might Acme Corporation's strategies be adapted to a different industry or company size? What challenges or opportunities might arise?

(Hint: Think about the specific characteristics of different industries and companies, such as their competitive landscape, workforce demographics, and organizational structure.)

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LEARNING OBJECTIVES

At the end of the unit, you will be able to

- Learn about the mechanisms used to monitor and control labor costs.
- Learn methods to calculate labor turnover rates.
- Identify key factors influencing turnover and strategies to mitigate it.
- Understand different techniques used for timekeeping.
- Explore different methods of Wage Payment
- Learn how separating overheads into fixed and variable categories aids in better control.
- Understand how manufacturing overheads are recorded and controlled.
- Explore the allocation of overheads across departments for accurate costing.
- Learn about overhead absorption techniques for cost allocation.
- Examine the different overhead rates and their applicability in cost control.

3.1 INTRODUCTION

Imagine you're running a company like Primewood Creations, a business that crafts high-quality wooden furniture. One of the key challenges for Primewood is managing labor costs—the amount spent on wages, salaries, and related expenses for the people who make the furniture. But why is labor cost such a big deal? It's because labor cost forms a major part of total production costs. If it's not controlled properly, it could make the products too expensive to compete in the market.

So, what is labor cost? Often called Employee Cost, it's not just the wages paid to workers. It includes everything from salaries, overtime pay, bonuses, and even the costs related to training and welfare. Calculating labor costs isn't as straightforward as counting raw materials. Why? Because unlike materials, people are involved, and people are unpredictable. To calculate labor costs accurately, companies like Primewood Creations need well-organized systems and processes. Proper records of work hours, performance, and productivity are essential to ensure every detail is captured.

Here's where it gets interesting. Let's say Primewood's workers are taking longer than expected to carve intricate designs on chairs. This increase in time directly raises labor costs. On the flip side, if workers learn faster methods to complete the carvings, labor costs per unit decrease. This shows how worker productivity affects overall labor costs. If productivity is high, Primewood can make more chairs with the same number of workers, reducing the cost per chair.

How can Primewood control labor costs? One effective way is to set clear targets and compare actual costs to those targets. For instance, if it's expected to take two hours to make one chair, but it's taking three hours, something is off. Primewood can investigate the reasons for this delay—maybe a worker needs more training, or perhaps a tool is malfunctioning. Tracking performance through labor performance reports is crucial. Reports highlight these issues, allowing supervisors to take timely corrective action. The production team can use labor budgets, performance standards, and production schedules as guides to ensure efficiency.

Another key concept for Primewood's management is labor turnover. Think about it this way: If employees keep leaving the company, it's like having a revolving door—new workers constantly need to be hired and trained. This process costs money and time, driving up labor costs. By understanding why employees leave—perhaps for better pay or work conditions—Primewood can take action to reduce turnover. For example, if offering a small performance bonus can keep skilled workers from leaving, it's often a cost-effective strategy.

Accurate timekeeping also plays a role. Imagine if Primewood's workers didn't record the exact hours they worked. How would the company know if they're being paid correctly? Timekeeping methods like biometric scanners, punch cards, or digital apps ensure that hours worked are tracked properly. With these records, Primewood's management can calculate wages accurately and spot any inefficiency.

Another useful tool for improving productivity is a work study. This involves analyzing how workers perform their tasks to find better, faster, and safer ways to do them. For instance, if Primewood's carpenters spend a lot of time walking back and forth to get tools, reorganizing the workshop layout might reduce wasted time. These changes directly lower labor costs by boosting efficiency.

Wage payment methods also affect labor cost control. At Primewood, workers could be paid in different ways—a flat hourly rate, a piece-rate system (paid per chair made), or an incentive-based wage where bonuses are awarded for exceeding production targets. Each system has its pros and cons. A flat hourly rate is simple but doesn't motivate extra effort. On the other hand, a piece-rate system might push workers to rush, which could impact quality. The right method depends on the type of work and the company's goals.

Now, let's turn to overhead costs. While labor costs are direct, overhead costs are a little trickier. Overheads include expenses like electricity, rent, supervisor salaries, and machine maintenance. These are costs that can't be linked to one specific product but still affect overall production costs. Think about it—Primewood's sanding machines run on electricity, but you can't say that one specific chair used a set amount of power. Instead, these costs are spread out across all the chairs produced.

Some people think that high overhead costs mean a company is being wasteful, but that's not always true. For instance, if Primewood buys a new automated carving machine, overhead costs might increase due to

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maintenance and depreciation expenses. But at the same time, production could speed up, reducing labor costs. So, higher overhead isn't always a bad thing—it could be the result of smart investments in technology.

Overhead costs can also rise with the scale of production. If Primewood decides to increase production from 100 chairs to 500, the costs for rent, utilities, and maintenance may rise. More machines may be needed, leading to higher depreciation expenses. However, if this larger production is more efficient, the company could achieve economies of scale—producing each chair at a lower overall cost.

3.2 THE ISSUES CONCERNED WITH LABOR COST CONTROL

It is imperative to concentrate on managing and reducing labor costs in the current competitive environment. It will take deliberate efforts to accomplish this goal. To successfully manage and lower labor costs, the following actions can be helpful:

3.2.1 CLASSIFICATION OF LABOR COST

Labor costs can be broadly categorized into two shown in Figure 3.1

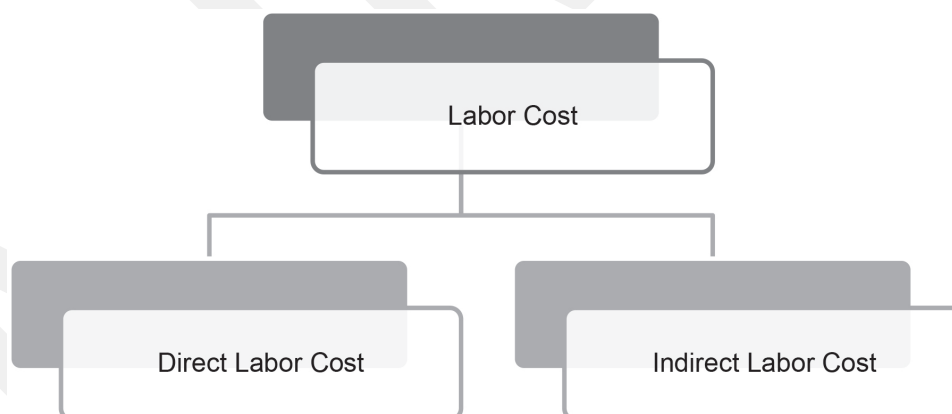


Figure 3.1 Classification of Labor Cost

Imagine you're part of a team running a bakery called *Freshly Baked Delights*. Every day, bakers, decorators, and support staff work together to produce delicious cakes, bread, and pastries. To understand how the bakery manages its labor costs, let's explore two key concepts:

Direct Labor Cost

Direct Labor Cost refers to the wages paid to employees who are actively involved in producing a specific product or service. Their work is clearly linked to the final output, making it easy to track and assign their wages to a specific product or task.

Think of it like this:

- ❑ The bakers kneading dough, decorating cakes, and baking bread—their wages are **Direct Labor Costs**.

- ❑ Why? Because you can easily trace their efforts to specific cakes, loaves of bread, or batches of pastries.

Other Examples of Direct Labor:

- ❑ Employees on an assembly line making smartphones.
- ❑ Carpenters building custom-made furniture pieces.
- ❑ Technicians assembling parts of a washing machine.

These workers' wages directly affect the cost of the product. If production increases, more bakers may be needed, and direct labor costs will increase too.

Indirect Labor Cost

These are wages paid to employees who support production but aren't directly involved in making the product. You can't link their efforts to one particular item, but their work is still crucial for the overall production process.

In the bakery, here's how it works:

- ❑ The store supervisor who oversees the entire bakery team—their salary is an **Indirect Labor Cost**.
- ❑ The cleaner who ensures hygiene standards are maintained in the bakery—another **Indirect Labor Cost**.
- ❑ The security guard at the entrance—you guessed it, **Indirect Labor Cost**.

Other Examples of Indirect Labor:

- ❑ Maintenance workers who fix bakery equipment.
- ❑ Clerks managing inventory for raw materials like flour, sugar, and baking powder.
- ❑ Warehouse staff responsible for unloading baking supplies.

These workers aren't directly making cakes or pastries, but without them, production would be disrupted. Unlike direct labor costs, indirect labor costs do not change directly with the level of production.

Key Differences between Direct and Indirect Labor Costs is shown in the Table 3.1:

Table 3.1 Difference between Direct Labor Cost and Indirect Labor Cost

Direct Labor Cost	Indirect Labor Cost
1. Labor costs directly involved in the production process.	1. Labor costs related to employees not directly engaged in production activities.
2. Direct labor costs are easily identifiable and allocated to cost units.	2. Indirect labor costs are allocated based on appropriate apportionment methods.

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Direct Labor Cost	Indirect Labor Cost
3. Direct labor costs fluctuate with the volume of production, showing a positive relationship with production. (more cakes = more wages)	3. Indirect labor costs do not necessarily vary with production volume.
4. Example: Bakers working on dough, cakes, and pastries	4. Example: Supervisors, security guards, and cleaners

Knowing the difference between Direct and Indirect Labor Costs helps businesses like *Freshly Baked Delights* to:

1. **Accurately Calculate Labor Costs:** When you know how much each cake costs to produce, you can price it correctly to ensure profit.
2. **Control Costs:** If you see labor costs rising, you can investigate whether bakers are taking too long or if production is inefficient.
3. **Measure Worker Efficiency:** Tracking direct labor performance helps managers identify top performers or areas for improvement.
4. **Allocate Overhead Costs:** Indirect labor is part of overhead, which must be shared across products. For example, how much of the supervisor's salary should be assigned to each cake?
5. **Incentivize Workers:** By tracking direct labor, workers can be rewarded for hitting production targets.
6. **Forecast and Compare Costs:** With clear labor cost data, managers can compare different bakery branches or production units.

3.2.2 LABOR COST CONTROLS

Labor costs are fundamentally linked to human resources. To effectively manage these costs, it's essential to understand human behavior. Labor cost control involves not just overseeing the total wage bill but also ensuring that the compensation for each worker is appropriately aligned with their productivity. The objective is to minimize wages per unit of output rather than simply reducing individual worker wages.

Achieving lower costs per unit necessitates motivating workers through suitable compensation strategies that enhance their efficiency and productivity. A highly motivated workforce can significantly boost organizational performance. Therefore, companies must continually focus on improving labor productivity right from the outset to ensure effective cost management in this area.

3.2.3 PROCESS AND PRODUCTION PLANNING

Efficient control of labor costs can be achieved through meticulous production planning. This includes key activities such as planning, scheduling, routing, machine loading, product and process engineering, as well as conducting work studies. Work studies, including time and motion analysis, help determine the standard time required for specific tasks. Consistent

comparisons between standard time and actual time allow for the identification of variances. If actual time frequently exceeds standard time, corrective actions can be taken to resolve the issue effectively.

3.2.4 LABOR BUDGET

Budgeting and budgetary control are valuable tools for effective cost management and achieving cost reduction. To compare actual labor costs with projected labor costs, a labor budget can be developed to set a target for labor expenses. The labor content for each manufacturing item, as specified in the production budget, is calculated based on the labor time required for each task, operation, and process, as well as the necessary worker grades and trades. Pay rates, bonuses, allowances, and other components of compensation for each worker category are then considered. The total labor cost for each budget center is obtained by multiplying the wage rate by the labor hours needed for the budgeted production units.

3.2.5 STANDARD LABOR COST

Standards are essential in managing labor costs, providing benchmarks for comparing actual labor expenditures. The standard labor cost denotes the anticipated expense expected to be incurred in producing a specific quantity of products. Setting this standard involves thorough analysis and planning, typically utilizing time and motion studies to establish the optimal time required for various production tasks.

By assessing the time and efficiency of various processes, organizations can develop realistic standards that accurately represent anticipated productivity levels. This facilitates effective comparisons between actual labor costs incurred during production and the predetermined standard labor costs. Such assessments help pinpoint variances, enabling management to implement corrective actions to enhance labor efficiency and maintain cost control.

3.2.6 JOB PERFORMANCE REPORT

A structured system for generating periodic labor efficiency and utilization reports is crucial. These reports offer valuable insights into workforce productivity and efficiency, allowing management to evaluate performance, identify areas needing improvement, and optimize the use of labor resources.

3.2.7 WORK PERFORMANCE INCENTIVES

Improving labor productivity is a vital strategy for lowering labor costs per unit. Employee motivation is a significant factor in enhancing productivity. Introducing both monetary and non-monetary incentives can drive substantial improvements. Regular reviews of these incentive programs are essential, requiring the preparation of scheduled reports on incentive schemes to evaluate their impact and make any necessary modifications.

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Activity

A manufacturing company wants to implement a labor budget to manage its costs effectively. They categorize their labor costs into direct and indirect labor costs. Identify two examples of direct labor costs and two examples of indirect labor costs for the company. Explain how the company can use the labor budget to compare budgeted costs against actual costs and what actions it might take if it finds a significant variance.

3.3 MECHANISM OF LABOR COST CONTROL

Labor cost control involves the effective management of workforce expenses by adhering to predetermined efficiency standards and regularly comparing actual costs against these benchmarks during production. Effective control is achieved through several key measures:

- (a) **Production Planning:** Production planning ensures that work is executed efficiently by establishing clear production schedules, defining responsibilities, and coordinating resources. This planning minimizes delays and prevents both overstaffing and understaffing, ensuring the optimal use of labor. By setting realistic timelines and targets, production planning creates a structured environment that reduces waste and aligns labor efforts with organizational objectives.
- (b) **Utilization of Labor Budgets and Standards:** Labor budgets act as a financial roadmap, outlining the maximum permissible costs for labor within a project or task. Coupled with efficiency standards that set measurable benchmarks, they provide a framework to evaluate productivity. These tools not only help allocate labor resources effectively but also foster accountability, ensuring that employees perform to their full potential while keeping labor expenses within the planned limits.
- (c) **Performance Reports:** By comparing actual labor costs and outputs to the established budgets and standards, these reports highlight variances that need attention. Timely reporting allows foremen and supervisors to identify inefficiencies, implement corrective actions, and optimize workflows. These insights are crucial for making informed decisions that enhance labor productivity and control costs effectively.
- (d) **Wage and Incentive Schemes:** Compensation structures like time-based pay, piece-rate wages, or performance bonuses link employee earnings to their productivity, fostering a sense of accomplishment. Incentive programs not only boost morale but also reduce absenteeism and turnover, creating a workforce that is both engaged and aligned with organizational goals.

3.3.1 PERSONNEL MANAGEMENT DEPARTMENT

The personnel department plays a crucial role in ensuring an efficient workforce and controlling labor costs. Its core function include managing recruitment, defining job roles, conducting job evaluations, and

implementing time and motion studies. Recruitment efforts may be aimed at replacing employees or expanding the workforce. The department oversees all aspects of hiring, including advertising vacancies, conducting interviews and tests, administering physical examinations, and facilitating employee induction and job assignments. Functions of personnel management department:

- (a) **Managing Recruitment:** The personnel department receives labor requisitions from various departments and identifies candidates with the required skills and qualifications. It ensures that all recruits meet the specific job requirements and possess the necessary competencies for their roles.
- (b) **Organizing Training Programs:** To maintain workforce efficiency, the department organizes training sessions for new employees and skill enhancement workshops for existing staff. These initiatives ensure that workers stay updated with the latest practices and technologies, contributing to higher productivity.
- (c) **Maintaining Employee Records:** Comprehensive personal and job-related records are maintained for every employee. These records, often referred to as service history cards or personal record cards, include detailed information about an employee's background, job performance, and service history. Accurate records support informed decisions regarding promotions, transfers, and other employment-related matters.
- (d) **Evaluating Employee Performance:** Periodic performance evaluations help assess employees' contributions to the organization. This process aids in identifying training needs, recognizing outstanding performers, and ensuring that employees remain aligned with organizational goals.

Additional Responsibilities in Labor Cost Control

Apart from managing permanent staff, the personnel department also addresses the need for temporary or casual workers. Effective labor cost control requires:

- ❑ Centralizing all recruitment activities through the personnel department to ensure consistency and efficiency.
- ❑ Conducting regular reviews of job placements to identify opportunities for promotions or transfers, which maximize employee potential.
- ❑ Implementing labor utilization reports in all departments to monitor workforce efficiency and resource allocation.
- ❑ Minimizing absenteeism and labor turnover through proactive employee engagement strategies and workplace incentives.

3.4 LABOR ATTRITION

Labor attrition or labor turnover refers to the rate at which employees leave and are replaced within an organization. A high labor turnover

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suggests instability in the workforce, often leading to low morale and increased costs due to the effects of demotivated staff. Keeping labor turnover as low as possible is crucial for maintaining a stable and efficient workforce. A low labor turnover rate typically indicates:

1. A well-managed organization.
2. Higher preventive costs being incurred by management to ensure employee satisfaction.
3. A strong and organized trade union that influences management decisions.
4. Significant state regulatory control over management policies regarding employment and retrenchment.
5. Limited opportunities for better employment alternatives.
6. High levels of unemployment, such as seen in India.

3.4.1 MEASUREMENT OF LABOR TURNOVER

Measuring labor turnover is crucial for any organization. It helps in understanding the turnover rate within the organization and allows for comparison between the current period and previous periods. The following methods can be used to calculate labor turnover:

1. **Additions Method:** This approach considers the number of employees added during a specific period to determine the labor turnover. The formula for calculation is:

$$\text{Labor Turnover} = \frac{\text{Number of additions}}{\text{Average number of workers during the period}} \times 100$$

2. **Separation Method:** This method accounts only for workers who have left during a specific period. The formula is:

$$\text{Labor Turnover Rate} = \frac{\text{No. of workers left during a period}}{\text{Average number of workers in the period}} \times 100$$

3. **Replacement Method:** This method focuses on the actual replacement of workers, excluding new hires due to business expansion. The formula is:

$$\text{Labor Turnover Rate} = \frac{\text{No. of workers replaced during a period}}{\text{Average number of workers in the period}} \times 100$$

4. **Flux Method:** This method calculates the total change in the workforce, including both separations and additions. The formula is:

$$\text{Labor Turnover Rate} = \frac{1/2(\text{No. of separations} + \text{No. of additions})}{\text{Average number of workers in the period}} \times 100$$

3.4.2 FACTORS AFFECTING LABOR TURNOVER

Labor turnover is influenced by a variety of factors, including dissatisfaction with work, inadequate wages, poor working conditions, and

a lack of essential amenities. These factors can be grouped into three main categories: personal causes, avoidable causes, and unavoidable causes.

1. Personal Causes: These are reasons that compel workers to leave their jobs due to personal circumstances, including:

- ☐ Changing jobs for better opportunities.
- ☐ Early retirement as a result of old age or health problems.
- ☐ Duties within the family, such taking care of aging parents.
- ☐ Discontentment with the job or the work environment.
- ☐ Marriage (especially among female workers) or childbirth.
- ☐ Seasonal departures, such as workers leaving for festivals, marriages, or agricultural harvesting.

2. Avoidable Causes: These causes arise from conditions that could be improved or rectified by the organization:

- ☐ Low wages, prompting workers to seek better-paying jobs.
- ☐ Dissatisfaction with the job.
- ☐ Adverse conditions at work.
- ☐ Extended and erratic workdays.
- ☐ Dissatisfactory interactions with managers.
- ☐ Disagreements with colleagues
- ☐ Lack of recreational facilities.
- ☐ Insufficient housing or medical services.
- ☐ Unfair promotion practices and lack of career advancement opportunities.
- ☐ Poor planning and foresight by management, seasonal work patterns, and supply issues like shortages of raw materials or power.

3. Unavoidable Causes: These factors are beyond the organization's control, including:

- ☐ The business's seasonal nature.
- ☐ The plant was relocated.
- ☐ Inadequate power, raw material, or product market demand.
- ☐ Workers who suffer illnesses or accidents that make them permanently unable to work.
- ☐ Dismissal or discharge due to insubordination, negligence, or inefficiency.
- ☐ Marriage, especially for female employees.

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3.4.3 LABOR TURNOVER AND ITS COST TO A FIRM

Increasing labor turnover is a twofold issue that negatively affects both productivity and costs. The costs associated with labor turnover are categorized into Preventive Costs and Replacement Costs.

Preventive Costs:

These are expenses incurred to maintain employee satisfaction and reduce the likelihood of workers leaving the organization. They can be classified under the following:

1. **Personnel Administration:** A portion of the Personnel Department's expenditure is allocated to maintaining good relationships between management and staff, ensuring workers' concerns are addressed, thus lowering labor turnover.
2. **Medical Services (Preventive and Curative):** Providing medical services for employees and their families increases job satisfaction, as workers are more inclined to stay with employers offering health care.
3. **Welfare Activities and Schemes:** Offering facilities such as subsidized canteens, housing schemes, transport, and education boosts employee morale, making them less likely to leave.
4. **Miscellaneous Schemes:** Pension plans, provident funds, bonuses, and other incentives like high wages also contribute to reducing turnover.

Replacement Costs:

These are costs associated with recruiting and training new workers to replace those who have left. They include:

1. **Loss of Output:** There may be delays in replacing workers, leading to reduced output. Extra costs may arise from employing surplus labor or paying overtime.
2. **Employment Department Expenses:** The recruitment process incurs additional administrative costs for selection, testing, and medical examinations.
3. **Induction Training:** New employees often require training, which increases costs, especially for skilled positions.
4. **Inefficiency of New Workers:** New employees are less efficient, resulting in decreased productivity and increased costs.
5. **Cost of Tool and Machine Breakage:** Inexperienced workers may damage tools or machines, adding to repair and replacement costs.
6. **Cost of Scrap and Defective Work:** New workers may spoil materials or produce defective work, leading to wasted resources.
7. **Cost of Accidents:** Inexperienced workers are more prone to accidents, increasing both direct and indirect costs.

Imagine you're managing a growing business, *Radha Textiles*, a company known for its high-quality fabrics. Over time, you notice that workers keep leaving, and new employees have to be hired and trained repeatedly. This constant cycle increases costs and disrupts production. To tackle this, you decide to implement some key remedial measures to reduce labour turnover:

1. Conducting Exit Interviews

When employees leave, it's essential to understand why they're leaving. *Radha Textiles* schedules a one-on-one chat with each outgoing worker before they leave.

Why it works:

- ❑ The employee might reveal key issues like low pay, lack of growth opportunities, or poor management practices.
- ❑ This feedback can be used to improve the work environment and reduce future resignations.

Example: If several employees say they're leaving because of better pay at a competitor, *Radha Textiles* might consider offering performance-based bonuses or wage hikes to retain its skilled workers.

2. Job Analysis and Evaluation

Before hiring new workers, it's important to be clear about the nature of each job. This step involves analyzing the role, tasks, and skills required.

How it helps:

- ❑ New employees know exactly what's expected of them, so there's less confusion or dissatisfaction later.
- ❑ It ensures the right person is hired for the right role, reducing the chances of misfit employees leaving.

Example: If a sewing machine operator is hired without knowing they'll be required to work night shifts, they may quit once they learn this. But with proper job analysis, the shift requirement would be made clear during recruitment.

3. Scientific Recruitment, Placement, and Promotion

Hiring and promoting employees should be done systematically, not randomly. A scientific approach to recruitment ensures that employees are placed in jobs that match their skills and potential for growth.

How it helps:

- ❑ Workers feel secure in their roles and are more likely to stay if they see fair promotions and career progression.
- ❑ It reduces the chances of hiring the wrong person for the job.

Example: If *Radha Textiles* promotes an employee to a supervisor role without assessing their leadership skills, they may struggle and quit.

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But if promotions are based on performance evaluations, employees feel they've earned their role and are more motivated to stay.

4. Creating a Positive Work Environment

A happy workplace keeps employees loyal. Management can create a positive atmosphere where employees feel valued, respected, and heard.

Key actions that work:

- ❑ **Clear Service Rules:** Set clear policies on work hours, leave, discipline, etc., and ensure workers are aware of these rules before implementation.
- ❑ **Training & Education:** Provide workers with training to improve their skills and advance in their careers.
- ❑ **Grievance Handling:** Create a system where workers can report problems (like bullying or unfair treatment) and get support.

Example: *Radha Textiles* notices complaints about unclear shift timings. The HR team discusses this issue with workers and establishes a fixed shift schedule. This action builds trust and reduces worker frustration.

5. Use of Committees

Instead of making decisions alone, *Radha Textiles* forms a committee with both management and worker representatives. This group discusses grievances and workplace issues.

How it helps:

- ❑ Workers feel included in decision-making, leading to better cooperation.
- ❑ Grievances are handled more quickly, reducing dissatisfaction.

Example: A worker files a complaint about favoritism in overtime assignments. The committee discusses this issue and introduces a rotation system for overtime. This move reduces frustration and prevents resignations.

3.4.4 EXAMPLE ON LABOR TURNOVER

Example 1: Company XYZ has the following labor statistics for the year 2023:

- ❑ Number of workers as of 01-01-2023: 5000
- ❑ Number of workers as of 31-12-2023: 6000
- ❑ Workers who voluntarily resigned during the year: 150
- ❑ Workers who were discharged during the year: 50
- ❑ Workers recruited during the year: 800
 - ◆ Workers recruited to replace resignations: 150
 - ◆ Workers recruited for expansion: 650

Calculate labour turnover with

- ❑ Separation Method ❑ Replacement Method ❑ Flux Method

Sol.

Average Number of Workers =

$$\begin{aligned} & \frac{\text{Number of workers at the beginning} + \text{Number of workers at the end}}{2} \\ &= \frac{5000 + 6000}{2} = 5,500 \end{aligned}$$

1. Separation Method:

$$\text{Labor Turnover Rate} = \frac{\text{No. of workers left during a period}}{\text{Average number of workers in the period}} \times 100$$

$$\begin{aligned} \text{Total workers who left} &= \text{Voluntary resignations} + \text{Discharged workers} \\ &= 150 + 50 = 200 \end{aligned}$$

$$\text{Labor Turnover Rate} = \frac{200}{5500} \times 100 = 3.64\%$$

Interpretation: A turnover rate of 3.64% indicates a moderate level of employee exits. This percentage shows that out of the average workforce of 5,500 employees, 3.64% either resigned or were discharged. It reflects the company's ability to retain employees or the necessity to let go of underperforming workers. A low turnover rate typically implies good employee retention, while a high rate could signal potential issues in employee satisfaction or performance.

2. Replacement Method:

$$\text{Labor Turnover Rate} = \frac{\text{No. of workers replaced during a period}}{\text{Average number of workers in the period}} \times 100$$

Total workers replaced = Workers recruited to replace resignations = 150 (since only replacements are considered)

$$\text{Labor Turnover Rate} = \frac{150}{5500} \times 100 = 2.73\%$$

Interpretation: A replacement rate of 2.73% shows the company's effort to fill the positions vacated by employees who left. It suggests that XYZ Company efficiently managed to recruit replacements for most of the workers who exited, specifically focusing on maintaining staffing levels. This turnover metric does not include workers added for expansion, so it shows how well the company has sustained its operations without disruption.

3. Flux Method:

$$\text{Labor Turnover Rate} = \frac{1/2(\text{No. of separations} + \text{No. of additions})}{\text{Average number of workers in the period}} \times 100$$

- ❑ Total separations = 200 (as calculated before)

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- ❑ Total additions = 800 (total recruits)

$$\text{Labor Turnover Rate} = \frac{(800 + 200)/2}{5500} \times 100 = 9.09\%$$

Interpretation: A high flux rate of 18.18% suggests significant changes in the workforce, driven both by employee exits and the company's expansion plans. The inclusion of 650 new workers recruited for expansion highlights the company's growth strategy and substantial adjustments in staffing needs to accommodate this expansion. While the high turnover rate may seem concerning, it is mainly due to the expansion, indicating a proactive approach toward business growth rather than excessive turnover.

3.5 RECORDING OF TIMINGS

The recording of time serves two key purposes: **time-keeping** and **time-booking**, and it is essential for both direct and indirect workers, even for those paid on a piece-rate basis. Accurate record-keeping requires that time-keeping and time-booking figures match, ensuring consistency in both attendance records and the utilization of labor.

1. Time-Keeping: Tracking Attendance and Discipline

Think of time-keeping as your library's check-in system. It records when you enter and leave the library to ensure you're making use of the facility within its operating hours. Similarly, in a workplace, the Time-Keeping Department plays a crucial role in managing labor costs by maintaining detailed records of each worker's entry and exit during regular working hours.

Purpose and Importance of Time-Keeping:

- ❑ **Tracks total hours worked:** Like how the library monitors the time you spend inside, this helps calculate wages for workers paid on a time-rate basis.
- ❑ **Encourages punctuality and discipline:** Imagine if library users came in late and disrupted the study environment—it's the same in a workplace!
- ❑ **Supports benefits calculation:** Just as regular library users might receive perks (like extended borrowing privileges), workers' attendance records are tied to benefits like pensions, paid leave, and promotions.
- ❑ **Helps with cost allocation:** Labor hours are often used to distribute overhead costs fairly, just like dividing library maintenance fees based on user activity.
- ❑ **Fulfills legal obligations:** Time-keeping is mandatory under labor laws, ensuring fairness for both employees and employers.

- ❑ **Enables further analysis:** Attendance data helps identify idle time, set standard times for tasks, and assess efficiency, benefiting the company just like tracking study patterns benefits frequent library-goers.

Real-Life Connection:

Imagine your library offers free coffee for users who clock in on time for a week straight. Similarly, workplaces encourage punctuality and attendance through time-keeping.

2. Time-Booking: Tracking Productive Time

While time-keeping records attendance, it doesn't show how that time is used. For example, knowing you spent three hours in the library doesn't reveal whether you were studying, reading a novel, or browsing your phone. This is where time-booking comes in—it focuses on how workers utilize their time on specific tasks.

Purpose and Importance of Time-Booking:

- ❑ **Tracks productive time:** Just like noting how much time you spend studying versus browsing, this ensures workers' time is used effectively.
- ❑ **Identifies idle time:** Helps managers address areas where workers may be unproductive.
- ❑ **Calculates the value of work done:** Imagine getting a badge for completing a research paper—companies do this by assigning value to completed jobs.
- ❑ **Supports fair wage distribution:** Workers' wages and bonuses are calculated based on the time booked for tasks.
- ❑ **Evaluates efficiency:** Just as finishing a project early shows your productivity, time-booking reflects how efficiently workers perform their tasks.

Real-Life Example:

If you track your library session and see that you spent one hour on research, one hour drafting an essay, and one hour chatting with friends, you know how your time was utilized. Similarly, time-booking tracks how much time workers dedicate to specific jobs or operations.

3.5.1 METHODS OF TIME RECORDING

Time Keeping Methods

There are two main approaches to time-keeping: manual and mechanical. The choice of method is influenced by the organization's needs and policies. Regardless of the system employed, it should accurately track workers' hours, minimize costs, and reduce the risk of wage fraud. List of methods of Time Keeping is shown in Figure 3.2

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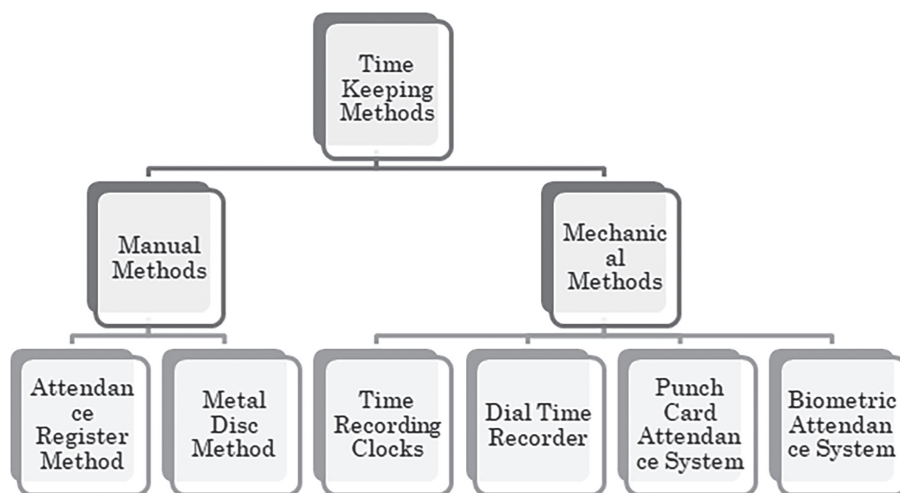


Figure 3.2 Time Keeping Methods

Manual Methods

- ❑ **Attendance Register Method:** This is the oldest form of time-keeping, where an attendance register is placed near the factory entrance or in each department. A time-keeper manually records the workers' arrival and departure times. The simplicity and low cost of this method make it suitable for smaller firms with fewer employees. However, it is vulnerable to dishonest practices, such as collusion between workers and the time-keeper. Despite these limitations, this method is often used when workers' time must be tracked at remote locations or customer premises. **Example:** Imagine a small bakery with 10 employees. The bakery owner places a notebook at the entrance where employees write down their arrival and departure times daily. This works well because the small team is easy to monitor. However, if an employee asks a friend to write down their time in case of tardiness, it could lead to dishonest records.
- ❑ **Metal Disc Method:** In this system, each worker is assigned a metal disc or token engraved with their identification number. Workers place their discs into a box when they enter the factory. The box is removed after the designated arrival time, and any late arrivals must hand their discs directly to the time-keeper for proper documentation. While this method is relatively straightforward, issues may arise if one worker removes another's disc, leading to disputes over attendance or allowing for dishonest recording, such as the inclusion of fictitious workers. **Example:** Think of a construction site where each worker is given a metal token with their ID. Workers drop their tokens into a box when they arrive. Latecomers must hand their tokens to the supervisor, ensuring accountability. However, if Worker A drops Worker B's token, it creates a false record of Worker B's presence.

Mechanical Methods

- ❑ **Time Recording Clocks:** This is a mechanical system that precisely records the time of arrival, departure, and breaks for each worker. Each employee receives a time card, which is inserted into a time clock that prints the exact time. This method eliminates disputes over hours worked, as there is little chance for time-keeper error or bias. However, challenges include the possibility of workers marking time for their absent colleagues, and mechanical failures can disrupt accurate time recording. **Example:** In a medium-sized garment factory, each worker has a time card. When they arrive, they insert their card into a machine that stamps the exact time. This ensures precise records for payroll. If a worker forgets their card or if the machine breaks down, it might disrupt time tracking.
- ❑ **Dial Time Recorder:** In this method, a machine with a dial featuring numbered holes is used, where each number corresponds to a worker's identification. Workers place a radial arm in their assigned hole and press it to record their time on a paper roll. This system is highly accurate and can directly calculate hours worked for wage computation. Despite its accuracy, the high installation cost and limited capacity make it less suitable for larger operations. **Example:** At a manufacturing plant, workers press a lever in their assigned number slot on a machine dial when they arrive and leave. This records their working hours on a paper roll, minimizing errors. However, the factory cannot afford the machine for all departments, so only a limited number of employees can use it.
- ❑ **Punch Card Attendance System:** In this system, workers use a punch card that is waved near a reader to log their in and out times. The system is automated, preventing proxy attendance and reducing manual errors. While the accuracy and reliability are high, the system requires complex software and expensive machinery, making it more suitable for larger firms with substantial budgets. **Example:** In a tech company, employees carry ID cards that they swipe at electronic readers when entering and leaving the office. The system automatically calculates attendance and working hours. If an employee forgets their card, their attendance must be manually updated, which could cause delays.
- ❑ **Biometric Attendance System:** To authenticate employees, this sophisticated technique relies on behavioral or physical characteristics like voice patterns, facial recognition, or fingerprints. Biometric systems, like fingerprint or face recognition, ensure secure and reliable attendance tracking by eliminating the possibility of buddy punching. While this method offers the highest accuracy and security, its installation is costly, though the long-term benefits in transparency and efficiency often outweigh the initial investment. **Example:** A multinational corporation installs fingerprint scanners at office entrances. Employees scan their fingerprints when they arrive, ensuring accurate attendance records. If an employee injures their finger, the system may fail to recognize them, requiring a backup process.

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Following table 3.2 summarizes the methods of time keeping

Table 3.2 Time Keeping Method

Category	Method	Description	Advantages	Limitations
Manual Methods	Attend-ance Reg-ister	Workers' arrival and departure times are recorded manually in a register.	Simple and low-cost; suitable for small firms.	Prone to dishonest practices; less reliable.
	Metal Disc Method	Workers drop metal discs with their ID numbers into a box upon entry.	Straightforward and inexpensive.	Susceptible to misuse, such as proxy attendance.
Mechanical Methods	Time Recording Clocks	Time cards are stamped with exact entry, exit, and break times by a clock.	Accurate; minimizes disputes.	Risk of misuse (proxy punching); vulnerable to mechanical issues.
	Dial Time Recorder	Workers use a dial with numbered holes linked to their ID to record time on a paper roll.	Highly accurate; facilitates wage calculation.	Expensive; limited capacity for large-scale use.
	Punch Card System	Workers log times using punch cards near automated readers.	Prevents proxy attendance; reliable.	High setup costs; requires advanced software.
	Biometric System	Uses physical or behavioral traits (e.g., fingerprints, facial recognition) for attendance tracking.	Eliminates buddy punching; highly secure.	Expensive to install; requires maintenance.

Time Booking Methods

The following methods shown in table 3.3 are commonly used for time booking, which record the time workers spend on different tasks:

Table 3.3 Time Booking Methods

Method	Description	Advantages	Limitations
Daily Time Sheet	Each worker records the time spent on their work daily.	Ensures high accuracy in records.	Generates significant paperwork for each worker.
Weekly Time Sheets	Similar to daily time sheets, but records are maintained weekly.	Reduces paperwork compared to daily sheets.	Risk of inaccuracies if not updated daily.

Method	Description	Advantages	Limitations
Job Ticket	Workers record start and finish times for each job on a ticket, with a new ticket provided for each task.	Allows precise tracking of time per job.	Requires frequent handling of tickets.
Labor Cost Card	A card passed among workers for jobs with multiple stages; workers record their time spent on the job.	Consolidates labor cost for the entire job.	Time entries depend on all workers' accuracy.
Time and Job Card	Combines records for job time and attendance in a single card.	Eliminates the need for separate records.	May be less detailed for specific time tracking.

- 1. Daily Time Sheet:** In this method, each worker is provided with a sheet to record the time spent on their work each day. The daily recording ensures accuracy, but it involves a significant amount of paperwork as daily sheets are maintained for each worker. *Example:* A worker in a textile factory records on a sheet that they worked from 9:00 a.m. to 12:00 p.m. stitching garments, took a lunch break, and then worked from 1:00 p.m. to 5:00 p.m. inspecting fabric. This record is submitted at the end of each day.
- 2. Weekly Time Sheets:** Similar to the daily time sheet, but the records are maintained on a weekly basis instead. This reduces the amount of paperwork. However, care must be taken to ensure that workers fill in their information daily to avoid inaccuracies. *Example:* A construction worker records their daily work hours (e.g., 8:00 a.m. to 5:00 p.m.) from Monday to Friday on one sheet. At the end of the week, they submit the completed sheet to the supervisor.
- 3. Job Ticket:** Workers are provided with job tickets to record the time they begin and finish each job. A separate job ticket is given for every task, allowing the time taken for each job to be tracked accurately. After completing one job, the worker receives a new ticket for the next assignment. *Example:* A technician repairing different appliances logs the start and end time for each repair. Once one appliance is fixed, they receive a new ticket for the next one, recording their work time separately for each task.
- 4. Labor Cost Card:** This card is used for jobs involving multiple stages or operations. Instead of assigning a separate card to each worker, one card is passed among all workers involved in the job. Each worker records the time they spend on the job, and the card shows the total labor cost for the job or product. *Example:* In an automobile manufacturing unit, different workers handle various tasks on a single car. One worker installs the engine, another

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paints the car, and a third works on the interiors. Each worker logs the time they spend on their part of the process on a shared labor cost card, which shows the total time and cost for the car's production.

5. **Time and Job Card:** This combined card records both the time taken to complete the job and the attendance time. As a result, separate records for job time and attendance are not required, making this method more efficient. *Example:* A furniture maker clocks in at 8:00 a.m., and this time is recorded on the time and job card. As they complete tasks like cutting wood, sanding, and assembly, the time spent on each task is also recorded on the same card. This system tracks both attendance and job performance simultaneously.

3.5.2 MAINTAINING OF TIME RECORDS

An effective time-recording system is essential for ensuring accurate attendance, wage calculation, and overall workforce management. To maintain precise time records, the system must meet several key criteria:

1. **Accuracy and Prevention of Proxy Attendance:** The system should be designed to prevent any worker from marking attendance on behalf of another, ensuring reliability and accuracy in attendance records.
2. **Inclusion of Piece Workers:** Even workers paid based on the number of units produced must have their time recorded to ensure regular attendance and maintain discipline. This helps in achieving a steady production flow.
3. **Recording Entry and Exit:** Both the arrival and departure times of workers should be documented to calculate total hours worked, which directly impacts wage determination.
4. **Mechanical or Automated System:** To minimize disputes and ensure accuracy, a mechanical or automated method should be employed for recording time, reducing human errors.
5. **Recording Late Arrivals:** Any late arrivals must be properly recorded to maintain discipline. Leniency from time-keepers in this regard can lead to indiscipline.
6. **Efficiency and Speed:** The system should operate smoothly and efficiently, avoiding long queues at the factory gate. Sufficient clocks or recording devices should be installed to accommodate all workers without causing delays.
7. **Supervision and Compliance:** A responsible officer should regularly monitor the time-recording process to ensure that all procedures are being followed correctly and efficiently.



SELF-ASSESSMENT QUESTIONS

1. What is the primary purpose of time-keeping in labor cost control?
 - a. To record how time is utilized for each job
 - b. To monitor the total number of hours spent on a job or process
 - c. To track attendance and ensure punctuality and discipline
 - d. To calculate the quantity and value of completed work
2. During April 2024, the following information is obtained from the Personnel Department of a manufacturing company. Labour force at the beginning of the month: 2,500 Labour force at the end of the month: 2,800. During the month, 40 people left the company and 60 persons were discharged. 350 workers were engaged, of which 50 were appointed to fill the vacancies created by separations, and the rest due to expansion. Calculate the Labour Turn-over by the Flux Method.
 - a. 9.00%
 - b. 8.49%
 - c. 16.00%
 - d. 15.49%

3.6 METHODS OF WORK STUDY

To effectively encourage workers, it is crucial to implement a well-designed incentive wage system. As money is a strong motivator, a monetary incentive structure becomes essential. In these systems, bonuses are awarded by comparing actual performance or production levels with predetermined standard levels. A bonus is granted when actual performance exceeds the established standard. To establish this standard, a fixed standard time should be defined, representing the time allotted to complete a specific task. The actual time spent is then compared to this standard. Work Study, including Job Study and Method Study, plays a vital role in determining the standard time for tasks, making it a key component in developing an effective incentive system.

3.6.1 METHOD STUDY

Method Study is undertaken to improve production techniques and ensure the efficient use of resources such as labor, machinery, and materials. The process involves several stages:

- ❑ **Job Selection:** The first step is to select the specific jobs or tasks that need to be analyzed. Jobs that are costly, complex, or time-consuming are prioritized since improving them can lead to significant cost savings and efficiency gains.

Example: In a manufacturing plant, the process of assembling a product might be chosen for analysis if it requires a lot of time and resources.

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- ❑ **Job Analysis:** Once a task is selected, a detailed examination of its current process is conducted. This involves analyzing aspects like the purpose of the job, how it's performed, its location, the sequence of operations, the tools or equipment used, and the skills required.

Example: For an assembly process, the current steps (like picking parts, fitting them together, and inspecting them) are observed and documented to understand how the task is performed.

- ❑ **Developing an Improved Method:** After analyzing the existing method, the next step is to develop a better approach. This might include changing the sequence of steps, using better tools, introducing new technology, or reorganizing the layout of the workspace. The goal is to make the process more efficient, less time-consuming, and cost-effective.

Example: If workers walk back and forth to collect parts, the new method might introduce a conveyor system that delivers parts directly to the worker's station, saving time and effort.

- ❑ **Implementation of the New Method:** Once a new method is developed, it is introduced on the production floor. This step may require training workers, modifying equipment, or rearranging the workspace to accommodate the new process.

Example: If the new method involves using an automatic tool instead of a manual one, workers may need to be trained on how to use the tool effectively.

- ❑ **Follow-up and Monitoring:** After implementation, it is essential to track and monitor the performance of the new method. This step ensures that workers are following the new process correctly and that the desired improvements are being achieved. If any issues arise, adjustments are made.

Example: Supervisors might observe workers using the new conveyor system to ensure that the parts are delivered on time and that workers are following the correct procedure.

3.6.2 EVALUATION OF JOB

Job evaluation is a systematic approach used to determine the relative worth of jobs within an organization. The process begins by identifying key factors for assessment, such as training, experience, intelligence, responsibility, and working conditions. Points are then assigned to each factor based on their significance, and the total score for a job is used to align its corresponding hourly wage.

Benefits of Job Evaluation

1. It helps establish an equitable wage and salary structure by linking pay rates to the specific attributes of each job.
2. It enhances employer-employee relationships by reducing personal biases in wage determination.

3. It ensures that jobs are assigned to individuals with the appropriate qualifications by emphasizing the unique requirements of each role.
4. Job evaluation often works in conjunction with job analysis, leading to advantages in recruitment, selection, placement, training, promotion, and transfer.
5. It facilitates job classification and streamlines work processes.
6. It ensures consistency in the wage structure across the organization.

There are several approaches to job evaluation, each of which is designed to assess job roles and determine appropriate wage structures:

1. **Point Ranking Method:** In this method, each job is assessed based on various factors or characteristics, such as the required skills, effort, working conditions, risks, and responsibilities. These factors represent the specific requirements necessary to perform the job effectively. Each factor is assigned a weight or point value based on its significance for the job. For example, some jobs may prioritize experience, while others may place more emphasis on education. After assigning points to the job factors, jobs are ranked according to the total points they earn. The wage structure is then developed based on these rankings. While this method is sound in theory, challenges can arise in accurately assigning the appropriate weight to each job factor. Example: In a hospital, jobs are evaluated based on factors like educational qualifications, experience, physical effort, working conditions, and level of responsibility. For example:

- ❑ A surgeon may earn higher points for educational qualifications (due to advanced degrees), experience, and responsibility (critical decisions impacting life and death).
- ❑ A nurse may score highly on physical effort and working conditions (long hours and exposure to health risks), but lower on educational qualifications compared to the surgeon.
- ❑ A receptionist might score lower overall, as the role involves less technical skill and responsibility.

Based on the total points, jobs are ranked, and corresponding wages are structured.

2. **Ranking Method:** In this approach, jobs are ranked in order of importance based on criteria such as required skills, experience, and working conditions. The jobs are arranged in a sequence, either from the lowest to the highest or in reverse order. Wages are then determined based on the job rankings. This method is simple, easy to understand, and cost-effective, making it suitable for smaller organizations with a limited number of well-defined jobs. However, in larger organizations with more complex roles, this method may

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not be as effective. Example: In a small retail business, jobs are ranked from least to most complex:

- ❑ A stock clerk may be ranked lowest, as the role primarily involves restocking shelves with limited decision-making.
- ❑ A cashier may be ranked higher, as it requires handling money and customer interactions.
- ❑ A supervisor may be ranked highest, as the role involves overseeing operations and staff performance.

Using this ranking, wages are set with supervisors earning the most and stock clerks earning the least.

3. Grading Method: The grading method improves upon the ranking method by categorizing jobs into predetermined grades or classes. Each job is analyzed according to job factors such as complexity, supervision, responsibility, and educational requirements. After this analysis, the job is assigned to a specific grade. This method provides a more structured approach than the ranking method and is helpful when a more detailed evaluation of job roles is required. Example: In a university, jobs are classified into grades based on factors like education, complexity of tasks, and responsibility:

- ❑ Grade A: Administrative assistants and data entry operators, requiring basic educational qualifications and limited responsibility.
- ❑ Grade B: Faculty members, requiring advanced degrees and a higher level of responsibility for teaching and research.
- ❑ Grade C: Department heads, who oversee faculty and administrative functions, requiring both leadership skills and substantial experience.

Jobs are placed into these grades based on shared characteristics, and wages are standardized within each grade.



SELF-ASSESSMENT QUESTIONS

3. What is the main purpose of Method Study in an incentive wage system?
 - a. To set a fair wage structure
 - b. To assess job performance
 - c. To improve production techniques and maximize efficiency
 - d. To evaluate workers' qualifications
4. Which method of job evaluation assigns a specific weight or point value to job factors such as skills and responsibilities?

a. Ranking Method	b. Grading Method
c. Point Ranking Method	d. Method Study

3.7 METHODS OF WAGE PAYMENT

A key element of labor cost control is the wage system. An effective wage payment system not only guarantees a minimum wage but also incentivizes efficient workers, significantly enhancing motivation. It's essential to recognize that high wages do not automatically equate to high labor costs. In many cases, higher wages can lead to increased worker productivity, ultimately reducing the per-unit cost of production. Conversely, paying lower wages may lead to diminished productivity, indicating that higher wages do not inherently result in higher costs.

3.7.1 FLAT TIME RATE SYSTEM

In this system, the hourly wage rate is pre-established, and payment is based solely on the number of hours worked, regardless of the output produced. Overtime pay is calculated according to statutory regulations. A key advantage for employees is the guarantee of a minimum income, even if their productivity is low. In cases where a worker is unable to perform their duties due to valid reasons, such as illness or physical disability, they are still entitled to receive wages for the time assigned to the task.

This method is particularly suitable under the following circumstances:

- (a) When the job requires specialized skills, prioritizing quality over quantity.
- (b) When output or services cannot be precisely measured or quantified.
- (c) When a worker's performance depends on another worker's, limiting individual control over output.
- (d) When production speed is determined by machine capabilities or processing times.
- (e) When employees are trainees or have limited experience.
- (f) When direct and continuous supervision is not feasible.

The primary benefit of this method is that it guarantees workers a minimum income, allowing them to concentrate on quality without the pressure of monetary incentives tied to increased output. However, a significant drawback is that it provides no incentive for high-performing workers. Both efficient and inefficient workers receive the same wage rate, which may lead to a situation where even efficient workers become less motivated due to the lack of incentives.

Example: Ramesh is a machine operator in a factory where the flat time rate system is followed. He is paid ₹200 per hour for an 8-hour workday. If Ramesh works 2 hours of overtime, for which he is compensated at 1.5 times his regular hourly rate, calculate his total earnings for the day, including overtime pay.

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Solution:**1. Regular Earnings:**

Regular hourly wage = ₹200 Regular work hours = 8 hours

Regular Earnings = Hourly Wage $\times$ Work Hours = ₹200 $\times$ 8 = ₹1,600

2. Overtime Earnings:

Overtime hourly rate = 1.5 $\times$ Regular hourly wage

Overtime Hourly Rate = 1.5 $\times$ ₹200 = ₹300

Overtime hours worked = 2 hours

Overtime Earnings = Overtime Hourly Rate $\times$ Overtime Hours
= ₹300 $\times$ 2 = ₹600

3. Total Earnings:

Total earnings = Regular earnings + Overtime earnings

Total Earnings = ₹1,600 + ₹600 = ₹2,200

3.7.2 TIME RATE AT HIGH DAY RATE PLAN

This plan was first introduced by the Ford Motor Company in the USA to motivate workers to put in extra effort in their jobs. Under this system, workers are paid a wage rate significantly higher than the average rate in the region or industry. In return, employees are expected to maintain a high level of performance, both in terms of quantity and quality. As a result, highly paid workers may not be as expensive as they initially appear. The high wage plan is particularly effective in environments where high-quality work and increased productivity are critical.

The advantages of implementing this plan include:

1. It is simple to administer and cost-effective.
2. It attracts skilled and efficient workers by providing strong incentives.
3. It reduces the need for extensive supervision.
4. Increased productivity can lead to lower unit labor costs.

3.7.3 GRADUATED TIME RATE PLAN

Under this method, payment is made at a time rate that varies according to the individual characteristics of the workers. Additionally, the rate is adjusted based on the official cost of living index. This approach benefits both employers and employees by offering a fair compensation system that takes into account the workers' personal qualities while also adjusting for fluctuations in living costs.

3.7.4 PIECE RATE METHOD

This method, often referred to as payment by results, compensates workers based on the output they produce. The more a worker produces, the

higher their earnings. Under the piece rate system, workers are paid a fixed rate for either the time worked or the units produced.

Advantages of the system include:

1. Workers are motivated to increase their output since their wages are directly linked to production levels.
2. Increased production reduces overhead costs per unit, even though total overheads may rise. However, the increase in overheads is usually smaller than the rise in turnover.
3. Wages tied to production make it easier for management to calculate labor costs per unit or job.
4. The system encourages efficiency, as less efficient workers earn less compared to their more productive peers.
5. Workers tend to take better care of tools and machinery, as this directly impacts their productivity and, consequently, their wages.
6. The system is easy to administer and implement.

However, there are also disadvantages:

1. Determining a fair piece rate can be difficult. If the rate is set too high and later adjusted downwards, it can cause dissatisfaction and conflict among workers.
2. Since labor cost per unit is fixed, workers may not see substantial gains from improved productivity unless there is a significant reduction in overheads.
3. A focus on increasing output might compromise quality, requiring stricter quality control, which could lead to higher costs.
4. In their pursuit of higher output, workers might misuse or damage materials.
5. The system could create dissatisfaction among slower workers or those on time-based wages.
6. Workers may handle machines roughly in an effort to increase production, potentially causing damage or breakdowns.

Example: A factory produces t-shirts, and the piece rate for each t-shirt made is ₹20. Two workers, Worker X and Worker Y, are given tasks for the day.

- ❑ Worker X produces 120 t-shirts in a day.
- ❑ Worker Y produces 150 t-shirts in a day.

Calculate the total earnings of Worker X and Worker Y based on the piece rate method.

Solution:

Calculate Worker X's Earnings

- ❑ Piece rate per t-shirt = ₹20
- ❑ Total t-shirts produced by Worker X = 120

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$$\begin{aligned}
 \text{Worker X's total earnings} &= \text{Piece rate} \times \text{Number of t-shirts produced} \\
 &= ₹20 \times 120 \\
 &= ₹2,400
 \end{aligned}$$

Calculate Worker Y's Earnings

- Piece rate per t-shirt = ₹20
- Total t-shirts produced by Worker Y = 150

$$\begin{aligned}
 \text{Worker Y's total earnings} &= \text{Piece rate} \times \text{Number of t-shirts produced} \\
 &= ₹20 \times 150 \\
 &= ₹3,000
 \end{aligned}$$

3.7.5 TAYLOR'S DIFFERENTIAL PIECE RATE SYSTEM

Taylor is recognized as the father of scientific management and proposed a system known as Differential Piece Rate. He classified workers into two categories: efficient and inefficient. His approach advocates for maximizing encouragement for efficient workers while penalizing inefficient ones. To implement this, he suggested two distinct rates based on performance: efficient workers should receive 120% of the normal piece rate, while inefficient workers should receive only 80% of that rate.

To measure efficiency, each worker is assigned a standard production quantity to complete within a specified timeframe. Actual production is then compared to this standard. Workers who exceed the standard are classified as efficient, while those who do not meet it are deemed inefficient. The advantages and disadvantages of this system are as follows:

Advantages of the system include:

- (a) The system offers employees a powerful incentive that promotes increased production;
- (b) The incentive structure draws in qualified personnel to the business.
- (c) Because of its simplicity, this approach is simple to comprehend.

However, there are also disadvantages:

- (a) Slow workers and beginners face significant penalties, as do those who encounter challenges beyond their control, such as medical issues or accidents. As a result, this system is often criticized for lacking a human element.
- (b) In the pursuit of increased output, there is a risk that quality may be compromised in favor of higher production quantities.

Example 1: Using Taylor's differential piece rate system, calculate the earnings of the workers John, Jack, and Jane with the following data:

- Standard time per piece: 15 minutes
- Jack produced: 35 units

- ❑ Normal rate per hour (in an 8-hour day): ₹120
- ❑ John produced: 40 units
- ❑ Jane produced: 50 units

Solution:**Earnings under Differential Piece Rate System**

Workers	John	Jack	Jane
Standard output per day (units)	32	32	32
Actual output per day (units)	40	35	50
Efficiency (%) (Actual output × 100) / Standard output	125%	108.33%	156.25%
Earning rate per unit	125% of the piece rate	125% of the piece rate	125% of the piece rate
Earning rate per unit (₹) (Refer to working note)	₹3.75	₹3.75	₹3.75
Earnings (₹)	₹150.00 (40 units × ₹3.75)	₹131.25 (35 units × ₹3.75)	₹187.50 (50 units × ₹3.75)

Working Note:

- ❑ Normal rate per hour = ₹120
- ❑ Normal rate per unit = ₹120 ÷ (60 minutes / 15 minutes) = ₹3.00 per unit

3.7.6 GANTT TASK BONUS PLAN

This system combines elements of both time and piecework compensation methods. A high standard or target is established, and workers are paid a time rate if their production falls below this set standard. However, if they meet or exceed the standard, they are compensated at a higher piece rate. This piece rate includes a bonus element, typically amounting to 20% of the time rate. Thus, the system fundamentally entails paying workers based on a time rate when they do not meet the standard, and a piece rate when they do.

The wages for workers under this plan are calculated as shown in Table 3.4:

Table 3.4 Gantt Task Bonus Plan

Output	Payment
(i) Output below standard	Guaranteed time rate.
(ii) Output at standard	Time rate plus a bonus of 20% (usually) of the time rate.
(iii) Output above standard	High piece rate for the worker's entire output, which includes a 20% bonus of the piece rate.

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Advantages:

1. By ensuring a time rate, the system protects less efficient workers while providing a strong incentive for efficient ones.
2. It is simple to comprehend and apply.
3. The method encourages better planning and oversight.

Drawbacks:

1. Slower workers may become less motivated to increase their productivity as a result of the guaranteed time rate.
2. While simple, the system might not account for unique challenges faced by individual workers.
3. Potential discrepancies in the quality of output may arise if workers prioritize speed over craftsmanship.

Example 1: In a factory, the standard time allowed for completing a given task (40 units) is 6 hours. The guaranteed time wages are ₹25 per hour. If a task is completed in less than the standard time, the high rate of ₹5 per unit is payable. Calculate the wages of a worker under the Gantt system, if he completes the task in: (i) 8 hours; (ii) 6 hours; and (iii) 4 hours. Also, ascertain the comparative rate of earnings per hour under the three situations.

Solution:

- (i) When the worker performs the task in **8 hours**:

Since the worker exceeds the standard time, he earns only the time wage rate.

$$\text{Earnings} = 8 \text{ hours} \times ₹25 \text{ per hour} = ₹200$$

- (ii) When the worker performs the task in **6 hours** (standard time):

The worker earns time wages plus a bonus of 20% on time wages.

$$\text{Time wages: } 6 \text{ hours} \times ₹25 \text{ per hour} = ₹150$$

$$\text{Bonus: } 20\% \times ₹150 = ₹30$$

$$\text{Total earnings: } ₹150 + ₹30 = ₹180$$

- (iii) When the worker performs the task in **4 hours**:

Since the worker completes the task in less than the standard time, he earns at the piece rate.

$$\text{Piece rate earnings: } 40 \text{ units} \times ₹5 \text{ per unit} = ₹200$$

Comparative Rate of Earnings Per Hour

- (i) ₹200/8 hours = ₹25 per hour (ii) ₹180/6 hours = ₹30 per hour
 (iii) ₹200/4 hours = ₹50 per hour

3.7.7 HALSEY PREMIUM PLAN

The American engineer F.A. Halsey created this plan. Bonuses are given under this approach according on the amount of time saved on a task. For a certain task, a standard time is set, and a worker is eligible for a bonus if they finish the job faster than the standard. 50% of the pay for the time saved is used to compute the bonus. In contrast, workers are assured of time wages if they work more than the required amount of time. The following formula is used to determine total earnings:

$$\text{Total Earnings} = H \times R + 50\% \times [S - H] \times R$$

Where:

- H = Hours worked
- R = Rate per hour
- S = Standard time

Example 1: Calculate the total earnings of a worker under the Halsey Premium Plan based on the following information:

- Standard Time: 15 hours
- Hourly Rate: ₹4
- Time Taken: 10 hours

Solution:

Earnings under the Halsey Premium Plan are calculated using the formula:

$$\text{Total Earnings} = H \times R + 50\% \times (S - H) \times R$$

Earnings Calculation under Halsey Premium Plan

Parameters	Value
Standard Time (S)	15 hours
Time Taken (H)	10 hours
Rate per Hour (R)	₹4
Time Saved (S - H)	5 hours (15 - 10)
Bonus (50% of Time Saved)	₹4 × 5 × 50% = ₹10
Total Earnings	₹40 + ₹10 = ₹50

3.7.8 HALSEY-WEIR PLAN

In this method, the primary distinction from the Halsey Plan is that the bonus for the time saved is set at 33⅓% instead of 50%. Consequently, the formula for calculating total earnings is adjusted as follows:

$$\text{Total Earnings} = H \times R + 33\frac{1}{3}\% \times [S - H] \times R$$

Where:

- H = Hours worked
- R = Rate per hour
- S = Standard time

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Benefits

1. This approach ensures a time rate while providing chances for higher profits via improved output.
2. The system is fair since the employer directly benefits from efforts to upgrade equipment and production techniques.

Drawbacks

1. The motivation is not as strong as it would be under the piece rate system; in fact, employees who work harder get paid less per piece.
2. Workers might not like the sharing principle that this approach entails.

Example 1: From the following details, calculate the total earnings of a worker under the Halsey-Weir Plan:

- Standard Time : 10 hours □ Time Taken : 8 hours
 □ Hourly Rate : ₹2

Solution:

Earnings under the Halsey-Weir Plan are calculated using the formula:

$$\text{Total Earnings} = H \times R + 33.33\% \times (S - H) \times R$$

Calculation:

Parameters	Value
Standard Time (S)	10 hours
Time Taken (H)	8 hours
Rate per Hour (R)	₹2
Time Saved (S - H)	2 hours (10 - 8)
Bonus (33.33% of Time Saved)	₹2 × 2 × 33.33% = ₹1.33
Total Earnings	₹16 + ₹1.33 = ₹17.33

3.7.9 ROWAN PLAN

In terms of time saved, the Halsey Plan and the premium bonus plan, which was created by Mr. James Rowan, are comparable. Bonus hours are paid at the time rate under this plan, but they are determined by dividing the amount of time saved by the amount of time consumed by the amount of time permitted. The following formula is used to calculate total earnings:

$$\text{Total Earnings} = H \times R + [(S - H) / S] \times H \times R$$

Where:

- H = Hours worked □ R = Rate per hour □ S = Standard time

Advantages

1. This system is often regarded as foolproof, ensuring that a worker can never double their earnings, even in cases of poor rate setting.

2. By providing a higher reward for intermediate efficiency than the Halsey Plan, it successfully encourages workers with moderate levels of efficiency.
3. Because it is seen as equal, employers find the sharing principle appealing.

Disadvantages

1. The method can be a little challenging to comprehend and use.
2. When the time saved surpasses 50% of the permitted time, the incentive becomes less strong at high production levels.
3. Employees often oppose the sharing idea, despite the fact that employers find it appealing.

Example 1:

From the following information, calculate the total earnings of a worker under the Rowan System:

- Standard Time (S): 10 hours
- Time Taken (H): 8 hours
- Rate per Hour (R): ₹3

Solution:

The formula for calculating total earnings under the Rowan System is:

$$\text{Total Earnings} = H \times R + \left(\frac{\text{Time Saved}}{\text{Standard Time}} \times H \times R \right)$$

Calculation:

Parameters	Value
Standard Time (S)	10 hours
Time Taken (H)	8 hours
Time Saved (S – H)	2 hours (10 – 8)
Rate per Hour (R)	₹3
Regular Earnings (H × R)	₹24 (8 × ₹3)
Bonus	₹4.8 (2/10 × 8 × ₹3)
Total Earnings	₹28.8 (₹24 + ₹4.8)

3.7.10 GROUP BONUS PLAN

The aforementioned plans are typically individual-based bonus systems. However, measuring individual performance can be difficult, and individual output is often dependent on the performance of the team. In such cases, a group bonus plan is introduced as an alternative. The bonus amount, calculated based on overall productivity, is either equally distributed or allocated in a mutually agreed proportion

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among the group members. The key objectives of the group bonus system include:

- (a) Fostering teamwork.
- (b) Reducing material and time waste.
- (c) Acknowledging collective efforts.
- (d) Enhancing productivity.

3.7.11 PRIESTMAN'S BONUS PLAN

In this system, production standards are defined in terms of units or points, and the actual production is compared with these standards. If actual production exceeds the set standard, workers receive additional compensation based on a percentage of the excess output. No bonus is given if actual production fails to meet or surpass the standard. This method is commonly applied in foundries.

Example:

In a foundry, the standard production is set at 100 units per day. Workers are paid at a rate of ₹20 per unit. If actual production exceeds the standard, a bonus of 10% of the rate per unit is awarded for the excess units. Calculate the total earnings of a worker under the following scenarios:

1. Actual production is 90 units.
2. Actual production is 100 units.
3. Actual production is 120 units.

Solution

Given Data:

- Standard production = 100 units
- Rate per unit = ₹20
- Bonus rate for excess units = 10% of ₹20 = ₹2 per unit

Case 1: Actual production = 90 units

Since the actual production is below the standard, no bonus is awarded.

$$\text{Earnings} = 90 \times 20 = ₹1,800$$

Case 2: Actual production = 100 units

The actual production equals the standard, so no bonus is awarded.

$$\text{Earnings} = 100 \times 20 = ₹2,000$$

Case 3: Actual production = 120 units

The actual production exceeds the standard by 20 units. A bonus is awarded on the excess production.

$$\text{Earnings} = (120 \times 20) + (20 \times 2)$$

$$\text{Earnings} = 2,400 + 240 = ₹2,640$$

3.7.12 TOWNE PROFIT SHARING PLAN

This method sets cost standards, mainly focusing on labor costs, and compares actual costs against these standards. If there are savings, they are shared between the workers and supervisory staff according to an agreed-upon ratio. The rationale behind this method is that cost savings stem from the combined efforts of both workers and supervisors, making it fair for both groups to share the benefits.

Example:

A company sets a standard labor cost for producing 1,000 units at ₹50,000. After production, the actual labor cost incurred is ₹45,000, resulting in a saving of ₹5,000. The company has agreed to share these savings between workers and supervisors in the ratio 3:2.

Calculate:

1. The amount of savings shared with workers.
2. The amount of savings shared with supervisors.
3. The total amount received by each group.

Solution

Given Data:

- Standard labor cost = ₹50,000
- Actual labor cost = ₹45,000
- Total savings = ₹50,000 - ₹45,000 = ₹5,000
- Savings sharing ratio (Workers : Supervisors) = 3:2

Step 1: Allocation of Savings

$$\text{Savings for Workers} = \frac{3}{3+2} \times ₹5,000 = ₹3,000$$

$$\text{Savings for Supervisors} = \frac{2}{3+2} \times ₹5,000 = ₹2,000$$

Step 2: Total Amount Earned

- **Workers' Total Earnings:** ₹45,000 (actual labor cost) + ₹3,000 (bonus) = ₹48,000
- **Supervisors' Total Earnings:** ₹2,000 (bonus only, as supervisory labor cost is part of the savings).

3.7.13 NON-MONETARY INCENTIVES

In addition to monetary rewards, non-monetary incentives are offered to boost employee morale and motivation. While these incentives do not result in direct financial compensation, they play a significant role

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in improving productivity by promoting a positive work environment. Examples of Non-monetary incentives are given in figure 3.3:

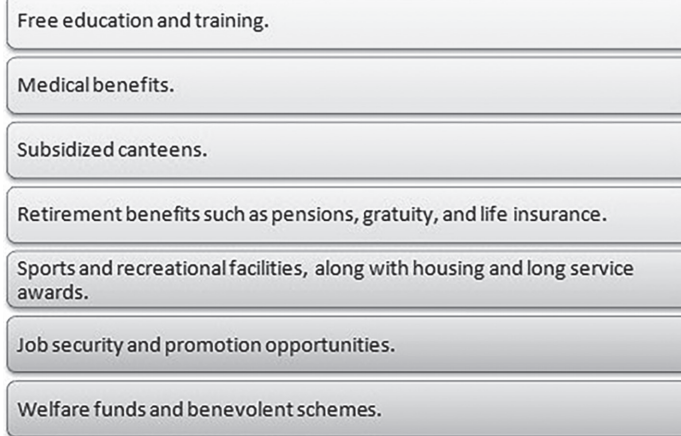


Figure 3.3 Non-Monetary Incentives

Below table 3.5 summarizes all the methods of payment:

Table 3.5 Methods of Payment

Method	Description	Advantages	Disadvantages	Formula
Flat Time Rate System	Payment based on pre-established hourly rates for the number of hours worked. Overtime is paid as per regulations.	Guarantees minimum income and emphasizes quality over quantity.	Lacks incentive for high performers; treats efficient and inefficient workers equally.	Total Earnings = Regular Earnings + Overtime Earnings
Time Rate at High Day Rate Plan	A wage rate significantly higher than the prevailing rate, ensuring high performance in terms of quality and quantity.	Simple to operate, attracts skilled workers, minimizes supervision, and reduces unit labor costs.	High wage costs if poorly implemented.	N/A
Graduated Time Rate Plan	Payment fluctuates based on workers' individual attributes and cost of living index.	Fair and flexible to workers and employers.	May complicate administration.	N/A
Piece Rate Method	Workers are paid based on their output, encouraging higher production.	Motivates efficiency, reduces unit overhead, rewards performance, and simplifies labor cost calculations.	Quality may be compromised; material and equipment may be mishandled; dissatisfaction among slower workers.	Total Earnings = Units Produced × Rate Per Unit

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Method	Description	Advantages	Disadvantages	Formula
Taylor's Differential Piece Rate System	Two rates are applied: 120% of normal rate for efficient workers and 80% for inefficient workers.	Strong incentive for efficient workers; straightforward to understand.	Penalizes slower workers excessively; lacks a human element.	Total Earnings = Efficiency % $\times$ Rate Per Unit
Gantt Task Bonus Plan	Combines time and piecework methods. Bonuses are given for meeting/exceeding standards, typically 20% of time rate.	Balances efficiency with worker protection; easy to understand and implement.	Guaranteed time rate may reduce motivation; may lead to uneven quality.	Based on output: (i) Below standard: Time rate, (ii) At standard: Time rate + Bonus, (iii) Above standard: High piece rate.
Halsey Premium Plan	Workers earn bonuses based on the time saved, calculated as 50% of wages for the time saved.	Guarantees minimum wage; motivates workers to save time.	Incentive weakens as efficiency increases; sharing principle may discourage employees.	Total Earnings = $H \times R + 50\% \times (S - H) \times R$
Halsey-Weir Plan	Similar to Halsey Plan, but bonus is 33⅓% of wages for time saved instead of 50%.	Guarantees time rate; equitable sharing of productivity gains.	Lower incentives compared to Halsey Plan.	Total Earnings = $H \times R + 33\frac{1}{3}\% \times (S - H) \times R$
Rowan Plan	Bonus is proportional to the time saved relative to time allowed, calculated at time rate.	Ensures fair sharing between employer and employee; encourages moderate efficiency.	Complex to understand and implement; weak incentive for very high efficiency.	Total Earnings = $H \times R + [(S - H) / S] \times H \times R$
Group Bonus Plan	Bonus distributed among a group based on collective productivity.	Promotes teamwork, reduces waste, and recognizes group efforts.	Individual contributions may not be rewarded fairly; may lead to conflicts.	Bonus = Total Group Bonus $\div$ Members
Priestman's Bonus Plan	Workers earn extra wages based on production exceeding standard output, expressed as a percentage.	Motivates higher productivity; simple to implement.	No bonus for not meeting standards; potential decline in quality.	Bonus = % of Output Above Standard
Towne Profit Sharing Plan	Cost savings are shared between workers and supervisors.	Encourages collaboration and cost efficiency.	Standards may be difficult to set; savings may vary across periods.	Savings = Standard Costs – Actual Costs; Bonus = Agreed % of Savings

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Method	Description	Advantages	Disadvantages	Formula
Non-Monetary Incentives	Rewards that are not financial but enhance morale and motivation (e.g., training, benefits, job security).	Improves employee satisfaction, loyalty, and productivity.	Benefits may not appeal equally to all employees; high administrative costs.	N/A

Self Assessment Questions

5. A factory produces t-shirts, and the piece rate for each t-shirt made is ₹25. Two workers, Worker A and Worker B, are assigned production tasks for the day.

- ☐ Worker A produces 130 t-shirts in a day.
- ☐ Worker B produces 170 t-shirts in a day.

Based on the piece rate method compute the combined total earning of Worker A and Worker B for the day?

- (a) ₹7,250 (b) ₹7,400 (c) ₹7,500 (d) ₹7,350

6. Shyam Ltd. follows Taylor's Differential Piece Rate System to calculate the earnings of its workers. The relevant details for the calculation are as follows:

- ☐ Standard time per piece: 12 minutes
- ☐ Normal rate per hour: ₹150 (for an 8-hour workday)
- ☐ Low piece rate: 80% of the normal rate (if production is below standard)
- ☐ High piece rate: 120% of the normal rate (if production meets or exceeds the standard)

The following workers produced the following number of units in a single day (8 hours):

- ☐ Aman produced 45 units ☐ Ravi produced 40 units
- ☐ Rohan produced 55 units

Who gets the highest wage and how much?

- (a) Aman, ₹1,620 (b) Ravi, ₹1,980
(c) Ravi, ₹1,440 (d) Rohan, ₹1,980

Activity

Design a simple wage incentive plan for a team of workers where their bonus is based on overall team performance. Consider factors such as productivity, quality, and team spirit. Explain how you would allocate bonuses fairly among team members.

3.8 OVERHEAD COST CONTROL

Overhead Cost Control refers to the practices businesses adopt to track, regulate, and minimize indirect expenses that are not directly linked to production. These costs include items such as rent, utilities, administrative salaries, and depreciation.

Controlling Administrative Overheads

Administrative overheads, being strategic in nature, are usually managed by top-level management. These costs often stem from activities like policy formulation and overall organizational decision-making. The process of control begins with classifying and departmentalizing these expenses. Actual expenditures are collected for each department and then benchmarked against a reference point. Deviations are examined, and the responsibility for addressing increased costs is assigned to the relevant departmental head.

Methods of Control:

1. **Comparison with Previous Year's Figures:** Expenses are compared to the prior year's data, and changes are evaluated. However, this approach may not always be effective due to varying conditions between years. **Example:** Suppose the rent for the corporate office was ₹2,00,000 in 2023 but increased to ₹2,50,000 in 2024. Upon investigation, it was found that the increase was due to an increase in market rent rates. If the management had signed a long-term rental contract at a fixed rate, they could have avoided this increase.
2. **Use of Budgets:** Budgets, which estimate current-year expenses while accounting for changes in conditions, are useful for systematic cost control. Comparing actual overheads with budgeted figures is an effective strategy. **Example:** If the budget for employee training expenses is set at ₹50,000 for the year but the actual expense reaches ₹70,000, the HR manager must justify the excess. It could be due to an unplanned software training program. The company may decide to control this by prioritizing essential training only.
3. **Use of Standards:** Though challenging to apply universally due to the subjective nature of administrative activities, standardization can be effective in specific areas, such as setting time benchmarks for voucher processing or payment approvals. **Example:** Suppose an organization sets a standard of 7 days for processing payment approvals. If it is observed that payments are being delayed to 12 days, the accounts payable team is asked to review the process and identify the bottlenecks. They may find that manual checks are causing the delay, and automation is recommended as a solution.

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Controlling Sales & Distribution (S&D) Expenses

S&D expenses relate to sales and distribution functions and are influenced by external factors like market trends, technology, and consumer behavior. Controlling these costs focuses on achieving maximum returns from every rupee spent, rather than restricting expenditures. Effective control requires proper classification and accurate tracking of these costs.

Methods of Analysis:

1. **Geographical Analysis:** Breaking down S&D expenses by location—such as regions, zones, or domestic versus international markets—helps identify performance variations. **Example:** A company selling fast-moving consumer goods (FMCG) finds that transportation expenses in North India are 20% higher than in South India. Upon investigation, it discovers that poor road infrastructure and longer delivery distances are the cause. To control costs, the company may establish a regional warehouse closer to key markets in the north.
2. **Customer Type Analysis:** Analyzing expenses by customer segments, such as institutional, government, or retail clients, provides insights into profitability across different groups. **Example:** A company notices that its costs to serve government clients are higher than those for retail clients because of the longer approval process and frequent documentation. To control costs, the company may establish a dedicated sales team for government clients or streamline the bidding process.
3. **Product or Service Analysis:** Evaluating costs across product categories, ranges, or brands allows for an understanding of each segment's contribution to overall sales. **Example:** A consumer electronics company finds that Product A (a new smartphone) requires higher advertising costs compared to Product B (an older model). While Product A has a profit margin of 20%, the higher advertising costs reduce its overall contribution to profit. The company decides to shift its focus to Product B, which generates stable returns at lower advertising costs.
4. **Salesmen Performance Analysis:** Reviewing expenses linked to individual sales personnel helps assess their efficiency and contribution to profitability. **Example:** If Salesperson A generates sales worth ₹10,00,000 and incurs ₹1,00,000 in travel expenses (10% of sales), while Salesperson B generates sales worth ₹8,00,000 but incurs ₹2,00,000 in travel expenses (25% of sales), the performance of Salesperson B is less efficient. The company may offer Salesperson B guidance on more cost-effective travel routes or switch to virtual client meetings where possible.
5. **Distribution Channel Analysis:** Examining costs by distribution channels, such as wholesalers, retailers, or commission agents, enables businesses to identify opportunities for optimization.

Example: A company observes that distribution costs through retailers are higher than those through e-commerce platforms. Retailers demand higher margins and require in-store promotions. As a control measure, the company may increase direct-to-consumer (D2C) e-commerce sales to reduce dependence on retail channels.

3.8.1 ADVANTAGES OF CLASSIFICATION OF OVERHEADS INTO FIXED AND VARIABLE

The primary goal of dividing semi-variable expenses into fixed and variable components is to determine marginal costs. The detailed explanation of dividing semi variable expenses into fixed and variable expense is given in unit-4. Additionally, this segregation offers the following benefits:

- (a) **Expense Control:** Categorizing costs into fixed and variable helps manage expenses more effectively. Fixed costs, often referred to as policy costs, are less flexible and are incurred regardless of output levels, making them harder to control. In contrast, variable costs fluctuate with production levels and can be managed by assigning responsibility and aligning allowances with output achieved.
- (b) **Budget Preparation:** Breaking down overheads into fixed and variable components facilitates the creation of a flexible budget. This allows firms to estimate costs at varying activity levels and compare them to actual expenditures.

For instance, consider a factory with the following expenses in October 2011, when the output was 800 units:

- Fixed costs: ₹4,000 □ Variable costs: ₹3,200
- Semi-variable costs (50% fixed): ₹5,000

Total: ₹12,200

If the output increases to 1,000 units in November 2011, the estimated expenses would be:

Fixed: ₹4,000

Variable: $₹3,200 \times 1,000/800 = ₹4,000$

Semi-variable:

- (i) Fixed portion (50% of ₹5,000): ₹2,500
- (ii) Variable portion $₹2,500 \times 1,000/800 = ₹3,125$

Total: ₹13,625

This calculation shows that assuming expenses increase proportionally to ₹15,250 due to higher output would be incorrect, leading to poor budgeting.

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(c) **Decision-Making:** Separating fixed and variable components of semi-variable costs aids in critical decision-making. For instance, pricing strategies during a recession, for exports, or evaluating make-or-buy scenarios can be guided by this segregation. Decisions regarding continuing or shutting down operations or changing manufacturing or distribution processes also require an understanding of fixed and variable costs.

Techniques like marginal costing, cost-volume-profit analysis, and break-even analysis heavily depend on this segregation to provide accurate insights into costs and revenue impacts when production changes are considered.

3.8.2 ACCOUNTING AND CONTROL OF MANUFACTURING OVERHEADS

When it comes to allocating overheads across products, imagine you're running a busy workshop making handmade furniture. Your team is crafting tables, chairs, and cabinets, but there's a catch: while it's easy to calculate the direct costs like materials and labor, what about shared costs like electricity, rent, or the supervisor's salary? These are your **overheads**, and figuring out how much each product should bear is both an art and a science.

The Challenge of Waiting for Actual Overheads

Think about this: if you wait until the end of the year to know your actual overheads, you'll be making decisions with outdated information. That's like driving with your eyes on the rear-view mirror—it doesn't help much when you need to decide now. To solve this, businesses use **pre-determined rates** for overhead absorption. It's like setting a budget based on an educated guess of what your costs will be. This approach ensures you can allocate overheads to products as soon as their **prime costs** (materials, labor, and other direct expenses) are ready.

How Do We Make It Work?

Here's a step-by-step breakdown, bringing clarity to what might feel like a puzzle:

1. Estimating Overheads

Picture yourself gathering all the data: electricity bills, maintenance costs, supervisor salaries—everything your factory incurs. You use past records as a starting point and adjust for expected changes. It's like planning for a family road trip by estimating fuel, snacks, and toll costs in advance.

2. Assigning Overheads to Departments

Once you have the overheads, they must be allocated. This is done in three ways:

- ❑ **Direct Allocation:** Some costs, like a machine's electricity bill tied to a specific department, go straight to that department. Simple!

- ❑ **Cost Apportionment:** Other costs, like the factory manager's salary, are shared across departments based on logical criteria (e.g., the size or output of each department).
- ❑ **Re-apportionment:** After assigning costs to production and service departments, the overheads of service departments (like maintenance or quality control) are reallocated to production departments. This ensures the production teams carry the full cost load.

3. Absorbing Overheads into Products

You calculate a **pre-determined absorption rate** based on your overhead estimate and production activity level. For example, if you expect to produce 1,000 tables and have ₹1,00,000 in overheads, each table will “absorb” ₹100 of those costs. Now, every product has its fair share of the factory overhead built into its price.

4. Handling Over or Under Absorption

Fast-forward to the end of the year. Reality hits: your actual overheads differ from your estimates. Maybe you saved on electricity, or production ran slower than expected. This difference is called **over-absorption** (you charged too much) or **under-absorption** (you charged too little). Adjustments are made to align your estimates with reality, ensuring your costing system stays accurate.

Example: A manufacturing company produces two products, X and Y. The following costs are incurred for each product:

- ❑ Direct Material Costs: ₹4,00,000 for X and ₹2,50,000 for Y.
- ❑ Direct Labor Costs: ₹2,50,000 for X and ₹1,00,000 for Y.
- ❑ Direct Expenses: ₹80,000 for X and ₹1,20,000 for Y.
- ❑ Overhead Expenses: ₹8,40,000.

Prepare the Statement of Cost for products X and Y, showing Prime Cost, Overheads, and Works Cost.

Solution:

Statement of Cost

Particulars	Product X	Product Y	Total
Direct Materials	₹4,00,000	₹2,50,000	₹6,50,000
Direct Labor	₹2,50,000	₹1,00,000	₹3,50,000
Direct Expenses	₹80,000	₹1,20,000	₹2,00,000
Prime Cost	₹7,30,000	₹4,70,000	₹12,00,000
Overheads (70% of Prime Cost)	₹5,11,000	₹3,29,000	₹8,40,000
Works Cost	₹12,41,000	₹7,99,000	₹20,40,000

Note:- Overhead absorption rate is calculated as – $8,40,000 / 12,00,000 \times 100 = 70\%$

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3.8.3 ALLOCATION OF OVERHEADS OVER VARIOUS DEPARTMENTS OR DEPARTMENTALIZATION OF OVERHEADS

In most manufacturing setups, processes are functionally distinct and distributed across various departments within the factory. These departments are categorized based on their roles—some focus on actual production, while others provide essential auxiliary services. When such functional divisions are in place, factory overheads that can be directly attributed to specific production or service departments are allocated directly to those departments for accurate cost tracking and management.

In certain cases, it becomes necessary to divide a manufacturing organization into several cost centers, enabling more precise expense distribution and detailed control. The primary goal of establishing cost centers is to facilitate the efficient collection of data for similar activities, reducing the need for extensive cost analysis. By setting up cost centers, several cost elements can be definitively determined, thus minimizing the need for estimation, such as allowances for normal idle time or spending on consumable stores.

Cost centers can be classified into two main types:

- ❑ **Machine Centers:** For processes dominated by equipment, like an automated assembly line.
- ❑ **Personal Centers:** For tasks run by people, such as hand-finishing or quality checks.

For ease of expenditure recording, cost centers are sometimes assigned a code number.

Example: A company has three production departments (P1, P2, and P3) and two service departments (S1 and S2). The overhead expenses and additional details for allocation are as follows:

Overhead Expenses (Primary Distribution):

- | | |
|--|---|
| ❑ Production Department P1:
₹1,80,000 | ❑ Service Department Maintenance
S1: ₹50,000 |
| ❑ Production Department P2:
₹1,60,000 | ❑ Service Department Personnel
S2: ₹40,000 |
| ❑ Production Department P3:
₹1,20,000 | |

Allocation Basis:

Basis	P1	P2	P3	S1	S2
Machine Hours	50	30	20	—	—
Number of Employees	40	30	20	—	—

Distribute the costs of the service departments (S1 and S2) to the production departments P1, P2, and P3.

Solution:**Statement Showing Allocation of Service Departments' Costs to Production Departments**

Particulars	P1	P2	P3	S1	S2	Total
Primary Distribution	₹1,80,000	₹1,60,000	₹1,20,000	₹50,000	₹40,000	₹5,50,000
Allocation of S1 (Machine Hours)	₹25,000	₹15,000	₹10,000	(₹50,000)	—	—
Allocation of S2 (No. of Employees)	₹16,000	₹12,000	₹8,000	—	(₹40,000)	—
Total Costs Allocated	₹2,21,000	₹1,87,000	₹1,38,000	—	—	₹5,50,000

The collection of overheads by department offers the following benefits:

- (a) **Improved Estimation of Expenses:** Certain expenses related to specific departments can be estimated with a high degree of accuracy, leading to better overall precision in overhead estimation. Imagine you're budgeting for next month. Knowing how much electricity Department A used last month helps you estimate next month's costs with accuracy. When every department has its own expense record, you can make smarter financial predictions.
- (b) **Enhanced Control:** For controlling departmental expenses, it is crucial to have separate figures for each department. Effective control requires knowledge of the expected versus actual expenses for each activity. If expense data is only available for the factory as a whole, identifying overspending by a specific department becomes impossible. Let's say Department B's expenses skyrocket. Without departmental breakdowns, you'd have no idea where the problem lies. With separate records, you can investigate and fix the issue—whether it's an inefficient process or unnecessary spending.
- (c) **Determination of Departmental Costs:** To accurately determine the cost of each job, it is necessary to know the expenses incurred in the departments that handle the job or product. This ensures the job or product is charged with its appropriate share of indirect expenses. Since not all jobs pass through every department, and the work required may vary across departments, it is essential to charge each job only for the work done in the relevant department. This is achievable only when the overheads for each department are known separately. If a product only passes through Departments A and C, it shouldn't share the overheads from Department B. Departmental allocation ensures each product pays only for the work done in its relevant departments.
- (d) **Appropriate Costing Methods:** Different departments can apply distinct costing methods, such as batch costing for manufacturing

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parts or single/output costing for assembling the product. For example:

- ❑ Department A might use batch costing for creating components.
- ❑ Department B might use single/output costing for assembling a final product.

3.8.4 APPORTIONING OVERHEAD EXPENSES OVER VARIOUS DEPARTMENTS

Apportionment is the process of dividing shared costs (like rent, utilities, or managerial salaries) among different departments based on some logical or fair criteria. Since certain costs can't be attributed directly to one department, apportionment ensures that each department pays its "fair share" of the cost.

Key Principles of Apportionment

When deciding how to apportion costs, we can follow these principles:

1. Actual Benefit Principle

Under this principle, you allocate costs based on the actual benefit a department receives from a service or resource. Imagine you're allocating rent in a factory with multiple departments. You would allocate rent based on how much floor space each department occupies. This way, departments using more space pay more. For example: Rent of ₹50,000 is apportioned based on floor area. If Department A occupies 50% of the total space, it will receive 50% of the rent cost (₹25,000).

2. Potential Benefit Principle

Sometimes measuring actual benefits is too complicated or not feasible. For example, how would you track how much electricity is being used by each department? Instead, you might estimate the potential benefit, such as using the number of light bulbs or power points in each department. For example: If Department A has 10 light points and Department B has 5, you could allocate lighting costs based on the number of light points. If the total lighting cost is ₹6,000, Department A would get ₹4,000 (10/15 of the total), and Department B would get ₹2,000 (5/15 of the total).

3. Specific Criteria (Survey Method)

Sometimes, it's not easy to identify an obvious basis for allocation. This is where a survey comes in handy! The survey helps to determine how much time or attention each department spends on a particular service. For example: If the works manager spends 60% of his time on Department A, 30% on Department B, and 10% on Department C, you can allocate his salary based on this ratio.

4. Ability to Pay Principle

This principle suggests that departments that contribute more revenue should bear a larger share of costs. It's a way to ensure that departments that benefit more financially from the factory's overall functioning contribute proportionally more to shared services. For example: If Department A consumes materials worth ₹50,000 and Department B consumes materials worth ₹30,000, you would allocate the cost of maintaining the stores based on the value of materials consumed. Department A would pay ₹50,000 / ₹80,000 of the total cost, while Department B would pay ₹30,000 / ₹80,000.

The following table 3.6 outlines some potential bases for apportioning these costs:

Table 3.6 Bases for Apportioning Overheads

Overhead Cost	Bases of Apportionment
1. (i) Rent and other building expenses (ii) Lighting and heating (iii) Fire precaution service (iv) Air-conditioning	Floor area, or volume of department
2. (i) Perquisites (ii) Labor welfare expenses (iii) Timekeeping (iv) Personnel office (v) Supervision	Number of workers
3. (i) Compensation to workers (ii) Holiday pay (iii) ESI and PF contribution (iv) Perquisites	Direct wages
4. General overhead	Direct labor hour, or direct wages, or machine hours
5. (i) Depreciation of plant and machinery (ii) Repairs and maintenance of plant and machinery (iii) Insurance of stock	Capital values
6. (i) Power/steam consumption (ii) Internal transport (iii) Managerial salaries	Technical estimates
7. Lighting expenses (light)	Number of light points, area, or metered units

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Overhead Cost	Bases of Apportionment
8. Electric power (machine operation)	Horsepower of machines, number of machine hours, value of machines, or units consumed
9. (i) Material handling (ii) Stores overhead	Weight, volume, value, or unit of materials

Example: Golden Yarn has two production departments (Spinning and Weaving) and two Service Departments (Stores and Time Keeping). The following information is supplied from the costing records of the company:

Expense	Spinning (Rs.)	Weaving (Rs.)	Stores (Rs.)	Time Keeping (Rs.)	Total (Rs.)
Rent	4,800	3,600	2,400	1,200	12,000
Maintenance	3,000	2,500	1,500	1,000	8,000
Depreciation	2,200	1,800	1,200	800	6,000
Lighting	600	400	300	100	1,400
Insurance	3,000	2,500	1,500	1,000	8,000
Employer's Contribution to Provident Fund	700	500	300	200	1,700
Energy	4,000	3,000	2,000	1,000	10,000
Supervision	8,000	6,000	4,000	2,000	20,000

The following additional details are provided:

- ❑ Floor space (sq. ft.): Spinning – 400, Weaving – 300, Stores – 200, Time Keeping – 100
- ❑ Number of workers: Spinning – 50, Weaving – 50, Stores – 30, Time Keeping – 30
- ❑ Total Direct Wages (Rs.): Spinning – 20,000, Weaving – 15,000, Stores – 10,000, Time Keeping – 5,000
- ❑ Cost of Machinery (Rs.): Spinning – 60,000, Weaving – 45,000, Stores – 30,000, Time Keeping – 15,000
- ❑ Stock of goods: Spinning – 40,000, Weaving – 25,000, Stores – 15,000

Prepare a statement showing the apportionment of costs to various departments

Solution:**Departmental Overheads Distribution Statement**

Expenses	Basis of Apportionment	Total Rs.	Spinning Rs.	Weaving Rs.	Stores Rs.	Time Keeping Rs.
Rent	Floor space (400:300:200:100)	12,000	4,800	3,600	2,400	1,200
Maintenance	Cost of Machinery (60,000:45,000:30,000:15,000)	8,000	3,200	2,400	1,600	800

Expenses	Basis of Apportionment	Total Rs.	Spinning Rs.	Weaving Rs.	Stores Rs.	Time Keeping Rs.
Depreciation	Cost of Machinery (60,000:45,000:30,000:15,000)	6,000	2,400	1,800	1,200	600
Lighting	Floor space (400:300:200:100)	1,400	560	420	280	140
Insurance	Stock of goods (40,000:25,000:15,000)	8,000	4,000	2,500	1,500	—
Employer's Contribution to Provident Fund	Direct wages (20,000:15,000:10,000:5,000)	1,700	680	510	340	170
Energy	Cost of Machinery (60,000:45,000:30,000:15,000)	10,000	4,000	3,000	2,000	1,000
Supervision	No. of workers (50:50:30:30)	20,000	6,250	6,250	3,750	3,750
Total Over-heads as per Primary Distribution		67,100	25,890	20,480	13,070	7,660

3.8.5 OTHER BASIS OF APPORTIONING OVERHEAD COSTS

We have already discussed that the primary basis for apportioning the costs of service departments or common expenses is generally the benefit received by each department. However, there are other bases of apportionment that may also be applied, as outlined below:

- Analysis or survey of existing conditions:** This involves studying the current operational conditions to determine a fair allocation of expenses.
- Ability to pay:** Costs may be apportioned based on the financial capability of the department, allocating more to those with higher profitability or resources.
- Efficiency or incentive:** Expenses can be allocated based on the department's performance or efficiency, offering incentives for better outcomes.

A business may adopt just one of these criteria as its primary approach or may utilize a combination of these (including the benefit received criterion) for various aspects of its operations.

Re-Appportionment of Service Department Costs:

Once the overheads have been distributed among production and service departments through allocation and apportionment, the next step is to re-apportion the total costs of service departments to the production departments.

Since the primary aim is to assign overhead costs to cost units, and no cost units directly pass through service departments, it becomes necessary to redistribute service department costs to production departments based on a fair and logical method. This process is referred to as **secondary distribution**.

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The typical bases for redistributing service department costs to production departments during secondary distribution are as follows:

Service Department	Basis for Re-Appportionment
Purchase Department	Number of purchase orders, purchase requisitions, or value of materials purchased.
Stores Department	Number of material requisitions or value/quantity of materials issued.
Time-Keeping/Payroll	Number of employees or total labour/machine hours.
Personnel, Canteen, Welfare, Medical, Security	Number of employees or total wages.
Repairs and Maintenance	Number of hours worked in each department.
Power House	Meter readings, H.P. hours for power, or floor space for lighting/heating.
Inspection	Inspection hours or value of inspected items.
Drawing Office	Number of drawings prepared or man-hours spent.
Accounts Department	Number of workers in each department or time spent.
Tool Room	Direct labour hours, machine hours, or wages.
Internal Transport	Weight and/or distance of goods transported.

Example: From the following information, prepare the departmental overhead distribution summary under the method of apportionment to production departments only.

Item	Production Dept.			Service Dept.	
	A	B	C	X	Y
Direct Wages (Rs.)	80,000	1,20,000	1,50,000	60,000	50,000
Direct Material (Rs.)	50,000	70,000	90,000	40,000	30,000
Staff Number	4,500	5,500	6,000	2,500	2,000
Electricity KWh	18,000	15,000	12,000	6,000	4,000
Asset Value (Rs.)	1,60,000	1,20,000	90,000	35,000	30,000
Light Points	30	25	20	10	8
Area (Sq. Yards)	400	350	250	150	100

The overheads for the period were as follows:

Overheads	Rs.
Power	3,000
Lighting	600
Stores	2,000
Welfare to Staff	7,500
Depreciation	70,000

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Overheads	Rs.
Repairs	16,000
General Overheads	28,000
Rent & Taxes	1,500

Additional Instructions:

Apportion the overheads of service department Y according to direct wages and those of service department X in the ratio of 5:3:2 among the production departments.

Solution:

Expenses	Basis of Apportionment	Total Rs.	A (Rs.)	B (Rs.)	C (Rs.)	X (Rs.)	Y (Rs.)
Power	KWh (18:15:12:6:4)	3,000	982	818	655	327	218
Lighting	Light Points (30:25:20:10:8)	600	194	161	129	65	51
Stores Overhead	Direct Material (5:7:9:4:3)	2,000	357	500	643	286	214
Welfare to Staff	Staff Number (45:55:60:25:20)	7,500	1,646	2,013	2,195	914	732
Depreciation	Asset Value (160:120:90:35:30)	70,000	25,747	19,310	14,483	5,632	4,828
Repairs	Asset Value (160:120:90:35:30))	16,000	5,885	4,414	3,310	1,288	1,103
General Overheads	Direct Wages (8:12:15:6:5)	28,000	4,870	7,304	9,130	3,652	3,044
Rent & Taxes	Area (40:35:25:15:10)	1,500	480	420	300	180	120
Direct Wages	Allocated	—	—	—	—	60,000	50,000
Direct Material	Allocated	—	—	—	—	40,000	30,000
Total (Primary Distribution)		1,28,600	40,161	34,940	30,845	1,12,344	90,310
Service Dept. Y	Direct Wages (8:12:15:6:5)	90,310	20,642	30,963	38,705		
Service Dept. X	Ratio (5:3:2)	1,12,344	56,172	33,703	22,467		
Total (Secondary Distribution)			1,16,975	99,606	92,017		

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3.8.6 DIFFERENCE BETWEEN ALLOCATION AND APPORTIONMENT

The distinction between allocation and apportionment is crucial, as both methods aim to assign costs to specific cost units or centers. However, the key differences between these methods are outlined below in table 3.7:

Table 3.7 Difference between Allocation and Apportionment

Basis	Allocation	Apportionment
Definition	Direct assignment of whole cost items to specific departments or cost centers.	Distribution of cost items across departments based on the proportion of the benefits received.
Process	A direct process of charging expenses to individual cost centers.	An indirect process that involves identifying the appropriate portion of an expense to be shared.
Example	Indirect wages are directly allocated to specific departments if they can be separately identified.	Service department costs (e.g., maintenance) are apportioned to production departments based on usage.
Nature of Cost	Costs that can be traced directly to a department or cost center.	Costs that cannot be directly traced and need to be divided among multiple departments or cost centers.
Applicability	Applied when the entire cost item can be clearly attributed to a specific cost center.	Applied when the cost item needs to be shared proportionally across multiple departments.
Scope	Allocation covers a wider scope as it deals with whole cost items.	Apportionment has a narrower scope as it deals with dividing costs among departments.

3.8.7 METHODS OF ABSORBING OVERHEADS TO VARIOUS PRODUCTS OR JOBS

The selected method for charging overheads to jobs or products must ensure:

1. The total overhead charged during a period aligns closely with actual expenses incurred.
2. Overheads allocated to specific jobs or products are fairly distributed. In the context of factory overheads, this includes:
 - ❑ Considering the time required for each job,
 - ❑ Differentiating between tasks handled by skilled and unskilled workers, and
 - ❑ Accounting for differences between manual and machine-assisted jobs.

Additionally, the method should be straightforward to implement and yield consistent results across periods. Variations should reflect genuine changes, such as shifts from manual labor to machine use.

Several methods are typically employed, either individually or in combination, to compute the appropriate overhead rate. The most common Methods of Absorbing Overheads are shown below in figure 3.4

Percentage of direct materials,

Percentage of prime cost,

Percentage of direct labor cost,

Labor hour rate,

Machine hour rate, and

Rate per unit of output.

Figure 3.4 Methods of Absorbing Overheads

1. **Percentage of Direct Material Cost:** This method uses the cost of direct materials consumed as the base for calculating the overhead absorbed. The overhead rate is computed using the following formula:

$$\text{Overhead Rate} = \frac{\text{Total Production Overheads of a Department}}{\text{Budgeted direct material of all products}} \times 100$$

Example: A company incurs production overheads of Rs. 1,20,000 and the direct material cost is Rs. 2,40,000.

1. Calculate the overhead rate as a percentage of the material cost.
2. If the direct material cost for a particular job or cost unit is Rs. 4,000, calculate the overhead to be absorbed by that job or cost unit.

Solution

1. Calculation of Overhead Rate

$$\text{Overhead Rate} = (\text{Production Overheads/Direct Material Cost}) \times 100$$

$$\text{Overhead Rate} = (1,20,000/2,40,000) \times 100$$

$$\text{Overhead Rate} = 50\%$$

Therefore, the overhead rate is 50% of the direct material cost.

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2. Overhead Absorption for a Cost Unit

If the direct material cost of a job or cost unit is Rs. 4,000, the overhead to be absorbed is calculated as:

$$\text{Overhead Absorbed} = (\text{Direct Material Cost} \times \text{Overhead Rate})/100$$

$$\text{Overhead Absorbed} = 4,000 \times 50/100$$

$$\text{Overhead Absorbed} = 2,000$$

Therefore, the overhead to be absorbed for the job or cost unit is Rs. 2,000

2. Percentage of Prime Cost Method: This method assumes that both materials and labor contribute to factory overheads. Therefore, the total prime cost is used as the base for absorbing the overhead. The formula for calculating the overhead rate is:

$$\text{Overhead Rate} = \frac{\text{Total Production Overheads of a Department}}{\text{Prime Cost}} \times 100$$

Example for Both Methods:

Suppose, for a given period, actual figures are as follows:

- ❑ Direct materials: ₹3,00,000
- ❑ Direct labor: ₹1,50,000
- ❑ Factory overheads: ₹1,20,000

The percentage of factory overheads to direct materials will be 40%, and to prime cost, it will be 25%.

Now, assume a job has a material cost of ₹12,000 and direct labor of ₹13,000. The total cost after absorbing factory overhead would be:

$$(i) \quad ₹20,000 + 40\% \text{ of } ₹12,000 = ₹20,000 + ₹4,800 = ₹24,800, \text{ and}$$

$$(ii) \quad ₹20,000 + 25\% \text{ of } ₹25,000 = ₹20,000 + ₹6,250 = ₹26,250.$$

Clearly, different methods yield different work costs. The first and second methods may be considered unsuitable for the following reasons:

- ❑ Manufacturing overhead expenses are time-based, meaning overheads are incurred based on time, and jobs that take longer will absorb more overheads. The first and second methods largely overlook this.
- ❑ Overheads do not closely correlate with prime cost or direct material cost, except to a small degree. Hence, two jobs requiring the same operational time but using different materials will have differing overhead costs, which is problematic.
- ❑ The method based on prime cost doesn't factor in the time factor and may be misleading when both labor and material costs are used for allocating overheads.
- ❑ Material prices often fluctuate, leading to wide fluctuations in overhead costs when based on material or prime costs.

- ❑ The skill level of workers and whether machinery was used are ignored in these methods.

However, the percentage of material cost can still be used to absorb material handling and storage overheads.

3. Percentage of Direct Labor Cost: The formula for this method is:

$$\text{Overhead Rate} = \frac{\text{Total Production Overheads of a Department}}{\text{Direct Labor Cost}} \times 100$$

Example: A company incurs production overheads of Rs. 2,00,000 and the direct labour cost is Rs. 2,50,000.

1. Calculate the overhead rate as a percentage of the direct labor cost.
2. If the direct labor for a specific job is Rs. 5,000, calculate the production overhead to be absorbed by that job.

Solution

1. Calculation of Overhead Rate

$$\text{Overhead Rate} = (\text{Production Overheads/Direct Labor Cost}) \times 100$$

$$\text{Overhead Rate} = (2,50,000/2,00,000) \times 100$$

$$\text{Overhead Rate} = 80\%$$

Therefore, the overhead rate is 80% of the direct labor cost.

2. Overhead Absorption for a Job

If the direct wages for a job are Rs. 5,000, the production overhead to be absorbed is calculated as:

$$\text{Overhead Absorbed} = (\text{Direct Wages} \times \text{Overhead Rate})/100$$

$$\text{Overhead Absorbed} = 5,000 \times 80/100$$

$$\text{Overhead Absorbed} = 4,000$$

Therefore, the production overhead to be absorbed for the job is Rs. 4,000.

Advantages:

1. The method is simple and economical to apply.
2. The time factor is recognized, albeit indirectly.
3. Total expenses recovered will not deviate significantly from the estimated figures, as wages tend not to fluctuate much.

Disadvantages:

1. It introduces inaccuracies due to insufficient consideration of the time factor.
2. It doesn't account for the use of machinery in the manufacturing process.
3. It does not reflect variations in worker skill levels.

N O T E S

4. Labor Hour Rate: This method improves upon the percentage of direct wages method by fully recognizing the importance of time in the incurrence and absorption of overheads. It is suitable for operations that don't involve significant use of machinery. To calculate the labor hour rate, the factory overhead is divided by the total number of direct labor hours.

The formula for this method is:

$$\text{Direct Labor Hour Rate} = \frac{\text{Total Production Overheads of a Department}}{\text{Direct Labor Hour}}$$

Example:

Given Data

- ❑ **Direct Labor Cost:** ₹2,40,000
- ❑ **Total Factory Overheads:** ₹1,20,000
- ❑ **Total Labor Hours:** 48,000 hours

1. Percentage of Direct Labor Cost Method

Using the formula:

$$\text{Overhead Rate} = \text{Total Production Overheads} / \text{Direct Labor Cost} \times 100$$

$$\text{Overhead Rate} = ₹1,20,000 / ₹2,40,000 \times 100 = 50\%$$

For a specific job with Direct Labor Cost = ₹15,000, the factory overhead absorbed would be:

$$\text{Factory Overhead} = ₹15,000 \times 50\% = ₹7,500$$

2. Labor Hour Rate Method

Using the formula:

$$\text{Direct Labor Hour Rate} = \text{Total Factory Overheads} / \text{Total Labor Hours}$$

$$\text{Labor Hour Rate} = ₹1,20,000 / 48,000 \text{ hours} = ₹2.50 \text{ per hour}$$

For a specific job requiring 600 labor hours, the factory overhead absorbed would be:

$$\text{Factory Overhead} = 600 \times ₹2.50 = ₹1,500$$

5. Machine Hour Rate: In this method, overheads are charged to production based on the hours machines are used. It is calculated as follows:

$$\text{Machine Hour Rate} = \frac{\text{Total Production Overheads of a Department}}{\text{No. of Machine Hours}}$$

There is a basic similarity between the machine hour and labor hour rate methods, both being time-based. The choice of method depends on the specific circumstances. For departments where machines play a dominant role and operators are less involved, the machine hour rate is more appropriate.

In such cases, the machine hour rate is the most accurate method for absorbing manufacturing overheads. Typically, the rate is predetermined based on estimated or normal expenses for the upcoming period. It is advisable to calculate a rate for each machine; however, for groups of similar machines, a single rate may suffice.

Methods of Computing Machine Hour Rate:

- (i) **Direct Machine Hour Rate:** This method only considers expenses directly related to machine operation, such as power, depreciation, repairs, and maintenance. The rate is computed by dividing the total estimated expenses by the estimated operational hours.
- (ii) **Comprehensive Machine Hour Rate:** This method also includes additional expenses like supervision charges, shop cleaning, lighting, rent, and rates. If such expenses are excluded from machine rates, they should be absorbed using a different method like the Productive Labor Hour Rate.

Some prefer to include the wages of the machine operator to calculate a comprehensive rate for machine operation.

Advantages:

1. Best suited for production where machines dominate.
2. Idle machine hours can be identified by under-absorbed overheads.
3. Particularly useful when multiple operators attend to machines.

Disadvantages:

1. Additional data on machine operation must be maintained.
2. Computing separate rates for each machine or group adds complexity.

Example:

A company uses machines in its production process and has the following expenses for the machine-related operations:

Expense Category	Amount (₹)
Power	₹25,000
Depreciation	₹15,000
Repairs and Maintenance	₹10,000
Supervision Charges	₹10,000
Shop Cleaning	₹5,000
Lighting	₹8,000
Rent	₹12,000

Estimated Machine Hours: 5,000 hours

The company wants to calculate the overhead absorbed for a specific job requiring 200 machine hours using two different methods:

1. Direct Machine Hour Rate
2. Comprehensive Machine Hour Rate

N O T E S

Solution:**1. Direct Machine Hour Rate**

First, calculate the Direct Machine Hour Rate using the formula:

Direct Machine Hour Rate = Direct Machine Expenses/Estimated Machine Hours

$$= (25,000 + 15,000 + 10,000)/5,000$$

$$= ₹10 \text{ per hour}$$

Now, calculate the overhead absorbed for the job that requires 200 machine hours:

$$\text{Overhead} = 200 \times ₹10 = ₹2,000$$

2. Comprehensive Machine Hour Rate

Next, calculate the Comprehensive Machine Hour Rate using the formula:

Comprehensive Machine Hour Rate = (Direct Machine Expenses + Additional Expenses)/Estimated Machine Hours

$$= (50,000 + 35,000)/5,000 = ₹17 \text{ per hour}$$

Now, calculate the overhead absorbed for the job that requires 200 machine hours:

$$\text{Overhead} = 200 \times ₹17 = ₹3,400$$

6. Rate per Unit of Output Method: This is the simplest method, where the overhead rate is determined using the formula:

$$\text{Overheads Rate} = \frac{\text{Amount of Overheads}}{\text{No. of Units}}$$

3.8.8 TYPES OF OVERHEAD RATES

The overhead rates may be of the following types:

1. Normal Rate:

This rate is derived by dividing the actual overheads by the actual base, commonly referred to as the actual rate. The formula used for this calculation is:

$$\text{Normal Rate} = \frac{\text{Actual Overheads}}{\text{Actual Base}}$$

2. Pre-determined Overhead Rate:

This rate is calculated ahead of time by estimating the overhead for the given period. The formula for pre-determined overhead rate is:

$$\text{Pre-determined Overhead Rate} = \frac{\text{Estimated Overheads}}{\text{Estimated Base}}$$

The overhead rate can be determined based on three approaches:

- ❑ The previous year's overhead rate is used, assuming the production level and overheads will remain constant or change proportionally.
- ❑ The overhead rate is calculated using estimated expenses and the anticipated level of production. For instance, if estimated expenses are ₹24,000 and the production output is 8,000 units, the rate would be ₹3 per unit.
- ❑ The overhead rate is set based on the normal business volume.

3. Blanket Overhead Rate:

This refers to a single overhead rate calculated for the entire factory. It differs from the departmental overhead rate, where each department has a separate rate. Blanket rates are most suitable for factories producing a single product in a continuous process or where the work across departments is fairly uniform or standardized.

The formula used to compute the blanket rate is:

$$\text{Blanket Rate} = \frac{\text{Total Factory Overheads}}{\text{Total Base for the Entire Factory}}$$

A blanket rate is appropriate under the following conditions:

- ❑ The factory produces only one primary product.
- ❑ If multiple products are produced: a. All products pass through all departments. b. All products are processed for the same duration in each department.

4. Departmental Overhead Rate:

This rate is specific to a particular department or production unit. It is preferred when product lines differ or machinery usage varies across departments, indicating non-uniform conditions within the factory.

The formula used for departmental overhead rate is:

$$\text{Departmental Overhead Rate} = \frac{\text{Departmental Overheads}}{\text{Total no. of units of base for that department}}$$



SELF-ASSESSMENT QUESTIONS

7. Which method is most suitable for controlling administrative overheads?
- | | |
|---------------------------|---------------------------|
| a. Use of labor hour rate | b. Geographical analysis |
| c. Use of budgets | d. Customer type analysis |

- | Expense Category | Amount (₹) |
|-------------------------|------------|
| Power | 30,000 |
| Depreciation | 20,000 |
| Repairs and Maintenance | 15,000 |
| Supervision Charges | 12,000 |
| Shop Cleaning | 8,000 |
| Lighting | 10,000 |
| Rent | 18,000 |

a. A4,707.50 **b.** A2,708.33 **c.** A5,416.66 **d.** A1,354.17

Calculate the Labor Turnover Rate for the factory by different methods

Sol.

Average Number of Workers =

$$\frac{\text{Number of workers at the beginning} + \text{Number of workers at the end}}{2}$$

$$= \frac{400 + 500}{2} = 450$$

1. Separation Method:

$$\text{Labor Turnover Rate} = \frac{\text{No. of workers left during a period}}{\text{Average number of workers in the period}} \times 100$$

Total workers who left = Voluntary resignations + Discharged workers
 = 30 + 40 = 70

$$\text{Labor Turnover Rate} = \frac{70}{450} \times 100 = 15.55\%$$

2. Replacement Method:

$$\text{Labor Turnover Rate} = \frac{\text{No. of workers replaced during a period}}{\text{Average number of workers in the period}} \times 100$$

Total workers replaced = Workers recruited to replace resignations = 35
 (since only replacements are considered)

$$\text{Labor Turnover Rate} = \frac{35}{450} \times 100 = 7.78\%$$

3. Flux Method:

$$\text{Labor Turnover Rate} = \frac{(\text{No. of separations} + \text{No. of additions}) / 2}{\text{Average number of workers in the period}} \times 100$$

□ Total separations = 40 (as calculated before)

□ Total additions = 35 (total recruits)

$$\text{Labor Turnover Rate} = \frac{37.5}{450} \times 100 = 8.33\%$$

Ques. 2. Calculate the total earnings and the effective rate of earnings per hour of three workers under the Rowan and Halsey systems, given the following details:

- The standard time allowed for producing 1 batch of items is 60 hours.
- The rate of wages is ₹2/- per hour.
- The actual time taken by three workers are:
 - ◆ Worker X: 55 hours ◆ Worker Y: 50 hours
 - ◆ Worker Z: 40 hours

Using both the Halsey (50%) and Rowan systems, calculate:

1. The total earnings for each worker.
2. The effective rate of earnings per hour for each worker.

N O T E S

Sol.**Halsey System**

Worker	Standard Time (hrs)	Actual Time (hrs)	Time Saved (hrs)	Wages for Actual Time (₹)	Bonus (50% of Time Saved) (₹)	Total Earnings (₹)	Effective Rate (₹/hr)
Worker A	50	45	5	$45 \times 1 = 45$	$50\% \times (5 \times 1) = 2.5$	$45 + 2.5 = 47.5$	$47.5 \div 45 = 1.06$
Worker B	50	40	10	$40 \times 1 = 40$	$50\% \times (10 \times 1) = 5$	$40 + 5 = 45$	$45 \div 40 = 1.125$
Worker C	50	30	20	$30 \times 1 = 30$	$50\% \times (20 \times 1) = 10$	$30 + 10 = 40$	$40 \div 30 = 1.33$

Rowan System

Worker	Standard Time (hrs)	Actual Time (hrs)	Time Saved (hrs)	Wages for Actual Time (₹)	Bonus ((Time Saved / Standard Time) Actual Wages) (₹)	Total Earnings (₹)	Effective Rate (₹/hr)
Worker A	50	45	5	$45 \times 1 = 45$	$(5/50) \times 45 = 4.5$	$45 + 4.5 = 49.5$	$49.5 \div 45 = 1.10$
Worker B	50	40	10	$40 \times 1 = 40$	$(10/50) \times 40 = 8$	$40 + 8 = 48$	$48 \div 40 = 1.20$
Worker C	50	30	20	$30 \times 1 = 30$	$(20/50) \times 30 = 12$	$30 + 12 = 42$	$42 \div 30 = 1.40$

Interpretation:**□ Halsey System:**

- ◆ In the Halsey system, workers receive a fixed wage for the actual time worked and a bonus of 50% of the time saved.
- ◆ **Worker C** has the highest effective hourly rate of ₹1.33 because they saved the most time (20 hours), which resulted in a higher bonus.
- ◆ **Worker A** has the lowest effective rate of ₹1.06 due to saving the least time (5 hours).

□ Rowan System:

- ◆ In the Rowan system, the bonus is calculated based on the proportion of time saved relative to the standard time.
- ◆ Here, **Worker C** again has the highest effective rate (₹1.40) because they saved the most time, but the bonus calculation is more conservative compared to Halsey.
- ◆ **Worker A** has a slightly higher rate (₹1.10) than in the Halsey system due to the proportional bonus approach.

Ques. 3. A, B, and C are workers at a factory, and their production during a specific 50-hour workweek was 150, 160, and 180 units, respectively. The guaranteed rate per hour is ₹12, the low piece rate is ₹5 per unit, and the high piece rate is ₹8 per unit. The high task is set at 170 units per week. Calculate the total earnings and labor cost per unit for A, B, and C under both the Taylor and Gantt Task Bonus Plans.

Sol.

Taylor Plan:

1. Worker A (Actual output: 150 units)

Since 150 units is less than the high task (170 units), Worker A is deemed inefficient and will get 80% of the low piece rate.

Piece Rate: 80% of ₹5 = ₹4.00 per unit.

Wages = ₹4.00 × 150 units = ₹600

2. Worker B (Actual output: 160 units)

Since 160 units is also less than the high task, Worker B is deemed inefficient and will get 80% of the low piece rate.

Piece Rate: ₹4.00 per unit.

Wages = ₹4.00 × 160 units = ₹640

3. Worker C (Actual output: 180 units)

Since 180 units are more than the high task, Worker C is considered efficient and will get 120% of the high piece rate.

Piece Rate: 120% of ₹8 = ₹9.60 per unit.

Wages = ₹9.60 × 180 units = ₹1,728

Gantt Task and Bonus Plan:

Worker A (Output: 150 units)

Since the output is below the high task, Worker A earns the guaranteed time rate.

Guaranteed Earnings = ₹12 × 50 hours = ₹600

Worker B (Output: 160 units)

Since the output is below the high task, Worker B earns the guaranteed time rate.

Guaranteed Earnings = ₹12 × 50 hours = ₹600

Worker C (Output: 180 units)

Since the output is above the high task, Worker C earns based on the high piece rate.

Earnings = ₹8 × 180 units = ₹1,440

Ques. 4. In a factory, wages are guaranteed at a rate of ₹2.00 per hour for a 50-hour work week. Based on a time and motion study, it is estimated that producing one unit of a particular product takes 25 minutes. The allowed time is increased by 20%. During the week, worker B produced 200 units of the product.

Calculate B's wages using the following methods:

1. Time Rate
2. Piece Rate.

N O T E S

Sol.**(a) Calculation of wages under Time Rate System**

Earnings under time wages = TR

$$= 50 \times 2.00 = ₹100$$

(b) Calculation of wages under Piece Rate with a Guaranteed Wage Rate

Normal time for one unit = 25 minutes

(+) Relaxation allowance @ 20% = 5 minutes

$$\text{Standard Time} = 25 + 5 = 30$$

No. of pieces per hour = $60/30 = 2$ pieces per hourPiece Rate = Hourly Rate $\div$ No. of pieces per hour

$$= 2.00 \div 2 = ₹1.00 \text{ per piece}$$

$$\text{Earnings under Piece Rate} = 200 \times 1.00 = ₹200$$

Ques. 5. From the following details, calculate the earnings for the week of a worker under:**(a)** Piece Rate**(b)** Taylor's Differential Piece Rate**(c)** Halsey Premium System**(d)** Rowan System☐ Number of working hours per week: 45 hours☐ Wages per hour: ₹4.00☐ Normal time per piece: 25 minutes☐ Normal output per week: 100 pieces☐ Actual output for the week: 140 pieces☐ Differential piece rate: 75% of the piece rate when output is below standard and 125% above standard.**Sol.****Computation of Earnings for the Week of a Worker****(a) Straight Piece Rate**

$$\text{Normal Piece Rate} = \frac{(45 \times 4.00)}{100} = ₹1.80$$

$$\text{Earnings under Straight Piece Rate} = 140 \times 1.80 = ₹252$$

(b) Differential Piece Rate

$$\text{Efficiency} = \frac{140}{100} \times 100 = 140\% (>100\%)$$

$$\text{Earnings under Differential Piece Rate} = 140 \times 1.80 \times \frac{125}{100} = 315$$

(c) Halsey Premium System

$$\text{Standard time for actual production} = 45 \times \frac{140}{100} = 63 \text{ hours}$$

$$\begin{aligned} \text{Earnings under Halsey Plan} &= (45 \times 4.00) + \left(\frac{50}{100} \times (63 - 45) \times 4 \right) \\ &= 180 + 36 = ₹216 \end{aligned}$$

(d) Rowan System

$$\begin{aligned} \text{Earnings under Rowan Plan} &= (45 \times 4.00) + \left[\left(\frac{63 - 45}{63} \right) \times 4.00 \times 45 \right] \\ &= 180 + 30 = ₹210 \end{aligned}$$

Ques. 6. A machine purchased for ₹ 80,00,000 is projected to operate for 12 years. At the end of its service life, the estimated scrap value will be ₹ 8,00,000. Total repair costs for the machine over its lifespan are anticipated to be ₹ 15,00,000, with an expected operational time of 4,800 hours per year. The machine consumes 20 units of electricity per hour, with each unit costing ₹ 6. It occupies one-fifth of the department's total area and has three lighting points out of fifteen available. The foreman allocates approximately one-seventh of his time to oversee the machine. The monthly rental cost for the department is ₹ 25,000, while monthly lighting expenses amount to ₹ 10,000. The foreman receives a monthly salary of ₹ 22,000.

CALCULATE the machine hour rate, assuming insurance costs are 1% per annum and oil expenses amount to ₹ 1,200 per month.

Sol.

Total number of hours per annum: 4,800

Total number of hours per month: 400

Computation of Machine Hour Rate

	Per month (₹)	Per hour (₹)
Fixed costs (Standing Charges)		
Depreciation (Refer working note-1)	72,000	180.00
Rent (₹25,000 × 1/5)	5,000	12.50
Lighting charges { (₹10,000 × 3 points) ÷ 15 points }	2,000	5.00
Foreman's salary (₹22,000 × 1/7)	3,143	7.86
Sundry expenses (oil, etc.)	1,200	3.00
Insurance { (1% of ₹80,00,000) ÷ 12 months }	6,667	16.67
Total Fixed Costs	89,010	225.53
Variable Costs:		
Repairs (Refer working note-2)		31.25
Electricity (20 units × ₹6)	120.00	30.00
Machine Hour Rate		286.78

N O T E S

Working Notes:**1. Depreciation per month:**

$$\text{Depreciation} = \frac{\text{Cost of Machine} - \text{Scrap Value}}{\text{Life of the Machine}} = \frac{₹80,00,000 - ₹8,00,000}{(12 \text{ years} \times 12 \text{ months})} = ₹72,000$$

The life of the machine is given as 12 years, and the machine is expected to run for 4,800 hours per annum. Depreciation is calculated based on time.

2. Repairs for the whole life of ₹15,00,000, linked to the activity level of 57,600 hours:

$$\text{Repairs cost per hour} = \frac{₹15,00,000}{(12 \text{ years} \times 4,800 \text{ hours})} = ₹31.25$$

Ques. 7. A machine shop has 10 identical Milling machines, each manned by 8 operators. The machines require an operator to work. The total cost of the machines is ₹10 lakhs. The following details are provided for a 12-month period:

Particulars	Figures (₹)
Normal available hours per machine/month	220 hours
Absenteeism (without pay) hours	20 hours
Leave (with pay) hours	25 hours
Normal idle time (unavoidable) hours	12 hours
Average rate of wages per worker (8 hrs/day)	₹25/hour
Production bonus estimated	12% on wages
Value of power consumed	₹10,500
Supervision and indirect labor	₹4,000
Lighting and electricity	₹1,500
Repairs and maintenance (including consumables)	4% of value of machines
Insurance	₹50,000
Depreciation (12% of original cost)	₹1,20,000
Sundry works expenses	₹15,000
General management expenses allocated	₹60,000

Solution:**Computation of Operator's Wages:**

Particulars	Figures
Average rate of wages per hour	₹25
Hours worked per month per operator	220 - (20 + 25 + 12) = 163 hours
Total wages for 8 operators for 12 months	163 hours * 8 operators * ₹25/hour * 12 months = ₹24,000

Computation of Overheads:

Particulars	Figures (₹)
Operator's wages	₹24,000
Production bonus (12% of wages)	₹2,880
Power consumed	₹10,500
Supervision and indirect labour	₹4,000
Lighting and electricity	₹1,500
Repairs and maintenance (4% of ₹10,00,000)	₹40,000
Insurance	₹50,000
Depreciation (12% of ₹10,00,000)	₹1,20,000
Sundry works expenses (50% of ₹15,000)	₹7,500
General management expenses (50% of ₹60,000)	₹30,000
Total Overheads	₹1,81,880

Computation of Total Machine Hours:

Particulars	Figures
Normal available hours per month/operator	220 hours
Less: Absenteeism hours	20 hours
Less: Leave hours	25 hours
Less: Idle time hours	12 hours
Utilizable hours per month per operator	$220 - (20 + 25 + 12) = 163$ hours
Total utilizable hours for 8 operators over 12 months	$163 \text{ hours} \times 8 \text{ operators} \times 12 \text{ months} = 15,624$ hours

Comprehensive Machine Hour Rate Calculation:

Particulars	Figures (₹)
Total Overheads	₹1,81,880
Total Machine Hours	15,624 hours
Machine Hour Rate	$\text{₹}1,81,880 / 15,624 = \text{₹}11.64$ per machine hour

Ques. 8. The production department of a factory provides the following details for the month of March, 2024:

Particulars	Amount (Rs.)	Hours
Materials used	72,000	—
Direct wages	50,000	—
Overheads	40,000	—
Labour hours worked	—	40,000
Machine hours worked	—	25,000

N O T E S

For a specific order completed by the department during March, the following information is available:

Particulars	Amount (Rs.)	Hours
Materials used	8,000	—
Direct wages	4,000	—
Labour hours worked	—	3,600
Machine hours worked	—	2,000

Required:

Calculate the overheads chargeable to the job using the following methods:

1. Direct Materials Cost Method
2. Direct Labour Cost Method
3. Labour Hour Rate Method
4. Machine Hour Rate Method

Solution**Step 1: Calculation of Overhead Absorption Rates**

1. Direct Materials Cost Method

$$\text{Overhead Rate} = \text{Overheads} / \text{Direct Material Cost} \times 100$$

$$\text{Overhead Rate} = 40,000 / 72,000 \times 100 = 55.56\%$$

2. Direct Labour Cost Method

$$\text{Overhead Rate} = \text{Overheads} / \text{Direct Labour Cost} \times 100$$

$$\text{Overhead Rate} = 40,000 / 50,000 \times 100 = 80\%$$

3. Labour Hour Rate

$$\text{Labour Hour Rate} = \text{Overheads} / \text{Labour Hours Worked}$$

$$\text{Labour Hour Rate} = 40,000 / 40,000 = \text{Rs. 1.00 per labour hour}$$

4. Machine Hour Rate

$$\text{Machine Hour Rate} = \text{Overheads} / \text{Machine Hours Worked}$$

$$\text{Machine Hour Rate} = 40,000 / 25,000 = \text{Rs. 1.60 per machine hour}$$

Step 2: Overheads Chargeable to the Job Using Different Methods

1. Direct Materials Cost Method

$$\text{Overhead for the Job} = \text{Direct Materials Cost} \times \text{Overhead Rate} / 100$$

$$\text{Overhead for the Job} = 8,000 \times 55.56 / 100 = 4,444.80 \approx \text{Rs. 4,445}$$

2. Direct Labour Cost Method

$$\text{Overhead for the Job} = \text{Direct Labour Cost} \times \text{Overhead Rate} / 100$$

$$\text{Overhead for the Job} = 4,000 \times 80 / 100 = 3,200$$

3. Labour Hour Rate Method

$$\text{Overhead for the Job} = \text{Labour Hours Worked} \times \text{Labour Hour Rate}$$

$$\text{Overhead for the Job} = 3,600 \times 1.00 = 3,600$$

4. Machine Hour Rate Method

Overhead for the Job = Machine Hours Worked $\times$ Machine Hour Rate

Overhead for the Job = $2,000 \times 1.60 = 3,200$

Step 3: Statement Showing Cost of the Job Under Different Methods

Particulars	Direct Materials Cost Method (Rs.)	Direct Labour Cost Method (Rs.)	Labour Hour Rate Method (Rs.)	Machine Hour Rate Method (Rs.)
Direct Materials	8,000	8,000	8,000	8,000
Direct Wages	4,000	4,000	4,000	4,000
Overheads	4,445	3,200	3,600	3,200
Total Cost of Job	16,445	15,200	15,600	15,200

Ques. 9. The Cost Accountant of Radha Ltd. has computed the labour turnover rates for the quarter ended 30th June, 2024, as 12%, 6%, and 4% under the Flux method, Replacement method, and Separation method, respectively. The number of workers replaced during that quarter is 36.

Find out:

1. The number of workers recruited and joined.
2. The number of workers who left and were discharged.

Solution:**1. Calculate the Average Number of Workers on Roll**

Use the formula for the replacement method:

$$\text{Labor Turnover Rate} = \frac{\text{No. of workers replaced during a period}}{\text{Average number of workers in the period}} \times 100$$

$$\text{Labor Turnover Rate} = \frac{\text{No. of workers replaced during a period}}{\text{Average number of workers in the period}} \times 100$$

Average number of workers in the period =

$$\frac{\text{No. of workers replaced during a period}}{\text{Labor Turnover Rate}} \times 100$$

$$\text{Average number of workers in the period} = \frac{36}{6} \times 100$$

$$\text{Average number of workers in the period} = 600$$

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2. Calculate the Number of Workers Who Left and Were Discharged

Use the formula for the separation method:

$$\text{Labor Turnover Rate} = \frac{\text{No. of workers left during a period}}{\text{Average number of workers in the period}} \times 100$$

$$4 = \frac{\text{No. of workers left during a period}}{600} \times 100$$

$$\text{No. of workers left during a period} = 24$$

3. Calculate the Number of Workers Recruited and Joined

Use the formula for the flux method:

$$\text{Labor Turnover Rate} = \frac{1/2(\text{No. of separations} + \text{No. of additions})}{\text{Average number of workers in the period}} \times 100$$

$$12 = \frac{1/2(\text{No. of separations} + \text{No. of additions})}{600} \times 100$$

$$\text{No. of separations} + \text{No. of additions} = \frac{12 \times 600 \times 2}{100}$$

$$\text{No. of separations} + \text{No. of additions} = 144$$

$$\text{No. of additions} = 144 - 24 = 120$$

3.10 SUMMARY

- ❑ Labor attrition, or turnover, measures how frequently employees leave and are replaced, with high turnover indicating instability and leading to decreased morale and increased costs for organizations. Reducing turnover is essential for a stable workforce.
- ❑ Timekeeping is crucial for labor cost control, as it accurately tracks employee attendance, ensuring discipline and punctuality while meeting statutory requirements. It provides essential data for wage calculations, benefits eligibility, and labor efficiency analysis.
- ❑ Time booking methods enhance timekeeping by tracking the productive time workers spend on specific tasks. This approach enables better management of idle time, earnings calculations, and overall labor efficiency assessments.
- ❑ Various time recording methods exist, including manual (e.g., attendance registers) and mechanical (e.g., biometric systems) approaches, each with pros and cons. Choosing the right method depends on organizational needs and the desired level of accuracy and efficiency.
- ❑ Job evaluation methods, such as point ranking, ranking, and grading, assess job roles for wage structuring. Effective wage payment systems, including flat time rates and piece rates, play a vital role in labor cost control by balancing worker motivation and productivity.
- ❑ Administrative overheads require control by top management due to their complex nature, involving tasks like policy formulation and

organizational direction; the control process begins with classifying these expenses and comparing them to benchmarks for analysis.

- ❑ Effective control methods include comparing current administrative expenses with previous years, using budgets to account for changing conditions, and applying standards to certain administrative activities, though standardization can be challenging.
- ❑ Controlling Sales & Distribution (S&D) expenses aims to maximize sales effectiveness rather than merely reducing costs; accurate classification and collection of S&D overheads are essential for this purpose.
- ❑ Analyzing S&D expenses can be done through geographical, customer type, product/service, sales personnel, and distribution channel analyses, providing insights into performance and profitability.
- ❑ Classifying overheads into fixed and variable categories aids in decision-making, effective cost control, preparing budgets, and understanding break-even points, ultimately leading to better absorption of overheads across various departments.

3.11 KEY WORDS

1. **Labor Turnover:** The rate at which employees leave and are replaced within an organization, indicating workforce stability.
2. **Timekeeping:** The process of recording employee attendance and hours worked, essential for wage calculation and discipline.
3. **Time Booking:** A method that tracks the amount of time workers spend on specific tasks, providing insights into productivity and efficiency.
4. **Piece Rate Method:** A wage system where workers are paid based on the quantity of work completed, incentivizing higher productivity.
5. **Gantt Task Bonus Plan:** A compensation approach that combines time and piecework, offering bonuses for meeting or exceeding production standards.
6. **Break-even Point:** The level of sales at which total revenues equal total costs, resulting in neither profit nor loss.
7. **Operating Expenses:** Regular expenses required for the day-to-day functioning of a business, excluding direct production costs.
8. **Variance Analysis:** The process of comparing actual financial performance to budgeted or expected figures to identify discrepancies.
9. **Direct Costs:** Expenses that can be directly attributed to a specific product, service, or project, such as materials and labor.
10. **Budgetary Control:** A financial management process that uses budgets to monitor and control an organization's operations and expenditures.

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3.12 CLOSING CASELET**OVERHEAD COST CONTROL AT ABC MANUFACTURING COMPANY**

ABC Manufacturing Company faced rising costs and increasing competition, prompting the need for better management of its administrative overheads. To tackle this, the company categorized its overheads into specific segments, such as sales and distribution expenses, manufacturing overheads, and general administrative costs. Implementing a robust budgeting system allowed departments to forecast expenses, fostering accountability and encouraging cost-saving initiatives. Regular budget reviews against actual expenses helped identify areas of overspending, leading to corrective actions and improved financial performance.

In managing sales and distribution (S&D) expenses, ABC conducted a comprehensive analysis by evaluating costs across regions, customer types, product lines, and distribution channels. This analysis revealed inefficiencies, particularly in certain distribution channels that incurred higher costs relative to sales generated. By optimizing its distribution strategy and reallocating resources towards more profitable channels, ABC effectively reduced S&D expenses while enhancing overall profitability.

For manufacturing overheads, ABC adopted a predetermined overhead rate based on direct labor hours, ensuring accurate allocation of indirect costs to products. This method facilitated informed pricing decisions and profit analysis. The establishment of cost centers for specific departments further improved the precision of overhead allocation, enabling ABC to identify resource-intensive products. Ultimately, these strategies allowed the company to control overhead costs effectively, positioning it for better financial health and competitiveness in the manufacturing sector.

**QUESTIONS**

1. How might ABC Manufacturing Company's approach to categorizing and analyzing its overhead costs influence its long-term strategic planning?

(Hint: Consider the impact of cost control on pricing strategies, resource allocation, and competitive advantage.)

2. What potential challenges could arise from implementing a predetermined overhead rate for manufacturing costs, and how might ABC address these challenges?

(Hint: Think about factors such as fluctuations in production volume, changes in labor costs, and the accuracy of overhead allocation methods.)

3.13 HIGHER ORDER THINKING SKILLS QUESTIONS (HOTS)

1. How does the implementation of effective labor cost control mechanisms, such as timekeeping and piece rate methods, impact overall organizational productivity and employee morale? Discuss the potential trade-offs between cost control and workforce satisfaction.
2. In what ways can an organization assess the effectiveness of its labor cost control measures? Consider the role of labor turnover rates and employee feedback in evaluating these measures' success.
3. Design a comprehensive labor cost control strategy that incorporates elements from timekeeping, labor turnover management, and incentive structures. How would you balance cost efficiency with maintaining employee motivation and engagement?
4. How can variance analysis influence the strategic decisions made by management regarding budget allocations, operational adjustments, and resource management within an organization? Provide examples of how significant variances might prompt changes in business strategy.
5. In what ways do operating expenses affect a company's overall profitability, and how might a business balance reducing these expenses without compromising quality or customer satisfaction? Discuss potential long-term consequences of aggressive cost-cutting measures.
6. The following figures are extracted from the records of a company with three production departments (A, B, C) and two service departments (D, E).

Overheads

Overhead	Amount (Rs.)
Rent and Rates	12,000
Indirect Wages	4,500
Depreciation	25,000
Lighting	1,500
Power	4,000
Sundries	22,000

Additional Details

Particulars	Total	A	B	C	D	E
Floor space (sq. ft.)	25,000	5,000	6,000	7,000	5,000	2,000
Light points	150	25	35	50	30	10
Direct wages (Rs.)	25,000	7,000	5,000	8,000	3,500	1,500
H.P. of machines	350	140	80	100	30	—
Value of machines (Rs.)	6,00,000	1,50,000	2,00,000	2,30,000	10,000	10,000

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Prepare the Departmental Overhead Distribution Summary to allocate and apportion the overheads among the production departments (A, B, C) and the service departments (D, E).

7. The Cost Accountant of Shyam Ltd. has computed the labour turn-over rates for the quarter ended 31st December, 2024, as 15%, 7%, and 5% under the Flux method, Replacement method, and Separation method, respectively. The number of workers replaced during that quarter is 42.

Find out:

- ☐ The number of workers recruited and joined.
- ☐ The number of workers who left and were discharged.

8. Gamma Industries operates three production departments (A1, A2, A3) and two service departments (Maintenance S1 and Cafeteria S2). The overhead expenses and additional details for allocation are as follows:

Overhead Expenses:

- ☐ Production Department A1: ₹2,50,000
- ☐ Production Department A2: ₹2,00,000
- ☐ Production Department A3: ₹1,50,000
- ☐ Service Department Maintenance S1: ₹80,000
- ☐ Service Department Cafeteria S2: ₹60,000

Allocation Basis for Distribution:

Basis	A1	A2	A3	S1	S2
Machine Hours	60	25	15	—	—
Number of Employees	50	30	20	—	—

Additional Information for Inter-Department Distribution:

- ☐ **Maintenance S1** provides services to **S2** (20%) and the production departments (A1: 40%, A2: 25%, A3: 15%).
- ☐ **Cafeteria S2** serves only the production departments (A1: 50%, A2: 30%, A3: 20%).

Distribute the overhead costs of service departments S1 and S2 to the production departments A1, A2, and A3 using the given allocation basis.

9. A factory has 4 production departments (A, B, C, D) and 3 service departments (X, Y, Z). The following overheads and allocation bases are provided for the month of July 2024:

Overheads:

Expense	Amount (₹)
Rent	8,000
Repairs	5,400

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Expense	Amount (₹)
Depreciation	3,600
Lighting	800
Supervision	12,000
Insurance on Stock	4,500
ESI Contribution	1,200
Power	7,200

Departmental Information:

Particulars	A	B	C	D	X	Y	Z
Area (sq. ft.)	500	400	350	300	200	150	100
Number of Workers	60	50	45	40	25	20	15
Wages (₹)	20,000	18,000	16,000	14,000	10,000	8,000	6,000
Value of Plant (₹)	80,000	60,000	50,000	40,000	20,000	15,000	10,000
Stock Value (₹)	50,000	35,000	25,000	20,000	—	—	—
Horsepower of Plant	700	500	400	300	150	100	50

Allocate or apportion the overheads among the various departments and redistribute service department overheads as follows:

- ❑ X: 40% to A, 30% to B, 20% to C, and 10% to D.
- ❑ Y: Equally among A, B, C, and D.
- ❑ Z: 50% to A, 30% to C, and 20% to D.

Prepare a detailed Overhead Distribution Summary.

3.14 ANSWER KEYS

Topic	Q. No.	Answer
Recording of Timings	1	c. To track attendance and ensure punctuality and discipline
	2	b. 8.49%
Methods of Work Study	3	c. To improve production techniques and maximize efficiency
	4	c. Point Ranking Method
Methods of Wage Payment	5	c. 7,500
	6	d. Rohan, ₹1,980
Overhead Cost Control	7	c. Use of budgets
	8	b. Number of workers
	9	c. ₹4,800
	10	d. ₹2,708.33

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3.15 SUGGESTED READINGS

- ❑ **Horngren, C. T., Datar, S. M., & Rajan, M. V. (2015).** *Cost Accounting: A Managerial Emphasis*. Pearson.
- ❑ **Drury, C. (2013).** *Management and Cost Accounting*. Cengage Learning.
- ❑ **O'Brien, J. T. (2016).** *Management Accounting: Tools for Creating Value*. Wiley.
- ❑ **Atkinson, A. A., Kaplan, R. S., Matsumura, E. M., & Young, S. M. (2012).** *Management Accounting*. Pearson.

3.16 REFERENCES

- ❑ **Bhimani, A. (2009).** *Management Accounting: In a Global Context*. Oxford University Press.
- ❑ **American Institute of CPAs (AICPA)**-www.aicpa.org
- ❑ **Institute of Management Accountants (IMA)**-www.imanet.org

4

U N I T

COST CONCEPTS, COST CLASSIFICATION AND UNIT COST ANALYSIS

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INTRODUCTORY CASELET

Alice's Bakery Expansion

Alice, a successful small business owner, is contemplating expanding her bakery by opening a second location in a nearby town. To finance this expansion, she plans to take out a loan. However, this decision involves several financial considerations. First, there are opportunity costs. The interest paid on the loan is a cost Alice incurs by choosing expansion over investing in other ventures that could generate returns, like stocks or bonds. Moreover, running a second location will require more time and effort, reducing her ability to focus on other aspects of her business or personal life. Additionally, there is the inherent risk of failure, which could result in financial loss if the new location does not succeed.

Alice also faces sunk costs related to the expansion. If she decides to renovate the new location before opening, the cost of renovations will be non-recoverable, regardless of the outcome of the new venture. This makes renovation expenses a sunk cost, as the money invested cannot be recouped if the second location underperforms or fails. Alice must carefully weigh this risk before committing to renovations.

Another factor is the differential costs that Alice will incur due to the increased production needs of the second location. These costs include hiring additional staff, purchasing new equipment, and increasing raw material orders. The differential cost, in this case, is the difference between the total costs of operating both locations and the cost of operating her current bakery alone. Additionally, the new location represents a new sales channel, and Alice will need to consider how much more it will cost to run both stores as opposed to just one.

Finally, there are joint and common costs to consider. Shared resources, such as a central bakery or delivery fleet, create joint costs that will need to be allocated between both locations fairly. Similarly, utilities like electricity and water are common costs shared by both stores, requiring a fair allocation method. By carefully analyzing these cost components, Alice can evaluate the financial feasibility of expanding her business and make an informed decision on whether to proceed with the second bakery location.

**QUESTIONS**

1. How can Alice classify the different costs involved in her bakery expansion to assist in effective management decision-making?

(Hint: Consider the classification of costs)

2. How should Alice allocate joint and common costs between her two bakery locations?

(Hint: Consider the challenges of fairly distributing these costs between the two locations and what methods Alice could use to ensure an equitable allocation.)

**LEARNING OBJECTIVES**

At the end of the unit, you will be able to

- Understand the various types of costs and their significance in management decision-making.
- Develop the ability to prepare a comprehensive cost sheet that outlines all cost components associated with production.
- Learn how to identify and treat specific cost components, in the cost sheet.
- Acquire skills in preparing a Statement of Cost and an Abridged Profit and Loss Account, ensuring accurate representation of financial performance.
- Understand the concept of work-in-progress (WIP) and its significance in production costing.

4.1 INTRODUCTION

One of the primary goals of a cost accounting system is the determination of costs associated with a specific cost object, such as a product or service. For example, imagine you are managing Shyam Manufacturing, a company that produces wooden tables. To determine the cost of producing a single table, the process begins with categorizing costs into Direct Materials (e.g., wood, nails, varnish), Direct Labor (e.g., wages of workers assembling the table), and Overheads (e.g., factory electricity, machine depreciation). Let's assume the cost of wood for one table is ₹500, labor costs ₹300, and factory overheads amount to ₹200.

The next step is to accumulate these costs systematically. First, calculate the Prime Cost by summing up Direct Materials and Direct Labor. In this case, Prime Cost = ₹500 + ₹300 = ₹800. Then, add Overheads to the Prime Cost to arrive at the Total Cost. Thus, Total Cost = ₹800 + ₹200 = ₹1000 for one table. This detailed breakdown can be organized in a cost sheet, where each element of cost is presented clearly under respective headings. For example, Direct Materials: ₹500, Direct Labor: ₹300, Prime Cost: ₹800, Factory Overheads: ₹200, and Total Cost: ₹1000.

Cost sheets are versatile and can be tailored to meet specific needs. While functional classification—such as production, selling, or administrative costs—is common, alternative classifications like fixed, variable, or semi-variable costs can also be used. In the above example, fixed costs (e.g., machine depreciation) remain unchanged regardless of production levels, while variable costs (e.g., wood and labor) fluctuate with the number of tables produced. This structured approach simplifies cost determination and aids in effective decision-making.

4.2 COST CLASSIFICATION

The different types of costing are tailored to meet the unique requirements of individual business environments. Despite this variation, the

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fundamental principles remain consistent across all methods: gathering and analyzing expenditures based on cost elements and determining the cost for each Cost Centre or Cost Unit.

Classification of cost refers to the systematic grouping of cost items into logical categories, depending on their nature or purpose. For clarity and efficiency, costs should be classified based on a single characteristic for each specific purpose, ensuring there is no ambiguity in the classification. The classification system should be comprehensive, ensuring that every cost item can be accurately categorized. Classification of cost on different basis are shown in figure 4.1 and discussed in detail below.

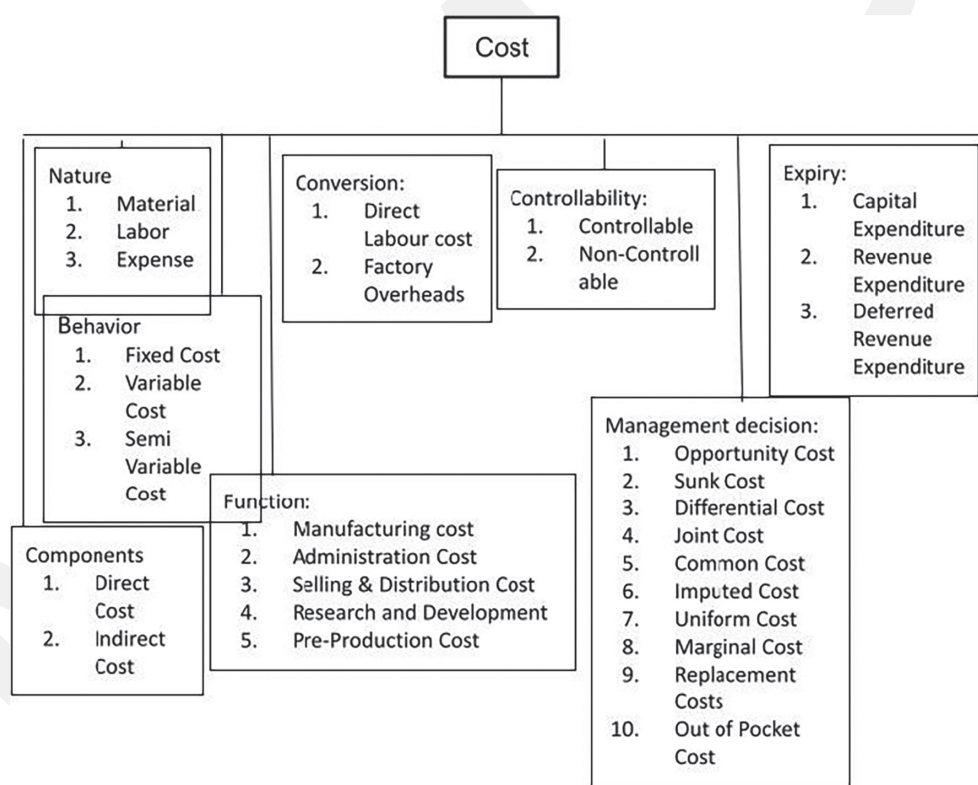


Figure 4.1 Classification of Cost

4.2.1 CLASSIFICATION OF COST ACCORDING TO ITS COMPONENTS

If expenditure can be assigned to a cost centre or cost object in a financially feasible manner, it is categorized as direct; otherwise, it is referred to as indirect (Shown in Figure 4.2). Based on this classification, material costs are split into direct material costs and indirect material costs, labor costs into direct labor and indirect labor costs, and expenses into direct expenses and indirect expenses.

Classification of Material, Labor, and Expenses Based on Relation to Cost Centre:

- 1. Direct Costs:** Direct costs are those costs that can be clearly and easily allocated to a specific cost centre (e.g., a department or

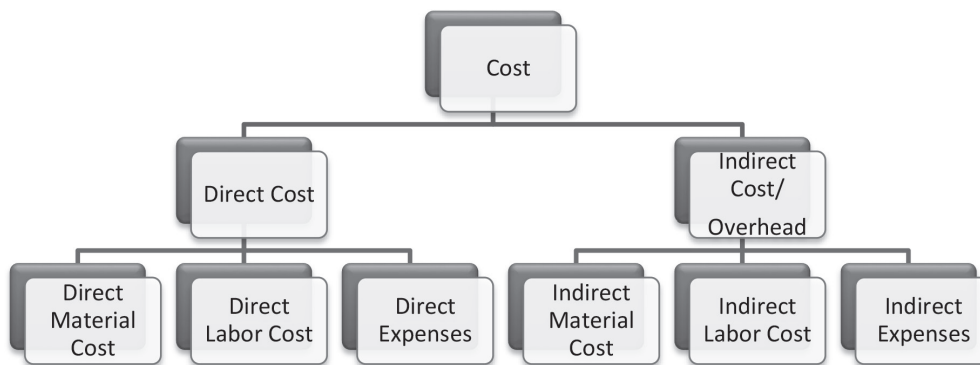


Figure 4.2 Classification of Cost According to its Components

process) or a cost object (e.g., a product or service). These costs are directly tied to the production of goods or services, ensuring they can be measured and attributed with accuracy.

Key Components:

- ❑ **Direct Material Cost:** The cost of materials that can be clearly attributed to a cost centre or cost object in a cost-effective manner. Examples: Timber used in furniture manufacturing, fabric in garment production, or metal sheets for car bodies.
- ❑ **Direct Labor Cost:** The wages of workers that can be directly linked or identified with a cost centre or cost object. Examples: Assembly line workers, machine operators, or craftsmen.
- ❑ **Direct Expenses:** Expenses, excluding direct material and labor, that can be linked to a cost centre or cost object. Examples: Special tools, royalties for production, or subcontracting charges specific to a product.

The sum of all direct costs is termed **Prime Cost**, representing the core costs involved in producing goods or services:

$$\text{Direct Material} + \text{Direct Labor} + \text{Direct Expenses} = \text{Prime Cost}$$

- 2. Indirect Costs:** Indirect costs are expenses that cannot be directly traced to a specific cost centre or cost object. Instead, they are shared across multiple cost centres or products and are collectively referred to as **overheads**. These costs are essential for operations but are not directly tied to the production process.

Key Components:

- 1. Indirect Material:** Materials that cannot be directly assigned to a particular cost centre or object. Examples: Lubricants, cleaning supplies, or nails used in various production processes.
- 2. Indirect Labor:** Wages of employees that cannot be directly assigned to a specific cost centre. Examples: Supervisors, maintenance staff, or quality control personnel.

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3. **Indirect Expenses:** Non-material and non-labor costs that cannot be directly assigned to a cost centre. Examples: Factory rent, utility bills, insurance, or depreciation of equipment.

The sum of all indirect costs forms the Overheads:

$$\text{Indirect Material} + \text{Indirect Labor} + \text{Indirect Expenses} = \text{Overheads}$$

4.2.2 CLASSIFICATION OF COST ACCORDING TO NATURE

Costs should be compiled in their natural categories, such as Material, Labor, and Other Direct Expenses. The nature of cost items varies. The elements of cost can be classified into three categories shown in figure 4.3:

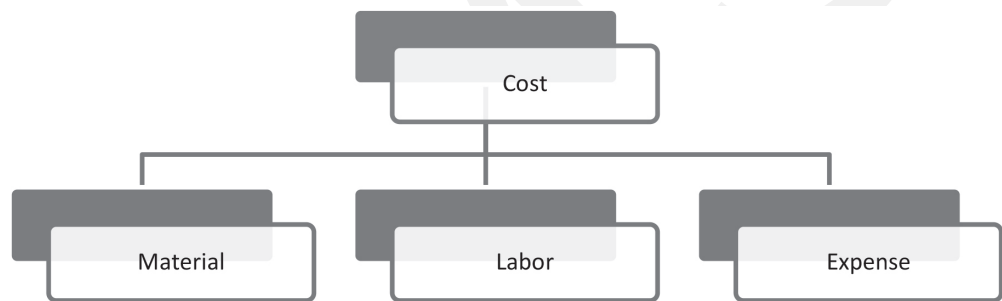


Figure 4.3 Cost Classification on the basis of Nature

1. **Material Cost:** Material cost refers to the expense incurred for any type of material used in the production of a product or service.

Key Components of Material Cost:

- ❑ **Raw Materials:** The basic substances used directly in production, such as wood for furniture or flour for bread.
- ❑ **Freight Inwards:** The transportation cost incurred to bring materials to the production site.
- ❑ **Taxes and Duties:** Costs like import duties or GST paid during material acquisition.
- ❑ **Insurance:** Premiums paid to insure materials in transit or storage.

Exclusions:

- ❑ **Trade Discounts:** Discounts received on purchases.
- ❑ **Duty Drawbacks and Refunds:** Rebates on excise duty, VAT, or other taxes.

Example:

For a bakery producing bread, material costs include the price of flour, sugar, yeast, and butter, along with transport charges for delivering these items to the bakery.

2. **Labor Cost:** Labor cost denotes the compensation paid to employees, whether permanent or temporary, for their services.

Key Components of Labor Cost:

- ❑ **Salaries and Wages:** Paid to both permanent staff and contractor employees.
- ❑ **Employee Benefits:** Contributions to provident funds, gratuity payments, and ESI (Employees' State Insurance).
- ❑ **Overtime and Incentives:** Additional payments for working beyond regular hours or for achieving specific targets.

Example:

In a textile factory, labor costs include the wages paid to machine operators, supervisors, and contract workers, along with overtime bonuses during high-demand periods.

3. **Expenses:** Expenses consist of all costs other than material or labor incurred in the execution of an activity.

Key Types of Expenses:

- ❑ **Factory Overheads:** Costs like electricity for running machines or depreciation of factory equipment.
- ❑ **Administrative Expenses:** Salaries of office staff and costs of office supplies.
- ❑ **Selling and Distribution Expenses:** Advertising, packing, and delivery charges.

4.2.3 CLASSIFICATION OF COST ACCORDING TO BEHAVIOR

Costs are categorized according to their behavior as fixed costs, variable costs, and semi-variable costs, based on how they respond to fluctuations in activity levels (shown in Figure 4.4).

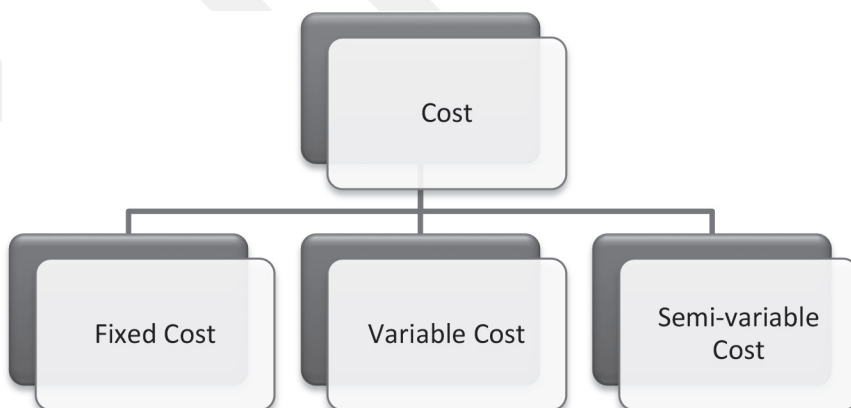


Figure 4.4 Classification of Cost According to Behavior

1. Fixed Costs

The Chartered Institute of Management Accountants (CIMA) in London defines fixed costs as “the cost which is incurred for a period and which,

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within certain output and turnover limits, tends to be unaffected by fluctuations in the levels of activity (output or turnover).” These costs are essential for providing the physical and human resources necessary for business operations. Fixed costs arise from contractual obligations and management decisions, and they accumulate over time rather than with production, typically being expressed in terms of time. Examples include rent, property taxes, insurance, and supervisors’ salaries.

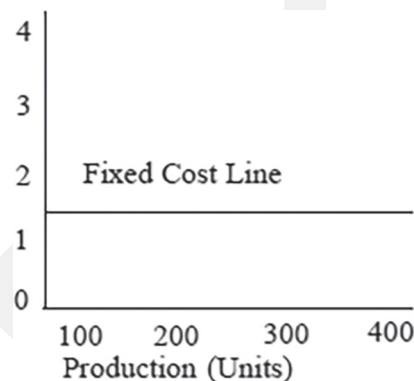


Figure 4.5 Fixed Cost Line

The graph in Figure 4.5 represents **fixed costs**, which remain constant regardless of production levels. The horizontal line shows that even as production increases, the cost stays unchanged.

For Instance, a manufacturing company pays Rs. 50,000 per month as rent for its factory, regardless of production levels. Even if no goods are produced, this cost remains constant.

It is important to note that fixed costs can change; however, they usually remain stable within a relevant range. The term “fixed” refers to their non-variability concerning that specific range. For analytical purposes, fixed costs can be classified into the following categories:

- (a) **Committed Costs:** These costs are necessary to maintain certain facilities and cannot be quickly eliminated. Management has limited or no discretion over these costs, which include items like rent and insurance. *Example:* Shyam Enterprises, a manufacturing firm, has signed a 5-year lease for its production facility. The monthly rent of ₹1,00,000 is a committed cost, as it must be paid regardless of the company’s production level. Similarly, the company’s annual insurance premium of ₹50,000 is a committed cost since it is required to safeguard the facility and its equipment. Even if production halts temporarily, these costs must still be paid.
- (b) **Policy and Managed Costs:** Policy costs are incurred to implement specific management strategies, such as executive development and housing, and are often discretionary. Managed costs are necessary for the company’s operational existence, such as staff services. *Example:* Radha Textiles, a clothing manufacturer, introduces a

policy to offer executive development training for senior managers at a cost of ₹2,00,000 annually. This is a policy cost, as it arises from a strategic decision made by management. On the other hand, Radha Textiles employs support staff for roles like customer service, maintenance, and security. The salaries paid to these staff are managed costs since they are necessary for the company's day-to-day operations. While policy costs are discretionary, managed costs are essential for the company to function smoothly.

- (c) **Discretionary Costs:** These costs are not essential to operations and can be controlled by management. They arise from special policy decisions or new research initiatives and can be eliminated or reduced at management's discretion. *Example:* Shyam Enterprises decides to launch a new advertising campaign costing ₹3,00,000 to promote its latest product line. Since this cost is based on a marketing decision and not essential to daily operations, it is a discretionary cost. If Shyam Enterprises faces financial pressure, it can cancel or reduce this advertising campaign to save costs. Similarly, research and development (R&D) expenses for a new product are discretionary since management can delay or eliminate the project.
- (d) **Step Costs:** These costs remain constant for a specific level of output but increase by a fixed amount when production exceeds that level. *Example:* Radha Textiles employs one supervisor to oversee production up to 500 units per day, with a salary of ₹40,000 per month. If production increases beyond 500 units per day, the company needs to hire a second supervisor at an additional ₹40,000 per month. Thus, for production between 1 to 500 units, the total supervision cost is ₹40,000. However, when production increases to 501 units, the cost "steps up" to ₹80,000. The cost remains constant at this level until production exceeds 1,000 units, at which point a third supervisor might be hired, further increasing the cost.

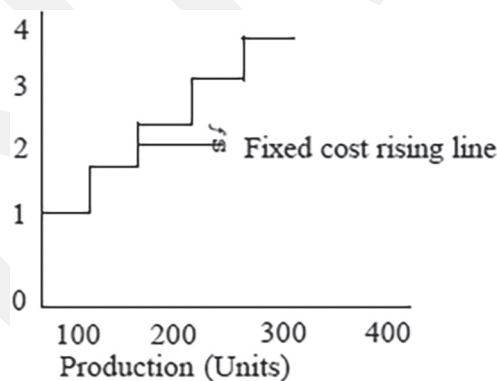


Figure 4.6 Step Cost

The graph in figure 4.6 represents a **step cost line**, where fixed costs increase in steps as production reaches specific levels. This occurs when additional resources or capacity are required at certain production thresholds.

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2. Variable Costs

Variable costs are those that change directly and proportionately with the level of output, such as direct materials and direct labor. While the variable cost per unit remains constant, the total cost fluctuates according to production levels. Variable costs are expressed in terms of units rather than time. Management decisions can influence cost behavior patterns, and the concept of variability is relative; any change in the conditions that determine variability necessitates a re-evaluation of that variability. Suppose, If the cost of raw material is Rs. 10 per unit, the total cost increases proportionately. Producing 100 units will cost Rs. 1,000, while producing 200 units will cost Rs. 2,000.

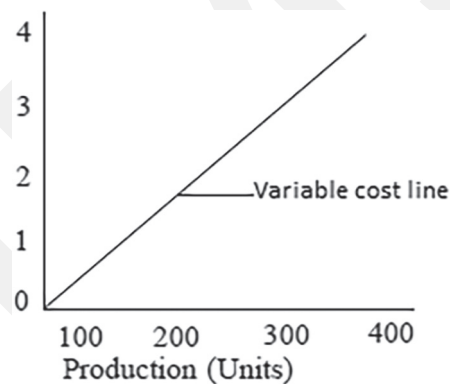


Figure 4.7 Variable Cost

The graph in figure 4.7 shows the **variable cost line**, which increases proportionally with production units. As production rises from 100 to 400 units, the variable cost increases linearly from 0 to 4 units on the cost axis.

3. Semi-Fixed (Semi-Variable) Costs

Semi-fixed or semi-variable costs comprise both fixed and variable components. Due to their variable element, these costs fluctuate with volume; however, because of the fixed element, they do not change in direct proportion to output. Semi-variable costs increase in the same direction as output but not necessarily at the same rate. For instance, depreciation can be semi-variable; for two shifts, the total depreciation may only be 50% higher than for a single shift. These costs can change with relatively small fluctuations in output but do not do so in a one-to-one ratio. For Instance, a telephone bill includes a fixed monthly charge of Rs. 500 and an additional cost of Rs. 1 per call. If 300 calls are made, the total bill is Rs. 800 (Rs. 500 fixed + Rs. 300 variable).

The graph in figure 4.8 shows the **semi-variable cost line**, which includes a fixed component (intercept at the cost axis) and a variable component

that increases with production. The cost starts above zero and rises linearly as production units increase.

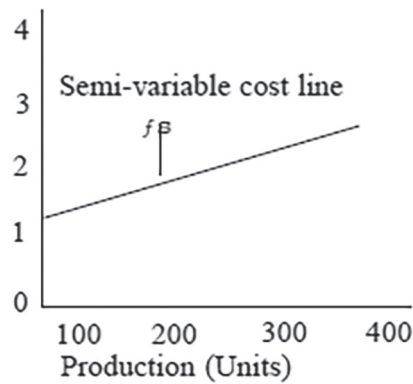


Figure 4.8 Semi Variable Cost

Example 1: A company has provided the following information for an output of 10,000 units. We are required to calculate the total cost for 12,000 units of output and the cost per unit.

Given Information:

- Variable Cost: ₹60,000
- Fixed Cost: ₹40,000
- Semi-variable Cost (60% fixed, 40% variable): ₹90,000

Statement of Cost for 12,000 Units

Particulars	Amount (₹)
Variable Cost @ ₹6 per unit ($\text{₹60,000} / 10,000 \text{ units} \times 12,000$ units)	₹72,000
Fixed Cost (Remains constant)	₹40,000
Semi-variable Cost:	
Fixed Component (60% of ₹90,000)	₹54,000
Variable Component ($\text{₹36,000} / 10,000 \text{ units} \times 12,000$ units)	₹43,200
Total Semi-variable Cost	₹97,200
Total Cost (Variable + Fixed + Semi-variable)	₹2,09,200
Cost per unit ($\text{₹2,09,200} / 12,000$)	₹17.43

Table 4.1: Comparison: Fixed Costs, Variable Costs, and Semi-Fixed (Semi-Variable) Costs

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Table 4.1 Comparison: Fixed Costs, Variable Costs, and Semi-Fixed (Semi-Variable) Costs

Criteria	Fixed Costs	Variable Costs	Semi-Fixed (Semi-Variable) Costs
Definition	Costs that remain unchanged within a relevant range of activity or time, regardless of output.	Costs that vary directly and proportionately with production or activity levels.	Costs with both fixed and variable components, changing with activity levels but not proportionately.
Nature	Incurred over time and not tied to production levels.	Tied directly to production; increase or decrease with changes in output.	Partially constant (fixed) and partially fluctuating (variable) with activity levels.
Examples	Rent, property taxes, insurance, supervisors' salaries.	Direct materials, direct labor, power costs tied to machine hours.	Utility bills (fixed charge plus usage cost), maintenance costs, depreciation.
Behavior with Output	Unaffected by changes in production within a certain range.	Changes in direct proportion to the level of output.	Changes in the same direction as output but not in the same proportion.
Expression	Typically expressed in terms of time (monthly, yearly).	Expressed in terms of units of production.	Expressed as a mix of time- and unit-based factors.
Predictability	Highly predictable within the relevant range of activity.	Predictable as long as production output is known.	Partially predictable due to mixed behavior.
Management Control	Can be discretionary (e.g., policy-related costs) or committed (e.g., rent).	Directly influenced by production levels and management decisions regarding volume.	Can be controlled to some extent by managing both fixed and variable components.
Key Characteristics	- Arise from contractual obligations or strategic decisions. - Remain stable within a range.	- Directly linked to production volume. - Per-unit cost is constant.	- Includes both fixed and variable elements. - Does not vary proportionally with output.
Relevance in Decision-Making	Important for calculating break-even points and long-term planning.	Critical for determining cost per unit and short-term production planning.	Useful for cost analysis in scenarios with fluctuating production levels.

4.2.4 CLASSIFICATION OF COST ACCORDING TO FUNCTION

A business enterprise carries out several functions such as manufacturing, selling, and research, each of which incurs specific costs. These costs can be categorized into different functional types:

1. PRODUCTION OR MANUFACTURING COSTS

Production costs refer to the expenses involved in producing a product or service. These costs are related to the manufacturing division and include all costs incurred from receiving raw materials, supplying labor, and providing services, up until the product is fully produced and packaged. The key components of Production or Manufacturing Costs include:

1. Direct Materials
 2. Direct Labor
 3. Direct Expenses
 4. Factory Overheads: These consist of indirect materials, indirect labor, and other factory-related expenses.
- Manufacturing costs are often considered as the sum of prime costs and factory overhead.

2. ADMINISTRATION COSTS

Administration costs are the expenses associated with the overall management of the organization. As indirect costs, they are commonly referred to as administrative overheads. Understanding these costs requires knowing the role of administrative functions, which mainly involve:

1. Policy Formulation
 2. Organizational Direction
 3. Operational Control
- However, administration costs do not include management activities related to production, selling, distribution, or R&D. Typically, administration costs cover indirect expenses such as:
4. Salaries of office staff, accountants, and directors
 5. Rent, rates, and depreciation of office buildings
 6. Postage, stationery, and telephone costs
 7. Office supplies and expenses
 8. General administration expenses

3. SELLING & DISTRIBUTION COSTS

Selling costs are indirect costs related to the sale of goods or services. They cover all expenses involved in sales management. Distribution costs refer to the costs incurred from the completion of the product until it reaches the consumer. The selling function focuses on creating demand for the products and securing orders, while distribution ensures goods are available to customers. Examples of selling and distribution costs include:

1. Salaries and commissions for sales personnel and managers
2. Advertising and insurance expenses
3. Rent, rates, depreciation, and insurance for sales offices and warehouses

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4. Costs of insurance, freight, export duties, packing, and shipping
5. Maintenance of delivery vans

4. RESEARCH & DEVELOPMENT COSTS

Research and development (R&D) costs are incurred for activities aimed at improving existing products, developing new ones, or enhancing manufacturing processes. R&D expenses include:

1. Development of new products
2. Improvement of existing products
3. Exploring new uses for existing products
4. Solving technical issues in manufacturing and applying products
5. Costs associated with commercializing or implementing research findings

5. PRE-PRODUCTION COSTS

Pre-production costs are the expenses incurred before actual production begins. These costs are usually associated with setting up a new factory, launching a new project, or introducing a new product line. Such costs are often treated as deferred revenue expenditure and charged to the future production costs.

Let's consider a company, Radha Textiles, that produces and sells custom clothing. Below are examples of how each functional cost type applies to their operations:

(i) Production or Manufacturing Costs

Radha Textiles receives an order for 1,000 T-shirts. The production costs include various components. The cost of fabric is ₹50 per T-shirt, amounting to ₹50,000, while threads, buttons, and dyes cost ₹5,000. Tailors are paid ₹40 per T-shirt, leading to ₹40,000 in direct labor expenses. Additionally, the company incurs ₹10,000 as a special embroidery design fee for the customer. Factory overheads, including machine maintenance at ₹2,000 and electricity costs of ₹3,000, further add to the total manufacturing cost, which sums up to ₹1,10,000.

(ii) Administration Costs

The management at Radha Textiles incurs costs for general administration activities. This includes salaries for administrative staff, which amount to ₹25,000 per month, and office rent of ₹10,000. Expenses for stationery, such as paper and ink, total ₹1,000, while internet and phone bills for communication cost ₹2,000. Altogether, the administration costs add up to ₹38,000 per month.

(iii) Selling & Distribution Costs

When Radha Textiles launches a marketing campaign for their new winter wear, selling and distribution costs are incurred. The company

spends ₹15,000 on social media advertisements and ₹10,000 on salaries and commissions for sales staff. Distribution costs include freight charges of ₹5,000 for delivering products to retail stores and ₹2,000 for packaging materials to ensure secure delivery. These activities result in a total cost of ₹32,000 for selling and distribution.

(iv) Research & Development Costs

Radha Textiles decides to innovate by developing a new eco-friendly fabric. To achieve this, they hire textile researchers at a cost of ₹50,000 and spend ₹20,000 on raw materials for testing. Additionally, market research to understand customer preferences costs ₹10,000. These R&D efforts lead to a total expenditure of ₹80,000.

(v) Pre-Production Costs

As Radha Textiles plans to set up a new factory to manufacture winter jackets, pre-production costs are incurred. Training new workers involves an expense of ₹30,000, while setting up machinery costs ₹50,000. A trial production run adds another ₹20,000 to the expenses. The total pre-production cost sums up to ₹1,00,000, which will be allocated to future production costs.

Table 4.2: Comparison of Functional Costs

Table 4.2 Comparison of Functional Cost

Criteria	Production Costs	Administration Costs	Selling & Distribution Costs	R&D Costs	Pre-Production Costs
Nature	Directly related to manufacturing activities.	Associated with overall business management.	Linked to selling products and delivering to customers.	Focused on innovation and product/process improvement.	Costs before regular production begins.
Examples	Raw materials, wages, factory rent.	Office staff salaries, rent, stationery.	Advertising, freight, sales commissions.	Product development, market research.	Installation, testing, and training costs.
Purpose	To produce finished goods or services.	To manage and organize operations.	To create demand and deliver goods.	To enhance quality and innovate.	To set up operations for future production.
Relevance in Cost Accounting	Forms the foundation for determining prime and factory costs.	Aids in calculating general overheads.	Used for pricing and distribution cost analysis.	Helps allocate R&D expenses effectively.	Necessary for planning new ventures.

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4.2.5 CLASSIFICATION OF COST BASED ON CONVERSION

Conversion cost refers to the costs incurred in converting raw materials into finished products, excluding direct material costs. It includes:

1. **Direct Labor Cost:** Wages of workers directly engaged in the production process.
2. **Factory Overheads:** All indirect costs involved in manufacturing, such as indirect materials, indirect labor, and other production-related expenses.

Formula to Calculate Conversion Cost:

Conversion Cost = Direct Labor + Factory Overheads

Example:

Suppose a furniture manufacturing company incurs the following costs:

- ❑ Direct labor wages: ₹50,000
- ❑ Indirect materials (polish, glue, etc.): ₹5,000
- ❑ Salaries of factory supervisors: ₹10,000
- ❑ Depreciation of machinery: ₹8,000
- ❑ Factory electricity bills: ₹7,000

The total conversion cost would be:

$$\text{Conversion Cost} = ₹50,000 + (₹5,000 + ₹10,000 + ₹8,000 + ₹7,000) = ₹80,000$$

4.2.6 CLASSIFICATION OF COST BY CONTROLLABILITY

Costs can be categorized into controllable and non-controllable (Shown in Figure 4.9) costs based on the ability of managers to influence them.

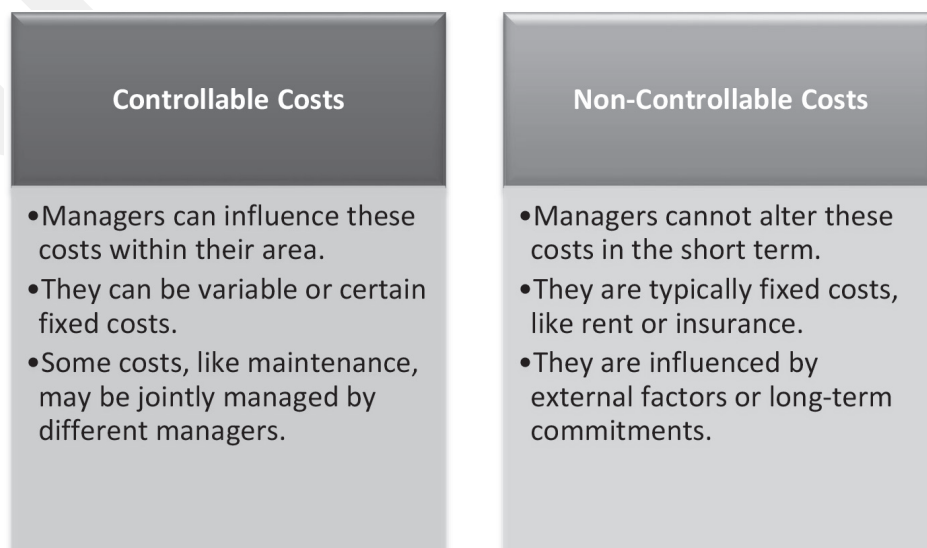


Figure 4.9 Classification of Cost by Controllability

1. CONTROLLABLE COSTS

The Chartered Institute of Management Accountants defines controllable costs as those “which can be influenced by its budget holder.” These costs are subject to management by individuals within specific levels of an organization, enabling more effective financial control and budgeting. Controllable costs have certain characteristics: they are overseen by designated individuals or managers within their respective areas of responsibility and can include both variable costs and, in some cases, fixed costs. Additionally, some costs may be jointly influenced by multiple managers, such as maintenance costs, which can be controlled by both production and maintenance managers.

Examples:

- ❑ **Raw Material Costs:** A production manager can decide how much material to use efficiently.
- ❑ **Overtime Wages:** A department head can approve or limit overtime based on workload.
- ❑ **Advertising Costs:** The marketing manager controls the budget for ad campaigns.
- ❑ **Utility Usage:** A facility manager can regulate electricity and water consumption to optimize costs.

2. NON-CONTROLLABLE COSTS

Non-controllable costs are expenses that cannot be influenced or modified by any managerial level within the organization. These costs remain constant, irrespective of managerial actions or decisions. Key characteristics of non-controllable costs include their association with long-term commitments or external factors, making them difficult to alter in the short term. They are generally considered fixed costs, as they do not fluctuate with changes in the level of business activity.

Examples:

- ❑ **Rent for Office Space:** A company cannot easily reduce rental expenses during a lease period.
- ❑ **Depreciation:** The systematic allocation of an asset's cost over its useful life cannot be modified.
- ❑ **Government Taxes:** Regulatory taxes or duties imposed are beyond managerial control.
- ❑ **Insurance Premiums:** The premiums agreed upon in a contract remain constant regardless of managerial actions.

4.2.7 CLASSIFICATION OF COST BY MANAGEMENT DECISIONS

1. OPPORTUNITY COSTS

Opportunity cost refers to the potential benefits lost when one course of action is chosen over an alternative. It represents the return that could

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have been gained if the asset had been used in its next best alternative. The Chartered Institute of Management Accountants defines it as “the benefits lost by rejecting the best competing alternative to the one chosen.” For example, if capital is invested in plant and machinery, the opportunity cost is the interest and dividends that could have been earned if the funds had been invested elsewhere. Similarly, if a business owner chooses not to take up another job opportunity, the income from that alternative is also considered an opportunity cost. Though opportunity costs do not appear in financial statements, they are vital for decision-making and evaluating options. *Example:* Radha has Rs. 2,00,000 and decides to invest it in a business instead of depositing it in a fixed deposit that offers an annual interest of 8%. The opportunity cost of investing in the business is Rs. 16,000 (8% of Rs. 2,00,000), which she forgoes by not choosing the fixed deposit option.

2. SUNK COSTS

A sunk cost is an expenditure that has already been incurred and cannot be recovered, making it irrelevant for future decisions. The National Association of Accountants (USA) defines it as “an expenditure for equipment or productive resources which has no economic relevance to the present decision-making process.” Since it pertains to past expenditures, it is often referred to as an unavoidable cost. The Chartered Institute of Management Accountants similarly states that sunk costs should not impact current decisions. It can also be understood as the difference between the cost of an asset and its salvage value. *Example:* Shyam purchased a machine for Rs. 1,50,000 two years ago, but due to technological advancements, the machine is now obsolete. The amount spent on the machine is a sunk cost since it cannot be recovered, and it should not influence future decisions about purchasing a new machine.

3. DIFFERENTIAL COSTS

Differential cost refers to the difference in total cost between two alternatives, helping decision-making by showing the increase or decrease in costs due to:

- ❑ (a) Producing more or fewer units of a product,
- ❑ (b) Changing production or distribution methods,
- ❑ (c) Adding or removing a product or territory, and
- ❑ (d) Choosing an additional sales channel.

This cost represents the variations in expenses between different production levels or scenarios, including changes in marginal and fixed costs. Differential costs are used for specific purposes and planning decisions. Incremental cost is a similar concept, calculating the additional cost per unit for increased output. It is expressed on a per-unit basis, unlike differential cost, which is calculated in total terms and applies to both increases and decreases in production. Differential costs help management plan for better resource utilization or potential investments. *Example:* A company currently produces 5,000 units

at a total cost of Rs. 2,00,000. If production increases to 6,000 units, the total cost rises to Rs. 2,40,000. The differential cost for producing an additional 1,000 units is Rs. 40,000 (Rs. 2,40,000 - Rs. 2,00,000).

4. JOINT COSTS

Joint costs arise when a single raw material is processed to produce multiple products. These costs are incurred up to the point where the products can be separately identified, called the split-off point. At this stage, joint products are recognized, and one product may be primary while others are by-products. According to Bierman and Djckman, joint costs “result from factors of production that produce two or more products,” stemming from the nature of the raw materials or production methods, not managerial decisions. The National Association of Accountants similarly defines joint costs as those incurred for products produced through a common process, requiring an appropriate basis for allocation among products. *Example:* A dairy processes raw milk into two products — butter and cream. The production cost incurred up to the point where milk is separated into butter and cream is Rs. 50,000. This Rs. 50,000 is a joint cost that needs to be allocated between butter and cream using an appropriate method, like physical units or relative sales value.

5. COMMON COSTS

Common costs are expenses shared by multiple products, jobs, or specific costing objects. These costs cannot be directly assigned to an individual product and must be allocated among them. The National Association of Accountants defines them as “the cost of services used in the creation of two or more outputs that cannot be clearly allocated to those outputs on a justified basis.” An example of a common cost is factory rent, which is shared by all departments in the facility. Management decisions often influence these costs, as in the case of allocating overhead or shared services. *Example:* A manufacturing plant produces multiple products (furniture, cabinets, and shelves) under one roof. The rent for the factory, Rs. 1,00,000 per month, is a common cost shared by all three product lines. Since the rent benefits all products, it must be allocated among them using a suitable allocation method.

6. IMPUTED COSTS

Imputed costs are hypothetical or theoretical expenses that do not involve actual cash transactions but are considered in decision-making. These costs are not typically recorded in financial accounts but play a role in economic evaluations. Examples include interest on internally generated funds, owner salaries in sole proprietorships or partnerships, and notional rent for using an owned space instead of leasing. *Example:* Shyam operates his own retail store and does not pay himself a salary. However, for decision-making purposes, he considers the imputed cost of his own salary as Rs. 50,000 per month, which he could have earned if he worked elsewhere as a store manager. This cost is relevant for calculating the profitability of his store.

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7. UNIFORM COSTS

Uniform costs are not distinct in themselves but refer to the standardized methods and procedures used by multiple firms for cost accounting. This approach is useful for comparing costs across firms and industries, providing a consistent basis for financial evaluation and analysis. *Example:* The textile industry adopts a uniform costing system, requiring all firms to follow the same cost accounting methods for calculating production costs. This allows for industry-wide comparison of costs, aiding industry associations and regulators in benchmarking.

8. MARGINAL COSTS

Marginal cost refers to the additional cost incurred for producing one more unit of output. It includes all variable costs, such as prime costs and variable overheads. This cost helps in analyzing how the total costs change with the production of additional units. *Example:* A company produces 1,000 units at a total variable cost of Rs. 80,000. When it produces 1,001 units, the total variable cost increases to Rs. 80,080. The marginal cost of producing one additional unit is Rs. 80 (Rs. 80,080 - Rs. 80,000).

9. REPLACEMENT COSTS

Replacement cost refers to the current market value of replacing an asset, focusing on the price required to substitute the asset with a new one in the present market conditions. This cost is often relevant for decision-making in asset management and insurance evaluations. *Example:* A company owns a delivery van that was purchased five years ago for Rs. 5,00,000. Due to wear and tear, the company decides to replace it. The current market price of a new van with similar specifications is Rs. 7,00,000. The replacement cost of the van is Rs. 7,00,000, which will influence the company's decision on whether to buy a new van or continue using the old one.

10. OUT OF POCKET COSTS

Out of pocket costs are actual expenses paid to external parties, resulting in a cash outflow. Unlike non-cash expenses like depreciation, out of pocket costs directly affect the cash flow of a business. These costs are particularly important when setting pricing strategies during economic downturns or deciding between manufacturing a product in-house or outsourcing it. *Example:* Shyam's company pays Rs. 20,000 in cash each month for utility bills (electricity, water, and internet) for the factory. This cash outflow is an out-of-pocket cost, as it directly impacts the company's cash flow. Unlike depreciation, this expense requires an actual payment to an external party.

4.2.8 CLASSIFICATION OF COST BASED ON EXPIRY

The classification of costs based on expiry categorizes costs according to when they are incurred and how long their benefits last, helping determine their recognition and impact on financial reporting.

1. CAPITAL EXPENDITURE (EXPIRED OVER TIME):

Capital expenditures are costs for acquiring or improving long-term assets like property or machinery, providing benefits over multiple periods. These costs are capitalized on the balance sheet and reduced over time through depreciation, amortization, or depletion. For example, a machine's cost is spread over its useful life rather than recognized fully in the year of purchase.

Example: Shyam Ltd. purchased new machinery for ₹10,00,000 on April 1, 2024. The machine has an estimated useful life of 10 years and a residual value of ₹1,00,000. Instead of charging the entire amount as an expense in the year of purchase, Shyam Ltd. will allocate the machine's cost over its useful life through depreciation.

Calculation

Depreciation is calculated as:

Depreciation per year = (Cost of Asset – Residual Value)/Useful Life
 Depreciation per year = (₹10,00,000 – ₹1,00,000)/10 = ₹90,000

This means ₹90,000 will be recorded as a depreciation expense every year for 10 years. The initial purchase of ₹10,00,000 is classified as Capital Expenditure, and its benefit is spread over multiple years.

2. REVENUE EXPENDITURE (EXPIRED WITHIN THE PERIOD):

Revenue expenditures include costs for day-to-day operations, providing benefits within the current accounting period. Examples are wages, rent, and raw materials. These costs are fully expensed in the profit and loss statement for the period they occur and do not contribute to long-term value.

Example: Radha Enterprises incurred the following costs during the month of November 2024:

1. Wages Paid to Workers = ₹50,000
2. Electricity Bill for the Factory = ₹12,000
3. Raw Materials Purchased = ₹1,00,000

Since these expenses are incurred for day-to-day operations and benefit only the current accounting period, they are classified as Revenue Expenditure. These amounts are fully charged as expenses in the profit and loss account for the month of November 2024, reducing the net profit for that period.

3. DEFERRED REVENUE EXPENDITURE (EXPIRED OVER A FEW PERIODS):

Deferred revenue expenditures are upfront costs whose benefits extend over several periods, not classified as capital expenses. Initially recorded as assets, these costs are gradually expensed to align with the periods benefiting from them. For instance, an advertising campaign paid for upfront may have benefits recognized over the following years.

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Example: Radha Ltd. launched an advertising campaign for ₹6,00,000 on January 1, 2024, to promote a new product line. The company expects the benefits of the campaign to last for 3 years. Instead of charging the entire cost as an expense in the first year, Radha Ltd. records it as a Deferred Revenue Expenditure in the balance sheet and amortizes it over 3 years.

Calculation

Amortization per year = Total Advertising Cost/Period of Benefit

Amortization per year = ₹6,00,000/3 = ₹2,00,000

Each year, ₹2,00,000 will be expensed in the profit and loss account. For the year ending December 31, 2024, the profit and loss account will show an Advertising Expense of ₹2,00,000, while the remaining ₹4,00,000 will be shown as a deferred revenue expenditure (asset) in the balance sheet.

Activity

Classify each of the following as Direct or Indirect Cost (D/I) and Fixed or Variable Cost (F/V):

Item	D/I	F/V
1. Fuel used in delivery trucks of a logistics company	()	()
2. Salaries of production supervisors in a textile factory	()	()
3. Raw wood used in the furniture manufacturing process	()	()
4. Depreciation of machinery in a plastic bottle factory	()	()
5. Rent for the corporate office of a beverage company	()	()
6. Packaging material used for shipping goods to customers	()	()
7. Utilities (electricity, water) in a manufacturing plant	()	()
8. Sugar used for producing sweets in a bakery	()	()
9. Lubricants used for maintaining assembly line machines	()	()
10. Advertising expenses for promoting a product	()	()
11. Salaries of accountants in a shoe production company	()	()
12. Ink used for printing labels in a packaging company	()	()
13. Rental cost of forklifts used in warehouses	()	()
14. Tires used on tractors in an agricultural equipment factory	()	()
15. Commission paid to sales agents for selling products	()	()

**SELF-ASSESSMENT QUESTIONS**

1. What is classified as a variable cost?
 - a. Rent for a factory building
 - b. Wages of permanent employees
 - c. Direct materials that change with production levels
 - d. Depreciation of machinery

2. A company is selling 2,000 units at Rs. 20 per unit. The variable cost per unit is Rs. 8, and the total fixed costs amount to Rs. 8,000. The company receives an export order to supply 400 units at Rs. 16 per unit. If this order is accepted, fixed costs will increase by Rs. 500. What is the incremental profit to the company?
 - a. 2,500 b. 3,700 c. 2,650 d. 2,700
3. What are opportunity costs?
 - a. Costs associated with past expenditures
 - b. The benefits lost by choosing one alternative over another
 - c. Costs incurred for future projects
 - d. Costs that have already been paid and cannot be recovered
4. Which type of costs can be classified as controllable?
 - a. Non-controllable costs b. Costs that can be influenced by its budget holder
 - c. Sunk costs d. Joint costs

4.3 COST SHEET

A Cost Sheet is a statement that details the different elements contributing to the cost of products or services. It breaks down these costs by their components, functions, and sometimes by their behavior. Typically, this statement is prepared for specific periods such as weekly, monthly, quarterly, half-yearly, or yearly, catering to the needs of the industry.

A Cost Sheet can be defined as a comprehensive statement that outlines the various costs incurred in the production process. These costs are arranged logically under different categories such as materials, labor, and overheads. The document is usually prepared at regular intervals, such as monthly, quarterly, or annually, to meet industry standards.

According to the Chartered Institute of Management Accountants (CIMA), a cost sheet is “a document used to assemble detailed costs of a cost center or cost unit.”

Objectives of a Cost Sheet

1. **Reveal Total and Per Unit Costs:** It provides insights into the total and per unit costs of products or services.
2. **Breakdown of Total Costs:** It dissects the total cost into its individual components.
3. **Comparative Analysis:** It facilitates a comparison of costs between the current and previous periods.
4. **Guidance for Pricing:** It aids management in setting appropriate selling prices and preparing tenders.

Features of a Cost Sheet

- ❑ The cost sheet is typically prepared using the output costing method, which focuses on calculating the per unit cost of production.

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- ❑ It is prepared for a defined period, such as monthly, quarterly, half-yearly, or yearly.
- ❑ A typical cost sheet includes the following details:
 - ◆ **Period:** The duration for which the cost is being calculated.
 - ◆ **Total Output:** The total quantity of goods produced during the period.
 - ◆ **Cost of Raw Materials:** The total amount spent on raw materials consumed during production.
 - ◆ **Direct Labor Cost:** Wages paid to workers directly involved in production.
 - ◆ **Chargeable Expenses:** Specific expenses related to production.
 - ◆ **Overheads:** Factory, office, administration, and distribution costs.
 - ◆ **Cost Components:** Includes Prime Cost, Works Cost, Office Cost, and Total Cost.

Purposes and Uses of a Cost Sheet**Purposes:**

1. **Determine Unit Cost:** Helps calculate the unit cost of a product or service.
2. **Selling Price Fixation:** Assists in determining the correct selling price for goods or services.
3. **Cost Comparison:** Allows for cost comparisons between different periods.
4. **Identify Inefficiencies:** Helps identify inefficiencies or weaknesses in the production process.

Uses:

1. **Presentation of Costs:** It displays both the total cost and the cost per unit, along with the breakdown of individual components.
2. **Pricing and Quotation:** Supports the process of setting prices or preparing quotations for products.
3. **Cost Control:** Aids in monitoring and controlling costs by comparing actual costs with standard costs.
4. **Production Policy Formulation:** Helps in formulating production policies and strategies.
5. **Management Communication:** Provides important cost-related information to various management levels.

Elements of Costs:

A cost consists of three main components: material, labor, and expenses. Each of these elements can be categorized as either direct or indirect refer figure 4.10 below.

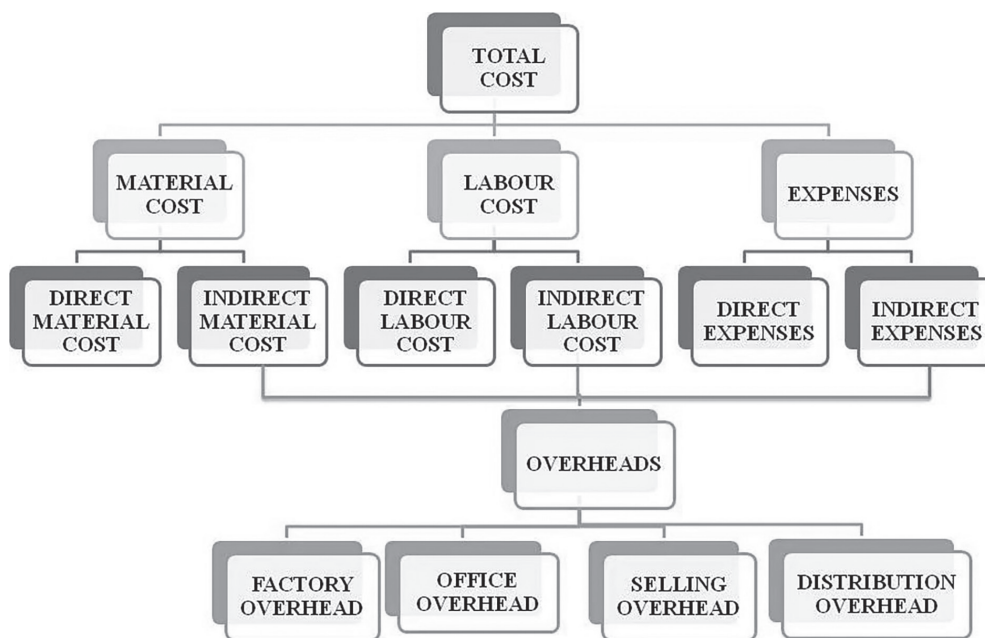


Figure 4.10 Elements of Cost

Material Cost: According to CIMA, London, material cost refers to “the cost of commodities supplied to an undertaking.” It includes expenses related to procurement, freight inwards, taxes, insurance, and other costs directly linked to acquisition. Any deductions such as trade discounts, rebates, duty drawbacks, and refunds (e.g., on modvat or sales tax) are subtracted when determining the material cost. Materials are classified as direct or indirect.

- ❑ **Direct Materials:** These are materials that can be clearly identified with and allocated to specific cost units. Typically, they form a part of the finished product. Examples include cotton in textiles, clay in bricks, leather in shoes, steel in machinery, cloth in garments, and timber in furniture.
- ❑ **Indirect Materials:** These are materials that cannot be directly linked to individual cost units. They are often minor in value or usage, such as small, inexpensive items (e.g., screws, nuts, bolts, thread) or materials not physically incorporated into the final product (e.g., coal, lubricating oil, grease, sandpaper, soap).

Labour Cost: As defined by CIMA, London, labor cost is “the cost of remuneration (wages, salaries, commissions, bonuses, etc.) of the employees of an undertaking.” This category also includes fringe benefits like provident fund contributions, gratuity, ESI, overtime, incentive bonuses, holiday wages, and idle time pay. Labor can be classified as direct or indirect.

- ❑ **Direct Labour:** This refers to wages paid to employees directly involved in converting raw materials into finished goods. These wages can be specifically identified with a product, job, or process. Examples include wages for machine operators, shoemakers, carpenters, weavers, or tailors.

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- ❑ **Indirect Labour:** This type of labor supports production but cannot be directly linked to a specific cost unit. Indirect labor includes roles such as supervisors, inspectors, cleaners, clerks, peons, and watchmen.

Expenses: All costs other than material and labor fall under the category of expenses. CIMA defines expenses as “the cost of services provided to an undertaking and the notional cost of the use of owned assets.” Expenses can also be divided into direct and indirect categories.

- ❑ **Direct Expenses:** These expenses can be directly attributed to specific cost centres or units and are incurred for particular jobs. Also known as chargeable expenses, examples include the hire of specialized plant for a job, costs related to securing contracts, patent fees, experimental costs, job-specific drawings and designs, royalties in mining, and depreciation or hire of plant used on-site.
- ❑ **Indirect Expenses:** Indirect expenses include all indirect costs apart from indirect materials and labor. These expenses are common across cost centres and cannot be directly allocated to a specific job or process. Examples include rent, depreciation, lighting, advertising, insurance, and repairs.

Overheads: Overheads are classified into three groups:

1. **Manufacturing (Works/Factory) Overheads:** Indirect costs related to factory operations, such as rent, rates, insurance for factory premises, power usage, and machinery upkeep.
2. **Office and Administrative Overheads:** Indirect costs related to the management and administration of the business, including office rent, lighting, postage, telecommunication, depreciation of office equipment, bank charges, legal fees, and audit fees.
3. **Selling Overheads:** These are indirect costs incurred to promote and secure orders for a product or service. They are primarily focused on marketing and sales activities. Examples include advertisement expenses, costs of preparing tenders, sales commission, and travelling expenses related to securing sales.
4. **Distribution Overheads:** These indirect costs are associated with the physical distribution of goods to customers and ensuring timely delivery. Examples include warehouse charges, packing costs, carriage outwards, loading and unloading expenses, and costs related to dispatching goods.

Information to Include in a Cost Sheet

1. **Direct Expenses:** Materials, direct wages, and other direct costs for the period.
2. **Indirect Expenses:** Estimated indirect expenses and their distribution under various heads.
3. **Expenditure Breakdown:** The percentage relationship of each expenditure to the total cost.
4. **Comparative Data:** Figures from previous periods for comparison and progress evaluation.

5. **Per Unit and Total Costs:** Enables the calculation of both per unit and total costs.
6. **Profit and Loss Statement:** Helps in preparing a profit and loss statement along with the cost sheet.
7. **Production Volume:** Details the number of units produced during the period.

Categories of Expenses Excluded from a Cost Sheet

Certain expenses are not part of production costs and should not be included in the cost sheet:

1. Financial Charges:

- ☐ Losses on the sale of investments or fixed assets.
- ☐ Interest on debentures, bank loans, and mortgages.
- ☐ Losses due to obsolescence or legal settlements.

2. Financial Incomes:

- ☐ Interest earned on bank deposits or investments.
- ☐ Commission, transfer fees, or other financial gains.
- ☐ Rent, dividends, and profits from asset sales.

3. Appropriation of Profits:

- ☐ Write-off of goodwill, preliminary expenses, or capital raising costs.
- ☐ Income tax, dividend payments, or charity donations.
- ☐ Appropriations to reserves or adjustments in depreciation methods.

4. Abnormal Gains and Losses:

- ☐ Losses of materials or idle labor time due to unforeseen circumstances.

Advantages of Cost Sheet

The key advantages of a cost sheet for a manufacturing business are outlined as follows:

1. **Cost Disclosure:** A cost sheet allows the manufacturer to determine the precise cost per unit in a systematic way. It also reveals how each component of expenditure contributes to the total cost.
2. **Selling Price Determination:** By presenting the actual cost of production, the cost sheet serves as a dependable foundation for setting a competitive selling price. This helps the manufacturer avoid losses caused by improperly setting prices without considering production costs.
3. **Cost Control:** A cost sheet enables the manufacturer to compare current costs with those of previous periods, helping identify cost variations. This comparison aids in eliminating unfavorable factors that contribute to increased costs.

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- 4. Tender Price Determination:** The cost sheet provides a solid basis for preparing tenders or quotations, enabling the manufacturer to propose a price for goods to a prospective customer for future delivery.

4.3.1 TREATMENT AND ADJUSTMENTS OF CERTAIN COST COMPONENTS

1. RAW MATERIAL

The treatment of raw material stocks is essential, as costs must be linked to the units produced. Accurate computation of material consumption, completion of previous period semi-finished goods, and finished goods sold is crucial.

Raw Material Consumed can be calculated using the formula:

Raw Material Consumed = Opening Stock + Purchases – Closing Stock

Understanding this formula requires knowing where the purchase figures originate. These figures are taken from supplier invoices for stockable materials and should also account for all incidental charges related to the purchase, such as carriage, insurance, customs duties, etc., that are directly associated with incoming materials.

Example: Given Information:

- ❑ Raw Material Consumed = ₹30,000
- ❑ Opening Stock = ₹2,500
- ❑ Closing Stock = ₹5,000

Calculation:

We know the formula for Purchases:

Raw Material Consumed = Opening Stock + Purchases – Closing Stock

Rearranging the formula to find Purchases:

Purchases = Raw Material Consumed + Closing Stock – Opening Stock

$$\text{Purchases} = ₹30,000 + ₹5,000 - ₹2,500 = ₹32,500$$

2. ABNORMAL COSTS

Any abnormal cost that is material and quantifiable should not be included in the cost of production, acquisition, or service provision. Examples of abnormal costs include:

- ❑ Costs related to a pandemic, such as COVID-19.
- ❑ Expenses linked to employees during a sudden lockdown.

3. SUBSIDY/GRANT/INCENTIVES

Payments received or expected in the form of subsidies, grants, or incentives should be deducted from the corresponding cost objects.

4. PENALTIES, FINES, DAMAGES, AND DEMURRAGE

These types of expenses are excluded from the overall cost.

5. INTEREST AND OTHER FINANCE COSTS

Interests, including any payments that resemble interest for non-equity funds, along with incidental costs incurred in securing those funds, are not considered part of the production costs. Interest and finance charges should be itemized separately in the cost statement as part of the cost of sales.

6. MATERIAL SCRAP OR WASTAGE

Scrap materials are those that are deemed unusable for their original purpose. When scrap is in its raw form, the cost of materials used should be reduced by the amount obtained from the sale of that scrap.

If the scrap is generated during the manufacturing process, the proceeds from its sale should be deducted from either the works overhead or the total works cost.

7. NON-COST ITEMS

Non-cost items are those expenses which are excluded from the computation of cost. E.g. – Donation, income tax, debenture interest, transfer to reserve, provisions, preliminary expense, discount on issue of shares, dividend, charity, cash discount, loss on sale of an asset, goodwill written-off, etc.

**SELF-ASSESSMENT QUESTIONS**

5. Given the following information, calculate the amount of purchases:

☐ Raw Material Consumed = ₹45,000

☐ Opening Stock = ₹3,000

☐ Closing Stock = ₹7,000

a. 48,000 **b.** 10,000 **c.** 41,000 **d.** 38,000

6. What is the primary purpose of a Cost Sheet?

a. To calculate the market price of finished goods

b. To provide detailed cost information for production and cost control

c. To determine the company's tax liabilities

d. To summarize sales revenue and profit margins

4.4 VALUATION OF CLOSING STOCK

Stocks are always valued at either cost or market price, whichever is lower. This principle should be applied to all items of material in stock, leading to a total valuation of stock. The stock ledger maintains records of

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all receipts and issues, detailing the quantity and rate of material items. Proper valuation of material issues must be based on an appropriately selected method of issue pricing.

4.5 CALCULATION OF COST AND SELLING PRICE

The calculation of cost and selling price using a cost sheet starts with determining the prime cost, which includes direct material, direct labor, and direct expenses. Adding factory overheads to this gives the factory cost, followed by adding administrative overheads to determine the cost of production. After adjusting for the opening and closing stock of finished goods, the cost of goods sold (COGS) is obtained. By adding selling and distribution overheads, the total cost is calculated. Finally, the selling price is determined by adding a desired profit margin to the total cost.

Example Calculation:

Let's say we have the following data:

- | | |
|--|--|
| ☐ Direct Material: ₹50,000 | ☐ Opening Stock of Finished Goods: ₹5,000 |
| ☐ Direct Labor: ₹30,000 | ☐ Closing Stock of Finished Goods: ₹7,000 |
| ☐ Direct Expenses: ₹10,000 | ☐ Selling & Distribution Overheads: ₹12,000 |
| ☐ Factory Overheads: ₹20,000 | ☐ Desired Profit: ₹20,000 |
| ☐ Administrative Overheads: ₹15,000 | |

Using the above formulas:

1. Prime Cost = ₹50,000 + ₹30,000 + ₹10,000 = ₹90,000
2. Factory Cost = ₹90,000 + ₹20,000 = ₹110,000
3. Cost of Production = ₹110,000 + ₹15,000 = ₹125,000
4. COGS = ₹125,000 + ₹5,000 - ₹7,000 = ₹123,000
5. Total Cost = ₹123,000 + ₹12,000 = ₹135,000
6. Selling Price = ₹135,000 + ₹20,000 = ₹155,000

So, the Selling Price is ₹155,000.

4.6 PREPARATION OF STATEMENT OF COST AND ABRIDGED PROFIT AND LOSS ACCOUNT

The Statement of Cost and the Abridged Profit and Loss Account are essential documents used for evaluating the financial performance of a business. Below is a guide to preparing both:

Preparation of the Statement of Cost:

A Statement of Cost provides a breakdown of all expenditures associated with producing goods or services. A specimen of cost sheet is given below in figure 4.11:

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Particulars	Total Cost (Rs.)	Cost Per Unit (Rs.)
1. Direct Material Consumed:		
- Opening Stock of Raw Material	xxx	
- Add: Purchases	xxx	
- Less: Closing Stock of Materials	xxx	
	xxx	
2. Direct Labour Cost	xxx	
3. Direct Expenses	xxx	
4. Prime Cost (1+2+3)	xxx	
5. Factory Overheads	xxx	
6. Gross Work Costs (4+5)	xxx	
7. Add: Opening Work in Progress	xxx	
8. Less: Closing Work in Progress	(xxx)	
9. Factory Cost (6+7+8)	xxx	
10. Quality Control Cost	xxx	
11. Research and Development Cost	xxx	
12. Administrative Overhead	xxx	
13. Less: Credit for Recoveries/Scrap	(xxx)	
14. Add: Packing Cost (Primary)	xxx	
15. Cost of Production (9+10+11+12+13+14)	xxx	
16. Add: Opening Cost of Finished Goods	xxx	
17. Less: Closing Stock of Finished Goods	(xxx)	
18. Cost of Goods Sold (15+16+17)	xxx	
19. Add: Administrative Overheads	xxx	
20. Add: Selling and Distribution Overheads	xxx	
21. Cost of Sales (18+19+20)	xxx	

Figure 4.11 Specimen of Cost Sheet

The costs classified based on functions are organized into several cost heads in a cost sheet, including:

- | | |
|-------------------------|--------------------------|
| (i) Prime Cost | (iii) Cost of Goods Sold |
| (ii) Cost of Production | (iv) Cost of Sales |

Prime Cost

Prime cost encompasses the total of direct material costs, direct employee (labor) costs, and direct expenses shown in Table 4.3. Each cost component must be calculated separately as shown in Table 4.4.

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Table 4.3 Prime Cost

Cost Component	Amount (xxx)
Direct Material Cost	XXX
Direct Employee (Labor) Cost	XXX
Direct Expenses	XXX
Total Prime Cost	XXXX

Direct Material Cost Calculation

Table 4.4 Direct Material

Calculation	Amount (xxx)
Opening Stock of Material	XXX
Add: Additions/Purchases	XXX
Less: Closing Stock of Material	(XXX)
Direct Materials Consumed	XXXX

Items to Include in Raw Material Cost

- (a) Freight inwards
- (b) Insurance and other procurement-related expenses
- (c) Trade discounts or rebates (to be deducted)
- (d) Duties & Taxes (if input tax credit is not available)

Direct employee (labor) cost includes payments made to employees engaged in production:

- (a) Wages and Salary
- (b) Allowances and Incentives
- (c) Payment for Overtime
- (d) Bonus/Ex-Gratia
- (e) Employer's Contributions to Welfare Funds

Direct expenses incurred in manufacturing a product or providing a service include:

- (a) Utilities (power, fuel)
- (b) Royalty paid for production
- (c) Hire charges for specific equipment
- (d) Fees for technical assistance
- (e) Amortized costs of molds, patterns, patents
- (f) Product-specific design or drawing
- (g) Software specific to a product/service
- (h) Other directly related expenses

Example: Calculate the Prime Cost for a bakery business based on the following details:

Particulars	Rs.
Cost of ingredients purchased	50,000
Opening stock of ingredients	8,000
Closing stock of ingredients	6,000
Wages paid to bakers	12,000
Rent for using a specialized oven	7,000

Solution:

Statement Showing Prime Cost

Details	Amount (Rs.)
Direct Material: Ingredients Consumed	
Opening stock of ingredients	8,000
Add: Ingredients purchased	50,000
Material available for consumption	58,000
Less: Closing stock of ingredients	6,000
Material consumed	52,000
Direct Labour: Wages Paid	12,000
Direct Expenses: Rent for specialized oven	7,000
Prime Cost	71,000

Factory Cost

In addition to the prime cost, it incorporates factory or works overheads. These overheads include the expenses for indirect materials, indirect labor, and other indirect costs incurred within the factory. Factory cost is alternatively referred to as works cost, production cost, or manufacturing cost.

Factory Cost = Prime Cost + Factory Overheads

Example: Calculate the Factory Cost for a garment production unit based on the following particulars:

Particulars	Rs.
Fabric consumed	80,000
Productive wages	25,000
Direct expenses (e.g., design fee)	10,000
Consumable supplies (e.g., sewing thread)	3,000
Maintenance supplies (e.g., machine oil)	800
Salary of the production supervisor	7,000
Unproductive wages	2,000
Factory rent	4,000
Repair and depreciation on sewing machines	1,200

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Solution:**Statement Showing Factory Cost**

Details	Amount (Rs.)
Direct Material: Fabric Consumed	80,000
Direct Labour: Productive Wages	25,000
Direct Expenses: Design Fee	10,000
Prime Cost	115,000
Add: Factory Overheads	
Indirect Material:	
Consumable supplies	3,000
Maintenance supplies	800
Total Indirect Material	3,800
Indirect Labour:	
Unproductive wages	2,000
Salary of the production supervisor	7,000
Total Indirect Labour	9,000
Indirect Expenses:	
Factory rent	4,000
Repair and depreciation on sewing machines	1,200
Total Indirect Expenses	5,200
Factory Cost	1,33,000

Cost of Production

In a conventional cost sheet, the total of prime costs and factory-related costs and overheads are calculated as shown in table 4.5:

Table 4.5 Cost of production

Cost Component	Amount (xxx)
Prime Cost	XXX
Add: Factory Overheads	XXX
Gross Works Costs	XXXX
Add: Opening Stock of Work-in-Process	XXX
Less: Closing Stock of Work-in-Process	(XXX)
Factory or Works Costs	XXXX
Add: Quality Control Costs	XXX
Add: R&D Costs (Process-related)	XXX
Add: Administrative Overheads (related to production)	XXX
Less: Credit for Recoveries (miscellaneous income)	(XXX)
Add: Packing Cost (Primary)	XXX
Cost of Production	XXXX

Example: From the following information, calculate the Total Cost of Production for a toy manufacturing unit:

Particulars	Rs.
Direct Material	1,10,000
Direct Labour	40,000
Direct Expenses	12,000
Factory Overheads	30,000
Office and Administration Overheads	22,000

Solution:

Statement Showing Total Cost of Production

Details	Amount (Rs.)
Direct Material	1,10,000
Direct Labour	40,000
Direct Expenses	12,000
PRIME COST	1,62,000
Add: Factory Overheads	30,000
FACTORY COST	1,92,000
Add: Office and Administration Overheads	22,000
TOTAL COST OF PRODUCTION	2,14,000

Cost of Goods Sold

This represents the cost of production for goods sold, calculated by adjusting the opening and closing stock values of finished goods are shown in Table 4.6:

Table 4.6 Cost of Goods Sold

Calculation	Amount (xxx)
Cost of Production	XXX
Add: Cost of Opening Stock of Finished Goods	XXX
Less: Cost of Closing Stock of Finished Goods	(XXX)
Cost of Goods Sold	XXXX

Example 1: From the following information, calculate the Cost of Goods Sold for a bakery business:

Particulars	Rs.
Total Cost of Production	2,40,000
Opening Stock of Finished Goods	18,000
Closing Stock of Finished Goods	25,000

Solution:

Cost of Goods Sold = Total Cost of Production + Opening Stock of Finished Goods – Closing Stock of Finished Goods

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Cost of Goods Sold = ₹2,40,000 + ₹18,000 – ₹25,000

Cost of Goods Sold = ₹2,33,000

Administrative Overheads

These costs pertain to the general administration of the entity and include:

- (a) Depreciation and maintenance of buildings
- (b) Salaries of administrative staff
- (c) Rent, insurance, office expenses, etc.
- (d) Indirect materials like printing and stationery
- (e) Legal charges and audit fees

Selling Overheads

These costs relate to the sale of products/services and include:

- (a) Salaries for sales department employees
- (b) Rent and maintenance for the sales department
- (c) Advertising costs and market research

Packing Cost (Secondary)

This refers to packing materials that aid in storing, transporting, and promoting the product to make it marketable.

Distribution Overheads

These are costs associated with delivering goods to customers, including:

- (a) Salaries of distribution employees
- (b) Transportation and insurance costs
- (c) Depreciation and maintenance for distribution vehicles

Example: From the following information, calculate the Total Cost for a furniture production unit:

Particulars	Rs.
Direct Material	1,20,000
Direct Labour	48,000
Direct Expenses	15,000
Factory Overheads	50,000
Office and Administration Overheads	25,000
Selling and Distribution Overheads	30,000

Solution:

Statement Showing Total Cost

Details	Amount (Rs.)
Direct Material	1,20,000
Direct Labour	48,000

Details	Amount (Rs.)
Direct Expenses	15,000
PRIME COST	1,83,000
Add: Factory Overheads	50,000
FACTORY COST	2,33,000
Add: Office and Administration Overheads	25,000
TOTAL COST OF PRODUCTION	2,58,000
Add: Selling and Distribution Overheads	30,000
TOTAL COST (Cost of Sales)	2,88,000

Illustrative Format of Statement of Cost in Table 4.7:**Table 4.7** Statement of Cost

Particulars	Amount (₹)
Direct Materials	XX
Direct Labor	XX
Prime Cost	XX
Factory Overheads	XX
Cost of Production	XX
Add: Opening Stock of Finished Goods	XX
Less: Closing Stock of Finished Goods	(XX)
Cost of Goods Sold (COGS)	XX
Administrative Overheads	
Selling and Distribution Overheads	XX
Total Cost	XX
Profit Margin	XX
Selling Price	XX

Sales

When a profit margin is added to the total cost, the sales figure is determined. The amount by which sales exceed the total cost is referred to as profit. Conversely, if the total cost surpasses sales, it results in a loss.

Sales = Total Cost + Profit

Calculation of Profit

In some cases, profit is calculated as a percentage of either cost or sales based on the provided information. When this occurs, the base amount (cost or sales) is assumed to be 100, and the percentage is applied accordingly. Profit can then be calculated as follows:

Case 1:

If the cost is ₹12,000 and the profit on cost is 15%, assume the cost is ₹100, with a profit of ₹15. The profit for a cost of ₹12,000 is:

$$12,000 \times \frac{15}{100} = ₹1,800$$

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Thus, the sales value is:

$$₹12,000 + ₹1,800 = ₹13,800$$

Case 2:

If the cost is ₹14,500 and the profit on the sales price is 12%, assume the sales price is ₹100. The cost price is ₹88 [100 – 12]. For a cost of ₹88, the profit is ₹12. The profit for a cost of ₹14,500 is:

$$14,500 \times \frac{12}{88} = ₹1,977.27$$

The sales value is:

$$₹14,500 + ₹1,977.27 = ₹16,477.27$$

Case 3:

If the sales price is ₹15,600 and the profit on cost is 20%, assume the cost price is ₹100 and the sales price is ₹120 [100 + 20]. For a sales price of ₹120, the profit is ₹20. The profit for a sales price of ₹15,600 is:

$$15,600 \times \frac{20}{120} = ₹2,600$$

Thus, the cost price is:

$$₹15,600 - ₹2,600 = ₹13,000$$

Example: From the following information, calculate the Sales Value of goods sold for a textile unit:

Particulars	Rs.
Total Cost of Production	2,10,000
Opening Stock of Finished Goods	30,000
Closing Stock of Finished Goods	15,000
Selling and Distribution Overheads	40,000
Profit	35,000

Solution:

Statement Showing Sales

Details	Amount (Rs.)
Total Cost of Production	2,10,000
Add: Opening Stock of Finished Goods	30,000
Goods Available for Sale	2,40,000
Less: Closing Stock of Finished Goods	(15,000)
Cost of Goods Sold	2,25,000
Add: Selling and Distribution Overheads	40,000
Total Cost	2,65,000
Add: Profit	35,000
Sales	3,00,000

Preparation of Abridged Profit and Loss Account:

The Abridged Profit and Loss Account is a concise version of a typical profit and loss statement, presenting a summary of the company's profitability for a given period. It focuses on the following components:

- ❑ Sales/Revenue: The total revenue earned from the sale of goods or services.
- ❑ Cost of Goods Sold (COGS): The cost associated with producing or acquiring the sold goods.
- ❑ Gross Profit: Calculated by subtracting COGS from sales revenue.
- ❑ Operating Expenses: Includes all indirect expenses like administrative, selling, and distribution costs.
- ❑ Operating Profit: The result of subtracting operating expenses from gross profit.
- ❑ Other Income/Expenses: Includes any non-operating income or expenses.
- ❑ Net Profit: This is the operating profit plus other income, minus non-operating expenses.

Illustrative Format of Abridged Profit and Loss Account in Table 4.8:**Table 4.8** Abridged Profit and Loss Account

Particulars	Amount (₹)
Sales/Revenue	XX
Less: Cost of Goods Sold	(XX)
Gross Profit	XX
Less: Operating Expenses	(XX)
Operating Profit	XX
Add: Other Income	XX
Less: Non-operating Expenses	(XX)
Net Profit	XX

Activity

Cost Sheet for January 2024

Particulars	Amount (₹)
Opening stock of raw material	25,000
Add: Purchases	1,20,000
Less: Closing stock of raw material	
Raw material consumed	1,25,000
Add: Direct wages	50,000
Add: Other direct expenses	5,000
(a) Prime Cost	
Add: Factory overhead:	

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Particulars	Amount (₹)
Indirect material	3,000
Indirect labour	
Indirect expenses	7,000
(b) Factory or Works Cost	2,00,000
Add: Office or administration overheads	15,000
(c) Cost of Production	
Add: Opening stock of finished goods	12,000
Less: Closing stock of finished goods	
Cost of goods sold	
Add: Selling and distribution overheads	10,000
(d) Total Cost	2,19,000
Sale of finished goods	2,50,000
Profit for the month	



SELF-ASSESSMENT QUESTIONS

7. Radha Ltd. has the following information for the month of November 2024:

- ☐ Opening Stock of Finished Goods = ₹40,000
- ☐ Closing Stock of Finished Goods = ₹25,000
- ☐ Cost of Goods Sold = ₹1,50,000

Calculate the Cost of Production.

- a. 1,35,000 b. 1,65,000 c. 15,000 d. 2,15,000

8. Calculate per unit factory cost for an engineering company which produces standard components in batches of 1000 pieces each. A batch passes through three processes viz. Casting, Finishing, and Packaging.

The materials used for a batch number 001 were: Foundry 1200 tonnes @ Rs.60 per tonne of which 40 tonnes were sent back to stores.

Process	Direct Labour	Overheads
Casting	250 Hrs @ ₹12	₹18 per Labour Hour
Finishing	150 Hrs @ ₹8	₹22 per Labour Hour
Packaging	150 Hrs @ ₹20	₹12 per Labour Hour

- a. 86.4 b. 84.6 c. 85.7 d. 87.5

4.7 CALCULATION OF WORK-IN-PROGRESS FIGURE BASED ON COST SHEET

Work-in-progress refers to goods that are partially completed and still undergoing the manufacturing process. The valuation of work-in-progress can be based on either Prime Cost or Works Cost. When work-in-progress is valued at Prime Cost, the total cost includes the expenses for direct materials consumed, direct wages, and any other direct costs incurred. Adjustments for work-in-progress valued at Prime Cost should be made during the calculation of the Prime Cost.

If work-in-progress is valued at Works Cost, the valuation comprises the costs of direct materials, direct wages, direct expenses, and a proportionate share of the total works overhead. Typically, work-in-progress is valued using the Works Cost method. If no specific basis for the valuation of work-in-progress is provided, it should be assumed that the valuation is based on Works Cost. Adjustments for work-in-progress valued at Works Cost should be made when determining the total Works Cost.

Example 1: From the following information, calculate the Works Cost for a furniture manufacturing unit:

Particulars	Rs.
Direct Material	90,000
Direct Labour	28,000
Direct Expenses (e.g., design consultation)	7,000
Factory Overheads	15,000
Work-in-progress: Opening Stock	10,000
Work-in-progress: Closing Stock	8,000

Solution:

Statement Showing Works Cost

Details	Amount (Rs.)
Direct Material	90,000
Direct Labour	28,000
Direct Expenses	7,000
Prime Cost	1,25,000
Add: Factory Overheads	15,000
Factory Cost (Gross)	1,40,000
Add: Opening Stock of WIP	10,000
Less: Closing Stock of WIP	(8,000)
Works or Factory Cost (Net)	1,42,000

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SELF-ASSESSMENT QUESTIONS

9. What does the valuation of work-in-progress (WIP) include when it is based on Prime Cost?

- a. Direct materials, direct wages, direct expenses, and total works overhead
- b. Direct materials, direct wages, and any other direct costs incurred
- c. Only direct materials and direct wages
- d. Direct materials and a proportionate share of total works overhead

Correct Answer: B) Direct materials, direct wages, and any other direct costs incurred

10. If no specific basis for the valuation of work-in-progress is provided, which method should be assumed for its valuation?

- a. Cost of Goods Sold
- b. Prime Cost
- c. Works Cost
- d. Marginal Cost

Correct Answer: C) Works Cost

4.8 ADDITIONAL SOLVED PROBLEMS

Ques. 1: Calculate the Prime Cost, Works Cost, Cost of Production, Total Cost, and Profit from the following details:

- | | |
|---|---|
| ☐ Direct Materials = ₹25,000 | ☐ Administration Expenses = ₹6,000 |
| ☐ Direct Labour = ₹15,000 | ☐ Selling Expenses = ₹9,000 |
| ☐ Factory Expenses = ₹8,000 | ☐ Sales = ₹75,000 |

Solution**1. Prime Cost**

Prime Cost = Direct Materials + Direct Labour

Prime Cost = ₹25,000 + ₹15,000 = ₹40,000

2. Works Cost

Works Cost = Prime Cost + Factory Expenses

Works Cost = ₹40,000 + ₹8,000 = ₹48,000

3. Cost of Production

Cost of Production = Works Cost + Administration Expenses

Cost of Production = ₹48,000 + ₹6,000 = ₹54,000

4. Total Cost (Cost of Sales)

Total Cost = Cost of Production + Selling Expenses

Total Cost = ₹54,000 + ₹9,000 = ₹63,000

5. Profit

Profit = Sales – Total Cost

Profit = ₹75,000 – ₹63,000 = ₹12,000

Ques. 2: The following data relates to the manufacture of a marketable product during the first quarter of 2024:

Particulars	(Rs.)
Raw materials	Rs. 2,50,000
Direct wages	Rs. 1,20,000
Machine hours worked (hours)	12,000
Machine hour rate (per hour)	Rs. 10
Administration overheads (general)	Rs. 40,000
Selling overheads (per unit)	Rs. 6
Units produced	5,000
Units sold	4,500
Selling price per unit	Rs. 170

You are required to PREPARE a cost sheet in respect of the above showing:

- (i) Cost per unit
- (ii) Profit for the month

Sol.

COST SHEET

Particulars	Total Cost(Rs.)	Cost per(unit)(Rs.)
Raw Materials	2,50,000	50
Direct Wages	1,20,000	24
Total Prime Cost	3,70,000	74
Machine Overheads (12,000 hrs @ Rs.10)	1,20,000	24
Total Cost of Production	4,90,000	98
Less: Closing Stock (5000-4500, i.e. 500@98)	49,000	
Total Cost of Goods Sold	4,41,000	98
Add: Administration Overheads	40,000	8.89
Add: Selling Overheads (4,500 units @ Rs.6)	27,000	6
Total Cost of Sales	5,08,000	112.89

N O T E S

(i) Statement of Profit

Particulars	Total Cost (Rs.)
Sales revenue (4,500 units @ Rs. 170)	7,65,000
Less: Cost of sales	5,08,000
Profit	2,57,000

Ques. 3: The following information has been obtained from the records of XYZ Industries for the period from May 1 to May 31.

	On May 1 (Rs.)	On May 31 (Rs.)
Cost of raw materials	80,000	70,000
Cost of work-in-process	15,000	20,000
Cost of stock of finished goods	1,20,000	1,50,000
Purchase of raw materials during July		5,00,000
Wages paid		3,00,000
Factory overheads		1,20,000
Administration overheads (related to production)		60,000
Selling & distribution overheads		30,000
Sales		12,00,000

PREPARE a statement giving the following information:

- ☐ Raw materials consumed
- ☐ Prime cost
- ☐ Factory cost
- ☐ Cost of goods sold
- ☐ Net profit

Sol.

Particulars	(Rs.)
Opening stock of raw materials	80,000
Add: Purchase of raw materials during the month of July	5,00,000
Less: Closing stock of raw materials	(70,000)
(a) Raw materials consumed	5,10,000
Add: Direct wages	3,00,000
(b) Prime cost	8,10,000
Add: Factory overheads	1,20,000
Works cost	9,30,000
Add: Opening work-in-process	15,000
Less: Closing work-in-process	(20,000)

Particulars	(Rs.)
(c) Factory cost	9,25,000
Add: Administration overheads (related to production)	60,000
Cost of production	9,85,000
Add: Opening stock of finished goods	1,20,000
Less: Closing stock of finished goods	(1,50,000)
(d) Cost of goods sold	9,55,000
Add: Selling & distribution overheads	30,000
Cost of sales	9,85,000
(e) Net Profit(b.f.)	2,15,000
Sales	12,00,000

Ques. 4: Following data is available from the cost records of a company for the month of April 2023:

(1) Opening stock of job as on 1st April 2023:

- ❑ **Job no. B 101:** Direct Material 100, Direct Wages 200, and Factory Overheads Rs.250
- ❑ **Job no. B 102:** Direct Material 500, Direct Wages 600, and Factory Overheads Rs.450

(2) Direct material issued during the month of April 2023 was:

- ❑ **Job no. B 101:** Rs.150
- ❑ **Job no. B 102:** Rs.350
- ❑ **Job no. B 103:** Rs.200
- ❑ **Job no. B 104:** Rs.400

(3) Direct labor details for April 2023 were:

Job no	Hours	Amount (₹)
B 101	500	800
B 102	300	600
B 103	400	900
B 104	150	300

- (4) Factory Overheads are applied to jobs on production according to direct labor hour rate which is Rs.3 per hour.
- (5) Factory Overhead incurred in April 2023 were Rs.2,700.
- (6) Job numbers B 101 & B 102 were completed during the month. They were billed to the customers at a price which included 12% of the price of the job for Selling & Distribution expenses and another 8% of the price for Profit.

Prepare:

- (a) Job cost sheet for job number B 101 and B 102.

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- (b) Determine the selling price for the jobs.
 (c) Calculate the value of work in process.

Sol.

Remarks:

- (1) The Factory Overheads actually incurred are 2,700. This amount will be apportioned based on labor hours. Thus, the rate to be considered is 2 per hour ($2,700 / 1,350$ hours), which differs from the earlier assumed rate of Rs. 3 per hour. The calculations for Job Sheets and the work in progress will change accordingly.
- (2) Work in progress is to be calculated for the incomplete jobs; hence Job no. B 103 and B 104 should only be included in the calculations of work in progress.

Job Cost Sheets for the month of April 2023

Cost Items	Job B 101	Job B 102
Direct Material issued	150	350
Direct Labor spent	800	600
Prime Cost	950	950
Factory Overheads @ Rs. 2 per hour	$(500 \times 2) = 1,000$	$(300 \times 2) = 600$
Add: Opening WIP (Material + Labor + Overheads)	$(100 + 200 + 250) = 550$	$(500 + 600 + 450) = 1,550$
Factory Cost	2,500	3,100
Add: Selling & Distribution Overheads (Note 1)	375 (12/80 of 2,500)	465 (12/80 of 3,100)
Cost of Sales	2,875	3,565
Profit (Note 1)	250 (8/80 of 2,500)	310 (8/80 of 3,100)
Billing price for the job	3,125	3,875

Note 1: Selling & Distribution (S&D) and profit are calculated indirectly.

Item	Calculation	Amount (₹)
Assuming Selling Price		100
Less: Selling & Distribution (S&D) @ 12%	$100 \times 12\%$	(12)
Less: Profit @ 8%	$100 \times 8\%$	(8)
Balance (Factory Cost)	$100 - 12 - 8$	80
S&D Price	$12 / 80$	
Profit	$8 / 80$	

Computation of Work in Process for April 2023

Items	Amount (₹)	(₹)
Opening balance as on 1st April		
Job B 101	200	

N O T E S

Items	Amount (₹)	(₹)
Job B 102	300	500
Material issued during the month		
Job B 101	150	
Job B 102	350	
Job B 103	200	
Job B 104	400	1,100
Direct Labor		
Job B 101	800	
Job B 102	600	
Job B 103	900	
Job B 104	300	2,600
Factory Overheads on 1,350 hours @ Rs. 2		2,700
Factory Cost		6,900
Less: Factory Cost of completed jobs		
Job B 101	1,000	
Job B 102	600	1,600
Closing Work in Process as on 30th April 2023		5,300

Ques. 5: From the following particulars, calculate:

- | | |
|------------------------|-------------------|
| (a) Prime Cost | (b) Works Cost |
| (c) Cost of Production | (d) Cost of Sales |
| (e) Profit | (f) Cost per unit |

Sundar Textiles manufactures Product X. The information available for the year ended 31st December 2023 is as follows:

Particulars	Rs.
Opening Stock of Raw Materials (1.1.23)	1,20,000
Direct Materials Purchased	9,50,000
Carriage Inwards	50,000
Closing Stock of Raw Materials	1,00,000
Direct Wages	4,00,000
Factory Overheads	3,00,000
Opening Work-in-Progress (1.1.23)	50,000
Closing Work-in-Progress (31.12.23)	80,000
Office and Administrative Overheads	90,000
Opening Stock of Finished Goods (1.1.23)	60,000
Closing Stock of Finished Goods (31.12.23)	70,000
Selling and Distribution Overheads	1,10,000
Sales (1,50,000 units)	30,00,000

N O T E S

Solution:**Cost Sheet for the Year Ending 31st December 2023**

Particulars	Amount (Rs.)	Total (Rs.)
Opening Stock of Raw Materials	1,20,000	
Add: Direct Materials Purchased	9,50,000	
Add: Carriage Inwards	50,000	
Less: Closing Stock of Raw Materials	(1,00,000)	
Raw Material Consumed		10,20,000
Add: Direct Wages	4,00,000	
PRIME COST (a)		14,20,000
Add: Factory Overheads	3,00,000	
Add: Opening Work-in-Progress	50,000	
Less: Closing Work-in-Progress	(80,000)	
WORKS COST (b)	2,70,000	
Add: Office and Administrative Overheads	90,000	
COST OF PRODUCTION (c)		17,80,000
Add: Opening Stock of Finished Goods	60,000	
Less: Closing Stock of Finished Goods	(70,000)	
COST OF GOODS SOLD (d)		17,70,000
Add: Selling and Distribution Overheads	1,10,000	
COST OF SALES		18,80,000
Profit (b.f.)		11,20,000
Sales		30,00,000

Cost Per Unit (f) = Total Cost ÷ Total Units Sold

= ₹17,80,000 ÷ 1,50,000 = ₹11.87 per unit

Ques. 6: The following information is provided to prepare the Cost Sheet for the period ended on 31st December 2023:

- | | |
|---|---|
| <ul style="list-style-type: none"> ❑ Consumable material: <ul style="list-style-type: none"> ◆ Opening stock: ₹15,000 ◆ Purchases: ₹1,00,000 ◆ Closing stock: ₹12,000 ❑ Direct wages: ₹50,000 ❑ Direct expenses: ₹30,000 ❑ Factory overheads: 40% of direct wages ❑ Office and administration overheads: 15% of works cost | <ul style="list-style-type: none"> ❑ Selling and distribution expenses: ₹4 per unit sold ❑ Units of finished goods: <ul style="list-style-type: none"> ◆ In hand at the beginning of the period (Value ₹10,000): 400 units ◆ Units produced during the period: 10,000 units ◆ In hand at the end of the period: 1,000 units |
|---|---|

Required:

1. Prepare a Cost Sheet for the period.
2. Find the selling price per unit if a 25% profit on selling price is desired. Assume there is no work-in-progress either at the beginning or at the end of the period.

Solution:**Cost Sheet for the period ended on 31st December 2023 (output 10,000 units)**

Particulars	Total Cost (₹)	Cost per Unit (₹)
Material Consumed:		
Opening Stock	15,000	
Add: Purchases	1,00,000	
Total (Material Available)	1,15,000	
Less: Closing Stock	12,000	
Cost of Raw Material Consumed	1,03,000	10.30
Direct Wages	50,000	5.00
Direct Expenses	30,000	3.00
Prime Cost	1,83,000	18.30
Factory Overheads (40% of Direct Wages)	20,000	2.00
Works Cost (Prime + Factory Overheads)	2,03,000	20.30
Office & Administration Overheads (15% of work cost)	30,450	3.05
Total Cost of Production	2,33,450	23.35
Add: Opening Stock of Finished Goods	10,000	
Cost of Goods Available for Sale (10,000 + 400 units)	2,43,450	
Less: Closing Stock of Finished Goods @ 23.41	23,410	
Cost of Goods Sold	2,20,040	22.00
Add: Selling & Distribution Expenses	40,000	4.00
Cost of Sales	2,60,040	26.00
Add: Profit (25% of Selling Price)	86,680	8.67
Sales (Total Cost + Profit)	3,46,720	34.67

Ques. 7: The following information pertains to the manufacturing of a product during the month of June 2024:

- ☐ Raw materials consumed: ₹25,000
- ☐ Direct wages: ₹15,000
- ☐ Machine hours worked: 1,500 hours

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- ❑ Machine hour rate: ₹3 per hour
- ❑ Office overhead: 25% on works cost
- ❑ Selling overhead: ₹0.50 per unit
- ❑ Units produced: 25,000 units
- ❑ Units sold: 22,000 units at ₹4 each

Required: Prepare a Cost Sheet and calculate:

- | | |
|------------------------|------------------------|
| (a) Prime Cost | (b) Works Cost |
| (c) Cost of Production | (d) Cost of Goods Sold |
| (e) Cost of Sales | (f) Profit |

Solution:

Cost Sheet for June 2024

Particulars	Amount (₹)
Raw Materials Consumed	25,000
Direct Wages	15,000
Prime Cost (a)	40,000
Add: Factory Overhead (1,500 × ₹3)	4,500
Works Cost (b)	44,500
Add: Office Overhead (25% of Works Cost)	11,125
Cost of Production or Cost of Goods Sold (c)	55,625
Add: Selling Overhead (22,000 × ₹0.50)	11,000
Cost of Sales (e)	66,625
Add: Profit(b.f.)	21,375
Sales Revenue (22,000 × ₹4)	88,000

Ques. 8: XYZ Ltd. manufactures and sells a line of sports water bottles under its own brand. The plant's installed capacity is 1,50,000 units per year, evenly distributed across the calendar year.

The Cost Accountant has provided the following cost structure:

- | | |
|--------------------------------------|---|
| ❑ Raw Material: ₹30 per unit | ❑ Semi-variable Overheads: |
| ❑ Direct Labour: ₹15 per unit | ◆ ₹10,000 per month up to 50% capacity. |
| ❑ Direct Expenses: ₹5 per unit | ◆ Additional ₹5,000 per month for every extra 25% capacity utilization or part thereof. |
| ❑ Variable Overheads: ₹20 per unit | |
| ❑ Fixed Overhead: ₹4,00,000 per year | |

The plant operated at 50% capacity for the first eight months, producing 50,000 units. For the remaining four months, the plant operated at 75% capacity, producing 37,500 units.

The selling price for the first eight months was ₹85 per unit. XYZ Ltd. aims to achieve an annual profit target of ₹10,00,000 and seeks to revise the selling price for the remaining months to meet this goal.

You are required to:

1. Suggest the selling price per unit for the period from 1st September to 31st December.
2. Prepare a cost sheet showing the total cost, per unit cost, and profit for:
 - ❑ 1st January to 31st August (50% capacity, 50,000 units).
 - ❑ 1st September to 31st December (75% capacity, 37,500 units).

Cost Sheet for the Period

Particulars	50% Capacity Utilization (50,000 units)	75% Capacity Utilization (37,500 units)
Raw Material (₹30/unit)	₹15,00,000	₹11,25,000
Direct Labour (₹15/unit)	₹7,50,000	₹5,62,500
Direct Expenses (₹5/unit)	₹2,50,000	₹1,87,500
Variable Overheads (₹20/unit)	₹10,00,000	₹7,50,000
Fixed Overheads (Annual: ₹4,00,000)**		
Allocated per period	₹2,66,667	₹1,33,333
Semi-variable Overheads		
Base (₹10,000/month)	₹80,000 (8 months)	₹40,000 (4 months)
Additional (₹5,000/month for extra capacity)	₹0	₹20,000
Total Semi-variable Overheads	₹80,000	₹60,000
Total Cost	₹38,96,667	₹28,18,333
Cost Per Unit	₹77.93/unit	₹75.16/unit

Profit Calculation

1. For 50% Capacity Utilization (1st Jan to 31st Aug)
 - ❑ Selling Price: ₹85/unit
 - ❑ Total Sales Revenue = 50,000 units × ₹85 = ₹42,50,000
 - ❑ Total Cost = ₹38,96,667

Profit for 50% Capacity Utilization:

Profit = Sales Revenue – Total Cost

Profit = ₹42,50,000 – ₹38,96,667 = ₹3,53,333

2. For 75% Capacity Utilization (1st Sept to 31st Dec)

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Annual Profit Target = ₹10,00,000

Profit already earned = ₹3,53,333

Profit required in the remaining period = ₹10,00,000 – ₹3,53,333 = ₹6,46,667

Profit Per Unit Required:

Profit Per Unit = Total Required Profit/Units Produced

Profit Per Unit = ₹6,46,667/37,500 = ₹17.24

New Selling Price for 75% Capacity:

Selling Price Per Unit = Cost Per Unit+Profit Per Unit

Selling Price Per Unit = ₹75.16+₹17.24 = ₹92.40

4.9 SUMMARY

- ❑ The primary goal of a cost accounting system is to determine the costs associated with specific cost objects, such as products or services, through systematic cost collection and organization into cost sheets.
- ❑ Cost classification involves grouping cost items into logical categories based on characteristics like components (direct vs. indirect), nature (material, labor, expenses), behavior (fixed, variable, semi-variable), and function (production, administration, selling, research).
- ❑ Direct costs, which can be directly allocated to a cost object, include direct material, direct labor, and direct expenses, while indirect costs, known as overhead, consist of indirect materials, labor, and expenses.
- ❑ Costs can also be classified based on controllability (controllable vs. non-controllable), management decisions (opportunity, sunk, differential, joint, common, imputed, uniform, marginal, replacement, out-of-pocket costs), and expiry (capital, revenue, deferred revenue expenditures).
- ❑ Each classification serves a specific purpose in cost analysis and management decision-making, enabling businesses to track expenses effectively and allocate resources efficiently.
- ❑ A cost sheet provides detailed cost information, typically based on functional classification, and is tailored to business type, level of detail required, and cost control needs.
- ❑ It includes direct material, direct labor, direct expenses, factory overheads, administrative overheads, and selling & distribution costs, leading to the calculation of the total cost and selling price.
- ❑ Specific treatments and adjustments are made for raw materials, abnormal costs, subsidies, penalties, finance costs, and scrap materials to ensure accurate cost representation.

- ❑ Work-in-progress is valued either at Prime Cost or Works Cost, incorporating direct material, labor, expenses, and overheads, and adjustments are made during cost calculations.
- ❑ The Abridged Profit and Loss Account includes sales revenue, cost of goods sold, gross profit, operating expenses, and net profit to provide a concise summary of profitability.

4.10 KEY WORDS

1. **Cost Object:** Any item, product, service, or project for which costs are measured and assigned.
2. **Direct Costs:** Expenses that can be directly traced to a specific cost object, including materials and labor.
3. **Indirect Costs:** Costs that cannot be directly traced to a specific cost object and are often allocated, such as utilities and administrative salaries.
4. **Fixed Costs:** Expenses that remain constant regardless of production levels, like rent and salaries.
5. **Variable Costs:** Costs that fluctuate with production volume, such as raw materials and direct labor.
6. **Prime Cost:** The sum of direct materials, direct labor, and direct expenses involved in production.
7. **Factory Overheads:** Indirect costs incurred in manufacturing, including utilities, maintenance, and depreciation.
8. **Work-in-Progress:** Goods that are partially completed but not yet ready for sale.
9. **Selling Price:** The total cost of production plus a desired profit margin.
10. **Cost Sheet:** A detailed statement that summarizes the total cost and cost per unit for a product.

4.11 CLOSING CASELET

XYZ Manufacturing Company, a leader in electronic component production, faces challenges in accurately assessing production costs. This case study emphasizes the importance of managing raw material costs and abnormal expenses to optimize cost management practices. With an annual production volume of 1 million units, precise calculations are vital for maintaining profitability.

The management of raw materials is crucial to XYZ's strategy. The company calculates Raw Material Consumed using the formula:

Raw Material Consumed = Opening Stock + Purchases – Closing Stock

For the fiscal year ending December 31, 2023, the company reported an opening stock of ₹2,00,000, purchases of ₹15,00,000, and a closing stock

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of ₹1,50,000, resulting in Raw Material Consumed of ₹15,50,000. This calculation is validated by the procurement department, ensuring all purchase figures include incidental charges, linking costs directly to units produced.

In 2023, the COVID-19 pandemic resulted in substantial abnormal costs, including employee health and safety expenditures. Management chose not to include these extraordinary costs in production expenses, allowing for a more accurate financial assessment. The company also received a government subsidy of ₹3,00,000, which was deducted from production costs, enhancing profitability.

XYZ Manufacturing strategically excludes non-cost items, such as penalties, donations, and interest expenses, from its cost calculations. For instance, a ₹50,000 penalty for regulatory non-compliance was excluded, emphasizing accurate representation of operational costs. By maintaining a clear distinction between operational expenses and non-cost items, XYZ enhances its financial reporting and positions itself for informed strategic planning and analysis.



QUESTIONS

1. How does the inclusion of incidental charges in raw material costs affect the overall calculation of production costs for XYZ Manufacturing Company?

(Hint: Consider how these charges impact the formula for calculating Raw Material Consumed and the relationship between costs and the units produced.)

2. Why did XYZ Manufacturing Company choose to exclude abnormal costs related to the COVID-19 pandemic from their production costs, and how does this decision affect their financial analysis?

(Hint: Think about the implications of accurately representing production costs and the potential impact on profitability and financial reporting.)

4.12 HIGHER ORDER THINKING SKILLS QUESTIONS (HOTS)

1. How would the classification of costs into direct and indirect categories impact a company's pricing strategy and overall financial decision-making? Provide examples to support your reasoning.
2. In what ways can understanding fixed and variable costs contribute to effective budget forecasting and financial planning for a business facing fluctuating market demands? Discuss the potential consequences of misclassifying these costs.
3. How can the treatment of abnormal costs in a cost sheet affect management's decision-making process and the overall financial performance evaluation?

4. In what ways can the cost sheet be customized to suit different industries, and how does this impact the accuracy of cost control and pricing strategies?
5. How does the method of valuing work-in-progress (Prime Cost vs. Works Cost) influence the final cost of goods sold and the financial reporting of a manufacturing company?
6. Prepare a statement of cost from the following data to show material consumed, Prime cost, factory cost, cost of goods sold, and profit.

Particulars	1-1-2021 (₹)	31-12-2021 (₹)
Raw material	70,000	55,000
Work-in-progress	28,000	35,000
Finished goods	1,40,000	1,20,000
Purchase of materials during the year	10,50,000	
Wages paid	5,50,000	
Factory overheads	2,50,000	
Administration overheads	60,000	
Selling and distribution overheads	40,000	
Sales	22,50,000	

7. A digital marketing agency has received an inquiry for designing a comprehensive campaign for a new product launch. You are required to prepare a quotation for the client.

The bill of materials prepared by the project department for the job specifies the following material requirements:

- ❑ Digital media tools and software licenses: ₹18,000
- ❑ Graphic elements (stock images, templates): ₹7,500
- ❑ Animation software add-ons: ₹5,500

The job also requires a freelance video editor as the agency does not have one on staff. The editor will be paid ₹15,000 for the assignment. The estimated project sheet specifies the services of the following employees:

- ❑ Graphic Designer (₹25,000 per month): 50 hours
- ❑ Content Strategist (₹20,000 per month): 45 hours
- ❑ Project Coordinator (₹18,000 per month): 30 hours

The project will also require additional cloud storage space costing ₹6,000. Production Overheads are estimated at 30% of direct costs, while Sales and Distribution Overheads are likely to be 15% of the Production Cost. The agency expects a profit margin of 20% on the quoted price.

8. The following information has been extracted from the records of Global Manufacturing Ltd. for the period August 1 to August 31:

N O T E S

Particulars	On August 1 (₹)	On August 31 (₹)
Cost of raw materials	80,000	70,000
Cost of work-in-progress (WIP)	20,000	25,000
Cost of stock of finished goods	1,20,000	1,40,000

Other Information for August:

Particulars	Amount (₹)
Purchases of raw materials	6,00,000
Wages paid	3,00,000
Factory overheads	1,50,000
Power and fuel expenses	40,000
Repairs and maintenance	30,000
Administrative overheads (related to production)	70,000
Selling & distribution overheads	35,000
Miscellaneous production expenses	20,000
Sales	12,50,000

Prepare a statement showing the following details:

- (a) Raw materials consumed (b) Prime cost
 (c) Factory cost (d) Cost of goods sold
 (e) Net profit

9. The following data relates to the production of a customized component during the month of July by Rama Industries Ltd.:

Particulars	Amount (₹)
Raw materials purchased	2,50,000
Direct wages	1,20,000
Machine hours worked	12,000
Machine hour rate (per hour)	12
Opening stock of raw materials	40,000
Closing stock of raw materials	30,000
Opening stock of work-in-progress (WIP)	25,000
Closing stock of work-in-progress (WIP)	35,000
General administrative overheads	50,000
Selling overheads (per unit)	10
Opening stock of finished goods (units)	500 units
Units produced during the month	6,000 units
Units sold during the month	5,500 units
Selling price per unit	180

- ❑ Closing stock of finished goods is valued at Factory Cost.

You are required to PREPARE a cost sheet showing the following:

- (i) Cost per unit (ii) Profit for the month

10. A manufacturing company produces a single product. The following information is available for the month of November:

Particulars	Amount (₹)
Raw Materials	
Opening Stock	25,000
Purchases	1,50,000
Closing Stock	(30,000)
Direct Wages	80,000
Factory Expenses	
Factory Rent	20,000
Depreciation on Factory Equipment	10,000
Supervisor's Salary	15,000
Power and Fuel	12,000
Office and Administrative Expenses	
Salary of Office Staff	30,000
Office Rent	5,000
Miscellaneous Office Expenses	3,000
Selling and Distribution Expenses	
Commission to Salesmen	8,000
Advertising	10,000
Distribution Expenses	7,000
Production Data	
Units Produced	10,000
Units Sold	8,000

Calculate the Cost of Goods Manufactured, Total Cost, Cost per Unit, and Profit @ 20% of the cost.

4.13 ANSWER KEYS

Topic	Q.No.	Answer
Cost Classification	1	c) Direct materials that change with production levels
	2	d) 2,700
	3	b) The benefits lost by choosing one alternative over another
	4	b) Costs that can be influenced by its budget holder
Cost sheet	5	c) 41,000

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Topic	Q. No.	Answer
	6	b) To provide detailed cost information for production and cost control
Preparation of Statement of Cost and Abridged Profit and Loss Account	7	a) 1,35,000
	8	a) 86.4
Calculation of Work-in-Progress Figure Based on Cost Sheet	9	b) Direct materials, direct wages, and any other direct costs incurred
	10	c) Works Cost

4.14 SUGGESTED READINGS

- ❑ **Drury, Colin. (2013).***Management and Cost Accounting*. 9th Edition. Cengage Learning.
- ❑ **Shillinglaw, George. (2010).***Accounting: A Management Approach*. 8th Edition. South-Western College Publishing.
- ❑ Article: **“Cost Classification in Management Accounting”** Published by the Institute of Management Accountants.
- ❑ Article: **“How to Create a Cost Sheet”** By Indeed.

4.15 REFERENCES

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5

U N I T

COST ANALYSIS: JOB ORDER, BATCH AND CONTRACT COSTING

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INTRODUCTORY CASELET

Custom Furniture Manufacturing

Artisan Custom Furniture, established in 2010 in Chicago, Illinois, specializes in creating high-quality, bespoke furniture tailored to the specific needs of individual clients. By operating on a job costing method of cost accounting, the company effectively tracks costs for each unique order, allowing for flexibility in production. Unlike continuous production, which is impractical for custom pieces, Artisan Custom Furniture treats each client order as a distinct job, ensuring that it meets the unique specifications of every customer.

The job costing methodology at Artisan Custom Furniture begins with job identification, where each order is assigned a unique job number to track all associated costs. Costs are categorized into direct materials, direct labor, and manufacturing overhead. For instance, a custom dining table's direct materials would include the wood and finishes chosen by the client, while direct labor encompasses the hours worked by craftsmen. Manufacturing overhead is allocated based on predetermined rates, ensuring accurate cost allocation across different jobs.

For example, when a client orders a custom dining table, the total job cost is calculated by summing the costs of direct materials, direct labor, and allocated overhead. This thorough cost calculation allows the company to determine the profitability of each order, enabling them to set transparent pricing for their clients. By applying a markup to the total job cost, Artisan Custom Furniture can establish a selling price that reflects both the costs incurred and the desired profit margin.

The job costing method provides several benefits, including customization and flexibility in meeting unique client requests, accurate costing that leads to informed pricing strategies, and improved financial reporting. By treating each job as a separate cost unit, Artisan can evaluate the performance of individual orders, enhancing decision-making related to resource allocation. This transparent approach fosters trust and satisfaction among clients, resulting in repeat business and referrals, ultimately contributing to the company's success in the bespoke furniture market.

**QUESTIONS**

1. How does Artisan Custom Furniture utilize job costing to manage the costs associated with each client order?

(Hint: Consider the components of job costing, such as direct materials, direct labor, and manufacturing overhead, and how they contribute to the overall cost of a specific job.)

2. What are the advantages of using a job order costing system for a company that specializes in custom products, like Artisan Custom Furniture?

(Hint: Think about how job costing advantages.)

**LEARNING OBJECTIVES**

At the end of the unit, you will be able to,

- Understand the steps involved in the job order costing process.
- Identify and articulate the advantages and limitations of job costing.
- Explain the concept of Economic Batch Quantity (EBQ) and its significance in minimizing production costs while meeting demand.
- Identify the key features of contract costing.
- Describe the principles of cost-plus contracts.
- Explain the concept of work certified in contract costing.

5.1 INTRODUCTION

Costing methods play a crucial role in accurately determining the expenses incurred in production and ensuring the financial success of an organization. Imagine you're running a business where every project is a little different—like a custom furniture shop. You have orders coming in, each unique to a customer's preferences. Here, you'd need to figure out how much each order costs to ensure profitability. The best approach? Job Costing (or Job Order Costing).

This method tracks all expenses—materials, labour, and overhead—for every individual order. Think of a Job Cost Sheet as your go-to ledger for each project. It's where you jot down every nail, plank of wood, and hour of labour dedicated to a specific customer's furniture set. By the end, you have a clear picture of how much the job cost and, more importantly, how much profit you made!

Now, let's think bigger—say you manufacture products in batches, like pharmaceutical tablets or T-shirts. Here, you'd use Batch Costing, treating each batch as a single cost unit. This method is perfect for industries where items are produced in lots, saving time and effort in cost calculation.

On the flip side, imagine you're building skyscrapers or highways—huge projects spanning months or even years. For this, you'd switch to Contract Costing. Each contract, like a skyscraper, becomes a unique cost unit. What's unique here? You often calculate profits even before completing the project, especially for long-term contracts.

Let's discuss:

- ❑ Can you think of businesses around you that might use Job Costing, Batch Costing, or Contract Costing?
- ❑ Which method would you choose for a custom wedding cake bakery? How about a car manufacturing unit?

Let's dive into how these methods differ and why choosing the right one matters!

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5.2 JOB ORDER COSTING

Industries that produce goods or provide services based on specific customer orders, rather than continuous production for stock or general sales, use the job costing or job order costing method of cost accounting. This approach is also referred to as specific order costing, production order costing, job lot costing, or lot costing. In this method, each order is treated as a separate job. It is not necessary for all jobs to use the same materials or follow identical manufacturing processes. However, similar products or orders can be grouped into lots or batches, with each batch treated as a job order. In this system, the job, unit, lot, or batch serves as the cost unit.

Job costing does not average costs, except when calculating the unit cost of a lot within a specific order. Jobs may span multiple accounting periods, so job costs are not tied to a single accounting period.

Types of organizations that use job costing include:

1. **Jobbing concerns:** These businesses produce a range of goods or services based on specific customer requirements. They do not focus on manufacturing a single product for the general market. Each job is unique, involving different specifications, materials, equipment, and labour hours, making job cost accounting essential for tracking costs accurately.
2. **Small firms:** Small manufacturers often produce specialized products in limited quantities. They may find it challenging to use process costing because of low production volumes and frequent product changes. Job costing is more suitable for determining the costs of each batch in these cases.
3. **Large enterprises with diverse products:** Large companies often handle multiple products in a single department without focusing on continuous production lines. When establishing specific process departments is impractical, job costing provides an effective way to allocate costs.

Job Costing Example: A Custom Wedding Cake

To illustrate job costing in an Indian context, consider a bakery specializing in custom wedding cakes. Each cake is unique and tailored to customer requirements, involving specific materials, labor, and overhead costs. Here's how the job costing process would work:

Step 1: Customer's Enquiry

A customer requests a wedding cake for 100 guests, specifying a three-tier design with floral decorations and a mix of chocolate and vanilla layers.

Step 2: Quotation for the Job

The bakery estimates the cost as follows:

- ❑ **Materials:** Ingredients such as flour, sugar, eggs, fondant, food colors, and decorations.

- ❑ **Labour:** Time spent by bakers and decorators to make the cake.
- ❑ **Overheads:** Costs for electricity, equipment, and bakery space.

For example, the estimated costs might be:

- ❑ Materials: ₹6,000 ❑ Labour: ₹3,000 ❑ Overheads: ₹1,000

the total estimated cost is ₹10,000. Adding a 30% profit margin, the final price quoted to the customer is ₹13,000.

Step 3: Customer's Order

If the customer agrees to the quotation, they place the order. A unique Job Order Number is assigned for tracking.

Step 4: Production Order

The bakery's production team receives the order, including customer specifications. A Job Cost Sheet is prepared to record costs incurred during production.

Step 5: Cost Accumulation

The costs are tracked throughout the production process:

1. **Direct Materials:** The bakery records the cost of specific ingredients. Example: Fondant for floral decorations costs ₹1,000, and the chocolate for layers costs ₹2,000.
2. **Direct Labour:** Bakers and decorators track their hours. Example: If the head baker works 6 hours at ₹500 per hour, the labour cost is ₹3,000.
3. **Overheads:** Indirect costs like electricity and oven depreciation are allocated at a predetermined rate. Example: ₹1,000 for overheads.

Step 6: Completion of the Job

After the cake is completed, the Job Cost Sheet summarizes the following:

- ❑ Direct materials: ₹6,000 ❑ Direct labour: ₹3,000
- ❑ Overheads: ₹1,000 ❑ Total actual cost: ₹10,000

Step 7: Profit or Loss on Job

The bakery compares the actual cost (₹10,000) to the price charged to the customer (₹13,000). The profit for this job is ₹3,000.

Why Job Costing Works for Custom Cakes

1. **Unique Orders:** Every wedding cake is customized, requiring separate cost tracking.
2. **Cost Control:** By comparing actual costs with estimates, the bakery can identify inefficiencies, such as reducing waste or optimizing labour.

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- 3. Customer Satisfaction:** Accurate costing ensures the bakery maintains profitability while delivering high-quality products at competitive prices.

5.2.1 PROCESS OF JOB ORDER COSTING

When a customer places an order or the sales department notifies the production team about a specific product's requirements, the production planning department takes charge of designing the product or job. This team identifies the materials needed and prepares a detailed operations list, outlining the tasks to be performed, their order, and the respective shops, departments, plants, or machines designated for each task. Figure 5.1 shows the steps under process of job order costing:

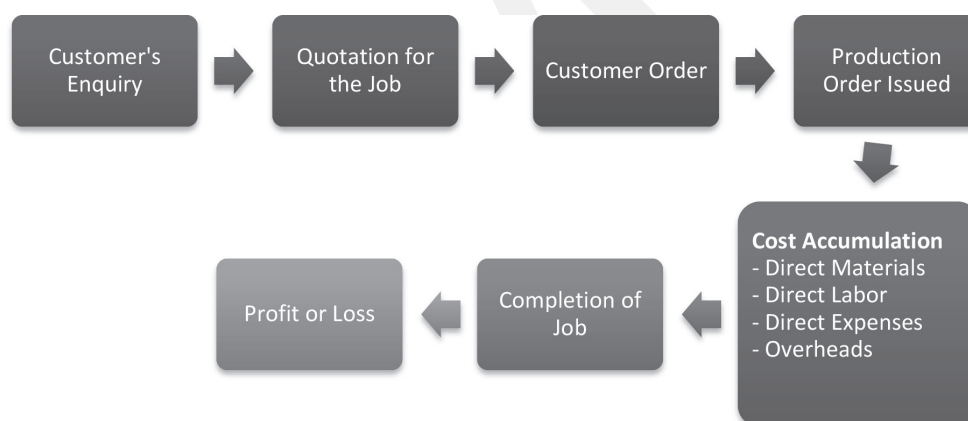


Figure 5.1 Process of Job Order Costing

Step 1: Customer Enquiry Process

The first step involves customers reaching out to the company with inquiries about products or services. These inquiries may focus on aspects such as estimated costs, desired quality, and production timelines. The primary objective at this stage is to evaluate the feasibility of the request and provide accurate information to assist in planning.

Step 2: Quotation for the Job

After understanding the customer's specifications, a responsible team member prepares a detailed cost estimate or quotation. The finalized price is then communicated to the customer. The goal of this step is to present a clear, competitive, and acceptable price tailored to the specific job.

Step 3: Customer's Order

If the quotation aligns with the customer's expectations, they place the order. This step formalizes the agreement and initiates the planning and execution phases of the project.

Step 4: Production Order Issued

A Production Order is subsequently issued, authorizing the shops to commence the manufacturing process. This order acts as the formal approval to initiate production. It typically includes all essential production details, such as the job or product specifications, the quantity or number of units to be produced, the production start date, the anticipated completion date, the required materials as per the bill of materials, the operations to be executed, the shops involved, and the job's routing.

In most cases, the production order specifies only the quantities of materials required and the time allocated for each operation. However, it may occasionally include the material and labor costs, in which case the production order doubles as both a manufacturing directive and a cost sheet for compiling the job's costs.

Additionally, the production order accounts for material and labor related to normal wastage or spoilage that may occur at various manufacturing stages or during the final phase of production.

Types of Production Orders

1. **Assembly Type Order:** This involves acquiring components that are then assembled into the final product within the factory. A production order is issued solely for the assembly process.

Example: A company that manufactures bicycles issues an assembly-type production order to assemble different parts like wheels, frames, and handlebars. The components, already produced or procured from suppliers, are brought to the factory, and the production order specifies only the final assembly of these parts into the finished bicycle.

2. **Sub-Assembly Type Order:** Components are procured, and both sub-assemblies and final assemblies are carried out within the factory. In such cases, production orders are issued for each sub-assembly as well as the final assembly.

Example: A company producing computers may issue a sub-assembly type order for the creation of a computer motherboard (a sub-assembly) and another for the final assembly of the computer. The motherboard is produced first and used in the final assembly, which involves connecting it with other components like RAM, CPU, and hard drives.

3. **Components or Parts Production Type Order:** Here, components are manufactured and subsequently sub-assembled. Separate production orders are issued for each component, sub-assembly, and the final assembly.

Example: A car manufacturing company may need a separate production order for creating specific components like an engine, transmission system, and chassis. Each of these components would have its own production order. Afterward, sub-assemblies such as the engine transmission assembly would be produced, and finally, a separate order would be issued for the final assembly of the entire car.

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Step 5: Cost Accumulation

Production orders are usually distributed in the following manner:

- ❑ A copy is sent to the stores to facilitate material provisioning and issuance as required.
- ❑ Each department or shop involved in the production process receives a copy to manage the demand for materials and allocate labor and machines for operations.
- ❑ One copy is forwarded to the cost department for job cost calculation. Separate job cost sheets are maintained for each job.

When a job involves multiple significant operations, distinct cost sub-sheets are maintained to track costs for each operation, with the total cost summarized in the main cost sheet.

Direct Material

Upon receiving a production order, the shop retrieves the required materials from the stores. Any surplus, excess, or incorrect materials are returned to the stores using material return notes. Scrap and waste generated during production are also returned in the same manner. The materials requisitions, return notes, and transfer notes are priced based on the methods followed by the organization.

Direct Labor

Labor summaries or wage analysis sheets are compiled for each accounting period, and the total is debited to the Work-in-Progress or Overhead Control Account, with a corresponding credit to the Wages Control Account. This summary may include overtime, idle time, shift differentials, and fringe benefits. Direct labor costs are recorded on the appropriate cost sheets, while indirect labor is handled similarly to indirect materials.

Manufacturing Overhead

Overhead costs are monitored against standing order numbers and cost centers. Each cost centre has predetermined or actual overhead rates established. The overhead applied to each job is calculated by multiplying the overhead rate by the actual variable base incurred for the job.

Step 6: Completion of Jobs

The posting of direct material, direct labor, direct expenses, and manufacturing overhead costs to the cost sheet for a job or production order is done periodically during the execution of the job. A completion report indicates the end of manufacturing activities, signaling that no further expenses should be recorded in order to close the cost sheet.

Step 7: Profit or Loss Analysis on Job

At this stage, the actual costs incurred are compared with the price charged to the customer. This comparison helps calculate the profit or loss for the job and assess overall efficiency. The primary objective is

to evaluate the job's performance and pinpoint areas that may require improvement for future projects.

Other aspects to be considered under job costing:

1. Work-in-Progress (WIP)

The cost associated with an incomplete job, where certain manufacturing processes or operations remain before the product is finished, is known as Work-in-Progress (WIP). If a production order is only partially completed by the end of an accounting period, it is essential to determine the closing stock of work-in-progress.

2. Cost Control in Job Order System

Cost control over job expenses can be achieved by comparing actual costs with estimated costs, which serve as the basis for determining job prices. Effective control can be applied to direct material and direct labor; however, managing overhead costs on a job-by-job basis is not feasible. Consequently, overhead control is limited to the department level, where the predetermined overhead rate has been established.

3. Treatment of Spoiled and Defective Work

- ❑ **Spoiled Work:** This refers to production that has been completely rejected and cannot be repaired.
- ❑ **Defective Work:** This describes production that fails to meet the standards of a saleable product but can be corrected with additional expenditure.

Manufacturing operations are seldom entirely successful, leading to some defective work over time. However, a normal rate of defects can be established for each manufacturing process, reflecting the anticipated number of defective items despite careful efforts. Defects may arise under various circumstances:

Treatment of Defects

1. **Normal Defective Rate:** If a normal rate of defects has been established and the actual number of defects is within or close to this limit, the cost of rectification is charged to the overall job and spread across the entire batch output. If defects exceed the normal rate significantly, the excess rectification costs are recorded as a loss in the Costing Profit and Loss Account.
2. **Bad Workmanship:** If defects are caused by poor workmanship, the rectification costs are classified as abnormal costs and are not considered legitimate expenses. These costs will be written off as a loss unless there is a penalty arrangement with the worker involved. However, if management anticipates a certain proportion of defects due to unavoidable poor workmanship, the rectification cost up to that provision will be treated as a normal cost and charged to the batch.

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3. **Poor Quality Incoming Material:** If defects result from the Inspection Department accepting poor-quality materials, the rectification costs will be charged to the department, rather than included in the manufacturing costs for the batch. These costs are classified as abnormal and written off to the Costing Profit and Loss Account.

5.2.2 ADVANTAGES OF JOB COSTING

Job order costing provides several benefits to organizations, making it an effective tool for managing costs and operations. Here are the key advantages, explained with real-world examples for better understanding:

1. **Identifies Profitable and Unprofitable Jobs**

By tracking costs and revenues for individual jobs, management can assess which projects are profitable. For instance, a custom furniture company can evaluate if designing high-end bespoke tables generates more profit than mass-producing standard chairs.

2. **Provides Data for Accurate Quotations**

Job order costing offers precise historical data, which is invaluable for preparing estimates for similar jobs in the future. For example, an event management firm can use past data from weddings to provide accurate quotes for new clients with similar requirements.

3. **Facilitates Effective Cost Control**

It allows management to analyze operational efficiency by comparing the estimated costs with the actual costs of each job. For instance, a construction company can review whether a specific house renovation stayed within budget and identify inefficiencies.

4. **Aids in Setting Selling Prices**

By understanding the total cost of a job, companies can set appropriate selling prices. For example, a custom jewellery designer can price each piece based on the cost of materials and the labour involved in crafting it.

5. **Tracks Spoilage and Defective Work**

Job order costing makes it easier to attribute spoilage or defective work to specific jobs or employees. For example, a bakery can pinpoint if a batch of cakes was ruined due to errors by a particular team during preparation.

6. **Enables Cost-Plus Pricing for Large Contracts**

The system supports applying the cost-plus pricing formula, commonly used in industries like construction or defense. For example, a contractor working on a government infrastructure project can justify costs and add a predetermined profit margin.

7. **Facilitates Budgetary Control**

Overheads are allocated on a predetermined basis, which simplifies budgeting and cost control. For instance, a software development

firm can allocate office rent and utilities proportionately to each project based on labor hours, helping maintain financial discipline.

5.2.3 LIMITATIONS OF JOB COSTING

While job order costing is helpful in many ways, it has some drawbacks that businesses should consider. Here's a breakdown of these limitations with relatable examples:

1. It Can Be Expensive

Tracking every detail for each job requires a lot of paperwork and effort, which can increase costs. For instance, a custom cabinet maker may need to spend extra on hiring an accountant or purchasing software to keep track of wood, labor, and other material costs for each order.

2. Higher Chance of Errors

The more clerical work involved, the more likely mistakes can happen. For example, in a small graphic design agency, misreporting the hours worked on a client's project could lead to incorrect billing or missed profits.

3. Costs Are Always Historical

Job costing calculates expenses after the job are completed, meaning the information isn't real-time. For example, a construction company might realize after finishing a project that they overspent on materials—when it's too late to fix the issue.

4. Costs May Lack a Scientific Basis

The costs recorded reflect what was spent, not necessarily what should have been spent. For example, a bakery using job costing might see high labour costs but won't get insight into whether that labour was used efficiently or could be reduced with better processes.

The contrast between advantages and limitation of job costing is shown in Table 5.1:

Table 5.1 Job Costing: Advantages and Limitations

Aspect	Advantages	Limitations
Profitability Analysis	Helps identify which jobs are profitable or unprofitable.	Does not provide real-time insights; profitability is assessed only after job completion.
Accurate Quotations	Offers historical data to prepare precise estimates for future jobs.	Gathering and maintaining historical data can be time-consuming and expensive.
Cost Control	Enables management to compare estimated and actual costs, highlighting inefficiencies.	Tracks actual costs but does not analyze whether they were the most efficient costs.

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Aspect	Advantages	Limitations
Pricing Assistance	Helps set appropriate selling prices by understanding total costs.	Calculated costs are post-completion and may not address price adjustments during the job.
Spoilage and Defects Tracking	Makes it easier to attribute defects or waste to specific jobs or employees.	Requires meticulous record-keeping, increasing administrative burden and the likelihood of errors.
Cost-Plus Pricing for Contracts	Supports cost-plus pricing for large projects.	May not suit small or fast-paced jobs where fixed pricing is preferred.
Budgetary Control	Allocates overheads on a predetermined basis, simplifying budgeting.	Overhead allocation methods may lack precision, leading to inefficiencies or disputes.
Administrative Cost	Useful for detailed job analysis and long-term improvement.	Involves significant clerical work, raising overall administrative costs.
Error Risk	Provides a structured system to analyze individual job efficiency.	Increased manual work raises the chance of errors in recording or analysis.

5.2.4 ASSESSMENT OF JOB COSTING

In the evaluation of job costing, the analysis of profits from completed jobs and the examination of cost variances are essential elements. Reports generated on these aspects provide valuable insights into financial performance and operational efficiency, enabling management to make informed decisions.

Report on Profits from Completed Jobs: A monthly statement can be generated to reflect the gross profit earned from all jobs completed within that month. This statement serves as a valuable tool for management to assess past performance. A net profit analysis can also be conducted similarly by including administration, selling, and distribution overheads related to the job in the statement.

When management reviews the **gross profit** from completed jobs, they gain an immediate understanding of how well each job or project is performing. Here's how this influences decision-making:

- ❑ **Identifying High-Performing Jobs:** If certain jobs generate higher profits, management can focus on understanding the factors contributing to this success—whether it's efficient use of materials, labor, or effective pricing strategies. This insight helps replicate best practices in future jobs.
- ❑ **Adjusting Pricing Strategies:** If the gross profit is lower than expected, it can signal the need to review pricing strategies. Management may decide to increase prices for similar future jobs or make adjustments to ensure profitability is maintained.

- ❑ **Resource Allocation:** Jobs with higher profit margins might suggest that more resources should be allocated to similar projects. Conversely, jobs that consistently show low profits could be re-evaluated, with decisions to either improve efficiency or discontinue them.

By including **net profit analysis**, which factors in administration, selling, and distribution costs, management gets a more comprehensive picture of profitability. This can lead to:

- ❑ **Cost-Control Measures:** If net profits are low despite high gross profits, it indicates that overhead costs might be eating into profits. Management could implement measures to reduce these overheads, such as negotiating with suppliers, improving labor efficiency, or streamlining distribution processes.

Sample Report on Profits from Completed Jobs

Company Name: Shyam Manufacturing Ltd.

Period: November 2024

Department: Custom Job Production

Gross Profit from Completed Jobs

Job No.	Job Description	Revenue (₹)	Material Cost (₹)	Labor Cost (₹)	Over-head (₹)	Gross Profit (₹)
101	Job 1 – Custom Furniture	50,000	15,000	20,000	5,000	10,000
102	Job 2 – Steel Railing	30,000	10,000	12,000	3,000	5,000
103	Job 3 – Office Desk	45,000	13,000	18,000	4,500	9,500
Total		₹125,000	₹38,000	₹50,000	₹12,500	₹24,500

Net Profit from Completed Jobs

Job No.	Admin Cost (₹)	Selling & Distribution Cost (₹)	Net Profit (₹)
101	2,000	1,000	7,000
102	1,500	500	3,000
103	2,500	1,500	5,500
Total	₹6,000	₹3,000	₹15,500

Interpretation of Profits Report:

- ❑ Job 1 (Custom Furniture) shows the highest gross profit of ₹10,000. However, after deducting admin and selling costs, the net profit is ₹7,000. Management might want to explore what contributed to the strong performance (e.g., efficient labor use or higher margin pricing) and replicate this in future jobs.
- ❑ Job 2 (Steel Railing) has a gross profit of ₹5,000, which is lower, and after administrative and selling costs, the net profit is ₹3,000. Management might analyze why this job had a lower profit margin and assess if pricing or material costs could be optimized for future similar jobs.

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- ❑ Based on the net profit analysis, Job 3 (Office Desk) has a balanced profit margin. Since it's a moderate performer, management could decide to streamline processes to increase profitability.

Report on Cost Variances: When cost estimates are established, a cost variance report can be created to highlight the discrepancies between actual costs and estimated costs. This report aims to reveal significant differences that warrant further investigation. It can be prepared for individual jobs or for a department, detailing variances for all jobs carried out by the department over a specified period.

The **cost variance report** is an essential tool for identifying discrepancies between **estimated costs** and actual costs. This report directly influences decision-making in the following ways:

- ❑ **Investigating Discrepancies:** Significant variances between estimated and actual costs often require investigation. Management might discover that certain jobs are running over budget due to factors like inefficient labor or unexpected material costs. Once identified, corrective measures can be taken, such as better planning for future jobs or improving the accuracy of estimates.
- ❑ **Improving Cost Estimates:** If certain types of jobs consistently show cost overruns, management can review and refine the estimating process. This could involve improving the accuracy of cost forecasting, factoring in potential risks more carefully, or using historical data to predict more realistic costs for similar jobs.
- ❑ **Departmental Performance Analysis:** By preparing cost variance reports at the department level, management can identify which departments are over or underperforming. For example, if one department consistently goes over budget, it might indicate a need for better training, revised workflows, or new technology to improve efficiency.
- ❑ **Strategic Adjustments:** If cost variances are significant across multiple jobs, management may decide to revise overall budgeting strategies. This could involve adjusting the overhead rate or establishing stricter cost controls for specific departments or job types.

Sample Report on Cost Variances

Company Name: Shyam Manufacturing Ltd.

Period: November 2024

Department: Custom Job Production

Cost Variance Report for Job No. 101 (Custom Furniture)

Cost Element	Estimated Cost (₹)	Actual Cost (₹)	Variance (₹)	Reason for Variance
Materials	14,000	15,000	+1,000	Increase in material prices
Labor	18,000	20,000	+2,000	Extended work hours for customization

Cost Element	Estimated Cost (₹)	Actual Cost (₹)	Variance (₹)	Reason for Variance
Overhead	4,500	5,000	+500	Increased utility costs during production
Total Variance	36,500	40,000	+3,500	

Cost Variance Report for Job No. 102 (Steel Railing)

Cost Element	Estimated Cost (₹)	Actual Cost (₹)	Variance (₹)	Reason for Variance
Materials	9,500	10,000	+500	Additional material waste due to cutting
Labor	11,000	12,000	+1,000	Labor inefficiencies due to new workers
Overhead	3,000	3,200	+200	Unexpected machine downtime
Total Variance	23,500	25,200	+1,700	

Interpretation of Cost Variance Report:

- ❑ Job 101 (Custom Furniture) has a total variance of ₹3,500, with the primary contributors being higher material costs and additional labor hours. Management can look into negotiating better material prices or adjusting labor schedules for future jobs to stay within budget.
- ❑ Job 102 (Steel Railing) shows a variance of ₹1,700. The increased material waste and labor inefficiencies are the primary culprits. Management might focus on improving cutting techniques or providing more training to reduce waste and increase worker efficiency on future jobs.

These variance reports highlight areas where cost control measures are necessary. By investigating the causes behind variances, management can develop corrective actions, such as better planning or process adjustments, to improve future cost estimates and job profitability.

Examples on Job Costing

Example 1: A particular Job No. 200 of a company passes through three departments: A, B, and C. The following information is given regarding this job:

Particulars	Department A	Department B	Department C
Materials issued to the job (₹)	12,000	4,000	3,000
Direct labour hours for job (No.)	800	1,200	3,000
Rate of direct labour per hour (₹)	2.50	3.00	4.00
Sale of scrap materials (₹)	2,000	500	300
Total overheads for departments (₹)	20,000	18,000	30,000
Total labour hours for departments (No.)	8,000	12,000	15,000

N O T E S

You are required to calculate the Job Cost Sheet for Job No. 200.

Solution:**Job Cost Sheet for Job No. 200**

Particulars	Dept. A	Dept. B	Dept. C	Total (₹)
Materials Issued	12,000	4,000	3,000	
Less: Sale of Scrap	(2,000)	(500)	(300)	
Net Materials	10,000	3,500	2,700	16,200
Direct Wages				
800 hours @ ₹ 2.50 per hour	2,000			
1,200 hours @ ₹ 3.00 per hour		3,600		
3,000 hours @ ₹ 4.00 per hour			12,000	
Total Direct Wages	2,000	3,600	12,000	17,600
Overheads				
20,000/8,000 × 800	2,000			
18,000/12,000 × 1,200		1,800		
30,000/15,000 × 3,000			6,000	
Total Overheads	2,000	1,800	6,000	9,800
Total Cost				43,600

**SELF-ASSESSMENT QUESTIONS**

- A company is working on Job No. 501. The following data is available:
 - ☐ Direct Materials: ₹20,000
 - ☐ Direct Labour: ₹15,000
 - ☐ Factory Overhead: 60% of Direct Labour
 Calculate the total works cost of Job No. 501.
 - a. ₹35,000 b. ₹44,000
 - c. ₹50,000 d. ₹45,000
- What happens to the costs of defective work in a job costing system?
 - a. They are absorbed by the company and not accounted for
 - b. They are written off as a profit loss without further investigation
 - c. They are directly charged to the cost of goods sold (COGS)
 - d. They are charged to the overall job and distributed across the entire batch
- A company is working on Job No. 808. The following details are available:

- ❑ Direct Materials: ₹60,000
- ❑ Direct Labour: ₹40,000
- ❑ Factory Overhead: 150% of Direct Labour cost
- ❑ Administrative Overhead: 25% of Works Cost
- ❑ Selling and Distribution Overhead: 20% of Production Cost

The company wants to achieve a 25% profit on the selling price. Calculate the selling price of Job No. 808.

- | | |
|--------------|--------------|
| a. ₹3,00,000 | b. ₹3,15,000 |
| c. ₹3,10,000 | d. ₹3,20,000 |

5.3 BATCH COSTING

Batch costing is a specific order costing method where each batch is considered a cost unit, and costs are separately accumulated and determined for each batch, which consists of multiple identical units.

Features of Batch Costing:

Think of **batch costing** as a way to track the cost of making products in large groups. Here are its main features:

1. Batch as a Cost Unit:

Imagine you're making 500 cupcakes for a local bakery. All 500 cupcakes belong to the same batch, so instead of tracking the cost for each cupcake individually, you track the cost for the entire batch and then divide it by the number of cupcakes to get the cost of each one. Easy, right?

2. Batch vs. Job Costing:

In **job costing**, products are made based on a customer's specific order. For example, if you're designing a custom logo for a company, that's job costing. But in **batch costing**, you produce identical items in groups—like printing 1,000 T-shirts with the same design. These batches are then tracked for cost.

3. Batch Numbering for Easy Tracking:

Each batch gets its own number. Think of it like giving each group of items a student ID number. If one batch of products has a problem, you can trace it back easily by checking the batch number. For example, if one batch of medicine has a problem, the batch number helps you recall all the products in that group.

4. Traceability of Issues:

If something goes wrong, like a mistake in production, batch costing helps to track and fix it fast. For instance, if 50 cupcakes from a batch were undercooked, you could identify them by their batch number and remove them from the shelves.

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Advantages of Batch Costing: Why It Works**1. Reduced Accounting Work:**

Instead of tracking each individual item, you focus on the whole batch, which makes your job easier. It's like tracking 10 groups of 100 cupcakes instead of each individual cupcake—less paperwork and fewer headaches!

2. Consistent Cost per Unit:

Since you're spreading the cost of the entire batch across all units, every product in the batch has a consistent cost. No surprises! Imagine making 100 custom T-shirts: the cost for each one is the same, no matter the design, because you averaged the cost across the batch.

3. Better Cost Control:

By producing large batches, businesses can lower the cost of each item—this is where the Economic Batch Quantity (EBQ) comes in. It's about finding the sweet spot where making more items reduces costs, but doesn't go overboard.

4. Easy Supervision:

It's easier to manage a group of 100 items at once than to handle each item individually. For example, a bakery manager can oversee the whole batch of cupcakes, ensuring everything is going smoothly.

5. Less Time Wasted:

With batch production, workers don't waste time moving between different jobs. Everything in one batch is handled together, cutting down on time spent switching tools or materials.

Disadvantages of Batch Costing: The Downsides**1. Difficulty in Homogeneity:**

Not every job is exactly the same. Sometimes, it's hard to create completely identical products in one batch. Imagine trying to bake 100 cupcakes but some of the ingredients aren't exactly the same for each one—batch costing assumes everything is the same!

2. Different Batch Sizes for Different Customers:

Not all customers want the same number of products. One customer might want 100 T-shirts, while another wants 1,000. Adjusting batch sizes for each customer can be tricky.

3. Risk of Batch Failure:

If one item in a batch is faulty, the whole batch might have to be thrown out. For example, if you bake 100 cupcakes but one batch of frosting is spoiled, you might have to discard all 100 cupcakes, leading to a loss.

Applications

Batch costing is commonly used in industries that produce similar articles in defined batches for either internal consumption in the production of finished goods or for sale to customers. Its applications typically include:

- ❑ Readymade Garment Manufacturing Industries
- ❑ Pharmaceutical and Drug Industries
- ❑ Spare Parts and Components Manufacturing Industries
- ❑ Toy Manufacturing Industries
- ❑ Tyre and Tube Manufacturing Industries

Example: A jobbing factory has undertaken to supply 240 pieces of a component per month for the next six months. Each month, a batch order is opened to track materials and labor hours at actual cost. Overheads are levied at a rate per labor hour. The selling price contracted is ₹10 per piece.

The following data is available for each month's production:

Month	Batch Output	Material Cost (₹)	Direct Wages (₹)	Direct Labour Hours
January	250	800	200	260
February	240	780	220	280
March	260	850	240	300
April	230	770	210	270
May	240	820	230	290
June	260	900	250	310

Other information regarding overheads is provided below:

Month	Chargeable Expenses (₹)	Direct Labour Hours
January	13,000	5,000
February	11,200	4,800
March	13,500	5,200
April	11,400	4,900
May	12,600	5,100
June	13,200	5,400

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Particulars	Jan	Feb	March	April	May	June	Total
Batch Output (units)	250	240	260	230	240	260	1,480
Sales Value (₹)	2,500	2,400	2,600	2,300	2,400	2,600	14,800
Material Cost (₹)	800	780	850	770	820	900	4,920
Direct Wages (₹)	200	220	240	210	230	250	1,350
Chargeable Expenses (₹)	676	654	779	627	717	757	4,210

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Particulars	Jan	Feb	March	April	May	June	Total
Total Cost (₹)	1,676	1,654	1,869	1,607	1,767	1,907	10,480
Profit per Batch (₹)	824	746	731	693	633	693	4,320
Total Cost per Unit (₹)	6.70	6.89	7.19	6.99	7.36	7.34	7.08
Profit per Unit (₹)	3.30	3.11	2.81	3.01	2.64	2.66	2.92

Notes: Chargeable Expenses = $\frac{\text{Chargeable Expenses for Month}}{\text{Total Direct Labour Hours for Month}} \times \text{Direct Labour Hours for Batch}$

5.3.1 ECONOMIC BATCH QUANTITY

EBQ refers to the optimal quantity of items that should be produced in a batch to balance and minimize two key costs: **setup costs** (costs associated with preparing and running the production process) and **carrying costs** (costs associated with storing inventory). The goal is to determine the most cost-effective batch size that minimizes the overall cost.

Imagine you're running a furniture manufacturing company, and you're deciding how many chairs to produce at once. If you produce too many, you may end up with high storage costs. But if you produce too few, you'll have to set up and re-set up your machines too often, which increases setup costs. EBQ helps you find the balance between these two extremes!

Formula

$$EBQ = \sqrt{\frac{2AS}{C}}$$

Where:

- **EBQ** = Economic Batch Quantity
- **A** = Annual Demand
- **S** = Setup Cost per batch
- **C** = Carrying Cost per unit per year

Let's Explore the Components:

1. **Annual Demand (A):** This is how many units of a product you need over the course of a year. If you're manufacturing 1,00,000 chairs a year, that's your annual demand.
2. **Setup Costs (S):** These are the costs incurred each time you prepare your production line. For example, setting up machinery, ordering materials, and paying workers for setup might cost ₹5,000 per batch.
3. **Carrying Costs (C):** These are the expenses for storing each unit of your product. For instance, if it costs ₹10 per chair per year to store a chair in your warehouse, this is your carrying cost.

Step-by-Step Calculation Example:

Let's assume:

- ❑ Annual Demand (A) = 1,00,000 units
- ❑ Setup Cost (S) = ₹5,000 per batch
- ❑ Carrying Cost (C) = ₹10 per unit per year

Now, plugging these values into the formula:

$$EBQ = \sqrt{\frac{2 \times 1,00,000 \times 5,000}{10}} = \sqrt{\frac{1,00,00,00,000}{10}} = 10,000 \text{ units}$$

So, the optimal batch size (EBQ) is 10,000 units. This is the batch size that minimizes your setup and carrying costs.

Setup and Processing Costs

Setup and processing costs pertain to the expenses incurred for preparing and executing operations prior to initiating the production of a batch. There exists an inverse relationship between batch size and setup and processing costs:

- ❑ **Larger Batch Size:** Results in lower setup costs due to the reduced number of batches required.
- ❑ **Smaller Batch Size:** Leads to higher setup costs as it necessitates more batches.

Carrying Costs

Carrying costs are the expenses associated with maintaining a specific level of inventory. There is a positive correlation between batch size and carrying costs:

- ❑ **Larger Batch Size:** Incurs higher carrying costs due to a greater average inventory level.
- ❑ **Smaller Batch Size:** Results in lower carrying costs because of a reduced average inventory.

Example of Setup and Carrying Costs:**Scenario 1: Large Batch Size (10,000 units)**

- ❑ **Lower Setup Costs:** Since you're producing a large batch, you need to set up your machinery only once for the 10,000 units.
- ❑ **Higher Carrying Costs:** You will have more inventory sitting in the warehouse, which means higher costs for storage and handling.

Scenario 2: Small Batch Size (1,000 units)

- ❑ **Higher Setup Costs:** You'll need to set up the production line more frequently, which means more setup costs.
- ❑ **Lower Carrying Costs:** Fewer units in storage means lower carrying costs, but there might be a risk of stockouts if demand exceeds expectations.

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Trade-off

The whole idea behind EBQ is to find the sweet spot between these two costs:

- ❑ Larger batches reduce the frequency of setups but increase the carrying cost of storing larger amounts of inventory.
- ❑ Smaller batches reduce carrying costs but increase the frequency of setups.

The optimal batch size is the point where the total cost (setup + carrying) is minimized.

5.3.2 ASSESSMENT OF ECONOMIC BATCH QUANTITY

In the Figure 5.2, costs are plotted on the Y-axis, while batch size (or quantity) is represented on the X-axis. As the batch size increases, carrying costs generally rise, while setup costs tend to decline. The point at which the two cost lines intersect indicates the lowest cost combination

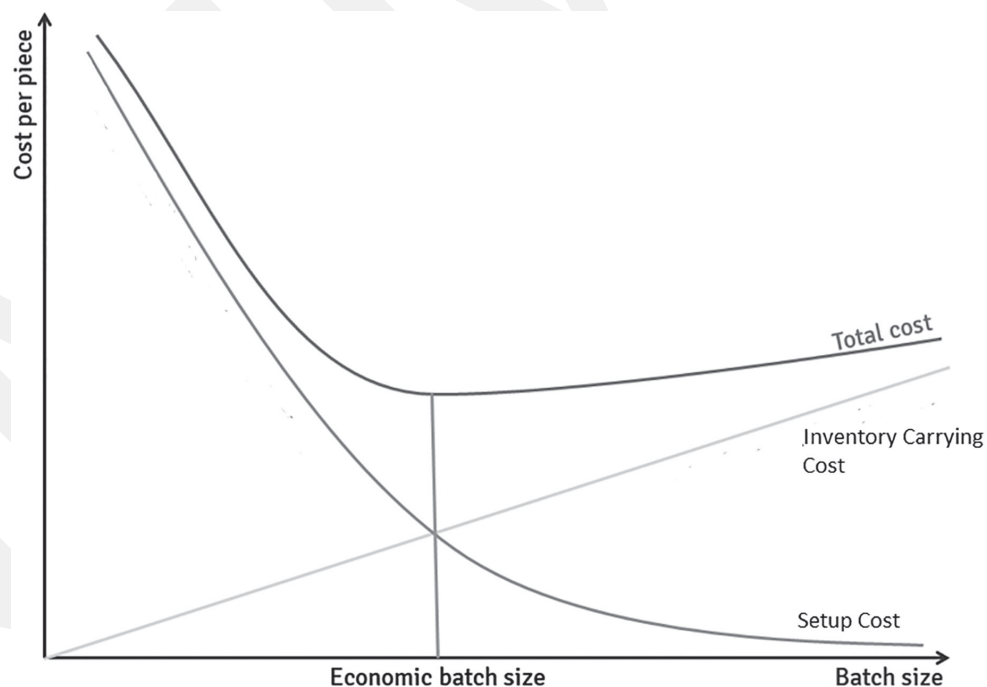


Figure 5.2 Economic Batch Quantity

1. **Cost per Unit (Y-Axis):** This shows how much it costs to produce each unit in the batch.
2. **Batch Size (X-Axis):** This shows how many units are produced in each batch.
3. **Total Cost Curve (Red Curve):** This represents the total cost of production, which includes both the **setup cost** (fixed costs for setting up the production process) and **inventory carrying cost** (the cost of storing the finished goods).

- 4. Inventory Carrying Cost:** This cost increases as the batch size increases. More units mean higher inventory costs, such as storage or maintenance fees.
- 5. Setup Cost:** This cost decreases as batch size increases. It includes the cost of setting up production for each batch. Larger batches mean fewer setup operations and thus a lower setup cost.

Let us understand the concept with following example:

From the following information, calculate the Economic Batch Quantity (EBQ) for a company using batch costing:

- ❑ Annual Demand for the components: 3,600 units
- ❑ Setting up cost per batch: ₹150
- ❑ Manufacturing cost per unit: ₹250
- ❑ Carrying cost per unit: 8% per annum

Using the data provided, determine the Economic Batch Quantity (EBQ) for the company.

Solution:

$$\begin{aligned}
 \text{EBQ} &= \sqrt{\frac{2AS}{C}} \\
 &= \sqrt{\frac{2 \times 3600 \times 150}{8\% \text{ of } 250}} \\
 &= \sqrt{\frac{10,80,000}{20}} \\
 &= 232.38
 \end{aligned}$$



SELF-ASSESSMENT QUESTIONS

- 4.** In the context of batch costing, what happens to setup costs as the batch size increases?
- They increase due to more frequent setups.
 - They decrease due to fewer batches needed.
 - They remain constant regardless of batch size.
 - They are eliminated entirely.
- 5.** A company produces 600 units in a single batch. The following information is available:
- ❑ Material Cost: ₹50,000
 - ❑ Direct Wages: ₹30,000 (paid at ₹10 per hour)
 - ❑ Overheads: Applied at 80% of direct wages
 - ❑ Selling Price per Unit: ₹200

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Calculate the total profit for the batch.

- | | |
|------------|------------|
| a. ₹10,000 | b. ₹12,000 |
| c. ₹14,000 | d. ₹16,000 |

6. A company has the following information for a particular product:

- Annual Demand = 60,000 units
- Set-up Cost per Batch = ₹800
- Holding Cost per Unit per Year = ₹40

Additional Data:

- Material Cost per Unit = ₹5
- Overhead Cost per Unit = ₹100 (excluding holding cost)

Calculate the total cost of production for one year using the Economic Batch Quantity (EBQ).

- | | |
|---------------|---------------|
| a. ₹63,20,200 | b. ₹62,40,200 |
| c. ₹64,61,200 | d. ₹62,91,200 |

5.4 CONTRACT COSTING

Contract Costing, also known as Terminal Costing, is a method used to track expenses and profits for large projects, such as constructing a school, bridge, or road. For instance, if you are a contractor hired to build a mall for ₹50 crore, contract costing enables you to monitor how much money has been spent and how much revenue has been earned at various stages of the project.

Key Terms with Examples in Rupees

- **Contractor:** The individual or company undertaking the construction work. Example: If “XYZ Constructions” is building a hospital for ₹20 crore, XYZ Constructions is the contractor.
- **Contractee:** The person or organization hiring the contractor. Example: If the government hires XYZ Constructions, the government is the contractee.
- **Contract Account:** A detailed record of all project-related expenses (debits) and income (credits). Example: If ₹8 crore has been spent on materials, wages, and overheads so far, this amount is recorded in the Contract Account.
- **Notional Profit:** Profit calculated for incomplete projects to assess financial performance and progress. Example: For a ₹40 crore road construction project where 50% is complete, a contractor might calculate notional profit based on the revenue earned to date.
- **Work Certified and Uncertified**
 - ◆ Certified Work: Work completed and approved by the client.

- ◆ **Uncertified Work:** Work completed but awaiting approval.

Example: For a ₹10 crore building project, if 70% is approved by the client, ₹7 crore represents certified work. An additional 10% completed but not yet approved represents ₹1 crore as uncertified work.

Table 5.2 highlighting the distinctions between Job and Contract Costing:

Table 5.2 Difference Between Job Costing and Contract Costing

Feature	Job Costing	Contract Costing
Cost Allocation	Costs are first allocated to cost centres and then to individual jobs.	Most expenses are direct; overheads represent a small percentage (e.g., head office expenses, central storage costs).
Pricing	Influenced by individual conditions and the general policy of the organization.	Influenced by specific clauses of the contract.
Cost Unit	Each job is a cost unit.	Each contract is a cost unit.
Location of Work	Work is often carried out at the contractee's own premises.	Work is usually carried out at a site other than the contractee's premises.

Example:

ABC Construction takes on a ₹5 crore contract to build a school. The *Contract Account* tracks all associated expenses and income.

Contract Account

Particulars	Amount (₹)	Interpretation
Expenses (Debits):		
Materials	1,50,00,000	Cost of raw materials such as cement, steel, and bricks.
Wages	50,00,000	Payments made to workers directly involved in the project.
Overheads	20,00,000	Indirect costs, including electricity and machine rentals.
Asset Depreciation	5,00,000	Depreciation on equipment like cranes and cement mixers.
Total Expenses	2,25,00,000	The cumulative cost of the project to date.
Income (Credits):		
Certified Work	3,00,00,000	Revenue from 60% of the work approved by the client.
Scrap Value	10,00,000	Recovery from the sale of unused materials such as steel.
Total Income	3,10,00,000	The total revenue generated to date.

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Particulars	Amount (₹)	Interpretation
Notional Profit	85,00,000	Income (₹3.1 crore) - Expenses (₹2.25 crore).
Profit Transferred	42,50,000	50% of the notional profit is recognized as profit, while the rest is reserved.

5.4.1 FEATURES OF CONTRACT COSTING**1. Each Contract is treated as an Individual Cost Unit**

Every contract is considered a separate project for costing.

Example: If a construction company is building both a hospital and a shopping mall, the costs for the hospital (e.g., materials, wages) will not be mixed with the costs of the mall. Each has its own records.

2. Contracts Typically Extend over Multiple Accounting Periods

Since contracts often last for years, costs and profits must be tracked annually.

Example: A contractor building a ₹200 crore bridge that takes 3 years will calculate profits or losses for each financial year, even before the project is completed.

3. Contract Work is performed at the Site, Not in Factory Premises

Unlike manufacturing, the work happens at the project site.

Example: If a road construction project is happening 50 km away from the contractor's office, all activities (e.g., paving, grading) will be done at the road site, not in a factory.

4. A Distinct Account is Created for Each Contract

Each contract is assigned a unique number for clear identification.

Example: A contractor handling three projects simultaneously—Project 101 (school), Project 102 (stadium), and Project 103 (apartment)—uses these numbers to differentiate and track costs for each project.

5. Two Parties in a Contract: Contractor and Contractee

❑ **Contractor:** The person or company executing the project.

❑ **Contractee:** The person or organization hiring the contractor.
Example: If the municipal corporation (contractee) hires “ABC Builders (contractor)” to build a flyover, the contract agreement defines their roles clearly.

6. Direct Costs Form the Bulk of the Total Contract Cost

Costs like materials, labor, and equipment rentals directly related to the project are the largest part of expenses.

Example: If ₹150 crore is spent on a ₹200 crore metro rail project, ₹140 crore could be for direct costs like steel, cement, and wages,

while only ₹10 crore might be for indirect costs (office rent, administrative salaries).

7. Materials Purchased Specifically for the Contract are Charged Directly

Materials bought for a specific project are charged based on the supplier's invoice, while those taken from inventory are charged based on requisition notes.

Example: For a ₹5 crore school construction project, cement purchased directly from the supplier (₹20 lakh) is charged directly, whereas bricks taken from stock are recorded using store requisition forms.

8. Labor Costs are directly Assigned to the Contract

Wages for workers are directly linked to the contract they're working on.

Example: If 50 workers are employed to construct a dam for ₹10 lakh/month, this labor cost is directly recorded in the Contract Account for the dam.

9. Most Expenses are Direct in Nature

The majority of costs in contract work (e.g., materials, wages) are directly tied to the project.

Example: For a residential building contract worth ₹50 crore, expenses like cement, steel, and on-site labor (₹45 crore) will be direct, while expenses like office rent or management salaries (₹5 crore) will be indirect.

5.4.2 PROCESS OF CONTRACT COSTING

The process of contract costing involves several essential steps (shown in Figure 5.3), which are outlined as follows:

1. **Cost Allocation:** Costs are assigned to specific contracts according to the resources utilized in each project. These costs encompass direct materials, direct labor, and direct expenses, in addition to indirect costs such as overheads and administrative expenses. The allocation of these indirect costs is done using a logical basis, such as labor hours, machine hours, or a percentage of direct costs.
2. **Cost Recording:** Throughout the duration of the contract, costs are systematically recorded and monitored via a dedicated cost ledger or a project management system. This process allows organizations to track expenses in real time and measure them against the predetermined budget or estimated costs.
3. **Cost Analysis:** The accumulated costs are periodically examined to assess the contract's progress, profitability, and cost-efficiency. This analysis includes comparing actual costs with budgeted figures, reviewing the expenses of specific activities or resources, and

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identifying areas where cost reductions or process improvements may be implemented.

4. **Cost Control:** Drawing on insights from cost analysis, organizations implement control measures to ensure the contract stays within the allocated budget and achieves its financial goals. Cost control measures may include renegotiating supplier contracts, optimizing resource allocation, or enhancing operational processes to minimize expenses.
5. **Final Account:** Upon completion of the contract, a final account is prepared, which provides a comprehensive summary of the total costs incurred, revenue earned, and overall profitability of the project. This final statement serves as a valuable tool for assessing the project's success and shaping future pricing and bidding strategies.

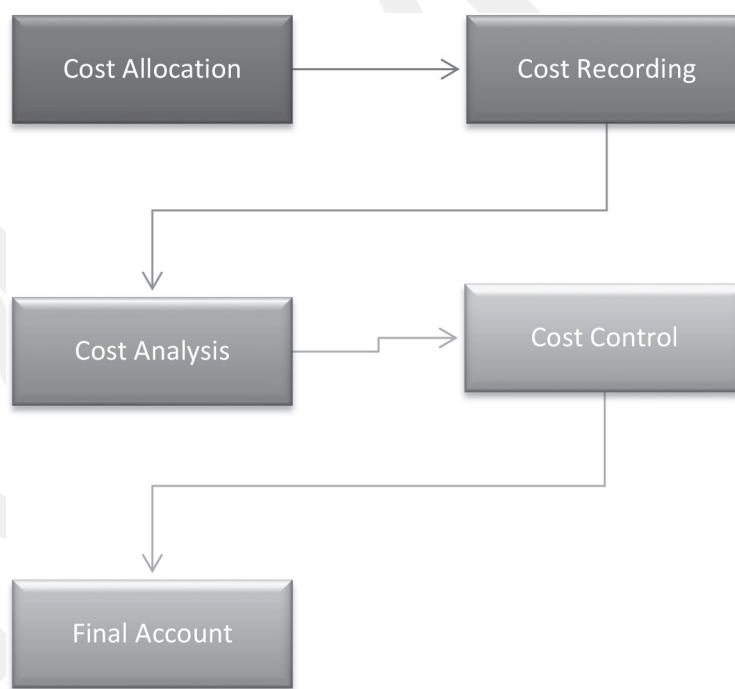


Figure 5.3 Process of Contract Costing

Example of Contract Costing

Let us consider a hypothetical example involving a construction company, Stellar Builders Ltd., that specializes in building commercial office buildings. Stellar Builders Ltd. has entered into a contract to construct a 10-story office building for a client at an agreed price of ₹250 crore. The company needs to carefully manage the costs associated with this project to ensure profitability and avoid overruns. In this case, contract costing helps monitor and allocate costs effectively.

1. Cost Allocation

Cost allocation is the first and most critical step in contract costing. Stellar Builders Ltd. identifies all costs directly related to the

construction project and allocates them to the contract. Direct material costs, such as steel, cement, concrete, and glass, are accounted for based on actual usage. Direct labour costs include the wages paid to site workers, machine operators, and technical staff working specifically on the project. Additionally, direct expenses like machinery rentals, site permits, and insurance for the project are carefully tracked. Indirect costs, such as site administration expenses, utilities, and general management costs, are also apportioned to the project using a fair method, such as a percentage of direct costs or total labour hours.

2. Cost Recording

Throughout the construction process, Stellar Builders Ltd. ensures that all project costs are recorded systematically. A **project management system** is used to maintain up-to-date records of material deliveries, usage, and payments to suppliers. Labour costs are tracked using timesheets that log daily work hours for employees and subcontractors. Expenses for rented equipment, machinery usage, and subcontractor fees are also documented to ensure transparency. By recording costs in real-time, the company can monitor spending against the budget and identify any deviations early in the process.

3. Cost Analysis

Periodic cost analysis allows Stellar Builders Ltd. to evaluate the project's financial progress and identify areas for improvement. The company compares the **actual costs** incurred with the **budgeted estimates** for different cost components, such as materials, labour, and overheads. If any activity, like foundation work or interior finishing, shows a significant cost overrun, the company investigates the cause and takes corrective action. Additionally, progress is measured using key performance indicators (KPIs), such as the percentage of work completed relative to the total costs incurred. This analysis helps ensure that the project stays on track both in terms of time and cost.

4. Cost Control

Based on the insights from cost analysis, Stellar Builders Ltd. implements various cost control measures to ensure the contract remains within budget. These measures include negotiating better terms with suppliers to secure discounts on bulk purchases of materials and reducing idle time for labour and equipment. The company also focuses on improving resource utilization by effectively managing work shifts and minimizing material wastage. Subcontractor performance is closely monitored to ensure they deliver work efficiently and within the agreed timelines. By applying these measures, the company minimizes unnecessary expenses and keeps the project on schedule.

5. Final Account

Once the construction of the 10-story office building is completed, Stellar Builders Ltd. prepares the final account for the project. This

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account details all the costs incurred, including materials, labour, direct expenses, and overheads, ensuring each expenditure is accounted for. The final account also includes the agreed contract price of ₹250 crore and calculates the overall profitability of the project. By analyzing the final account, the company can assess the success of the project and gather insights to improve cost management, pricing strategies, and bidding accuracy for future contracts.

In this example, contract costing enables Stellar Builders Ltd. to efficiently manage the costs of constructing a commercial office building. By tracking costs, analyzing progress, and implementing cost control measures, the company ensures that the project stays within budget and achieves the desired profitability. This systematic approach enhances project management, supports financial success, and strengthens the company's competitive edge in the construction industry.

Treatment of Important Items:

Material Cost:

All materials supplied from the stores or purchased directly for the contract are debited to the specific contract account. In cases where excess materials are transferred from one contract to another, their costs are adjusted using a material transfer note signed by both the transferee and transferor foreman. If returning surplus materials is uneconomical due to high transportation costs, the materials are sold, and the contract account is credited with the sale price. Any resulting profit or loss is transferred to the Costing Profit and Loss Account. If materials are stolen or destroyed by fire, the loss is also transferred to the Costing Profit and Loss Account. If stores items are used to manufacture tools, their cost is charged to the work expenses account. If the contractee supplies materials without affecting the contract price, no accounting entries are made in the contract account, but a note should be added.

Example: In a ₹20 crore highway construction project, ₹3 crore worth of cement purchased from a supplier is recorded in the contract account. If excess cement worth ₹2 lakh is sold due to high transportation costs, the sale amount is credited back to the contract account, reducing overall material expenses.

The procedures of recording materials in a contract account are shown in Table 5.3:

Table 5.3 Material Treatment

Item	Treatment
Stock of Materials	Opening stock is debited, and closing stock is credited.
Purchase of Materials	Materials purchased specifically for the contract are debited to the contract account.
Transfer of Materials	Materials transferred <i>to</i> the contract from other contracts are debited, while materials transferred <i>out</i> are credited.

Item	Treatment
Sale of Materials	The sale of materials from the contract is credited at the selling price.
Profit/Loss on Sale	Profit on the sale of materials is credited; loss is debited to the contract account.
Loss of Material	Losses due to theft, fire, or damage are debited; any insurance claim accepted is credited similarly to sales.

Labor Cost:

Labor employed directly on the contract site, regardless of the type of work performed, is considered direct labor. The wages for these workers are charged directly to the contract account or allocated through a wage analysis sheet if the laborers work on multiple contracts concurrently.

Example: For a ₹50 crore shopping mall project, ₹12 lakh is spent monthly on workers' wages. This amount is directly debited to the mall's contract account as a labor expense, ensuring accurate tracking of personnel costs.

Direct Expenses:

Any direct expenses incurred are charged directly to the contract account.

Indirect Expenses:

Indirect expenses, such as those for engineers, surveyors, and supervisors, are distributed across multiple contracts based on a percentage of the cost of materials, wages, or prime cost. For large contracts, the labor-hour method can be used to allocate these expenses.

Plant and Machinery:

The value of plant and machinery used in a contract may be debited to the contract account, and the written-down value at year-end is credited when closing the account. Alternatively, only a depreciation charge for the use of the plant may be debited to the contract account.

The treatments of plant and machinery in a contract account under different conditions have been presented below in Table 5.4:

Table 5.4 Plant and Machinery Treatment

Item	Treatment
Plant at Beginning	The value of the plant at the beginning is debited, while the plant at the end is credited.
Purchase of Machinery	Machinery purchased specifically for the contract is debited to the contract account.
Transfer of Machinery	Machinery transferred <i>to</i> the contract from other contracts is debited, while machinery transferred <i>out</i> is credited.
Sale of Machinery	The sale of machinery is credited at its selling price or market value.

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Item	Treatment
Profit/Loss on Sale	Profit on the sale of machinery is credited; loss on sale is debited to the contract account.
Loss of Machine	Loss due to theft, fire, or damage is debited, while any insurance claim accepted is credited similarly to sales.

Sub-Contract

Sub-contracting, which often involves a portion of the overall work, is a significant aspect commonly encountered in contract work. Sub-contracting may be warranted under several circumstances:

- (a) When specialized work is required that cannot be performed internally within the organization, it may be outsourced to a sub-contractor.
- (b) It may be more economical to acquire a specific part or component externally rather than manufacturing it in-house if production costs are higher.
- (c) The consideration of opportunity cost may lead management to decide against investing capital that could be better utilized in more lucrative ventures.
- (d) Limited capacity within the firm may necessitate outsourcing work to sub-contractors to adhere to project timelines.

Methods of Costing: Payments made to sub-contractors are recorded in total as direct expenses in the respective Contract Account, without the need for detailed breakdowns of the sub-contract amounts for cost accounting purposes.

For instance, ABC Constructions Ltd. gets a ₹50 crore contract to build a shopping mall. To complete the work efficiently and on time, the company decides to hire other companies to handle specific tasks.

(a) Specialized Work

The mall needs a cooling system that ABC cannot install due to a lack of expertise. They hire CoolTech Pvt. Ltd. for ₹3 crore to handle this, ensuring the system works perfectly.

(b) Cost Saving

ABC needs steel beams for the building. Making these beams themselves would cost ₹5 lakh each, but SteelMasters Pvt. Ltd. offers them for ₹4 lakh each. ABC hires SteelMasters to supply 100 beams for ₹4 crore, saving ₹1 crore.

(c) Better Opportunity

ABC could do the landscaping for ₹1 crore themselves, but this would tie up workers and machines needed for another project worth ₹10 crore. They hire GreenScape Pvt. Ltd. for the landscaping, allowing them to take on the more profitable project.

(d) Meeting Deadlines

The project must finish in 18 months, but ABC's team can only handle 70% of the work on time. They hire DesignBuild Pvt. Ltd. for ₹2 crore to design and build the mall's front, helping them finish the project on schedule and avoid penalties.

Surveyor's Certificate and Retention Money:

For long-duration contracts, it is standard practice for the contractor's firm to receive 'on account' payments based on the work completed. The payment amount depends on the work certified by the technical assessor, referred to as surveyor's certificates. Generally, these payments are not made for the entirety of the work completed; a small percentage is retained as retention money, which is payable upon contract completion. This retention money acts as a safeguard for the contractee should the contractor fail to meet one or more contract conditions.

Example: In a ₹100 crore bridge project, the surveyor certifies that 40% of the work, valued at ₹40 crore, is complete. The contractor is paid ₹36 crore, while 10% (₹4 crore) is retained by the client as retention money for quality assurance.

Defective Work

The contractee will not pay for defective work, but the costs associated with such work should be charged to the Contract Account. In some instances, the contractor may be required to rectify defective work at their expense; the costs of such rectification should also be charged to the Contract Account but presented separately.

Example: If a contractor uses substandard concrete in a building project, requiring ₹10 lakh to rebuild defective walls, this expense is separately debited to the project's contract account.

Work-in-Progress

In Contract Accounts, the value of work-in-progress comprises:

- (a) the costs of completed work, both certified and uncertified,
- (b) the costs of work not yet finished, and
- (c) the amount of profit recognized as a credit.

On the Balance Sheet, work-in-progress is typically classified into two categories: certified and uncertified. The costs of completed and certified work, along with credited profit, are recorded under 'certified' work-in-progress, while completed work that has yet to be certified, as well as costs associated with labor, materials, and expenses for work not yet completed, are recorded under 'uncertified' work-in-progress.

Example: A company undertakes a contract to construct a building for ₹50,00,000. The following information is available at the end of the year:

- Work Certified = ₹20,00,000

N O T E S

- Progress Payments Received = ₹15,00,000
- Materials at Site = ₹2,50,000
- Total Costs Incurred = ₹18,00,000

What is the value of Work-in-Progress to be shown in the Balance Sheet at year-end?

Sol.

Work-in-Progress = Work Certified + Work Uncertified - Progress Payments Received

$$\text{Work-in-Progress} = ₹20,00,000 + ₹2,50,000 - ₹15,00,000$$

$$\text{Work-in-Progress} = ₹7,50,000 - ₹2,50,000 = ₹5,00,000$$

Profit on Incomplete Contracts

To determine the portion of notional profit to be transferred to the Profit and Loss Account, contracts are categorized as follows:

(A) Contracts that have just started: For these contracts, no portion of the notional profit will be transferred to the Profit and Loss Account, and the entire amount will be reserved. There are no strict criteria to determine whether a contract is just started, reasonably advanced, or nearly complete. However, generally, contracts with less than one-fourth of the work completed are considered to be in the initial stages.

(B) Contracts that have reasonably advanced: For these contracts, the profit transferred to the Profit and Loss Account from notional profit is determined by the contract's completion degree. This degree of completion is assessed by comparing the work certified with the contract price.

- (a) If the completion degree is less than or equal to one-fourth, no portion of the notional profit will be transferred to the Profit and Loss Account, and the entire amount will remain reserved.
- (b) If the completion degree is greater than one-fourth but less than one-half, one-third of the notional profit will be transferred to the Profit and Loss Account, with the remainder kept as reserve.
- (c) If the completion degree is one-half or more, two-thirds of the notional profit will be transferred to the Profit and Loss Account, and the remaining amount will be reserved. The profit calculated in this manner will further be adjusted according to the ratio of cash received to work certified. The formula is as follows:

$$(\text{Notional Profit} \times \frac{2}{3} \text{ or } \frac{1}{3} \text{ (as applicable)}) \times \frac{\text{Cash received}}{\text{Work certified}}$$

(C) Contracts those are nearly complete: For these contracts, the profit portion transferred to the Profit and Loss Account is calculated using the estimated total profit, which is derived by subtracting the total costs incurred to date and any additional costs required to complete the contract

from the contract price. The various formulas for calculating profit in this context include:

- (i) Estimated Profit $\times \frac{\text{Work certified}}{\text{Contract price}}$
- (ii) Estimated Profit $\times \frac{\text{Work certified}}{\text{Contract price}} \times \frac{\text{Cash received}}{\text{Work certified}}$
- (iii) Estimated Profit $\times \frac{\text{Total cost to date}}{\text{Total cost}}$
- (iv) Estimated Profit $\times \frac{\text{Total cost to date}}{\text{Total cost}} \times \frac{\text{Cash received}}{\text{Work certified}}$

Example 1: Given Data

- Total Contract Price: ₹2,50,000
- Expenditure to Date: ₹1,20,000
- Estimated Further Expenditure to Complete the Contract (including contingencies): ₹30,000
- Work Certified: ₹1,60,000
- Work Uncertified: ₹12,000
- Cash Received: ₹1,44,000

Step 1: Calculate Estimated Profit on the Contract

1. Total Contract Price = ₹2,50,000
2. Total Estimated Expenditure = Expenditure to Date + Estimated Further Expenditure
 $= ₹1,20,000 + ₹30,000$
 $= ₹1,50,000$
3. Estimated Profit = Total Contract Price – Total Estimated Expenditure
 $= ₹2,50,000 - ₹1,50,000$
 $= ₹1,00,000$

Step 2: Methods to Compute Profit to be Transferred to the Profit & Loss Account

1st Method: Based on Work Certified

$$\text{Profit to be Transferred} = \text{Estimated Profit} \times \frac{\text{Work certified}}{\text{Contract price}} = 1,00,000 \times \frac{₹1,60,000}{₹2,50,000}$$

$$\text{Profit} = ₹64,000$$

N O T E S

2nd Method: Based on Work Certified and Cash Received

$$\text{Profit to be Transferred} = \text{Estimated Profit} \times \frac{\text{Work certified}}{\text{Contract price}} \times \frac{\text{Cash received}}{\text{Work certified}}$$

$$= 1,00,000 \times \frac{1,60,000}{2,50,000} \times \frac{1,44,000}{1,60,000}$$

$$\text{Profit} = ₹57,600$$

3rd Method: Based on Cost of Work to Date

$$\text{Profit to be Transferred} = \text{Estimated Profit} \times \frac{\text{Total cost to date}}{\text{Total cost}} = 1,00,000 \times \frac{1,20,000}{1,50,000}$$

$$\text{Profit} = ₹80,000$$

4th Method: Based on Cost of Work to Date and Cash Received

$$\text{Profit to be Transferred} = \text{Estimated Profit} \times \frac{\text{Total cost to date}}{\text{Total cost}} \times \frac{\text{Cash received}}{\text{Work certified}}$$

$$= 1,00,000 \times \frac{1,20,000}{1,50,000} \times \frac{1,44,000}{1,60,000}$$

$$\text{Profit} = ₹72,000$$

5.4.3 COST PLUS CONTRACT

A **Cost Plus Contract** is a type of contract where the contract's value is determined by adding a specified profit margin over the total cost of the project. This system is typically used when it is challenging for the contractor to provide a fixed contract price due to a lack of precedent. The profit for the contractor may be a fixed sum or a percentage of the cost or capital employed. Such contracts are usually used for the production of specialized items not typically manufactured and are often employed when the contractee is the government. In these contracts, there is usually a clear agreement between the contractor and contractee regarding the costs to be included, the type of materials, labor rates, allowable wastages, and the profit margin.

Advantages of Cost Plus Contract:

1. Ensures the contractor earns a reasonable profit, even in high-risk projects.
2. Simplifies the process of tendering and quoting.
3. Incorporates escalation clauses to protect the contractor from fluctuations in costs and production elements.

4. Guarantees the customer pays only a fair profit margin.
5. Provides the customer with the right to conduct cost audits to ensure transparency and fairness.

Disadvantages of Cost Plus Contract:

1. The contractor, assured of a profit margin, may lack the motivation to reduce costs or improve efficiency by minimizing wastages.
2. The final price cannot be determined until the project is completed, which delays the customer's budget planning.
3. The customer ends up paying not only for the high costs incurred but also for the corresponding high profit margin, resulting in significant expenses due to the contractor's lack of cost-consciousness and efficiency.

Escalation Clause: An Escalation Clause is a provision in a contract that allows for an adjustment in the contract price if, during the execution of the contract, the cost of materials, labor, or other factors rises beyond a specified threshold. When this happens, the contract price is increased by a predetermined amount or percentage. The purpose of the Escalation Clause is to protect the contractor from unforeseen cost increases during the project.

5.4.4 WORK CERTIFIED

When a contract remains incomplete at the end of an accounting period, the progress of the work must be evaluated by an architect. The work-in-progress for the incomplete contract is divided into two categories: Value of Work Certified and Cost of Work Uncertified.

Work Certified refers to the portion of the completed work that has been approved by the contractee's architect and is valued at the contract price.

The **Value of Work Certified** is calculated as:

$$\text{Value of Work Certified} = \frac{\text{Cash Received}}{\text{Cash Received as a \% of work certified}}$$

OR

$$\text{Value of Work Certified} = \text{Value of Contract} \times \text{Work certified (\%)}$$

OR

$$\text{Cost of Work Certified} = \text{Cost of work to date} - (\text{Cost of work uncertified} + \text{Material in hand} + \text{Plant at site})$$

This value is debited to the Contractee's Account or Work-in-progress Account and credited to the Contract Account. It is carried forward to the next accounting year on the debit side of the Contract Account and is also reported as an asset on the balance sheet.

N O T E S

Example: Shyam Constructions Ltd. undertook a contract for the construction of a commercial building. The contract work commenced on 1st April 2022, and the following information was available for the financial year ended 31st March 2023:

Particulars	Amount (₹'000s)
Contract Price	40,000
Work Certified	22,000
Progress Payment Received	16,000
Materials Issued to Site	9,000
Planning and Estimating Costs	1,200
Direct Wages Paid	5,500
Materials Returned from Site	300
Plant Hire Charges	2,100
Wages-Related Costs	750
Site Office Costs	850
Head Office Expenses (Apportioned)	450
Direct Expenses Incurred	1,100
Work Not Certified	250

Shyam Constructions Ltd. used its own plant for the contract, which originally cost ₹25 lakhs. The plant has been in use throughout the year, and its expected residual value after 5 years is ₹5 lakhs. Depreciation is calculated using the straight-line method. As of 31st March 2023, direct wages due and payable amounted to ₹3,00,000, and the materials at the site were valued at ₹2,50,000.

Solution**Dr. Contract A/c for the Year Ended 31st March 2023 Cr.**

Particulars	Amount (₹'000s)	Particulars	Amount (₹'000s)
To Materials Issued	9,000	By Materials Returned	300
To Direct Wages Paid	5,500	By Materials at Site	250
To Wages-Related Costs	750	By Work-in-Progress A/c	
To Direct Expenses	1,100	Work Certified	22,000
To Plant Hire Charges	2,100	Work Uncertified	250

N O T E S

Particulars	Amount (₹'000s)	Particulars	Amount (₹'000s)
To Planning and Estimation Costs	1,200		
To Site Office Costs	850		
To Head Office Expenses	450		
To Depreciation of Plant (See below)	400		
To Direct Wages Accrued	300		
To Notional Profit c/d	4,900		
Total	25,650	Total	25,650

Calculation of Depreciation on Plant

Original cost of plant = ₹25,00,000

Residual value = ₹5,00,000

Useful life = 5 years

Depreciation per year = $(₹25,00,000 - ₹5,00,000)/5 = ₹4,00,000$

Calculation of Notional Profit

Particulars	Amount (₹'000s)
Total Revenue (Work Certified + Work Uncertified)	22,250
Total Costs Incurred (Sum of all expenses)	20,750
Notional Profit (Revenue – Costs)	4,900

Calculation of Profit to be Transferred to Profit & Loss A/c

Since the contract is between **50% and 90% complete**, we use the following formula to calculate the amount of profit to be transferred to the Profit & Loss A/c:

$$\text{Profit to P\&L A/c} = \text{Notional Profit} \times \frac{2}{3} \times \frac{\text{Cash received}}{\text{Work certified}}$$

$$\text{Profit to P\&L A/c} = 4,900 \times \frac{2}{3} \times \frac{16,000}{22,000} = 2,375$$

The remaining profit is transferred to Reserve.

Particulars	Amount (₹'000s)
Profit Transferred to P&L A/c	2,375
Transferred to Reserve	2,525
Total Notional Profit	4,900

N O T E S



SELF-ASSESSMENT QUESTIONS

7. Radha Constructions Ltd. has a contract for ₹75,00,000. The following information is provided for the financial year:

- | | |
|--|---|
| ☐ Work Certified = ₹45,00,000 | ☐ Overheads = ₹5,00,000 |
| ☐ Work Uncertified = ₹3,00,000 | ☐ Materials at Site = ₹1,50,000 |
| ☐ Direct Wages = ₹18,00,000 | ☐ Direct Wages Outstanding = ₹1,00,000 |
| ☐ Materials Issued = ₹15,00,000 | ☐ Plant Hire Charges = ₹2,50,000 |
| ☐ Cash Received = ₹40,00,000 | |

The company uses its own plant worth ₹30,00,000 with a residual value of ₹5,00,000 and a useful life of 5 years. Depreciation is calculated using the straight-line method. Calculate the profit to be transferred to the Profit & Loss Account.

- a. ₹88,999 b. ₹77,888 c. ₹88,888 d. ₹99,888

8. A construction company takes on a contract for ₹1,00,00,000. The following details are available for the year:

- ☐ Total cost incurred = ₹60,00,000
- ☐ Work Certified = ₹75,00,000
- ☐ Work Uncertified = ₹5,00,000
- ☐ Cash received = ₹60,00,000

If 60% of the Notional Profit is transferred to the Profit & Loss A/c, what is the amount to be transferred?

- a. ₹12,00,000 b. ₹12,60,000 c. ₹13,00,000 d. ₹13,80,000

9. What does “work certified” refer to in the context of incomplete contracts?

- a. The portion of work that has been approved and valued by the contractee’s architect
- b. The entire project completed
- c. The estimated total profit of the contract
- d. Unapproved work that has not yet been finished

10. What happens to the profit transferred to the Profit & Loss Account for contracts that are just started?

- a. A portion of the notional profit is transferred
- b. The entire notional profit is transferred
- c. No portion of the notional profit is transferred
- d. All profits are held until the contract is completed

5.5 ADDITIONAL SOLVED PROBLEM

Ques 1. A furniture manufacturing company has received an order to produce custom-designed office desks for a client. You are required to prepare a cost estimate for the job. The production department provides the following material requirements:

- ❑ Wood (premium quality): 50 square feet @ ₹500 per square foot
- ❑ Steel frames and other fittings: ₹20,000
- ❑ Varnish and finishing material: ₹8,000

Since the job requires detailed craftsmanship, the company decides to hire a freelance carpenter for the job at a cost of ₹18,000. The internal workforce required for the job includes:

- ❑ Senior carpenter (₹25,000 per month): 100 hours
- ❑ Junior carpenter (₹18,000 per month): 120 hours
- ❑ Supervisor (₹15,000 per month): 50 hours

Packaging materials for safe delivery will cost ₹6,500. Production Overheads are expected to be 50% of the direct cost, and Selling & Distribution Overheads are estimated at 15% of the Production Cost. The company expects a profit margin of 20% on the quoted price. The company operates 26 days per month and 8 hours per day. You are required to prepare a quotation for the given job.

Sol.

Items	Amount (₹)
Direct Material Required	
Wood (50 sq. ft. @ ₹500 per sq. ft.)	25,000
Steel frames and other fittings	20,000
Varnish and finishing material	8,000
Primary packing material	6,500
Total Direct Material Cost	59,500
Direct Labor Spent	
Freelance carpenter	18,000
Senior carpenter: $[\text{₹}25,000 / (26 \times 8)] \times 100$	12,019
Junior carpenter: $[\text{₹}18,000 / (26 \times 8)] \times 120$	10,385
Supervisor: $[\text{₹}15,000 / (26 \times 8)] \times 50$	3,606
Total Direct Labor Cost	44,010
Prime Cost	1,03,510
Factory Overheads (50% on Direct Cost)	51,755
Production Cost	1,55,265
S&D Overheads (15% on Production Cost)	23,290

N O T E S

Items	Amount (₹)
Total Cost	1,78,555
Profit (20% on price, i.e., 25% on cost)	44,640
Price to be Quoted	2,23,195

Ques. 2. Elite Stationers (ES) is a company that manufactures customized notebooks. ES has a production setup where a maximum of 100 notebooks can be produced in a single batch. A customer has placed an order for 1,000 customized notebooks. To process a batch of 100 notebooks, the following costs would be incurred:

- ❑ Direct materials: ₹1,200
- ❑ Direct wages: ₹200
- ❑ Machine setup cost: ₹300

ES applies production overheads at a rate of 30% of direct wages cost. Additionally, 12% is added to the total production cost of each batch to cover selling, distribution, and administration overheads. The company requires a profit margin of 20% of the sales value.

Determine the selling price for 1,000 customized notebooks.

Sol.

No. of batches = 1,000 units ÷ 100 units = 10 batches

Statement of Cost per Batch and per Order

Particulars	Cost per Batch (₹)	Total Cost (₹)
Direct Material Cost	1,200.00	12,000
Direct Wages	200.00	2,000
Machine Setup Cost	300.00	3,000
Add: Production Overheads (30% of Direct Wages)	60.00	600
Total Production Cost	1,760.00	17,600
Add: S&D and Administration Overheads (12% of Total Production Cost)	211.20	2,112
Total Cost	1,971.20	19,712
Add: Profit (1/4 of Total Cost)	657.07	6,570.70
Selling Price	2,628.27	26,282.70

Selling Price per Unit = 2,628.27 ÷ 100
= ₹26.28

Ques. 3. M/s. XYZ Components Ltd. is committed to supply 60,000 gears per annum to M/s. ABC Machinery on a steady daily basis. It is estimated that it costs ₹1.50 as inventory holding cost per gear per month, and the setup cost per run of gear manufacturing is ₹4,000.

1. Determine the optimum run size of gear manufacturing.

2. State what would be the interval between two consecutive optimum runs.

3. Find out the minimum inventory holding cost.

Sol. (i) Optimal batch size or Economic Batch Quantity (EBQ):

$$EBQ = \sqrt{\frac{2AS}{C}}$$

Where:

A = Annual demand = 60,000 gears

S = Setup cost per run = ₹4,000

C = Holding cost per unit per month = ₹1.50 × 12 = ₹18 per annum

Substituting the values:

$$EBQ = \sqrt{\frac{2 \times 60000 \times 4000}{18}} = \sqrt{\frac{48,00,00,000}{18}} = \sqrt{2,66,666.67} = 5,163.97 \approx$$

5,165 units

$$(ii) \text{ Number of Optimum runs} = \frac{60,000}{5,165} \approx 11.6 \text{ or } 12 \text{ runs}$$

$$\text{Interval between 2 runs (in days)} = \frac{365 \text{ days}}{12} \approx 30.4 \text{ days}$$

(iii) Minimum Inventory Cost = Average Inventory × Inventory Carrying Cost per unit per annum

$$\text{Average Inventory} = \frac{5,165}{2} \approx 2,582.5 \text{ units}$$

Carrying Cost per unit per annum = ₹18

Minimum Inventory Holding Costs = 2,582.5 × 18 ≈ ₹46,485

Ques. 4. A Company has an annual demand from a single customer for 80,000 liters of a cleaning solution. The total demand can be made up of various formulations to be produced in a continuous production run, after which a set-up of the machinery will be required to accommodate the formulation change. The total output of each formulation will be stored and then delivered to the customer as a single load immediately before the production of the next formulation commences.

The setup costs are ₹150 per setup. The service is supplied by an outside company as required. The holding costs are incurred on rented storage space, which costs ₹60 per square meter per annum. Each square meter can hold 300 liters suitably stacked.

You are required to:

- ❑ Calculate the total cost per year where batches may range from 8,000 to 12,000 liters in multiples of 1,000 liters, and hence choose the production batch size which will minimize the cost.

N O T E S

- ❑ Use the economic batch size formula to calculate the batch size which will minimize total cost.

(i)

Production Batch Size (Lt.)	Set-up Costs per Annum (₹)	Holding Costs per Annum (₹)	Total Costs per Annum (₹)
8,000	1500	800	2,300
9,000	1333	900	2,233
10,000	1200	1,000	2,200
11,000	1090	1,100	2,190
12,000	1000	1,200	2,200

As the total cost is least at 11,000 liters, i.e., ₹2,190, thus the economic production lot would be 11,000 liters.

(ii) Economic Batch Quantity (EBQ):

$$EBQ = \sqrt{\frac{2AS}{C}}$$

Where,

D = Annual demand for the product = 80,000 liters

S = Setup cost per batch = ₹150 per setup

H = Carrying cost per unit of production = ₹60/300 liters = ₹0.20 per liter per annum

Substituting the values:

$$EBQ = \sqrt{\frac{2 \times 80000 \times 150}{0.2}} = \sqrt{\frac{2,40,00,000}{0.2}} = \sqrt{12,00,00,000} = 10,954.45$$

It can be seen that the EBQ determined with the mathematical formula (10,954 liters) slightly varies from the one determined by the trial and error method (11,000 liters).

Ques. 5. Compute a conservative estimate of profit on a contract (which has been 85% complete) from the following particulars:

Particulars	Amount (₹)
Total expenditure to date	26,75,000
Estimated further expenditure to complete the contract (including contingencies)	3,00,000
Contract price	38,50,000
Work certified	32,00,000
Work uncertified	2,10,000
Cash received	24,00,000

Required: Compute a conservative estimate of profit based on the above details.

Sol.

Calculation of Conservative Estimate of Profit (₹)

Particulars	Amount (₹)
Total expenditure to date	26,75,000
Estimated further expenditure to complete the contract (including contingencies)	3,00,000
Total estimated cost	29,75,000
Contract price	38,50,000
Estimated profit on contract	8,75,000

Profit to be transferred to Contract Profit and Loss A/c:

$$\text{Estimated Profit} \times \frac{\text{Cash Received}}{\text{Contract Price}} = 8,75,000 \times \frac{24,00,000}{38,50,000} = ₹ 5,45,455$$

Therefore, the estimate of profit is ₹5,45,455.

Ques. 6. The following expenses were incurred on a contract:

Particulars	Amount (₹)
Materials purchased	7,50,000
Materials drawn from stores	1,50,000
Wages	3,00,000
Plant issued	1,00,000
Chargeable expenses	90,000
Apportioned indirect expenses	35,000

The contract was for ₹25,00,000 and commenced on January 1, 2024. The value of work completed and certified up to 30th November, 2024 was ₹16,00,000, of which ₹12,80,000 was received in cash, with the balance being held back as retention money by the contractee. The value of work completed subsequent to the architect's certificate but before 31st December, 2024 was ₹75,000. Materials worth ₹55,000 were lying on the site, and the value of plant as at 31st December, 2024 was estimated to be ₹40,000.

Required:

Prepare the Contract Account, calculate the notional profit, and determine the amount to be transferred to the Profit and Loss Account based on cash received.

N O T E S

Sol.

Contract Account

Dr.

Cr.

Particulars	Amount (₹)	Particulars	Amount (₹)
To Materials purchased	7,50,000	By Work-in-progress	
To Stores issued	1,50,000	Work certified	16,00,000
To Wages	3,00,000	Work uncertified	75,000
To Plant	1,00,000	Materials unused	55,000
To Chargeable expenses	90,000	Plant	40,000
To Indirect expenses	35,000		
To Profit and Loss Account, 2/3rd of profit on cash basis	1,84,000*		
To Work-in-progress			
Balance of profit c/d	1,61,000		
Total	17,70,000	Total	17,70,000

Computation of Profit:

Particulars	Amount (₹)
Notional profit (17,70,000-14,25,000)	3,45,000
2/3rd of that since 65% of the work is complete	2,30,000
80% of that on cash basis(12,80,000/16,00,000)	1,84,000

5.6 SUMMARY

- ❑ Job costing, or specific order costing, is a method used by industries to account for costs associated with manufacturing products or providing services based on specific customer orders, treating each order as a separate cost unit.
- ❑ Various organizations utilize job costing, including jobbing concerns that create customized products, small firms that don't produce uniform lines, and large enterprises that manufacture multiple products simultaneously, such as in construction, engineering, and repair services.
- ❑ The job costing process begins with receiving an order and creating a detailed production order that specifies necessary materials and operations. Production orders are categorized into assembly or sub-assembly types and distributed to relevant departments for tracking costs related to direct materials, labor, and manufacturing overhead.
- ❑ Advantages of job costing include timely cost tracking, quick profitability assessments, and the identification of spoilage and defects, while limitations involve high clerical costs, potential errors in allocation, and reliance on historical data. Effective assessment relies

on analyzing profits from completed jobs and monitoring cost variances for informed management decisions.

- ❑ Batch costing involves treating each batch as a distinct cost unit, with costs separately accumulated for every batch containing multiple identical units, commonly used in industries producing similar articles in defined batches.
- ❑ Each batch has a unique Batch Cost Sheet, and the cost per unit is determined by dividing the total cost of the batch by the number of items produced. Economic Batch Quantity (EBQ) is the optimal batch size that minimizes the combined setup and carrying costs, calculated using the formula $EBQ = \sqrt{\frac{2AS}{C}}$, where A is annual demand, S is setup cost per batch, and C is carrying cost per unit per year.
- ❑ Contract costing, or terminal costing, is a job costing variant used for construction projects, where each contract is treated as an individual cost unit with distinct accounts to track income and expenses.
- ❑ The profit or loss from a contract is determined through a 'Contract Account,' which debits all expenses and credits income, allowing for profit assessment even when contracts span multiple accounting periods.
- ❑ Incomplete contract work is assessed by an architect, with the "Work Certified" value representing the portion of completed work approved for payment, calculated based on cash received and the contract price.

5.7 KEY WORDS

1. **Job Costing:** A cost accounting method that tracks expenses associated with specific customer orders, treating each order as a separate cost unit.
2. **Production Order:** A document that authorizes the start of manufacturing a specific product, detailing necessary materials, operations, and timelines.
3. **Work-in-Progress (WIP):** The cost associated with incomplete jobs, representing the value of materials, labor, and overhead incurred up to the reporting date.
4. **Cost Variance:** The difference between estimated costs and actual costs incurred, used to analyze performance and identify areas needing improvement.
5. **Batch Costing:** A costing method that treats each batch of identical units as a separate cost unit, with costs accumulated for each batch.
6. **Economic Batch Quantity (EBQ):** The optimal batch size that minimizes the total costs of setup and carrying by balancing production and inventory holding expenses.

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7. **Contract Costing:** A costing technique used for construction projects, where each contract is treated as a unique cost unit with separate accounts for tracking expenses and income.
8. **Work Certified:** The value of completed work approved by the contractee's architect, which is recognized for payment and reflects progress on an incomplete contract.
9. **Setup Costs:** The expenses incurred to prepare for production, including labor, materials, and equipment, which decrease as batch size increases.
10. **Carrying Costs:** The costs associated with storing and maintaining inventory, which rise with larger batch sizes due to increased average inventory levels.

5.8 CLOSING CASELET

Construction of the Riverside Bridge

Skyline Contractors, established in 2015 in Riverside, California, specializes in infrastructure projects, including roads and bridges. The company employs Contract Costing, or Terminal Costing, to effectively manage the accounting processes associated with its contracts. In 2023, Skyline secured a \$5 million contract from the Riverside City Council to construct the Riverside Bridge. The contract included an escalation clause to address potential material price fluctuations and established staged payments based on work completed, certified by an overseeing architect. To ensure quality, the city council retained 10% of each payment as retention money for six months after project completion.

For the Riverside Bridge project, Skyline Contractors created a distinct contract account, assigning it a unique identification number to track all related costs. Direct material costs, such as the \$1 million spent on steel beams and \$600,000 on labor, were recorded directly against the contract. Additionally, overhead costs, amounting to \$200,000, were included but represented a smaller portion of the total expenses. As costs were incurred, Skyline meticulously maintained records to facilitate accurate profit and loss calculations at the end of each accounting period.

By the end of the first year, Skyline Contractors had incurred total costs of \$1.8 million, with certified work valued at \$2 million, resulting in a notional profit of \$200,000. A portion of this profit was reserved for contingencies, while the remainder was transferred to the Profit & Loss Account. This systematic tracking of costs not only allowed Skyline to manage its financial performance effectively but also fostered a strong relationship with the Riverside City Council, culminating in the successful completion of the Riverside Bridge within budget and on schedule.

**QUESTIONS**

1. What are the key components included in the contract account for the Riverside Bridge project, and how do they contribute to calculating the overall project cost?

(**Hint:** Consider the direct material costs, labor costs, overhead expenses, and how each category is recorded in the contract account.)

2. In what ways did the use of Contract Costing benefit Skyline Contractors in managing the Riverside Bridge project?

(**Hint:** Think about advantages of contract costing.)

5.9 HIGHER ORDER THINKING SKILLS QUESTIONS (HOTS)

1. Considering the advantages and limitations of job costing, how might a manufacturing firm's choice of job costing affect its strategic decision-making regarding product pricing, profitability assessment, and operational efficiency? Provide examples to support your analysis.
2. Compare and contrast the applicability and effectiveness of job costing in a construction company versus a small-scale furniture manufacturer. What specific challenges might each industry face in implementing job costing, and how could these challenges influence their overall cost control strategies?
3. How would a significant increase in carrying costs affect the decision-making process for determining the Economic Batch Quantity (EBQ) in a manufacturing setting? Discuss the potential trade-offs that management might face.
4. Compare the advantages and disadvantages of using batch costing versus contract costing in a manufacturing organization that also engages in project-based work. In what scenarios would one method be preferred over the other, and why?
5. Given a situation where a construction company consistently exceeds its budget on fixed-price contracts, what strategies could the company implement to improve its cost estimation and management practices in future projects? Discuss how these strategies might impact overall profitability and client satisfaction.
6. A factory uses a job costing system. The following data are available from the books at the year ending on 31st March 2023:

Particulars	Amount (₹)
Direct Materials	2,00,000
Direct Wages	1,60,000
Profit	1,25,000

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Particulars	Amount (₹)
Selling and Distribution Overheads	1,10,000
Administrative Overheads	90,000
Factory Overheads	1,00,000

Required:

A. Prepare a job cost sheet

B. For the year 2023-24, the factory receives an order for several jobs. It is estimated that:

□ Direct Materials = ₹3,00,000 □ Direct Wages = ₹2,00,000

Assuming that the factory intends to earn the same rate of profit on sales as in the previous year and that selling and distribution overheads increase by 10%, calculate the price for the jobs. Factory overheads are recovered as a percentage of direct wages. Administrative and selling & distribution overheads are recovered as a percentage of works cost, based on the cost rates prevalent in the previous year.

7. A particular Job No. 200 of a company passes through three departments: A, B, and C. The following information is given regarding this job:

Particulars	Departments
Materials issued to the job (₹)	A: 10,000; B: 2,000; C: 1,000
Direct labour hours for job (No.)	A: 800; B: 1,500; C: 4,000
Rate of direct labour per hour (₹)	A: 1.50; B: 2.00; C: 2.50
Sale of scrap materials arising from job (₹)	A: 800; B: 200; C: 150
Total overheads for the departments (₹)	A: 12,000; B: 18,000; C: 28,000
Total labour hours for the departments (No.)	A: 12,000; B: 25,000; C: 35,000

Prepare the **Job Cost Sheet** for Job No. 200

8. Zenith Bearings Ltd. has an annual commitment to supply 30,000 bearings to Omega Industries on a steady basis. The company incurs an inventory holding cost of ₹0.15 per bearing per month, and the set-up cost per production run is ₹400.

Required:

1. Determine the optimal run size for bearing production.
2. If Zenith Bearings Ltd. follows a policy of manufacturing 7,500 bearings per run, calculate the additional cost incurred compared to the optimal run size calculated in part (1).
3. Calculate the minimum inventory holding cost.

9. The following expenses were incurred on a contract:

Particulars	Amount (₹)
Materials purchased	9,00,000
Materials drawn from stores	2,00,000
Wages	4,00,000
Plant issued	1,50,000
Chargeable expenses	1,20,000
Apportioned indirect expenses	50,000

The contract was for ₹30,00,000 and commenced on January 1, 2025. The following additional details are provided:

- ❑ The value of work completed and certified up to 30th November, 2025, was ₹18,00,000, of which ₹14,40,000 was received in cash, with the balance being retained as retention money.
- ❑ The value of uncertified work completed before 31st December, 2025, was ₹1,00,000.
- ❑ Materials worth ₹80,000 were lying on the site as of 31st December, 2025.
- ❑ The value of the plant at the site as of 31st December, 2025, was estimated to be ₹60,000.

Required:

1. Prepare the Contract Account for the year ending 31st December, 2025, showing:
 - ❑ Total cost of the contract.
 - ❑ Work-in-progress.
 - ❑ Profit to be transferred to the Profit and Loss Account, considering retention money.
2. Calculate the percentage of completion of the contract as of 31st December, 2025.

5.10 ANSWER KEYS

Topic	Q. No.	Answer
Job Order Costing	1	b) ₹44,000
	2	d) They are charged to the overall job and distributed across the entire batch
	3	d) ₹3,20,000
Batch Costing	4	b) They decrease due to fewer batches needed.

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Topic	Q. No.	Answer
	5	d) ₹16,000
	6	c) ₹63,61,200
Contract Costing	7	c) ₹88,888
	8	a) ₹12,00,000
	9	a) The portion of work that has been approved and valued by the contractee's architect
	10	c) No portion of the notional profit is transferred

5.11 SUGGESTED READINGS

- ❑ Anthony, R. N., & Govindarajan, V. (2016). *Management Control Systems*. McGraw-Hill Education.
- ❑ Bhimani, A., Horngren, C. T., Datar, S. M., & Rajan, M. V. (2019). *Management Accounting*. Pearson.
- ❑ Hilton, R. W., & Platt, D. E. (2019). *Managerial Accounting: Creating Value in a Dynamic Business Environment*. McGraw-Hill Education.

5.12 REFERENCES

- ❑ Institute of Management Accountants (IMA) - www.imanet.org
- ❑ Chartered Institute of Management Accountants (CIMA) - www.cima-global.com
- ❑ American Institute of Certified Public Accountants (AICPA) - www.aicpa.org

6

U N I T

INCOME RECOGNITION UNDER MARGINAL AND ABSORPTION COSTING

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INTRODUCTORY CASELET

Understanding Marginal Costing: A Practical Case Study

XYZ Electronics, a mid-sized consumer electronics manufacturer, implemented marginal costing to improve decision-making. By analyzing their cost structure, the company distinguished between variable costs (such as direct materials and labor) and fixed costs (including rent and salaries). This distinction allowed them to calculate the contribution margin for each product, ensuring that pricing strategies covered both fixed and variable costs, while also determining the break-even point.

With the selling price of each smartphone set at ₹10,000 and a contribution margin of ₹3,000, XYZ Electronics used marginal costing to assess production levels. For example, they determined that selling 67 smartphones per month was necessary to cover their ₹2,00,000 monthly fixed costs. This analysis also helped the company decide to increase production during peak demand, as each additional unit contributed directly to covering fixed costs and improving profitability.

The company further used marginal costing to control costs by identifying opportunities to reduce variable costs, such as negotiating better material prices. These efforts increased the contribution margin, enhancing profitability while ensuring that the business could respond dynamically to changes in production volume. Marginal costing, in this case, proved to be a vital tool for making informed pricing, production, and cost-control decisions.

**QUESTIONS**

1. What is the primary difference between marginal costing and absorption costing in terms of how fixed costs are treated?

(Hint: Consider how each method assigns fixed costs to products and how that affects profit calculations when inventory levels fluctuate.)

2. In what scenarios might a business prefer to use marginal costing over absorption costing for internal decision-making? Provide specific examples.

(Hint: Consider situations where marginal costing can be practically applied.)

**LEARNING OBJECTIVES**

At the end of this unit, you will be able to

- Identify and explain the key characteristics of marginal costing
- Evaluate the benefits of using marginal costing for decision-making
- Identify the limitations of marginal costing
- Recognize the advantages of absorption costing
- Identify the limitations of absorption costing
- Compare and contrast the fundamental principles of marginal costing and absorption costing

6.1 MARGINAL COSTING

Marginal Cost is described as “the amount by which total costs change when the volume of output is increased or decreased by one unit.” It can also be defined as the sum of Prime Cost and Variable Overheads. Marginal Cost is a constant figure, typically expressed as the cost per unit of output. In contrast, fixed costs, which cannot generally be attributed to individual units, represent a fixed expenditure over an accounting period. For this reason, fixed costs are also referred to as time costs, period costs, standby costs, capacity costs, or constant costs. Variable costs or marginal costs may also be known as direct costs, activity costs, volume costs, or out-of-pocket costs. From this definition and analysis, we can see that marginal cost fluctuates depending on changes in output volume. However, by definition, marginal cost represents the variation in total cost that occurs when one additional unit is produced. Economists might distinguish between marginal cost and variable cost, but for accountants, these terms are considered identical and used interchangeably. Therefore, in this context, marginal cost and variable cost are treated as the same.

Let’s say your business produces customized notebooks. The paper and ink used are variable costs—they increase with each unit produced. On the other hand, the rent for your workshop remains fixed. Using marginal costing, you’d price your notebooks by focusing on the contribution margin, ensuring each unit contributes to covering fixed costs and eventually generating profit.

Marginal costing involves “determining marginal costs and assessing the impact on profit from changes in the volume or type of output by differentiating between fixed and variable costs.” Several other terms, such as direct costing, contributory costing, variable costing, comparative costing, differential costing, and incremental costing, are used similarly to marginal costing.

Marginal costing involves classifying costs as either fixed or variable, and this distinction plays a crucial role in many managerial decisions. The core concept of marginal costing lies in separating total costs into

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fixed and variable components. Variable costs fluctuate with production volume, while fixed costs remain constant regardless of output changes. It's important to understand that unit variable cost stays consistent at different output levels, and total variable cost changes proportionally with the number of units produced. Meanwhile, total fixed costs remain unchanged regardless of output, but fixed cost per unit has an inverse relationship with the number of units produced. The key facts about marginal costing:

- ❑ **Not a Distinct Method:** Marginal costing is not a separate method of costing like job costing, process costing, or operating costing. Instead, it is a special technique used to aid managerial decision-making. Marginal costing provides a framework for interpreting cost data to evaluate the profitability of different products, processes, and cost centers during decision-making. It can be applied alongside various costing methods, such as job or process costing, and even in combination with other techniques like standard costing or budgetary control.
- ❑ **Cost Ascertainment:** Marginal costing determines costs based on their nature and behavior. It focuses on how costs behave, reflecting their impact on profitability. This technique is built on a specific understanding of the nature and behavior of costs and how they affect an organization's profitability.
- ❑ **Decision-Making:** Traditional or total cost methods classify costs functionally, unlike marginal costing. Under traditional methods, total cost is the sum of direct material, direct labor, direct expenses, manufacturing overheads, administration overheads, and selling and distribution overheads. In this system, the total cost per unit remains constant only if the output level or product mix stays the same from period to period. However, these factors fluctuate, leading to varying total costs over time. For example, the costing department may report that an item costs Rs. 20 one day and Rs. 18 the next. These changes arise from fluctuations in output volume and the behavior of fixed costs within total cost. Such variations make it difficult for management to make sound decisions. As a result, marginal costing is widely recognized as a valuable tool for decision-making.

DETERMINATION OF COST AND PROFIT UNDER MARGINAL COSTING

Under marginal costing, costs are classified into variable and fixed categories. All variable costs are considered part of the product or service, while fixed costs are charged against the contribution margin. Specimen of Cost and Profit Statement under Marginal Costing is shown in Figure 6.1:

1. **Product (Inventoriable) Costs:** For merchandise inventory, these costs relate to the purchase and sale of goods. In production, they refer to the costs of acquiring and converting materials, along with all other manufacturing inputs, into a finished product for sale. Under marginal costing, only variable manufacturing costs are considered inventoriable or product costs.

Particulars	Amount (Rs.)	Amount (Rs.)
Revenue (A)		xxx
Product Cost:		
- Direct Materials	xxx	
- Direct Employee (Labor)	xxx	
- Direct Expenses	xxx	
- Variable Manufacturing Overheads	xxx	
Product (Inventoriable) Costs (B)		xxx
Product Contribution Margin (A – B)		xxx
- Variable Administration Overheads	xxx	
- Variable Selling & Distribution Overheads	xxx	
Contribution Margin (C)		xxx
Period Cost (D):		
- Fixed Manufacturing Expenses	xxx	
- Fixed Non-Manufacturing Expenses	xxx	
Total Period Cost (D)		xxx
Profit/(Loss) (C – D)		xxx

Figure 6.1 Cost and Profit Statement under Marginal Costing

Finished goods are valued at product cost, while work-in-process (WIP) inventories are also valued based on the percentage of completion (refer to the Process & Operation Costing chapter).

- Contribution:** Contribution, or contribution margin, is the difference between sales revenue and total variable costs, whether manufacturing or non-manufacturing. It is obtained by subtracting variable costs from sales revenue.

$$\text{Contribution (C)} = \text{Sales Revenue (S)} - \text{Total Variable Cost (V)}$$

Example:

Variable Cost = Rs.60,000,

Fixed Cost = Rs. 25,000,

Selling Price = Rs. 95,000

Contribution = Selling Price – Variable Cost = 95,000 – 60,000 = 35,000

Profit = Contribution – Fixed Cost = 35,000 – 25,000 = Rs. 10,000

Since the contribution exceeds the fixed cost, the profit is of the magnitude of Rs.10,000. Suppose the fixed cost is Rs. 45,000, then the position shall be:

$$\begin{aligned} \text{Contribution} - \text{Fixed Cost} &= \text{Profit or,} \\ &= 35,000 - 45,000 = - \text{Rs.10,000} \end{aligned}$$

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The amount of 10,000 represents the extent of loss since the fixed costs are more than the contribution. At the level of fixed cost of 35,000, there shall be no profit and no loss.

Contribution represents the excess of sales revenue over variable costs. The concept assumes that the profit and fixed expenses of a business are 'joint costs,' which cannot be easily allocated to different business segments. Therefore, contribution is used as a measure of operational efficiency across various business segments.

- 3. Period Costs:** These costs are not assigned to products but are treated as expenses for the period in which they are incurred. In marginal costing, all fixed costs, whether related to manufacturing or non-manufacturing, are recognized as period costs.

Example

Shyam Industries Ltd. produces three products: X, Y, and Z. Each of these products requires the same raw material, but the available supply of raw material is limited to 72,000 kg. The following information is provided from the company's books and records:

Particulars	Product X	Product Y	Product Z
Selling Price per Unit (₹)	150	180	120
Variable Cost per Unit (₹)	110	140	90
Raw Material Requirement per Unit (kg)	6	9	4
Market Demand (Units)	6,000	4,000	5,000

Task:

Advise Shyam Industries Ltd. on the most profitable product mix, and compute the resulting profit if the fixed costs amount to ₹2,00,000.

Solution:

Since the raw material is limited to 74,000 kg, it is not possible to meet the full market demand for all three products. Therefore, the company must prioritize production based on the contribution per kg of raw material. The following statement is prepared to determine the priority of the products.

I. Statement Showing the Contribution per Unit and per Kg of Raw Material

Particulars	Product X	Product Y	Product Z
Selling Price per Unit (₹)	150	180	120
Less: Variable Cost per Unit (₹)	110	140	90
Contribution per Unit (₹)	40	40	30
Raw Material Requirement per Unit (kg)	6	9	4
Contribution per kg of Raw Material (₹)	$40 \div 6 = 6.67$	$40 \div 9 = 4.44$	$30 \div 4 = 7.50$
Priority	II	III	I

II. Statement Showing the Optimum Product Mix and the Resulting Profit

Product (In order of priority)	Number of Units to be Produced	Raw Material Requirement per Unit (kg)	Total Raw Material Consumption (kg)	Contribution per Unit (₹)	Total Contribution (₹)
Z	5,000	4 kg	20,000 kg	30	1,50,000
X	6,000	6 kg	36,000 kg	40	2,40,000
Y	2,000** (Balance raw material)	9 kg	18,000 kg	40	80,000
Total	-	-	74,000 kg	-	₹4,70,000

Explanation:

- ❑ Product Z is produced to its full market demand, consuming 20,000 kg of raw material.
- ❑ Product X is produced to its full market demand, consuming 36,000 kg of raw material.
- ❑ After producing Z and X, the remaining raw material is 74,000 kg – (20,000 kg + 36,000 kg) = 18,000 kg.
- ❑ Product Y requires 9 kg of raw material per unit. With the remaining 18,000 kg, only 2,000 units of Product Y can be produced ($18,000 \div 9 = 2,000$ units).

III. Calculation of Profit

The total contribution is calculated as:

$$\text{Total Contribution} = ₹1,50,000 \text{ (Z)} + ₹2,40,000 \text{ (X)} + ₹80,000 \text{ (Y)} = ₹4,70,000$$

The profit is calculated as:

$$\text{Profit} = \text{Total Contribution} - \text{Fixed Cost}$$

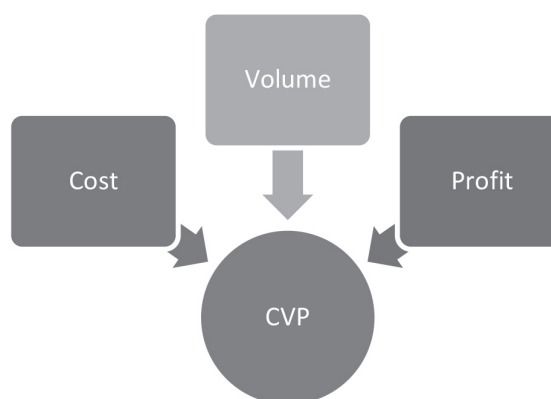
$$\text{Profit} = ₹4,70,000 - ₹2,00,000$$

$$\text{Profit} = ₹2,70,000$$

COST-VOLUME-PROFIT (CVP) RELATIONSHIP

CVP analysis (figure 6.2) examines the relationship between variable costs, production volume, and profit. It focuses on understanding how costs, revenue, and activity levels interact to influence a company's profit. The analysis helps identify how changes in cost and volume affect profitability.

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**Figure 6.2** CVP Analysis

The key factors influencing cost, volume, and profit include:

- | | |
|---|---|
| ☐ Volume of production | ☐ Raw material costs |
| ☐ Employee efficiency | ☐ Price levels |
| ☐ Plant size | ☐ Other external factors |
| ☐ Production methods | |

Volume of production is influenced by:

- | | |
|---|--|
| ☐ Product cost | ☐ Price levels |
| ☐ Plant size | ☐ Other factors |
| ☐ Availability of inputs | |

Profits depend on both cost and volume. CVP analysis aids in assessing how changes in cost and production volume impact the firm's profits.

Uses of CVP Analysis:

1. Estimating profits at different production levels.
2. Developing flexible budgets that adapt to changes in volume.
3. Supporting decision-making by providing insights on cost and profit relationships.
4. Evaluating the impact of costs on overall profitability.
5. Setting appropriate selling prices based on cost and profit analysis.

Let's consider a company, XYZ Ltd., that manufactures and sells widgets. The company is evaluating how changes in production volume will impact its profits.

Key Information:

- ☐ Selling price per widget: ₹100
- ☐ Variable cost per widget: ₹40
- ☐ Fixed costs: ₹3,00,000 per year
- ☐ Current production and sales: 50,000 widgets per year

Step 1: Calculate Contribution Margin

The contribution margin is the difference between the selling price and the variable cost per unit.

Contribution Margin per widget = Selling Price – Variable Cost

Contribution Margin per widget = ₹100 – ₹40 = ₹60

Step 2: Break-even Point

The break-even point is the level of sales at which total revenue equals total costs, resulting in no profit or loss.

Break-even Point (in units) = Fixed Costs / Contribution Margin per unit

Break-even Point = ₹3,00,000 / ₹60 = 5,000 widgets

This means XYZ Ltd. needs to sell 5,000 widgets to cover its fixed costs.

Step 3: Profit at Different Sales Levels

Now, let's evaluate the profit at different production levels.

1. At current sales of 50,000 widgets:

- ❑ Total Revenue = 50,000 widgets × ₹100 = ₹50,00,000
- ❑ Total Variable Costs = 50,000 widgets × ₹40 = ₹20,00,000
- ❑ Total Fixed Costs = ₹3,00,000
- ❑ Profit = Total Revenue – Total Variable Costs – Fixed Costs
- ❑ Profit = ₹50,00,000 – ₹20,00,000 – ₹3,00,000 = ₹27,00,000

2. At increased sales of 60,000 widgets:

- ❑ Total Revenue = 60,000 widgets × ₹100 = ₹60,00,000
- ❑ Total Variable Costs = 60,000 widgets × ₹40 = ₹24,00,000
- ❑ Total Fixed Costs = ₹3,00,000
- ❑ Profit = ₹60,00,000 – ₹24,00,000 – ₹3,00,000 = ₹33,00,000

Step 4: Impact of Volume Increase on Profit

By increasing production from 50,000 to 60,000 widgets, the company's profit increases by ₹6,00,000. This illustrates how volume directly impacts profitability.

Step 5: Decision-Making

If XYZ Ltd. wants to increase its profit, it can either increase production or find ways to reduce costs. CVP analysis shows that increasing production will result in higher profits due to the contribution margin from each additional widget sold.

Example 1: Shyam Ltd. sells a product for ₹50 per unit. The variable cost per unit is ₹30, and the total fixed costs for the period amount to ₹40,000.

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Required:

1. Calculate the break-even point (BEP) in units.
2. Calculate the profit if the company sells 3,000 units.

Solution:**1. Break-Even Point (BEP) in Units**

$\text{BEP (units)} = \text{Fixed Costs} / (\text{Selling Price per unit} - \text{Variable Cost per unit})$

$$\text{BEP (units)} = ₹40,000 / (₹50 - ₹30) = ₹40,000 / ₹20 = 2,000 \text{ units}$$

2. Profit Calculation for 3,000 units

$\text{Profit} = \text{Total Sales} - \text{Total Variable Costs} - \text{Fixed Costs}$

$$\text{Total Sales} = 3,000 \times ₹50 = ₹1,50,000$$

$$\text{Total Variable Costs} = 3,000 \times ₹30 = ₹90,000$$

$$\text{Profit} = ₹1,50,000 - ₹90,000 - ₹40,000 = ₹20,000$$

6.1.1 FEATURES OF MARGINAL COSTING

The technique of marginal costing is based on the distinction between product costs and period costs. Only variable costs are regarded as the costs of the products, while fixed costs are treated as period costs incurred during the period regardless of the volume of output. The key features of marginal costing are as follows:

- ❑ All elements of cost are classified into fixed and variable components, and semi-variable costs are also analyzed into fixed and variable elements.
- ❑ Marginal or variable costs, such as direct materials, direct labor, and variable factory overheads, are treated as the cost of the product.
- ❑ Under marginal costing, the value of finished goods and work-in-progress consists solely of marginal costs. Variable selling and distribution costs are excluded from the valuation of these inventories, and fixed costs are not considered for the valuation of closing stock of finished goods and closing work-in-progress.
- ❑ Fixed costs are treated as period costs and charged to the profit and loss account for the period in which they are incurred.
- ❑ Prices are determined with reference to marginal costs and contribution margins.
- ❑ Profitability of departments and products is assessed based on their contribution margins.

6.1.2 ADVANTAGES OF MARGINAL COSTING

1. **Simplicity of Operation:** Marginal costing is straightforward to implement, as it avoids the complexities involved in overhead apportionment and recovery.

2. **Clarity in Overhead Absorption:** This system eliminates the need to explain the purpose and basis of overhead absorption, making it easier to attribute profit fluctuations to cost-volume interactions rather than inventory valuation changes.
3. **Facilitated Decision-Making:** Marginal costing simplifies decision-making by clearly identifying which products contribute positively and which fail to cover avoidable (variable) costs.
4. **Effective Cost Analysis and Control:** Marginal costing is a valuable tool for cost analysis and presentation, helping management at various levels to control costs effectively.
5. **Future Profit Planning:** The marginal cost approach aids in profit planning by leveraging contribution and marginal cost ratios to assess changes in selling prices, variable costs, and more.
6. **Performance Evaluation:** In organizations with diverse units and products, marginal costing helps evaluate the performance of each segment effectively.
7. **Beneficial for Forecasting:** Marginal costing is a powerful tool for forecasting future business conditions and outcomes.
8. **Optimum Product Mix:** This method assists in determining the optimal production quantity for each product, ensuring the most profitable product mix.
9. **Optimum Sales Mix:** Marginal costing also helps establish the ideal sales levels for each product to maximize overall profits.
10. **Support for Managerial Decisions:** Marginal costing supports various managerial decisions, including:
 - ☐ Make or buy decisions,
 - ☐ Exploring foreign markets,
 - ☐ Deciding whether to accept an order,
 - ☐ Determining selling prices under various conditions,
 - ☐ Replacing one product with another,
 - ☐ Optimizing the use of labor or machine hours,
 - ☐ Evaluating alternative choices,
 - ☐ Deciding whether to subcontract certain production processes,
 - ☐ Making decisions about business expansion,
 - ☐ Considering diversification,
 - ☐ Evaluating whether to shut down or continue operations.

6.1.3 LIMITATIONS OF MARGINAL COSTING

1. **Separation of Fixed and Variable Costs:** The division of costs into fixed and variable categories presents technical difficulties, as no cost is purely variable or completely fixed.

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2. **Understatement of Stock Valuations:** In the marginal cost system, finished goods and work-in-progress stocks tend to be understated because fixed costs, which are integral to product manufacturing, are excluded from product costs.
3. **Impact on Financial Statements:** Excluding fixed overheads from inventories affects the Profit and Loss Account, leading to an unrealistic and overly conservative Balance Sheet unless adjusted at the period's end.
4. **Profit Fluctuations:** Profits under marginal costing vary with sales volume, especially during seasonal sales, causing monthly operating statements to be less accurate compared to absorption costing.
5. **Early Recession Misinterpretation:** Low profits or losses reflected during early recessions may unnecessarily alarm management, leading to counterproductive decisions that exacerbate market conditions.
6. **Limited Information Scope:** Marginal costing does not account for factors such as equipment overuse, resource expansion, or labor replacement by machines, which influence production and sales.
7. **Suitability for Short-Term vs. Long-Term Analysis:** While marginal costing helps in short-term profitability assessments, long-term profit determination requires a full cost basis for accuracy.
8. **Issues with Variable Overhead:** Though marginal costing minimizes fixed overhead complications, similar challenges persist with the apportionment and absorption of variable overhead.
9. **Importance of Fixed Costs in Automation:** With increasing automation, fixed costs often outweigh variable costs in significance, making the exclusion of fixed costs a limitation in marginal costing systems.
10. **Lack of Performance Standards:** Marginal costing does not provide performance benchmarks. Budgetary control and standard costing systems offer more robust control and evaluation.

Table 6.1 below contrasts the limitations of marginal costing with the advantages of absorption costing:

Table 6.1 Difference between Marginal Costing and Absorption Costing

Aspect	Marginal Costing - Limitations	Absorption Costing - Advantages
Cost Allocation	Excludes fixed costs from product costs, leading to understated inventory and work-in-progress values.	Includes both fixed and variable costs, ensuring a comprehensive allocation of production costs.
Inventory Valuation	Understates inventory value as fixed costs are not included, affecting balance sheet accuracy.	Accurately values inventory by including a portion of fixed overhead, aligning with accounting standards.

Aspect	Marginal Costing - Limitations	Absorption Costing - Advantages
Profit Calculation	Marginal contribution and profits fluctuate significantly with sales volume, making profit analysis less stable.	Provides a stable view of profitability by spreading fixed costs across units, regardless of sales fluctuations.
Short-Term Focus	Primarily useful for short-term decision-making; lacks comprehensive data for long-term planning.	Suitable for long-term financial planning as it captures all production-related costs.
Performance Evaluation	Does not set benchmarks for performance; limited for evaluating efficiency and cost control.	Facilitates better control and evaluation when used with standard costing and budgetary control systems.
Regulatory Compliance	Not accepted for external reporting as it does not comply with accounting regulations.	Meets external reporting and regulatory requirements, ensuring transparency in financial statements.
Treatment of Fixed Costs	Ignores the impact of fixed costs on production decisions, potentially leading to flawed decision-making.	Allocates fixed costs to units produced, ensuring they are absorbed and reflected fairly across products.
Reaction to Market Changes	May cause panic in early recession stages due to low profits, leading to hasty decisions.	Provides a realistic profit view by evenly distributing fixed costs, reducing overreaction to market conditions.
Automation and Fixed Costs	Increasing automation emphasizes fixed costs, which marginal costing overlooks.	Accounts for fixed costs, making it more relevant in automated and technology-driven production environments.
Decision-Making	Offers limited insights into long-term strategies, such as investments or expansions.	Supports strategic decision-making by providing a complete picture of production costs and profitability.
Simplicity	Simplifies cost analysis by focusing on variable costs, but this may lead to oversights.	Straightforward and accessible, particularly for businesses requiring detailed cost allocations.

Case Study: A Manufacturing Company Facing Challenges with Marginal Costing

Excel Furniture Ltd., a furniture manufacturing company, specializes in modular office desks and chairs. While the management relies on marginal costing for decision-making, the company has faced challenges in accurately representing costs and evaluating long-term profitability. This reliance on marginal costing has highlighted its limitations, particularly in cost classification and inventory valuation.

The company classifies its costs into variable and fixed categories under marginal costing. Variable costs include raw materials, direct labor, and

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transportation, while fixed costs cover factory rent, staff salaries, and machinery depreciation. Marginal costing has helped in pricing decisions but has introduced significant challenges.

Challenges Faced

1. Technical Challenges in Cost Classification

Classifying costs as purely fixed or variable has proven difficult. For instance, while factory rent is treated as fixed, relocation of machinery occasionally increases rent. Similarly, direct labor costs show semi-variable behavior during periods of overtime, which complicates accurate classification.

2. Inventory Valuation Issues

Marginal costing excludes fixed costs from inventory valuations, leading to undervalued work-in-progress and finished goods. For example, finished desks were reported at ₹50,000 (based only on variable costs), whereas absorption costing, which includes fixed costs, would value them at ₹60,000. This undervaluation impacts the Balance Sheet and gives a conservative profit figure.

3. Fluctuations in Profitability

Seasonal sales cycles caused the marginal contribution to fall, creating apparent losses in certain months. Fixed costs, however, remained constant, exaggerating the reported losses in the Profit and Loss Account.

4. Misleading Decisions

Marginal costing suggested reducing production volumes to control costs during low-demand periods. However, this decision increased per-unit fixed cost allocations under absorption costing, leading to higher costs per unit and reduced profitability.

Analysis of Limitations

Exclusion of Fixed Costs in Inventory

By excluding fixed costs, marginal costing undervalued inventory on the Balance Sheet, creating a conservative profit figure. This reduced stakeholder confidence and misrepresented the company's actual financial position.

Profit Fluctuations Due to Marginal Contribution

Seasonal demand resulted in inconsistent profits under marginal costing. Absorption costing, which includes fixed costs, would have smoothed these fluctuations, providing a more stable profit picture.

Impact of Automation

With increasing automation, fixed costs like machinery depreciation became significant. Ignoring these under marginal costing gave an incomplete cost picture, particularly as automation investments grew.

Resolution and Learning**1. Transition to Absorption Costing for Financial Reporting**

The management decided to adopt absorption costing for inventory valuation. This included fixed costs in inventory figures, presenting a more accurate and realistic profit and Balance Sheet.

2. Hybrid Approach for Decision-Making

Marginal costing was retained for short-term decision-making, particularly to evaluate contribution margins. For long-term profitability assessments, the company adopted full costing to include all relevant costs.

3. Investment in Cost Analysis Tools

To address cost classification issues, the company invested in software to analyze and allocate semi-variable costs accurately. This improved understanding of cost behavior and facilitated better managerial decisions.

**SELF-ASSESSMENT QUESTIONS**

1. In marginal costing, which costs are considered as period costs?
 - a. Variable costs
 - b. Direct labor costs
 - c. Direct material costs
 - d. Fixed costs
2. A company incurs fixed costs of ₹1,00,000 and has a contribution per unit of ₹40. How many units must the company sell to break even?
 - a. 2,000 units
 - b. 2,500 units
 - c. 1,500 units
 - d. 3,000 units
3. A company has the following data:

☐ Selling price per unit: ₹60	☐ Fixed costs: ₹1,20,000
☐ Variable cost per unit: ₹40	☐ Desired profit: ₹80,000

How many units must be sold to achieve the desired profit?

 - a. 8,000 units
 - b. 10,000 units
 - c. 5,000 units
 - d. 6,000 units
4. A company has fixed costs of ₹5,00,000 and variable costs of ₹30 per unit. The selling price is ₹50 per unit. The company is operating at 80% of its capacity, producing and selling 20,000 units. If the company increases capacity utilization to 100% and sells all units, what will be the additional profit?
 - a. ₹1,50,000
 - b. ₹2,00,000
 - c. ₹1,00,000
 - d. ₹2,50,000

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Activity

A company produces two products, X and Y. The manager needs to decide whether to increase the production of Product X or Product Y. Analyze the impact on profit if the manager increases the production of one of the products. Discuss which factors the manager should consider when making this decision, such as contribution margin, fixed costs, and demand for each product. How would understanding marginal costing help in making the best decision for the company?

6.2 ABSORPTION COSTING

Absorption Costing, also called the **Full Cost Method**, is a way of determining the total cost of a product by including **both variable and fixed costs**. Think of it like baking a cake. The cost of flour, sugar, and eggs (variable costs) changes depending on how many cakes you bake. But the rent for the kitchen and the salary of the baker (fixed costs) remain the same whether you bake one cake or a hundred.

In this method, fixed costs are distributed across all the products made during a specific period, so each product “absorbs” its share of the cost.

Imagine This Scenario:

You and your friends decide to start a small business selling custom-made T-shirts. To calculate the price for each T-shirt, you need to consider all the costs that go into making them. This is where Absorption Costing comes into play, just like a sponge absorbing everything around it!

Here’s how you’d calculate the cost of each T-shirt:

Step 1: Identify the Variable Costs (Direct Costs)

These are costs that change with production volume. For one T-shirt, you have:

- ❑ Fabric cost = ₹50 per T-shirt
- ❑ Printing cost = ₹30 per T-shirt
- ❑ Wages for workers = ₹20 per T-shirt

Total Variable Cost per T-shirt = ₹50 + ₹30 + ₹20 = ₹100 per T-shirt

Step 2: Identify the Fixed Costs (Indirect Costs)

These are costs that don’t change, no matter how many T-shirts you produce. For example:

- ❑ Rent for the workshop = ₹10,000 per month
- ❑ Machine depreciation = ₹2,000 per month
- ❑ Manager’s salary = ₹5,000 per month

Total Fixed Cost = ₹10,000 + ₹2,000 + ₹5,000 = ₹17,000 per month

Step 3: Calculate the Total Production Volume

Suppose you produce 1,000 T-shirts in a month.

Step 4: Calculate Fixed Cost Per T-shirt

To distribute the fixed costs among all the T-shirts, divide total fixed costs by the number of T-shirts produced.

Step 5: Calculate the Total Cost Per T-shirt

Add the variable cost and the fixed cost per T-shirt:

This means that each T-shirt costs ₹117 to make under the Absorption Costing method.

Reasons why businesses use Absorption Costing:

- ❑ **Full Cost Tracking:** It includes all production costs, helping you understand the true cost of each product.
- ❑ **Pricing Decisions:** Knowing the total cost allows you to set a selling price that covers expenses and earns a profit.
- ❑ **Profitability Analysis:** By tracking fixed costs, you'll know how many T-shirts you need to sell to cover all your expenses.
- ❑ **Financial Reporting:** Companies use absorption costing to prepare financial statements because accounting standards (like GAAP) require it.

Think of planning a birthday party. To calculate the cost per guest, you consider:

- ❑ **Variable costs:** Food and drinks (these increase if more people join).
- ❑ **Fixed costs:** Renting the party hall, hiring a DJ (these don't change even if only a few people attend). If you invite 50 guests, you'll divide the fixed costs across 50 people, but if only 25 people attend, each person's share of the fixed cost increases. This is the idea behind Absorption Costing.

6.2.1 ADVANTAGES OF ABSORPTION COSTING

Absorption costing is a method used to determine inventory value, providing a more accurate accounting of ending inventory by including both variable and fixed costs. Here are some of its key advantages:

1. Comprehensive Cost Allocation

Absorption costing encompasses all product costs, both variable and fixed, offering a more precise reflection of the total cost of manufacturing a product or delivering a service. This enhances decision-making. For instance, if a company producing 1,000 chairs incurs a fixed cost of ₹1,00,000 and a variable cost of ₹500 per chair, the fixed cost per chair would be ₹100 ($₹1,00,000 \div 1,000$). Adding this to the variable cost, the total cost per chair would be ₹600 ($₹500 + ₹100$).

2. Meets Requirements for External Reporting

Widely accepted for external reporting, absorption costing complies with regulatory requirements, ensuring transparency in financial

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statements. For example, a manufacturer with a total cost (fixed + variable) of ₹50,00,000 for producing 10,000 units would value inventory at ₹50 per unit ($₹50,00,000 \div 10,000$). This ensures regulatory compliance and accurate reporting.

3. Simple to Understand

This method is straightforward and accessible to businesses of all sizes, assigning all costs to units produced. It provides a clear view of the total costs per unit. Consider a bakery producing 5,000 loaves of bread with fixed costs of ₹20,000 and variable costs of ₹10 per loaf. The fixed cost per loaf is ₹4 ($₹20,000 \div 5,000$), and the total cost per loaf is ₹14 ($₹10 + ₹4$). This clarity helps businesses of all sizes understand their cost structures.

4. Reflects Fixed Costs

Absorption costing allocates fixed costs to the units produced, ensuring these essential costs are absorbed by the products. This promotes a fair distribution of costs across the product range. For example, a car manufacturer with fixed costs of ₹50,00,000 and a variable cost of ₹3,00,000 per car produces 500 cars. The fixed cost per car is ₹10,000 ($₹50,00,000 \div 500$), and the total cost per car is ₹3,10,000 ($₹3,00,000 + ₹10,000$).

5. Facilitates Revenue Calculation

By incorporating both variable and fixed costs, this method enables accurate profit calculations. All expenses are assigned to units produced, offering a comprehensive view of product profitability. For instance, a clothing company incurs fixed costs of ₹2,00,000 and variable costs of ₹800 per jacket for 1,000 jackets. The fixed cost per jacket is ₹200 ($₹2,00,000 \div 1,000$), making the total cost per jacket ₹1,000 ($₹800 + ₹200$). If the selling price is ₹1,500, the profit per jacket is ₹500 ($₹1,500 - ₹1,000$).

6. Supports Inventory Valuation

Absorption costing includes a portion of fixed overhead in the ending inventory, aligning with general accounting principles. This provides a more accurate reflection of the total production costs in inventory value. For example, a company with 2,000 unsold units at a total cost of ₹250 per unit ($₹200$ variable + $₹50$ fixed) would value its inventory at ₹5,00,000 ($2,000 \times ₹250$), ensuring a fair reflection of production costs in financial statements.

6.2.2 LIMITATIONS OF ABSORPTION COSTING

1. Impact of Inventory Fluctuations:

Absorption costing ties fixed overheads to inventory levels, causing profits to fluctuate with stock changes. For example, a garment manufacturer producing winter jackets might incur fixed overheads of ₹10,00,000 monthly. If the company increases inventory from

1,000 units to 2,000 units, the fixed overhead per unit decreases from ₹1,000 to ₹500. This increase in inventory artificially inflates profits even if no additional sales occur. Conversely, if inventory decreases, the fixed cost per unit rises, lowering reported profits despite high sales.

2. Difficulty in Decision-Making:

Since absorption costing includes fixed overheads in unit costs, it may lead to rejecting profitable orders. For instance, a bakery may decline a bulk order for cakes priced at ₹300 per unit because the absorption cost is ₹350, even though the variable cost is only ₹200. Accepting the order would add ₹100 per unit to profits, but the inclusion of fixed overheads in unit costs obscures this potential.

3. Complexity in Overhead Allocation:

Allocating fixed overheads requires a predetermined rate, often based on labor or machine hours. Suppose a furniture manufacturer allocates ₹5,00,000 of fixed overheads based on labor hours. If one type of chair takes 10 hours and another takes 5 hours, the cost allocation may disproportionately burden the first chair, even if both chairs use similar machine time. This can result in inflated costs for some products and mislead profitability analysis.

4. No Distinction Between Fixed and Variable Costs:

Absorption costing combines fixed and variable costs, making it hard to distinguish cost behaviors. For example, a factory incurs ₹5,00,000 in fixed costs monthly and ₹100 in variable costs per unit. Producing 5,000 units results in a unit cost of ₹200 (₹100 fixed + ₹100 variable), but if production drops to 2,500 units, the unit cost increases to ₹300. This blending hides the fact that only ₹100 per unit changes with production levels.

5. Unsuitability for Short-Term Decision-Making:

Absorption costing emphasizes total costs, which can mislead managers in short-term decisions. For instance, a factory may reject a special order for T-shirts at ₹200 per unit because the absorption cost is ₹250, even though the variable cost is ₹150. Since fixed costs (e.g., ₹1,00,000 per month) are already incurred, accepting the order would still yield a ₹50 profit per unit, but absorption costing masks this benefit.

6. Inaccuracy in Cost Control:

Since fixed costs are spread over production, changes in output alter unit costs. For example, a car manufacturer with monthly fixed costs of ₹1 crore might produce 500 cars in one month, resulting in a fixed cost per car of ₹20,000. If production increases to 1,000 cars, the fixed cost per car drops to ₹10,000. This change could give the false impression of improved efficiency, even though the total fixed cost remains unchanged.

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6.2.3 DIFFERENCE BETWEEN MARGINAL COSTING AND ABSORPTION COSTING

When it comes to costing methods, marginal costing and absorption costing (Figure 6.3) offer distinct advantages depending on the context. Have you ever wondered how fluctuations in inventory might affect profitability? Marginal costing helps avoid such distortions by focusing only on variable costs, making it ideal for short-term decision-making, like pricing or product line adjustments. It provides a clear view of the contribution margin, which can drive better decisions in the moment.

On the other hand, absorption costing takes a broader approach, including both fixed and variable costs. This method is especially useful for long-term financial planning because it reflects the true cost of production and ensures compliance with accounting standards like GAAP and IFRS. By considering all costs, absorption costing gives a clearer picture of profitability, especially for external financial reporting.

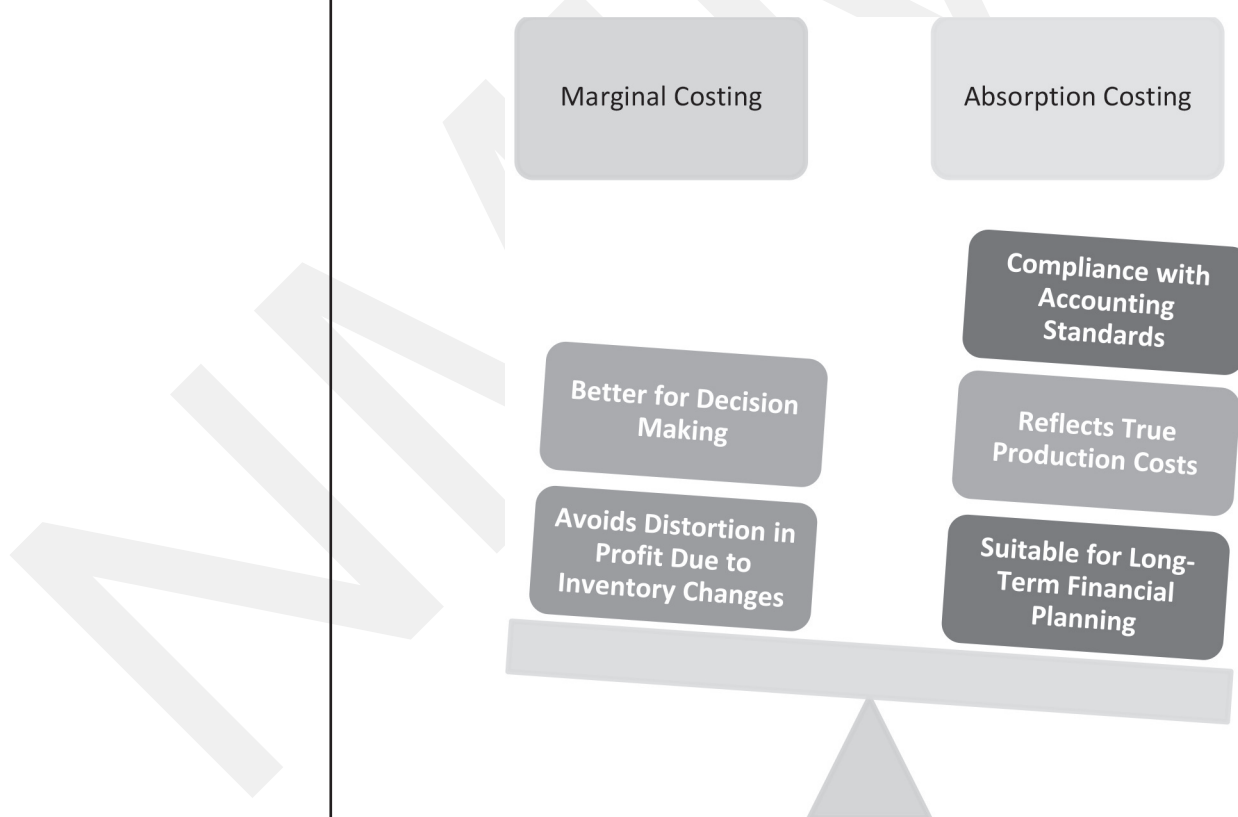


Figure 6.3 Strong Points of Marginal costing and Absorption Costing

The primary distinctions between marginal costing and absorption costing are outlined below in Table 6.2:

Table 6.2 Difference Between Marginal Costing and Absorption Costing

Aspect	Marginal Costing	Absorption Costing
Cost Consideration	Only variable costs are included in product costing and inventory valuation.	Both fixed and variable costs are included in product costing and inventory valuation.
Treatment of Fixed Costs	Fixed costs are treated as period costs; profitability is assessed using the Profit/Volume (P/V) ratio.	Fixed costs are allocated to production costs, affecting profitability based on apportionment.
Cost Data Presentation	Highlights the total contribution of each product.	Presented in a conventional manner; net profit is determined after deducting fixed and variable costs.
Impact of Stock Levels on Unit Cost	The difference in opening and closing stock does not affect unit cost of production.	The difference in opening and closing stock affects unit cost due to related fixed costs.
Cost per Unit	Remains constant, irrespective of production levels; valued at variable cost.	Decreases as production increases; fixed costs are spread over more units.

The profit difference between marginal and absorption costing arises from variations in stock valuation, explained as follows:

- 1. No Opening and Closing Stock:** In this scenario, profit or loss will be equal under both methods.
- 2. When Opening Stock Equals Closing Stock:** Profit or loss will be the same for both approaches, provided the fixed cost element in both stocks is identical.
- 3. When Closing Stock Exceeds Opening Stock:** If production during a period surpasses sales, profit under absorption costing will be higher than under marginal costing. This is due to a portion of fixed overhead included in the closing stock value being carried forward to the next accounting period.
- 4. When Opening Stock Exceeds Closing Stock:** If production is less than sales, profit shown by marginal costing will be higher than that shown by absorption costing. This occurs because a portion of fixed costs from the previous period is added to the current year's cost of goods sold as part of the opening stock.



SELF-ASSESSMENT QUESTIONS

- A company manufactures 1,200 units of Product Y. The following costs are incurred during production:
 - ☐ Direct Material = ₹30,000
 - ☐ Direct Labour = ₹40,000
 - ☐ Variable Overhead = ₹12,000
 - ☐ Fixed Overhead = ₹48,000

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Absorption costing is extensively utilized for external reporting and financial accounting purposes. It encompasses the allocation of all

Absorption Costing:

Absorption costing is extensively utilized for external reporting and financial accounting purposes. It encompasses the allocation of all

manufacturing costs, including both fixed and variable, to the production cost. Notable practical applications are:

1. **Financial Reporting:** Absorption costing is required by various accounting standards when preparing financial statements. It presents a more detailed view of total costs by incorporating both variable and fixed overheads, thus delivering a comprehensive cost picture for external reporting.
2. **Pricing Decisions:** In long-term pricing strategies, absorption costing assists businesses in establishing prices that account for both fixed and variable costs, thereby ensuring long-term profitability. By factoring in fixed costs, it helps avoid under-pricing that could arise from using marginal costing.
3. **Stock Valuation:** Absorption costing allocates fixed overheads to inventory valuation, which is crucial for adherence to accounting standards. This practice results in higher inventory values, consequently leading to increased profits when inventories are not sold immediately.
4. **Profitability Analysis:** This approach aids in assessing product profitability by ensuring that all costs (including overheads) are factored into cost calculations. It proves especially beneficial for businesses with varying production volumes, as fixed overheads are distributed across the total units produced.

Versatile Applications of Absorption Costing Across Industries

Case Study 1: IT Services Company

Scenario: A mid-sized IT services company specializes in software development and implementation projects for clients. The company adopts absorption costing to allocate indirect costs such as office rent, utilities, and administrative salaries to different projects.

Application: The company incurs fixed overheads for maintaining infrastructure and providing software tools to employees. These overheads are absorbed into project costs based on the number of hours spent by developers on each project.

Result: By using absorption costing, the company ensures that project bids include all associated costs, preventing underpricing. Additionally, the method helps identify projects with lower profitability margins, allowing the management to negotiate better terms or reduce costs in future projects.

Case Study 2: Retail Chain

Scenario: A nationwide retail chain operates multiple stores selling a wide range of consumer goods. The company uses absorption costing to allocate indirect costs, such as logistics, marketing, and store maintenance, to individual product categories.

Application: The company distributes these overheads based on product turnover rates and shelf space occupied by each category. For example,

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grocery items, which are fast-moving, absorb a higher portion of logistical costs, while luxury goods are allocated a greater share of store rent due to their premium positioning.

Result: Absorption costing enables the company to determine the true cost and profitability of each product category. This helps in strategic decisions like pricing, promotions, and discontinuation of underperforming items.

Case Study 3: Manufacturing Company

Scenario: A furniture manufacturing company produces chairs, tables, and wardrobes. It faces challenges in accurately distributing factory overheads, such as machinery depreciation, electricity, and supervisor salaries, across its product lines.

Application: Absorption costing is implemented by allocating overheads based on machine hours used by each product line. For instance, wardrobes require longer machining times and therefore absorb a larger share of the overheads.

Result: This approach provides a clearer picture of production costs for each product line, ensuring that pricing decisions reflect both direct and indirect costs. The company avoids underestimating the cost of complex products and identifies opportunities for cost optimization in high-overhead areas.

Marginal Costing:

Marginal costing emphasizes the variable costs of production and is particularly beneficial for decision-making and internal reporting. Practical applications include:

1. **Determining Production Volume:** Marginal costing plays a crucial role in identifying the optimal output level that maximizes profitability for an ongoing business. By leveraging production capacity to its fullest extent, it establishes a profitable connection between cost, price, and volume, which assists management in setting competitive selling prices for their products. Consequently, this leads to profit maximization, a concept explored in greater detail in a dedicated unit.
2. **Selecting Production Lines:** The marginal costing approach aids in pinpointing the most lucrative production lines by analyzing the profitability of various products. Over time, certain products or activities may become less profitable. Discontinuing the production of these items while focusing on more profitable alternatives can be beneficial. This method also facilitates the introduction of new products and serves as an effective guide for determining the optimal product mix based on available capacity and resources.
3. **Deciding Between Production and Procurement:** The decision regarding whether to manufacture a product in-house or source it externally can be made by comparing the external procurement price with the marginal production cost. If the procurement price is lower than the marginal cost, it is advisable to acquire the product externally instead of manufacturing it.

4. **Choosing Manufacturing Methods:** When a product can be produced using multiple methods, calculating the marginal cost for each method is essential in deciding which manufacturing process to pursue.
5. **Evaluating Production Continuation:** Marginal costing is particularly valuable during economic downturns as it helps determine whether to temporarily halt production or to continue operations despite low demand for the company's products.
6. **Profit Planning:** Marginal costing helps businesses plan for optimal profits by analyzing how changes in costs and sales impact profitability. Example: A clothing retailer analyzes its sales data. By using marginal costing, it identifies that increasing sales by offering a discount leads to higher contributions per unit due to reduced idle capacity and increased volume, thereby maximizing profits.
7. **Pricing of Products:** Businesses may price products based on marginal costs to penetrate markets or utilize spare capacity. Example: A hotel offering rooms during the off-season at prices covering marginal costs (like utilities and cleaning) ensures that rooms are occupied while fixed costs like rent are already covered.
8. **Introduction of a New Product:** Marginal costing evaluates the profitability of launching new products without incurring extra fixed costs. Example: A bakery introduces a new cookie flavor by using idle oven capacity. As the fixed costs (e.g., rent) are unchanged, the extra contribution from sales adds to the bakery's profits.
9. **Problem of Key/Scarce Factor:** When resources are limited, marginal costing guides the allocation of resources to maximize contribution. Example: A factory facing labor shortages decides to prioritize producing product A, which has a higher contribution per labor hour, over product B.
10. **Make-or-Buy Decisions:** Companies use marginal costing to decide whether to manufacture components or purchase them from suppliers. Example: An automotive company compares the marginal cost of manufacturing a gearbox (₹10,000/unit) with the supplier's price (₹12,000/unit). It chooses in-house production, saving ₹2,000/unit.
11. **Closing or Suspending Operations:** Marginal costing helps evaluate whether shutting down operations will save costs or result in greater losses. Example: A factory with low seasonal demand calculates that operating at reduced capacity provides a higher contribution than closing the facility and incurring fixed costs without revenue. The factory remains operational.

How Absorption Costing Complies with GAAP and IFRS

1. Comprehensive Cost Allocation:

Absorption costing incorporates all manufacturing costs—both fixed and variable—into the cost of goods sold (COGS) and inventory valuation. This approach aligns with the matching principle under

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GAAP and IFRS, which dictates that costs associated with generating revenue should be recognized in the same accounting period as the revenue itself. *Example:* IFRS requires inventory to reflect the total production cost, including fixed overheads. Absorption costing fulfills this requirement, ensuring compliance with standards like IAS 2 (Inventories).

2. Accurate Inventory Valuation:

GAAP and IFRS mandate that inventory valuation include full production costs. Marginal costing, which excludes fixed overheads, often leads to understated inventory values and may fail to meet these standards. *Example:* A company manufacturing 10,000 units incurs a fixed cost of ₹500,000 and a variable cost of ₹30 per unit. Under absorption costing, the inventory cost includes the ₹50 fixed cost per unit alongside variable costs, ensuring a true economic valuation of inventory in compliance with IAS 2.

3. Presentation of Fixed Costs:

Absorption costing allocates fixed manufacturing overheads to products, treating them as part of product costs rather than period costs. This method adheres to the accrual basis of accounting under GAAP and IFRS, ensuring that fixed costs are matched with the revenue they contribute to generating. Marginal costing on contrast treats fixed costs as period expenses, which can distort profit figures in periods with fluctuating production or sales volumes.

4. Support for External Reporting:

Financial reports prepared under absorption costing offer a complete representation of production costs in inventory and COGS, meeting the requirements of external reporting guidelines under GAAP and IFRS. Marginal costing, although useful for internal decision-making, does not fully comply with these external reporting standards. *Example:* If 2,000 units remain unsold, absorption costing ensures these units are valued at their full cost (₹80 per unit, including ₹50 fixed cost), whereas marginal costing would only account for the ₹30 variable cost per unit, leading to an understated inventory value on the balance sheet.

Income Measurement in Marginal Costing and Absorption Costing

In marginal costing, only variable factory overheads that vary directly with production levels are included in product costs along with prime costs. For inventory valuation, only direct materials, direct labor, and variable factory overheads are considered as product costs. Fixed factory overheads, on the other hand, are treated as period costs and charged to revenue in the period they are incurred, rather than being included in inventory valuation under marginal costing.

In contrast, absorption costing, also known as full costing or conventional costing, assigns all manufacturing costs—both fixed and variable—to product costs. This method ensures that the unit cost of a product includes not only variable expenses but also a portion of fixed production

overheads. Under absorption costing, profit is realized only after accounting for all costs, including fixed and variable expenses. While marginal costing also recognizes profit by offsetting fixed costs against contribution, the treatment of fixed costs differs significantly between the two methods.

Key Concepts in Marginal Costing

Contribution, which is the difference between the selling price and variable costs, is central to marginal costing. Fixed costs are deducted from the contribution to calculate profit. The relationships are represented by the following equations:

1. Selling Price – Variable Costs = Contribution
2. Contribution – Fixed Costs = Profit

Alternatively, if profit and fixed costs are known:

Fixed Costs + Profit = Contribution

This leads to the fundamental equation:

Sales – Marginal Costs = Contribution = Fixed Costs + Profit

Comparative Analysis of Profit

When there is no inventory (opening or closing), profits calculated under both absorption and marginal costing align. However, when production and sales volumes differ, discrepancies in net profit emerge. Absorption costing allows fixed manufacturing costs to be deferred in closing inventory, potentially inflating profits when production exceeds sales. Conversely, under marginal costing, fixed costs are fully expensed in the period incurred, leading to a lower profit in such scenarios.

For example, if fixed factory overheads are ₹100 per unit, production is 10,000 units, and sales are 8,000 units, the difference in inventory valuation under the two methods can be calculated as:

Fixed Factory Overhead × (Production – Sales)

₹100 × (10,000 – 8,000) = ₹2,00,000

This ₹2,00,000 in fixed costs deferred to closing inventory under absorption costing results in a higher profit compared to marginal costing. Similarly, if sales exceed production, absorption costing may show lower profits than marginal costing.

Applicability and Challenges

Absorption costing considers both production and sales in its profit determination, whereas marginal costing emphasizes sales. Marginal costing assumes that variable costs remain constant per unit within a defined output range, even though costs may vary at different production levels. While variable costs effectively illustrate price-cost-volume relationships, reliance solely on marginal costing might lead to the misconception that the business can sustain profitability with low contribution margins, potentially leading to underperformance.

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SELF-ASSESSMENT QUESTIONS

9. Radha Ltd. produces 2,000 units of Product A. The following cost information is available for the period:

- ☐ Selling Price per unit = ₹200
- ☐ Direct Material = ₹40 per unit
- ☐ Direct Labour = ₹30 per unit
- ☐ Variable Overhead = ₹20 per unit
- ☐ Fixed Overhead = ₹1,20,000

Out of the 2,000 units produced, 1,500 units are sold. Calculate the difference in profit between the Marginal Costing and Absorption Costing methods.

- a. ₹30,000
- b. ₹24,000
- c. ₹36,000
- d. ₹40,000

10. How does marginal costing assist in decision-making regarding production?

- a. By calculating total manufacturing costs for each product
- b. By comparing the external procurement price with marginal production costs
- c. By determining the total fixed costs associated with production
- d. By providing a detailed analysis of all variable costs in inventory

6.4 ADDITIONAL SOLVED PROBLEMS

Ques. 1. A manufacturing plant operates at 60% capacity, producing 12,000 units. At 70% capacity, the selling price decreases by 4%. At 100% capacity, the selling price drops by 6%, along with a similar reduction in the costs of raw materials.

Current Cost Structure:

- ☐ Material: ₹8
- ☐ Labour: ₹5
- ☐ Overheads: ₹6 (70% fixed)

The selling Price per unit is ₹25. Estimate the profit of the company at 70% and 100% capacity production.

Sol.

Statement Showing Computation of Profit at 60%, 70%, and 100% Capacity:

Particulars	60%		70%		100%	
	Unit	Total	Unit	Total	Unit	Total
I. Selling Price	25.00	3,00,000	24.00	3,36,000	23.50	4,70,000
II. Variable Cost						
- Material	8.00	96,000	8.00	1,12,000	7.50	1,50,000

- Labour	5.00	60,000	5.00	70,000	5.00	1,00,000
- Variable OH	1.80	21,600	1.80	25,200	1.80	36,000
Total Variable Cost	14.80	1,77,600	14.80	2,07,200	14.30	2,86,000
III. Contribution	10.20	1,22,400	9.20	1,28,800	9.20	1,84,000
IV. Fixed Cost		42,000		42,000		42,000
V. Profit		80,400		86,800		1,42,000

Ques. 2. NYU Ltd. operates at a production capacity of 2,00,000 units annually, with normal capacity utilization considered at 85%. Standard variable production costs are ₹12 per unit, and fixed production costs total ₹4,50,000 per year. Additionally, variable selling costs amount to ₹4 per unit, while fixed selling expenses are ₹3,00,000 annually. The selling price per unit is set at ₹25.

During the fiscal year ending 31st March, 2024, the company produced 1,70,000 units and sold 1,60,000 units. Closing inventory was 30,000 units. Actual variable production costs were ₹40,000 higher than standard.

Required:

- Calculate the profit for the year:
 - Using the Absorption Costing Method
 - Using the Marginal Costing Method
- Explain the difference in the profits between the two costing methods.

Sol.

Income Statement (Absorption Costing) for the year ending 31st March, 2024

Particulars	Amount (Rs.)
Sales (1,60,000 units @ ₹25)	40,00,000
Production Costs:	
Variable (1,70,000 units @ ₹12)	20,40,000
Add: Additional variable costs	40,000
Fixed (1,70,000 units @ ₹2.65)*	4,50,000
Total Cost of Goods Produced	25,30,000
Add: Opening stock (20,000 units @ ₹14.65)**	2,93,000
Total Cost of Goods Available	28,23,000
Less: Closing stock (30,000 units @ ₹14.65)***	4,39,500
Cost of Goods Sold	23,83,500
Add: Non-production Costs:	
Variable selling costs (1,60,000 units @ ₹4)	6,40,000
Fixed selling costs	3,00,000
Total Cost	33,23,500
Profit (Sales – Total Cost)	6,76,500

N O T E S

Working Notes:

- 1. Fixed Production Overhead Rate:** Fixed production overheads are absorbed based on normal capacity (1,70,000 units).

Fixed Production Overhead per Unit = ₹4,50,000/ 1,70,000 units = ₹2.65 per unit

- 2. Opening Stock Valuation:**

Opening Stock = 20,000 units × (₹12 + ₹2.65) = ₹2,93,000

- 3. Closing Stock Valuation:**

Closing Stock under Absorption Costing = 30,000 units × (₹12 + ₹2.65) = ₹4,39,500

Income Statement (Marginal Costing) for the year ending 31st March, 2024

Particulars	Amount (Rs.)
Sales (1,60,000 units @ ₹25)	40,00,000
Variable Production Costs:	
Variable cost (1,70,000 units @ ₹12)	20,40,000
Add: Additional variable costs	40,000
Variable selling costs (1,60,000 units @ ₹4)	6,40,000
Total Variable Costs	27,20,000
Add: Opening stock (20,000 units @ ₹12)	2,40,000
Total Variable Cost of Goods Available	29,60,000
Less: Closing stock (30,000 units @ ₹12)	3,60,000
Total Variable Costs of Goods Sold	26,00,000
Contribution (Sales – Variable Costs)	14,00,000
Less: Fixed Costs	
Fixed production costs	4,50,000
Fixed selling costs	3,00,000
Profit (Contribution – Fixed Costs)	6,50,000

Explanation of Profit Difference:

Particulars	Amount (Rs.)
Profit as per Absorption Costing	6,76,500
Add: Opening stock under-valued in marginal costing	53,000
Less: Closing stock under-valued in marginal costing	79,500
Profit as per Marginal Costing	6,50,000

Ques. 3. A company manufactures and sells a single product. At the beginning of period 1, there is no opening stock of the product. The variable production cost is ₹8, and the selling price is ₹12 per unit. The fixed costs amount to ₹3,000 per period, of which ₹2,500 are fixed production costs. The following data is available:

Period	Sales	Production
Period 1	1,500 units	2,000 units
Period 2	2,000 units	2,000 units

Required:

1. Compute the profit for each period using:
 - (a) Absorption costing (Assume normal output is 2,000 units per period)
 - (b) Marginal costing

Sol.**(a) Absorption Costing Method**

The absorption rate for fixed production overhead is calculated as follows:

$$₹2,500/2,000 \text{ units} = ₹1.25 \text{ per unit}$$

Particulars	Period 1	Period 2	Total
Sales	₹18,000	₹24,000	₹42,000
Production Costs			
Variable @ ₹8	₹16,000	₹16,000	₹32,000
Fixed @ ₹1.25	₹2,500	₹2,500	₹5,000
Total Production Cost per Unit @ ₹9.25	₹18,500	₹18,500	₹37,000
Add: Opening Stock	—	₹4,625	—
Total Production Cost	₹18,500	₹23,125	₹40,700
Less: Closing Stock (500 units @ ₹9.25)	₹4,625	₹4,625	₹4,625
Production Cost of Sales	₹13,875	₹18,500	₹32,375
Other Fixed Costs	₹500	₹500	₹1,000
Total Cost of Sales	₹14,375	₹19,000	₹33,375
Profit	₹3,625	₹5,000	₹8,625

(b) Marginal Costing Method

Particulars	Period 1	Period 2	Total
Sales	₹18,000	₹24,000	₹42,000
Variable Production Cost	₹16,000	₹16,000	₹32,000
Add: Opening Stock (500 units @ ₹8)	—	₹4,000	—
Total Variable Cost	₹16,000	₹20,000	₹32,000
Less: Closing Stock (500 units @ ₹8)	₹4,000	₹4,000	₹4,000
Variable Production Cost of Sales	₹12,000	₹16,000	₹28,000
Contribution	₹6,000	₹8,000	₹14,000
Fixed Costs	₹3,000	₹3,000	₹6,000
Profit	₹3,000	₹5,000	₹8,000

N O T E S

Ques. 4. In an effort to boost sales volume, Dynamic Products Ltd. is considering reducing the price of its product by 15%. The company anticipates no changes in the total fixed costs or variable costs per unit. The management wants to ensure that the current level of profit is maintained after the price reduction.

The following data is provided:

- **Sales:** 40,000 units at ₹800,000
- **Variable Costs:** ₹10 per unit
- **Fixed Costs:** ₹150,000

Advise management on the sales volume required to maintain the current profit level, using the information given.

Solution

Marginal Cost Statement (Current Scenario):

- **Sales Revenue:** ₹800,000
- **Variable Costs:** ₹400,000 (40,000 units × ₹10 per unit)
- **Contribution:** ₹400,000 (Sales – Variable Costs)
- **Less: Fixed Costs:** ₹150,000
- **Profit:** ₹250,000

Present Profit Volume Ratio (PV Ratio):

$$\text{PV Ratio} = (\text{Contribution/Sales}) \times 100 = (\text{₹}400,000/\text{₹}800,000) \times 100 = 50\%$$

Future Selling Price per Unit:

$$\text{Current selling price per unit} = \text{₹}800,000 \div 40,000 = \text{₹}20$$

$$\text{Reduced selling price} = \text{₹}20 \times (1 - 15/100) = \text{₹}17$$

Future Contribution per Unit:

$$\text{Contribution per Unit} = \text{Selling Price per Unit} - \text{Variable Cost per Unit} = \text{₹}17 - \text{₹}10 = \text{₹}7$$

Future Profit Volume Ratio:

$$\text{Future PV Ratio} = (\text{Contribution per Unit/Selling Price per Unit}) \times 100 = (\text{₹}7/\text{₹}17) \times 100 = 41.18\%$$

Sales Required to Maintain Current Profit:

$$\begin{aligned} \text{Required Sales} &= (\text{Fixed Costs} + \text{Desired Profit})/\text{Future PV Ratio} \\ &= (\text{₹}150,000 + \text{₹}250,000)/41.18\% \\ &= \text{₹}400,000 \div 0.4118 \\ &= \text{₹}971,345 \end{aligned}$$

Units Required to Maintain Profit:

Required Units = Required Sales/Reduced Selling Price = ₹971,345/₹17 ≈ 57,137 units

Ques. 5. ABC Ltd. operates at a production capacity of 2,50,000 units per year, with a normal capacity utilization rate of 85%. The standard variable production cost is ₹12 per unit, and the fixed costs amount to ₹4,50,000 annually. The company incurs variable selling costs of ₹4 per unit, and fixed selling costs of ₹3,00,000 per year. The unit selling price is ₹25.

For the year ending on 31st March 2024, ABC Ltd. produced 2,00,000 units and sold 1,80,000 units. The closing inventory as of 31st March 2024 was 20,000 units. The actual variable production costs were ₹48,000 higher than the standard cost.

Required:

- (i) Calculate the profit for the year:
 - (a) by absorption costing method, and
 - (b) by marginal costing method.
- (ii) Explain the difference in profits between the two methods.

Solution:**Income Statement (Absorption Costing) for the Year Ending 31st March 2024**

Particulars	₹
Sales (1,80,000 units @ ₹25)	45,00,000
Production Costs:	
Variable (2,00,000 units @ ₹12)	24,00,000
Add: Increase in Variable Production Costs	48,000
Total Variable Production Costs	24,48,000
Fixed (4,50,000 × (1,80,000/2,00,000))	4,05,000
Cost of Goods Produced	28,53,000
Less: Closing Stock (₹31,83,000 × 20,000 units / 2,00,000 units)	2,85,300
Cost of Goods Sold	26,08,200
Add: Under-absorbed Fixed Production Overhead (₹4,50,000 – ₹4,05,000)	45,000
Non-production Costs:	
Variable Selling Costs (1,80,000 units @ ₹4)	7,20,000
Fixed Selling Costs	3,00,000
Total Costs	36,32,700
Profit (Sales – Total Costs)	8,67,300

N O T E S

Income Statement (Marginal Costing) for the Year Ending 31st March 2024

Particulars	₹
Sales (1,80,000 units @ ₹25)	45,00,000
Variable Production Costs (2,00,000 units @ ₹12 + ₹48,000 increase)	24,48,000
Variable Selling Costs (1,80,000 units @ ₹4)	7,20,000
Total Variable Costs	31,68,000
Less: Closing Stock (₹24,48,000 × 20,000 units / 2,00,000 units)	2,44,800
Variable Cost of Goods Sold	29,23,200
Contribution (Sales – Variable Cost of Goods Sold)	15,76,800
Less: Fixed Costs:	
Fixed Production Costs	4,50,000
Fixed Selling Costs	3,00,000
Profit	8,26,800

Reasons for the Difference in Profit:

Particulars	₹
Profit as per Absorption Costing	8,67,300
Less: Closing Stock Under-Valued in Marginal Costing (₹2,85,300 – ₹2,44,800)	40,500
Profit as per Marginal Costing	8,26,800

6.5 SUMMARY

- ❑ Marginal cost refers to the change in total costs when the output increases or decreases by one unit. It includes prime cost and variable overheads and remains consistent per unit of output, unlike fixed costs, which remain constant over a period.
- ❑ In marginal costing, costs are divided into fixed and variable categories. Fixed costs remain unchanged regardless of output, while variable costs fluctuate with production volume, directly impacting managerial decisions.
- ❑ Marginal costing focuses on the contribution margin, which is the difference between sales revenue and total variable costs. It helps evaluate operational efficiency and is used in decision-making processes.
- ❑ Marginal costing aids managerial decisions, including pricing, production levels, and assessing profitability. It is not a distinct costing method but a technique used alongside others like job or process costing.

- ❑ Absorption costing, or full cost method, includes all manufacturing costs—both fixed and variable—providing a complete view of product costs for accurate pricing and profitability assessments.
- ❑ Absorption costing complies with external reporting standards, ensuring transparency in financial statements by allocating fixed and variable costs appropriately.
- ❑ Absorption costing method is simple to understand, making it accessible to various businesses and helping them recognize total costs per unit, including the reflection of fixed costs in product pricing.
- ❑ Key advantages of absorption costing include improved decision-making for pricing and inventory valuation, which is crucial for long-term business strategies and profitability analysis.
- ❑ Limitations of absorption costing involve challenges in classifying costs, potential misinterpretation of marginal costs leading to incorrect pricing, and risks of overvaluing inventory or work-in-progress, impacting financial assessments.
- ❑ Marginal costing simplifies decision-making and cost control but may understate inventory values and profits, leading to inaccurate financial reporting. It is more effective for short-term decisions but may overlook long-term fixed costs.

6.6 KEY WORDS

1. **Marginal Cost:** The cost incurred when producing one additional unit of output.
2. **Prime Cost:** The direct costs of manufacturing, including direct materials and direct labor.
3. **Fixed Costs:** Costs that remain constant regardless of the level of production.
4. **Variable Costs:** Costs that fluctuate in direct proportion to changes in output volume.
5. **Contribution Margin:** The difference between sales revenue and total variable costs, used to assess profitability.
6. **Absorption Costing:** A costing method that includes both fixed and variable manufacturing costs in the calculation of product costs, ensuring all expenses are accounted for when determining profitability.
7. **Fixed Costs:** Costs that do not vary with production levels, such as rent and salaries, which are allocated to products under absorption costing to reflect the total cost of goods produced.
8. **Variable Costs:** Costs that fluctuate with production volume, such as direct materials and labor, directly assigned to products in both absorption and marginal costing methodologies.

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9. **Gross Profit:** The profit remaining after deducting total production costs (both fixed and variable) from sales revenue, used to determine the financial performance of a business.
10. **Inventory Valuation:** The process of assigning a monetary value to a company's inventory, which under absorption costing includes both fixed and variable costs, resulting in a higher valuation compared to marginal costing.

6.7 CLOSING CASELET

Absorption Costing at XYZ Manufacturing Ltd.

XYZ Manufacturing Ltd. is a mid-sized company that produces home appliances, including refrigerators and washing machines. In response to growing complexity in product lines, management implemented absorption costing to obtain a more accurate understanding of product costs. This method encompasses both variable costs, such as direct materials and labor, and fixed costs, including factory rent and equipment depreciation. By allocating fixed overhead to each unit produced, management aimed to achieve a comprehensive view of total production costs, facilitating informed pricing decisions.

The adoption of absorption costing yielded several advantages for XYZ Manufacturing. Firstly, it allowed for a more precise calculation of total product costs, which improved pricing strategies and clarified the profitability of various product lines. Employees found the method straightforward, fostering better communication between production and sales teams. Additionally, absorption costing enabled compliance with external reporting requirements, enhancing the company's credibility with stakeholders by providing transparent financial statements.

Despite its benefits, XYZ Manufacturing faced limitations with absorption costing. Management struggled to accurately classify certain expenses as fixed or variable, leading to potential mispricing issues. The sales team occasionally misinterpreted absorption costs, which created risks for profitability. Furthermore, fluctuations in fixed costs due to external factors complicated financial projections. Overall, while absorption costing improved cost allocation and compliance, the company recognized the need for ongoing training and adjustments to address its limitations effectively.



QUESTIONS

1. What are the key advantages that XYZ Manufacturing Ltd. experienced by implementing absorption costing in their accounting practices?

(**Hint:** Consider aspects such as pricing strategy, cost visibility, and compliance with financial reporting standards.)

2. What challenges did XYZ Manufacturing Ltd. encounter when applying absorption costing, particularly regarding expense classification and profitability?

(Hint: Think about the difficulties in distinguishing between fixed and variable costs and how this affected the sales team's understanding of costs.)

6.8 HIGHER ORDER THINKING SKILLS QUESTIONS (HOTS)

1. How can the concept of marginal costing be applied to optimize product mix decisions in a company with multiple product lines, considering both short-term and long-term profitability?
2. In what ways could excluding fixed costs from inventory valuation under marginal costing affect the accuracy of financial statements, and how might this influence management's decision-making process during periods of fluctuating sales?
3. How does the use of absorption costing versus marginal costing impact managerial decision-making regarding pricing strategies and inventory management in a manufacturing company?
4. In what ways can the limitations of absorption costing, such as the potential for over- or under-absorption of fixed overheads, affect the accuracy of financial reporting and profitability analysis?
5. How might the choice between absorption costing and marginal costing influence the strategic direction of a company in terms of cost control, product pricing, and overall financial health, especially in times of economic uncertainty?
6. A company manufactures and sells three products — A, B, and C. All three products are produced using the same set of machines. The production is constrained by the available machine capacity. Using the information provided below, determine the priority order for manufacturing A, B, and C to maximize profits.

Particulars	Product A	Product B	Product C
Raw Material per unit (₹)	12.00	18.00	22.50
Direct Labour per unit (₹)	3.00	3.50	2.80
Variable Overheads (₹)	2.50	3.00	4.20
Selling Price (₹)	28.00	36.00	42.00
Machine time required per unit (in minutes)	30	25	15

Required:

- ☐ Calculate the contribution per unit for each product (A, B, and C).
- ☐ Rank the products in order of priority for production to maximize the company's profit.

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7. Production costs of Dynamic Ventures Ltd. are as follows:

Levels of Activity	Output (in Units)	Direct Materials (₹)	Direct Labour (₹)	Factory Overheads (₹)	Works Cost (₹)
60%	3,000	30,000	9,000	15,000	54,000
70%	3,500	35,000	10,500	16,000	61,500
80%	4,000	40,000	12,000	17,000	69,000

The management is considering increasing production to the **90% level of activity**, which corresponds to 4,500 units. The fixed factory overheads will remain unchanged.

You are required to prepare a statement showing:

- ☐ Prime Cost
- ☐ Total Marginal Cost
- ☐ Total Factory Cost at 90% level of activity.

8. Evergreen Appliances Ltd. operates at a production capacity of 3,00,000 units per year, with a normal capacity utilization rate of 80%. The standard variable production cost is ₹15 per unit, and the fixed production costs amount to ₹6,00,000 annually. The company also incurs variable selling costs of ₹5 per unit, and fixed selling costs of ₹4,00,000 per year. The unit selling price is ₹32.

For the year ending on 31st March 2025, Evergreen Appliances Ltd. produced 2,40,000 units and sold 2,20,000 units. The closing inventory as of 31st March 2025 was 20,000 units. The actual variable production costs were ₹72,000 higher than the standard cost.

Required:

1. Calculate the profit for the year using:
 - (a) Absorption costing method
 - (b) Marginal costing method
2. Explain the difference in profits between the two methods.

6.9 ANSWER KEYS

Topic	Q. No.	Answer
Marginal Costing	1	d. Fixed costs
	2	b. 2,500 units
	3	b. 10,000 units
	4	c. ₹1,00,000
Absorption Costing	5	a. ₹71,667
	6	c. Allocated to the cost of goods produced
	7	d. ₹29,500
	8	d. Absorption costing allocates fixed costs to production, while marginal costing does not.

Practical Application
of Absorption and
Marginal Costing

9 a. ₹30,000

10 b. By comparing the external procurement
price with marginal production costs.

6.10 SUGGESTED READINGS

- ❑ Kaplan, R. S., & Atkinson, A. A. (1998). *Advanced Management Accounting* (3rd ed.). Prentice Hall.
- ❑ Banerjee, B. (2020). *Cost Accounting: Theory and Practice* (13th ed.). PHI Learning Pvt. Ltd.
- ❑ Article: “Absorption Costing vs. Variable Costing” (The CPA Journal)
- ❑ Article: “The Role of Costing in Strategic Decision Making” (Harvard Business Review)

6.11 REFERENCES

- ❑ Drury, C. (2021). *Management and Cost Accounting* (11th ed.). Cengage Learning.
- ❑ Horngren, C. T., Datar, S. M., & Rajan, M. V. (2019). *Cost Accounting: A Managerial Emphasis* (16th ed.). Pearson.
- ❑ Corporate Finance Institute (CFI)-<https://corporatefinanceinstitute.com/>

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7

U N I T

PROCESS AND JOINT COSTING

CONTENTS

- 7.1 Introduction
- 7.2 Features of Process Costing
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INTRODUCTORY CASELET

Process Costing in a Dairy Manufacturing Company

Mooo Dairy Co. is a major dairy processing company that produces milk, butter, cheese, and yogurt. Their operations involve continuous processing, where raw milk is the primary input that goes through several stages like pasteurization and fermentation. Each product undergoes its specific process, with large volumes being produced simultaneously, making cost allocation a challenge for the company.

The company's process flow starts with pasteurizing raw milk, which is then split across different production lines for yogurt, butter, and cheese. Each production process incurs various costs, such as labor for churning butter or fermenting yogurt. To manage the costs across these processes, Mooo Dairy Co. applies **process costing**, a system that assigns costs to each stage rather than individual units. For example, the cost of milk in the yogurt production process includes raw material costs and additional process costs like fermentation.

The main challenge Mooo Dairy Co. faces is the complexity of allocating costs accurately across multiple products. By using process costing, they can distribute both fixed and variable costs across all products and production processes. This method ensures that production costs are fairly divided, simplifying the tracking process and leading to better pricing decisions. With insights into cost per liter or kilogram for each product, the company maintains competitiveness and efficiency.

**QUESTIONS**

1. What are the benefits of using process costing for XYZ Dairy Co., and how does it impact their pricing decisions for various products?

(**Hint:** Think about how process costing provides accurate cost information, which helps the company in determining the pricing of its products.)

2. What are the key challenges that Mooo Dairy Co. faces in assigning costs to individual products, and how does process costing help solve these challenges?

(**Hint:** Focus on the complexity of continuous production and the simultaneous manufacturing of different products).



LEARNING OBJECTIVES

After reading this unit, you will be able to:

- Understand the characteristics of process costing
- Distinguish between process costing and job costing systems
- Learn the structure and format of a process account.
- Identify how costs are allocated across different processes.
- Define normal loss and learn how to account for normal loss in process accounts.
- Understand the treatment of abnormal loss in financial reporting.
- Define abnormal gain and its unusual occurrence
- Understand the concepts of joint products and by-products.
- Distinguish between joint products and by-products
- Learn different methods of costing of joint products

7.1 INTRODUCTION

Process costing is a cost accounting method applied in industries where production is continuous, and the output from one process serves as the input for the next. It is commonly used in sectors such as paint manufacturing, chemical processing, food production, oil refining, textile mills, sugar factories, fruit canning, and dairies. In these industries, raw materials are typically introduced at the start of the process, with the final product emerging after multiple stages. Calculating the cost of a specific quantity, such as 300 kg of sugar or 400 kg of cement, is complex due to the bulk and ongoing nature of production. The cost per unit is therefore derived by dividing the total production cost by the total output for a given time period.

This costing method determines the cost of production at each stage, especially when products pass through distinct and sequential manufacturing processes. While most products undergo multiple operations before becoming finished goods, the costs associated with each stage may be relatively minor compared to the total cost. Thus, computing costs for each operation individually is often impractical. Process costing is well-suited for scenarios where the output of one process serves both as input for the subsequent process and as a marketable product. Procedure of process costing is illustrated in **Figure 7.1**.

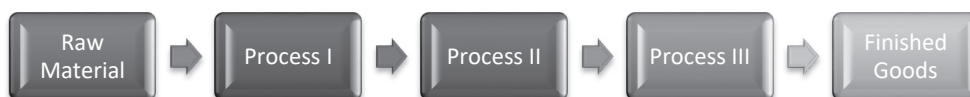


Figure 7.1 Process Costing.

N O T E S

For example, in a cotton textile mill, cloth production involves three main processes: spinning, weaving, and finishing. The spinning process produces yarn, which is sent to the weaving process as raw material but can also be sold if excess stock exists. Similarly, the weaving process produces coarse cloth, which is passed to the finishing process, with any surplus sold to other mills. By computing the costs for spinning, weaving, and finishing processes separately, the textile mill can determine the costs of yarn, coarse cloth, and finished cloth, facilitating cost comparisons with market prices.

Industries Using Process Costing

1. **Industries Producing Identical Products:** This method is ideal for manufacturers producing standardized items. For instance, in mass-produced televisions, where all units share identical components, consistent prices are assigned based on total production costs.
2. **Industries with Multiple Departments:** Businesses operating across multiple departments often rely on process costing to evaluate departmental expenses. For example, one department may refine raw materials, another assembles components, and a third tests finished goods. Additional costs incurred during inter-departmental transfers can also be tracked. If production costs rise unexpectedly, process costing helps identify the department responsible for the increase.
3. **Industries Producing Interchangeable Parts:** Process costing is applied in factories producing identical parts that can be used across various products. For instance, a computer manufacturing plant producing interchangeable components benefits from knowing the exact production costs, enabling the sale of individual parts.
4. **Industries with Variants of Products:** When manufacturers produce products with varying features, process costing aids in cost determination. For example, a coffee pot may be available in two versions—one with a timer and one without. Process costing helps compute the additional cost of the timer, allowing the manufacturer to price both versions accurately.
5. **Innovation-Driven Industries:** In industries reliant on innovation, process costing is essential for determining the total production cost of new products. This information helps businesses price products appropriately and assess their profitability by comparing production costs with potential selling prices. For example, a tech start-up develops a new wearable fitness tracker. Using process costing, the company calculates the cost of each unit produced, ensuring they set a price that covers costs and generates profit, considering the market's willingness to pay.

**SELF-ASSESSMENT QUESTIONS**

1. Which of the following industries is most likely to use process costing?
 - a. Custom furniture manufacturing
 - b. Law firms
 - c. Chemical plants
 - d. Interior design companies
2. Which of the following is a key benefit of process costing for industries with multiple departments?
 - a. It allows for better customization of products
 - b. It helps pinpoint which department is responsible for increased production costs
 - c. It simplifies calculating taxes on raw materials
 - d. It reduces the overall time taken to complete production

Activity

A company manufactures a product that passes through three distinct processes: Grinding, Assembling, and Packaging. At each process, costs are incurred for raw materials, labor, and overheads. The product can be sold after any process or continue through the remaining processes. Analyze how process costing would be applied in this situation, where the output from each process can be sold as a finished product or transferred to the next process.

7.2 FEATURES OF PROCESS COSTING

The features of process costing include:

1. **Continuous Production Flow:** Process costing is used in industries like chemicals, oil refining, or cement production where manufacturing involves a continuous and uninterrupted flow of operations.
2. **Homogeneous Products:** The products produced are identical in nature and follow a standardized design or specification. For instance, all units of paint or sugar are alike and undistinguishable.
3. **Loss of Individual Unit Identity:** Since production is continuous and standardized, individual units merge into the process and lose their unique identity, making it impractical to track costs for each specific unit.
4. **Output of One Process as Input for the Next:** The production process is organized into sequential steps. The finished product of one process becomes the raw material for the subsequent process. For example, in textile manufacturing, yarn produced in one stage is used in the weaving process at the next stage.

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5. **Transfer to Finished Stock:** The output from the last process is transferred to finished goods inventory, making it ready for sale or further distribution.
6. **Accumulation of Costs:** Costs are grouped under material, labor, direct expenses, and overheads for each process to ascertain the total expenditure incurred at each stage.
7. **Average Cost Per Unit:** By dividing the total cost of a process by the number of units produced, an average cost per unit is determined, which simplifies cost allocation and pricing decisions.
8. **Cumulative Cost Transfer:** As the product progresses through processes, the costs accumulated in earlier processes are carried forward and added to the costs incurred in the next process. This cumulative approach ensures accurate cost tracking.
9. **Pre-determined Sequence of Operations:** The manufacturing process is structured and follows a specific sequence of operations, ensuring uniformity and consistency in production.
10. **Unavoidable Material Loss:** Inherent in many production processes, some material loss due to spoilage, evaporation, or waste is expected. Process costing accounts for this loss and spreads the cost over the remaining production.

7.3 DIFFERENCE BETWEEN PROCESS AND JOB COSTING

The type of cost accumulation chosen depends on the nature of the manufacturing operations. The primary distinction between the two methods lies in how the costing is carried out. In both methods, product cost is determined through an averaging process, but the denominator used in the calculation varies. Job costing applies costs to specific jobs, which may consist of a single or multiple units, whereas process costing is used for a large volume of units. The key differences between job costing and process costing can be summarized as shown in **Table 7.1** below:

TABLE 7.1 DIFFERENCE BETWEEN JOB COSTING AND PROCESS COSTING

Aspect	Job Costing	Process Costing
Costing Basis	Specific order costing based on customer requirements.	Standard costing applicable where homogeneous goods are produced.
Cost Unit	The job is the cost unit; costs are collected for each job.	Costs are collected by process or department and divided by output for an average cost per unit.
Treatment of Losses	Losses are generally not segregated; normal losses are predetermined, and abnormal losses are segregated.	Normal losses are accounted for, and abnormal losses are segregated.

(Continued)

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Aspect	Job Costing	Process Costing
Overhead Allocation	Overheads allocated and apportioned to cost centres, absorbed by jobs based on time taken.	Overheads apportioned to processes on a suitable basis; sometimes predetermined rates are used.
Joint Products/By-products	Joint products/by-products does not usually arise.	Joint products/by-products may arise, necessitating joint cost apportionment.
Applicability of Standard Costing	Standard costing is generally unsuitable for jobbing work.	Standardized nature of products allows for the adoption of standard costing.
Work-in-Progress Valuation	Specific and obtained from analysis of outstanding jobs.	Operating costs spread over completed output and partially complete products using equivalent units.
Product Identity	Each job is separate and independent; costs are computed when a job is complete.	Products lose individual identity; costs are calculated at the end of the cost period.
Transfer of Costs	No transfers from one job to another unless surplus work exists.	Transfer of costs occurs as products move from one process to another.
Work-in-Progress Presence	May or may not exist at the beginning or end of the accounting period.	Always present at the beginning and end of the accounting period.
Ease of Control	Proper control is comparatively difficult due to product uniqueness.	Proper control is easier as production is standardized and stable.
Documentation Requirement	Requires more forms and detailed documentation.	Requires fewer forms and less detailed documentation.

To illustrate the differences between job costing and process costing, let's consider two scenarios:

Scenario 1: Job Costing

Imagine a furniture workshop where a customer places an order for a custom dining table. This specific table will require unique materials, labor, and design specifications as per the customer's request. The **cost unit** here is the specific job—"Job No. 123," for example. Costs are gathered for this individual job, including materials (wood, varnish), labor (craftsmen's time), and overheads (workshop electricity, equipment depreciation). Overheads are allocated based on the time the craftsmen spend on this job. Since the dining table is unique, **work-in-progress valuation** reflects the costs incurred on incomplete jobs at the period's end. If the workshop has unused materials or defective outputs, these are accounted for separately but are directly related to this job.

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Scenario 2: Process Costing

Now, consider a chocolate factory producing identical bars. Production involves multiple stages, such as mixing, molding, and packaging. Here, the cost unit is the batch or process. All costs incurred (materials like cocoa, sugar; labor; and overheads) are collected for each process. For example, the molding department calculates costs for the entire batch, then averages the costs per chocolate bar. Any **normal losses** (e.g., spillage or wastage) are expected and factored into the process cost. Abnormal losses, such as a machinery breakdown, are accounted for separately. Since production is continuous and standardized, **work-in-progress** (partially completed bars) is a constant and calculated using equivalent units at the end of each period.

Conclusion

In job costing, each job is **unique**, with its costs calculated separately, suitable for industries like construction, interior design, or custom manufacturing. In contrast, process costing is for **standardized products**, spreading costs across units, ideal for industries like chemicals, food processing, or textiles. The method you use depends on the production nature and the level of cost control required.

7.4 PREPARATION OF PROCESS ACCOUNT

The primary steps in the costing process are as follows:

1. Debit the cost of basic raw materials to the first process account, including both the quantity and the total amount involved.
2. Record the costs of other materials, direct labor, and direct expenses relevant to each process in their respective accounts.
3. Debit each process account for production overheads, either as specified or based on a fair allocation method.
4. Credit the process account with the realizable value of scrap and containers of materials returned or sold, if applicable. Alternatively, these amounts can be deducted from the raw materials cost.
5. Determine the total cost of the process and compute the average cost per unit of output.
6. If all output from a process has been transferred to the next process, reflect the total cost on the credit side as a transfer to the next process. This amount should also be recorded on the debit side of the next process account.
7. If a portion of the output is designated for sale or has been sold, record its cost as a transfer to stock and the remaining amount as a transfer to the next process. Note that when a part of the output is sold, the process account should be credited only with its cost, not the sale value.
8. Debit the cost of containers used for packaging the finished goods to the last process account.

9. The total cost of the last process should be transferred to the Finished Stock Account.
10. The Finished Stock Account functions similarly to a Trading Account. Therefore, sales will be credited to this account, and gross profit will be calculated.

The costs associated with each process include:

- i. Materials
- ii. Employee Costs (Labor)
- iii. Direct Expenses
- iv. Production Overheads

Materials: The materials and supplies needed for each process are obtained from the stores based on Material Requisition Notes. Each process that utilizes these materials is charged with the cost of materials consumed, as informed by the Cost Accounting department. Typically, the finished product from the first process serves as the raw materials for the second process (shown in **Figure 7.2**); in this case, the account for the second process is debited with the transfer cost from the first process, as well as any additional materials used.

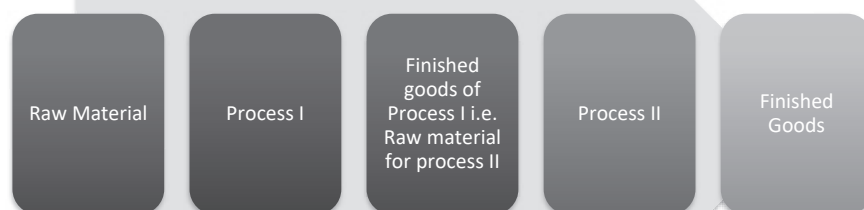


Figure 7.2 Material Under Process Costing.

Employee Costs (Labor): Each process account must reflect the labor costs or wages paid to workers for performing processing tasks. Occasionally, wages may be allocated across different processes based on an appropriate criterion.

Direct Expenses: Each process account should include direct expenses such as depreciation, repairs, maintenance, and insurance that are directly related to that process.

Production Overheads: Expenses such as rent, power, lighting, gas, and water are classified as production overheads. Since these expenses cannot

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be directly allocated to a specific process, they are typically apportioned among various processes using a suitable basis. Generally, these costs are estimated in advance, and processes are charged based on a predetermined allocation method.

Specimen of Process Account is shown in **Figure 7.3**.

Particulars	Units	Amount in ₹	Particulars	Units	Amount in ₹
To Basic material	XX	XXX	By Normal Loss	XX	XXX
To Direct Material	XX	XXX	By Abnormal Loss	XX	XXX
To Direct labor	XX	XXX	By Process II (Output transferred to next processes)		
To Direct Expenses	XX	XXX	By Finished stock Account	XX	XXX
To Production overhead					
To Cost of rectification of defective material	XX	XXX		XX	XXX
To Abnormal gains	XX	XXX			
	XX	XXX			
	XX	XXX		XX	XXX

Figure 7.3 Process Account Specimen.



SELF-ASSESSMENT QUESTIONS

3. In process costing, what is the first step when assigning costs to a process account?
 - a. Debit the cost of basic raw materials to the first process account
 - b. Credit the realizable value of scrap
 - c. Debit the cost of containers used for packaging
 - d. Calculate the average cost per unit of output
4. What is typically debited to the second process account in process costing?
 - a. The cost of materials purchased from suppliers
 - b. The cost of containers used in packaging
 - c. The sale value of the product
 - d. The cost of the finished product from the first process

7.5 PROCESS ACCOUNTS WITH SCRAP

Scrap refers to materials that hold some residual value but are not part of the finished product. This material can either be sold for its salvage value or, in some cases, reused within the production process. When scrap is sold, the proceeds from the sale are credited to the process account, reducing the overall cost of production. However, if the scrap has no market value, it is considered a normal loss, and no monetary value is assigned to it. In such cases, the cost of the scrap is distributed across the remaining good units, which absorbs the associated cost.

Example: A product undergoes processing in two stages: Process A and Process B. For Process A, the following details are provided:

- ❑ Raw materials issued: ₹90,000
- ❑ Wages incurred: ₹70,000
- ❑ Manufacturing overheads: ₹60,000
- ❑ Input quantity: 12,000 units
- ❑ Units produced and transferred to Process B: 11,520 units
- ❑ Normal loss: 4% of the input quantity
- ❑ Scrap value: ₹5 per unit
- ❑ No opening stock

Requirements

1. Calculate the value of normal loss.
2. Calculate the value of units transferred to Process B.

Solution:

1. Calculation of Value of Normal Loss

- ❑ Input quantity = 12,000 units
- ❑ Normal loss = 4% of input quantity Normal loss units = $12,000 \times 4\% = 480$ units
- ❑ Scrap value = ₹5 per unit Value of normal loss = $480 \times 5 = ₹2,400$

2. Calculation of Value of Units Transferred to Process B

- ❑ Total cost of production = Raw materials + Wages + Manufacturing overheads Total cost of production = $₹90,000 + ₹70,000 + ₹60,000 = ₹2,20,000$
- ❑ Net cost after adjusting for normal loss Net cost = Total cost – Scrap value of normal loss
- ❑ Net cost = $₹2,20,000 - ₹2,400 = ₹2,17,600$
- ❑ Effective units (input after accounting for normal loss) Effective units = $12,000 - 480 = 11,520$ units
- ❑ Value per unit Value per unit = $₹2,17,600 / 11,520 = ₹18.88$ per unit (approx.)
- ❑ Units transferred to Process B = 11,520 units
- ❑ Value of units transferred to Process B Value of units transferred = $11,520 \times ₹18.88 = ₹2,17,600$

7.6 NORMAL LOSS, ABNORMAL LOSS AND ABNORMAL GAIN

Material loss is a natural occurrence during processing operations. This loss can result from factors such as evaporation or changes in moisture content. Process loss is specifically defined as the reduction in material

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that occurs during a processing operation, calculated as the difference between the input quantity of the material and the output quantity.

7.6.1 NORMAL LOSS

Certain losses are inherent in the production process and cannot be avoided due to the very nature of the materials or the production methods used. These losses may arise from factors such as evaporation, chemical reactions, scrap, or unavoidable spoilage. The reduction in output caused by these factors is referred to as “normal process loss” or “normal wastage.” Since such losses are expected under normal conditions, they can often be estimated in advance based on historical data.

Accounting Treatment: A fundamental principle in accounting is that the cost associated with any normal loss should be absorbed by the cost of producing good units. Therefore, to determine the cost per unit of output, the total cost should be divided by the number of good units produced (normal output). If the wastage has a realizable value, that amount should be credited to the process account and adjusted in the overall cost of output.

Example: Radha’s biscuit factory produces “Crunchy Delight” biscuits, which pass through two processes: Mixing and Packaging. The data for the month just ended is as follows:

Process	Input (kg)	Cost (₹)
Process 1	Ingredients: 6,000 kg	₹9,000
	Labour & Overheads:	₹7,000
Process 2	Packaging:	₹12,000
	Labour & Overheads:	₹10,000

Radha estimates that 8% of the biscuits will crumble (normal loss) during the Mixing stage. There is no work in progress at the end of the month.

Task: Prepare the process accounts for both stages and calculate the cost per kg.

Process 1 Account (Mixing)

Particulars	kg	₹	Particulars	kg	₹
To Ingredients	6,000	₹9,000	By Normal Loss (W1)	480	—
To Labour & Overheads		₹7,000	By Transfer to Process 2 (W2)	5,520	₹16,000
Total	6,000	₹16,000	Total	6,000	₹16,000

Workings

1. Normal Loss Calculation:

$$\text{Normal loss} = 6,000 \times 8\% = 480 \text{ kg.}$$

2. Quantity Transferred to Process 2:

$$\text{Input} - \text{Normal Loss} = 6,000 - 480 = 5,520 \text{ kg}$$

Process 2 Account (Packaging)

Particulars	kg	₹	Particulars	kg	₹
To Transfer from Process 1	5,520	₹16,000	By Finished Goods (balancing figure)	5,520	₹38,000
To Packaging		₹12,000			
To Labour & Overheads		₹10,000			
Total	5,520	₹38,000	Total	5,520	₹38,000

Cost per kg Calculation:

Cost per kg = Total Costs/Expected Output = ₹38,000/5,520 kg = ₹6.88 per kg

7.6.2 ABNORMAL LOSS

It is also referred to as abnormal wastage. This type of loss is defined as any loss that exceeds the predetermined normal loss (normal process loss). Abnormal loss may arise due to factors such as worker carelessness, poor plant design or operation, sabotage, etc. Such losses cannot be estimated in advance; however, they can be managed through appropriate measures.

Abnormal Losses = Actual Loss – Normal Loss **Accounting Treatment:**

The cost associated with an abnormal process loss unit is equivalent to the cost of a good unit. The total cost of abnormal process loss is credited to the process account from which it originates. Unlike normal losses, the cost of abnormal process loss is not included as part of the product cost. Instead, the total cost of abnormal process loss is debited to the costing profit and loss account.

To address the issue of abnormal loss in process costing, you should follow this procedure:

1. Calculate the amount of normal loss and record it on the credit side of the respective process account, including its realizable value.
2. Assuming there is no abnormal loss, determine the cost per unit of output using the following formula:

$$\frac{\text{Cost of Production}}{\text{Normal Output}}$$

3. Determine the cost of abnormal loss units based on the cost per unit calculated in the previous step.
4. Debit the Abnormal Loss Account and credit the respective process account with the quantity and amount of the abnormal wastage.
5. The remaining balance in the Process Account will reflect the cost of the actual output, which should be transferred to the next process account.

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6. Create an Abnormal Loss Account, recording the cost of abnormal loss units on the debit side and their scrap value on the credit side. The balance in the Abnormal Loss Account is then transferred to the Costing Profit and Loss Account.

Example: Prepare a Process Account, Abnormal Loss Account, and Normal Loss Account from the following information.

Particulars	Units	Cost per Unit (₹)	Total Cost (₹)
Input of Raw Material	1200	25	30,000
Direct Material			5,000
Direct Wages			8,000
Production Overheads			7,500
Actual Output (Transferred to Process II)	1000		
Normal Loss	120 (10% of 1200)	10 (Scrap value)	1,200

Solution:

Dr. Process - I A/c. Cr.

Particulars	Units	₹	Particulars	Units	₹
To Raw Material @ 25	1200	30,000	By Normal Loss (10% of 1200)	120	1,200
To Direct Material		5,000	By Abnormal Loss A/c.	80	3,652
			$\frac{50,500 - 1200}{1200 - 120} \times 80$		
To Direct Wages		8,000	By Process - II A/c.	1000	45,648
To Production Overheads		7,500			
Total	1200	50,500	Total	1200	50,500

Dr. Abnormal Loss A/c. Cr.

Particulars	Units	₹	Particulars	Units	₹
To Process - I A/c.	80	3,652	By Bank A/c.	80	800
			By Costing P & L A/c.	80	2,852
Total	80	3,652	Total	80	3,652

Dr. Normal Loss A/c. Cr.

Particulars	Units	₹	Particulars	Units	₹
To Process - I A/c.	120	1,200	By Bank	120	1,200
Total	120	1,200	Total	120	1,200

Working Notes**1. Cost of Abnormal Loss:**

$$\begin{aligned}\text{Cost of abnormal Loss} &= \frac{\text{Total Cost increased} - \text{Sales value of Scrap}}{\text{Input units} - \text{Normal Loss Units}} \\ &\quad \times \text{abnormal units} \\ &= \frac{50,500 - 1200}{1200 - 120} \times 80 = 3,652\end{aligned}$$

- 2. Assumption:** It has been assumed that the abnormal loss units were sold at the same scrap rate of ₹10 per unit as the normal loss.

7.6.3 ABNORMAL GAIN

Sometimes, the loss in a process is lower than the anticipated normal figure, meaning that actual production exceeds expected levels. In such cases, the difference between actual and expected loss, or between actual and expected production, is referred to as abnormal gain or yield. Abnormal gain can be defined as an unexpected increase in production under normal conditions. This may occur due to overestimation of process loss, improvements in worker efficiency, or the use of better technology in production.

Accounting Treatment: The process account where the abnormal gain occurs is debited with the amount of the abnormal gain and credited to the Abnormal Gain Account, which will eventually be closed by transferring to the Costing Profit and Loss Account. The cost of abnormal gain is calculated based on the normal production levels.

**SELF-ASSESSMENT QUESTIONS**

- 5.** A company processes 1,200 units during a period. Normal loss is 10% of the input. Abnormal loss during the period was 20 units. The cost per unit is ₹50.

What is the value of **abnormal loss**?

- | | |
|------------------|------------------|
| a. ₹1,111 | b. ₹1,110 |
| c. ₹1,000 | d. ₹1,100 |
- 6.** What defines an abnormal gain in process costing?
- A loss exceeding the normal loss.
 - A predetermined process loss.
 - A decrease in production below expected levels.
 - The excess production over the expected normal output.

7.7 JOINT PRODUCTS AND BY-PRODUCTS

Imagine you walk into a chocolate factory (like in the movie *Charlie and the Chocolate Factory*). You see a conveyor belt carrying blocks of delicious chocolate. But wait—you also see cocoa butter and cocoa powder being produced at the same time from the same cocoa beans! Here's the

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interesting part: All these products come from the same source but serve different purposes.

This is the essence of joint products and by-products. Just like how cocoa beans give us chocolate, cocoa powder, and cocoa butter, many industries deal with multiple products coming from a single process.

Real-Life Industries Involved in Joint and By-Products

Consider these industries where multiple products come from a single source:

- ❑ **Oil Refineries:** Crude oil produces petrol, diesel, kerosene, and lubricating oil.
- ❑ **Sugar Industry:** Sugar production also creates molasses, which is used in alcohol production.
- ❑ **Chemical Industry:** While making soap, companies may also produce glycerin, which is used in cosmetics.
- ❑ **Mining Industry:** Mining for copper also yields other metals like silver as by-products.

Think About This: Have you ever seen petrol and diesel being produced from the same crude oil? That's because the same raw material (crude oil) is processed into different valuable products.

Joint Products vs. By-Products

It's important to understand the difference. Let's break it down with a fun comparison:

Criteria	Joint Products	By-Products
Intention	Products are produced intentionally	By-products are unintended but useful outcomes
Economic Value	Products have similar economic value	By-products have a lower value than main products
Examples	Petrol, diesel, and kerosene from crude oil	Molasses from sugar production
Use of Costs	Joint costs are shared by all joint products	By-products are usually treated as additional revenue
Production	Products emerge from the same process	Products result from the same process but are secondary outcomes

Features of Joint Products

Let's say you're managing a factory that processes crude oil. Here's what you'd observe about joint products:

1. **Single Raw Material, Multiple Products:** Crude oil produces petrol, diesel, kerosene, and lubricating oil—all from the same raw material.

2. **Similar Value:** No single product (like petrol or diesel) is drastically more valuable than the others.
3. **Planned Production:** You plan to produce all the products, not just one.
4. **Further Processing:** Some products may need more processing after being separated from the main product.
5. **Shared Costs:** The costs incurred before separation (like refining costs) are shared among all products.
6. **No Control Over Quantity:** The output of each product depends on the production process and can't be controlled precisely.

Features of By-Products

Now, think of the **sugar industry**. The primary focus is on making sugar, but guesses what? During production, **molasses** is also created. Let's see what makes it a by-product:

1. **Incidental Creation:** By-products aren't the main goal. Molasses happens naturally during sugar production.
2. **Lower Value:** Molasses has value, but it's far less than sugar.
3. **Not Always Planned:** By-products aren't always planned, but they can be used later for additional income (like selling molasses to alcohol manufacturers).
4. **Potential for Reuse:** Sometimes, by-products like coke oven gas are reused as energy sources within the same industry.

Classification of By-Products: By-products are classified based on their marketable condition at the split-off point:

- a. Those sold in their original form.
- b. Those that undergo further processing before being sold.

Example: Wish Ltd. manufactures product X which yields two by-products Y and Z.

The actual joint expenses of manufacturing for a period were ₹10,500. The profits on each product as a percentage of sales are 40%, 30%, and 10%, respectively. Subsequent expenses are as follows:

Particulars	X	Y	Z
Direct Material	120	90	40
Direct Labour	250	150	80
Overheads	180	140	100
Separate expenses	550	380	220
Sales	7,500	5,000	3,000

Apportion the joint expenses.

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Solution:**Statement Showing Apportionment of Joint Expenses**

Serial	Particulars	X	Y	Z	Total
1	Sales	7,500	5,000	3,000	15,500
2	Less: Profit (40%, 30%, 10%)	3,000	1,500	300	4,800
3	Total Cost (Joint & Separate cost) (1 – 2)	4,500	3,500	2,700	10,700
4	Less: Separate Expenses (Given)	550	380	220	1,150
5	Share of Joint Expenses (3 – 4)	3,950	3,120	2,480	10,500

Accounting Treatment: By-products are jointly produced items with minor importance and do not have individual costs until the split-off point. They arise as secondary results of the main manufacturing process. The methods for accounting for by-products fall into two broad categories: Non-Cost Methods and Cost Methods.

(A) Non-Cost Methods

- 1. Other Income or Miscellaneous Income Method:** The sales value of the by-product is credited to the Profit and Loss Account as other or miscellaneous income without crediting the Cost Accounts. This method doesn't ascertain the cost of the by-product or value inventory, with all costs charged to the main product. It is the least scientific method, used when the by-product's sales value is negligible. For example, company manufacturing furniture may produce small amounts of sawdust as a by-product. Since the sales value of sawdust is insignificant, the company credits the revenue from its sale directly to the Profit and Loss Account as "Other Income" without involving the cost accounts. All production costs, including those related to the by-product, are charged entirely to the main product (furniture), making this method less precise and often used for low-value by-products.
- 2. Total Sales Less Total Cost Method:** The sales value of the by-product is added to that of the main product. The total cost of the main product, including the by-product, is deducted from the combined sales revenue. For instance, in a leather tanning process, the main product is leather, and the by-product might be leather scraps. If the total sales revenue from the leather is ₹50,000 and from the scraps is ₹5,000, the combined sales revenue is ₹55,000. Then, the total cost of producing both the leather and scraps is subtracted from the combined sales revenue to determine the overall profit. This method combines all costs and revenues, including those of the by-product, into one calculation.
- 3. Total Cost Less Sales Value of By-Product Method:** In this method, the total production cost is reduced by the by-product's sales value. This method is preferred as by-product revenue, like scrap, reduces the cost of the main product. For example, a sugar refinery produces

sugar as the main product and molasses as a by-product. If the total production cost of the sugar is ₹100,000 and the sales value of the molasses is ₹10,000, then the production cost of sugar would be reduced by ₹10,000 (the sales value of molasses). This method is preferred as it directly decreases the cost of the main product, making it more cost-effective.

4. **Total Cost Less Sales Value of By-Products after Deducting Selling and Distribution Overheads:** Here, the by-product's sales value, less its selling and distribution expenses, is deducted from the total cost. These overheads are charged against sold by-products. For example, a refinery may produce both petrol and gas as by-products. If the gas is sold for ₹5,000, but the selling and distribution costs total ₹1,000, the net sales value of the gas is ₹4,000. This amount is then deducted from the total production cost of petrol and gas to determine the effective cost of the main product (petrol). This method ensures that the costs associated with selling the by-product are properly accounted for.
5. **Reverse Cost Method:** This method assumes the by-product's sales value includes a profit margin. The by-product's cost is calculated by working backward, reducing the sales value by the estimated profit margin, selling expenses, and post-split-off costs. The main product's cost is then reduced by this net amount. For example, a paper mill may produce both paper and wood chips. The sales value of the wood chips is ₹8,000, and it is assumed that this includes a profit margin. The company works backward to calculate the cost of the wood chips by subtracting the estimated profit margin, selling expenses, and any additional processing costs. Suppose the estimated profit margin is ₹2,000, selling expenses are ₹500, and processing costs are ₹1,000. The cost of the by-product is ₹4,500 (₹8,000 sales value minus ₹2,000 profit margin, ₹500 selling costs, and ₹1,000 processing costs). The cost of the main product (paper) is then reduced by this ₹4,500. This method allows for more precise cost allocation, assuming the by-product's sales value includes profit.

(B) Cost Methods

1. **Replacement or Opportunity Cost Method:** When by-products are used within the factory, they are valued at their opportunity or replacement cost—the amount that would have been spent if the by-product were purchased externally. For instance, in the sugar industry, bagasse, a by-product of sugar extraction, is often used as a fuel for boilers. Instead of considering its direct production cost, bagasse is valued at the market price, which reflects the cost the company would have incurred if it had bought fuel externally. This method ensures that the value of by-products used within the factory is aligned with external market conditions, helping the company evaluate its internal resource utilization more accurately.
2. **Standard Cost Method:** By-products are valued at a pre-determined standard cost, based on a technical assessment. The process account

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for the main product is credited with the by-product's standard cost, facilitating effective cost control for the main product. For example, a paper manufacturing company may set a standard cost for the by-product, such as waste paper, based on prior production experiences. The value of this by-product is credited to the process account of the main product (in this case, paper) using this standard cost. This helps in maintaining cost control, as the company has an expected cost for the by-product, making it easier to track variances and manage costs related to the main product.

- 3. Joint Cost Proration:** For significant by-products, joint costs are apportioned between the main products and by-products using an appropriate method. In industries where by-products are important, like petroleum refining, where gas is now treated as a joint product, the joint cost is prorated between the joint and by-products. For example, if a refinery produces both gasoline and gas, the joint costs of refining crude oil might be split between these two products, depending on the proportion of their respective values or volumes.

Split-Off Point: This is the stage in a production process where joint products become separately identifiable. The significance of the split-off point in joint product costing is that joint costs incurred up to this stage must be shared among the products resulting from the common process. Any costs incurred after the split-off point are considered product-specific costs and are allocated to the respective products.

Example: In the oil refining industry, crude oil undergoes the same refining processes to produce multiple products, such as petrol, diesel, kerosene, fuel oil, and lubricating oil. These products generally have comparable economic value, with none being significantly more valuable than the others. The production of these joint products is deliberate, as the organization aims to produce all the resulting goods rather than focusing on just one.

At the *split-off point*, where the products are separated, some may require additional processing (e.g., lubricating oil), while others, such as kerosene or fuel, might be ready for sale. Costs incurred up to this stage are known as *joint costs* and are allocated among the products based on methods like physical quantity, market value, or other relevant criteria. The proportions of each product are dictated by the refining process and raw materials, giving the company limited control over the exact output mix.

Figure 7.4 gives the pictorial representation of the example.

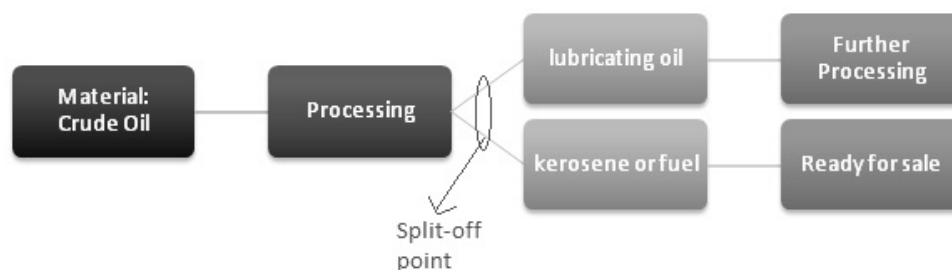


Figure 7.4 Split-off Point.

Example: During the production of a primary product, two by-products, 'A' and 'B', are also generated, producing 300 units and 100 units, respectively. At the *split-off point*, the saleable value is ₹120 per unit for A and ₹200 per unit for B. If these by-products are further processed, additional costs of ₹40,000 for A and ₹25,000 for B are incurred. After further processing, the saleable value increases to ₹200 per unit for A and ₹400 per unit for B.

Determine whether the management should sell these by-products at the *split-off stage* or process them further based on incremental profit/loss.

Solution:

Statement of Incremental Profit/Loss:

Details	Product A	Product B
Output (units)	300	100
Incremental Revenue		
After split-off process:		
	$(₹200 - ₹120) \times 300 = ₹24,000$	$(₹400 - ₹200) \times 100 = ₹20,000$
Less: Additional Costs	₹20,000	₹25,000
Incremental Profit/Loss	₹4,000 (gain)	-₹5,000 (Loss)

Recommendation:

- ❑ Product A should be processed further as it yields an incremental profit of ₹4,000.
- ❑ Product B should be sold at the split-off stage, as further processing leads to a loss of ₹5,000.



SELF-ASSESSMENT QUESTIONS

7. A company manufactures two joint products, Product A and Product B. Joint production costs amount to ₹60,000. The output and selling price per unit are as follows:
 - ❑ Product A: 300 units, ₹40 per unit
 - ❑ Product B: 200 units, ₹20 per unit
 Costs are apportioned based on sales value. What is the joint cost allocated to Product A?
 - a. ₹36,000 b. ₹30,000
 - c. ₹24,000 d. ₹45,000
8. How are by-products defined in process costing?
 - a. Main products with equal sale value produced simultaneously.
 - b. Incidental products with minor realizable value compared to the main product.
 - c. The primary products produced from a single process.
 - d. Products that require no further processing after production.

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7.8 DIFFERENCE BETWEEN JOINT PRODUCT AND BY-PRODUCT

Distinguishing between joint products, main products, and by-products can be quite challenging. However, there are a few criteria that can be used to identify whether a product is classified as a by-product, joint product which is given in **Table 7.2** below:

TABLE 7.2 DIFFERENCE BETWEEN JOINT PRODUCTS AND BY-PRODUCTS

Criteria	Joint Products	By-Products
Economic Value	Joint products have comparable value and are of equal importance.	By-products hold much lower economic value compared to the main products.
Production Process	Joint products are produced simultaneously as a part of the planned manufacturing process.	By-products are generated incidentally during the production of the main products.
Main Product Status	Joint products are considered significant outputs of the production process.	By-products are typically considered secondary or unwanted outputs resulting from the main production.
Distinction Rigidity	There is usually a clear and fixed distinction between joint products.	The distinction between by-products and main products is flexible and may vary depending on the circumstances.

Activity

A company operates in the chemical industry, producing multiple products through a single production process. These products are separated at the split-off point. The company faces challenges in distinguishing between joint products, main products, and by-products. Discuss the implications of misclassifying a product (e.g., treating a joint product as a by-product) on financial reporting, cost allocation, and profitability analysis.

7.9 JOINT COSTS

Joint product costs are prevalent in various industries, including petroleum, oil refining, textiles, dairy, food processing, and other process industries. Business management requires accurate and dependable cost information related to joint products to make informed decisions, such as whether to process further or sell at the split-off stage. To reach either conclusion, it is essential to determine the allocation of joint costs among the different joint products.

Joint costs are the expenses incurred up to the point of separation, known as the split-off point. These costs represent the resources expended by a manufacturer to produce multiple products from a common input. They

encompass raw materials, labor, power, fuel, depreciation, and overhead costs associated with the production of joint products.

The primary challenge in dealing with joint products and by-products is the apportionment of these joint costs to the respective products. For costs incurred after the split-off point, the allocation is straightforward, as these costs can be directly assigned to individual joint products or by-products.

7.10 METHODS OF COSTING OF JOINT PRODUCTS

Properly allocating joint costs among joint products is crucial, as it impacts (a) the valuation of closing inventory; (b) product pricing; and (c) the profit or loss from the sale of different products. Given the complex and unobservable relationships between materials, processes, and joint products, accurately determining the cost of the various production factors for each joint product is challenging. Thus, costs incurred in manufacturing each joint product cannot be precisely identified and must instead be apportioned using rational methods.

Common methods for apportioning total process costs up to the split-off point among joint products are illustrated in **Figure 7.5**:

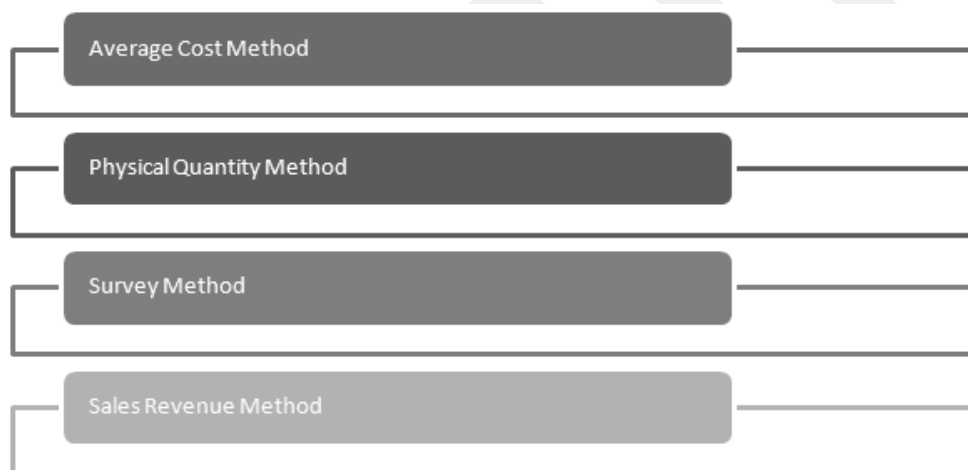


Figure 7.5 Methods of Costing of Joint Products.

7.10.1 AVERAGE COST METHOD

Under this method, the total process cost (up to the point of separation) is divided by the total units of joint products produced, resulting in the average cost per unit of production.

$$\text{Average unit cost} = \frac{\text{Total process cost (upto the point of separation)}}{\text{Total units of joint product produced}}$$

This method is straightforward, leading to a uniform cost per unit for all joint products. If this approach is used as a basis for pricing, all products may end up with similar prices. This method benefits customers purchasing higher-quality items, as they pay a lower price for their purchases.

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The joint products P, Q, R, and S are produced at a total joint production cost of ₹1,50,000. Quantities produced are:

- **P:** 25,000 units
- **Q:** 18,000 units
- **R:** 12,000 units
- **S:** 10,000 units

The products are sold at the split-off point for:

Solution:**Average Cost per Unit**

Total Joint Production Cost = ₹1,50,000

Total Units Produced = 25,000 (P) + 18,000 (Q) + 12,000 (R) + 10,000 (S)
= 65,000 units

Average Cost per Unit = Total Joint Costs ÷ Total Units Produced

= ₹1,50,000 ÷ 65,000 units

= ₹2.31 per unit

Apportionment of Joint Costs

Product	Units Produced	Average Cost per Unit (₹)	Apportioned Joint Cost (₹)
P	25,000	2.31	25,000 × 2.31 = 57,750
Q	18,000	2.31	18,000 × 2.31 = 41,580
R	12,000	2.31	12,000 × 2.31 = 27,720
S	10,000	2.31	10,000 × 2.31 = 23,100
Total	65,000		1,50,000

7.10.2 PHYSICAL QUANTITY METHOD

This method is based on the premise that joint products can be measured in the same units. Joint costs are apportioned according to a physical base, such as weight or number. Essentially, the allocation of joint costs to the joint products is determined by their physical volume at the split-off point. Any losses incurred during the joint production process are also allocated based on this criterion. When physical units differ, joint products must be converted to a common unit of measurement; if this is not feasible, the method cannot be applied. A significant drawback of this method is that it assigns equal importance and value to all joint products. This method is often used when the sale price of all products is uniform.

Example: A chemical manufacturing company produces the following products using 100 tons of raw material:

Product	Production (in Tons)
Product A	40
Product B	30
Product C	10
Product D	15
Product E	5
Total	100

The prices for the joint costs are as follows:

- ❑ **Price of Raw Material:** ₹300 per ton
- ❑ **Direct Labour Cost:** ₹150 per ton
- ❑ **Overheads:** ₹200 per ton

Using the **physical unit method**, apportion the joint costs to each joint product.

Solution:

Apportionment of Joint Costs based on Physical Unit Method:

Total Joint Costs

- ❑ Raw Material = 100 tons × ₹300 = ₹30,000
- ❑ Direct Labour = 100 tons × ₹150 = ₹15,000
- ❑ Overheads = 100 tons × ₹200 = ₹20,000

Total Joint Costs: ₹65,000

Apportionment Based on Physical Unit Method

Joint Product	Production (tons)	% of Total Production	Apportioned Raw Material Cost	Apportioned Labour Cost	Apportioned Overheads	Total Apportioned Cost
Product A	40	40%	₹12,000	₹6,000	₹8,000	₹26,000
Product B	30	30%	₹9,000	₹4,500	₹6,000	₹19,500
Product C	10	10%	₹3,000	₹1,500	₹2,000	₹6,500
Product D	15	15%	₹4,500	₹2,250	₹3,000	₹9,750
Product E	5	5%	₹1,500	₹750	₹1,000	₹3,250
Total	100	100%	₹30,000	₹15,000	₹20,000	₹65,000

7.10.3 SURVEY METHOD

This method is often referred to as the point value method. It relies on a technical survey of all factors related to the production and distribution of products. In this approach, joint costs are allocated to the joint products based on percentage or point values assigned to each

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product according to their relative significance. Typically, management computes these percentages or points with assistance from technical advisors. This method is regarded as more equitable compared to other allocation methods.

Example: A factory produces three products, X, Y, and Z, in the following quantities:

- **X:** 3,500 units
- **Y:** 800 units
- **Z:** 1,200 units

The joint cost of these three products is ₹80,000. Based on a survey and technical assessment, the following weights are allotted to each product:

- **X:** 5 per unit
- **Y:** 7 per unit
- **Z:** 9 per unit

Required: Apportion the joint cost using the Survey Method.

Solution:

Calculate the weighted units for each product

- **X:** 3,500 units $\times$ 5 = 17,500 weighted units
- **Y:** 800 units $\times$ 7 = 5,600 weighted units
- **Z:** 1,200 units $\times$ 9 = 10,800 weighted units

Calculate the total number of weighted units

Total weighted units = 17,500 + 5,600 + 10,800 = 33,900 weighted units

Apportion the joint cost based on the weighted units

The joint cost is ₹80,000, so the apportionment of joint cost is calculated as follows:

Apportionment of Joint Cost for each Product = (Weighted Units of each/ Product Total Weighted Units) $\times$ Total Joint Costs

- **X:** $(17,500/33,900) \times 80,000 = 41,430$
- **Y:** $(5,600/33,900) \times 80,000 = 13,200$
- **Z:** $(10,800/33,900) \times 80,000 = 25,370$

Product	Production Units	Weights Allotted	Weighted Units	Apportioned Joint Cost (₹)
X	3,500	5	17,500	41,430
Y	800	7	5,600	13,200
Z	1,200	9	10,800	25,370
Total	5,500		33,900	80,000

7.10.4 SALES REVENUE METHOD

Under this method, the joint cost is allocated based on the sales value at the split-off point. The rationale is that each product should bear a share of the joint costs proportional to its sales price. If a product's sales price is higher than that of others, it should be allocated a larger share of the joint costs, and conversely, if its sales price is lower, it should bear a smaller share. While this method appears logically sound, in practice, assigning a higher share of joint costs to products with higher sales values may not be justified, as these products may require less effort or resources to manufacture.

Example: The Joint Products P, Q, R, and S are produced at total joint production costs of ₹1,00,000. The quantities produced and their respective sale prices at the split-off point are as follows:

Product	Units Produced	Sale Price per Unit (₹)
P	25,000	12
Q	18,000	6
R	12,000	10
S	20,000	5

Required: Show the apportionment of joint costs using the Sale revenue method.

Solution:

Calculate the total sale value for each product

- **P:** 25,000 units × ₹12 = ₹3,00,000
- **Q:** 18,000 units × ₹6 = ₹1,08,000
- **R:** 12,000 units × ₹10 = ₹1,20,000
- **S:** 20,000 units × ₹5 = ₹1,00,000

Total Sale Value of all products = ₹3,00,000 + ₹1,08,000 + ₹1,20,000 + ₹1,00,000 = ₹6,28,000

Apportion joint costs based on the total sale value of each product

The formula for apportioning joint costs is:

Apportioned Cost for each Product = (Total Sale Value of each Product / Total Sale Value of all Products) × Total Joint Costs

The joint costs are ₹1,00,000. Let's calculate the apportionment:

- **P:** (3,00,000/6,28,000) × 1,00,000 = 47,800
- **Q:** (1,08,000/6,28,000) × 1,00,000 = 17,200

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$$\square \text{ R: } (1,20,000/6,28,000) \times 1,00,000 = 19,100$$

$$\square \text{ S: } (1,00,000/6,28,000) \times 1,00,000 = 15,900$$

Product	Units Produced	Sale Price per Unit (₹)	Total Sale Value (₹)	Apportioned Joint Cost (₹)
P	25,000	12	3,00,000	47,800
Q	18,000	6	1,08,000	17,200
R	12,000	10	1,20,000	19,100
S	20,000	5	1,00,000	15,900
Total	60,000		6,28,000	1,00,000



SELF-ASSESSMENT QUESTIONS

9. What is a significant drawback of the Physical Quantity Method for allocating joint costs?
 - a. It requires the use of historical data for cost estimation.
 - b. It assigns equal importance and value to all joint products, regardless of their sales value.
 - c. It is complicated and time-consuming to implement.
 - d. It cannot account for joint costs before the split-off point.
10. Which method of apportioning joint costs allocates a higher share of costs to products with a higher sales price at the split-off point?
 - a. Physical Quantity Method
 - b. Average Cost Method
 - c. Survey Method
 - d. Sales Revenue Method

7.11 ADDITIONAL SOLVED PROBLEMS

1. The following particulars for Process III are given:

Particulars	Units	Amount (₹)
Transfer to Process III at cost	5,000	15,000
Direct wages		4,000
Direct material		5,000
Transfer to Finished Stock	4,500	
Factory overheads (absorbed at 300% of direct wages)		
Allowance for Normal Loss (15% of units worked)		
Scrap value per unit		10

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Evaluate the cost of transfer to finished stock.

Solution:

Dr. PROCESS-III Account Cr.

Particulars	Units	Amount (₹)	Particulars	Units	Amount (₹)
To Transfer from Process II	5,000	15,000	By, Normal Loss A/c (5,000 × 15%) × 10	750	7,500
To Direct Wages A/c		4,000			
To Direct Material A/c		5,000	By, Transfer to Finished Stock A/c (4,500 units @ ₹6.71 per unit)	4,500	30,176
To Factory Overheads (300% of wages)		12,000			
To Abnormal Gain A/c	250	1,676			
<u>36,000 – 7,500</u>					
<u>5,000 – 750</u>					
= 6.71					
	5,250	37,676		5,250	37,676

Total Cost Transferred to Finished Stock = ₹37,676

2. Product-Y is produced after undergoing three sequential processes. You are required to prepare process accounts for each process based on the following data:

Process Details

Particulars	Process A	Process B	Process C	Total
Materials (₹)	6,500	4,300	5,800	16,600
Direct Wages (₹)	5,000	6,500	8,500	20,000
Production Overheads (₹)	-	-	-	30,000

1,200 units were introduced in Process A at ₹8 per unit. Production overheads are to be apportioned at 150% of direct wages.

Actual Output and Loss Details

Process	Actual Output (Units)	Normal Loss (%)	Scrap Value (₹/Unit)
Process A	1,150	4.17%	20
Process B	1,100	7%	15
Process C	1,050	4%	12

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Required

Prepare the Process Accounts for Process A, Process B, and Process C including the normal loss and scrap value for each process.

Solution:

Dr. PROCESS-A Account Cr.

Particulars	Units	₹	Particulars	Units	₹
To Material introduced	1,200	9,600	By Normal Loss (4.17% of 1,200) × 20	50	1,000
To Additional Material		6,500			
To Direct Wages A/c		5,000	By Transfer to Process-B A/c @ ₹24 per unit	1,150	27,600
To Production Overheads A/c		7,500			
	1,200	28,600		1,200	28,600

Dr. PROCESS-B Account Cr.

Particulars	Units	₹	Particulars	Units	₹
To Transfer from Process-A A/c	1,150	27,600	By Normal Loss (7% of 1,150) × 15	80	1,200
To Direct Material A/c		4,300	By Transfer to Process-C A/c @ ₹43.88 per unit	1,100	48,266
To Direct Wages A/c		6,500			
To Production Overheads A/c		9,750			
To Abnormal Gain $\frac{48,150 - 1200}{1,150 - 80} = 43.88$	30	1316			
	1,180	49,466		1,180	49,466

Dr. PROCESS-C Account Cr.

Particulars	Units	₹	Particulars	Units	₹
To Transfer from Process-B A/c	1,100	48,266	By Normal Loss (4% of 1,050) × 12	42	504
			By Abnormal loss	8	566
			$\frac{75,316 - 504}{1,100 - 42} = 70.71$		

(Continued)

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Particulars	Units	₹	Particulars	Units	₹
To Direct Material A/c	5,800		By Transfer to Finished Stock A/c @ ₹ 70.71 per unit	1,050	74,246
To Direct Wages A/c	8,500				
To Production Overheads A/c	12,750				
	1,100	71,050		1,155	82,457

3. ABC Ltd. manufactures a primary product X, which results in two by-products, Y and Z. The total joint manufacturing expenses for the period amounted to ₹10,000. The profits on each product as a percentage of sales are 35%, 20%, and 10%, respectively. The subsequent expenses for each product are as follows:

Particulars	X	Y	Z
Material	600	200	60
Direct	200	200	40
Overheads	400	100	100
Total Costs	1,200	500	200
Sales (₹)	8,000	5,000	3,000

Apportion the joint expenses of ₹10,000 among the three products using a suitable method and determine the profit for each product.

Solution: Statement Showing Apportionment of Joint Expenses

Particulars	X	Y	Z	Total
Sales	8,000	5,000	3,000	16,000
(-) Profit	2,800	1,000	300	4,100
Total Cost (Joint & Separate Cost)	5,200	4,000	2,700	11,900
Separate Expenses	1,200	500	200	1,900
Share of Joint Expenses	4,000	3,500	2,500	10,000

4. A vegetable oil manufacturing company processes 10,000 kilograms of seeds at a cost of ₹20 per kilogram into various products. The outputs from this process are as follows:

- ❑ **Refined Oil:** 6,000 kg
- ❑ **Cake Meal:** 3,000 kg
- ❑ **Glycerin:** 800 kg
- ❑ **Waste:** 200 kg

Apportion the joint cost among the products using the physical unit method.

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Solution:

Products	Refined Oil	Cake Meal	Glycerin	Wastage	Total
Output (in tonnes)	6,000	3,000	800	200	10,000
Wastage (in tonnes)	123	61	16		200
Total weight (in tonnes)	6,123	3,061	816		10,000
Joint Cost (in ₹)@ ₹20	₹1,22,460	₹61,220	₹16,320		₹2,00,000

Apportionment of wastage of 200 kgs over the three products is as follows

□ **Refined Oil:** $\frac{200}{9,800} \times 6,000 = 123$

□ **Cake Meal:** $\frac{200}{9,800} \times 3,000 = 61$

□ **Glycerin:** $\frac{200}{9,800} \times 800 = 16$

5. Find the cost of joint products X, Y, and Z using the average unit cost method from the following data:

a. Pre-separation Joint Cost: ₹90,000

b. Production data:

Products	Units Produced
X	400
Y	300
Z	600
Total	1,300

Solution:

$$\text{Average cost per unit} = \frac{\text{Total joint costs}}{\text{Units produced}} = \frac{₹90,000}{1,300 \text{ units}} = ₹69.23$$

The joint costs apportioned @ ₹ 69.23 are as follows:

Products	Units	Costs per unit (₹)	Value (₹)
X	400	69.23	27,692
Y	300	69.23	20,770
Z	600	69.23	41,538
			90,000

6. A manufacturing process yields 70% of the input material as main Product X, 20% as by-product Y, and 8% as by-product Z, with the remaining 2% being wastage.

The allocation of raw material and labor costs among the products is as follows:

1. One unit of product Z requires one-third of the raw material required for one unit of product Y. One unit of product X requires twice the raw material required for one unit of product Y.
2. Product X requires three times the labor time needed for one unit of product Y. One unit of product Z requires one-fourth of the labor time needed for one unit of product Y.
3. Overheads are to be absorbed in the ratio of 8:1:1 for products X, Y, and Z, respectively.

4. Cost Data:

- ❑ **Input:** 1,200 units at a cost of ₹8,400
- ❑ **Direct Labor:** ₹6,800
- ❑ **Overheads:** ₹7,500

Calculate the distribution of costs between products X, Y, and Z using the data provided.

Solution:

Determine the units of each product:

- ❑ **Product X (70%):** $1,200 \times 70\% = 840$ units
- ❑ **Product Y (20%):** $1,200 \times 20\% = 240$ units
- ❑ **Product Z (8%):** $1,200 \times 8\% = 96$ units
- ❑ **Wastage (2%):** $1,200 \times 2\% = 24$ units

Statement Showing Apportionment of Joint Cost

Element	Basis of Apportionment	Total	Main Product X	By-Product Y	By-Product Z
Material	6:3:1	8,400	5,040	2,520	840
Labour	12:4:1	6,800	4,800	1,600	400
Overheads	8:1:1	7,500	6,000	750	750

7.12 SUMMARY

- ❑ Process costing is a method used in industries where production is continuous, and the product passes through various stages or processes, such as in textiles, chemicals, and food production.
- ❑ Costs are collected by process or department, and the cost per unit is calculated by dividing the total cost of a process by its total output during a specific period.
- ❑ Normal loss, abnormal loss, and abnormal gain are common occurrences in process costing. Normal loss is expected, abnormal loss

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exceeds expectations, and abnormal gain refers to unexpected production efficiency.

- ❑ Joint products and by-products are accounted for separately when multiple outputs are derived from a single production process, with joint costs allocated based on methods like average cost or sales revenue method.
- ❑ Process accounts include all costs such as materials, labor, overheads, and scrap value, and these accounts track the movement and cost of products as they flow from one process to another.
- ❑ Joint products are produced simultaneously and hold similar value, whereas by-products are secondary and have lesser value compared to the main product.
- ❑ The split-off point is where joint products become individually identifiable, and joint costs incurred up to this point must be allocated among them.
- ❑ Joint costs include expenses like raw materials, labor, and overhead, and they need to be shared among products produced before the split-off point.
- ❑ Methods for allocating joint costs include the Average Cost, Physical Quantity, Survey, and Sales Revenue methods, each with distinct approaches for cost distribution.
- ❑ Joint products are equally important in production, while by-products are incidental, with their distinction often varying based on economic value and production context.

7.13 KEY WORDS

1. **Process Costing:** A costing method used in continuous production industries where products pass through multiple processes before completion.
2. **Normal Loss:** An expected and unavoidable loss that occurs naturally during the production process.
3. **Abnormal Loss:** A loss exceeding the anticipated normal loss due to inefficiency or other unexpected factors.
4. **Joint Products:** Two or more products that are produced simultaneously from a common input in the same production process.
5. **By-Products:** Secondary products that are produced incidentally during the manufacturing of a main product, often with lesser value.
6. **Joint Costs:** Expenses incurred up to the split-off point, shared among all products resulting from a common process.
7. **Average Cost Method:** Allocation method dividing total process costs equally among all joint products.

8. **Physical Quantity Method:** Allocates joint costs based on the physical volume or weight of each product.
9. **Survey Method:** Uses technical evaluations to assign joint costs based on the relative importance of each product.
10. **Sales Revenue Method:** Allocates joint costs according to the sales value of each product at the split-off point.

7.14 CLOSING CASELET

JOINT PRODUCTS AND BY-PRODUCTS IN THE SUGAR INDUSTRY

In the sugar industry, companies like Sweet Sugar Mills face challenges with joint and by-products during sugarcane processing. Alongside refined sugar, by-products such as molasses and bagasse are produced, each requiring appropriate cost allocation. The primary issue is how to allocate joint costs incurred before the split-off point, where the main and by-products separate, for accurate inventory valuation, pricing, and income determination.

Sweet Sugar Mills adopted the Survey Method to allocate joint costs based on the market value of each product at the split-off point. This ensured fair distribution of costs between refined sugar and by-products like molasses and bagasse. The company also evaluated the profitability of further processing by-products, deciding to convert molasses into ethanol for higher revenue while selling bagasse to reduce processing costs.

By using a well-considered cost allocation method and strategically deciding to process molasses further, Sweet Sugar Mills enhanced profitability. Accurate income reporting and cost management allowed the company to maximize revenue from by-products that would otherwise be undervalued, demonstrating the importance of thoughtful decision-making in industries with joint products.



QUESTIONS

1. How does the survey method help in allocating joint costs between main products and by-products in the sugar industry? What are the other methods of allocating joint cost?

(Hint: Consider how the market value of each product at the split-off point plays a role in distributing costs and ensuring fair valuation.)

2. What factors should Sweet Sugar Mills consider before deciding to further process by-products?

(Hint: Think about the costs involved in further processing versus the potential increase in the sale value of the by-products, along with market demand.)

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7.15 HIGHER ORDER THINKING SKILLS QUESTIONS (HOTS)

1. How might process costing improve decision-making in industries with continuous production flows, and what challenges might arise in allocating joint costs for products like oil and chemical by-products?
2. In what ways could an abnormal gain signal improvements in production efficiency, and how should management adjust their future cost estimates and operational strategies based on this occurrence?
3. How do the methods of costing joint products (e.g., average cost method, sales revenue method) impact pricing strategies and profitability, particularly when market prices for joint products fluctuate?
4. In what situations would the Sales Revenue Method of cost allocation lead to distorted profit margins, and how could these affect long-term strategic decisions for product development and marketing?
5. Considering the differences between joint products and by-products, how can a business determine when it is more profitable to further process by-products rather than selling them at the split-off point?
6. Assume 600 units were worked on during a period, and a total of 550 good units were completed. Normal spoilage consisted of 40 units, and abnormal spoilage consisted of 10 units. The total production costs were ₹3,000. The company accounts for abnormal spoilage separately on the income statement as a **loss due to abnormal spoilage**. Normal spoilage is included in the cost of good units produced. What is the cost of the **good units produced**?
7. Product B passes through three distinct processes, and the following data for the week ending 31st March 2006 are provided:

Particulars	Total Amount	Process I	Process II	Process III
Direct Material	₹7,542	₹2,600	₹1,980	₹2,962
Direct Wages	₹9,000	₹2,000	₹3,000	₹4,000
Production Overheads	₹9,000			

- ❑ 1,000 units were introduced into Process I at ₹3 each.
- ❑ There was no opening stock of materials or work in progress, and the output from each process passes directly to the next process and finally to the finished goods store.

The following additional information is provided:

- ❑ **Process I:** 950 units were completed, with a normal loss of 5%. Scrap value per unit = ₹2
- ❑ **Process II:** 840 units were completed, with a normal loss of 10%. Scrap value per unit = ₹4
- ❑ **Process III:** 750 units were completed, with a normal loss of 15%. Scrap value per unit = ₹5

Production Overheads are recovered on 100% of direct wages.

Required:

1. Prepare the Process Cost Accounts for Process I, Process II, and Process III.
 2. Prepare the Abnormal Loss and Abnormal Gain Accounts.
8. Alpha Chemicals purchases raw material A and processes it into three refined products: Product P, Product Q, and Product R. In the month of August, XYZ Chemicals purchased raw material A for ₹50,000. Conversion costs of ₹70,000 were incurred up to the split-off point, at which time three sellable products were produced. Product Q can be further processed into Product R. The August production and sales information is as follows:

Product	Production (tonnes)	Sales Quantity (tonnes)	Selling Price (per tonne)
Product P	1,500	1,500	₹60
Product Q	1,000	—	—

XYZ Chemicals could have sold all its August production of Product Q at ₹100 per tonne.

Calculate how the joint cost of ₹1,20,000 would be apportioned between Product P and Product Q under each of the following methods:

- a. Sales value at split-off point method
 - b. Physical measure (tonnes) method
9. In the course of manufacturing the main product P, by-products A and B also emerge. The joint expenses of manufacturing total ₹1,19,550. All the products are processed further after separation and sold according to the following details:

Particulars	Main Product P	By-Product A	By-Product B
Sales	₹90,000	₹60,000	₹40,000
Cost beyond split-off point	₹6,000	₹5,000	₹4,000
Profit as percentage of sales	25%	20%	15%

Additionally, selling and administration overheads are absorbed as a percentage of the cost of sales.

Required:

- ❑ Apportion the joint costs to the main product P and the by-products A and B.
- ❑ Prepare the Main Product P Account, showing the cost of production, the profit, and the sales.

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7.16 ANSWER KEY**SELF-ASSESSMENT QUESTIONS**

Topic	Q. No.	Answer
Introduction	1.	c. Chemical plants
	2.	b. It helps pinpoint which department is responsible for increased production costs
Preparation of Process Account	3.	a. Debit the cost of basic raw materials to the first process account
	4.	d. The cost of the finished product from the first process
Normal Loss, Abnormal Loss and Abnormal Gain	5.	a. ₹1,111
	6.	d. The excess production over the expected normal output.
Joint Products and By-Products	7.	d. ₹45,000
	8.	b. Incidental products with minor realizable value compared to the main product.
Methods of Costing of Joint Products	9.	b. It assigns equal importance and value to all joint products, regardless of their sales value.
	10.	d. Sales Revenue Method

7.17 SUGGESTED READINGS

- ❑ Horngren, C. T., Datar, S. M., & Rajan, M. (2021). *Cost Accounting: A Managerial Emphasis* (17th ed.). Pearson.
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- ❑ Article: Gupta, P., & Yadav, N. (2020). *Effective Apportionment of Joint Costs: Challenges and Solutions*. *International Journal of Financial and Accounting Studies*, 15(1), 45–59.
- ❑ Smith, R. (2019). *Cost Allocation Practices for By-Products: An Empirical Study in the Chemical Industry*. *Global Journal of Management and Business Research: Accounting and Auditing*, 19(3), 45–54.

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8

U N I T

STANDARD COSTING

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INTRODUCTORY CASELET

A well-known case of successful cost control and reduction is that of Dell Technologies, particularly during its early years of growth in the 1990s. Dell adopted a direct-to-consumer sales model and lean production practices to streamline operations and minimize costs. The company focused on maintaining predetermined cost standards across its production processes, specifically aiming to reduce inventory and overhead expenses through a build-to-order approach.

Dell's standard costing system was integral to this success. By setting clear cost targets for materials, labor, and overheads, the company ensured that actual costs were closely monitored against these benchmarks. Any discrepancies between standard and actual costs were promptly addressed, with management identifying areas for improvement, such as eliminating excess inventory or reducing supplier lead times. This method of continuous cost evaluation allowed Dell to avoid unnecessary expenses while maintaining product quality and responsiveness to customer demands.

As a result of this rigorous cost control strategy, Dell was able to significantly reduce its cost-per-unit, pass these savings on to customers, and maintain a competitive pricing advantage in the market. By using standard costing not only as a cost-monitoring tool but also as a philosophy for operational efficiency, Dell remained profitable during periods of rapid expansion, ultimately revolutionizing the personal computer industry with its cost-effective and customer-centric business model.



QUESTIONS

1. How did Dell Technologies implement cost control through its direct-to-consumer sales model? Discuss the role of standard costing in reducing inventory and overhead costs.

(Hint: Focus on how Dell reduced excess inventory and improved supply chain efficiency by setting predetermined cost levels for materials, labor, and overheads, and how this approach aligned with its build-to-order production model.)

2. Explain how standard costing allowed Dell to maintain a cost-per-unit advantage while growing rapidly in a competitive market. What key processes helped identify variances and achieve cost reduction?

(Hint: Explore how comparing actual costs with standard costs helped Dell address discrepancies, reduce production waste, and improve operational efficiency, allowing the company to offer competitive prices without compromising quality.)



LEARNING OBJECTIVES

After reading this unit, you will be able to:

- Learn the primary goals of implementing standard costing in an organization
- Grasp the importance of establishing cost standards
- Understand the step-by-step processes of standard costing.
- Explore the key advantages of standard costing
- Identify potential drawbacks of standard costing
- Learn how standard costing complements budgetary control

8.1 STANDARD COSTING

In any organization, especially those dealing with limited resources, the goal should always be to control and reduce costs. While it may seem acceptable for a company to focus on selling its products or services at high prices, it's equally important to minimize the use of resources without lowering the quality of the final product or service. Think of it like managing a personal budget—even if additional funds are available, being careful with expenses allows more room for savings or other needs.

One way companies achieve this is through a practice called standard costing. Imagine planning a work schedule. You aim to complete 6 tasks daily for a week, but one day you only complete 4 tasks because of an unexpected event. If you're tracking your work progress, you'd recognize you're 2 tasks short and figure out how to make up for it. Businesses do something similar. They set predetermined cost levels—their “work plan”—and then track whether actual expenses stay within these limits. If they don't, they investigate the reasons and make adjustments.

To understand standard costing better, think about a bakery. Suppose a bakery sets a target of producing 100 loaves of bread in 8 hours at an ingredient cost of ₹200. Ideally, the cost per loaf should be ₹2. However, if only 80 loaves are produced due to a machine breakdown, it may seem like the cost per loaf has risen to ₹2.50 (₹200 divided by 80). But with the logic of standard costing, the bakery still considers the target cost per loaf as ₹2. The cost of the 20 unproduced loaves is recorded as a “loss” of ₹40, separate from the cost of the 80 loaves actually produced. This process helps the bakery's management identify where they're losing money and take action to prevent it in the future.

This concept might feel similar to budgeting for a group activity. Let's say you plan to print 50 color copies for a total cost of ₹100 (which makes the cost per copy ₹2). But if you're only able to print 40 copies because of a paper shortage, the total cost of ₹100 still stands, but the cost per copy seems to rise to ₹2.50. Instead of simply accepting this, you'd identify the “loss” of 10 copies and figure out how to avoid it next time.

The heart of standard costing is setting clear cost targets, trying to meet them, and comparing actual expenses to these targets. If there's a

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difference—called a variance—you investigate to find out why. Was it due to a mistake, equipment failure, or something else? This process ensures that management has a clear picture of financial performance. It's like receiving feedback on a project—knowing which parts went wrong allows you to focus on improving those areas.

According to the Chartered Institute of Management Accountants (CIMA), standard costing is a control tool that helps measure performance by comparing standard (target) costs with actual results. It's a method that offers valuable insights into where the organization's processes can be improved. Much like tracking a work schedule, a business that tracks its costs can make better decisions to achieve success.

Definition of Standard Costs

Standard costs are scientifically predetermined costs associated with manufacturing a single unit or multiple units of a product or providing a service during a specified future period. The Chartered Institute of Management Accountants (CIMA), London, defines standard cost as “a standard expressed in money. It is built up from an assessment of the value of cost elements. Its main uses are providing bases for performance measurement, control by exception reporting, valuing stock, and establishing selling prices.” This definition indicates that standard costs are planned costs of a product based on current or anticipated operating conditions.

The dictionary meaning of “standard” refers to a “thing serving as a basis for comparison” or a “thing recognized as a model for imitation.” It is important to note that “standard” is a relative term; what is standard for one may be substandard for another. However, within an organization, it represents a desirable target.

Components of Standard Cost

The term “standard cost” comprises two parts: ‘standard’ and ‘cost.’ Standards can be established concerning quantities and qualities, such as materials and labor, while cost involves expressing the established standard in monetary values.

CIMA's Definition of Standard Costing

CIMA defines standard costing as “a control technique that compares standard costs and revenues with actual results to obtain variances, which are used to stimulate improved performance.” The technique of standard costing can be summarized as follows:

1. **Pre-Determination of Technical Data:** This involves determining the technical details related to production, including materials and labor operations required for each product, expected losses, and levels of activity.
2. **Pre-Determination of Standard Costs:** Standard costs are determined in detail for each cost element, including materials, labor, and overhead.

3. **Comparison of Actual Performance with Standards:** This step involves comparing actual performance and costs against the established standards and calculating variances, which are the differences between actuals and standards.
4. **Analysis of Variances:** This step aims to understand the reasons behind the deviations of actual performance from the standards.
5. **Presentation of Information:** The final step involves presenting the analyzed information to the appropriate management level for suitable action.

Standard Cost and Estimated Costs

Estimates are predetermined costs that rely on historical data and are often not based on scientific analysis. They are usually compiled from loosely gathered information, making them unreliable for performance measurement. In contrast, standard costs are predetermined costs that reflect what the cost should be rather than what it will be. Both standard costs and estimated costs are utilized to determine prices in advance, with the aim of controlling costs. However, there are key differences between these two types of costs, as outlined below in **Table 8.1**:

TABLE 8.1 DIFFERENCES BETWEEN STANDARD COSTS AND ESTIMATED COSTS

Aspect	Standard Cost	Estimated Cost	Example
Focus	Represents what the cost <i>should be</i> under ideal conditions.	Indicates what the cost is <i>likely to be</i> in the future.	A manufacturer sets a standard cost of ₹50 per unit, but the estimated cost is ₹55 per unit due to expected raw material price hikes.
Determination	Calculated using technical analysis and efficiency factors.	Based on past data with adjustments for expected changes.	The standard cost to produce a chair is set at ₹1,000, but the estimated cost is ₹1,200, considering past increases in material costs.
Efficiency Measurement	Used to measure and evaluate operational efficiency.	Does not measure efficiency; it predicts future costs.	If the actual production cost of a table is ₹900 compared to a standard cost of ₹1,000, efficiency is achieved. Estimated costs, however, do not offer this insight.
Purpose of Cost Control	Helps control costs by setting cost benchmarks.	Not directly used for cost control.	A bakery sets a standard ingredient cost of ₹200 for 100 loaves of bread, and any deviation (like ₹220) is investigated. Estimated costs are only used as forecasts.

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Aspect	Standard Cost	Estimated Cost	Example
Accounting Process	Forms part of formal cost accounting records.	Used for informal planning and not recorded in ledgers.	Variances between standard and actual costs are recorded in the accounts, while estimated costs are only referenced in planning documents.
Recognition	Widely accepted as a formal cost accounting tool.	Informal, often used in budgeting and forecasts.	Standard costs are recognized by accounting bodies like CIMA, while estimated costs are primarily used for future planning.
Application	Used by firms that employ standard costing systems.	Used for forecasting in all types of organizations.	A car manufacturer uses standard costs for each vehicle, while a retailer forecasts rent expenses using estimated costs.

8.1.1 OBJECTIVES OF STANDARD COSTING

1. **Cost Control:** The primary objective of standard cost is to assist management in cost control. It serves as a benchmark against which actual costs can be compared to assess efficiency. Management can compare actual costs with standard costs at regular intervals and take corrective measures to maintain cost control.

Example: A clothing manufacturer sets a standard cost of ₹500 per shirt. If actual production costs rise to ₹550 per shirt, management identifies reasons like wastage of fabric or labor inefficiency and takes corrective action.

2. **Management by Exception:** Another objective of standard cost is to enable management to exercise cost control through the principle of exception. Standard cost helps establish benchmarks, and management's attention is drawn only when actual performance deviates from these standards, allowing a focus on variations.

Example: If the standard production cost of a table is ₹1,000 and the actual cost remains the same, no action is required. However, if the actual cost increases to ₹1,200, management investigates the cause of the deviation, such as higher raw material prices or machine breakdowns.

3. **Develops Cost-Conscious Attitude:** A further objective of standard cost is to cultivate a cost-conscious culture throughout the organization. It encourages employees to recognize the importance of efficient operations, fostering collaboration to reduce costs.

Example: In a food processing company, employees are aware of the standard material usage of 2 kg of fruit to produce 1 liter of juice. If they notice wastage, they report it to management to prevent future losses.

- 4. Fixation of Prices:** Standard cost aids management in formulating production policies and setting price quotations, as well as in submitting tenders for various products. Standard costs provide greater accuracy compared to actual costs and facilitate effective production planning by eliminating the impact of abnormal price fluctuations.

Example: A construction company prepares a tender for a government project. Using standard costs, it estimates the production cost of concrete at ₹2,000 per cubic meter, providing a reliable price quote that is unaffected by temporary fluctuations in cement prices.

- 5. Determining Prices and Formulating Policies:** Another goal of standard cost is to assist management in determining prices and developing production policies. It supports areas such as profit planning, product pricing, and inventory valuation.

Example: A bakery uses standard costs to price its cakes. If the standard cost of ingredients and labor per cake is ₹200, the selling price is set at ₹250, ensuring a profit margin of ₹50. This approach supports product pricing and production planning.

- 6. Management Planning:** Budget planning is conducted by management at various levels periodically to maximize profits through different product mixes. Standard costing is more convenient for this purpose than actual costs, as it is based on a scientific and rational approach that considers all technical aspects.

Example: A toy manufacturer uses standard costs to prepare a production budget. It sets a standard cost of ₹300 per toy, helping management decide how many units to produce and how to optimize the production mix for maximum profitability.



SELF-ASSESSMENT QUESTIONS

1. Which of the following best differentiates standard costs from estimated costs?
 - a. Standard costs are based on historical data; estimated costs are predetermined.
 - b. Standard costs reflect what the cost should be; estimated costs reflect what the cost will be.
 - c. Standard costs are unreliable for performance measurement; estimated costs are scientifically analyzed.
 - d. Standard costs can only be used in manufacturing; estimated costs can be used in any industry.
2. What is one of the objectives of standard costing related to management control?
 - a. Management should focus on all performance metrics at all times.
 - b. Management should increase standard costs annually.
 - c. Management should ignore variances from standard costs.
 - d. Management should be alerted only when actual performance deviates from established standards.

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8.2 NEED FOR FIXING STANDARDS

The standard costing system is widely embraced due to its ability to meet various organizational needs. Here are several reasons for its preference:

- (a) **Future Cost Prediction for Decision Making:** Standard costs are established by considering all potential future scenarios, making them useful as future costs. They play a critical role in assessing the profitability of proposed projects, orders, or activities, thereby enhancing decision-making processes.
- (b) **Establishing Targets to Achieve:** Standard costs serve as target costs that should not be exceeded, presenting challenging benchmarks for responsibility centers. Management continuously monitors performance against these established standards, allowing for immediate corrections of any deviations.
- (c) **Application in Budgeting and Performance Evaluation:** Standard costs are instrumental in creating budgets, which in turn facilitate the evaluation of managerial performance. This approach offers two key benefits: it ensures that managers of responsibility centers maintain quality while meeting budgeted quantities, and it allows for tracking variances back to the responsible department or individual.
- (d) **Interim Profit Measurement and Inventory Valuation:** Actual profits are typically determined only at the end of an accounting period. Some organizations, however, prepare interim profitability statements as needed by management. To calculate profitability figures during these periods, standard costs are subtracted from revenue.

**SELF-ASSESSMENT QUESTIONS**

- 3. How do standard costs contribute to interim profit measurement?
 - a. By allowing the calculation of profitability figures during accounting periods
 - b. By providing an estimate of future costs
 - c. By eliminating the need for variance analysis
 - d. By focusing solely on fixed costs
- 4. Which of the following is NOT a component of standard costing?
 - a. Elimination of all variances
 - b. Pre-determination of technical data
 - c. Comparison of actual performance with standards
 - d. Analysis of variances

8.3 PROCESS OF STANDARD COSTING

The standard costing process (shown in **Figure 8.1**) involves the following steps:



Figure 8.1 Steps Under Standard Costing.

1. **Setting of Standards:** The initial step involves establishing the standards to be achieved. This process of standard setting has been detailed previously.
2. **Determining Actual Costs:** The actual costs for each cost component are determined. These costs are derived from financial records, such as accounting books, material invoices, wage sheets, and charge slips.
3. **Comparison of Actual Costs and Standard Costs:** The actual costs are then compared with the established standard costs, allowing for the identification of variances.
4. **Investigation of Variances:** Any variances that arise are thoroughly investigated to determine the underlying causes. This analysis informs performance evaluation, leading to the implementation of necessary corrective actions.
5. **Disposition of Variances:** The identified variances are addressed by transferring them to the relevant accounts (costing profit and loss account) in accordance with the adopted accounting method or plan.

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Hypothetical Scenario: Automobile Manufacturing

An automobile company, such as Toyota, sets production standards for manufacturing a car model. For instance, it establishes that the standard material cost for producing a car is ₹1,00,000, the standard labor cost is ₹30,000, and the standard overhead cost is ₹20,000, totaling ₹1,50,000 per car. These standards are based on historical data, industry benchmarks, and efficiency targets. During a specific period, the company produces 1,000 cars. It **records actual costs** from its accounting system:

- ❑ **Material Cost:** ₹1,10,000 per car.
- ❑ **Labor Cost:** ₹28,000 per car.
- ❑ **Overhead Cost:** ₹22,000 per car.

The total actual cost per car is ₹1,60,000.

The company **compares actual costs to standard costs** for each component:

- ❑ **Material Variance:** ₹1,10,000 – ₹1,00,000 = ₹10,000 unfavorable per car.
- ❑ **Labor Variance:** ₹28,000 – ₹30,000 = ₹2,000 favorable per car.
- ❑ **Overhead Variance:** ₹22,000 – ₹20,000 = ₹2,000 unfavorable per car.

Overall, there is an unfavorable variance of ₹10,000 per car (₹1,60,000 – ₹1,50,000).

The company **investigates the unfavorable material variance** and discovers:

- ❑ An increase in material prices due to supply chain disruptions.
- ❑ Higher wastage in the production process caused by defective raw materials.

Corrective actions are planned, such as sourcing alternative suppliers and training workers to reduce wastage.

The company **allocates the variances** to relevant accounts. For instance:

- ❑ Unfavorable material variance of ₹10,000 per car is charged to the costing profit and loss account.
- ❑ Favorable labor variance is credited to reflect cost savings in the labor component.

Types of Standards

Standards represent the level of performance recognized by management as the foundation for determining standard costs. They are primarily categorized into four types:

- (i) **Ideal Standards:** These standards reflect the level of performance achievable under the most favorable conditions, such as optimal material and labor prices, highest output with advanced equipment, and maximum resource efficiency, resulting in minimal costs. However, these standards face criticism for several reasons:

- a. Such unattainable standards may not be taken seriously by individuals.

- b. Variances calculated against these standards would not represent realistic and practically avoidable differences.
- c. There is no logical basis for addressing or managing these variances effectively.

for instance, XYZ Ltd., a manufacturing company producing electric fans, sets an ambitious goal of manufacturing 1,000 fans per month under ideal conditions. This assumes material costs at Rs. 500 per fan, representing the lowest possible market price. The labor force is expected to produce 10 fans daily without delays or breakdowns, and the machines are expected to operate 24/7 without any downtime. While this target is theoretically possible, the employees find it unattainable and demotivating, as real-world challenges like price fluctuations and equipment maintenance make it unrealistic. The large variances between actual and ideal performance offer little actionable insight for improving efficiency.

- (ii) **Normal Standards:** These are based on what can be accomplished under regular operating conditions. The concept of normal activity refers to “the number of standard hours required to produce sufficient goods at normal efficiency to meet average sales demand over a period of years.” While these standards are practical, they are challenging to establish due to the need for accurate forecasting. Variances under this system represent deviations from normal efficiency, sales volume, or production volume. If actual performance is abnormal, significant variances may arise, requiring revision of these standards.

For instance, Based on past production data, XYZ Ltd. sets a realistic goal of producing 800 fans per month under normal operating conditions. This accounts for material costs of Rs. 550 per fan, assuming average market prices, and labor productivity of 8 fans per worker daily, accounting for regular breaks and minor delays. Machine uptime is estimated at 90% of the planned hours. Employees find this target achievable with reasonable effort, and the management monitors weekly progress to ensure alignment with goals. Normal standards provide a balanced approach, allowing the company to analyze manageable variances and improve efficiency within realistic parameters.

- (iii) **Basic or Bogey Standards:** These standards are set for extended periods and remain constant. A base year is selected as a reference point, similar to price indices used by statisticians. Since basic standards represent long-term benchmarks, current standards must also be prepared for meaningful comparisons. These standards are particularly suitable for businesses with a limited product range and long production cycles. Basic standards are rarely revised, and variances are not computed. Instead, actual costs are expressed as percentages of basic costs, and these percentages are compared with current and previous periods to identify trends.

For example, XYZ Ltd. sets a benchmark of 700 fans per month based on historical performance from five years ago, using this figure as a base for long-term comparisons. This basic standard serves as a reference point to evaluate growth and trends over time. While the

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management appreciates its utility in trend analysis, employees find the standard outdated, given that the company now operates with upgraded machinery and improved processes. As a result, while basic standards are useful for identifying long-term improvements, they are not practical for setting operational goals in the present period.

- (iv) **Current Standards:** These standards reflect management's expectations of actual costs for the current period. They account for anticipated prices for goods and services and the required usage to achieve planned output. Variances from these standards indicate the efficiency of resource usage, price fluctuations, and differences in production volume. For example, For the current month, XYZ Ltd. anticipates that material prices will rise to Rs. 600 per fan due to increased market demand, machine uptime will decrease by 5% due to a planned power outage, and labor efficiency may slightly decline during peak summer due to heat-related challenges. These anticipated conditions form the basis for current standards, which reflect the company's realistic expectations for costs and production. Employees appreciate this practical approach, as it aligns closely with real-time challenges and allows for better planning. Current standards help identify variances, such as unexpected material price increases, and enable the company to take immediate corrective actions.



SELF-ASSESSMENT QUESTIONS

5. Which step in the standard costing process involves comparing actual costs with established standard costs?
 - a. Setting of Standards
 - b. Determining Actual Costs
 - c. Comparison of Actual Costs and Standard Costs
 - d. Investigation of Variances
6. What type of standard is established under perfect conditions but is often unattainable in practice?

a. Expected Standard	b. Normal Standard
c. Ideal Standard	d. Basic Standard

8.4 FACTORS DETERMINING STANDARDS UNDER EACH COST COMPONENT

The determination of standard costs relies on management's estimates. These estimates are based on technical details provided by the engineering department or other specialists, such as production engineers. Typically, when setting standards, factors like historical data, the current production schedule, and anticipated future conditions are taken into account. To ensure effective analysis and control, standard costs are determined for each cost component, including materials, labor, variable overheads, and fixed overheads. Additionally, standards are established for sales volume and sales value, commonly referred to as budgeted sales.

Standards are specified in terms of both quantity (units or hours) and cost (price or rate), serving as benchmarks for the quantities, hours, and costs associated with production inputs. Standard costs are categorized into three key components:

- a. Direct Material Cost b. Direct Labor Cost c. Overheads**

For each cost component, standards are defined in both physical and monetary terms, as follows:

- (i) Physical Standards** Physical standards represent the expression of benchmarks in terms of units or hours. At this stage, standard quantities and hours are determined for a specific product or service. The purpose of setting these standards is to achieve production efficiencies and facilitate accurate pricing for quotations.

In manufacturing organizations, the responsibility for defining physical standards typically falls to the industrial engineering department. Several factors are considered during this process, including:

- ❑ The company's operational plan (e.g., budgets)
- ❑ The planned production output
- ❑ Material specifications for quality and quantity provided by the engineering team
- ❑ The mix of materials used when multiple inputs are required
- ❑ The production methodology, such as fully automated, semi-automated, or manual processes
- ❑ The skills and availability of workers
- ❑ Internal working conditions and influences
- ❑ External factors (e.g., labor laws, the Factory Act, and government policies)

Material Quantity Standards: The procedure generally followed for setting material quantity standards is shown in **Figure 8.2**.

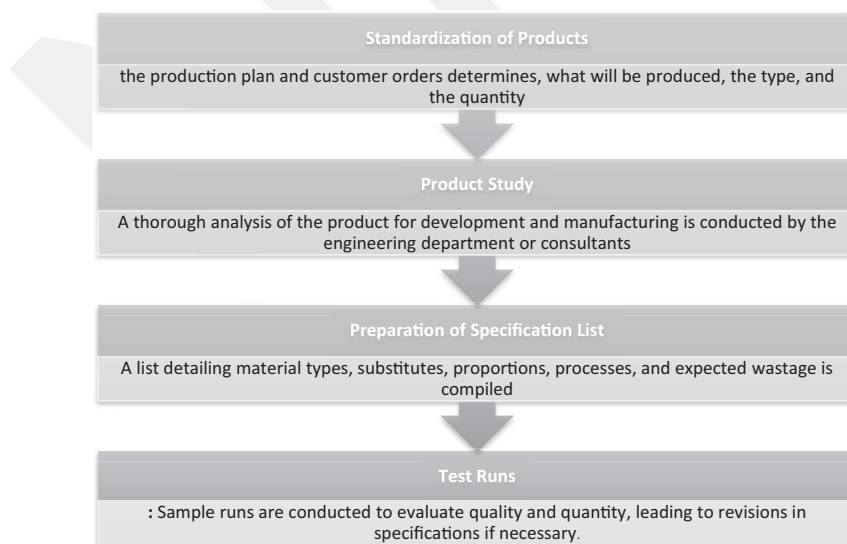


Figure 8.2 Procedure for Setting Material Quantity Standards.

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Labor Time Standards: The steps involved in establishing labor quantity standards are shown in **Figure 8.3**:

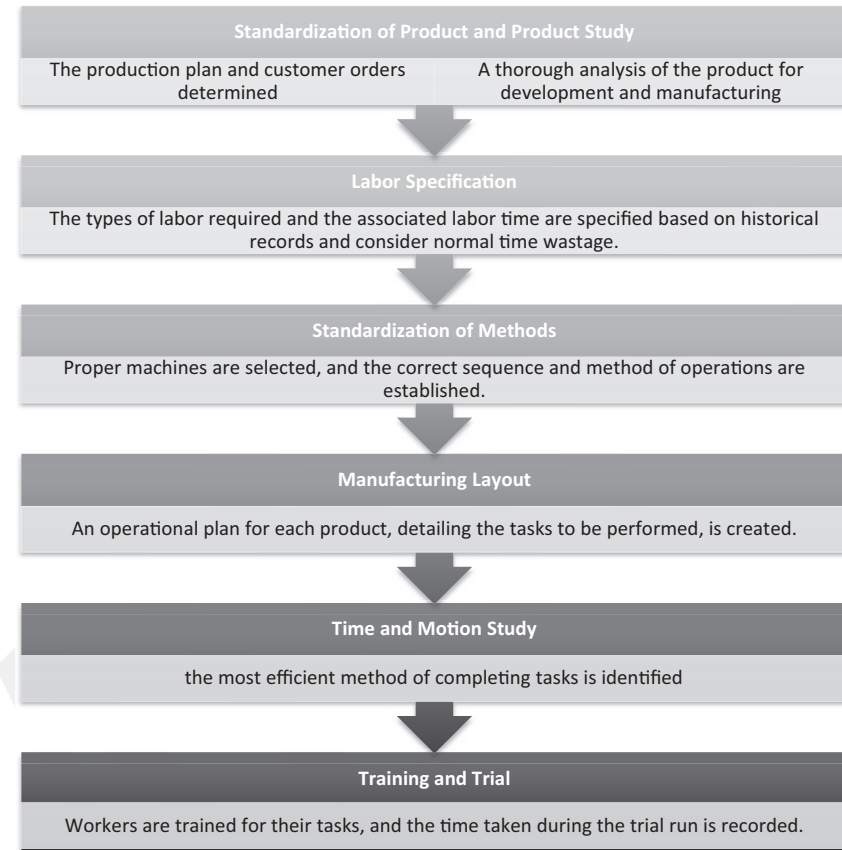


Figure 8.3 Procedure for Setting Labor Time Standard.

Overheads Time/Quantity Standards: Variable overhead time/quantity is estimated based on specifications provided by the engineering departments. Variable overheads may depend on direct material quantity or labor hours, typically based on the labor time worked. Fixed overhead time is based on budgeted production volume.

Challenges in Setting Physical Standards: The challenges encountered when establishing physical standards can vary across industries and may include:

- ❑ A scenario may arise where a company introduces a new product line. In this case, it may be necessary to hire workers without prior experience, complicating the standard time-setting process as adjustments must be made for the inexperience of the workers. Imagine a company introducing a new product line. To produce it, they hire workers with no prior experience. Setting a realistic standard time for tasks becomes tricky because the workers may take longer to perform their duties initially. Over time, as they gain experience, adjustments to these standards may be necessary.

- ❑ Technological changes may require the installation of advanced machinery. When such machines are introduced, accurately estimating output and the achievable standard of efficiency will be challenging until operational conditions stabilize. Consequently, establishing standards for these machines and estimating standard costs will require significant effort. Picture this: A company installs advanced machinery to improve efficiency. At first, it's hard to determine how much the machine can produce or how efficiently it can run until operations stabilize. During this period, setting accurate production standards is a challenge that requires ongoing monitoring and fine-tuning.
- ❑ Manufacturers often pursue product diversification to enhance profitability. A key issue arising from such product changes is the need to reconfigure production facilities. For instance, when transitioning from an old copper component to a new bronze one, special care must be taken to order new tools, creating challenges in setting standard times for the new equipment. Manufacturers often diversify their products to stay competitive. For instance, transitioning from producing copper components to bronze ones may require reconfiguring production lines and ordering new tools. This creates challenges in defining standard times for new operations, as each change introduces uncertainties.
- ❑ Established material specifications may not be attainable if the required materials are unavailable. Imagine a scenario where production standards rely on specific materials, but those materials become unavailable. The company must adjust its standards based on alternative materials, which could have different properties and impact production efficiency.
- ❑ Cost accountants frequently face challenges in determining which type of standards to adopt. For instance, the industrial engineer may provide standard times for all direct labor operations, including:
 1. Maximum efficiency achievable by the best operations is 2 hours per unit of product, inclusive of allowances for personal fatigue and delays.
 2. Attainable performance for the average trained operator is 2.10 hours per unit of product.
 3. Average past performance is 2.60 hours per unit.

The dilemma is whether to base direct labor standard hours on maximum efficiency, attainable good performance, or average past performance. If costs reflect maximum efficiency, the unit cost applied to the selling price will be lower, but a significant debit variance may arise if the standard efficiency is not realized. Alternatively, if the standard cost is grounded in attainable good performance, variances might remain minimal. If efficiency needs to be evaluated, setting the maximum efficiency standard will highlight off-standard performance, thus enabling departmental heads to maintain control.

N O T E S

Similar challenges, as noted above, may arise when establishing waste standards. For example, it may be debated whether only unavoidable waste should be accounted for, or if past average waste levels should be considered. Each approach will impact the standard cost of production differently.

(ii) Price or Rate Standards

Price or rate standards can generally be established based on either of the following criteria:

- a. The actual average or mean price expected to prevail over the forthcoming period, typically one year; or
- b. Normal prices anticipated to prevail throughout a seasonal cycle, which may span several years.

Material Price Standards: While material prices are not entirely within the manufacturer's control, the purchasing department should be able to estimate reasonable prices for constituent items based on its awareness of current market conditions and trends, once informed of the required production quantities. The standards for material prices should be determined considering the following factors depicted in **Figure 8.4**, especially when price fluctuations are minimal and manageable:

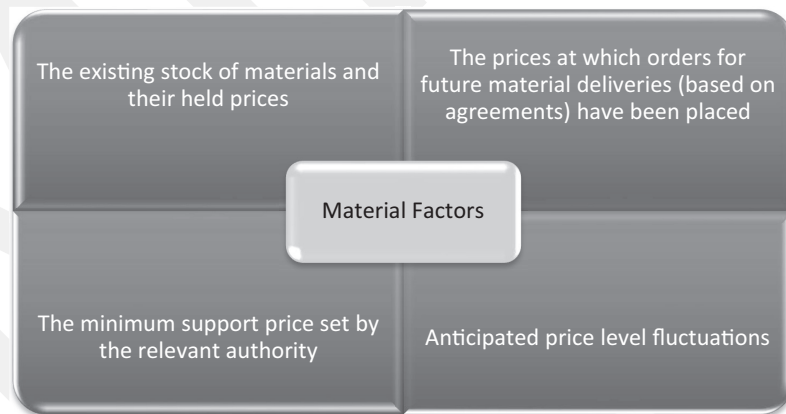


Figure 8.4 Factors for Material Price Standard.

When faced with unpredictable fluctuations in market prices, determining standard material costs can be complex. Price changes can vary from month to month, between seasons, or across years. There may also be an overarching trend that pushes prices up or down. The nature of the challenges in setting standard material costs will differ in each case. Furthermore, the company's purchasing policy and its objectives for cost accounting will influence the situation.

To address difficulties in establishing standard material costs in such scenarios, the following approaches can be utilized:

- (a) If prices fluctuate monthly, an average price for the year, adjusted for known trends and anticipated changes, can serve as the standard price for the upcoming year.

- (b) In cases of seasonal fluctuations where annual material requirements are procured in one go, the weighted average of anticipated prices should be treated as the standard price. However, if purchases are spread throughout the year, the weighted average of prices for the entire year should be considered the standard price.
- (c) If prices vary significantly from year to year, a careful estimate of the likely price for the upcoming year, grounded in statistical analysis, should be established as the standard price.

Wage Rate Standards: The nature of labor necessary for executing a particular job is crucial for determining wage rates. Standard wage rates for skilled and unskilled labor are established based on the following criteria depicted in **Figure 8.5**:

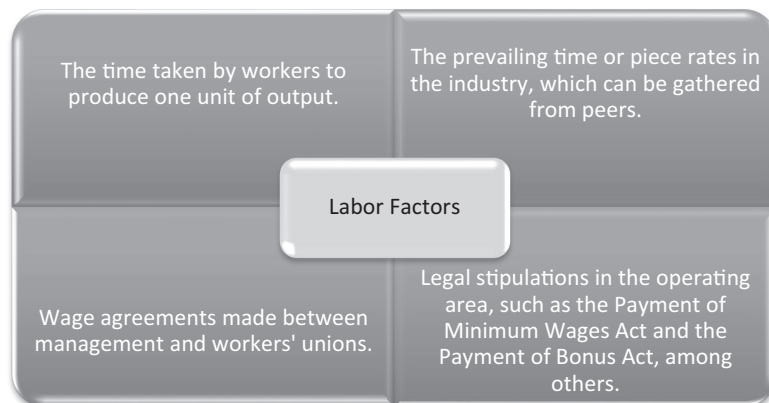


Figure 8.5 Factors for Wage Rate Standards.

Overhead Expense Standards: Determining overhead expense standards requires considering the production level and budgeted expenses. The basis for setting these standards may rely on practical manufacturing capacity, average sales capacity, or the budgeted capacity for the next year. Expenses are classified as fixed or variable, and a budget is prepared for each category based on the selected output level. Overhead expense standards are then calculated by optimizing production department output levels and drafting a budget for related fixed and variable expenses.

Other Cost Components

- 1. Standard Administration Costs:** To minimize administrative costs while ensuring high-quality services, a detailed analysis of administrative functions is essential. Contributions from the Operations and Management (O&M) team can aid in streamlining office operations and standardizing procedures. The standard quantity of administrative work can be determined using:
 - ☐ Historical performance data.
 - ☐ Inputs from the O&M team.
 - ☐ Time and motion studies.
 - ☐ Identifying suitable work units and assigning standard costs per unit.

N O T E S

Administrative costs should be categorized as fixed, variable, or semi-variable before determining standard rates.

2. **Standard Selling and Distribution Costs:** Selling and distribution costs, being closely tied to sales volume, require accurate sales forecasts to set appropriate standards. These expenses should also be classified into fixed, variable, and semi-variable categories. A comprehensive review of related functions and the identification of standard operating units are necessary steps in determining these standards.



SELF-ASSESSMENT QUESTIONS

7. What is the primary focus when establishing wage rate standards?
 - a. Setting the lowest possible wage
 - b. Estimating historical wage increases
 - c. Evaluating the performance of the sales team
 - d. Considering the time taken by workers and industry rates
8. Which of the following methods can be used to determine standard administration costs?
 - a. Projecting future sales
 - b. Time and motion studies
 - c. Estimating market demand
 - d. Analyzing competitor pricing

8.5 USES OF STANDARD COSTING

Standard costing is highly beneficial for management in various functions, such as planning, controlling, and particularly in decision-making and performance evaluation. Standard costing can be employed for the following purposes:

1. Projecting the profit level of the business at any given production level.
2. Assisting in the effective execution of management functions, namely planning and cost control.
3. Analyzing the effects of cost when sales volume increases or decreases by a certain percentage.
4. Measuring production efficiency.
5. Evaluating the performance of each segment.
6. Identifying and measuring variances between standards and actual results.
7. Designing performance measurement systems that encourage employee participation for the improvement of the organization.

Application of Standard Costing in Different Industries

Manufacturing Industry: In the manufacturing sector, profit projections play a crucial role by helping manufacturers estimate profit margins at varying production levels. This aids in setting realistic production goals

and pricing strategies. Efficiency measurement becomes essential as it allows for a comparison of actual costs against standard costs for materials, labor, and overheads, ensuring streamlined operations. Through variance analysis, manufacturers can identify discrepancies in material usage or labor efficiency, enabling corrective measures to reduce waste and enhance productivity.

Retail Industry: The retail industry benefits from sales volume analysis, which helps retailers, assess the impact of sales volume fluctuations on overall costs and profitability. Cost control mechanisms are also vital for monitoring operational overheads, such as inventory carrying costs and utilities, to ensure expenditures remain within acceptable limits and profitability is maximized.

Healthcare Industry: For the healthcare industry, performance evaluation is integral to assessing departmental efficiency by comparing standard costs of treatments and procedures with actual expenditures. Additionally, resource allocation processes are enhanced, ensuring cost-effective distribution of medical supplies and staffing based on accurate projections and real-time data.

Construction Industry: In construction, budgeting and planning enable precise cost projections for labor, materials, and equipment required for specific projects. This ensures resource optimization and financial preparedness. Additionally, cost monitoring helps track deviations caused by delays, material price changes, or labor inefficiencies, facilitating timely interventions.

IT and Software Industry: In the IT and software industry, project costing establishes benchmarks for development expenses, including labor, technology, and infrastructure. This ensures financial discipline across projects. Furthermore, variance management tracks deviations in resource usage, helping maintain budgetary control and project timelines.

Food and Beverage Industry: The food and beverage industry relies on production efficiency to optimize ingredient and labor usage, minimizing food waste while maximizing output. For pricing decisions, standard costing assists in determining menu prices by accounting for anticipated costs and profitability targets, ensuring a balanced financial approach.

Automotive Industry: The automotive industry uses performance measurement to evaluate the efficiency of vehicle assembly by comparing standard costs for materials and labor with actual expenditures. Additionally, cost-impact analysis provides insights into how production level changes influence manufacturing costs and overall profitability.

Energy and Utilities Industry: In the energy and utilities sector, cost control ensures that production expenses, such as those for energy generation or water treatment, remain aligned with budgeted standards. Variance identification highlights inefficiencies, such as energy losses or operational delays, enabling swift corrective actions to enhance efficiency and reduce costs.

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8.6 BENEFITS OF STANDARD COSTING

The implementation of a Standard Costing system can provide numerous benefits, though these may vary between businesses. The following advantages can be derived from standard costing, considering the different objectives of the system:

1. To establish standards that are both practical and indicative of efficient performance, management must be fully informed about available resources and the most effective work methods (such as employing time and motion studies for accurate labor standards). This ongoing assessment will help the organization identify numerous sources of waste and loss, allowing them to mitigate these issues effectively.
2. Individuals are often motivated to work hard to achieve attainable standards; therefore, the establishment of realistic targets naturally leads to enhanced operational efficiency. When the reward and punishment systems are aligned with achieving these targets, it further encourages this mindset among employees.
3. Periodic reassessment of standards will foster a culture of continuous improvement in efficiency.
4. Standard costing not only encompasses pre-established quantity standards but also includes benchmarks for prices and rates. This means that all materials used and labor applied will be valued according to standard prices and rates, thereby reducing clerical workload. In general, standard costing tends to be more cost-effective compared to conventional costing systems, where quantities and prices fluctuate frequently.
5. The implementation of standard costing allows for objective evaluations of personnel, leading to a more accepted and transparent system for promotions and other advancements within the organization.
6. Management can save significant time by focusing only on issues that genuinely require their attention, based on the analysis of deviations between standard costs and actual costs. This “management by exception” approach ensures that the focus remains on areas that impact efficiency.
7. In price-setting, standard costs are invaluable as they eliminate the daily fluctuations caused by inefficient resource use and price variations. Standard costs represent long-term estimates, allowing for stable long-term pricing strategies.
8. For inventory valuation, standard costs provide a sound basis. If actual costs are inflated due to resource wastage, it is inappropriate to incorporate these losses into inventory values. Instead, inventory should be valued based on standard costs, which reflect true worth.
9. Ultimately, if a firm implements standard costing effectively, with standards that do not impose excessive burdens on workers or the organization, it can foster a culture of ambition among employees, leading to enhanced efficiency.

10. The process of establishing standards simplifies and standardizes production methods and operations, eliminating time and material waste. This streamlining aids managerial planning for more efficient operations across all divisions.
11. The costing procedure becomes more straightforward, resulting in reduced paperwork and fewer required forms and records. This leads to substantial savings in clerical time and costs associated with the costing system.
12. Standard costing facilitates the delegation of authority and the assignment of responsibility within departments or to individual employees.
13. When consistently reviewed, standards provide a pathway to achieving cost reductions through improved product quality, superior materials and labor, and effective capital resource management.
14. Standard costs enhance performance analysis by providing convenient tools for preparing and interpreting information.
15. Finally, this approach promotes the integration of accounts, thereby eliminating the need for reconciliation between cost accounts and financial accounts.

8.7 LIMITATIONS OF STANDARD COSTING

Despite the numerous advantages, standard costing also has certain drawbacks:

1. The establishment of standard costs is challenging in practice.
2. Over time, and sometimes even within a short duration, standards can become inflexible.
3. Standards that are inaccurate, unreliable, or outdated can cause more harm than benefit.
4. Occasionally, standards can have negative psychological effects. If a standard is set too high, failing to achieve it may lead to frustration and resistance.
5. Due to the influence of random factors, variances can sometimes be difficult to explain, making it hard to differentiate between controllable and non-controllable expenses.
6. Standard costing may not always be suitable for small organizations. When production cannot be precisely scheduled and frequent changes in production conditions occur, the resulting variances render detailed analysis meaningless, redundant, and expensive.
7. Standard costing may also be unsuitable and costly in industries dealing with non-standardized products or for repair jobs that change according to customer specifications.
8. A lack of interest in standard costing from management can render the system largely ineffective. This limitation applies equally to any system that management does not fully embrace.

Despite these limitations, standard costing remains a valuable tool for cost control and performance evaluation, particularly in industries that produce standardized, repetitive products.

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8.8 RELATIONSHIP BETWEEN STANDARD COSTING AND BUDGETARY CONTROL

Like Budgetary Control, Standard Costing operates under the premise that costs can be managed within specific lines of supervision and accountability. It seeks to achieve managerial control by comparing actual performance against suitable predetermined benchmarks. The fundamental principles of cost control, such as establishing targets or standards, measuring performance, comparing actual results with those targets, and analyzing and reporting variances, are consistent across both standard costing and budgetary control systems. Each technique is significant in its own field and complements the other. Thus, conceptually, there is little difference between standard costs and budgets, and for many organizations, the terms “budgeted performance” and “standard performance” often refer to the same concept.

Budgets are generally based on historical costs that have been adjusted for anticipated future changes, while standard costs are useful in preparing production cost budgets. In fact, standards are frequently essential in establishing budgets. Conversely, when setting standard overhead rates for standard costing purposes, the budgets prepared for overhead costs can be used with modifications as needed. Hence, standard costs and budgets are interrelated, though not interdependent.

Despite the similarities in the foundational principles of Standard Costing and Budgetary Control, the two systems differ in scope and in the detailed techniques employed. The differences can be summarized in the **Table 8.2** below:

TABLE 8.2 DIFFERENCES BETWEEN STANDARD COSTING AND BUDGETING

Aspect	Standard Costing	Budgeting
Basis of Formation	Based on technical data and scientifically determined.	Formulated using standard costs, historical data, and future estimates.
Scope of Application	Primarily used in manufacturing and occasionally in marketing and administration; does not need functional coordination.	Applied across various functional departments (sales, purchase, production, finance, personnel), requiring coordination.
Focus on Cost Control	Aims at identifying and reducing cost levels.	Ensures costs do not exceed predetermined limits.
Variance Analysis	Variances are analyzed through accounting records.	Variances are evaluated by comparing budgeted figures with actual results without reliance on accounts.

(Continues)

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Aspect	Standard Costing	Budgeting
Analysis of Variances	Requires detailed analysis in case of variances.	No further analysis needed if costs remain within budget.
Control of Costs	Sets realistic benchmarks, making it effective for cost control and reduction.	Sets maximum expenditure limits without focusing on cost efficiency.
Frequency of Revisions	Revised only when there is a significant change in assumptions or conditions.	Prepared before each accounting period begins.
Basis of Standards	Standards are set by management.	Based on the current level of operational efficiency.
Flexibility	Must be applied comprehensively across all cost elements.	Can be implemented either fully or partially.
Projection Type	Represents a projection of cost accounts.	Represents a projection of financial accounts.



SELF-ASSESSMENT QUESTIONS

9. Which of the following best describes the focus of budgets in relation to cost levels?
 - a. Budgets emphasize cost levels that should be reduced.
 - b. Budgets emphasize cost levels that should not be exceeded.
 - c. Budgets focus solely on maximizing profits.
 - d. Budgets emphasize actual costs incurred.
10. How often are standard costs compared to budgets?
 - a. Standard costs are revised quarterly, while budgets are annual.
 - b. Standard costs are revised when there is a change in basic assumptions, while budgets are set before each accounting period.
 - c. Both are revised at the end of each month.
 - d. Standard costs are not revised at all.

8.9 STANDARD COSTING AND VARIANCE ANALYSIS

The primary objective of standard costing is to identify the difference between actual costs and standard costs. In standard costing, a 'variance' refers to the deviation of actual costs from standard costs. Variances across different cost items are crucial for cost control, as they indicate whether the established standards have been met and to what degree. This information allows management to address adverse trends effectively.

Once standard costs are determined, the next step involves calculating the actual costs for each component and comparing them to the standard

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costs. The difference between these two figures is known as cost variance. Cost variance represents the difference between a standard cost and the corresponding actual cost incurred over a specific period.

The Chartered Institute of Management Accountants in London defines variance as “the difference between planned, budgeted, or standard cost and actual cost; and similarly for revenue.” Imagine you plan to spend ₹500 on groceries this week, but you end up spending ₹550. That extra ₹50 is a **variance**. The same concept applies in a business like Shyam’s Furniture. The company sets a standard production cost (like ₹1,000 to produce one chair), but the actual cost might differ.

Variance analysis can be defined as “the analysis of performance through variances.” It involves calculating the amount of variances and determining their causes by comparing actual costs to standard costs.

Establishment of Cost Centers

A cost centre refers to a location, individual, or piece of equipment (or a group of these) for which costs can be identified and attributed to cost units. A cost centre associated with individuals is termed a personal cost centre, while one linked to a location or equipment is known as an impersonal cost centre. The primary purposes of establishing cost centers are cost ascertainment and cost control. When creating cost centers, it is crucial to identify who is accountable for each centre. Often, each department or function naturally forms a distinct cost centre, but multiple cost centers can exist within a single department or function. For instance, in a manufacturing department with six machines, each machine could be designated as a separate cost centre. Cost centers play a vital role in setting standards and analyzing variances. Think of your household budget. Instead of tracking one big “total expenses” figure, you split it into categories like groceries, rent, utilities, and entertainment. If you overspend on groceries, you know exactly which part of your budget went off track. Cost centers work the same way.

Example: Shyam’s Furniture has two main departments: **Cutting** and **Assembly**. Each department is a cost center.

Cost Center	Standard Cost (₹)	Actual Cost (₹)	Variance (₹)
Cutting	1,00,000	1,10,000	+10,000 (Unfavorable)
Assembly	80,000	75,000	–5,000 (Favorable)

Analysis

- ❑ In the **Cutting** department, there is a ₹10,000 unfavorable variance. The manager investigates and finds that machine repairs cost ₹7,000 more than expected, and labor hours exceeded targets by ₹3,000.
- ❑ In the **Assembly** department, costs are ₹5,000 below the target. This favorable variance occurred because the team completed production ahead of schedule, saving on wages.

8.10 SUMMARY

- ❑ Organizations should prioritize cost reduction as a means of optimizing scarce resources, ensuring reduced resource usage while maintaining product or service quality. This requires ongoing evaluation of potential cost savings and the establishment of predetermined cost levels.
- ❑ Standard costing involves setting target costs as benchmarks for efficiency. By meticulously identifying wastages and losses, organizations can prevent them from distorting production costs and accurately assess performance against set standards.
- ❑ Standard costs are scientifically predetermined expenses for producing a product or providing a service within a specified timeframe. They are expressed in monetary values and help measure performance, control costs, and establish selling prices.
- ❑ The primary goals include facilitating cost control, promoting management by exception, fostering a cost-conscious culture, assisting in price fixation and policy formulation, and enabling effective budget planning.
- ❑ The standard costing process encompasses several steps, including setting standards, determining actual costs, comparing actual costs with standard costs, investigating variances, and addressing these variances appropriately to improve performance and decision-making.
- ❑ Standard costs are established based on management estimates, technical specifications from the engineering department, and considerations of historical data and future conditions, encompassing elements like materials, labor, variable and fixed overheads, as well as budgeted sales quantities and values.
- ❑ Physical standards involve setting standards in units or hours, aiming for economies of scale in production, with responsibility for establishing these typically falling to the industrial engineering department, taking into account the company's operational plan, final output, material specifications, production methods, and labor availability.
- ❑ Material quantity standards are established through product standardization, product study, preparation of specification lists, and test runs, while labor time standards involve labor specifications, standardization of methods, and time and motion studies to enhance efficiency in labor utilization.
- ❑ Overhead standards are determined based on variable overheads linked to labor hours and fixed overheads based on budgeted production volumes, with challenges including the introduction of new product lines, technological changes, and availability of materials affecting standard setting.
- ❑ Despite its advantages, standard costing faces challenges such as inflexibility over time, potential negative psychological effects of high standards, difficulties in establishing accurate standards, and its applicability to small or non-standardized production environments.

N O T E S

8.11 KEY WORDS

1. **Cost Control:** The process of managing and regulating expenses to maintain financial efficiency and reduce waste within an organization.
2. **Standard Costing:** A costing method that involves setting predetermined costs for products or services to compare against actual performance and identify variances.
3. **Variance Analysis:** The process of examining the differences between actual and standard costs to understand the reasons for discrepancies and inform decision-making.
4. **Efficiency Measurement:** The assessment of how effectively resources are utilized in production or service delivery, often compared against established standards.
5. **Cost Centre:** A specific location, department, or piece of equipment within an organization where costs can be tracked and assigned for better financial management.
6. **Material Standards:** Specifications set for the quantity and quality of materials required for production, ensuring consistency and cost control.
7. **Labor Standards:** Benchmarks established for the amount of labor time needed to complete a task, aiding in efficiency assessments and labor cost management.
8. **Overhead Standards:** Predetermined rates for variable and fixed overhead costs associated with production, used for cost allocation and budgeting.
9. **Efficiency:** The ratio of actual output to standard output, measuring how effectively resources are utilized in the production process.
10. **Budgetary Control:** The process of comparing actual financial performance against budgeted figures to monitor and control costs, ensuring that resources are used efficiently and effectively.

8.12 CLOSING CASELET**RADHE MANUFACTURING CO. - IMPLEMENTING STANDARD COSTING AND VARIANCE ANALYSIS**

Radhe Manufacturing Co., specializing in electronic components, recognized the need for an effective cost management strategy as it expanded. To address this, the company implemented standard costing, establishing benchmarks for material, labor, and overhead costs. They determined that the standard cost for producing their flagship product, the E-Component, would be ₹300 per unit, broken down into ₹150 for materials, ₹100 for labor, and ₹50 for overheads. This system aimed to streamline cost control and enhance profitability.

At the end of the fiscal quarter, Radhe Manufacturing calculated the actual costs for the E-Component, which amounted to ₹340 per unit.

Upon conducting a variance analysis, the management identified unfavorable variances in material, labor, and overhead costs. Specifically, the material costs exceeded standards by ₹20, labor by ₹10, and overheads by ₹10. The total unfavorable variance of ₹40 highlighted discrepancies between expected and actual expenditures, prompting further investigation.

Through this analysis, management pinpointed the root causes of the variances, which included a supplier price hike for materials, unplanned overtime due to machine downtimes, and increased utility expenses from recent plant upgrades. Armed with these insights, Radhe Manufacturing took corrective actions such as renegotiating supplier contracts and optimizing production schedules, leading to improved cost control. This case illustrates the value of standard costing and variance analysis as essential tools for driving operational efficiency and profitability in manufacturing.



QUESTIONS

1. What is the significance of variance analysis in standard costing, and how can it impact management decision-making?

(Hint: Consider how identifying variances helps management understand performance deviations)

2. Explain the process of setting standard costs and how they are used in a manufacturing company. Include the role of historical data in this process.

(Hint: Think about the steps involved in determining standard costs)

8.13 HIGHER ORDER THINKING SKILLS QUESTIONS (HOTS)

1. How can the implementation of standard costing and variance analysis improve decision-making in resource allocation and operational efficiency within an organization?
2. In what ways can an organization balance the pursuit of cost reduction through standard costing with the need to maintain product quality and customer satisfaction?
3. How can a company use variance analysis to identify underlying issues in its production processes, and what steps should be taken to address these issues effectively?
4. Assess the impact of setting unrealistic standards in standard costing. How might these standards influence employee morale and operational efficiency?
5. Design a comprehensive plan for implementing standard costing in a manufacturing organization that currently uses a traditional costing system. What key factors should be considered to ensure successful adoption and integration?

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8.14 ANSWER KEYS**SELF-ASSESSMENT QUESTIONS**

Topic	Q. No.	Answer
Standard Costing	1.	b. Standard costs reflect what the cost should be; estimated costs reflect what the cost will be.
	2.	d. Management should be alerted only when actual performance deviates from established standards.
Need for Fixing Standards	3.	a. By allowing the calculation of profitability figures during accounting periods.
	4.	a. Elimination of all variances.
Process of Standard Costing	5.	c. Comparison of Actual Costs and Standard Costs
	6.	c. Ideal Standard
Factors Determining Standards Under Each Cost Component	7.	d. Considering the time taken by workers and industry rates
	8.	b. Time and motion studies
Relationship Between Standard Costing and Budgetary Control	9.	b. Budgets emphasize cost levels that should not be exceeded.
	10.	b. Standard costs are revised when there is a change in basic assumptions, while budgets are set before each accounting period.

8.15 SUGGESTED READINGS

- ❑ Weil, R. L., Schipper, K., & Francis, J. (2021). *Management accounting: Tools for business decision making*. Wiley.
- ❑ Blocher, E., Stout, D. E., & Juras, P. (2019). *Cost management: A strategic emphasis*. McGraw-Hill Education.
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8.16 REFERENCES

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- ❑ Corporate Finance Institute (CFI): <https://corporatefinanceinstitute.com/>
- ❑ Institute of Company Secretaries of India: <https://www.icsi.edu/home/>

9

U N I T

VARIANCE ANALYSIS

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INTRODUCTORY CASELET

NOTES

Material Cost Variance at Shyam Manufacturing

Shyam Manufacturing Company, which produces plastic containers, set a standard cost for its raw materials (plastic resin) to maintain cost control. The standard material quantity is 1 kilogram of resin per container at a standard price of ₹400 per kilogram. In the last quarter, Shyam produced 50,000 containers but experienced material cost variances due to differences in both the price and usage of materials. The company consumed 52,000 kilograms of resin and paid ₹440 per kilogram, which exceeded the standard price.

The Material Cost Variance (MCV) is calculated as the difference between the standard and actual costs. Using the formula $(SQ \times SP) - (AQ \times AP)$, where $SQ = 50,000$ kg, $SP = ₹400$, $AQ = 52,000$ kg, and $AP = ₹440$, the MCV is $(50,000 \times 400) - (52,000 \times 440) = ₹2,00,00,000 - ₹2,28,80,000 = -₹28,80,000$. This negative variance indicates that Shyam spent ₹28,80,000 more than expected on raw materials.

Breaking it down, the Material Price Variance (MPV) accounts for ₹20,80,000 of the difference. The formula $(SP - AP) \times AQ$ gives $(400 - 440) \times 52,000 = -₹40 \times 52,000 = -₹20,80,000$. This excess cost was primarily due to rising market prices, emergency purchases, and the inability to buy in bulk. The Material Usage Variance (MUV) reveals that the company used more resin than expected, with a variance of ₹8,00,000. Calculated as $SP \times (AQ - SQ)$, it is $400 \times (52,000 - 50,000) = 400 \times 2,000 = ₹8,00,000$, which likely resulted from production inefficiencies and material wastage.

To address these variances, Shyam should consider entering long-term contracts to stabilize prices and review production processes to reduce wastage. Additionally, improving the quality of materials and training staff could help prevent excess material usage. By focusing on these areas, Shyam Manufacturing can better manage its material costs in the future.



QUESTIONS

1. What factors contributed to Shyam Manufacturing's negative Material Price Variance (MPV) of ₹20,80,000, and how can the company address these factors in the future?

(Hint: Review the causes of price variance, such as market price changes and purchase strategies. Consider solutions like long-term contracts and bulk buying.)

2. How did the excess material usage at Shyam Manufacturing result in a Material Usage Variance (MUV) of ₹8,00,000, and what steps can be taken to improve material usage efficiency?

(Hint: Think about production inefficiencies, wastage, and material quality. Focus on solutions like improving production processes and training staff.)

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LEARNING OBJECTIVES

At the end of the unit, you will be able to:

- Grasp the purpose and importance of variance analysis in cost control and performance evaluation.
- Understand how to calculate material yield variance and its significance in measuring material efficiency.
- Identify factors that contribute to material yield variances.
- Learn the different types of labor cost variances, including wage rate variance and labor efficiency variance.
- Understand how to calculate and interpret labor cost variances to assess workforce efficiency.
- Understand the concept of idle time variance
- Learn to calculate overhead variances, including variable and fixed overhead variances.
- Understand the different types of sales variances, such as sales price variance and sales volume variance.
- Learn how to calculate and use control ratios (such as efficiency ratio, capacity ratio) to monitor and control performance.

9.1 VARIANCE ANALYSIS-INTRODUCTION

Imagine you're a manager of a company and it's time to review the monthly performance report. You notice that the cost of raw materials was higher than expected, while labor costs were lower. Curious to understand why, you turn to **variance analysis** — a vital tool for assessing performance through the differences, or variances, between actual and standard costs.

Variance analysis involves two key steps:

1. **Calculating individual variances** for each cost element, such as materials, labor, and overhead.
2. **Identifying the cause(s) of each variance** to understand why the difference occurred.

To make it more relatable, think of planning a home renovation project. You set a budget for materials, labor, and other expenses. If you end up spending more on tiles but less on labor due to a shorter project timeline, you'd want to understand why. Was it because tile prices increased or because you chose higher-quality tiles? Similarly, in a business setting, management looks at variances to understand the “why” behind the numbers.

When actual costs are higher than standard costs, it's called an **unfavorable or adverse variance**, indicating inefficiency. This could be due to higher material costs, longer production times, or increased labor rates. Unfavorable variances reduce profit, so they're a red flag for management.

On the flip side, when actual costs are lower than standard costs, it's a **favorable variance**. This might result from better purchasing deals, faster production, or other efficiency gains, leading to increased profit. But be cautious—not all favorable variances are good. Sometimes, they arise from using lower-quality materials or setting standards too loosely.

To visualize this, suppose your company planned to produce 1,000 units at a standard material cost of ₹5 per unit, expecting total material costs to be ₹5,000. However, at the end of the month, the actual cost was ₹5,500. This ₹500 difference is an unfavorable material variance, and you'd want to know if the supplier raised prices, if there was excessive waste, or if employees used more materials than expected. If, instead, the cost was only ₹4,500, you'd have a ₹500 favorable variance. But it's essential to check if the materials were of sufficient quality to avoid future production issues.

Controllable and Uncontrollable Variances: Variances caused by internal factors like labor efficiency or material usage are called controllable variances. Management can take corrective action to address them. On the other hand, uncontrollable variances result from external factors, such as market changes or shifts in supply and demand, which are beyond the control of the company.

Favorable and Unfavorable Variances: When actual costs are lower than standard costs, the difference is a favorable variance (sometimes called a positive or credit variance) and is marked as **F or Cr**. This increases profit. Conversely, when actual costs exceed standard costs, it's an unfavorable variance (negative or debit variance) and is marked as **A or Dr**, reducing profit.

Revision Variance: This variance occurs when standard costs are not updated to reflect changes in wage rates, fiscal policies, or other factors. Since these changes are often beyond the company's control, the resulting variance is isolated to avoid distorting the analysis of other variances.

Method Variance: This variance arises from changes in the production method. For instance, if the company switches from manual production to automation, the production costs might change, leading to a method variance. Such changes are tracked to ensure proper variance analysis.

Variance analysis typically begins after adjusting for **revision and method variances**. Before diving into specific variances, it's important to recognize how variances can be useful for control purposes.

Key Aspects of Variance Analysis:

- ❑ **Variance as an Indicator:** Variances act as signals for areas where costs may have escalated. It's up to management to investigate if these increased costs are justified or if corrective action is required. In some cases, higher costs may be due to factors beyond the control of the responsible individual, so it's important not to impose undue blame.
- ❑ **Comparability of Activity Levels:** When comparing actual costs with standards, ensure the level of activity is consistent. If the actual production level differs from the standard production level, comparing

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costs without adjustment would be misleading. Revised standards should be used for accurate analysis, and the difference between the original and revised standards is known as **Revision Variance**.

- ❑ **Fixed Costs and Predetermined Rates:** Fixed costs, like rent or equipment depreciation, are often calculated using predetermined rates. Variance analysis takes into account the difference between the actual expense and the recovery amount, typically based on production output or standard hours.

Illustration 1

Given the following data for standard and actual costs for a production period, how can the variances in material, labor, and overhead expenses help identify areas for improvement in cost management and efficiency within the factory's operations?

- ❑ Material: (6 units @ ₹3 each) = ₹18
- ❑ Labour: (15 hours @ ₹2 per hour) = ₹30
- ❑ Overhead expenses = ₹12

Actual Cost for 10,000 units:

- ❑ Material: 62,000 units @ ₹3.50 each = ₹2,17,000
- ❑ Labour: 1,60,000 hours @ ₹2.20 each = ₹3,52,000
- ❑ Overhead expenses = ₹1,40,000

Analyze the variances in the table below and discuss how these variances could impact the factory's cost control and future production planning.

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Particulars	Standard cost (₹)	Actual cost (₹)	Variance (₹)
Material	1,80,000	2,17,000	37,000 (A)
Labour	3,00,000	3,52,000	52,000 (A)
Overhead expenses	1,20,000	1,40,000	20,000 (A)
Total	6,00,000	7,09,000	1,09,000 (A)

9.1.1 MATERIAL COST VARIANCE

Direct materials cost variance represents the difference between the actual direct material cost incurred and the standard direct material cost set for the achieved production. Material cost variances occur due to differences in either the price or the usage of materials. Accordingly, material cost variances are categorized into two types as shown in Figure 9.1:

Material Price Variance: This variance arises when there is a deviation between the actual price paid for materials and the standard price set.

Material Usage Variance: This variance occurs due to differences between the actual quantity of material used and the standard quantity specified for production.

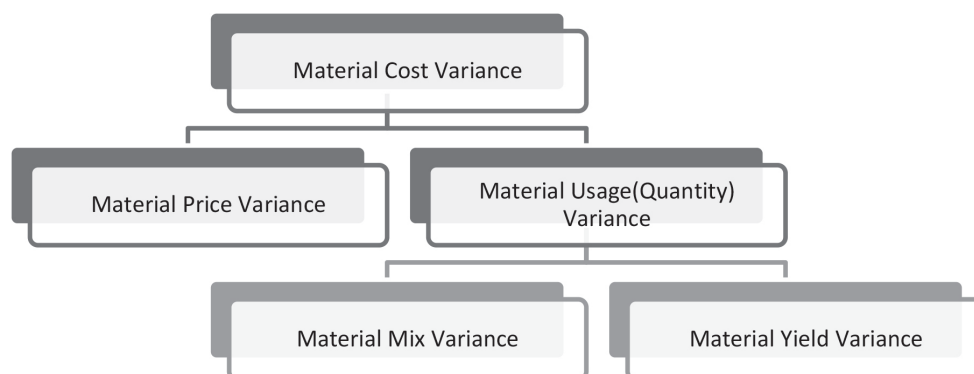


Figure 9.1 Material Cost Variance

The calculation of material cost variance is as follows:

$$\begin{aligned}\text{Material Cost Variance} &= \text{Standard Cost} - \text{Actual Cost} \\ &= [(SQ \times SP) - (AQ \times AP)]\end{aligned}$$

SQ = Standard Quantity = Expected Consumption for Actual Output

AQ = Actual Quantity of Material Consumed

SP = Standard Price per Unit

AP = Actual Price per Unit

9.1.2 MATERIAL PRICE VARIANCE

Material price variance refers to the portion of material cost variance that arises due to the difference between the standard price set for materials and the actual price paid. It reflects the additional or reduced price paid for the units purchased, and is an “incurring” variance. The standard consumption of units is not considered in this calculation. It can be computed using the following formula:

$$\begin{aligned}\text{Material Price Variance} &= (\text{Actual Quantity} \times \text{Standard Price}) - \\ &\quad (\text{Actual Quantity} \times \text{Actual Price})\end{aligned}$$

or

$$\text{Material Price Variance} = (SP - AP) \times AQ$$

AQ = Actual Quantity of Material Consumed

SP = Standard Price per Unit

AP = Actual Price per Unit

In simple terms, it shows the difference between what the materials actually cost and what they would have cost if the actual quantity had been paid for at the standard price.

Causes of Material Price Variance:

- ❑ Material price variances may arise due to several factors, including:

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- ❑ Changes in market price of materials.
- ❑ Variations in purchase quantities or uneconomical order sizes.
- ❑ Failure to obtain expected cash or trade discounts.
- ❑ Rush orders to meet supply shortages.
- ❑ Missed opportunities to buy during off-season or at lower prices.
- ❑ Emergency purchases requested by production or sales managers.
- ❑ Variations in issue prices due to storage, handling, or inward carriage costs.
- ❑ Changes in taxes and duties.
- ❑ Purchase of materials with different quality or specifications.
- ❑ Use of substitute materials with higher or lower prices.
- ❑ Shifts in tax patterns or amounts.

While the purchase manager is typically responsible for material price variances, external factors such as market fluctuations may also contribute beyond their control.

9.1.3 MATERIAL USAGE (QUANTITY) VARIANCE

Material usage variance refers to the portion of material cost variance that arises due to the difference between the standard quantity of materials specified for production and the actual quantity used, both valued at the standard price. The difference between the actual quantity of materials used and the standard quantity set, multiplied by the standard price, is known as the material usage variance. It can be calculated using the formula:

$$\text{Material Usage Variance} = \text{Standard Price} \times (\text{Actual Quantity} - \text{Standard Quantity})$$

Or

$$\text{Material Usage Variance} = \text{SP} \times (\text{AQ} - \text{SQ})$$

SQ = Standard Quantity = Expected Consumption for Actual Output

AQ = Actual Quantity of Material Consumed

SP = Standard Price per Unit

Causes of Material Usage Variance:

- ❑ Material usage variance may arise from a variety of factors, including:
- ❑ Lack of proper care in material usage.
- ❑ Defective production that requires additional materials for correction.
- ❑ Abnormal wastage due to pilferage or other losses.

- ❑ Inefficiency in production caused by improper methods or lack of necessary worker skills.
- ❑ Using a material mix different from the standard mix.
- ❑ Variations in the yield from materials compared to the standard yield.
- ❑ Purchase of inferior materials or changes in material quality.
- ❑ Strict technical specifications and inspections leading to more rejections, requiring additional materials.
- ❑ Substitution of materials leading to lower quality.
- ❑ Poor maintenance of machinery, causing breakdowns and increased material usage.
- ❑ Inadequate inspection of raw materials.

A favorable variance in material usage doesn't always imply an advantage for the company. For example, reducing material wastage by slowing down work may lead to increased labor and overhead costs, which can offset the benefits of the favorable material usage variance.

9.1.4 MATERIAL MIX VARIANCE

One of the causes of material usage variance is a change in the composition of the material mix. This occurs when the actual mix of materials used in production differs from the standard mix. For instance, if a higher proportion of more expensive materials is used than specified in the standard mix, the material usage will result in a higher cost than expected. Conversely, using a larger proportion of cheaper materials will result in a lower material cost compared to the standard.

This specific variance, known as material mix variance, arises from the difference between the standard and actual composition of the material mixture. In other words, it reflects the variation in the ratio of materials used compared to the standard ratio. The material mix variance is calculated by finding the difference between the standard price of the standard mix and the standard price of the actual mix.

The formula for calculating the Material Mix Variance is:

Material Mix Variance = [Standard Cost of Actual Quantity in Standard Proportion – Standard Cost of Actual Quantity]

$[(RAQ - AQ) \times SP]$

Or

$[(RAQ \times SP) - (AQ \times SP)]$

AQ = Actual Quantity of Material Consumed

RAQ = Revised Actual Quantity = Actual Quantity Rewritten in Standard Proportion

SP = Standard Price per Unit

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Example 1:

The standard material requirement per unit of actual output is 5 kgs. The following are the relevant figures:

Material	A	B	C	D
Standard mix %	25%	35%	30%	10%
Price per kg. (₹)	2.00	1.80	4.00	3.50
Actual quantity used (Kg.)	1,500	1,850	1,320	480
Actual price per kg. (₹)	2.10	2.00	3.80	3.60
Actual output: 1,200 units				

Calculate the following variances for material costs:

- ☐ Price variance
- ☐ Mix variance
- ☐ Usage variance
- ☐ Total material cost variance

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Statement showing computation of Standard Cost/Actual Cost/ Revised Actual Quantity

Material	Standard Cost of 1,000 Units			Actual Cost of 1,000 Units			Revised Actual Quantity [RAQ] (Kg.)
	Quantity	Price	Amount	Quantity	Price	Amount	
	[SQ] (Kg.)	[SP] (₹)	[SQ × SP] (₹)	[AQ] (Kg.)	[AP] (₹)	[AQ × AP] (₹)	
A	1,250	2.00	2,500	1,500	2.10	3,150	1,287.5
B	1,750	1.80	3,150	1,850	2.00	3,700	1,802.5
C	1,500	4.00	6,000	1,320	3.80	5,016	1,545
D	500	3.50	1,750	480	3.60	1,728	515
	5,000		13,400	5,150		13,594	5,150

Note:

SQ = Standard Quantity = Expected Consumption for Actual Output

AQ = Actual Quantity of Material Consumed

RAQ = Revised Actual Quantity = Actual Quantity Rewritten in Standard Proportion

SP = Standard Price per unit

AP = Actual Price per unit

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Statement showing variances (₹)

Particulars	Material A	Material B	Material C	Material D	Total
Material Cost Variance = $SQ \times SP - AQ \times AP$	(2,500 – 3,150) = 650 (A)	(3,150 – 3,700) = 550 (A)	(6,000 – 5,016) = 984 (F)	(1,750 – 1,728) = 22 (F)	194 (A)
Material Price Variance = $AQ \times (SP - AP)$	1,500 × (2.00 – 2.10) = 150 (A)	1,850 × (1.80 – 2.00) = 370 (A)	1,320 × (4.00 – 3.80) = 264 (F)	480 × (3.50 – 3.60) = 48 (A)	304 (A)
Material Usage Variance = $SP \times (SQ - AQ)$	(1,250 – 1,500) × 2.00 = 500 (A)	(1,750 – 1,850) × 1.80 = 180 (A)	(1,500 – 1,320) × 4.00 = 720 (F)	(500 – 480) × 3.50 = 70 (F)	110 (F)
Material Mix Variance = $SP \times (RAQ - AQ)$	(1,287.5 – 1,500) × 2.10 = 446.25 (A)	(1,802.5 – 1,850) × 2.00 = 95 (A)	(1,545 – 1,320) × 3.80 = 855 (F)	(515 – 480) × 3.60 = 126 (F)	439.75 (F)



SELF-ASSESSMENT QUESTIONS

- Which of the following best describes an adverse variance?
 - Actual costs exceed standard costs, leading to a decrease in profit.
 - Actual costs are less than standard costs, leading to an increase in profit.
 - Standard costs exceed actual costs, resulting in higher profit.
 - Standard and actual costs are equal, resulting in no variance.
- A product uses two types of materials (X and Y). The standard and actual data are as follows:

Material	Standard Mix (%)	Standard Price (₹/kg)	Actual Usage (kg)	Actual Price (₹/kg)
X	60%	20	120	22
Y	40%	25	80	24

Calculate the Material Price Variance.

- ₹160 (Adverse)
- ₹150 (Adverse)
- ₹170 (Adverse)
- ₹180 (Adverse)

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9.2 MATERIAL YIELD VARIANCE

Yield variance represents the difference between the standard yield specified and the actual yield achieved. In other words, it measures the deviation between the actual yield of materials in production and the standard yield, i.e., the expected yield from a given standard input, valued at the standard output price. This variance is particularly important in processing industries, where the output of one process becomes the input for the next, leading up to the final product. Analyzing yield variance enables effective control over material usage.

A low actual yield indicates an unfavorable yield variance, suggesting that material consumption exceeded the standard. Conversely, a high actual yield signifies efficiency, although a consistently high yield may signal the need for standard revision.

The formula for calculating the yield variance can be expressed as:

Yield Variance = Standard Cost of Standard Quantity for Actual Production – Standard Cost of Actual Quantity in Standard Proportion

This represents the difference between the standard quantity specified for actual production and the actual quantity in standard proportion, valued at the standard purchase price.

The formula can be written as:

$$\text{Yield Variance} = (\text{SQ} - \text{RAQ}) \times \text{SP}$$

or

$$\text{Yield Variance} = (\text{SQ} \times \text{SP}) - (\text{RAQ} \times \text{SP})$$

Where:

SQ = Standard Quantity specified for actual production

RAQ = Actual Quantity in standard proportion

SP = Standard Purchase Price

Continuing example 1:

Particulars	Material A	Material B	Material C	Material D	Total
Material Yield Variance	(1,250 – 1,287.5) × 2.00 = 75 (A)	(1,750 – 1,802.5) × 1.80 = 94.5 (A)	(1,500 – 1,545) × 4.00 = 180 (A)	(500 – 515) × 3.50 = 52.5 (A)	402 (A)

Causes of Yield Variance may include:

- ☐ Defective methods of operation
- ☐ Substandard quality of purchased materials
- ☐ Carelessness in handling materials
- ☐ Lack of proper supervision

Example 2: MNO Ltd. manufactures a single product, the standard mix of which is as follows:

- ❑ Material X: 70% at ₹15 per kg
- ❑ Material Y: 30% at ₹25 per kg

Normal loss in production is 10% of input. Due to a shortage of Material X, the standard mix was changed, and the actual mix was as follows:

- ❑ Material X: 140 kg at ₹16 per kg
- ❑ Material Y: 60 kg at ₹24 per kg

The actual loss was 30 kg, while the actual output was 170 kg.

Calculate all material variances.

Solution:

Material	Standard Qty (kg)	Standard Price (₹)	Standard Cost (₹)	Actual Qty (kg)	Actual Price (₹)	Actual Cost (₹)
X	70	15	1050	140	16	2240
Y	30	25	750	60	24	1440
Total	100		1800	200		3680

- ❑ Normal Loss: 10% of 200 = 20 kg
- ❑ Standard Output: 200 – 20 = 180 kg
- ❑ Actual Loss: 30 kg
- ❑ Actual Output: 170 kg

Variance Calculations:

1. Material Cost Variance (MCV):

Standard Cost for Actual Production – Actual Cost

Standard cost for actual production = $(1800/180) \times 170 = ₹1700$

MCV = ₹1700 – ₹3680 = ₹1980 (A)

2. Material Price Variance (MPV):

Actual Quantity × (Standard Price – Actual Price)

❑ Material X = $140 \times (₹15 - ₹16) = ₹140$ (A)

❑ Material Y = $60 \times (₹25 - ₹24) = ₹60$ (F)

Total MPV = ₹140 (A) – ₹60 (F) = ₹80 (A)

3. Material Quantity Variance (MQV):

Standard Price × (Standard Quantity – Actual Quantity)

❑ Standard Quantity for Actual Production:

♦ Material X = $(170/180) \times 70 = 66.11$ (or 66 kg)

♦ Material Y = $(170/180) \times 30 = 28.33$ (or 28 kg)

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$$\diamond \text{ Material X} = ₹15 \times (66 - 140) = ₹1110 \text{ (A)}$$

$$\diamond \text{ Material Y} = ₹25 \times (28 - 60) = ₹800 \text{ (A)}$$

$$\text{Total MQV} = ₹1110 \text{ (A)} + ₹800 \text{ (A)} = ₹1910 \text{ (A)}$$

4. Material Mix Variance (MMV):

Standard Cost of Standard Mix – Standard Cost of Actual Mix

□ Standard Cost of Standard Mix:

$$\diamond \text{ Material X} = 70\% \times 200 = 140 \times ₹15 = ₹2100$$

$$\diamond \text{ Material Y} = 30\% \times 200 = 60 \times ₹25 = ₹1500$$

$$\diamond \text{ Total} = ₹3600$$

□ Standard Cost of Actual Mix:

$$\diamond \text{ Material X} = 140 \times ₹15 = ₹2100$$

$$\diamond \text{ Material Y} = 60 \times ₹25 = ₹1500$$

$$\diamond \text{ Total} = ₹3600$$

$$\text{MMV} = ₹3600 - ₹3600 = ₹0$$

5. Material Yield Variance (MYV):

Standard Yield Rate $\times$ (Actual Yield – Standard Yield)

$$\text{Standard yield} = 200 - 20 = 180 \text{ kg}$$

$$\text{Standard yield rate} = ₹1800/180 = ₹10 \text{ per kg}$$

$$\text{MYV} = ₹10 \times (170 - 180) = ₹100 \text{ (A)}$$

Summary Table 9.1 for Material Cost Variance

Table 9.1 Material Cost Variances

Variance Type	Formula	Explanation	Causes
Material Cost Variance	Material Cost Variance = (SQ $\times$ SP) – (AQ $\times$ AP)	Difference between the standard cost of materials and the actual cost incurred for production.	Caused by a deviation in either price or usage of materials.
Material Price Variance	Material Price Variance = (SP – AP) $\times$ AQ	Difference between the standard price and the actual price paid for materials, multiplied by the actual quantity of materials used.	- Changes in market price - Uneconomical order sizes - Rush orders - Missed discounts

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Variance Type	Formula	Explanation	Causes
Material Usage Variance	Material Usage Variance = $SP \times (AQ - SQ)$	Difference between the actual quantity of materials used and the standard quantity specified, valued at the standard price per unit.	- Defective production - Abnormal wastage - Inefficiency in production methods
Material Mix Variance	Material Mix Variance = $[(RAQ \times SP) - (AQ \times SP)]$	Difference in the composition of materials used compared to the standard mix. Reflects the variance in the ratio of materials used.	- Change in composition of materials - Using a different mix of materials than the standard mix
Material Yield Variance	Yield Variance = $(SQ - RAQ) \times SP$	Measures the difference between the expected yield (standard quantity) and the actual yield achieved, valued at the standard purchase price.	- Inefficiency in production - Lower than expected output or excess waste

9.3 LABOR COST VARIANCES

Labor cost variance, also known as direct wage variance, refers to the difference between the standard direct wages set for the accomplished activity and the actual direct wages disbursed. The formula for calculating labor cost variance is:

Labor Cost Variance = Standard Cost – Actual Cost

This represents the difference between the Standard Direct Labor Cost and the Actual Direct Labor Cost incurred for the production achieved.

The formula can be expressed in detail as:

$$\text{Labor Cost Variance} = (SH \times SR) - (AH \times AR)$$

Where:

SH = Standard Hours (Expected time allowed for actual output)

AH = Actual Hours (Time paid for)

SR = Standard Rate per Labor Hour

AR = Actual Rate per Labor Hour Paid

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Since labor costs are influenced by both labor time and wage rates, the labor cost variance can be divided into two components shown in Figure 9.2: the labor rate variance and the labor efficiency variance. This breakdown allows for a more detailed analysis of the factors contributing to the overall labor cost variance.

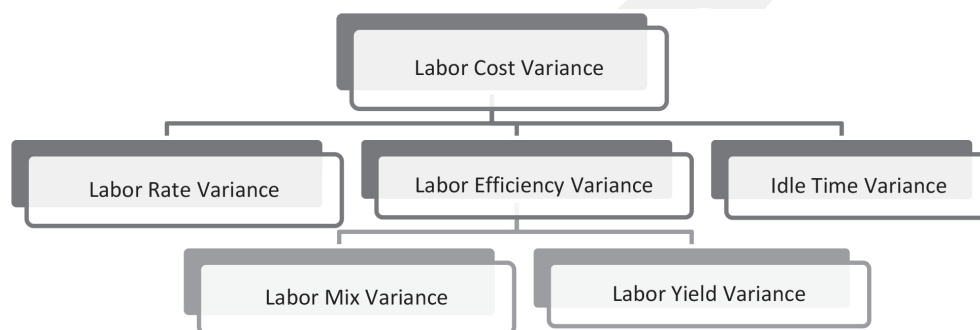


Figure 9.2 Labor Cost Variance

9.3.1 LABOR RATE VARIANCE

This segment of the wages variance is attributable to the discrepancy between the actual rate and the standard rate of any specified labor. It is calculated similarly to the materials price variance.

Labor Rate Variance = Standard Cost of Actual Time – Actual Cost

(The difference between the Standard Rate per hour and the Actual Rate per hour for the Actual Hours paid)

$$\text{Labor Rate Variance} = (\text{SR} - \text{AR}) \times \text{AH}$$

Or

$$\text{Labor Rate Variance} = (\text{SR} \times \text{AH}) - (\text{AR} \times \text{AH})$$

Where:

AH = Actual Hours (Time paid for)

SR = Standard Rate per Labor Hour

AR = Actual Rate per Labor Hour Paid

Causes of Labor Rate Variance

- ❑ Wage rate variance arises from several factors, including:
- ❑ Changes in the basic wage structure or variations in piecework rates.
- ❑ Overtime work exceeding the amount accounted for in the standard rate.
- ❑ The hiring of one or more workers of a different grade than specified in the standards.
- ❑ Payment of guaranteed wages to workers unable to earn their normal wages, if these guaranteed wages are included in direct labor costs.

- ❑ New workers not receiving full normal wage rates.
- ❑ Utilization of a different payment method, such as paying day rates while standards are based on piecework remuneration.
- ❑ Higher wages paid for overtime due to urgent work requirements.
- ❑ The composition of a workgroup differing in skill and wage rates from what is outlined in the standard.

Wage rates are typically influenced by factors beyond the control of the personnel department, such as labor market conditions and wage awards from governing bodies. Consequently, wage rate variances are often largely uncontrollable, except for the portion resulting from deploying the wrong grade of labor, for which the departmental executive may be held accountable.

9.3.2 LABOR EFFICIENCY VARIANCE

Labor efficiency variance refers to the segment of the direct wages variance attributable to the discrepancy between the standard labor hours set and the actual labor hours used. Clearly, this variance serves as an important indicator for managing worker efficiency and labor costs. Essentially, it represents a usage variance. The calculation of the variance is as follows:

Labor Efficiency Variance = Standard Wage Rate $\times$ (Standard Hours of Production – Actual Hours Worked)

OR

$LEV = SR \times (SH - AHW)$

Where,

SH = Standard Hours = Expected time (Time allowed) for Actual Output

AHW = Actual Hours worked

SR = Standard Rate per Labor Hour

The factors contributing to labor efficiency variance include:

- (i) Insufficient or excessively strict supervision;
- (ii) Unsatisfactory working conditions;
- (iii) Faulty machinery and equipment;
- (iv) Worker dissatisfaction stemming from poor personnel relations;
- (v) An increase in labor turnover;
- (vi) Usage of substandard materials that demand more or less time for operations.

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Example:

From the following details, calculate the direct labor variances:

- Direct Labor Rate: ₹ 2 per hour □ Hours set per unit: 9 hours

Actual data:

- Units produced: 400 □ Actual Direct Labor Cost: ₹ 7,800
□ Hours worked: 4,000

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Basic Calculations:**1. Standard Time**

$$= 9 \text{ hours} \times 400 \text{ units} = 3,600 \text{ hours}$$

2. Standard Cost

$$= \text{Standard Rate} \times \text{Standard Time} = ₹ 2 \times 3,600 \text{ hours} = ₹ 7,200$$

Computation of Variances:**1. Labor Cost Variance (LCV)**

$$= \text{Standard Cost} - \text{Actual Cost} = ₹ 7,200 - ₹ 7,800 = ₹ 600 \text{ (Adverse)}$$

2. Labor Rate Variance (LRV)

$$= \text{Actual Time} \times (\text{Standard Rate} - \text{Actual Rate})$$

$$= 4,000 \text{ hours} \times (₹ 2 - ₹ 1.95)$$

$$= ₹ 200 \text{ (Favorable)}$$

3. Labor Efficiency Variance (LEV)

$$= \text{Standard Rate} \times (\text{Standard Time} - \text{Actual Time})$$

$$= ₹ 2 \times (3,600 \text{ hours} - 4,000 \text{ hours})$$

$$= ₹ 800 \text{ (Adverse)}$$

Verification:

$$\text{LCV} = \text{LRV} + \text{LEV}$$

$$₹ 600 \text{ (A)} = ₹ 200 \text{ (F)} + ₹ 800 \text{ (A)} = ₹ 600 \text{ (Adverse)}$$

**SELF-ASSESSMENT QUESTIONS****3. Given the following data:**

- Standard labor hours per unit: 5 hours
□ Standard labor rate: ₹20 per hour
□ Actual production: 200 units
□ Actual labor hours worked: 1,100 hours

Calculate the Labor Efficiency Variance.

- a. ₹2,000 (Adverse) b. ₹1,000 (Adverse)
 - c. ₹1,000 (Favorable) d. ₹2,000 (Favorable)
4. Which of the following factors contributes to labor rate variance?
- a. Poor supervision
 - b. Overhead inefficiency
 - c. Hiring workers of a different grade than specified
 - d. Reduced material wastage

9.4 IDLE TIME VARIANCE

This variance, which constitutes a part of the wages efficiency variance, is expressed by the standard cost of the actual hours during which workers are idle due to abnormal circumstances.

Idle Time refers to a period when a workstation is available for production but remains unused, such as when there is a shortage of tools, materials, or operators. During Idle Time, direct labor wages are still paid, yet no output is generated. The cost of this idle time can be tracked separately through an Idle Time Variance to prevent it from being obscured within an unfavorable Labor Efficiency Variance.

In some organizations, Idle Time occurs regularly. In such cases, the standard labor rate may factor in an allowance for the expected idle time cost, and only the impact of unexpected or abnormal Idle Time would be reflected in the Idle Time Variance. The calculation of the variance is as follows:

Idle Time Variance = Standard Rate per Hour $\times$ Actual Idle Hours

i.e. $(AH - AHW) \times SR$

Or

$(AH \times SR) - (AHW \times SR)$

Where,

AH = Actual Hours paid for

AHW = Actual Hours worked

SR = Standard Rate per Labor Hour

Example:

A firm provides the following data related to labor:

Standard time per unit: 3 hours

Actual hours (paid): 2,500 hours

Standard rate of pay: ₹3 per hour

N O T E S

Idle time: 20% of the actual hours (paid) have been lost as idle time.

Actual output: 900 units

Actual wages (paid): ₹6,000

Calculate the Idle Time Variance.

Sol.

Basic Calculations:

Standard Cost = (Standard time per unit × Actual output × Standard rate per hour)

$$= 900 \text{ units} \times 3 \text{ hours} \times ₹3$$

$$= ₹8,100$$

Actual wages paid: ₹6,000

Actual wage rate per hour = ₹6,000 ÷ 2,500 hours = ₹2.40

Standard wage rate per hour: ₹3.00

Abnormal idle time = 20% of 2,500 hours = 500 hours

Computation of Variances:

$$\square \text{ Rate Variance} = \text{Actual Time} \times (\text{Standard Rate} - \text{Actual Rate})$$

$$= 2,500 \text{ hours} \times (₹3 - ₹2.40) = ₹1,500 \text{ (Favorable)}$$

$$\square \text{ Efficiency Variance} = \text{Standard Rate} \times (\text{Standard Time} - \text{Actual time worked})$$

$$= ₹3 \times (2,700 \text{ hours} - 2,000 \text{ hours}) = ₹2,100 \text{ (Favorable)}$$

$$\square \text{ Idle Time Variance} = \text{Idle Time} \times \text{Standard Rate}$$

$$= 500 \text{ hours} \times ₹3 = ₹1,500 \text{ (Adverse)}$$

$$\square \text{ Total Variance} = \text{Standard Labor Cost} - \text{Actual Labor Cost}$$

$$= ₹8,100 - ₹6,000 = ₹2,100 \text{ (Favorable)}$$

9.5 LABOR YIELD VARIANCE

This variance arises from the difference between the specified standard output and the actual output achieved. It is calculated as follows:

Labor Yield Variance = [Standard Cost of Standard Time for Actual Production – Standard Cost of Actual Time Worked in Standard Proportion]

$$(SH - RAH) \times SR$$

Or

$$(SH \times SR) - (RAH \times SR)$$

Where,

SH = Standard Hours = Expected time (Time allowed) for Actual Output

SR = Standard Rate per Labor Hour

RAH = Revised Actual Hours = Actual Hours (worked) rewritten in Standard Proportion

If the actual output exceeds the standard output, it results in a favorable variance, and if the actual output is less than the standard, it leads to an unfavorable variance.

9.6 LABOR MIX VARIANCE

This variance, also referred to as Gang Composition Variance, occurs due to changes in the composition of the standard gang or labor force mix. It is calculated as:

Labor Mix Variance = Standard Cost of Actual Time Worked in Standard Proportion – Standard Cost of Actual Time Worked

$$= [(RAH - AHW) \times SR]$$

Or

$$= [(RAH \times SR) - (AHW \times SR)]$$

Where,

SH = Standard Hours = Expected time (Time allowed) for Actual Output

AHW = Actual Hours worked

SR = Standard Rate per Labor Hour

RAH = Revised Actual Hours = Actual Hours (worked) rewritten in Standard Proportion

Revised labor hours = (Standard time / Total standard time) × Total actual time

Example: The standard output of product ZYN is 30 units per hour in a manufacturing department employing 80 workers. The standard wage rate per labor hour is ₹7.

In a 40-hour week, the department produced 1,200 units of the product despite losing 4% of the paid time due to abnormal reasons. The hourly wage rate actually paid was ₹7.50, ₹7.20, and ₹6.80 to 15, 25, and 40 workers, respectively.

Compute the various relevant labor variances.

N O T E S

Step 1: Prepare the Table for Calculations

Particulars	Standard	Actual
Output	30 units/hour for 80 workers	1,200 units
Standard Man-Hours per Unit	80 workers / 30 units per hour = 2.67 hrs	—
Total Hours	40 hrs × 15 workers = 600 hrs	Hourly rate: ₹7.50 × 600 hrs = ₹4,500
	40 hrs × 25 workers = 1,000 hrs	Hourly rate: ₹7.20 × 1,000 hrs = ₹7,200
	40 hrs × 40 workers = 1,600 hrs	Hourly rate: ₹6.80 × 1,600 hrs = ₹10,880
Total Hours	—	Actual Hours: 3,200 hrs
Idle Time (4% of 3,200)	—	128 hrs
Actual Hours Worked	—	3,200 hrs – 128 hrs = 3,072 hrs
Standard Hours for Actual Production	1,200 × 2.67 hrs = 3,204 hrs	—
Standard Cost for Actual Production	3,204 hrs × ₹7 = ₹22,428	—
Actual Cost	—	₹4,500 + ₹7,200 + ₹10,880 = ₹22,580

Step 2: Calculate Labor Variances**1. Labor Cost Variance (LCV)**

LCV = Standard Cost for Actual Production – Actual Cost

$$\text{LCV} = ₹22,428 - ₹22,580 = ₹152 \text{ [A]}$$

2. Labor Rate Variance (LRV)

LRV = Actual Hours × (Standard Rate – Actual Rate) LRV = (600 × (₹7 – ₹7.50)) + (1,000 × (₹7 – ₹7.20)) + (1,600 × (₹7 – ₹6.80))

$$\text{LRV} = 600 \times (-₹0.50) + 1,000 \times (-₹0.20) + 1,600 \times ₹0.20$$

$$\text{LRV} = -₹300 - ₹200 + ₹320 = ₹180 \text{ [F]}$$

3. Labor Efficiency Variance (LEV)

LEV = Standard Rate × (Standard Hours – Actual Hours Worked)

$$\text{LEV} = ₹7 \times (3,204 - 3,072) \text{ LEV} = ₹7 \times 132 = ₹924 \text{ [F]}$$

4. Labor Idle Time Variance (LITV)

LITV = Abnormal Idle Time × Standard Rate

$$\text{LITV} = 128 \text{ hrs} \times ₹7$$

$$\text{LITV} = ₹896 \text{ [A]}$$

Summary Table 9.2 for Labor Cost Variances**Table 9.2** Labor Cost Variance

Variance Type	Formula	Explanation	Causes
Labor Cost Variance	Labor Cost Variance = $(SH \times SR) - (AH \times AR)$	Difference between the standard labor cost (standard hours $\times$ standard rate) and the actual labor cost (actual hours $\times$ actual rate).	Caused by discrepancies in labor hours and wage rates, affecting overall labor costs.
Labor Rate Variance	Labor Rate Variance = $(SR - AR) \times AH$	Difference between the standard rate and the actual rate paid for labor, multiplied by the actual hours worked.	- Changes in wage structure - Overtime - Hiring of different-grade workers - Different pay methods
Labor Efficiency Variance	Labor Efficiency Variance = $SR \times (SH - AHW)$	Difference between the standard hours allowed for the actual output and the actual hours worked, valued at the standard wage rate.	- Insufficient supervision - Faulty machinery - Worker dissatisfaction - Substandard materials
Idle Time Variance	Idle Time Variance = $(AH - AHW) \times SR$	The cost of idle time when workers are paid for hours not worked due to abnormal conditions, calculated using the standard rate.	- Unused workstations - Delays due to external factors (lack of materials, tools, etc.)
Labor Yield Variance	Labor Yield Variance = $(SH - RAH) \times SR$	Difference between the expected output (standard hours for actual output) and the actual output achieved (actual hours worked in standard proportion).	- Higher/lower output than expected - Inefficiencies in the production process
Labor Mix Variance	Labor Mix Variance = $[(RAH - AHW) \times SR]$	Difference caused by changes in the composition of the labor force, reflecting how the actual labor hours worked in different grades compare to the standard mix.	- Changes in workforce composition - Use of workers with different skill levels

N O T E S

9.7 OVERHEAD VARIANCES

Overhead variances occur due to the difference between actual overheads and absorbed overheads. Actual overheads are determined at the end of the accounting period, once the expense accounts are finalized. Absorbed overheads, on the other hand, are the overheads allocated to each unit of production using a pre-determined overhead rate. This pre-determined rate is also referred to as the standard overhead recovery rate, standard overhead absorption rate, or standard burden rate.

To calculate the standard overhead recovery rate, an estimate of the expected overhead expenses for each department for the upcoming year is made. These estimated overheads are divided into fixed and variable components. Then, an estimate of normal capacity utilization is made, based on production units, machine hours, or direct labor hours. The estimated overheads are divided by the projected capacity level to compute the pre-determined overhead absorption rate, as follows:

$$\text{Standard Fixed Overhead Rate} = \frac{\text{Budgeted Fixed Overheads}}{\text{Normal Volume}}$$

Overhead variances are categorized into two main types: a) Fixed Overhead Variances b) Variable Overhead Variances

9.7.1 VARIABLE OVERHEAD VARIANCE

Variable overhead variances are based on the assumption that variable overheads fluctuate directly with production levels. As production changes, the associated costs vary accordingly. Any change in variable overhead expenditure impacts costs, and this is generally classified as an expenditure variance. Some experts argue that inefficiencies may lead to variances in variable overheads, but since the cost per unit is usually small, these inefficiencies are often overlooked, and the variance is typically attributed solely to changes in expenditure.

9.7.2 FIXED OVERHEAD VARIANCE

Fixed overhead variances, on the other hand, are more complex as they depend on the volume or level of activity. Unlike variable overheads, fixed costs do not change with production levels, but the overhead rate is influenced by changes in activity levels. Determining the standard overhead rate for fixed costs requires careful consideration of the activity volume. If the level of activity changes without a corresponding adjustment to the fixed overheads, this will result in a change in the overhead rate. This issue is unique to fixed overheads, as changes in activity levels do not impact the standard rates for materials or labor. To resolve this challenge, budgets are typically used instead of standard rates when setting fixed overhead rates.

Example:

Budgeted Output: 12,000 units

Budgeted Overheads: Rs. 15,000

□ Fixed: Rs. 9,000

□ Variable: Rs. 6,000

Actual Overheads: Rs. 17,400

□ Fixed: Rs. 9,500 □ Variable: Rs. 7,900

Actual Output: 10,000 units

Calculate the following overhead variances:

- Total Overhead Cost Variance (OCV)
- Variable Overhead Cost Variance (VOCV)
- Fixed Overhead Cost Variance (FOCV)

Would you like the solution to this question as well?

Step 1: Calculate Standard/Budgeted Overhead Rates

1. Standard/Budgeted Overhead Rate per Unit:

Standard Overhead Rate = Budgeted Overheads/Budgeted Output =
 $15,000/12,000 = \text{Re. } 1.25 \text{ per unit}$

2. Standard/Budgeted Fixed Overhead Rate per Unit:

Fixed Overhead Rate = Budgeted Fixed Overheads/Budgeted Output
 $= 9,000/12,000 = \text{Re. } 0.75 \text{ per unit}$

3. Standard/Budgeted Variable Overhead Rate per Unit:

Variable Overhead Rate = Budgeted Variable Overheads/Budgeted Output
 $= 6,000/12,000 = \text{Re. } 0.50 \text{ per unit}$

Step 2: Calculate Overhead Variances

(a) Total Overhead Cost Variance (OCV):

OCV = Recovered Overheads – Actual Overheads

Recovered Overheads = Standard Overhead Rate × Actual Output

Recovered Overheads = $1.25 \times 10,000 = 12,500$

OCV = $12,500 - 17,400 = -4,900$ (Adverse)

(b) Variable Overhead Cost Variance (VOCV):

VOCV = Recovered Variable Overheads – Actual Variable Overheads

Recovered Variable Overheads = Standard Variable Overhead Rate
 × Actual Output

Recovered Variable Overheads = $0.50 \times 10,000 = 5,000$

VOCV = $5,000 - 7,900 = -2,900$ (Adverse)

(c) Fixed Overhead Cost Variance (FOCV):

FOCV = Recovered Fixed Overheads – Actual Fixed Overheads

Recovered Fixed Overheads = Standard Fixed Overhead Rate ×
 Actual Output

Recovered Fixed Overheads = $0.75 \times 10,000 = 7,500$

FOCV = $7,500 - 9,500 = -2,000$ (Adverse)

N O T E S

Step 3: Verification

$$\text{OCV} = \text{VOCV} + \text{FOCV} - 4,900 = -2,900 + (-2,000)$$

**SELF-ASSESSMENT QUESTIONS**

5. What is the primary cause of variable overhead variance?
 - (a) Changes in activity levels.
 - (b) Fluctuations in fixed overheads.
 - (c) Differences in material usage.
 - (d) Changes in variable overhead expenditure.
6. A company uses the following data for overheads:
 - ☐ Budgeted overhead cost: ₹50,000
 - ☐ Actual overhead cost: ₹55,000
 What is the Overhead Spending Variance?

(a) ₹5,000 (Favorable)	(c) ₹50,000 (Adverse)
(b) ₹5,000 (Adverse)	(d) ₹50,000 (Favorable)

9.8 ANALYSIS OF VARIANCES

Overhead cost variance refers to the difference between the standard cost of overhead absorbed for the actual output and the actual overhead cost incurred. In simpler terms, it is the difference between the total standard overheads absorbed and the total actual overheads incurred. The formula for calculating overhead cost variance is as follows:

$$\text{Overhead Cost Variance (OHCV)} = \text{Total Standard Overheads} - \text{Total Actual Overheads}$$

9.8.1 FIXED OVERHEAD (FO) VARIANCES

Fixed overhead includes all expenditures that tend to remain relatively constant regardless of the level of output or the number of hours worked. Classification of fixed overhead variance is shown in figure 9.3:

Fixed overhead cost variance refers to the difference between the standard fixed overhead costs allowed for the actual output achieved and the actual fixed overheads incurred. It is calculated as:

$$\text{FOCV} = (\text{Actual output} \times \text{Standard fixed overhead rate}) - \text{Actual fixed overheads}$$

OR

$$(\text{Standard hours produced} \times \text{Standard fixed overhead rate per hour}) - \text{Actual fixed overheads}$$

OR

$$\text{Recovered fixed overhead} - \text{Actual fixed overheads}$$

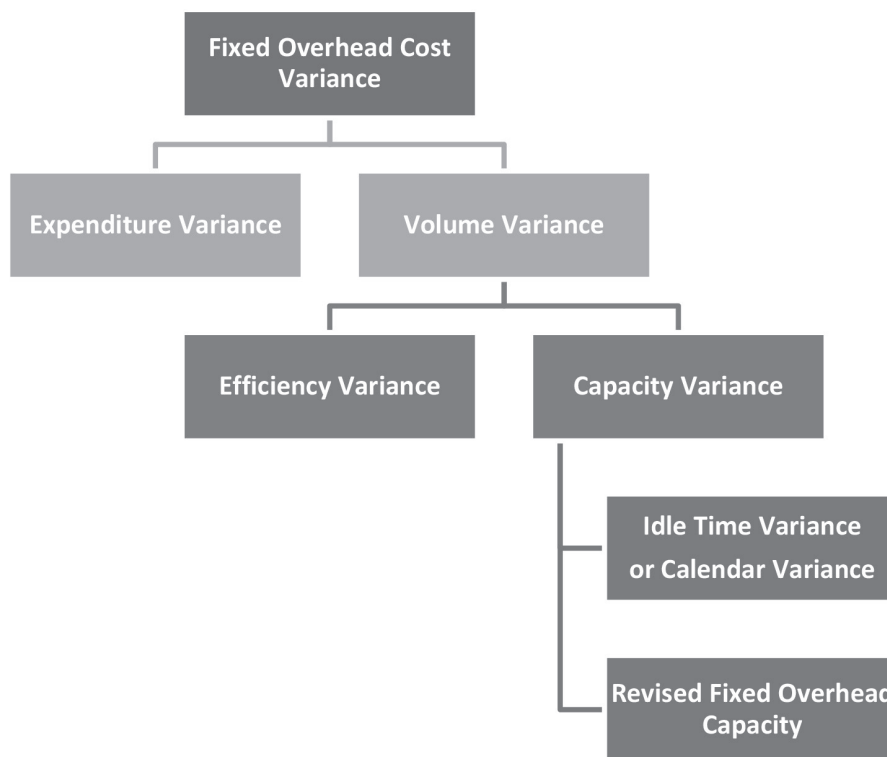


Figure 9.3 Classification of Fixed Overhead Variances

Here, “standard overhead produced” refers to the hours that should have been used for the actual output.

Fixed overhead variance can be broadly divided into:

1. Expenditure variance
2. Volume variance

(i) Expenditure Variance

Also known as the budget variance, it is calculated by comparing the actual total overhead cost incurred against the budgeted overhead cost:

Expenditure Variance = Budgeted fixed overhead – Actual fixed overhead

OR

(Budgeted hours × Standard fixed overhead rate) – Actual fixed overhead

If actual overheads are higher than budgeted, it results in an adverse variance, and vice versa. This variance measures the efficiency of expenditure.

(ii) Volume Variance

This represents the difference between the overhead absorbed for actual output and the overhead based on budgeted output. It indicates whether there was over- or under-absorption of fixed overheads during a specific period. When actual output exceeds the standard output, the variance is favorable, and vice versa if actual output falls short.

N O T E S

Volume Variance (FOVV) = (Actual output $\times$ Standard rate) – Budgeted fixed overheads

OR

Standard rate (Actual output – Standard output)

OR

Standard rate per hour (Standard hours produced – Budgeted hours)

OR

Absorbed overhead – Budgeted overhead

Volume variance can be further divided into:

- (a) **Efficiency Variance:** This variance arises from the difference between the actual output achieved and the output that should have been achieved in the actual hours worked. The variance is favorable if actual production exceeds standard production during actual hours.

Fixed Overhead Efficiency Variance (FOEfV) = Standard Fixed Overhead Rate per hour $\times$ (Standard Production – Actual Production)

- (b) **Capacity Variance:** This portion of the volume variance occurs when the plant operates at a higher or lower capacity than the standard capacity. It reflects under- or over-utilization of plant and equipment. If capacity utilization exceeds the budgeted capacity, the variance is favorable, and vice versa.

Fixed Overhead Capacity Variance = Standard rate $\times$ (Standard quantity – Budgeted quantity)

- (c) **Revised Capacity Variance:** This variance reflects differences in capacity utilization due to working more or fewer days than budgeted. It is calculated as:

Fixed Overhead Revised Capacity Variance (FORCV) = Standard Rate $\times$ (Standard Quantity – Revised Budgeted Quantity)

- (d) **Calendar (Idle Time) Variance:** Calendar Variance occurs because the estimated fixed overheads remain constant for each month or period, regardless of the actual number of working days. This variance is a part of the volume variance and arises due to the difference between the number of working days anticipated in the budget period and the actual working days in the period to which the budget is applied. The budgeted working days are determined by dividing the total number of annual working days by twelve.

FO Calendar Variance = Standard rate $\times$ (Revised budgeted units – Budgeted units)

OR

(The increase or decrease in production due to working more or fewer days at the revised capacity rate) $\times$ (standard rate per unit)

Example: You are provided the following data:

	Budgeted	Actual
Fixed overhead for March	₹15,000	₹15,400
Units of production in March	3,500	3,600
Standard time for one unit	5 hours	
Actual hours worked		17,800 hours

Calculate all variances relating to fixed overheads.

Sol.

Basic Calculations:

1. Absorbed Fixed Overheads

$$\begin{aligned}
 &= \text{Standard Rate per Hour} \times \text{Standard Hours for Actual Output} \\
 &= ₹ 0.8571^{\#} \times (5 \text{ Hours} \times 3,600 \text{ Units}) \\
 &= ₹ 15,428.57
 \end{aligned}$$

OR

$$\begin{aligned}
 &= \text{Standard Rate per Unit} \times \text{Actual Output} \\
 &= ₹ 4.2857^* \times 3,600 \text{ Units} \\
 &= ₹ 15,428.57
 \end{aligned}$$

Standard Rate per Hour

$$\begin{aligned}
 &= ₹ 15,000 (\text{Budgeted Fixed Overheads}) / [3,500 \text{ Units (Budgeted Output)} \times 5 \text{ Hours (Budgeted Hours)}] \\
 &= ₹ 0.8571
 \end{aligned}$$

*** Standard Rate per Unit**

$$\begin{aligned}
 &= ₹ 15,000 (\text{Budgeted Fixed Overheads}) / 3,500 \text{ Units (Budgeted Output)} \\
 &= ₹ 4.2857
 \end{aligned}$$

2. Budgeted Fixed Overheads

$$= ₹ 15,000$$

3. Actual Fixed Overheads

$$= ₹ 15,400$$

4. Budgeted Fixed Overheads for Actual Hours

$$= ₹ 0.8571 \times 17,800 \text{ Hours} = ₹ 15,256.38$$

Computation of Variances:

□ Fixed Overhead Cost Variance

$$\begin{aligned}
 &= \text{Absorbed Fixed Overheads} - \text{Actual Fixed Overheads} \\
 &= ₹ 15,428.57 - ₹ 15,400 \\
 &= ₹ 28.57 \text{ (F)}
 \end{aligned}$$

N O T E S

❑ **Fixed Overhead Expenditure Variance**

$$\begin{aligned}
 &= \text{Budgeted Fixed Overheads} - \text{Actual Fixed Overheads} \\
 &= ₹ 15,000 - ₹ 15,400 \\
 &= ₹ 400 \text{ (A)}
 \end{aligned}$$

❑ **Fixed Overhead Volume Variance**

$$\begin{aligned}
 &= \text{Absorbed Fixed Overheads} - \text{Budgeted Fixed Overheads} \\
 &= ₹ 15,428.57 - ₹ 15,000 \\
 &= ₹ 428.57 \text{ (F)}
 \end{aligned}$$

❑ **Fixed Overhead Capacity Variance**

$$\begin{aligned}
 &= \text{Budgeted Fixed Overheads for Actual Hours} - \text{Budgeted Fixed Overheads} \\
 &= ₹ 15,256.38 - ₹ 15,000 \\
 &= ₹ 256.38 \text{ (F)}
 \end{aligned}$$

❑ **Fixed Overhead Efficiency Variance**

$$\begin{aligned}
 &= \text{Absorbed Fixed Overheads} - \text{Budgeted Fixed Overheads for Actual Hours} \\
 &= ₹ 15,428.57 - ₹ 15,256.38 \\
 &= ₹ 172.19 \text{ (F)}
 \end{aligned}$$

9.8.2 VARIABLE OVERHEAD (VO) VARIANCES

The difference between the standard variable overhead cost allowed for the actual output achieved and the actual variable overheads incurred is referred to as this variance. Typically, this variance is identified as an expenditure (cost) variance because variable overheads fluctuate in direct proportion to production, meaning that only changes in expenditure can lead to this variance.

The calculation method is as follows: The variable overhead cost variance is usually determined as a total, since variable overheads change based on output rather than time. Consequently, only one variance is typically considered. However, some accountants argue that certain variable overheads might also fluctuate with time, leading to the emergence of a variable overhead efficiency variance, similar to labor efficiency variance. This can be calculated if information on actual time taken and time allowed is provided. In such cases, the variable overhead variance can be divided into two components.

Variable Overhead (VO) Variances = (Standard Variable Overheads for Production – Actual Variable Overheads)

Classification of Variable Overhead variances is as shown in figure 9.4:

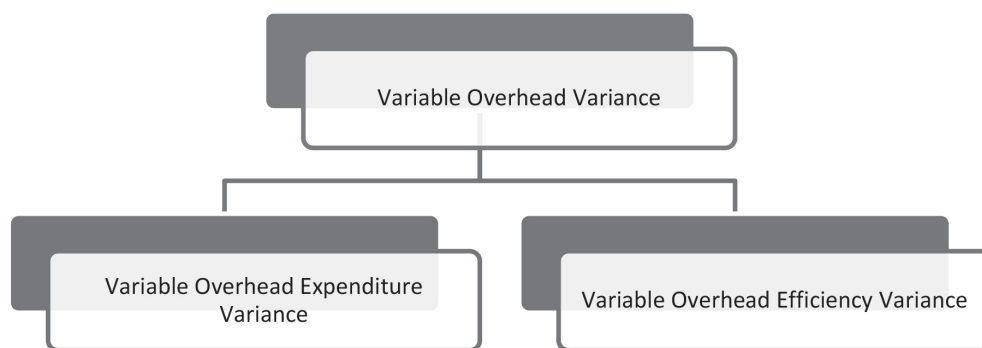


Figure 9.4 Variable Overhead Variance

Variable Overhead Expenditure Variance: This portion of the variable overhead variance occurs due to the difference between the budgeted variable overhead and the actual variable overhead incurred. It is calculated as follows:

(Budgeted Variable Overheads for Actual Hours) – (Actual Variable Overheads)

Or

(Standard Rate per Hour × Actual Hours) – (Actual Rate per Hour × Actual Hours)

Or

Actual Hours × (Standard Rate per Hour – Actual Rate per Hour)

Variable Overhead Efficiency Variance: This component of the variable overhead variance arises from the difference between the standard hours required for the actual output and the actual hours worked. It is calculated by multiplying the difference between the standard hours and actual hours worked by the standard variable overhead rate per hour. If the standard hours exceed the actual hours worked, the variance is favorable; otherwise, it is unfavorable. It is calculated as follows:

(Standard Variable Overheads for Production) – (Budgeted Overheads for Actual Hours)

Or

(Standard Variable Overhead Rate per Hour × Standard Hours for Actual Output) – (Standard Variable Overhead Rate per Hour × Actual Hours)

Or

Standard Variable Overhead Rate per Hour × (Standard Hours for Actual Output – Actual hours)

N O T E S

Illustration 6

Future Company has established the following standards for variable factory overhead:

- Standard hours per unit: 6 hours
- Variable overhead per hour: ₹ 2

The actual data for the month are as follows:

- Actual variable overheads incurred: ₹ 2,00,000
- Actual output (units): 20,000 units
- Actual hours worked: 1,12,000 hours

Calculate the following variable overhead variances:

- Variable Overhead Cost Variance
- Variable Overhead Efficiency Variance
- Variable Overhead Expenditure Variance

Sol.

Basic Calculations:**1. Standard Variable Overheads for Production**

$$\begin{aligned}
 &= \text{Standard Rate per Hour} \times \text{Standard Hours for Actual Output} \\
 &= ₹ 2 \times (6 \text{ Hours} \times 20,000 \text{ Units}) \\
 &= ₹ 2,40,000
 \end{aligned}$$

2. Actual Variable Overheads

$$= ₹ 2,00,000$$

3. Budgeted Variable Overheads for Actual Hours

$$\begin{aligned}
 &= \text{Standard Rate per Hour} \times \text{Actual Hours} \\
 &= ₹ 2 \times 1,12,000 \text{ hours} \\
 &= ₹ 2,24,000
 \end{aligned}$$

Computation of Variances:**1. Variable Overhead Cost Variance**

$$\begin{aligned}
 &= \text{Standard Variable Overheads for Production} - \text{Actual Variable Overheads} \\
 &= ₹ 2,40,000 - ₹ 2,00,000 \\
 &= ₹ 40,000 \text{ (Favorable)}
 \end{aligned}$$

2. Variable Overhead Expenditure Variance

$$\begin{aligned}
 &= \text{Budgeted Variable Overheads for Actual Hours} - \text{Actual Variable Overheads} \\
 &= ₹ 2,24,000 - ₹ 2,00,000 \\
 &= ₹ 24,000 \text{ (Favorable)}
 \end{aligned}$$

3. Variable Overhead Efficiency Variance

= Standard Variable Overheads for Production – Budgeted Variable Overheads for Actual Hours

= ₹ 2,40,000 – ₹ 2,24,000

= ₹ 16,000 (Favorable)

9.9 SALES VARIANCES

Sales are influenced by two main factors:

- (i) the selling price and (ii) the quantity of sales.

Variations between the standards set and the actual figures may be primarily due to changes in market trends. Generally, if the selling price increases, the sales volume may fall below the standard, potentially resulting in a favorable price variance and an unfavorable quantity variance. It is important to note that in the case of sales, a higher price is considered a favorable variance, unlike in materials, where a higher price leads to an adverse variance. Similarly, a lower sales volume is seen as unfavorable, whereas lower material usage results in a favorable variance.

The dynamics of demand and supply in the market largely determine both the quantity of sales and the selling price. Variations may arise from both controllable and uncontrollable factors. While changes in market conditions and customer demand are beyond management's control, certain factors like unnecessarily high prices are controllable. Efforts should be made to mitigate adverse variances caused by such factors.

9.9.1 SALES VALUE VARIANCE

The variance between budgeted and actual sales leads to Sales Value Variance. The formula is:

$$\text{Sales Value Variance} = \text{Budgeted Sales} - \text{Actual Sales}$$

If actual sales exceed budgeted sales, a favorable variance is reported, and if actual sales are lower, an adverse variance is noted. The difference in value could stem from variations in price or sales volume, which is further analyzed.

9.9.2 SALES PRICE VARIANCE

This is calculated similarly to material price variance. It arises from the difference between the actual selling price and the standard selling price for the actual quantity sold. The formula is:

$$\text{Actual Quantity Sold} \times (\text{Standard Price} - \text{Actual Price})$$

OR

$$\text{Price Variance} = \text{Standard Sales} - \text{Actual Sales}$$

N O T E S

9.9.3 SALES VOLUME VARIANCE

Similar to material usage variance, this is calculated based on the difference between budgeted and standard sales. In other words, the budgeted quantity at the standard price may differ from the actual quantity sold at standard prices. The variance results from the difference in budgeted and actual sales quantities. The formula is:

$$\text{Standard Price} \times (\text{Budgeted Quantity} - \text{Actual Quantity})$$

OR

$$\text{Volume Variance} = \text{Budgeted Sales} - \text{Standard Sales}$$

If standard sales exceed budgeted sales, a favorable variance occurs, and if the reverse is true, it leads to an adverse variance.

Illustration 7

The budgeted and actual sales of a firm, manufacturing and marketing a product, are furnished below:

Details	Budgeted	Actual
Sales Units	12,000 units @ ₹7 per unit	6,000 units @ ₹5 per unit
Sales Units		9,000 units @ ₹6 per unit

Calculate the following:

- Sales Price Variance
- Sales Volume Variance
- Sales Value Variance

Sol.

Basic Calculations:**□ Standard Sales**

$$\begin{aligned} &= \text{Standard price} \times \text{Actual quantity} \\ &= ₹7 \times 15,000 \text{ units} \\ &= ₹1,05,000 \end{aligned}$$

□ Actual Sales

$$\begin{aligned} &= \text{Actual price} \times \text{Actual quantity} \\ &= ₹5 \times 6,000 \text{ units} + ₹6 \times 9,000 \text{ units} \\ &= ₹30,000 + ₹54,000 \\ &= ₹84,000 \end{aligned}$$

- Sales Price Variance** = Standard Sales – Actual Sales
 $= ₹1,05,000 - ₹84,000$
 $= ₹21,000 \text{ (Favorable)}$

b. Sales Volume Variance

= Budgeted Sales – Standard Sales

= ₹84,000 – ₹1,05,000

= ₹21,000 (Adverse)

c. Sales Value Variance

= Budgeted Sales – Actual Sales

= ₹84,000 – ₹84,000

= ₹Nil

Example: Given Information:

Product	Number of Units (Standard)	Standard Rate Per Unit (Rs.)	Standard Sales (Rs.)	Actual Number of Units	Actual Rate Per Unit (Rs.)	Actual Sales (Rs.)
A	4000	8	32,000	4500	9	40,500
B	3500	10	35,000	4000	9	36,000
C	5000	7	35,000	5500	8	44,000

Calculate:

- I. Total Sales Variance
- II. Sales Price Variance
- III. Sales Volume Variance

Solution:**I. Total Sales Variance**

This is calculated as the difference between actual sales and standard sales.

Total Sales Variance = Actual Sales – Standard Sales

□ Actual Sales = Rs. 40,500 (A) + Rs. 36,000 (B) + Rs. 44,000 (C) = Rs. 120,500

□ Standard Sales = Rs. 32,000 (A) + Rs. 35,000 (B) + Rs. 35,000 (C) = Rs. 102,000

Total Sales Variance = Rs. 120,500 – Rs. 102,000 = Rs. 18,500 [Favorable]

II. Sales Price Variance

This is calculated as the difference between actual price and standard price, multiplied by the actual quantity.

Sales Price Variance = Actual Quantity × (Actual Price – Standard Price)

□ Product A: 4500 × (Rs. 9 – Rs. 8) = Rs. 4500 [Favorable]

□ Product B: 4000 × (Rs. 9 – Rs. 10) = Rs. 4000 [Unfavorable]

□ Product C: 5500 × (Rs. 8 – Rs. 7) = Rs. 5500 [Favorable]

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Total Sales Price Variance = Rs. 4500 [F] – Rs. 4000 [A] + Rs. 5500 [F] = Rs. 5000 [Favorable]

III. Sales Volume Variance

This is calculated by multiplying the standard price by the difference between the actual quantity and the standard quantity.

Sales Volume Variance = Standard Price × (Actual Quantity – Standard Quantity)

- Product A: Rs. 8 × (4500 – 4000) = Rs. 4000 [Favorable]
- Product B: Rs. 10 × (4000 – 3500) = Rs. 5000 [Favorable]
- Product C: Rs. 7 × (5500 – 5000) = Rs. 3500 [Favorable]

Total Sales Volume Variance = Rs. 4000 [F] + Rs. 5000 [F] + Rs. 3500 [F] = Rs. 12,500 [Favorable]

Summary Table 9.3 for Sales Variances

Table 9.3 Sales Variance

Variance Type	Formula	Explanation	Causes
Sales Value Variance	Sales Value Variance = Budgeted Sales – Actual Sales	Difference between the budgeted sales and actual sales. It reflects the overall performance in sales.	- Differences in both price and volume factors.
Sales Price Variance	Sales Price Variance = Actual Quantity Sold × (Standard Price – Actual Price)	Difference between the actual selling price and the standard selling price for the actual quantity sold. Measures price changes.	- Changes in market conditions or pricing strategies affecting the selling price.
Sales Volume Variance	Sales Volume Variance = Standard Price × (Budgeted Quantity – Actual Quantity)	Difference between budgeted sales volume at standard price and actual sales volume at standard price. Measures the impact of sales volume changes.	- Variations in market demand, customer preferences, or other external factors affecting sales volume.



SELF-ASSESSMENT QUESTIONS

7. The following data is available:

- Budgeted production: 1,000 units
- Actual production: 900 units
- Budgeted fixed overhead cost: ₹40,000

Calculate the Fixed Overhead Volume Variance.

- a. ₹4,000 (Adverse) c. ₹36,000 (Adverse)
b. ₹4,000 (Favorable) d. ₹36,000 (Favorable)

8. A company sells two products, X and Y, with the following data for the current period:

Product	Budgeted Quantity (Units)	Actual Quantity (Units)	Budgeted Sales Price (Rs.)	Actual Sales Price (Rs.)	Standard Cost (Rs.)
X	5000	5500	20	18	10
Y	4000	4500	30	32	15

Calculate the total sales margin variance.

- a. ₹10,500 [Favorable] c. ₹15,000 [Unfavorable]
b. ₹10,500 [Unfavorable] d. ₹20,000 [Favorable]

9.10 VARIANCES BASED ON PROFITS

The objective of using the Variances Based on Profits is to assess how changes in sales quantities and selling prices impact the company's profits. It is important to analyze both quantity and mix variances together since the sales manager is responsible for managing both. In companies that produce and sell multiple products, variances between budgeted and actual sales may occur due to several factors:

- ❑ Changes in unit price and cost.
- ❑ Changes in the physical volume of each product sold, which results in quantity variance.
- ❑ Changes in the volume of more profitable or less profitable products sold.

There are five key variables that may cause actual performance to deviate from budgeted performance:

1. Direct substitution of products.
2. Differences between the actual and budgeted quantities of sales components.
3. Variations in actual total quantity versus budgeted total quantity.
4. Differences between actual and budgeted unit costs.
5. Differences between actual and budgeted unit selling prices.

Sales management should particularly consider the interaction of multiple variables when making decisions. For example, a decrease in selling price, combined with a favorable product quantity variance, can help assess price elasticity of demand.

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9.10.1 SALES MARGIN VARIANCE DUE TO SELLING PRICE

This variance occurs due to the difference between the budgeted price of the quantity sold and the actual price. It is calculated as:

$$\text{Sales Margin Price Variance} = \text{Actual Quantity} \times (\text{Actual Margin per Unit} - \text{Budgeted Margin per Unit})$$

9.10.2 SALES MARGIN VARIANCE DUE TO VOLUME OF SALES

This arises due to the difference between actual and budgeted quantities for each product, both evaluated at the budgeted margin. It is computed as:

$$\text{Sales Margin Volume Variance} = \text{Budgeted Margin per Unit} \times (\text{Actual Units} - \text{Budgeted Units})$$

Illustration 8

Given:

- Budgeted Margin per Unit = ₹30
- Actual Margin per Unit = ₹25
- Budgeted Units = 10,000 units
- Actual Units = 12,000 units
- Actual Quantity Sold = 12,000 units

Calculate: Sales Margin Price Variance & Sales Margin Volume Variance

Sol.

1. Sales Margin Price Variance = Actual Quantity $\times$ (Actual Margin per Unit – Budgeted Margin per Unit)

Substituting the given values:

$$\text{Sales Margin Price Variance} = 12,000 \times (25 - 30) = -₹60,000$$

Interpretation: The Sales Margin Price Variance is ₹60,000 unfavorable, indicating that the actual margin per unit is lower than the budgeted margin.

Sales Margin Variance Due to Volume of Sales

This arises due to the difference between actual and budgeted quantities, evaluated at the budgeted margin. The formula is:

2. Sales Margin Volume Variance = Budgeted Margin per Unit $\times$ (Actual Units – Budgeted Units)

Substituting the given values:

$$\text{Sales Margin Volume Variance} = 30 \times (12,000 - 10,000) = ₹60,000$$

Interpretation: The Sales Margin Volume Variance is ₹60,000 favorable, indicating that the actual sales volume was

Summary Table 9.4 for Sales Margin Variances**Table 9.4** Sales Margin Variance

Variance Type	Formula	Explanation	Causes
Sales Margin Variance Due to Selling Price	Sales Margin Price Variance = Actual Quantity × (Actual Margin per Unit – Budgeted Margin per Unit)	Difference between actual and budgeted sales margin due to changes in the selling price for the quantity sold. Measures how price changes affect overall profit margin.	- Changes in the selling price per unit leading to higher or lower margins.
Sales Margin Variance Due to Volume of Sales	Sales Margin Volume Variance = Budgeted Margin per Unit × (Actual Units – Budgeted Units)	Variance caused by the difference in actual and budgeted sales quantities, evaluated at the budgeted margin per unit. Measures how volume changes affect the profit margin.	- Changes in the quantity of units sold, reflecting either a higher or lower sales volume compared to the budgeted quantity.

9.11 CONTROL RATIOS

Control ratios are key performance indicators used to assess the efficiency and effectiveness of an organization's operations by comparing actual performance to standards or budgets. They help management identify variances and take corrective actions where necessary. Some common control ratios include:

1. Capacity Ratio

The Capacity Ratio is vital for understanding how effectively a company is utilizing its resources in relation to what was planned. It measures the extent to which available production capacity is being used. This ratio is particularly important in industries where capacity constraints—such as machine hours or labor availability—are critical to production efficiency. This ratio is useful for capacity planning and for managers to understand whether they need to scale up or down production capabilities based on demand and current resource utilization.

$$\text{Capacity Ratio} = \frac{\text{Actual Hours Worked}}{\text{Budgeted Hours}} \times 100$$

- ❑ **Above 100%:** Indicates better utilization of capacity.
- ❑ **Below 100%:** Indicates under-utilization of capacity.

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Example: ABC Ltd., a toy manufacturing company, evaluates its production efficiency using the Capacity Ratio. For the month of November, the company planned to operate for 2,400 machine hours to meet its budgeted production target. However, due to increased demand and additional shifts, the company actually operated for 2,700 machine hours during the same period.

Step 1: Formula for Capacity Ratio

$$\text{Capacity Ratio} = (\text{Actual Hours Worked} / \text{Budgeted Hours}) \times 100$$

Step 2: Substitute Values

❑ Actual Hours Worked = 2,700 hours

❑ Budgeted Hours = 2,400 hours

$$\text{Capacity Ratio} = (2,700 / 2,400) \times 100$$

$$\text{Capacity Ratio} = 1.125 \times 100 = 112.5\%$$

Interpretation:

The Capacity Ratio of 112.5% indicates that ABC Ltd. utilized its production capacity better than planned. This efficiency was achieved through additional machine hours, possibly due to higher demand or better management of resources.

2. Efficiency Ratio

The Efficiency Ratio gauges how well an organization is converting input (in terms of labor or machine hours) into output (finished goods). It is a reflection of operational efficiency and how well resources are being used to generate the planned output. This ratio is essential for productivity analysis, helping management to identify areas where processes can be optimized to save time and reduce costs.

$$\text{Efficiency Ratio} = \frac{\text{Standard Hours for Actual Output}}{\text{Actual Hours Worked}} \times 100$$

❑ **Above 100%:** Indicates more efficiency than planned.

❑ **Below 100%:** Indicates lower efficiency.

Example: XYZ Ltd., a furniture manufacturing company, evaluates its operational efficiency using the Efficiency Ratio. The company produces tables, and it has determined that the standard time required to manufacture one table is 2 hours.

In the month of October:

❑ The company produced 1,200 tables.

❑ The actual time spent by the workforce was 2,500 hours.

Step 1: Formula for Efficiency Ratio

$$\text{Efficiency Ratio} = (\text{Standard Hours for Actual Output} / \text{Actual Hours Worked}) \times 100$$

Step 2: Calculate Standard Hours for Actual Output

Standard Hours for Actual Output = Standard Time per Unit $\times$ Actual Output

Standard Hours for Actual Output = $2 \times 1,200 = 2,400$ hours

Step 3: Substitute Values in the Efficiency Ratio Formula

□ Standard Hours for Actual Output = 2,400 hours

□ Actual Hours Worked = 2,500 hours

Efficiency Ratio = $(2,400/2,500) \times 100$

Efficiency Ratio = $0.96 \times 100 = 96\%$

Interpretation:

The Efficiency Ratio of 96% indicates that XYZ Ltd. operated at slightly less efficiency than planned. Workers required more time than the standard time to produce the planned output, suggesting a need to review processes or address potential inefficiencies.

3. Production Volume Ratio

This ratio helps in assessing whether the company is producing the number of units it initially planned to produce, considering both production planning and resource availability. It is an indicator of overall production effectiveness in relation to the budget. This ratio helps managers to adjust production plans and ensures that the business is on track to meet its production goals, or if corrective action needs to be taken to boost production.

$$\text{Production Volume Ratio} = \frac{\text{Standard Hours for Actual Output}}{\text{Budgeted Hours}} \times 100$$

□ **Above 100%:** Indicates higher production than budgeted.

□ **Below 100%:** Indicates lower production than budgeted.

Example: ABC Textiles Ltd. is evaluating its production effectiveness using the Production Volume Ratio. The company manufactures cotton shirts. For the month of November, the following details are provided:

□ Budgeted production: 5,000 shirts

□ Standard time required to produce one shirt: 1.5 hours

□ Actual production: 5,600 shirts

□ Budgeted machine hours: 7,500 hours

Step 1: Formula for Production Volume Ratio

Production Volume Ratio = $(\text{Standard Hours for Actual Output} / \text{Budgeted Hours}) \times 100$

Step 2: Calculate Standard Hours for Actual Output

Standard Hours for Actual Output = Standard Time per Unit $\times$ Actual Output

Standard Hours for Actual Output = $1.5 \times 5,600 = 8,400$ hours

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Step 3: Substitute Values in the Formula

- ❑ Standard Hours for Actual Output = 8,400 hours
- ❑ Budgeted Hours = 7,500 hours

$$\text{Production Volume Ratio} = (8,400 / 7,500) \times 100$$

$$\text{Production Volume Ratio} = 1.12 \times 100 = 112\%$$

Interpretation:

The Production Volume Ratio of 112% indicates that ABC Textiles Ltd. produced more units than planned. This positive variance suggests effective resource utilization and an increase in production efficiency.

4. Activity Ratio

The Activity Ratio evaluates how well the company's actual production activity matches up with the production plan. It considers the operational activity in terms of time spent on production and relates it to the budgeted hours for that activity. The Activity Ratio is a useful metric for operational managers to track if actual production levels are in line with the company's strategic goals and budgeted plans.

$$\text{Activity Ratio} = \frac{\text{Standard Hours for Actual Production}}{\text{Budgeted Hours}} \times 100$$

- ❑ **Above 100%:** Indicates more activity than planned.
- ❑ **Below 100%:** Indicates less activity than planned.

Example: XYZ Electronics Ltd. is evaluating how well its actual production matches its planned production using the Activity Ratio. The company manufactures smartphones, and for the month of December, the following details are provided:

- ❑ Budgeted production: 3,000 smartphones
- ❑ Standard time required to assemble one smartphone: 2 hours
- ❑ Actual production: 3,200 smartphones
- ❑ Budgeted machine hours: 6,000 hours

Find:

- ❑ If the company had only produced 2,800 smartphones, calculate the Activity Ratio and interpret the result.
- ❑ What steps can the company take to optimize its production process if the Activity Ratio consistently exceeds 100%?
- ❑ How would the result change if the company had operated within the budgeted hours of 6,000 but produced 3,500 smartphones?

Sol.

Step 1: Formula for Activity Ratio

$$\text{Activity Ratio} = (\text{Standard Hours for Actual Production} / \text{Budgeted Hours}) \times 100$$

Step 2: Calculate Standard Hours for Actual Production

Standard Hours for Actual Production = Standard Time per Unit $\times$ Actual Production

Standard Hours for Actual Production = $2 \times 3,200 = 6,400$ hours

Step 3: Substitute Values in the Formula

□ Standard Hours for Actual Production = 6,400 hours

□ Budgeted Hours = 6,000 hours

Activity Ratio = $(6,400/6,000) \times 100$

Activity Ratio = $1.067 \times 100 = 106.7\%$

Interpretation:

The Activity Ratio of 106.7% indicates that XYZ Electronics Ltd. has achieved more production activity than planned. The company used 6,400 hours of labor and machine time, which is higher than the 6,000 hours that were budgeted for the production of 3,000 smartphones. This indicates that more resources were utilized to produce 3,200 smartphones, possibly due to increased production or inefficiencies in the process.

1. If the company has only produced 2,800 smartphones:**Step 1:** Calculate Standard Hours for Actual Production

Standard Hours for Actual Production = Standard Time per Unit $\times$ Actual Production

Standard Hours for Actual Production = $2 \times 2,800 = 5,600$ hours

Step 2: Calculate the Activity Ratio

□ Standard Hours for Actual Production = 5,600 hours

□ Budgeted Hours = 6,000 hours

Activity Ratio = $(5,600/6,000) \times 100 = 0.933 \times 100 = 93.33\%$

Interpretation:

The Activity Ratio of 93.33% indicates that the company has performed less activity than planned. The actual production (2,800 smartphones) used fewer hours (5,600 hours) than the 6,000 budgeted hours. This suggests underutilization of resources or that the production process was more efficient than planned.

2. Steps the company can take to optimize its production process if the Activity Ratio consistently exceeds 100%:

- **Identify inefficiencies:** Review the production process for any areas that cause delays or require extra time. This could include slow machine setups, unproductive shifts, or idle time.
- **Process Improvement:** Implement techniques such as Lean Manufacturing, Six Sigma, or Kaizen to eliminate waste and reduce time spent on non-value-added activities.

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- ❑ **Increase automation:** If possible, invest in automation to reduce reliance on labor and machine hours.
- ❑ **Improve workforce training:** Ensure workers are fully trained and efficient in performing their tasks.
- ❑ **Adjust production scheduling:** Review whether the scheduled shifts and work hours are optimized for the available resources and the production goal.

3. If the company has operated within the budgeted hours of 6,000 but produced 3,500 smartphones:

Step 1: Calculate Standard Hours for Actual Production

Standard Hours for Actual Production = $2 \times 3,500 = 7,000$ hours

Step 2: Calculate the Activity Ratio

- ❑ Standard Hours for Actual Production = 7,000 hours
- ❑ Budgeted Hours = 6,000 hours

Activity Ratio = $(7,000/6,000) \times 100 = 1.167 \times 100 = 116.7\%$

Interpretation:

The Activity Ratio of 116.7% indicates that the company used more time than budgeted to produce 3,500 smartphones. While the actual production exceeded the budgeted output, the company had to spend more time (7,000 hours) than originally planned (6,000 hours), suggesting that more resources were required for production than expected. This could be due to inefficiencies, machine breakdowns, or longer than planned setup times.

4. Sales Volume Ratio

This ratio measures the difference between actual sales volume and the budgeted or planned sales volume, providing insight into sales performance. It is critical for determining how effectively the company is converting its products into revenue. Sales Volume Ratio is crucial for sales management and financial planning, as it provides an early indication of whether the company will meet its revenue targets. If sales volume is lower than expected, it may require adjustments in pricing, marketing, or sales strategy.

$$\text{Sales Volume Ratio} = \frac{\text{Actual Sales Volume}}{\text{Budgeted Sales Volume}} \times 100$$

- ❑ **Above 100%:** Indicates higher sales than anticipated.
- ❑ **Below 100%:** Indicates lower sales than anticipated.

Example: XYZ Electronics Ltd. is assessing its sales performance using the Sales Volume Ratio for the month of December. The company sells laptops and provides the following details:

- ❑ Budgeted sales volume: 4,000 laptops
- ❑ Actual sales volume: 4,500 laptops

- ❑ Budgeted selling price per laptop: Rs. 30,000
- ❑ Actual selling price per laptop: Rs. 31,000 (Note: This is not necessary for the Sales Volume Ratio, but could be used for a more detailed analysis such as Revenue Variance.)

Step 1: Formula for Sales Volume Ratio

Sales Volume Ratio = (Actual Sales Volume/Budgeted Sales Volume) × 100

Sales Volume Ratio = (4,500/4,000) × 100

Sales Volume Ratio = 1.125 × 100 = 112.5%

Interpretation:

The Sales Volume Ratio of 112.5% indicates that XYZ Electronics Ltd. has achieved higher sales than planned. The actual sales of 4,500 laptops exceeded the budgeted sales volume of 4,000 laptops, representing a positive variance. This higher sales performance may be attributed to effective marketing, increased demand, or a successful sales strategy.

Purpose of Control Ratios:

- ❑ **Identify Performance Gaps:** Control ratios help highlight areas where actual performance deviates from standards, allowing management to focus on underperforming areas.
- ❑ **Decision-Making:** By understanding the causes of variances, management can make informed decisions to improve operational efficiency.
- ❑ **Benchmarking:** These ratios can serve as benchmarks for future performance or industry comparisons.

Activity

For each situation described below, identify the type of variance (e.g., material price variance, labor efficiency variance) and indicate whether the variance is *favourable* or *unfavourable*.

1. **Scenario:** Ramesh, a senior worker in the assembly line, resigned and was replaced by Mohan, a junior employee. Ramesh's wages were Rs. 300 per day, while Mohan is being paid Rs. 250 per day.
2. **Scenario:** Due to defective tools in the machining section, only 50 units were produced from a batch of raw materials, whereas the standard output should have been 60 units.
3. **Scenario:** The implementation of a new energy-saving system reduced electricity expenses in the factory by Rs. 25,000 per month.
4. **Scenario:** The cost of steel sheets increased by 10%, higher than the anticipated rise of 6% used to compute standard material cost.
5. **Scenario:** A supplier's delay led to the use of a substitute raw material, which was 8% more expensive than the standard material.
6. **Scenario:** The standard time for assembling a product is 2 hours. A newly recruited worker, while still in training, took 2.5 hours to complete the assembly.

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7. **Scenario:** The actual production in 2023 was 15% lower than the estimated level, while total fixed manufacturing overhead costs remained as budgeted.



SELF-ASSESSMENT QUESTIONS

9. What does a Capacity Ratio above 100% indicate?
- Under-utilization of capacity
 - Efficient use of available capacity
 - Better utilization of capacity than planned
 - Inefficiency in operations
10. Which control ratio evaluates how well an organization is converting input (labor or machine hours) into output?
- | | |
|-----------------------|----------------------------|
| a. Activity Ratio | c. Efficiency Ratio |
| b. Sales Volume Ratio | d. Production Volume Ratio |

9.12 SUMMARY

- ❑ Variance analysis involves calculating variances between actual and standard costs and identifying their causes to assess performance.
- ❑ Unfavorable variances indicate inefficiency and reduce profit, while favorable variances suggest improved efficiency and increase profit.
- ❑ Variances are analyzed to focus on controllable factors, adhering to the principle of management by exception, while uncontrollable variances are influenced by external factors.
- ❑ Material cost variance is divided into material price variance and material usage variance, each highlighting different causes of cost deviations.
- ❑ Labor cost variance includes labor rate and labor efficiency variances, allowing management to investigate discrepancies in wage rates and worker efficiency.
- ❑ Overhead variances arise from the difference between actual and absorbed overheads, with actual overheads determined at the end of the accounting period and absorbed overheads allocated using a pre-determined rate.
- ❑ The standard overhead recovery rate is calculated by estimating expected overhead expenses, dividing them into fixed and variable components, and dividing by normal capacity utilization.
- ❑ Overhead variances are classified into two main types: variable overhead variances (due to fluctuations in production levels) and fixed overhead variances (dependent on activity levels).
- ❑ Fixed overhead variances are further categorized into expenditure variance (difference between actual and budgeted fixed overheads)

and volume variance (over- or under-absorption of fixed overheads based on output).

- ❑ Sales variances, including sales price and volume variances, are influenced by changes in market trends, selling prices, and sales quantities, with price increases leading to favorable variances and lower volumes potentially causing adverse variances.

9.13 KEY WORDS

1. **Variance Analysis:** A method used to identify the difference between actual and standard costs to analyze performance and control inefficiencies.
2. **Favorable Variance:** Occurs when actual costs are lower than standard costs, resulting in increased profits.
3. **Adverse Variance:** A variance where actual costs exceed standard costs, leading to decreased profits and inefficiencies.
4. **Material Price Variance:** The difference between the standard price of materials and the actual price paid for them.
5. **Labor Efficiency Variance:** The discrepancy between the standard labor hours allowed for production and the actual hours worked.
6. **Overhead Variance:** The difference between actual overhead costs and the standard overheads allocated to production.
7. **Standard Overhead Rate:** A pre-determined rate used to allocate overheads based on estimated costs and capacity utilization.
8. **Expenditure Variance:** The variance arising from the difference between actual and budgeted overhead expenditures.
9. **Volume Variance:** The difference between overheads absorbed for actual output and those budgeted, indicating over- or under-utilization.
10. **Sales Price Variance:** The effect of deviations in actual selling price compared to the standard price, impacting sales revenue.

9.14 CLOSING CASELET

Sales Variance at Radha Enterprises

Radha Enterprises, a company selling consumer electronics, set a budget to sell 10,000 units of a popular smart phone model at a standard price of ₹20,000 per unit, expecting total revenue of ₹2,00,00,000. However, due to changes in market conditions and competition, the company was able to sell only 9,000 units at an average price of ₹19,500. This led to variances in both the sales price and the sales volume, which impacted overall performance.

The Sales Value Variance for Radha Enterprises is calculated as the difference between the budgeted and actual sales revenue. Using the

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formula Budgeted Sales – Actual Sales, the variance is ₹2,00,00,000 – ₹1,75,50,000 = –₹24,50,000. This negative variance of ₹24,50,000 indicates that the company's actual sales fell short of its budgeted target.

Breaking it down, the Sales Price Variance shows the impact of the price change on the sales value. The formula (Standard Price – Actual Price) × Actual Quantity Sold gives (₹20,000 – ₹19,500) × 9,000 = ₹500 × 9,000 = –₹4,50,000. This adverse variance of ₹4,50,000 reflects the fact that the company had to sell its smartphones at a lower-than-expected price due to increased competition and price pressure.

The Sales Volume Variance explains the remaining ₹20,00,000 of the total variance. Using the formula Standard Price × (Budgeted Quantity – Actual Quantity), the variance is ₹20,000 × (10,000 – 9,000) = ₹20,000 × 1,000 = ₹20,00,000. This unfavorable variance indicates that Radha Enterprises sold 1,000 fewer units than expected, primarily due to a decline in demand and competitive market forces.

To improve its sales performance, Radha Enterprises should evaluate pricing strategies and consider promotional campaigns to boost sales volume. Additionally, the company may explore product differentiation to reduce price competition and improve margins in the long run.



QUESTIONS

1. How did Radha Enterprises' Sales Value Variance of ₹24,50,000 arise, and what does this figure suggest about the company's performance?

(Hint: Review the formula for Sales Value Variance and focus on the difference between budgeted and actual sales.)

2. What factors contributed to Radha Enterprises' unfavorable Sales Price Variance of ₹4,50,000, and how can the company address this issue in the future?

(Hint: Consider the role of competition and market conditions in pricing strategies, and think about solutions such as product differentiation.)

9.15 HIGHER ORDER THINKING SKILLS QUESTIONS (HOTS)

1. How can variance analysis be used not only to identify inefficiencies but also to predict future trends in production costs?
2. In what ways could adverse variances be a strategic investment rather than a sign of inefficiency?
3. How might a company use labor efficiency variance data to redesign its workflow for both short-term gains and long-term sustainability?

4. How can understanding overhead variances, such as expenditure and volume variances, assist management in making more informed decisions regarding cost control and resource allocation in production?
5. In what ways can analyzing standard overhead rates and variances contribute to identifying inefficiencies in a company's production process, and how can this analysis influence future budgeting and capacity planning strategies?
6. The standard material cost to produce one unit of Product Y is as follows:

- 200 kg of Material P at ₹12 per kg
- 300 kg of Material Q at ₹8 per kg
- 400 kg of Material R at ₹10 per kg

During a particular period, 150 units of Product Y were produced using the following quantities and prices:

- 32,000 kg of Material P at ₹13,500 per ton
- 46,000 kg of Material Q at ₹9,000 per ton
- 60,000 kg of Material R at ₹11,000 per ton

Required:

1. Calculate the Material Cost Variance.
 2. Calculate the Material Price Variance.
 3. Calculate the Material Usage Variance.
 4. Calculate the Material Mix Variance.
7. The following was the composition of a gang of workers in a factory during a particular month in one of the production departments.

Standard Composition of Workers and Wage Rates

- Skilled: 3 workers at a standard rate of ₹25 per hour each
- Semi-Skilled: 3 workers at a standard rate of ₹15 per hour each
- Unskilled: 2 workers at a standard rate of ₹10 per hour each

The standard output of the gang was 5 units per hour of the product.

During a specific month, the actual composition of the gang and hourly wage rates paid were as follows:

- Skilled: 2 workers at ₹28 per hour
- Semi-Skilled: 4 workers at ₹16 per hour
- Unskilled: 2 workers at ₹12 per hour

The gang was engaged for 180 hours, which included 10 hours of downtime due to machine breakdowns.

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The gang produced 840 units of the product during the month.

Calculate the following labour variances:

1. Labor Cost Variance (LCV)
 2. Labor Rate Variance (LRV)
 3. Labor Efficiency Variance (LEV)
8. The following information is available from the records of a manufacturing company that uses a standard costing system.

Particulars	Standard	Actual
Production	5,000 units	4,800 units
Working days	25 days	26 days
Fixed overheads	₹50,000	₹52,000
Variable overheads	₹15,000	₹16,200

Required: Calculate the following overhead variances

1. Variable Overhead Variance
 2. Fixed Overhead Cost Variance
 3. Fixed Overhead Expenditure Variance
 4. Fixed Overhead Volume Variance
 5. Fixed Overhead Efficiency Variance
 6. Fixed Overhead Calendar Variance
9. Given Information:

Product	Budgeted Quantity (Units)	Actual Quantity (Units)	Budgeted Sales Price (Rs.)	Actual Sales Price (Rs.)	Standard Cost (Rs.)
A	300	350	60	55	35
B	200	250	40	38	25

Compute the following variances:

- I. Total Sales Margin Variance
- II. Sales Margin Volume Variance
- III. Sales Margin Price Variance

9.16 ANSWER KEYS

Topic	Q. No.	Answer
Variance Analysis- Introduction	1	a. Actual costs exceed standard costs, leading to a decrease in profit.
	2	a. ₹160 (Adverse)
Labor Cost Variances	3	a. ₹2,000 (Adverse)
	4	c. Hiring workers of a different grade than specified.

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Topic	Q. No.	Answer
Overhead Variances	5	d. Changes in variable overhead expenditure.
	6	b. ₹5,000 (Adverse)
Sales Variances	7	a. ₹4,000 (Adverse)
	8	a. Rs. 10,500 (Favorable)
Control Ratios	9	c. Better utilization of capacity than planned
	10	c. Efficiency Ratio

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10

U N I T

MANAGEMENT ACCOUNTING IN GLOBAL PERSPECTIVE

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INTRODUCTORY CASELET

UPS and Activity-Based Management (ABM)

United Parcel Service (UPS) faced increasing operational costs and growing customer demands for faster, more affordable delivery services. With heightened competition from companies like FedEx and e-commerce platforms, UPS needed a solution to control costs while maintaining its service quality. To address this challenge, UPS adopted Activity-Based Management (ABM), which helped the company analyze its cost drivers and streamline operations.

Using ABM, UPS identified and assigned costs to various activities in its logistics network, such as sorting, transportation, and last-mile delivery. This allowed the company to pinpoint which activities were consuming the most resources and focus on reducing non-value-added activities like inefficient delivery routes. ABM helped UPS optimize its processes by understanding the cost implications of each activity.

One major outcome of ABM implementation was the development of UPS's ORION system, an algorithm that optimized delivery routes to cut fuel consumption and driving time. By using ABM insights, UPS was able to significantly reduce unnecessary miles driven and fuel usage, leading to substantial cost savings and environmental benefits.

As a result of these efforts, UPS not only lowered its operational costs but also improved delivery efficiency. The company maintained its competitive pricing and continued to meet customer expectations for reliable, on-time deliveries. ABM enabled UPS to balance cost control, service quality, and sustainability goals, proving its effectiveness in a large, complex organization.

**QUESTIONS**

1. How did Activity-Based Management (ABM) help UPS improve its operational efficiency and cost control?

(Hint: Think about how ABM helped UPS identify non-value-added activities and optimize its logistics processes.)

2. What role did the ORION system play in UPS's implementation of ABM, and how did it contribute to cost savings?

(Hint: Consider how ABM data informed the development of ORION and how optimizing delivery routes reduced expenses.)

**LEARNING OBJECTIVES**

After reading this unit, you will be able to:

- Analyze how management accounting practices can influence strategic decision-making in competitive environments.
- Evaluate the impact of globalization on management accounting practices and techniques
- Explore the implications of e-business on management accounting practices and reporting.
- Assess the impact of global competition on pricing strategies and cost management practices.
- Examine the influence of product life cycles on cost management and pricing strategies.
- Evaluate the integration of Total Quality Management (TQM) principles with management accounting practices.
- Assess the features and benefits of modern cost management systems in today's business environment.

10.1 INTRODUCTION

Imagine you're the manager of a company navigating a competitive market where every decision counts. How do you ensure your company stays ahead? This is where **management accounting** comes in, acting as your compass for strategic decision-making and sustainable growth.

Picture this: Your company is launching a new product. To price it competitively, you need to understand production costs, supplier expenses, and expected profit margins. Management accounting helps you analyze these factors, enabling you to make informed decisions. It goes beyond traditional accounting by incorporating modern tools and methodologies tailored to today's globalized business environment.

Now, think of the shift towards **developed costing systems**. Imagine using a system that not only tracks expenses but also provides insights into the efficiency of each production stage. These systems factor in e-business models, the growing service sector, and the forces of global competition. For example, as global demand fluctuates, these systems help you adapt your strategy and stay ahead of competitors.

One of the most exciting aspects of modern management accounting is its integration of **cross-functional collaboration**. Imagine a scenario where your finance, production, and marketing teams come together to analyze the profitability of a new product. Management accounting facilitates this collaboration by offering insights into product life cycles, inventory management, and total quality management. This means every department is aligned, reducing waste and boosting efficiency.

Consider the power of **modern cost management systems**. Techniques like **Activity-Based Costing (ABC)** help you identify which activities

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add value and which do not. Imagine you're running a manufacturing unit where one process takes twice as long as expected. By applying ABC, you spot the bottleneck and address it, improving productivity. Similarly, **Lean Accounting** helps identify waste, streamline operations, and cut unnecessary costs, ensuring the organization stays competitive in a rapidly changing market.

Now, step into the shoes of a professional aiming to master management accounting. By understanding these advanced concepts, you're not just tracking numbers—you're influencing strategy. You're equipped to tackle the challenges of a globalized world, from managing product diversity to optimizing total quality management processes.

10.2 ACTIVITY AS A FOCUS

Activity-based management (ABM) is a managerial approach that leverages activity-based costing (ABC) data to serve various objectives, such as cost control, modeling, and analyzing customer profitability. Essentially, ABM applies insights derived from ABC analysis to enhance cost management. The main components of ABM include:

1. **Emphasis on Activities:** ABM centers on the role of activities in business processes, distinguishing between value-added and non-value-added activities. Value-added activities contribute to delivering customer satisfaction, while non-value-added activities represent inefficiencies or waste that should ideally be eliminated.
2. **Management Focus:** ABM encourages a continuous improvement mindset, supporting decision-making that revolves around activities. Collaboration is essential in identifying cost pools, cost drivers, and performance indicators to better manage costs and boost efficiency.
3. **Key Outputs of the ABM System:**
 - ❑ **Activity Costs:** ABM delivers critical cost insights tied to the core activities of the business, helping managers understand what drives costs.
 - ❑ **Non-Value-Added Activity Costs:** It identifies wasteful activities, giving management a focal point for eliminating inefficiencies.
 - ❑ **Activity-Based Performance Indicators:** ABM involves not only evaluating activity costs but also incorporating metrics like quality, cycle time, productivity, and customer service, which are essential for driving continuous improvement.
 - ❑ **Accurate Costing of Products/Services:** By accurately allocating costs to products and services, ABM ensures proper recognition that different offerings consume resources at varying rates.
 - ❑ **Cost Drivers:** ABM provides detailed information on cost drivers, helping management control and understand what influences activity levels.

4. Strategic Decision-Making with ABM: ABM supports strategic decisions regarding whether to maintain specific activities, compare cost structures with competitors, and assess how changes in activities impact suppliers and the broader value chain.

5. Benefits of ABM:

- ❑ Enhances communication and teamwork across the organization.
- ❑ Promotes informed, high-quality decision-making.
- ❑ Encourages continuous improvement and quality control.
- ❑ While ABM might not directly lead to cost reduction, it enhances management's understanding of cost behavior, aiding in better resource optimization.



SELF-ASSESSMENT QUESTIONS

1. Which of the following is a key component of Activity-Based Management (ABM)?
 - a. Focusing only on product costs
 - b. Reducing workforce size to cut costs
 - c. Allocating overheads based on traditional costing methods
 - d. Identifying and managing activities, distinguishing between value-added and non-value-added activities
2. What is the primary benefit of Activity-Based Management (ABM) in strategic decision-making?
 - a. Immediate reduction in operational costs
 - b. Simplification of accounting processes
 - c. Enhancing communication, promoting informed decision-making, and continuous improvement
 - d. Eliminating the need for cost drivers in cost management

10.3 CHANGE IN FOCUS

The focus in management accounting has progressively moved away from activity-based methods like Activity-Based Costing (ABC) toward more comprehensive and strategic approaches. Key shifts in focus include:

- 1. From Activity-Based Costing to Comprehensive Cost Management:** While ABC was once a key method for allocating costs, the focus has broadened to encompass more extensive cost management strategies. These strategies consider various elements, such as market trends, value chain analysis, and overall resource allocation, rather than just examining the costs of individual activities.

Example: Procter & Gamble (P&G)

While P&G initially relied on ABC to allocate costs across its wide range of products, the company has shifted to broader cost

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management strategies. By analyzing its value chain, P&G optimizes resource allocation across R&D, marketing, and production, balancing cost efficiency with innovation and market trends. This strategic approach allows P&G to identify profitable product lines and streamline less productive ones.

2. **From Isolated Activities to Holistic Process Management:** The focus has shifted from managing the costs and performance of individual activities to overseeing entire business processes. This wider approach promotes optimizing workflows across departments, improving efficiency, and reducing bottlenecks through more integrated methods.

Example: Toyota Production System (TPS)

Toyota moved beyond activity-specific costing to adopt Lean Manufacturing principles, focusing on entire processes. For instance, its “Just-In-Time” inventory system minimizes waste by synchronizing production with demand. Instead of tracking individual costs, Toyota evaluates workflows across departments, ensuring efficiency throughout the manufacturing process.

3. **From Activity-Level Metrics to Strategic Performance Indicators:** Management accounting has transitioned from tracking activity costs and processes to focusing on key performance indicators (KPIs) that align with the organization’s strategic objectives. These KPIs include both financial and non-financial metrics, such as customer satisfaction, market share, and innovation, to provide a more well-rounded view of performance.

Example: Amazon

Amazon has evolved its management accounting practices by tracking strategic KPIs like customer satisfaction (measured through Prime membership retention), delivery efficiency, and technological innovation. These metrics help Amazon align operational costs with its goal of providing fast, reliable services and dominating market share.

4. **From Internal Activity Control to External Market Considerations:** Previously, the emphasis was on controlling internal activities to manage costs. Now, the focus has expanded to include external factors like market dynamics, customer demands, and supplier relationships, enhancing decision-making by considering external and internal elements.

Example: Apple Inc.

Apple’s management accounting incorporates external factors like market trends and customer preferences. By investing in supplier relationships and monitoring customer demand for premium devices, Apple balances cost management with strategic market positioning, enabling it to set higher prices while maintaining a loyal customer base.

5. **From Short-Term Activity Management to Long-Term Value Creation:** The shift has moved from managing activities with a

short-term operational outlook to focusing on long-term strategic objectives. Modern management accounting prioritizes sustainable growth, innovation, and value creation over short-term cost reduction achieved through activity management.

Example: Tesla, Inc.

Tesla focuses on long-term value creation by prioritizing sustainability and innovation over short-term cost reductions. For instance, investments in gigafactories and battery technology are expensive upfront but align with Tesla's long-term goals of reducing production costs and expanding electric vehicle adoption.

10.4 MANAGEMENT ACCOUNTING IN A COMPETITIVE WORLD

Imagine being part of a business navigating a world of constant change. From intensified global competition to rapidly evolving technologies, companies today face unprecedented challenges. Picture your business dealing with **shorter product lifecycles**—like smartphones that need constant upgrades—or adapting to new **environmental regulations** that demand eco-friendly production processes. Each shift in this dynamic environment influences how businesses operate, prompting significant changes in **management accounting practices**.

Now, imagine you're in a strategy meeting. The discussion centers on how to stay ahead in this competitive landscape. Someone suggests focusing on customer satisfaction. You realize they're right—customers now hold more power than ever. This shift forces businesses to become **customer-driven**, prioritizing satisfaction and identifying the **key success factors** required to thrive. This is where **management accounting** steps in as a powerful decision-making tool. As your team assesses goals in a **VUCA (volatile, uncertain, complex, and ambiguous)** world, management accounting provides the insights needed to make strategic choices.

Consider this: your company wants to launch a new product. Instead of guessing market demand, you turn to management accounting. By analyzing customer trends, evaluating competitors' moves, and predicting future market shifts, you gain the strategic edge. No longer limited to **cost control**, management accounting now focuses on gathering crucial data for evaluating **market trends, pricing strategies, and investment opportunities**—all of which position your company ahead of competitors.

Fast forward to a long-term planning session. Your company's leadership discusses how to stay relevant in the next five years. They need to align company capabilities with emerging opportunities. But how do they identify which opportunities to pursue? This is where management accounting's role in supporting **long-term strategy** becomes evident. Imagine having access to a roadmap showing which products, markets, and investments are most likely to yield success. **Strategy** isn't just a buzzword; it's a plan that aligns a company's resources with its future goals. Management accounting helps break down this plan, tracking progress and ensuring actions remain aligned with the company's vision.

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Companies typically follow one of two broad strategies:

Cost Leadership Strategy	Product Differentiation Strategy
Objective: Deliver quality at the lowest cost.	Objective: Offer unique, high-value products or services.
Focus Areas: Efficiency, cost management, competitive pricing.	Focus Areas: Innovation, branding, premium pricing.
Examples: Budget airlines like Ryanair.	Examples: Luxury brands like Tesla and Apple.

Managers are tasked with choosing between these strategies, a decision that has long-term implications for a company's profitability and growth. Management accountants collaborate with managers across departments to help formulate strategies by providing valuable information on the sources of competitive advantage. This may include:

- ❑ The company's cost, productivity, or efficiency advantages over competitors.
- ❑ The ability to charge premium prices due to the unique features that make its products or services stand out.

The best-designed strategies and the strongest capabilities are ineffective without proper execution, which depends heavily on the information provided by management accountants. The connection between the successful implementation of strategy and the information generated by management accounting is a key aspect of strategic cost management. Management accountants play a pivotal role in providing insights that help execute strategies, ensuring that the company remains competitive and responsive to changes in the business environment.

Comparison between traditional and modern role of management accounting:

Aspect	Traditional Role	Modern Role
Primary Objective	Focused on cost control and operational efficiency.	Emphasizes strategic decision-making , linking financial insights to overall business goals.
Scope of Analysis	Limited to internal cost analysis to manage production and operational costs.	Broadens to include market trend evaluation and external economic factors influencing business.
Time Horizon	Predominantly short-term, emphasizing immediate gains like reducing waste or cutting costs.	Long-term perspective, ensuring alignment with the business's strategic goals and sustainability.
Tools and Techniques	Relied on traditional tools like budgetary control and variance analysis.	Incorporates modern tools such as predictive analytics, activity-based costing (ABC), and balanced scorecards.

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Aspect	Traditional Role	Modern Role
Decision-Making Role	Supportive, primarily providing cost data to assist operational managers.	Proactive, playing a key role in strategic planning , mergers, investments, and competitive positioning.
Interaction with Departments	Minimal collaboration, confined to financial or production teams.	Cross-functional involvement, integrating insights from marketing, operations, and human resources for holistic decision-making.
Role of Technology	Limited use of technology; manual processes for data gathering and reporting.	Heavy reliance on technology such as ERP systems, AI, and big data analytics for real-time and accurate information.



SELF-ASSESSMENT QUESTIONS

3. Which of the following strategies focuses on offering quality products at the lowest possible cost to ensure long-term sustainability?
 - a. Product Differentiation Strategy
 - b. Cost Leadership Strategy
 - c. Premium Pricing Strategy
 - d. Market Penetration Strategy
4. How do management accountants support strategic decision-making in a highly competitive business environment?
 - a. By only focusing on cost control measures
 - b. By providing information about market trends, cost advantages, and strategic pricing
 - c. By managing day-to-day cash flows
 - d. By eliminating all non-value-added activities

10.5 MANAGEMENT ACCOUNTING AND GLOBAL ENVIRONMENT

Management Accounting in the global environment refers to the role management accounting plays in helping businesses navigate challenges and opportunities in an interconnected world. With globalization, companies must adapt to various market conditions, regulations, cultural differences, and economic factors.

Globalization, first proposed by Theodore Levitt in 1983, refers to the convergence of markets worldwide. The International Monetary Fund (IMF) defines it as the growing economic interdependence of countries through increasing cross-border transactions in goods and services, free international capital flows, and widespread diffusion of technology. These three factors—free trade, capital mobility, and technological diffusion—are considered the pillars of globalization.

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As globalization advanced, companies began operating across national borders, leading to a shift in the role of management accountants. Traditional management accounting, which was adequate in earlier stages, did not provide the strategic financial information necessary for globalized business environments. This shift marked the fourth stage in the evolution of management accounting, focusing more on strategic analysis.

Criticism of traditional management accounting practices arose, with many practices having been developed by 1925 and a stagnation in innovation for nearly 60 years. This period, referred to as the “period of lost relevance,” highlighted the need for modernization in management accounting practices. In response, the Chartered Institute of Management Accountants (CIMA) identified strategic management accounting as a key area for development. Strategic management accounting aims to provide the financial analysis necessary for competitive strategy formulation and better support business decision-making in the global era.

Globalization has thus significantly impacted the business environment, requiring management accountants to adopt a more strategic, market-focused approach, reporting on both internal financials and external factors like market conditions and competitor activities.

Key Interactions between Management Accounting and the Global Environment

Aspect	Traditional Role	Modern Challenges/Role	Impact
Globalization and Decision-Making	Focused on domestic markets and internal operations.	Manages global supply chains, diverse regulations, and currency fluctuations.	Incorporates cross-border cost comparisons, tax planning, and risk management for foreign investments.
Diverse Regulatory Environments	Local regulatory compliance.	Ensures adherence to international accounting standards (e.g., IFRS). Strategizes for tax structuring and transfer pricing.	Maintains compliance while optimizing tax efficiency across regions.
Technological Advancements	Limited to local data systems.	Utilizes blockchain, cloud computing, and AI for real-time, global data sharing and analysis.	Enables predictive analytics and integrates multinational financial data for informed decisions.

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Aspect	Traditional Role	Modern Challenges/Role	Impact
Sustainability and ESG	Focus on profit maximization.	Develops ESG metrics (e.g., carbon footprint, resource usage) and aligns them with financial goals.	Ensures sustainable practices while achieving profitability.
Currency Risks and Economic Volatility	Limited exposure to currency fluctuations.	Manages foreign exchange risks and conducts scenario analysis for economic volatility.	Mitigates financial risks associated with global markets.
Cultural and Workforce Diversity	Uniform financial strategies.	Encourages cultural sensitivity and adapts localized metrics for performance evaluation.	Aligns financial strategies with regional operations and diverse teams.
Competition and Strategic Planning	Analyzed local competitors.	Conducts global competitive benchmarking and market analysis.	Identifies competitive advantages through cost and profitability analysis in diverse markets.



SELF-ASSESSMENT QUESTIONS

5. Which of the following factors is NOT considered one of the pillars of globalization as defined by the IMF?
 - a. Free trade
 - b. Capital mobility
 - c. Technological diffusion
 - d. Taxation policies
6. What was the primary criticism of traditional management accounting practices in the era of globalization?
 - a. They focused too much on short-term cost control
 - b. They did not provide strategic financial information for global business environments
 - c. They overemphasized the role of international trade
 - d. They were developed with an excessive focus on modern technology

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10.6 MANAGEMENT ACCOUNTING AND DEVELOPED COSTING SYSTEMS

Management accounting has evolved alongside the development of more sophisticated costing systems to meet the demands of complex business environments. Developed costing systems are essential for accurately tracking costs, improving financial performance, and supporting strategic decision-making. These systems help management accountants allocate costs more precisely and provide insights that enable better control over operations.

10.6.1 E-BUSINESS

Think of e-business as the engine running your car. It's not the destination (sales), but it ensures everything works smoothly to get there. Unlike **e-commerce**, which focuses on buying and selling online, **e-business** enhances the internal workings of a company by using online tools to improve processes like inventory management, procurement, and supply chain optimization.

How Does E-Business Work?

Imagine you run a small handmade jewelry shop. Right now, you're keeping track of inventory with pen and paper, calling suppliers manually, and struggling with piles of invoices. Here's how e-business can transform your operations:

- ❑ **Online Inventory Management:** Know in real time how many gold chains you have in stock.
- ❑ **Automated Procurement:** Receive alerts when supplies are low and place orders instantly online.
- ❑ **Supplier Performance Tracking:** Monitor how quickly suppliers deliver and ensure you're working with the best.
- ❑ **Streamlined Accounting:** Generate invoices and manage payments automatically.

These changes won't generate revenue directly, but they save time, reduce errors, and help you focus on creating value for your customers.

Where to Start?

If you're a small business struggling with inefficiencies, look for areas where paperwork slows you down. For example:

- ❑ Replace paper order forms with an online ordering system.
- ❑ Use an intranet to share updates with employees.
- ❑ Automate restocking with a supplier extranet.

Why E-Business Matters for Small Businesses

E-business spans the entire business cycle—production, sales, fulfillment, and beyond. It helps build relationships with customers, employees,

suppliers, and even external partners like banks and accountants. For example, you can use e-business tools to send marketing emails, restock inventory faster, or simplify payments for customers.

Components of E-Business

E-business encompasses several core components: business intelligence (BI), customer relationship management (CRM), supply chain management (SCM), enterprise resource planning (ERP), e-commerce, conducting electronic transactions, and collaboration among businesses.

- ❑ **Business intelligence (BI)** refers to activities where a small business collects, stores, accesses, and analyzes market or competitive information to aid decision-making. Online BI processes are efficient, providing timely data to identify trends and make quicker, better decisions.
- ❑ **Customer relationship management (CRM)** focuses on building long-term customer relationships that add value for both the customer and the business. CRM consolidates data from various sources within and sometimes outside the organization, providing a holistic, real-time view of each customer. This strategy aims to increase profitability while lowering costs and improving customer satisfaction.
- ❑ **Supply chain management (SCM)** handles the network of vendors that supply the raw materials needed for production or service delivery. SCM focuses on streamlining this process to improve efficiency and ensure smooth delivery from raw materials to final product.
- ❑ **Enterprise resource planning (ERP)** integrates all departments and functions across an organization into a single system, enabling quick and efficient access to data. ERP systems help small businesses optimize their operations by streamlining information flow between departments.
- ❑ **E-commerce** differs from e-business in that it generates revenue through the marketing, selling, and buying of goods and services online. It is often linked with e-marketing but also includes the operational aspects of conducting business online.
- ❑ **Electronic transactions** within a company can be facilitated by intranets, email, and instant messaging. Intranets are private networks used for sharing, processing, and communicating information within an organization, aiming to streamline workflows.
- ❑ **Collaboration** can be internal or external, often involving business partners. Tools such as email, instant messaging, virtual team rooms, and wikis help teams or partners collaborate more efficiently, reducing travel costs and enhancing productivity.
- ❑ **Online business activities** between companies focus on communication and information sharing through extranets, which are parts of an intranet accessible to external partners. Extranets allow for secure, 24/7 access to collaborative tools, increasing productivity and speeding up response times.

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Although e-business is integral for many, small businesses like local shops and delis may choose not to adopt e-business practices, depending on their scale and operations. However, as the Internet continues to transform businesses, it is becoming increasingly difficult for small businesses to operate entirely outside the scope of e-business.

CASE STUDY: AMAZON - E-BUSINESS AND ITS IMPACT ON COST MANAGEMENT ACCOUNTING

Background

Amazon, a global leader in e-commerce, has revolutionized the retail industry with its technology-driven business model. Originally starting as an online bookseller, Amazon has expanded into various sectors, offering products ranging from consumer goods to cloud computing services through Amazon Web Services (AWS). This transition has had a profound impact on the company's cost management accounting practices. The adoption of e-business has fundamentally altered the way Amazon manages its costs and allocates resources, providing a glimpse into how digital platforms shape modern accounting systems.

Impact of E-Business on Cost Management Accounting

1. **Cost Structure Transformation:** Traditionally, cost management in retail was focused on physical store expenses, inventory management, and labor costs. However, Amazon's e-business model has drastically transformed its cost structure.
 - ❑ **Elimination of Physical Store Costs:** Amazon no longer incurs costs related to maintaining physical stores, which are typical for traditional retailers.
 - ❑ **Reduction in Labor Costs:** Automation, such as the use of robotic warehouses, has significantly lowered labor costs.
 - ❑ **Introduction of New Costs:** While physical store costs have been eliminated, Amazon has new expenses related to maintaining its website, digital advertising, and infrastructure for AWS.
 - ❑ **Sophisticated Cost Allocation:** The need for sophisticated allocation methods has arisen, as the company now needs to allocate costs accurately across its vast digital sales channels.
2. **Real-Time Cost Analysis:** Managing costs for millions of transactions globally presents a major challenge. Amazon has addressed this through advanced analytics and real-time cost tracking.
 - ❑ **Shipping and Inventory Management:** By tracking shipping costs, inventory holding costs, and customer acquisition costs in real-time, Amazon ensures optimal operational efficiency.
 - ❑ **Demand Forecasting:** Algorithms predict demand and help minimize excess inventory, reducing storage costs significantly.
3. **Activity-Based Costing (ABC):** Amazon employs Activity-Based Costing (ABC) to allocate costs based on specific activities such as order processing, packaging, and delivery.

- ❑ **Prime vs. Non-Prime Costs:** A notable example of ABC is the differentiation of costs for Prime members, who receive free shipping, compared to non-Prime customers.
- ❑ **Pricing Strategies:** This cost allocation method provides precise data that helps Amazon develop effective pricing strategies, ensuring profitability across different product categories.
- 4. Logistics and Distribution Costs:** The complexity of e-business operations also extends to logistics and distribution, with costs associated with last-mile delivery, reverse logistics (returns), and global distribution centers.
 - ❑ **In-House Logistics Network:** By using its own logistics network, Amazon Logistics, the company reduces its dependency on third-party carriers, lowering transportation costs.
 - ❑ **Warehouse Placement:** Cost management practices are enhanced by analyzing and determining the most efficient placement of warehouses globally to minimize shipping times and associated expenses.
- 5. Technology-Driven Overheads:** Amazon Web Services (AWS), a significant revenue driver for Amazon, requires substantial investments in data centers, servers, and energy to operate.
 - ❑ **Allocation of Overheads:** Management accountants at Amazon allocate overhead costs, such as data center cooling and server maintenance, to cloud service clients using advanced cost drivers like server usage hours or bandwidth consumption. This helps ensure that the costs of running AWS are accurately attributed to its services.
- 6. Data-Driven Decision-Making:** Amazon's e-business model heavily relies on big data and artificial intelligence (AI) to make data-driven decisions regarding pricing, inventory management, and promotions.
 - ❑ **Cost-Saving Opportunities:** Predictive analytics help identify cost-saving opportunities in inventory restocking by analyzing historical trends and demand patterns.
 - ❑ **Dynamic Pricing:** Amazon uses dynamic pricing to adjust product prices based on competitor activity, demand fluctuations, and associated costs. This ensures that prices remain competitive while maintaining profitability.

Results of E-Business Transformation

- 1. Lower Operating Costs:** The integration of automation and data-driven insights has allowed Amazon to achieve economies of scale and minimize inefficiencies across its operations. By automating processes such as order fulfillment and inventory management, Amazon has significantly reduced operating costs.
- 2. Enhanced Cost Transparency:** With the use of advanced cost tracking systems, Amazon has gained a more granular

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understanding of its spending. This transparency enables better control over costs and more accurate decision-making.

3. **Global Scalability:** Amazon's e-business model supports rapid, cost-effective expansion into international markets. Its digital infrastructure, supported by AWS and integrated logistics systems, allows the company to scale operations across borders without incurring prohibitive costs.
4. **Competitive Advantage:** Efficient cost management has given Amazon a competitive edge, enabling the company to offer lower prices and superior services compared to traditional retailers. The company's ability to manage costs effectively reinforces its market dominance and strengthens its position as a global e-commerce leader.

10.6.2 INCREASED ROLE OF SERVICE SECTOR

The growing prominence of the service sector has significantly influenced management accounting practices. Historically, management accounting was largely focused on manufacturing industries, where managing costs, controlling inventory, and optimizing production efficiency were central priorities. However, with the expansion of the service sector, which now contributes substantially to global economies, management accounting has adapted to meet the specific needs and challenges of service-oriented businesses.

Key areas of transformation include:

1. **Intangible Costs and Value Creation:** Unlike manufacturing, where tangible products are produced, the service sector primarily deals with intangible outputs such as knowledge, expertise, and customer experiences. Management accounting in this sector places greater emphasis on tracking and analyzing the costs tied to these intangible services, such as labor, technology, and client relationships, while also focusing on how value is delivered to customers.
2. **Customer-Focused Metrics:** The service industry is largely driven by customer demand, requiring management accountants to develop metrics that are more customer-centric. These include measures of customer satisfaction, retention, service quality, and lifetime value. The objective is to strengthen customer relationships and ensure long-term profitability, as customer experience is a critical factor for success in service industries.
3. **Activity-Based Costing (ABC):** Service organizations often utilize activity-based costing to better allocate costs to specific activities or processes that impact service delivery. This method helps identify value-adding activities and those that do not, allowing businesses to improve the efficiency and cost-effectiveness of their services.
4. **Labor and Time Management:** Labor frequently represents the largest cost driver in service industries. Consequently, management accountants focus on managing labor costs, time, and productivity. This includes tracking employee performance, measuring service

delivery times, and optimizing staffing levels to enhance operational efficiency while maintaining high service quality.

5. **Performance Metrics and KPIs:** Key performance indicators (KPIs) in service-oriented businesses extend beyond financial measures to include non-financial metrics such as customer feedback, service efficiency, employee satisfaction, and innovation. Management accounting systems must incorporate these broader performance metrics to offer a comprehensive view of business performance.
6. **Service Profit Chain:** Management accountants play a key role in analyzing the “service profit chain,” which links employee satisfaction and productivity to customer loyalty and profitability. By examining data on employee engagement, training, and service delivery, management accountants help refine operations to improve both employee and customer outcomes.
7. **Sustainability and Social Responsibility:** With increasing attention on corporate social responsibility (CSR) and sustainability in the service sector, management accounting now also focuses on monitoring environmental and social performance. Service businesses, particularly in sectors like healthcare, education, and hospitality, are often under scrutiny for their societal impact, making sustainability reporting and cost analysis critical components of management accounting.

CASE STUDY: STARBUCKS - THE IMPACT OF THE SERVICE SECTOR ON MANAGEMENT ACCOUNTING PRACTICES

Background

Starbucks, a global leader in the coffeehouse industry, exemplifies the growing influence of the service sector on management accounting. The company is renowned for its focus on providing an exceptional customer experience alongside high-quality products. This service-oriented business model has reshaped traditional management accounting practices, highlighting the unique challenges and opportunities faced by businesses in the service sector.

Impact of the Service Sector on Management Accounting

1. **Shift from Product-Centric to Customer-Centric Costing:** In traditional manufacturing settings, management accounting typically focuses on direct costs such as materials, labor, and overhead associated with producing physical products. However, Starbucks has shifted this focus to prioritize customer experience as a key cost driver.
 - ❑ **Service Focus:** For Starbucks, costs are centered around elements that enhance customer experience, including store ambiance, employee training, and the use of premium ingredients.
 - ❑ **Cost Tracking:** Management accountants at Starbucks track and evaluate these indirect costs to assess their impact on customer satisfaction and retention, ensuring that investments in service quality are appropriately managed.

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2. **Emphasis on Non-Financial Metrics:** In the service industry, financial measures alone cannot fully capture the quality of customer service or customer satisfaction. Starbucks has adapted by integrating non-financial metrics into its management accounting processes.
 - ❑ **Key Performance Indicators (KPIs):** Starbucks uses KPIs such as customer satisfaction scores, employee engagement levels, and average transaction time to measure service performance.
 - ❑ **Holistic Approach:** These non-financial metrics are integrated with traditional financial performance indicators, providing a more comprehensive view of the company's overall success and ensuring that customer satisfaction is prioritized alongside profitability.
3. **Activity-Based Costing (ABC) in Service Operations:** Starbucks employs Activity-Based Costing (ABC) to allocate costs associated with store operations. ABC allows the company to assign costs to specific activities, providing deeper insights into operational efficiency.
 - ❑ **Store Operations Analysis:** For instance, Starbucks distinguishes costs between high-traffic urban locations and smaller suburban stores. Activities such as brewing coffee, preparing food, and maintaining store ambiance are analyzed to identify cost drivers.
 - ❑ **Actionable Insights:** This detailed analysis helps the company develop more accurate pricing strategies and identify cost-saving opportunities, tailored to the needs of different store formats.
4. **Dynamic Pricing and Cost-Volume-Profit (CVP) Analysis:** Pricing in the service sector is complex, especially for businesses like Starbucks, where demand patterns fluctuate based on location, time of day, and customer demographics. Starbucks employs dynamic pricing models and CVP analysis to address these complexities.
 - ❑ **Dynamic Pricing:** During peak hours or special events, Starbucks uses dynamic pricing models to optimize revenue, adjusting prices based on demand.
 - ❑ **Cost-Volume-Profit Analysis:** CVP analysis helps Starbucks forecast the profitability of new product launches, such as seasonal holiday beverages, by analyzing fixed and variable costs against expected sales volumes.
5. **Human Capital as a Key Cost Driver:** In the service industry, employee-customer interaction is a critical factor in delivering value. Starbucks views its employees as essential to the customer experience, and therefore, human capital is a major cost driver in its management accounting system.
 - ❑ **Employee Investment:** Starbucks invests heavily in its employees, offering benefits like the "Starbucks College Achievement Plan" and other development programs.

- ❑ **Tracking ROI:** Management accountants evaluate the return on investment (ROI) for these programs, analyzing their impact on employee satisfaction and retention, as well as their overall effect on customer service quality.

6. Environmental and Sustainability Costs: As a global service business, Starbucks is increasingly accountable for its environmental impact. The company has integrated sustainability considerations into its management accounting practices, aligning them with broader ethical and business goals.

- ❑ **Tracking Sustainability Costs:** Starbucks tracks costs related to sourcing ethically produced coffee beans, reducing single-use plastics, and other sustainability initiatives.
- ❑ **Long-Term Brand Value:** These sustainability metrics are tied to long-term brand value and customer loyalty, ensuring that ethical practices contribute not only to environmental goals but also to the company's financial success.

Results of Service-Oriented Management Accounting Practices

- 1. Improved Decision-Making:** By integrating both financial and non-financial data, Starbucks has enhanced its ability to make informed decisions on store expansions, product offerings, and promotional campaigns. The combination of customer satisfaction data, cost analysis, and financial metrics allows for more effective and strategic decision-making.
- 2. Enhanced Customer Focus:** Starbucks' management accounting practices prioritize service quality, ensuring consistent customer satisfaction across its global network of stores. By aligning cost management with customer experience, the company fosters long-term brand loyalty and enhances its competitive position in the marketplace.
- 3. Sustainable Profitability:** Starbucks has successfully aligned its financial performance with ethical and sustainable business practices. Management accounting ensures that sustainability initiatives, such as environmentally friendly sourcing and waste reduction, contribute to long-term profitability while maintaining customer trust.
- 4. Operational Efficiency:** The use of ABC and CVP analysis has allowed Starbucks to optimize store operations, from pricing strategies to inventory management. These tools help Starbucks manage costs efficiently, ensuring that its service offerings remain both high-quality and cost-effective.

10.6.3 INCREASED GLOBAL COMPETITION

Increased global competition has had a profound impact on management accounting, compelling it to evolve in response to the demands of a more interconnected marketplace. As companies expand globally and encounter competitors worldwide, management accounting practices must adapt

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to meet these new challenges. The role of management accountants is more critical than ever, as they help businesses enhance cost management, measure performance, and make informed strategic decisions.

1. **Cost Efficiency and Pricing in a Competitive Landscape:** Global competition forces companies to improve cost efficiency and adopt competitive pricing strategies. Management accountants play a key role in identifying cost reduction opportunities without affecting product quality. Advanced costing methods such as **Activity-Based Costing (ABC)** and **Target Costing** have become essential in ensuring competitive pricing while protecting profit margins.
 - ❑ **ABC** accurately traces overhead costs to specific activities, enabling better decision-making on pricing and cost control.
 - ❑ **Target Costing** is particularly relevant in markets where prices are dictated by customer expectations. Accountants use this method to work backward from the target price, ensuring production costs remain manageable.
2. **Global Benchmarking and Best Practices:** With competition arising from different regions, companies must constantly compare their performance with global benchmarks. Management accountants facilitate this process by collecting and analyzing data on competitors' financial and operational performance.
 - ❑ **Benchmarking** helps businesses compare key performance indicators (KPIs) and cost structures, uncovering areas for improvement.
 - ❑ By adopting global best practices, companies can enhance efficiency, improve customer service, and maintain competitiveness on an international scale.
3. **Innovation and R&D Cost Management:** In an environment of global competition, continuous innovation is key to staying ahead. Research and development (R&D) costs must be carefully managed, and management accountants are tasked with tracking and assessing these costs to ensure innovation leads to profitability.
 - ❑ Accountants provide financial assessments of R&D projects, analyzing potential returns and ensuring alignment with market trends and consumer demands, often shaped by global competition.
4. **Strategic Management Accounting:** With increased competition, management accounting has shifted from its traditional focus on cost control to a more strategic role. **Strategic Management Accounting (SMA)** emphasizes both financial and non-financial data to support the development of competitive strategies.
 - ❑ **SMA** considers external factors such as market dynamics, competitor behavior, and customer preferences, providing insights that help businesses adapt to global changes.

- ❑ Management accountants also evaluate the profitability of market segments, regions, and products, ensuring a focus on areas where the company holds a competitive advantage.
- 5. **Risk Management and Currency Volatility:** Operating globally exposes companies to risks like currency fluctuations, political instability, and regulatory shifts. Management accountants are vital in identifying and managing these risks.
 - ❑ **Currency risk management** is essential, as exchange rate volatility can affect costs and revenues. Accountants assist in implementing hedging strategies and forecasting the impact of currency changes on profitability.
 - ❑ Management accountants also evaluate risks related to supply chains, tariffs, and global trade regulations that could affect operations across borders.
- 6. **Sustainability and Ethical Standards:** Global competition increases the importance of corporate responsibility, including sustainability and ethical practices. Companies must comply with international **Environmental, Social, and Governance (ESG)** standards, which management accountants help monitor and report.
 - ❑ Management accountants track metrics such as carbon emissions and energy usage, ensuring adherence to global environmental regulations and positioning companies as responsible corporate citizens.
 - ❑ Businesses that prioritize sustainability often gain a competitive edge in global markets, where ethical practices are increasingly valued by consumers and investors.
- 7. **Global Supply Chain Cost Management:** Global supply chains add complexity, requiring precise cost management to maintain competitiveness. Management accountants assist in optimizing supply chain costs, evaluating supplier performance, and managing inventory across different regions.
 - ❑ **Supply chain costing** analyzes logistics, procurement, and production expenses across multiple locations, helping businesses identify the most efficient and cost-effective solutions.
 - ❑ Managing working capital is crucial, as companies must balance cash flow with the need for inventory control in long global supply chains.
- 8. **Technological Advances and Their Impact:** Advances in technology, including automation, artificial intelligence (AI), and data analytics, have transformed management accounting in response to global competition. Businesses now have access to vast amounts of data that can be analyzed to support better decision-making.
 - ❑ Management accountants leverage **data analytics** to gain insights into market trends, customer preferences, and operational efficiency, helping companies respond swiftly to global shifts.

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- ❑ **Automation** streamlines routine accounting tasks, freeing management accountants to focus on strategic activities like forecasting and long-term financial planning.

9. Global Performance Measurement and KPIs: Increased global competition necessitates more advanced performance measurement systems. Management accountants track **Key Performance Indicators (KPIs)** that reflect business performance across multiple regions, accounting for currency fluctuations, local market conditions, and diverse regulatory environments.

- ❑ These KPIs are aligned with the company's global strategy, helping managers assess whether financial and operational targets are being met on an international level.

10.6.4 STRESS ON CROSS-FUNCTIONAL GROUPS

Cross-functional groups bring together members from different departments, including finance, marketing, operations, R&D, and HR. By working collaboratively, these groups enhance decision quality, streamline processes, and promote innovation, all of which are vital for achieving business success. For instance: Imagine a company called Shyam Electronics that plans to launch a new line of smart home devices. To ensure a successful launch, the company forms a cross-functional group with representatives from different departments:

1. **R&D Team:** Designs and develops the product's technical features.
2. **Marketing Team:** Develops a strategy to promote the product, identify the target audience, and plan advertising campaigns.
3. **Finance Team:** Estimates the production cost, sets the budget, and forecasts potential revenue.
4. **Operations Team:** Ensures the production process is efficient and cost-effective.
5. **HR Team:** Manages the recruitment of skilled labor and provides training for employees working on the project.

By collaborating, each department contributes its specialized knowledge to the project. For instance, the R&D team might design a cutting-edge feature, but without input from marketing, the feature may not align with customer preferences. Similarly, the operations team ensures production feasibility, while finance ensures that the project stays within budget. The combined efforts result in a high-quality product that meets market demand, is produced efficiently, and is launched successfully.

For management accountants, cross-functional collaboration offers numerous benefits, including:

1. **Improved Decision-Making:** Collaboration across functions enhances decision-making by integrating diverse viewpoints. In management accounting, relying solely on financial data may not provide a complete understanding of the situation. Engaging with other teams allows management accountants to incorporate operational

insights, market trends, and customer preferences into the decision-making process.

- ❑ For instance, when creating a **budget** or **forecast**, management accountants collaborate with marketing to gauge customer demand, with operations to assess production capabilities, and with R&D to account for innovation costs.
- ❑ This comprehensive approach leads to better resource allocation and more accurate performance projections, ensuring that decisions are well-informed and aligned with broader business goals.

2. Enhanced Cost Control and Efficiency: Cross-functional collaboration helps identify cost-saving opportunities and areas for operational efficiency. Management accountants working closely with other departments can gain deeper insights into cost drivers and inefficiencies, which allows for more effective cost management.

- ❑ In **Activity-Based Costing (ABC)**, management accountants team up with operations and engineering departments to understand the activities driving overhead costs, leading to more accurate cost allocation and better cost control.
- ❑ This type of collaboration not only improves operational efficiency but also helps to cut down on unnecessary expenses, which is crucial in competitive markets.

3. Alignment of Financial and Operational Goals: Cross-functional collaboration is key to ensuring that financial goals are aligned with operational priorities. One of the challenges businesses face is the misalignment between these objectives.

- ❑ By working together, management accountants and operations teams can ensure that **performance metrics** and **key performance indicators (KPIs)** are aligned with both financial and operational objectives. This promotes a unified approach to achieving strategic goals and avoids conflicts between departments.
- ❑ For example, while finance might prioritize profit maximization, operations may focus on production efficiency. Cross-functional collaboration helps balance these goals for sustainable business growth.

4. Strategic Management and Competitive Advantage: In an increasingly global and competitive market, companies must constantly innovate and adapt. Cross-functional teams help management accountants contribute to **Strategic Management Accounting (SMA)** by providing financial insights that consider external market and competitive dynamics.

- ❑ Management accountants collaborate with marketing, R&D, and operations to analyze external factors like competitor behavior, market trends, and customer preferences. This information helps them offer financial analysis that supports competitive strategy formulation.

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- ❑ By working with marketing, for example, management accountants may discover shifting customer preferences, prompting adjustments to pricing strategies or profitability evaluations for different market segments.
- 5. **Product Development and Innovation:** Cross-functional teams are critical to driving innovation, especially in areas such as product development. Management accountants collaborate with R&D and marketing to assess the financial feasibility of new products, monitor development costs, and project potential returns.
 - ❑ By partnering with these teams, management accountants ensure that **cost management** aligns with market expectations and profitability goals, ensuring that innovation efforts are both technically sound and financially viable.
 - ❑ This collaboration helps businesses maintain innovation while also keeping an eye on long-term financial sustainability.
- 6. **Risk Management:** Cross-functional groups contribute to a comprehensive view of organizational risks. Management accountants, by working alongside other departments like operations and legal, can identify and manage risks that may not be apparent from a financial perspective alone.
 - ❑ In managing **supply chain risks**, for example, management accountants can collaborate with procurement and operations to assess the financial implications of disruptions like supplier delays or currency fluctuations, developing contingency plans to mitigate these risks.
 - ❑ This collaborative risk management approach ensures the organization is well-prepared to address global business challenges.
- 7. **Improved Communication and Accountability:** Cross-functional collaboration also fosters better communication and accountability throughout the organization. When management accountants collaborate with other teams, they promote a culture of transparency and shared responsibility for achieving financial and operational goals.
 - ❑ Cross-functional discussions enable management accountants to explain financial implications more clearly to non-financial managers, ensuring everyone is aligned towards common objectives.
 - ❑ Improved communication reduces the risk of misunderstandings and silos, enhancing the organization's ability to execute strategies effectively.
- 8. **Sustainability and Corporate Social Responsibility (CSR):** Sustainability and CSR initiatives have become important aspects of modern business strategy, and cross-functional collaboration is vital for implementing these programs effectively. Management accountants work with environmental, social, and operational teams to assess the financial impact of sustainability efforts.

- ❑ Management accountants track the costs and savings associated with initiatives like energy efficiency and waste reduction, ensuring these programs align with financial performance goals.
- ❑ Collaborating across departments ensures sustainability is integrated into the company's overall strategy, positioning the business as a responsible corporate citizen while maintaining competitiveness.

10.6.5 PRODUCT LIFE CYCLES AND DIVERSITY

The **product life cycle** refers to the various stages a product goes through, from its introduction to its decline. These stages—**introduction**, **growth**, **maturity**, and **decline** (shown in Figure 10.1)—affect how costs and revenues are managed.

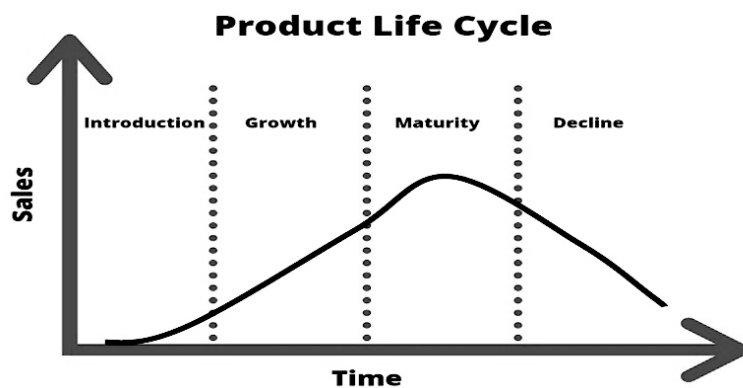


Figure 10.1 Product Life Cycle.

Management accounting helps in monitoring and adjusting strategies as a product moves through these stages:

1. **Introduction Stage:** In this phase, costs are typically high due to research and development, marketing, and production setup. Revenues may be low as the product is new and market awareness is limited. Management accounting provides information on initial costs and helps in pricing strategies that may focus on covering costs or penetrating the market to build a customer base.
2. **Growth Stage:** As the product gains market acceptance, sales increase, and economies of scale start to lower unit costs. Marketing efforts continue to drive awareness, but the focus shifts towards optimizing production and distribution. Management accounting tracks profitability and operational efficiency, allowing businesses to allocate resources effectively, manage cash flow, and optimize production processes.
3. **Maturity Stage:** At this point, sales growth slows down, and competition increases. Companies may experience price pressures, which affect profit margins. Management accounting becomes essential for cost control and margin analysis. Businesses need to focus on maintaining efficiency, possibly reducing costs, and exploring strategies like product differentiation to sustain profitability.

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4. **Decline Stage:** In this final stage, sales decrease, and companies may need to decide whether to discontinue the product or reinvent it. Management accounting helps in determining the profitability of maintaining the product and provides data to support decisions regarding product discontinuation or reallocation of resources to more profitable ventures.

Throughout the product life cycle, management accountants provide critical information for pricing decisions, cost management, and capital investment to align with the product's current stage.

Example of a New Energy Drink

Introduction Stage: The company launches its energy drink with innovative packaging and health claims. Costs are high due to research on the formula, designing the packaging, and promotional campaigns to create market awareness. Revenue is minimal as the product is new, and customers are just beginning to notice it. Management accounting tracks these expenses and helps the company decide on a pricing strategy—either setting a low price to attract customers quickly or a premium price to position the drink as a high-value product.

Growth Stage: The energy drink gains popularity among fitness enthusiasts and young professionals, leading to a sharp increase in sales. Bulk production reduces per-unit costs, while marketing focuses on expanding distribution to gyms and convenience stores. Management accounting ensures the company efficiently allocates resources, monitors profitability, and optimizes production processes to meet growing demand. It might also highlight the need for expanding production capacity or improving supply chain efficiency.

Maturity Stage: As competitors introduce similar energy drinks, the market becomes saturated. Sales growth slows, and price competition squeezes profit margins. The company relies on management accounting to analyze costs and maintain profitability through measures like improving production efficiency or offering discounts. Additionally, strategies like introducing new flavors or bundling the drink with other products are explored to sustain market share.

Decline Stage: Sales drop as the market shifts to newer trends, such as plant-based or organic beverages. The company uses management accounting to decide whether to discontinue the energy drink or reposition it as a budget-friendly option. If profitability is unsustainable, resources may be reallocated to more promising products. Alternatively, the drink could be reformulated to meet changing customer preferences, keeping it relevant in the market.

Product diversity refers to the range of products or services a company offers. It poses unique challenges for management accounting due to the variations in production costs, market demands, and pricing strategies across different products. Companies with a diverse product portfolio require more sophisticated cost allocation and management strategies.

1. **Cost Allocation:** When a company produces multiple products, it faces the challenge of allocating shared or indirect costs (such as

overhead, utilities, or management costs) accurately across different products. Management accounting utilizes techniques like **activity-based costing (ABC)** to allocate these costs based on the actual consumption of resources by each product.

2. **Product Profitability Analysis:** Not all products contribute equally to a company's profitability. Management accounting helps analyze the profitability of each product or product line, considering factors like cost structure, pricing, and sales volume. This analysis enables businesses to focus on the most profitable products and decide whether to discontinue or reposition underperforming ones.
3. **Pricing Strategies:** Diverse product lines may require different pricing strategies depending on factors such as market demand, competition, and production costs. Management accountants provide data to support pricing decisions that maximize profitability while considering customer willingness to pay, competitive pricing, and cost constraints.
4. **Investment Decisions:** Product diversity often leads to varying levels of investment in research, development, and marketing. Management accounting helps prioritize investments by providing insights into the expected returns from different product lines, ensuring that resources are allocated to the most promising and profitable products.
5. **Risk Management:** A diverse product portfolio spreads risk across different markets and product categories. Management accounting supports this by analyzing how changes in market conditions, raw material prices, or consumer preferences affect the various products, helping companies manage risk and ensure long-term profitability.

Example: A company producing a range of **kitchen appliances**—blenders, ovens, coffee makers, and slow cookers—faces unique challenges because of its product diversity. Shared costs like factory overhead, electricity, and marketing need to be divided among these products. Management accountants use activity-based costing (ABC) to allocate these costs more accurately. For example, blenders might consume more assembly labor due to their intricate design, while ovens incur higher utility costs during testing. This ensures each product is assigned costs based on actual resource usage.

By conducting a **profitability analysis**, the company learns that coffee makers yield the highest profit margin because they are cheaper to produce and enjoy strong market demand. On the other hand, slow cookers have low sales and high storage costs, making them less profitable. Based on this information, management decides to increase marketing efforts for coffee makers and scale back production of slow cookers, reallocating resources to more profitable lines.

Additionally, management accounting supports pricing decisions. For instance, blenders, being premium items, are priced higher to

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reflect their production complexity, while coffee makers, aimed at mass appeal, follow a competitive pricing strategy. These insights help the company optimize profits while managing the risks of a diverse product portfolio.

CASE STUDY: APPLE INC. - PRODUCT LIFE CYCLES, DIVERSITY, AND MANAGEMENT ACCOUNTING

Background

Apple Inc., a global technology leader, demonstrates how management accounting adapts to the challenges of product life cycles and a diverse product portfolio. With offerings ranging from iPhones and MacBooks to services like iCloud and Apple Music, Apple operates in varied markets, each with different customer demands and competitive pressures. This diversity, along with the rapid evolution of product life cycles in the tech industry, has a significant impact on Apple's management accounting practices.

Impact of Product Life Cycles on Management Accounting

1. **Understanding Product Life Cycle Stages:** Apple's products go through four life cycle stages: Introduction, Growth, Maturity, and Decline. For example, iPhones incur high R&D costs in the Introduction phase, achieve peak revenues during Growth, and experience price reductions in Maturity and Decline to stay competitive.
 - ❑ **Management Accounting Response:** Apple tracks and allocates costs at each stage to guide pricing, marketing, and resource allocation. Target Costing is used during the early stages to ensure profitability by aligning product design with cost objectives.
2. **Cost Management Across Life Cycle Stages**
 - ❑ **Introduction Phase:** High R&D and marketing costs are closely monitored through detailed budgets.
 - ❑ **Growth Phase:** Apple focuses on increasing production and distribution efficiencies while maintaining margins.
 - ❑ **Maturity Phase:** Price competition and reduced margins lead Apple to focus on cost control and value engineering.
 - ❑ **Decline Phase:** Management accounting aids in decisions to discontinue or reposition products, minimizing losses.

Impact of Diversity in Products on Management Accounting

1. **Diverse Product Portfolio:** Apple's product lines, including hardware, software, and services, each have distinct cost structures and revenue models. A challenge arises in allocating shared costs, such as R&D, across these diverse offerings.
 - ❑ **Management Accounting Response:** Apple uses Activity-Based Costing (ABC) to allocate costs based on actual resource consumption by each product line, such as differentiating R&D costs between iPhones and Apple Watches.

2. **Pricing Strategies:** Apple uses premium pricing for flagship products, like the iPhone, and bundle pricing for services, such as Apple One.
 - ❑ **Management Accounting Response:** Apple uses Cost-Volume-Profit (CVP) Analysis to assess the profitability of various pricing strategies. It also tracks customer lifetime value for subscription services to ensure long-term profitability.
3. **Sustainability and Product Diversity:** Apple emphasizes sustainability across its product lines, using recycled materials in devices. A challenge is quantifying and managing the costs of these sustainable practices.
 - ❑ **Management Accounting Response:** Apple develops Sustainability Accounting Metrics to track environmental and social costs, such as carbon footprint tracking for MacBooks and iPads.

Integrating Product Life Cycles and Diversity

1. **Decision-Making Tools:** Apple uses Life Cycle Costing (LCC) to evaluate total costs over a product's lifespan, such as assessing the profitability of long-term software updates for older iPhones.
2. **Innovation Investments:** Apple heavily invests in innovation to maintain its competitive edge.
 - ❑ **Management Accounting Response:** Apple tracks the ROI of R&D investments to prioritize high-potential projects, such as evaluating the success of AirPods as a new product category.
3. **Strategic Product Mix Decisions:** Apple balances its portfolio by managing high-growth products (e.g., Apple Watch) alongside cash cows (e.g., iPhones).
 - ❑ **Management Accounting Response:** Portfolio Analysis helps Apple allocate resources optimally across its product lines.

Results of Apple's Management Accounting Adaptations

- ❑ **Enhanced Profitability Across Life Cycles:** Apple sustains high margins by aligning cost management with the product life cycle stages.
- ❑ **Efficient Resource Allocation:** ABC and LCC ensure accurate cost allocation across diverse products, preventing over- or under-investment.
- ❑ **Informed Strategic Decisions:** Management accounting supports Apple in making decisions about which products to expand, maintain, or phase out.
- ❑ **Sustainability Integration:** Incorporating sustainability metrics enhances Apple's brand image and meets regulatory requirements.

10.6.6 FOCUS ON INVENTORY MANAGEMENT

Inventory management is a vital aspect of management accounting, guiding businesses in determining the optimal amount of stock to order and maintain at any given time. By tracking inventory from purchase to sale, it ensures that companies have sufficient stock to meet customer demand while avoiding overstocking, which can tie up valuable cash.

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Effective inventory management allows businesses to monitor stock levels, recognize trends, and make informed decisions that enhance operational efficiency.

Key Functions of Inventory Management

- ❑ **Tracking Inventory Levels:** Inventory management allows businesses to monitor their stock throughout its lifecycle, from procurement to sale. By keeping track of stock levels, companies can ensure they have the right amount of inventory to meet customer orders, reducing the risk of stock outs or overstocking. Example: A retail store starts the month with 500 units of a product in stock. During the month:

- ◆ **Purchases:** 300 units are bought at ₹800/unit.

- ◆ **Sales:** 700 units are sold at ₹1,200/unit.

- ❑ By tracking inventory, the store calculates its closing stock:

$$\begin{aligned} &\text{Opening Stock (500) + Purchases (300) – Sales (700)} \\ &= \text{Closing Stock (100 units).} \end{aligned}$$

The value of the closing stock is:

$$100 \text{ units} \times ₹800/\text{unit} = ₹80,000.$$

This ensures the store avoids both overstocking and stockouts by matching inventory levels to demand.

Inventory as an Asset: While inventory is recorded as an asset on the balance sheet, it represents capital tied up and unavailable for use elsewhere until sold. Effective inventory management helps maintain a balance between available stock and liquidity, preventing excessive stockpiling that can impact cash flow. Example: The company's balance sheet shows inventory as an asset:

$$\text{Inventory Value: } 100 \text{ units} \times ₹800/\text{unit} = ₹80,000.$$

This ₹80,000 is tied up capital, unavailable for use elsewhere until the inventory is sold. Effective inventory management ensures a balance between available stock and liquidity, avoiding excessive stockpiling.

- ❑ **Inventory Turnover:** A key metric in inventory management is inventory turnover, which measures how frequently a company sells and replaces its stock over a given period. A high inventory turnover ratio indicates efficient management, while a low ratio may suggest the risk of dead stock (unsold or obsolete inventory). By evaluating this metric, management accountants can provide insights on purchasing, production planning, and pricing strategies. Example: Inventory turnover is calculated as: $\text{Inventory Turnover Ratio} = \text{Cost of Goods Sold (COGS)} / \text{Average Inventory}$

- ◆ **COGS:** 700 units sold $\times ₹800/\text{unit} = ₹5,60,000.$

- ◆ **Average Inventory:** $(\text{Opening Stock} + \text{Closing Stock})/2 = (500 + 100)/2 = 300 \text{ units}$

- ◆ Value of Average Inventory: $300 \text{ units} \times ₹800/\text{unit} = ₹2,40,000.$

- ◆ Inventory Turnover Ratio $= ₹5,60,000 / ₹2,40,000 = 2.33$

An inventory turnover of 2.33 indicates the company sells and replaces its inventory about twice in the period. If this ratio were lower, it could suggest overstocking or slow-moving inventory requiring corrective action.

Importance of Inventory Management

Effective inventory management is crucial to maintaining a business's overall health by ensuring optimal stock levels. It helps minimize the risks of stock outs (insufficient stock to meet demand) and overstocking (excess inventory that ties up resources). Benefits of effective inventory Management

- ❑ **Cost Savings:** Understanding stock trends enables businesses to minimize excess inventory and reduce storage costs. By accurately forecasting demand and managing stock levels, companies can reduce the risk of unsold or obsolete inventory, improving profitability.
- ❑ **Improved Cash Flow:** Proper inventory management ensures that businesses invest in stock that sells, leading to better cash flow management. When cash isn't tied up in excess inventory, it can be reinvested in other areas of the business.
- ❑ **Customer Satisfaction:** Effective inventory management enhances customer satisfaction by ensuring products are always available when needed. This reliability helps foster customer loyalty and boosts the company's reputation.

Challenges in Inventory Management

Despite its advantages, inventory management presents several challenges:

- ❑ **Overstocking or Stock outs:** Striking the right balance between too much and too little inventory is one of the primary challenges. Overstocking results in dead stock and reduced cash flow, while stock-outs lead to missed sales opportunities and unhappy customers.
- ❑ **Inaccurate Stock Data:** Without accurate stock level data, businesses struggle to determine when to replenish inventory or which products are performing well. This lack of information leads to inefficiencies and poor decision-making.
- ❑ **Outdated Systems:** Manual or outdated inventory management processes can introduce errors and slow down operations. Automation and modern inventory management software help resolve these issues by providing real-time updates and reducing human error.
- ❑ **Changing Customer Demand:** Customer preferences evolve over time, and without effective tracking systems, businesses may struggle to adjust stock levels accordingly. This can result in either excess inventory of unpopular items or stockouts of high-demand products.
- ❑ **Warehouse Space Utilization:** Inefficient use of warehouse space can complicate the process of locating products, leading to delays in fulfilling customer orders. Proper inventory management optimizes warehouse operations by organizing stock to maximize efficiency.

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10.6.7 TOTAL QUALITY MANAGEMENT CONCEPT

Total Quality Management (TQM) and management accounting share a common goal—enhancing organizational efficiency, refining processes, and driving superior outcomes. By embedding TQM principles into management accounting, organizations can ensure the accuracy, relevance, and timeliness of financial data, thereby enabling better decision-making and improved overall performance.

Key Aspects of TQM in Management Accounting

- 1. Customer Focus** Imagine the “customers” of management accounting—they’re not just external stakeholders like investors and regulators but also internal users such as managers and department heads. TQM emphasizes delivering high-quality financial information that meets these users’ needs. Accountants play a pivotal role in ensuring that reports and performance metrics are precise, timely, and aligned with the specific objectives of these internal and external “customers.”
- 2. Continuous Improvement** Consider how an organization strives for ongoing refinement in its processes. Management accountants contribute by identifying areas for improvement in financial operations. Techniques like variance analysis and benchmarking are applied to pinpoint inefficiencies, enhance data accuracy, and streamline processes for better cost allocation and budgeting. The result? Leaner, more efficient financial workflows.
- 3. Employee Involvement** TQM thrives on employee engagement, and this extends to finance teams. Imagine a cross-functional team of accountants working with production, marketing, and supply chain departments. Their collaboration can streamline reporting, improve cost control, and ensure that performance metrics are aligned with overall business goals. This team effort fuels a shared commitment to continuous quality enhancement.
- 4. Process-Oriented Approach** Picture how a process-oriented approach comes to life in management accounting. By tracking costs, revenues, and financial performance across key business processes, accountants can spot inefficiencies in areas like production and supply chain management. This approach helps businesses refine their processes, reduce waste, and boost profitability.
- 5. Fact-Based Decision Making** Imagine a management team needing to decide on a new investment. They’ll need accurate financial insights. TQM’s emphasis on data-driven decisions means management accountants provide precise financial forecasts and analysis. Tools like variance analysis, cost-volume-profit analysis, and activity-based costing equip managers with the data required to make sound, evidence-based choices.
- 6. Strategic Integration** Think of an organization’s broader quality goals aligning with its financial strategy. Under TQM, financial strategies aren’t isolated—they’re integrated with quality improvement initiatives. Management accountants play a key role in ensuring budgets, performance metrics, and resource allocation all

support continuous quality improvement, creating a cohesive and unified organizational strategy.

TQM and Cost Management

Imagine a company aiming to reduce costs while maintaining quality. TQM's focus on process enhancement and waste elimination dovetails perfectly with management accounting's cost management objectives. Techniques like activity-based costing (ABC) help identify cost drivers and reveal inefficiencies. By addressing these cost drivers, organizations can cut expenses associated with poor quality, such as rework, scrap, and warranty claims.

Performance Measurement and TQM

How do organizations track the success of TQM initiatives? The answer lies in effective performance measurement. Management accountants provide tools like key performance indicators (KPIs), cost variances, and balanced scorecards to monitor the financial and non-financial impact of TQM efforts. For instance, a balanced scorecard goes beyond financial outcomes to assess process efficiency, quality, and customer satisfaction—ensuring a holistic view of performance.

Role of Management Accountants in TQM

- 1. Providing Cost Data for Quality Decisions** Picture a company deciding whether to invest in new technology or process improvements. Accountants supply financial insights on the costs and benefits of these quality initiatives, helping decision-makers evaluate the return on investment and make well-informed choices.
- 2. Supporting Quality Improvement Initiatives** Imagine an operations team working on a quality improvement project. Management accountants support this initiative by analyzing quality control costs, tracking savings from process enhancements, and ensuring that quality goals are met within budget constraints. Their financial expertise ensures that the initiative stays on track, both financially and operationally.
- 3. Evaluating Supplier and Customer Relationships** Consider the long-term value of working with high-quality suppliers. Management accountants analyze the financial impact of these relationships. They assess how partnerships with reliable suppliers affect production costs, customer satisfaction, and profitability. Their insights support sustainable supplier partnerships that benefit the company's bottom line.
- 4. Facilitating Cross-Departmental Collaboration** Picture an organization where finance, production, and marketing departments work together toward shared quality goals. Management accountants facilitate this collaboration by providing clear, actionable financial data. Their insights support initiatives to reduce waste, improve production efficiency, and enhance customer satisfaction—all while ensuring that performance metrics are aligned with TQM objectives.

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SELF-ASSESSMENT QUESTIONS

7. What is the primary focus of e-business compared to e-commerce?
 - a. E-business primarily generates revenue through online transactions.
 - b. E-business enhances operational efficiency and value for customers without necessarily generating revenue.
 - c. E-business focuses solely on marketing and selling goods and services online.
 - d. E-business does not involve any stakeholder interactions.
8. Which of the following best describes the shift in management accounting due to increased global competition?
 - a. Management accounting has become less relevant in a global context.
 - b. Management accountants now focus only on financial data without considering external factors.
 - c. Management accounting practices have adapted to support competitive pricing, risk management, and strategic decision-making.
 - d. Management accountants primarily concentrate on traditional cost control without addressing global dynamics.

10.7 MODERN COST MANAGEMENT SYSTEMS

Modern cost management systems have advanced considerably to cater to the intricate demands of today's dynamic business landscape. These systems integrate innovative techniques, technologies, and methodologies to deliver more accurate cost information, support strategic decision-making, and enhance operational efficiency. The move from traditional cost management methods to modern systems highlights an increased emphasis on improving profitability, boosting competitiveness, and aligning cost control with broader organizational objectives. Modern Cost Management Systems:

1. **Activity-Based Costing (ABC):** ABC is a key modern cost management approach that assigns overhead costs based on the actual activities driving those costs, rather than relying on generalized averages. By identifying the specific cost drivers within a business, ABC offers a clearer picture of resource consumption, aiding businesses in improving cost control, pricing strategies, and profitability analysis.

Example: In a company that manufactures furniture, overhead costs like electricity and machine maintenance can be spread across multiple products. However, ABC identifies that the cost of machine maintenance is primarily driven by the complexity of the design, not just the quantity of furniture produced. For instance, a complicated design for a luxury chair may require more frequent machine maintenance than a simpler design for a basic chair. By using ABC, the company can assign the appropriate share of maintenance costs to each product based on the actual use of the machinery, offering

a more accurate cost structure that can improve pricing strategies and profitability analysis.

2. **Lean Accounting:** Lean accounting supports organizations that adopt lean manufacturing principles by simplifying financial reporting and emphasizing value stream costing. It focuses on reducing waste, optimizing workflow, and delivering value to the customer. This approach utilizes both financial and non-financial data to track the performance of lean processes, promoting continuous improvement and cost reduction.

Example: A car manufacturer adopts lean manufacturing principles to streamline its production. They focus on eliminating waste by optimizing the assembly line, reducing the time spent on non-value-added activities, and improving employee efficiency. Lean accounting tracks these improvements through both financial and non-financial metrics, such as cost savings from reduced waste and faster production times, as well as employee satisfaction and defect rates. This approach enables the company to continuously reduce costs, improve quality, and deliver value to customers, all while simplifying financial reporting.

3. **Target Costing:** Target costing entails determining a product's price based on market conditions, then designing the product to meet that price while maintaining profitability. This method ensures effective cost management from the product development stage and is particularly popular in industries where market-driven pricing is critical, such as automotive and electronics.

Example: In the consumer electronics industry, a smartphone company faces intense price competition. The company determines that the maximum price point it can charge for a new model is Rs.40,000 based on market research. To ensure profitability, they design the phone to meet this price by carefully selecting materials, manufacturing processes, and design features that allow them to stay within a target cost of Rs. 30,000, leaving Rs.10,000 as a profit margin. Target costing ensures that the design and production processes are aligned with market-driven pricing from the very beginning.

4. **Just-in-Time (JIT) Costing:** JIT costing aligns with inventory management strategies that aim to minimize inventory levels and cut carrying costs. By producing goods only when needed, businesses can reduce waste, prevent overproduction, and lower storage costs. JIT costing is particularly effective in manufacturing sectors where efficiency and cost control are vital for profitability.

Example: A toy manufacturer produces action figures but faces high inventory holding costs. By implementing JIT costing, the company produces action figures only when they receive an order, which helps eliminate the need to store excess inventory. This minimizes storage costs and reduces the risk of overproduction. For instance, if a particular toy is in high demand during the holiday season, JIT enables the company to adjust production rates quickly to match the market demand, lowering waste and excess inventory costs.

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- 5. Kaizen Costing:** Kaizen costing focuses on ongoing cost reduction throughout a product's life cycle. Unlike target costing, which sets cost limits before production, kaizen costing operates during production, promoting incremental cost reductions through process improvements and employee involvement. It favors consistent, small improvements over drastic changes.

Example: A packaging company uses kaizen costing to continuously reduce costs during production. Employees are encouraged to propose small improvements, such as adjusting the layout of the production line for better workflow or fine-tuning machinery to use less energy. Each small change helps lower overall production costs, and the cumulative effect over time results in substantial cost savings. Unlike target costing, which sets a fixed cost goal before production, kaizen costing focuses on constant, incremental improvements throughout the product's life cycle.

- 6. Life Cycle Costing:** Life cycle costing (LCC) examines the total cost of owning a product, covering not only initial production costs but also maintenance, operational, and disposal costs over the product's lifespan. This holistic view helps businesses make informed decisions about product development, pricing, and resource allocation.

Example: A company that manufactures washing machines looks at the total cost of ownership over the life of the product, not just the production costs. LCC takes into account the cost of repairs, energy consumption, and disposal when calculating the overall cost of a washing machine. This helps the company set the right price for the product and decide whether to offer extended warranties or promote energy-efficient features that appeal to cost-conscious consumers, all while maintaining profitability throughout the product's lifespan.

- 7. Benchmarking:** Benchmarking involves comparing a company's cost performance against industry standards or best practices. This method enables organizations to identify gaps in their cost structures, assess their performance relative to competitors, and adopt cost management strategies that foster continuous improvement.

Example: A retail chain compares its supply chain costs with those of a leading competitor, using industry standards for shipping, warehousing, and inventory management. Through benchmarking, the company identifies that its shipping costs are 15% higher than the industry average. By implementing strategies that align with best practices, such as optimizing its delivery routes or negotiating better shipping rates, the company can reduce its logistics costs and improve overall profitability.

- 8. Integrated Financial and Non-Financial Data:** Modern cost management systems merge financial data with non-financial metrics like production statistics, customer satisfaction, and quality indicators. This comprehensive view enables businesses to understand the broader effects of their cost structures on overall performance, beyond financial outcomes alone.

Example: A hotel chain uses an integrated cost management system that combines financial data, such as revenue and operating costs, with non-financial metrics, such as guest satisfaction scores and room occupancy rates. By combining these data points, the hotel chain gains a deeper understanding of how customer experience impacts profitability. For example, it may find that a slight increase in room rates during peak seasons does not reduce occupancy if guest satisfaction remains high, leading to better decision-making regarding pricing strategies and customer service investments.

9. **Digital Transformation and Automation:** With technological advancements, modern cost management systems leverage tools such as enterprise resource planning (ERP) systems, artificial intelligence (AI), machine learning, and big data analytics. These tools automate processes, improve data accuracy, and provide real-time cost information, supporting quicker decision-making, predictive cost analysis, and better resource management.

Example: A manufacturing plant adopts an enterprise resource planning (ERP) system powered by artificial intelligence (AI) and big data analytics. The system automatically tracks production costs in real-time, predicts potential cost overruns, and suggests adjustments to improve efficiency. For instance, if the AI detects a sudden increase in raw material prices, it can alert the finance team, allowing them to adjust pricing strategies or look for alternative suppliers, ultimately improving cost management and resource allocation.

Benefits of Modern Cost Management Systems

- ❑ **Improved Accuracy:** Techniques like ABC and life cycle costing yield more precise cost information, aiding businesses in making well-informed decisions.
- ❑ **Enhanced Decision-Making:** These systems offer in-depth insights into cost drivers, enabling management to make strategic decisions related to pricing, product design, and resource allocation.
- ❑ **Better Cost Control:** Modern systems emphasize ongoing improvement and waste reduction, helping businesses manage costs effectively without compromising quality or efficiency.
- ❑ **Increased Competitiveness:** By aligning costs with market conditions and customer expectations, businesses can remain agile and competitive in fast-changing environments.
- ❑ **Greater Transparency:** Integrating financial and non-financial data offers a clearer understanding of cost impacts across the value chain, boosting transparency and accountability.

Challenges of Modern Cost Management Systems

Despite the benefits, modern cost management systems present certain challenges:

- ❑ **Implementation Complexity:** Establishing systems such as ABC, lean accounting, or target costing requires significant effort in terms of training, process redesign, and system integration.

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- ❑ **Data Requirements:** These systems need comprehensive and accurate data from various sources, which can be difficult to collect and maintain without appropriate technology.
- ❑ **Cultural Change:** Implementing continuous improvement methods, such as kaizen costing, may require a cultural shift within the organization, encouraging employees to embrace incremental changes and innovation.



SELF-ASSESSMENT QUESTIONS

9. What is the primary focus of Activity-Based Costing (ABC) in modern cost management systems?
 - a. Simplifying financial reporting and emphasizing value stream costing
 - b. Assigning overhead costs based on actual activities driving those costs
 - c. Minimizing inventory levels and carrying costs
 - d. Reducing waste through ongoing incremental improvements
10. Which of the following best describes the concept of Target Costing?
 - a. Setting cost limits before production based on historical data
 - b. Determining a product's price based on market conditions and designing to meet that price while maintaining profitability
 - c. Comparing a company's cost performance against industry standards
 - d. Examining the total cost of owning a product over its lifespan

10.8 SUMMARY

- ❑ Activity-Based Management (ABM) is a managerial approach that uses Activity-Based Costing (ABC) data to improve cost control, analyze customer profitability, and enhance cost management by focusing on value-added activities.
- ❑ ABM identifies cost drivers, helps manage activity-based performance indicators, and ensures accurate costing of products and services, aiding in strategic decision-making.
- ❑ Management accounting has shifted from activity-based methods like ABC to broader strategies focused on holistic process management, long-term value creation, and external market dynamics.
- ❑ In a competitive environment, management accounting helps businesses focus on customer satisfaction, cost efficiency, product differentiation, and strategic alignment with market trends for sustainable growth.
- ❑ With globalization, management accounting has evolved to emphasize strategic financial analysis, incorporating service sector challenges, sustainability, and social responsibility into decision-making.
- ❑ Management accounting supports strategy adjustments throughout the product life cycle, helping optimize costs, pricing, and production at each stage, from introduction to decline.

- ❑ Product diversity necessitates advanced cost allocation techniques like activity-based costing (ABC) and profitability analysis to manage different cost structures and pricing strategies across various products.
- ❑ Effective inventory management is enhanced by management accounting through optimized stock levels, improved cash flow, and key metrics like inventory turnover, supporting better purchasing and production decisions.
- ❑ Total Quality Management (TQM) principles in management accounting focus on continuous improvement, process optimization, and accurate financial reporting using methods like variance analysis and activity-based costing.
- ❑ Modern cost management systems, such as ABC, lean accounting, target costing, and digital transformation, provide businesses with tools to achieve cost efficiency, improve profitability, and streamline decision-making processes.

10.9 KEY WORDS

1. **Activity-Based Management (ABM):** A managerial approach that uses activity data to improve cost control and decision-making by focusing on value-added activities.
2. **Cost Drivers:** Factors that cause changes in the cost of activities, helping managers allocate resources more efficiently.
3. **Holistic Process Management:** Managing business processes as a whole, rather than focusing on isolated activities, to optimize efficiency and reduce bottlenecks.
4. **Strategic Decision-Making:** The process of making long-term decisions that align with a company's goals and market conditions to ensure sustainable growth.
5. **Service Profit Chain:** A concept linking employee satisfaction and productivity to customer loyalty and overall profitability in service-oriented businesses.
6. **Activity-Based Costing (ABC):** A cost allocation method that assigns overhead costs based on actual activities driving costs, improving cost accuracy.
7. **Inventory Turnover:** A ratio that measures how often a company sells and replaces its inventory over a period, indicating inventory management efficiency.
8. **Total Quality Management (TQM):** A management approach focused on continuous improvement of processes, quality, and customer satisfaction across an organization.
9. **Target Costing:** A pricing method where product cost is determined by market-driven prices, ensuring profitability from the design stage.
10. **Lean Accounting:** A cost management approach that simplifies financial reporting and focuses on value stream costing to support lean manufacturing principles.

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10.10 CLOSING CASELET**COCA-COLA'S ACTIVITY-BASED COSTING (ABC) IMPLEMENTATION**

Coca-Cola, one of the world's leading beverage companies, faced challenges in accurately determining the profitability of its diverse product portfolio due to limitations in its traditional costing system. The company relied on broad averages for overhead allocation, which skewed cost estimates and led to inefficient decision-making regarding pricing, product lines, and market strategies. As Coca-Cola expanded globally, the need for more precise cost information became crucial to maintain profitability and market competitiveness.

To address this, Coca-Cola implemented **Activity-Based Costing (ABC)**, which allowed the company to allocate overhead costs more accurately by focusing on specific activities like production, packaging, and distribution. By identifying key cost drivers such as machine hours, labor usage, and distribution distance, Coca-Cola could assign costs to products and services based on the actual resources consumed. This methodology provided a clearer picture of where inefficiencies were occurring.

The results were transformative. Coca-Cola gained improved visibility into its true cost structure, enabling more informed decisions about pricing and product discontinuation. The company identified high-cost activities, streamlined operations, and reduced waste, ultimately leading to significant cost savings. Additionally, ABC provided insights into customer profitability, allowing Coca-Cola to refine its market strategies and focus on high-margin products.

In conclusion, ABC gave Coca-Cola the tools to optimize resource allocation and enhance profitability. By understanding the true costs behind each activity, Coca-Cola improved its pricing strategies, reduced operational inefficiencies, and strengthened its competitive edge in the global market.

**QUESTIONS**

1. How did Coca-Cola's transition to Activity-Based Costing (ABC) improve their decision-making regarding product lines and market strategies?

(Hint: Focus on how ABC provided more accurate cost allocation and insights into the profitability of specific activities and products.)

2. What are the key benefits that Coca-Cola experienced by identifying cost drivers such as machine hours, labor, and distribution?

(Hint: Think about how understanding these drivers helped the company reduce inefficiencies and optimize resources for profitability.)

10.11 HIGHER ORDER THINKING SKILLS QUESTIONS (HOTS)

1. How can the shift from activity-based methods like ABC to holistic process management improve long-term organizational efficiency and competitiveness in a rapidly evolving market environment?
2. In what ways can the service profit chain model influence management accounting practices to ensure both employee engagement and customer satisfaction contribute to sustainable profitability in service industries?
3. How can management accounting strategies be adapted to respond to unexpected changes in the product life cycle, such as rapid technological advancements or sudden shifts in consumer demand?
4. In what ways can modern cost management systems, like activity-based costing (ABC) and lean accounting, enhance a company's ability to stay competitive in industries with fluctuating market conditions?
5. How might an organization's approach to inventory management impact its long-term financial sustainability, especially in industries with high variability in demand and product life cycles?

10.12 ANSWER KEY**SELF-ASSESSMENT QUESTIONS**

Topic	Q. No.	Answer
Activity as a Focus	1.	c. Identifying and managing activities, distinguishing between value-added and non-value-added activities
	2.	d. Enhancing communication, promoting informed decision-making, and continuous improvement
Management Accounting in a Competitive World	3.	b. Cost Leadership Strategy
	4.	b. By providing information about market trends, cost advantages, and strategic pricing
Management Accounting and Global Environment	5.	d. Taxation policies
	6.	b. They did not provide strategic financial information for global business environments
Management Accounting and Developed Costing Systems	7.	b. E-business enhances operational efficiency and value for customers without necessarily generating revenue.

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Topic	Q. No.	Answer
	8.	c. Management accounting practices have adapted to support competitive pricing, risk management, and strategic decision-making.
Modern Cost Management Systems	9.	b. Assigning overhead costs based on actual activities driving those costs
	10.	b. Determining a product's price based on market conditions and designing to meet that price while maintaining profitability

10.13 SUGGESTED READINGS

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